

GWTC-5 Compact Objects in CPTG

A 259-Event Compact-State, Core-Time, Geometric-Rotation, and Population Validation

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Abstract

This paper tests the registered Curvature Polarization Transport Gravity (CPTG) compact-state map across the 259-event black-hole-binary analysis set used by the GWTC-5.0 population study, extends the solved states to finite-core time and geometric frame-dragging readouts, and propagates the official selection-corrected `default_mmax` population posterior. Event luminosity-distance posteriors are inverted through the registered CPTG affine luminosity observation coordinate to recover the native π -branch phase; the population redshift coordinate is first decoded to luminosity distance with its documented GWTC convention and then inverted through the same map. The compact mass–spin handoff is applied component by component, with Structural Mode N_{BH} evaluated only after closure.

The finite continuation exposes interior quantities absent from the usual gravitational-wave catalog description. For every solved compact component, it determines the physical compact-source mass M_c , geometric mass-equivalent contribution M_{geom} , compact-source spin, and GR/CPTG matching radius r_m . The same geometry yields Structural Mode N_{BH} and organization shell r_N , together with the inner trapping surface r_- , local lapse field, and geometric frame-dragging field. These quantities distinguish the exterior handoff boundary, internal organization scale, and inner trapping structure rather than treating the compact object as a single unresolved radius.

All 259 events close with no hard rejection, yielding 288 retained release-label rows and 260 unique derived sample streams. Across retained component medians, the Structural- N clock spans 11.8526–19.9856 hours per 24 source-local exterior hours, while determinant-averaged Structural- N frame dragging spans 389.688–16330.834 rad s^{-1} ; these are geometric ZAMO rates, not material rotation. The population calculation propagates all 8200 hyperposterior rows with 512 realizations each. Median component-1 mass changes from 10.243 $M_\odot$ in the released source coordinate to 9.623 $M_\odot$ in the CPTG exterior frame and 6.507 $M_\odot$ for the physical compact source. Together, these calculations establish catalog-wide numerical closure.

1 Introduction

Black holes are usually inferred from the gravitational field they present to the outside world. Gravitational-wave observations constrain masses, spins, luminosity distance, and the dynamics of a binary through the waveform measured at the detector, while the compact interior itself remains hidden behind an exterior relativistic description. Curvature Polarization Transport Gravity (CPTG) changes what can be asked of the same observations. Once an exterior mass–spin state is reconstructed in the CPTG native coordinate, the registered compact relation continues that state into a finite interior and separates quantities that are normally compressed into a single exterior mass parameter. This makes it possible to use a gravitational-wave catalog not only as a population of exterior black-hole measurements, but also as a systematic probe of finite compact-state structure.

CPTG retains the General Relativistic gravitational sector and the ordinary Kerr or Schwarzschild exterior wherever the additional CPTG response is negligible [1, 2]. Ordinary physical stress–energy remains the material source. The difference appears in the compact continuation: curvature polarization and curvature transport permit the conserved exterior gravitational charge to be resolved into a localized physical compact-source contribution and a completed geometric mass-equivalent response. In the minimum conditional compact realization used in this paper,

$$M_\infty = M_c + M_{\text{geom}}, \quad (1)$$

so the mass measured through the exterior field need not be identified one-to-one with the physical compact-source mass. Angular-momentum conservation then links the exterior and compact-source spin states and fixes the finite matching radius at which the CPTG compact interior joins the retained Kerr exterior [3].

That finite continuation exposes a set of interior quantities that are not part of the usual gravitational-wave catalog description. For every solved compact component, the calculation can determine the physical compact-source mass M_c , the geometric mass-equivalent contribution M_{geom} , the compact-source spin parameter, and the GR/CPTG matching radius r_m . The same solved geometry supports the post-solution Structural Mode N_{BH} , which measures how the compact source–polarization–transport organization is distributed through the finite interior and locates its organization shell r_N [4, 3]. It also fixes an inner trapping surface r_- , a local lapse field, and the geometric frame-dragging field. These quantities distinguish the exterior handoff boundary, the internal organization scale, and the inner trapping structure rather than treating the black hole as a single unresolved radius.

The clock and rotation sectors provide two especially direct views of the finite core. The locally nonrotating lapse gives the proper time accumulated on an interior shell relative to source-local asymptotic exterior Kerr time. We therefore report, for each component, how much proper time accumulates at the Structural- N shell during a fixed 24-hour exterior interval and how that clock changes as the inner trapping surface is approached. Independently, the shift of the same rotating geometry fixes the local zero-angular-momentum-frame angular velocity. The paper resolves that geometric frame dragging on the axis, at the equator, through the determinant-measure angular average, and in the trapping-surface limit. These are geometric rotation readouts rather than an assumed material or fluid rotation law.

The result is a compact-object description with several physically distinct layers. Outside r_m , the retained Kerr exterior supplies the ordinary far-field relativistic description. At r_m , the exterior gravitational charge and the finite CPTG compact state are matched. Within the finite compact state, N_{BH} and r_N describe the organization of the solved source–transport–polarization structure; the lapse measures local clock behavior; the frame-dragging operator measures internal geometric rotation; and r_- marks the inner trapping surface of the declared rotating realization. The purpose of the present work is not to assume these interior quantities independently. They are all evaluated downstream of one fixed compact mass–spin closure, so the same posterior draw must satisfy the mass accounting, structural, clock, and rotation descriptions simultaneously.

The observational laboratory for this test is the black-hole-binary sample used by the GWTC-5.0 population analysis. That study defines a 259-event BBH-only set with false-alarm rate no greater than 1 yr^{-1} and both released component source masses at or above $3 M_\odot$ at 99% credibility under the default parameter-estimation priors [8]. The present analysis uses the released parameter-estimation products themselves as the event-level authority. Across the 259 events, the frozen source registry contains 288 retained release-label realizations. Numerical identity auditing reduces these to 260 unique within-event derived sample streams because 28 paired release labels are exact numerical duplicates, while the two retained reconstructions of GW231123_135430 remain genuinely distinct. The complete finite-core calculation represents 7,626,239 posterior draws.

The observational inputs are kept separate from quantities that were derived under an external cosmological convention. For individual events, the measured luminosity-distance posterior, detector-frame component masses, and exterior spin magnitudes are read from the parameter-estimation release. The observed luminosity distance is held fixed and inverted through the registered CPTG affine luminosity observation coordinate to recover the native π -branch phase before source-frame masses are formed [5]. Released source-frame masses and redshift are retained only as consistency coordinates

for the event-level audit; they are not substituted for the CPTG-native reconstruction. For the official GWTC-5.0 selection-corrected population posterior, the released redshift is first decoded back to luminosity distance with its documented reporting convention, and that recovered distance is then passed through the same CPTG observation map [9]. This prevents an externally encoded source coordinate from being treated as though it were a native CPTG observable.

The scale of the dataset allows the compact construction to be studied object by object rather than only through catalog averages. The revised presentation therefore treats each event as an individual compact system. Candidates are organized first by observational properties that exist before the CPTG solve, while the quantities revealed by CPTG—mass accounting, Structural Mode, interior clock behavior, geometric rotation, and the GR/CPTG boundary—are reported as results rather than used to define the classes. Each candidate discussion will retain the full component accounting and uncertainty support, with distinct waveform reconstructions kept separate whenever they produce distinct physical results.

This organization serves two purposes. First, it tests whether one registered CPTG compact-state interface remains numerically closed across the complete declared GWTC-5.0 BBH set without object-specific retuning. Second, it turns the catalog into a comparative map of finite black-hole interiors. Systems that appear similar in exterior mass can be compared in physical-source mass, geometric contribution, Structural Mode, clock rate, organization-shell radius, trapping radius, and frame dragging; systems that differ strongly in mass or spin can be examined for the way those exterior differences propagate into the finite core. The resulting paper is therefore both a catalog-wide validation of the compact construction and a systematic study of the interior quantities that the construction makes accessible.

The present calculation extends the CPTG compact-state law from individual benchmark black holes to the complete declared 259-event GWTC-5.0 BBH analysis set, resolving 520 component states across 7,626,239 posterior draws with one fixed compact construction and no object-specific retuning.

The remainder of the paper proceeds from observation to interior structure. The methodology first defines the released data authority and coordinate reconstruction, then the compact mass–spin closure, finite-core geometry, Structural Mode, clock extraction, geometric-rotation extraction, and validation rules. The candidate results are then presented by observational class, with a dedicated subsection for each event and branch-sensitive comparison where required. The later synthesis sections compare mass and mass-equivalent accounting, Structural Mode, interior time, geometric rotation, and the GR/CPTG matching structure across the catalog before the selection-corrected population result is revisited at the statistical level.

2 Methodology

The calculation is organized as an extraction chain. Each retained posterior draw begins as a released observational realization and is carried through the same sequence without object-specific retuning: observational authority, CPTG native-coordinate reconstruction, compact mass–spin closure, finite-core geometry, post-solution Structural Mode, interior clock extraction, and geometric frame-dragging extraction. The population calculation follows the same coordinate and compact-state map but remains a parallel statistical push-forward rather than consuming the event-level Structural-Mode, clock, or rotation outputs.

The parent strong-field organization is

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (2)$$

where ordinary physical stress–energy remains the material source [1]. The compact construction retains the General Relativistic Kerr or Schwarzschild exterior where the additional CPTG response is negligible and requires the finite interior to match that exterior at the registered compact boundary [2, 3]. The operational order used throughout this paper is

$$\begin{aligned} &\text{released observational draw} \longrightarrow \text{fixed } D_L \longrightarrow z_\pi \\ &\quad \Downarrow \\ &(M_\infty, \hat{a}_\infty) \longrightarrow \text{compact mass–spin closure} \longrightarrow (M_c, M_{\text{geom}}, \hat{a}_c) \\ &\quad \Downarrow \\ &N_{\text{BH}}, r_N, r_- \longrightarrow \text{clock} + \text{geometric rotation}, \\ &\text{population push-forward} \longrightarrow \text{population diagnostics}. \end{aligned} \quad (3)$$

No Structural-Mode, clock, rotation, or population diagnostic is allowed to flow backward into the compact solve.

2.1 Observational authority and candidate extraction

The event-level authority is the 259-event BBH-only analysis set defined by the GWTC-5.0 population study [8]. Its released selection is

$$\text{FAR} \leq 1 \text{ yr}^{-1}, \quad P(m_1 \geq 3M_\odot) \geq 0.99, \quad P(m_2 \geq 3M_\odot) \geq 0.99, \quad (4)$$

under the default parameter-estimation priors. This criterion defines the observational sample only; CPTG physical compact-source masses are solved later and are not used to select the catalog.

For each event, the parameter-estimation HDF5 product is treated as the source authority [6]. The frozen registry spans the O1/O2/O3a, O3b, O4a, and O4b releases and records the release identifier, official MD5, locally verified SHA-256, retained posterior labels, and finite sample counts [7]. The event solve reads luminosity distance, detector-frame component masses, and exterior spin magnitudes from the retained posterior samples. Released source-frame mass and redshift fields are also required for the detector/source identity audit, but they are not substituted for the CPTG-native coordinate reconstruction.

Released **Mixed** posterior labels are retained when available. If no mixed label exists, the best available IMRPhenomXPHM-family branch with the required fields is used. The retained **C00:IMRPhenomXPHM-SpinTaylor** branch is abbreviated **XPHM-ST**. Release labels remain separate for provenance until their transformed numerical identities are tested. The frozen catalog contains 288 retained release-label realizations: 230 events have one retained label and 29 have two. Twenty-eight paired-label cases are exact numerical duplicates under the declared transformation; GW231123_135430 retains two distinct numerical streams. Candidate-level descriptive statistics therefore use 260 unique within-event numerical streams without averaging independent reconstructions.

A branch is rejected only for a genuine data or coordinate failure: no finite posterior samples, nonphysical mass or distance support, spin magnitude outside $[0, 1]$, detector/source identity failure, or failure of the compact closure. Posterior width is retained as measurement uncertainty rather than converted into a generic rejection rule.

The official selection-corrected **default_mmax** population posterior is a separate observational authority [9]. It contains 8200 hyperposterior rows. Its released redshift coordinate is treated as a model-dependent reporting coordinate and is decoded before the CPTG population transformation described below.

2.2 Native π -branch coordinate reconstruction

The first transformation reconstructs the CPTG native phase while keeping the observational distance fixed. The geometric π branch contains one native cosmological history [5]. Define

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad (5)$$

with normalized shape

$$E_\pi(z)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi(1+z)^\pi, \quad (6)$$

and native normalization

$$H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (7)$$

The registered affine luminosity normalization is

$$H_{0,L}^{\text{aff}} = H_0^{(\pi)} \sqrt{\frac{10}{9}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (8)$$

with $E_L^{\text{aff}}(z) = E_\pi(z)$. The native and luminosity distances at the same native phase are

$$D_{L,\pi}^{\text{CPTG}}(z) = (1+z) \frac{c}{H_0^{(\pi)}} \int_0^z \frac{dz'}{E_\pi(z')}, \quad (9)$$

$$D_{L,L}^{\text{CPTG}}(z) = (1+z) \frac{c}{H_{0,L}^{\text{aff}}} \int_0^z \frac{dz'}{E_\pi(z')}. \quad (10)$$

For an individual event draw, the released luminosity distance is held fixed and Eq. (10) is inverted for z_π . The detector-frame component masses then give the CPTG exterior source-frame masses,

$$M_{\infty,i} = \frac{M_{\text{det},i}}{1 + z_\pi}. \quad (11)$$

The population product begins from a released redshift rather than an event-level D_L posterior. That coordinate is first decoded using the GWTC reference convention carried by the released product,

$$E_{\text{GWTC}}(z)^2 = 0.3065(1 + z)^3 + 0.6935, \quad (12)$$

$$D_{L,\text{GWTC}}^{\text{dec}}(z) = (1 + z) \frac{c}{67.9 \text{ km s}^{-1} \text{ Mpc}^{-1}} \int_0^z \frac{dz'}{E_{\text{GWTC}}(z')}. \quad (13)$$

The recovered luminosity distance is then held fixed and inverted through Eq. (10). The population mass-coordinate factor is therefore

$$\frac{M_\infty}{M_{\text{source,GWTC}}} = \frac{1 + z_{\text{GWTC}}}{1 + z_\pi}. \quad (14)$$

The external 67.9 normalization and $\Omega_m = 0.3065$ appear only in the decoder of the released GWTC coordinate. They are not CPTG-native inputs. After the physical CPTG solve, the same native phase can be projected back into a GWTC-style comparison coordinate,

$$H_{0,\text{GWTC}}^{\text{eq},\pi}(z_\pi) = 67.9 \frac{D_{L,\text{GWTC}}(z_\pi; 67.9, 0.3065)}{D_{L,\pi}^{\text{CPTG}}(z_\pi)}, \quad (15)$$

which is used only as a downstream reporting coordinate.

2.3 Compact mass–spin closure and component identity

The second extraction solves the compact mass accounting and compact-source spin for each binary component. The posterior-inferred exterior spin magnitude $\hat{a}_{\infty,i}$ fixes the compact polarization factor through

$$\hat{a}_{\infty,i}^2 A_{\text{pol},i}^3 + A_{\text{pol},i} - 2 = 0, \quad (16)$$

with physical root

$$1 \leq A_{\text{pol},i} \leq 2. \quad (17)$$

The exterior gravitational charge is then decomposed as

$$M_{c,i} = \frac{M_{\infty,i}}{A_{\text{pol},i}}, \quad M_{\text{geom},i} = M_{\infty,i} - M_{c,i}, \quad (18)$$

and angular-momentum conservation gives

$$\hat{a}_{c,i} = A_{\text{pol},i}^2 \hat{a}_{\infty,i}. \quad (19)$$

The minimum matching radius between the finite CPTG compact state and the retained Kerr exterior is

$$r_{m,i} = \frac{GM_{\infty,i}}{c^2} = \frac{GM_{c,i}}{c^2} \left(1 + \sqrt{1 - \hat{a}_{c,i}^2} \right). \quad (20)$$

Thus the exterior mass parameter remains the conserved exterior gravitational charge, while CPTG resolves it into physical compact-source and completed geometric-response contributions.

Both raw source states are solved before component labels are changed. The physical compact-source masses and every attached quantity are then reordered together so that CPTG component 1 is the component with larger M_c . This prevents a change in component order from breaking the association between mass, exterior spin, compact spin, Structural Mode, clock, and rotation.

The event-level numerical closure includes the compact cubic residual, the released detector/source identity audit, and the sample-wise mass identity

$$M_{\infty,i} = M_{c,i} + M_{\text{geom},i}. \quad (21)$$

Released source-frame masses and redshifts enter only the identity audit and never replace the native CPTG quantities above.

2.4 Finite-core geometry and the GR/CPTG matching structure

The third extraction builds the minimum Kerr-compatible finite-core realization from the already solved exterior state. Write

$$F(x) = \frac{1}{8} (63x^5 - 90x^7 + 35x^9), \quad x = \frac{r}{r_g}, \quad r_g = \frac{GM_{\infty}}{c^2}, \quad (22)$$

with $q = |\hat{a}_{\infty}|$. Because Eq. (20) gives $r_m = r_g$, the outer edge of this normalized finite interior is $x = 1$. The GR/CPTG handoff radius r_m and the inner trapping radius introduced below are therefore different surfaces and are kept distinct throughout the analysis.

Define

$$\delta(x, q) = x^2 - 2xF(x) + q^2, \quad (23)$$

$$\sigma(x, \theta, q) = x^2 + q^2 \cos^2 \theta, \quad (24)$$

$$\mathcal{A}_d(x, \theta, q) = (x^2 + q^2)^2 - q^2 \delta(x, q) \sin^2 \theta. \quad (25)$$

The inner CPTG trapping surface is the physical root $0 < x_- < 1$ of

$$\delta(x_-, q) = 0, \quad (26)$$

or, with $y = x^2$,

$$35y^5 - 90y^4 + 63y^3 - 4y - 4q^2 = 0. \quad (27)$$

Its physical radius is

$$r_-^{\text{CPTG}} = x_- r_g. \quad (28)$$

The calculation does not identify r_- with the GR/CPTG matching radius. The former is an interior trapping surface of the declared finite rotating realization; the latter is the outer boundary at which the finite CPTG compact state joins the retained exterior Kerr description.

2.5 Structural Mode and the organization shell

The fourth extraction measures the organization of the already solved compact state. Structural Mode is a downstream fingerprint, not a fit parameter, branch selector, or input to the compact mass closure [4]. For the registered Kerr-compatible equal-measure construction used here [3],

$$N_{\text{BH}}(\hat{a}_{\infty}) = \frac{\int_0^1 w(x) dx}{\int_0^1 \xi(x, \hat{a}_{\infty}) w(x) dx}, \quad (29)$$

with rotating structural coordinate

$$\xi(x, \hat{a}_{\infty}) = \left[\frac{x(x^2 + \hat{a}_{\infty}^2)}{1 + \hat{a}_{\infty}^2} \right]^{1/3}. \quad (30)$$

The compact structural weight is constructed from the same finite radial profile,

$$U(x) = \frac{1}{8} (63x^3 - 90x^5 + 35x^7), \quad (31)$$

$$\mathcal{C}(x) = U(x)^2 + U'(x)^2 + 2 \left[\frac{U(x)}{x} \right]^2, \quad w(x) = |\mathcal{C}'(x)|. \quad (32)$$

Mass, r_m , and c cancel from Eq. (29). The spherical limit is $N_{\text{BH}}(0) = 1.912233171442$, and the registered mode decreases monotonically with exterior spin magnitude [3]. Numerically, the absolute-value integral is split at the three internal roots of $\mathcal{C}'(x)$,

$$x = 0.565199642579, \quad 0.832620276257, \quad 0.955199547280, \quad (33)$$

and each smooth interval is evaluated independently with 64-point Gauss–Legendre quadrature before summation.

The Structural-Mode organization shell is defined from the solved N_{BH} by

$$x_N (x_N^2 + q^2) = \frac{1 + q^2}{N_{\text{BH}}^3}, \quad r_N = x_N r_g. \quad (34)$$

The relative ordering of r_N , r_- , and r_m is checked on the posterior draws rather than imposed as a classification rule.

2.6 Interior clock extraction

The fifth extraction measures proper-time accumulation inside the finite compact state relative to source-local asymptotic exterior Kerr time. The locally nonrotating, or ZAMO, lapse is

$$\alpha(x, \theta; q) = \sqrt{\frac{\delta \sigma}{\mathcal{A}_d}}, \quad (35)$$

and the rotating shell readout uses the determinant-measure angular average

$$\bar{\alpha}(x; q) = \frac{\int_0^\pi \alpha(x, \theta; q) \sigma(x, \theta, q) \sin \theta \, d\theta}{\int_0^\pi \sigma(x, \theta, q) \sin \theta \, d\theta}. \quad (36)$$

For a 24-hour interval of source-local asymptotic exterior Kerr coordinate time,

$$T_{24} = 24 \bar{\alpha} \text{ h}. \quad (37)$$

Two clock profiles are retained for every compact component. The first is the Structural-Mode shell clock,

$$T_{24,N} = 24 \bar{\alpha}(x_N; q) \text{ h}. \quad (38)$$

The second samples the same lapse at Boyer–Lindquist radial-coordinate offsets of 1 km, 500 m, 100 m, 10 m, 1 m, 10 cm, and 1 cm inward from r_-^{CPTG} , followed by the trapping-surface limit $\bar{\alpha} \rightarrow 0$. These offsets are radial-coordinate offsets, not proper distances. No whole-core average, compact-to-exterior mass ratio, or geometric mass fraction is substituted for the lapse.

Every clock value is calculated draw by draw before posterior quantiles are formed, so the clock uncertainty remains attached to the same mass, spin, trapping-radius, and Structural-Mode realization. The clock result is specific to the minimum rotating compact realization used here; it is not asserted to be a uniquely action-derived rotating interior.

2.7 Internal geometric rotation extraction

The sixth extraction uses the same finite geometry to determine the angular velocity of the local zero-angular-momentum frame. The magnitude of the geometric frame-dragging rate is

$$\omega(x, \theta; q) = \frac{c}{r_g} \frac{2q x F(x)}{\mathcal{A}_d(x, \theta, q)}. \quad (39)$$

The retained GWTC inputs provide spin magnitudes, so the calculation reports the magnitude of ω and assigns no rotation sign. At each declared shell, three angular readouts are kept: the axis, the equator, and the determinant-measure average

$$\bar{\omega}(x; q) = \frac{\int_0^\pi \omega(x, \theta; q) \sigma(x, \theta, q) \sin \theta d\theta}{\int_0^\pi \sigma(x, \theta, q) \sin \theta d\theta}. \quad (40)$$

The associated coordinate-time period is

$$P_\omega = \frac{2\pi}{\omega}. \quad (41)$$

In the inner trapping-surface limit, $\delta(x_-, q) \rightarrow 0$ and the angular dependence collapses to

$$\Omega_- = \frac{c}{r_g} \frac{q}{x_-^2 + q^2}. \quad (42)$$

The rotation operator is evaluated at r_N and at the same near-boundary offsets used for the clock. Periods are calculated sample by sample before posterior quantiles are taken. The angular integral is evaluated in closed form in the implementation and independently checked against high-order Gauss–Legendre quadrature. Validation requires the ordering

$$\omega_{\text{axis}} \leq \bar{\omega} \leq \omega_{\text{equator}} \quad (43)$$

away from the trapping-surface limit, angle independence in that limit, finite rates on every retained draw, and the pointwise period identity. No radial-monotonicity condition is imposed.

These quantities measure geometric frame dragging. They do not determine a material angular velocity, rigid-body rotation law, differential fluid rotation, tangential matter speed, or equation of state.

2.8 Selection-corrected population push-forward

The seventh extraction applies the same coordinate and compact-state map statistically to the official selection-corrected population posterior. Each of the 8200 hyperposterior rows is propagated with 512 joint realizations. Primary mass and redshift are sampled by deterministic stratified inverse CDF from the released row grids. The conditional secondary-mass law is

$$p(m_2 \mid m_1) \propto m_2^\beta S(m_2 \mid m_{\text{low}}, \Delta_m), \quad (44)$$

where S is the released Planck taper. The conditional law is sampled by direct inverse CDF using a tabulated taper integral and analytic power-law tail, while spin magnitudes are drawn from the released truncated-Gaussian law.

Each released population redshift draw is decoded to $D_{L, \text{GWTC}}^{\text{dec}}$ with Eq. (13). That distance is held fixed and inverted through Eq. (10) to obtain z_π . The mass-coordinate factor of Eq. (14) is then applied before the same compact handoff used on the event side. The equivalent-standard coordinate of Eq. (15) is evaluated only after the physical push-forward.

With 512 draws per hyperposterior row, one realization corresponds to

$$\frac{1}{512} = 0.1953125\%. \quad (45)$$

Rare-tail fractions are therefore reported as the mean row fraction together with the 5th–95th percentile range across the 8200 rows. A zero sampled row median does not erase finite-draw tail information.

2.9 Posterior uncertainty, diagnostics, and numerical closure

All physical quantities are evaluated on posterior draws before medians or 5th–95th percentile intervals are reported. Broad posterior support is retained as uncertainty. A separate resolved/unresolved label is used only when a mass-anchor diagnostic requires a component to lie clearly on one side of a threshold M_t . The resolved-above criterion is

$$P(M_c \geq M_t) \geq 0.95, \quad (46)$$

and the resolved-below criterion is

$$P(M_c \geq M_t) \leq 0.05. \quad (47)$$

The 10, 35, 40, and 45 $M_\odot$ anchors are therefore downstream readouts only and never enter the solve.

The full event-level validation checks source identity, finite sample support, detector/source identity, native-coordinate inversion, compact cubic closure, mass decomposition, component attachment after reordering, inner trapping-root closure, Structural-Mode quadrature and shell closure, sample-wise clock identity, angular ordering, trapping-surface angular collapse, and sample-wise $P_\omega = 2\pi/\omega$. The complete calculation retains 7,626,239 posterior draws over the 288 release-label realizations. Numerical identity auditing is then used only to distinguish exact duplicate release labels from genuinely distinct within-event reconstructions.

Element	Status in this paper	Role
Observed D_L , detector-frame masses, exterior spins	observational input	fixed quantities taken directly from the released posterior samples
GWTC redshift decoder	external-coordinate decoder	recovers the luminosity distance encoded in the released population coordinate
CPTG affine luminosity observation map	registered comparison coordinate	maps fixed observed or decoded D_L to native z_π
Compact mass–spin handoff	conditional compact closure	separates physical compact source and completed geometric response
GR/CPTG matching radius r_m	compact boundary	outer handoff between the finite compact state and retained Kerr exterior
Structural Mode N_{BH}	registered post-solution readout	measures solved interior organization; never used to fit or classify
Inner trapping surface r_-	finite-core geometric readout	interior trapping boundary of the declared rotating realization
Finite-core ZAMO clock	minimum rotating compact realization	local proper-time accumulation relative to source-local asymptotic Kerr time
Geometric ZAMO rotation	minimum rotating compact realization	local frame dragging; not a material rotation law
10, 35, 40, and 45 $M_\odot$ anchors	downstream diagnostics	resolved population structure after the CPTG map

Table 1: Authority and role of the principal quantities used by the GWTC-5 compact-state analysis.

This sequence fixes the extraction rules before any candidate result is interpreted. The next section applies the same procedure to the complete event set.

3 M1: Lower Observed-Mass Black-Hole Binaries

The first candidate class contains 82 unique event-level numerical streams with detector-frame total masses constructed from released component marginal medians spanning 13.911–55.527 $M_\odot$. The class boundary was fixed from the observational distribution before any CPTG output was examined. Within M1, 10 systems fall in the strongly asymmetric mass-ratio cross-cut, 43 are moderately asymmetric, and 29 are near-equal in the same observational screen. None of the M1 systems carries the predeclared broad-posterior flag. Candidate subsections are ordered by detector-frame total mass from lower to higher values; neither Structural Mode nor any compact-state, clock, or rotation quantity controls that order.

Each subsection reports the observational state first and then the complete headline CPTG compact-state readout for both post-handoff components. The radial profile table preserves every registered near-inner-boundary location while keeping the full posterior support for those profile points in the evidence archive. This format makes the physical compact-source mass, completed geometric mass-equivalent contribution, GR/CPTG matching radius, Structural-Mode organization shell, inner trapping surface, proper-time accumulation, and geometric frame dragging visible for each binary individually.

3.1 GW241109_115924

GW241109_115924 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 8.142 M_\odot$ and $m_2 = 5.769 M_\odot$, while the luminosity-distance median is 602.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,582
Observed luminosity distance	602.2 Mpc (median)
Detector-frame masses	$m_1 = 8.142 M_\odot$, $m_2 = 5.769 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.709$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1247$; $z_\pi = 0.1342$ (medians)
Post-handoff reorder fraction	15.81%

Table 2: Observational and coordinate summary for GW241109_115924. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	7.135 [5.403, 10.004]	5.156 [3.848, 6.767]
Physical compact-source mass $M_c (M_\odot)$	4.532 [3.376, 6.776]	3.193 [2.331, 4.463]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	2.871 [0.433, 4.298]	2.057 [0.352, 3.164]
Geometric share M_{geom}/M_∞ (%)	38.45 [7.26, 49.86]	40.52 [8.02, 49.92]
Exterior spin magnitude $ \hat{a}_\infty $	0.296 [0.026, 0.857]	0.259 [0.020, 0.843]
Polarization factor A_{pol}	1.625 [1.078, 1.995]	1.681 [1.087, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.781 [0.104, 0.997]	0.732 [0.081, 0.996]
GR/CPTG matching radius r_m (km)	10.536 [7.979, 14.773]	7.614 [5.682, 9.992]
Structural Mode N_{BH}	1.7752 [1.4883, 1.9107]	1.8000 [1.4929, 1.9113]
Organization-shell radius r_N (km)	5.549 [4.205, 7.779]	4.012 [2.983, 5.259]
Inner trapping-surface radius r_- (km)	6.724 [5.356, 9.441]	4.827 [3.670, 6.287]
Structural- N lapse $\bar{\alpha}_N$	0.5959 [0.4842, 0.8206]	0.5755 [0.4837, 0.8169]
Structural- N clock $T_{24,N}$ (h)	14.300 [11.621, 19.695]	13.812 [11.610, 19.606]
Structural- N axis ω (rad s $^{-1}$)	9549.22 [1958.86, 15414.44]	13497.05 [2068.79, 21826.72]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	11006.96 [1959.72, 17155.00]	15550.66 [2069.19, 24195.27]
Structural- N equatorial ω (rad s $^{-1}$)	12748.53 [1960.15, 18969.08]	17928.19 [2069.39, 26551.84]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.5708 [0.3663, 3.2062]	0.4040 [0.2597, 3.0365]
Trapping-surface limit Ω_- (rad s $^{-1}$)	15631.82 [2135.85, 24785.96]	21918.88 [2254.13, 34252.61]
Trapping-surface period P_- (ms)	0.4019 [0.2535, 2.9418]	0.2867 [0.1834, 2.7874]

Table 3: CPTG compact-state, structural, clock, and geometric-rotation results for GW241109_115924. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	14.3054	16.7903	13101.09	16296.38
500 m inward	10.2465	12.1904	14499.95	19600.68
100 m inward	4.6055	5.5176	15428.18	21551.76
10 m inward	1.4570	1.7474	15612.44	21877.20
1 m inward	0.4607	0.5526	15630.04	21914.66
10 cm inward	0.1457	0.1748	15631.65	21918.46
1 cm inward	0.0461	0.0553	15631.81	21918.84
surface limit	0.0000	0.0000	15631.82	21918.88

Table 4: Near-inner-boundary profile for GW241109_115924. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $4.532 M_\odot$ and $3.193 M_\odot$, with completed geometric mass-equivalent shares of 38.45% and 40.52%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $5.549 < 6.724 < 10.536$ km, and component 2 gives $4.012 < 4.827 < 7.614$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7752$ and 1.8000; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.300 and 13.812 h. The determinant-averaged frame-dragging rates at those shells are 11006.96 and 15550.66 rad s $^{-1}$, approaching surface-limit values of 15631.82 and 21918.88 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 14.305/16.790 h at 1 km inward to 0.0461/0.0553 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.2 GW240413_022019

GW240413_022019 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 8.811 M_\odot$ and $m_2 = 6.207 M_\odot$, while the luminosity-distance median is 572.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,179
Observed luminosity distance	572.0 Mpc (median)
Detector-frame masses	$m_1 = 8.811 M_\odot$, $m_2 = 6.207 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.704$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1189$; $z_\pi = 0.1280$ (medians)
Post-handoff reorder fraction	15.63%

Table 5: Observational and coordinate summary for GW240413_022019. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	7.783 [5.852, 10.639]	5.578 [4.186, 7.381]
Physical compact-source mass M_c ($M_\odot$)	4.938 [3.668, 7.649]	3.516 [2.558, 4.835]
Geometric mass-equivalent M_{geom} ($M_\odot$)	2.969 [0.565, 4.409]	2.167 [0.346, 3.475]
Geometric share M_{geom}/M_∞ (%)	36.97 [8.66, 49.65]	39.51 [7.43, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.322 [0.042, 0.830]	0.277 [0.026, 0.854]
Polarization factor A_{pol}	1.587 [1.095, 1.986]	1.653 [1.080, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.810 [0.167, 0.995]	0.757 [0.105, 0.997]
GR/CPTG matching radius r_m (km)	11.493 [8.641, 15.709]	8.236 [6.182, 10.899]
Structural Mode N_{BH}	1.7576 [1.4970, 1.9082]	1.7878 [1.4893, 1.9107]
Organization-shell radius r_N (km)	6.054 [4.549, 8.293]	4.341 [3.247, 5.731]
Inner trapping-surface radius r_- (km)	7.311 [5.804, 10.521]	5.280 [4.016, 6.828]
Structural- N lapse $\bar{\alpha}_N$	0.6101 [0.4861, 0.8137]	0.5855 [0.4842, 0.8198]
Structural- N clock $T_{24,N}$ (h)	14.643 [11.667, 19.528]	14.052 [11.622, 19.675]
Structural- N axis ω (rad s $^{-1}$)	9602.02 [2895.34, 14417.53]	12615.31 [2364.74, 19891.78]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	10847.05 [2974.49, 15961.79]	14568.11 [2365.84, 22044.68]
Structural- N equatorial ω (rad s $^{-1}$)	12246.03 [2975.61, 17638.23]	16947.61 [2366.39, 24265.57]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.5793 [0.3936, 2.1124]	0.4313 [0.2850, 2.6558]
Trapping-surface limit Ω_- (rad s $^{-1}$)	15047.42 [3243.99, 23003.77]	21058.80 [2578.60, 31307.00]
Trapping-surface period P_- (ms)	0.4176 [0.2731, 1.9369]	0.2984 [0.2007, 2.4367]

Table 6: CPTG compact-state, structural, clock, and geometric-rotation results for GW240413_022019. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	13.7646	16.0978	12816.82	16168.05
500 m inward	9.8372	11.6379	14055.30	19030.90
100 m inward	4.4146	5.2546	14873.16	20724.79
10 m inward	1.3967	1.6634	15030.56	21029.58
1 m inward	0.4417	0.5261	15045.94	21057.57
10 cm inward	0.1397	0.1664	15047.27	21058.67
1 cm inward	0.0442	0.0526	15047.40	21058.78
surface limit	0.0000	0.0000	15047.42	21058.80

Table 7: Near-inner-boundary profile for GW240413_022019. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $4.938 M_\odot$ and $3.516 M_\odot$, with completed geometric mass-equivalent shares of 36.97% and 39.51%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $6.054 < 7.311 < 11.493$ km, and component 2 gives $4.341 < 5.280 < 8.236$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7576$ and 1.7878 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.643 and 14.052 h. The determinant-averaged frame-dragging rates at those shells are 10847.05 and 14568.11 rad s $^{-1}$, approaching surface-limit values of 15047.42 and 21058.80 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 13.765/16.098 h at 1 km inward to 0.0442/0.0526 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.3 GW230627_015337

GW230627_015337 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 8.964 M_\odot$ and $m_2 = 6.095 M_\odot$, while the luminosity-distance median is 305.4 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	32,382
Observed luminosity distance	305.4 Mpc (median)
Detector-frame masses	$m_1 = 8.964 M_\odot$, $m_2 = 6.095 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.680$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0658$; $z_\pi = 0.0709$ (medians)
Post-handoff reorder fraction	8.75%

Table 8: Observational and coordinate summary for GW230627_015337. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	8.376 [6.453, 10.094]	5.741 [4.802, 7.373]
Physical compact-source mass $M_c (M_\odot)$	4.630 [3.768, 5.986]	3.376 [2.701, 4.540]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.855 [1.627, 4.699]	2.487 [0.909, 3.473]
Geometric share M_{geom}/M_∞ (%)	46.23 [23.93, 49.96]	43.82 [17.61, 49.93]
Exterior spin magnitude $ \hat{a}_\infty $	0.148 [0.014, 0.549]	0.198 [0.018, 0.663]
Polarization factor A_{pol}	1.860 [1.315, 1.998]	1.780 [1.214, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.511 [0.058, 0.949]	0.626 [0.073, 0.977]
GR/CPTG matching radius r_m (km)	12.369 [9.528, 14.905]	8.478 [7.091, 10.887]
Structural Mode N_{BH}	1.8686 [1.6171, 1.9118]	1.8397 [1.5615, 1.9115]
Organization-shell radius r_N (km)	6.501 [5.038, 7.843]	4.463 [3.737, 5.725]
Inner trapping-surface radius r_- (km)	7.342 [6.081, 8.943]	5.221 [4.373, 6.603]
Structural- N lapse $\bar{\alpha}_N$	0.5183 [0.4834, 0.7201]	0.5424 [0.4836, 0.7631]
Structural- N clock $T_{24,N}$ (h)	12.440 [11.602, 17.283]	13.018 [11.607, 18.315]
Structural- N axis ω (rad s $^{-1}$)	7948.69 [972.70, 13281.89]	12617.99 [1649.42, 18846.84]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8355.64 [972.80, 14743.21]	13983.12 [1649.70, 20604.27]
Structural- N equatorial ω (rad s $^{-1}$)	8477.73 [972.85, 16338.51]	15174.35 [1649.84, 22220.13]
Structural- N period P_ω (ms)	0.7520 [0.4262, 6.4589]	0.4493 [0.3049, 3.8087]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9431.88 [1059.55, 21093.88]	17297.60 [1796.94, 28047.86]
Trapping-surface period P_- (ms)	0.6662 [0.2979, 5.9300]	0.3632 [0.2240, 3.4966]

Table 9: CPTG compact-state, structural, clock, and geometric-rotation results for GW230627_015337. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	14.1227	16.3677	8072.34	13213.66
500 m inward	10.1642	11.8887	8812.28	15504.65
100 m inward	4.5918	5.3922	9317.85	16981.55
10 m inward	1.4547	1.7089	9421.14	17265.54
1 m inward	0.4601	0.5405	9430.90	17294.31
10 cm inward	0.1455	0.1709	9431.78	17297.27
1 cm inward	0.0460	0.0541	9431.87	17297.57
surface limit	0.0000	0.0000	9431.88	17297.60

Table 10: Near-inner-boundary profile for GW230627_015337. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $4.630 M_\odot$ and $3.376 M_\odot$, with completed geometric mass-equivalent shares of 46.23% and 43.82%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $6.501 < 7.342 < 12.369$ km, and component 2 gives $4.463 < 5.221 < 8.478$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8686$ and 1.8397 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 12.440 and 13.018 h. The determinant-averaged frame-dragging rates at those shells are 8355.64 and 13983.12 rad s $^{-1}$, approaching surface-limit values of 9431.88 and 17297.60 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 14.123/16.368 h at 1 km inward to 0.0460/0.0541 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.4 GW190924_021846

GW190924_021846 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 9.776 M_\odot$ and $m_2 = 5.673 M_\odot$, while the luminosity-distance median is 549.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	34,848
Observed luminosity distance	549.3 Mpc (median)
Detector-frame masses	$m_1 = 9.776 M_\odot$, $m_2 = 5.673 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.580$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1147$; $z_\pi = 0.1232$ (medians)
Post-handoff reorder fraction	6.55%

Table 11: Observational and coordinate summary for GW190924_021846. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	8.710 [6.283, 13.013]	5.074 [3.599, 6.859]
Physical compact-source mass $M_c (M_\odot)$	5.131 [3.838, 9.490]	3.203 [2.232, 4.549]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.618 [1.017, 4.942]	1.919 [0.362, 3.141]
Geometric share M_{geom}/M_∞ (%)	40.75 [13.46, 49.89]	38.73 [8.55, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.255 [0.024, 0.740]	0.291 [0.025, 0.833]
Polarization factor A_{pol}	1.688 [1.156, 1.996]	1.632 [1.094, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.726 [0.094, 0.988]	0.775 [0.101, 0.996]
GR/CPTG matching radius r_m (km)	12.861 [9.278, 19.215]	7.492 [5.315, 10.129]
Structural Mode N_{BH}	1.8028 [1.5296, 1.9110]	1.7785 [1.4963, 1.9108]
Organization-shell radius r_N (km)	6.767 [4.898, 10.157]	3.949 [2.798, 5.328]
Inner trapping-surface radius r_- (km)	7.865 [6.056, 13.003]	4.793 [3.439, 6.431]
Structural- N lapse $\bar{\alpha}_N$	0.5732 [0.4840, 0.7880]	0.5931 [0.4841, 0.8142]
Structural- N clock $T_{24,N}$ (h)	13.757 [11.616, 18.911]	14.235 [11.620, 19.542]
Structural- N axis ω (rad s $^{-1}$)	8133.99 [1543.95, 13313.98]	14238.28 [2555.19, 22445.34]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	9060.61 [1548.88, 14833.09]	16330.83 [2556.12, 25079.45]
Structural- N equatorial ω (rad s $^{-1}$)	9807.27 [1549.11, 16422.99]	18660.89 [2556.58, 27862.01]
Structural- N period P_ω (ms)	0.6935 [0.4236, 4.0566]	0.3847 [0.2505, 2.4581]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11645.57 [1687.58, 21278.10]	23276.01 [2786.67, 35535.53]
Trapping-surface period P_- (ms)	0.5395 [0.2953, 3.7232]	0.2699 [0.1768, 2.2547]

Table 12: CPTG compact-state, structural, clock, and geometric-rotation results for GW190924_021846. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	13.4955	16.7992	10205.58	17273.57
500 m inward	9.6585	12.1821	10991.02	20845.80
100 m inward	4.3443	5.5027	11524.08	22883.15
10 m inward	1.3750	1.7415	11633.45	23237.16
1 m inward	0.4349	0.5507	11644.51	23272.43
10 cm inward	0.1375	0.1742	11645.51	23275.70
1 cm inward	0.0435	0.0551	11645.57	23275.98
surface limit	0.0000	0.0000	11645.57	23276.01

Table 13: Near-inner-boundary profile for GW190924_021846. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $5.131 M_\odot$ and $3.203 M_\odot$, with completed geometric mass-equivalent shares of 40.75% and 38.73%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $6.767 < 7.865 < 12.861$ km, and component 2 gives $3.949 < 4.793 < 7.492$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8028$ and 1.7785 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.757 and 14.235 h. The determinant-averaged frame-dragging rates at those shells are 9060.61 and 16330.83 rad s $^{-1}$, approaching surface-limit values of 11645.57 and 23276.01 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 13.496/16.799 h at 1 km inward to 0.0435/0.0551 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.5 GW240925_005809

GW240925_005809 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 9.626 M_\odot$ and $m_2 = 7.471 M_\odot$, while the luminosity-distance median is 355.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	40,000
Observed luminosity distance	355.5 Mpc (median)
Detector-frame masses	$m_1 = 9.626 M_\odot$, $m_2 = 7.471 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.776$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0760$; $z_\pi = 0.0819$ (medians)
Post-handoff reorder fraction	14.07%

Table 14: Observational and coordinate summary for GW240925_005809. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	8.900 [7.167, 10.931]	7.021 [5.688, 8.684]
Physical compact-source mass $M_c (M_\odot)$	5.335 [4.211, 7.586]	4.111 [3.293, 5.766]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.744 [1.096, 4.881]	2.973 [0.820, 4.060]
Geometric share M_{geom}/M_∞ (%)	41.22 [13.98, 49.88]	43.04 [13.15, 49.93]
Exterior spin magnitude $ \hat{a}_\infty $	0.246 [0.025, 0.730]	0.213 [0.018, 0.746]
Polarization factor A_{pol}	1.701 [1.162, 1.995]	1.756 [1.151, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.713 [0.098, 0.987]	0.655 [0.073, 0.988]
GR/CPTG matching radius r_m (km)	13.142 [10.583, 16.141]	10.368 [8.399, 12.823]
Structural Mode N_{BH}	1.8084 [1.5333, 1.9109]	1.8304 [1.5273, 1.9115]
Organization-shell radius r_N (km)	6.921 [5.586, 8.511]	5.457 [4.423, 6.750]
Inner trapping-surface radius r_- (km)	8.111 [6.863, 10.441]	6.378 [5.290, 8.043]
Structural- N lapse $\bar{\alpha}_N$	0.5686 [0.4841, 0.7850]	0.5503 [0.4836, 0.7897]
Structural- N clock $T_{24,N}$ (h)	13.645 [11.618, 18.840]	13.206 [11.607, 18.953]
Structural- N axis ω (rad s $^{-1}$)	8342.17 [1549.02, 12069.21]	10212.24 [1405.98, 15479.52]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	9346.31 [1549.50, 13401.96]	11439.19 [1406.29, 16996.56]
Structural- N equatorial ω (rad s $^{-1}$)	10293.70 [1549.74, 14756.85]	12724.59 [1406.44, 18431.54]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.6723 [0.4688, 4.0550]	0.5493 [0.3697, 4.4679]
Trapping-surface limit Ω_- (rad s $^{-1}$)	12285.25 [1688.29, 19040.22]	14950.97 [1531.95, 23446.94]
Trapping-surface period P_- (ms)	0.5114 [0.3300, 3.7216]	0.4203 [0.2680, 4.1014]

Table 15: CPTG compact-state, structural, clock, and geometric-rotation results for GW240925_005809. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	13.3247	14.9865	10669.82	12160.44
500 m inward	9.5365	10.8120	11556.76	13707.01
100 m inward	4.2932	4.8878	12155.36	14726.47
10 m inward	1.3591	1.5484	12273.21	14928.78
1 m inward	0.4298	0.4897	12284.19	14948.85
10 cm inward	0.1359	0.1549	12285.12	14950.76
1 cm inward	0.0430	0.0490	12285.24	14950.95
surface limit	0.0000	0.0000	12285.25	14950.97

Table 16: Near-inner-boundary profile for GW240925_005809. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $5.335 M_\odot$ and $4.111 M_\odot$, with completed geometric mass-equivalent shares of 41.22% and 43.04%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $6.921 < 8.111 < 13.142$ km, and component 2 gives $5.457 < 6.378 < 10.368$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8084$ and 1.8304 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.645 and 13.206 h. The determinant-averaged frame-dragging rates at those shells are 9346.31 and 11439.19 rad s $^{-1}$, approaching surface-limit values of 12285.25 and 14950.97 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 13.325/14.987 h at 1 km inward to 0.0430/0.0490 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.6 GW200202_154313

GW200202_154313 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 11.023 M_\odot$ and $m_2 = 7.986 M_\odot$, while the luminosity-distance median is 413.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	43,480
Observed luminosity distance	413.6 Mpc (median)
Detector-frame masses	$m_1 = 11.023 M_\odot$, $m_2 = 7.986 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.725$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0879$; $z_\pi = 0.0945$ (medians)
Post-handoff reorder fraction	17.45%

Table 17: Observational and coordinate summary for GW200202_154313. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	9.951 [7.534, 13.594]	7.394 [5.596, 9.742]
Physical compact-source mass $M_c (M_\odot)$	6.222 [4.701, 9.383]	4.525 [3.328, 6.204]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.023 [0.779, 5.740]	3.005 [0.686, 4.610]
Geometric share M_{geom}/M_∞ (%)	38.67 [9.50, 49.81]	41.53 [10.86, 49.91]
Exterior spin magnitude $ \hat{a}_\infty $	0.292 [0.031, 0.814]	0.241 [0.021, 0.789]
Polarization factor A_{pol}	1.631 [1.105, 1.992]	1.710 [1.122, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.776 [0.124, 0.994]	0.704 [0.083, 0.993]
GR/CPTG matching radius r_m (km)	14.693 [11.126, 20.073]	10.919 [8.263, 14.385]
Structural Mode N_{BH}	1.7778 [1.5024, 1.9100]	1.8121 [1.5114, 1.9112]
Organization-shell radius r_N (km)	7.737 [5.853, 10.591]	5.750 [4.350, 7.563]
Inner trapping-surface radius r_- (km)	9.241 [7.501, 13.031]	6.870 [5.250, 8.850]
Structural- N lapse $\bar{\alpha}_N$	0.5937 [0.4847, 0.8094]	0.5655 [0.4838, 0.8023]
Structural- N clock $T_{24,N}$ (h)	14.249 [11.633, 19.425]	13.571 [11.611, 19.255]
Structural- N axis ω (rad s $^{-1}$)	7286.94 [1679.21, 11099.63]	9658.14 [1462.89, 15250.95]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8284.94 [1687.32, 12339.86]	11012.82 [1463.30, 16800.62]
Structural- N equatorial ω (rad s $^{-1}$)	9310.17 [1690.27, 13684.41]	12464.38 [1463.50, 18362.77]
Structural- N period P_ω (ms)	0.7584 [0.5092, 3.7238]	0.5705 [0.3740, 4.2939]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11280.17 [1842.66, 17827.88]	15041.43 [1594.28, 23449.34]
Trapping-surface period P_- (ms)	0.5570 [0.3524, 3.4098]	0.4177 [0.2679, 3.9411]

Table 18: CPTG compact-state, structural, clock, and geometric-rotation results for GW200202_154313. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	12.3363	14.3566	10034.07	12472.94
500 m inward	8.7885	10.3079	10719.56	13909.31
100 m inward	3.9425	4.6444	11178.70	14840.74
10 m inward	1.2470	1.4701	11269.74	15019.81
1 m inward	0.3943	0.4649	11279.08	15039.36
10 cm inward	0.1247	0.1470	11280.06	15041.26
1 cm inward	0.0394	0.0465	11280.16	15041.41
surface limit	0.0000	0.0000	11280.17	15041.43

Table 19: Near-inner-boundary profile for GW200202_154313. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.222 M_\odot$ and $4.525 M_\odot$, with completed geometric mass-equivalent shares of 38.67% and 41.53%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $7.737 < 9.241 < 14.693$ km, and component 2 gives $5.750 < 6.870 < 10.919$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7778$ and 1.8121 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.249 and 13.571 h. The determinant-averaged frame-dragging rates at those shells are 8284.94 and 11012.82 rad s $^{-1}$, approaching surface-limit values of 11280.17 and 15041.43 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 12.336/14.357 h at 1 km inward to 0.0394/0.0465 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.7 GW231224_024321

GW231224_024321 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 10.986 M_\odot$ and $m_2 = 8.670 M_\odot$, while the luminosity-distance median is 948.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	29,098
Observed luminosity distance	948.3 Mpc (median)
Detector-frame masses	$m_1 = 10.986 M_\odot$, $m_2 = 8.670 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.789$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1888$; $z_\pi = 0.2030$ (medians)
Post-handoff reorder fraction	18.28%

Table 20: Observational and coordinate summary for GW231224_024321. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	9.094 [7.328, 11.317]	7.323 [5.933, 9.018]
Physical compact-source mass $M_c (M_\odot)$	6.057 [4.512, 8.626]	4.410 [3.358, 6.242]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.014 [0.480, 5.082]	2.996 [0.812, 4.234]
Geometric share M_{geom}/M_∞ (%)	33.14 [5.74, 49.72]	41.59 [12.17, 49.91]
Exterior spin magnitude $ \hat{a}_\infty $	0.388 [0.038, 0.887]	0.240 [0.022, 0.764]
Polarization factor A_{pol}	1.496 [1.061, 1.989]	1.712 [1.139, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.869 [0.149, 0.998]	0.702 [0.087, 0.990]
GR/CPTG matching radius r_m (km)	13.428 [10.821, 16.711]	10.813 [8.760, 13.316]
Structural Mode N_{BH}	1.7132 [1.4792, 1.9090]	1.8129 [1.5204, 1.9112]
Organization-shell radius r_N (km)	7.070 [5.699, 8.806]	5.695 [4.611, 7.013]
Inner trapping-surface radius r_- (km)	8.795 [7.163, 11.543]	6.750 [5.419, 8.547]
Structural- N lapse $\bar{\alpha}_N$	0.6455 [0.4855, 0.8278]	0.5648 [0.4838, 0.7952]
Structural- N clock $T_{24,N}$ (h)	15.492 [11.651, 19.868]	13.556 [11.612, 19.084]
Structural- N axis ω (rad s $^{-1}$)	7702.99 [2197.37, 11736.05]	10030.14 [1608.85, 14988.39]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	9114.48 [2204.05, 13016.82]	11273.23 [1609.26, 16375.70]
Structural- N equatorial ω (rad s $^{-1}$)	10660.08 [2205.13, 14285.01]	12585.39 [1609.46, 17755.99]
Structural- N period P_ω (ms)	0.6894 [0.4827, 2.8507]	0.5574 [0.3837, 3.9044]
Trapping-surface limit Ω_- (rad s $^{-1}$)	13776.15 [2404.95, 18472.86]	15217.69 [1753.21, 22526.25]
Trapping-surface period P_- (ms)	0.4561 [0.3401, 2.6126]	0.4129 [0.2789, 3.5838]

Table 21: CPTG compact-state, structural, clock, and geometric-rotation results for GW231224_024321. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	12.5156	14.5458	12272.71	12589.68
500 m inward	8.8952	10.4575	13131.13	14082.04
100 m inward	3.9805	4.7179	13663.03	15017.88
10 m inward	1.2587	1.4939	13766.44	15196.86
1 m inward	0.3980	0.4725	13775.06	15215.58
10 cm inward	0.1259	0.1494	13776.01	15217.48
1 cm inward	0.0398	0.0472	13776.13	15217.67
surface limit	0.0000	0.0000	13776.15	15217.69

Table 22: Near-inner-boundary profile for GW231224_024321. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.057 M_\odot$ and $4.410 M_\odot$, with completed geometric mass-equivalent shares of 33.14% and 41.59%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $7.070 < 8.795 < 13.428$ km, and component 2 gives $5.695 < 6.750 < 10.813$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7132$ and 1.8129 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.492 and 13.556 h. The determinant-averaged frame-dragging rates at those shells are 9114.48 and 11273.23 rad s $^{-1}$, approaching surface-limit values of 13776.15 and 15217.69 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 12.516/14.546 h at 1 km inward to 0.0398/0.0472 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.8 GW240910_103535

GW240910_103535 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 12.528 M_\odot$ and $m_2 = 7.245 M_\odot$, while the luminosity-distance median is 739.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,426
Observed luminosity distance	739.0 Mpc (median)
Detector-frame masses	$m_1 = 12.528 M_\odot$, $m_2 = 7.245 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.578$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1505$; $z_\pi = 0.1620$ (medians)
Post-handoff reorder fraction	10.34%

Table 23: Observational and coordinate summary for GW240910_103535. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.816 [7.452, 13.882]	6.271 [5.009, 8.989]
Physical compact-source mass $M_c (M_\odot)$	6.675 [4.807, 10.113]	4.331 [2.918, 5.809]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.004 [1.072, 5.640]	2.222 [0.308, 4.186]
Geometric share M_{geom}/M_∞ (%)	36.71 [12.42, 49.32]	35.54 [5.43, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.326 [0.059, 0.759]	0.347 [0.031, 0.893]
Polarization factor A_{pol}	1.580 [1.142, 1.973]	1.551 [1.057, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.815 [0.230, 0.990]	0.834 [0.122, 0.998]
GR/CPTG matching radius r_m (km)	15.970 [11.004, 20.498]	9.260 [7.396, 13.274]
Structural Mode N_{BH}	1.7545 [1.5221, 1.9045]	1.7408 [1.4775, 1.9101]
Organization-shell radius r_N (km)	8.424 [5.800, 10.835]	4.881 [3.891, 6.986]
Inner trapping-surface radius r_- (km)	9.937 [7.425, 13.805]	6.291 [4.656, 8.287]
Structural- N lapse $\bar{\alpha}_N$	0.6127 [0.4890, 0.7938]	0.6236 [0.4847, 0.8292]
Structural- N clock $T_{24,N}$ (h)	14.704 [11.736, 19.051]	14.968 [11.632, 19.902]
Structural- N axis ω (rad s $^{-1}$)	7560.57 [2927.56, 11096.10]	10719.28 [2362.38, 17334.05]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8424.38 [3090.78, 12317.46]	12701.36 [2363.33, 19060.36]
Structural- N equatorial ω (rad s $^{-1}$)	9145.56 [3093.47, 13717.97]	14996.15 [2363.72, 20699.78]
Structural- N period P_ω (ms)	0.7458 [0.5101, 2.0329]	0.4947 [0.3296, 2.6586]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11106.49 [3377.78, 17998.62]	19467.16 [2575.12, 26459.37]
Trapping-surface period P_- (ms)	0.5657 [0.3491, 1.8602]	0.3228 [0.2375, 2.4400]

Table 24: CPTG compact-state, structural, clock, and geometric-rotation results for GW240910_103535. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.9199	14.6664	9964.40	15935.25
500 m inward	8.4763	10.4935	10586.18	18068.98
100 m inward	3.7964	4.7093	11011.27	19244.09
10 m inward	1.2006	1.4897	11096.78	19447.42
1 m inward	0.3797	0.4710	11105.55	19466.65
10 cm inward	0.1201	0.1490	11106.40	19467.04
1 cm inward	0.0380	0.0471	11106.48	19467.15
surface limit	0.0000	0.0000	11106.49	19467.16

Table 25: Near-inner-boundary profile for GW240910_103535. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.675 M_\odot$ and $4.331 M_\odot$, with completed geometric mass-equivalent shares of 36.71% and 35.54%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.424 < 9.937 < 15.970$ km, and component 2 gives $4.881 < 6.291 < 9.260$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7545$ and 1.7408 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.704 and 14.968 h. The determinant-averaged frame-dragging rates at those shells are 8424.38 and 12701.36 rad s $^{-1}$, approaching surface-limit values of 11106.49 and 19467.16 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.920/14.666 h at 1 km inward to 0.0380/0.0471 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.9 GW170608_020116

GW170608_020116 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 11.406 M_\odot$ and $m_2 = 8.383 M_\odot$, while the luminosity-distance median is 340.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	41,269
Observed luminosity distance	340.2 Mpc (median)
Detector-frame masses	$m_1 = 11.406 M_\odot$, $m_2 = 8.383 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.735$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0730$; $z_\pi = 0.0785$ (medians)
Post-handoff reorder fraction	15.25%

Table 26: Observational and coordinate summary for GW170608_020116. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.488 [8.155, 14.582]	7.859 [5.839, 10.054]
Physical compact-source mass $M_c (M_\odot)$	6.843 [5.030, 10.593]	4.752 [3.459, 6.485]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.822 [0.843, 5.838]	3.179 [0.812, 4.766]
Geometric share M_{geom}/M_∞ (%)	35.04 [9.09, 49.53]	41.38 [12.15, 49.91]
Exterior spin magnitude $ \hat{a}_\infty $	0.355 [0.049, 0.822]	0.243 [0.022, 0.764]
Polarization factor A_{pol}	1.539 [1.100, 1.981]	1.706 [1.138, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.842 [0.192, 0.995]	0.708 [0.087, 0.990]
GR/CPTG matching radius r_m (km)	15.487 [12.041, 21.532]	11.605 [8.622, 14.846]
Structural Mode N_{BH}	1.7350 [1.4997, 1.9069]	1.8103 [1.5202, 1.9112]
Organization-shell radius r_N (km)	8.163 [6.347, 11.371]	6.116 [4.537, 7.806]
Inner trapping-surface radius r_- (km)	9.942 [7.970, 14.519]	7.274 [5.473, 9.181]
Structural- N lapse $\bar{\alpha}_N$	0.6282 [0.4871, 0.8115]	0.5670 [0.4838, 0.7953]
Structural- N clock $T_{24,N}$ (h)	15.078 [11.691, 19.477]	13.607 [11.612, 19.087]
Structural- N axis ω (rad s $^{-1}$)	7153.69 [2320.51, 10518.84]	9457.42 [1465.37, 14563.09]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8225.87 [2490.17, 11649.38]	10666.93 [1465.76, 16077.03]
Structural- N equatorial ω (rad s $^{-1}$)	9274.96 [2505.26, 12816.02]	11892.17 [1465.97, 17495.59]
Structural- N period P_ω (ms)	0.7638 [0.5394, 2.5232]	0.5890 [0.3908, 4.2866]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11545.63 [2734.88, 16598.28]	14293.95 [1597.04, 22296.77]
Trapping-surface period P_- (ms)	0.5442 [0.3785, 2.2974]	0.4396 [0.2818, 3.9343]

Table 27: CPTG compact-state, structural, clock, and geometric-rotation results for GW170608_020116. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.8066	14.0302	10376.58	12015.91
500 m inward	8.3827	10.0696	11019.97	13297.44
100 m inward	3.7518	4.5369	11446.13	14116.01
10 m inward	1.1862	1.4363	11534.97	14276.77
1 m inward	0.3751	0.4542	11544.55	14292.64
10 cm inward	0.1186	0.1436	11545.54	14293.72
1 cm inward	0.0375	0.0454	11545.62	14293.92
surface limit	0.0000	0.0000	11545.63	14293.95

Table 28: Near-inner-boundary profile for GW170608_020116. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.843 M_\odot$ and $4.752 M_\odot$, with completed geometric mass-equivalent shares of 35.04% and 41.38%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.163 < 9.942 < 15.487$ km, and component 2 gives $6.116 < 7.274 < 11.605$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7350$ and 1.8103 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.078 and 13.607 h. The determinant-averaged frame-dragging rates at those shells are 8225.87 and 10666.93 rad s $^{-1}$, approaching surface-limit values of 11545.63 and 14293.95 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.807/14.030 h at 1 km inward to 0.0375/0.0454 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.10 GW191129_134029

GW191129_134029 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 12.292 M_\odot$ and $m_2 = 7.804 M_\odot$, while the luminosity-distance median is 789.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	24,169
Observed luminosity distance	789.5 Mpc (median)
Detector-frame masses	$m_1 = 12.292 M_\odot$, $m_2 = 7.804 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.635$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1602$; $z_\pi = 0.1720$ (medians)
Post-handoff reorder fraction	9.76%

Table 29: Observational and coordinate summary for GW191129_134029. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.544 [7.738, 14.654]	6.701 [4.986, 9.018]
Physical compact-source mass $M_c (M_\odot)$	6.466 [4.702, 9.935]	4.290 [3.021, 5.813]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.249 [1.316, 6.005]	2.533 [0.479, 4.243]
Geometric share M_{geom}/M_∞ (%)	39.03 [15.06, 49.76]	38.74 [8.49, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.286 [0.035, 0.710]	0.291 [0.026, 0.834]
Polarization factor A_{pol}	1.640 [1.177, 1.990]	1.632 [1.093, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.768 [0.139, 0.984]	0.775 [0.103, 0.996]
GR/CPTG matching radius r_m (km)	15.569 [11.427, 21.639]	9.894 [7.363, 13.317]
Structural Mode N_{BH}	1.7821 [1.5414, 1.9095]	1.7786 [1.4959, 1.9107]
Organization-shell radius r_N (km)	8.203 [6.034, 11.444]	5.216 [3.870, 7.004]
Inner trapping-surface radius r_- (km)	9.672 [7.432, 13.951]	6.385 [4.735, 8.273]
Structural- N lapse $\bar{\alpha}_N$	0.5902 [0.4852, 0.7787]	0.5931 [0.4842, 0.8145]
Structural- N clock $T_{24,N}$ (h)	14.164 [11.644, 18.689]	14.233 [11.621, 19.549]
Structural- N axis ω (rad s $^{-1}$)	7255.92 [1889.20, 11007.05]	10664.69 [1950.15, 16820.71]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8106.79 [1892.22, 12239.85]	12350.39 [1950.65, 18700.77]
Structural- N equatorial ω (rad s $^{-1}$)	8798.34 [1892.77, 13495.57]	14299.51 [1950.85, 20516.74]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.7751 [0.5133, 3.3205]	0.5087 [0.3360, 3.2211]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10510.51 [2063.29, 17461.25]	17833.20 [2124.95, 26310.30]
Trapping-surface period P_- (ms)	0.5978 [0.3598, 3.0452]	0.3523 [0.2388, 2.9569]

Table 30: CPTG compact-state, structural, clock, and geometric-rotation results for GW191129_134029. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	12.1129	14.7108	9403.53	14563.50
500 m inward	8.6275	10.5590	9999.54	16452.70
100 m inward	3.8704	4.7497	10416.96	17596.15
10 m inward	1.2243	1.5031	10502.32	17810.95
1 m inward	0.3872	0.4754	10509.69	17830.79
10 cm inward	0.1224	0.1503	10510.42	17832.96
1 cm inward	0.0387	0.0475	10510.50	17833.18
surface limit	0.0000	0.0000	10510.51	17833.20

Table 31: Near-inner-boundary profile for GW191129_134029. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.466 M_\odot$ and $4.290 M_\odot$, with completed geometric mass-equivalent shares of 39.03% and 38.74%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.203 < 9.672 < 15.569$ km, and component 2 gives $5.216 < 6.385 < 9.894$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7821$ and 1.7786 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.164 and 14.233 h. The determinant-averaged frame-dragging rates at those shells are 8106.79 and 12350.39 rad s $^{-1}$, approaching surface-limit values of 10510.51 and 17833.20 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 12.113/14.711 h at 1 km inward to 0.0387/0.0475 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.11 GW230630_234532

GW230630_234532 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 12.141 M_\odot$ and $m_2 = 8.039 M_\odot$, while the luminosity-distance median is 1086.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	32,446
Observed luminosity distance	1086.0 Mpc (median)
Detector-frame masses	$m_1 = 12.141 M_\odot$, $m_2 = 8.039 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.662$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2131$; $z_\pi = 0.2292$ (medians)
Post-handoff reorder fraction	15.46%

Table 32: Observational and coordinate summary for GW230630_234532. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	9.838 [7.117, 13.512]	6.604 [4.929, 9.097]
Physical compact-source mass $M_c (M_\odot)$	6.200 [4.723, 9.280]	4.267 [2.940, 6.046]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.708 [0.570, 5.952]	2.505 [0.411, 4.246]
Geometric share M_{geom}/M_∞ (%)	37.37 [7.08, 49.81]	38.60 [7.15, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.315 [0.031, 0.861]	0.293 [0.028, 0.860]
Polarization factor A_{pol}	1.597 [1.076, 1.993]	1.629 [1.077, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.803 [0.122, 0.997]	0.778 [0.111, 0.997]
GR/CPTG matching radius r_m (km)	14.528 [10.509, 19.952]	9.752 [7.279, 13.433]
Structural Mode N_{BH}	1.7623 [1.4872, 1.9101]	1.7769 [1.4876, 1.9105]
Organization-shell radius r_N (km)	7.654 [5.526, 10.509]	5.137 [3.828, 7.069]
Inner trapping-surface radius r_- (km)	9.234 [7.220, 12.870]	6.326 [4.663, 8.460]
Structural- N lapse $\bar{\alpha}_N$	0.6063 [0.4847, 0.8215]	0.5944 [0.4844, 0.8211]
Structural- N clock $T_{24,N}$ (h)	14.552 [11.632, 19.715]	14.266 [11.625, 19.707]
Structural- N axis ω (rad s $^{-1}$)	7175.24 [1608.57, 11605.89]	10620.68 [2149.49, 17208.97]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8349.95 [1613.33, 12932.40]	12246.87 [2151.16, 18983.71]
Structural- N equatorial ω (rad s $^{-1}$)	9668.85 [1613.76, 14380.33]	14092.41 [2152.00, 20770.66]
Structural- N period P_ω (ms)	0.7525 [0.4858, 3.8946]	0.5130 [0.3310, 2.9208]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11982.09 [1758.70, 18843.80]	17633.43 [2345.97, 26673.20]
Trapping-surface period P_- (ms)	0.5244 [0.3334, 3.5726]	0.3563 [0.2356, 2.6783]

Table 33: CPTG compact-state, structural, clock, and geometric-rotation results for GW230630_234532. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	12.3024	14.7455	10656.08	14548.53
500 m inward	8.7601	10.5740	11396.08	16319.30
100 m inward	3.9273	4.7546	11876.77	17419.86
10 m inward	1.2424	1.5043	11972.03	17612.46
1 m inward	0.3929	0.4757	11981.10	17632.01
10 cm inward	0.1242	0.1504	11981.99	17633.22
1 cm inward	0.0393	0.0476	11982.08	17633.40
surface limit	0.0000	0.0000	11982.09	17633.43

Table 34: Near-inner-boundary profile for GW230630_234532. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.200 M_\odot$ and $4.267 M_\odot$, with completed geometric mass-equivalent shares of 37.37% and 38.60%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $7.654 < 9.234 < 14.528$ km, and component 2 gives $5.137 < 6.326 < 9.752$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7623$ and 1.7769 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.552 and 14.266 h. The determinant-averaged frame-dragging rates at those shells are 8349.95 and 12246.87 rad s $^{-1}$, approaching surface-limit values of 11982.09 and 17633.43 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 12.302/14.745 h at 1 km inward to 0.0393/0.0476 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.12 GW250118_055802

GW250118_055802 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 12.183 M_\odot$ and $m_2 = 8.281 M_\odot$, while the luminosity-distance median is 936.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,516
Observed luminosity distance	936.1 Mpc (median)
Detector-frame masses	$m_1 = 12.183 M_\odot$, $m_2 = 8.281 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.680$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1866$; $z_\pi = 0.2007$ (medians)
Post-handoff reorder fraction	15.93%

Table 35: Observational and coordinate summary for GW250118_055802. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	10.057 [7.371, 17.516]	6.926 [4.328, 9.396]
Physical compact-source mass M_c ($M_\odot$)	7.131 [4.934, 13.483]	4.471 [2.798, 6.416]
Geometric mass-equivalent M_{geom} ($M_\odot$)	3.118 [0.434, 5.413]	2.396 [0.331, 4.369]
Geometric share M_{geom}/M_∞ (%)	28.29 [4.89, 48.23]	36.57 [5.82, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.472 [0.097, 0.903]	0.329 [0.028, 0.885]
Polarization factor A_{pol}	1.395 [1.051, 1.932]	1.576 [1.062, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.919 [0.364, 0.999]	0.817 [0.111, 0.998]
GR/CPTG matching radius r_m (km)	14.851 [10.883, 25.864]	10.228 [6.391, 13.874]
Structural Mode N_{BH}	1.6606 [1.4744, 1.8920]	1.7528 [1.4797, 1.9105]
Organization-shell radius r_N (km)	7.827 [5.719, 13.706]	5.390 [3.365, 7.304]
Inner trapping-surface radius r_- (km)	10.014 [7.501, 18.196]	6.658 [4.241, 8.890]
Structural- N lapse $\bar{\alpha}_N$	0.6866 [0.4990, 0.8317]	0.6140 [0.4844, 0.8274]
Structural- N clock $T_{24,N}$ (h)	16.479 [11.977, 19.962]	14.735 [11.625, 19.858]
Structural- N axis ω (rad s $^{-1}$)	6826.29 [2800.59, 11163.59]	10189.01 [2077.39, 18011.08]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8205.11 [3745.22, 12427.41]	11882.33 [2078.83, 20188.71]
Structural- N equatorial ω (rad s $^{-1}$)	9679.28 [4384.82, 13832.95]	13818.98 [2079.55, 22587.96]
Structural- N period P_ω (ms)	0.7658 [0.5056, 1.6777]	0.5288 [0.3112, 3.0225]
Trapping-surface limit Ω_- (rad s $^{-1}$)	12456.06 [5349.94, 18100.93]	17526.35 [2266.01, 29373.46]
Trapping-surface period P_- (ms)	0.5044 [0.3471, 1.1744]	0.3585 [0.2139, 2.7728]

Table 36: CPTG compact-state, structural, clock, and geometric-rotation results for GW250118_055802. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.2891	14.3639	11208.61	14682.23
500 m inward	7.9579	10.2709	11904.80	16328.90
100 m inward	3.5376	4.6104	12361.21	17328.17
10 m inward	1.1166	1.4582	12446.26	17507.64
1 m inward	0.3530	0.4611	12455.19	17524.04
10 cm inward	0.1116	0.1458	12455.97	17526.12
1 cm inward	0.0353	0.0461	12456.05	17526.33
surface limit	0.0000	0.0000	12456.06	17526.35

Table 37: Near-inner-boundary profile for GW250118_055802. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.131 M_\odot$ and $4.471 M_\odot$, with completed geometric mass-equivalent shares of 28.29% and 36.57%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $7.827 < 10.014 < 14.851$ km, and component 2 gives $5.390 < 6.658 < 10.228$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6606$ and 1.7528 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.479 and 14.735 h. The determinant-averaged frame-dragging rates at those shells are 8205.11 and 11882.33 rad s $^{-1}$, approaching surface-limit values of 12456.06 and 17526.35 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.289/14.364 h at 1 km inward to 0.0353/0.0461 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.13 GW241102_124058

GW241102_124058 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 11.783 M_\odot$ and $m_2 = 8.694 M_\odot$, while the luminosity-distance median is 389.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	12,190
Observed luminosity distance	389.9 Mpc (median)
Detector-frame masses	$m_1 = 11.783 M_\odot$, $m_2 = 8.694 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.738$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0830$; $z_\pi = 0.0894$ (medians)
Post-handoff reorder fraction	20.74%

Table 38: Observational and coordinate summary for GW241102_124058. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	10.654 [8.066, 13.986]	8.138 [6.327, 10.732]
Physical compact-source mass M_c ($M_\odot$)	6.688 [5.187, 9.446]	5.048 [3.809, 6.799]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.226 [0.761, 6.197]	3.296 [0.639, 5.051]
Geometric share M_{geom}/M_∞ (%)	38.69 [8.87, 49.67]	40.93 [9.37, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.292 [0.041, 0.826]	0.252 [0.024, 0.817]
Polarization factor A_{pol}	1.631 [1.097, 1.987]	1.693 [1.103, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.776 [0.162, 0.995]	0.721 [0.095, 0.995]
GR/CPTG matching radius r_m (km)	15.732 [11.910, 20.652]	12.017 [9.342, 15.847]
Structural Mode N_{BH}	1.7779 [1.4983, 1.9085]	1.8049 [1.5015, 1.9110]
Organization-shell radius r_N (km)	8.284 [6.267, 10.888]	6.334 [4.909, 8.334]
Inner trapping-surface radius r_- (km)	9.921 [8.177, 13.127]	7.598 [6.030, 9.719]
Structural- N lapse $\bar{\alpha}_N$	0.5936 [0.4859, 0.8126]	0.5715 [0.4840, 0.8101]
Structural- N clock $T_{24,N}$ (h)	14.246 [11.662, 19.503]	13.715 [11.616, 19.441]
Structural- N axis ω (rad s $^{-1}$)	7034.78 [2033.96, 10407.18]	8853.77 [1507.62, 13379.00]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7947.19 [2038.55, 11524.64]	10095.78 [1508.00, 14842.58]
Structural- N equatorial ω (rad s $^{-1}$)	8966.28 [2039.39, 12685.46]	11534.56 [1508.19, 16285.39]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.7906 [0.5452, 3.0822]	0.6224 [0.4233, 4.1666]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10909.58 [2224.69, 16507.84]	13949.08 [1642.89, 20943.78]
Trapping-surface period P_- (ms)	0.5759 [0.3806, 2.8243]	0.4504 [0.3000, 3.8245]

Table 39: CPTG compact-state, structural, clock, and geometric-rotation results for GW241102_124058. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.9215	13.6565	9767.99	11835.43
500 m inward	8.4864	9.7756	10382.10	13010.07
100 m inward	3.8074	4.3971	10813.61	13787.90
10 m inward	1.2042	1.3915	10901.86	13932.02
1 m inward	0.3809	0.4401	10908.81	13947.38
10 cm inward	0.1204	0.1392	10909.50	13948.91
1 cm inward	0.0381	0.0440	10909.57	13949.07
surface limit	0.0000	0.0000	10909.58	13949.08

Table 40: Near-inner-boundary profile for GW241102_124058. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.688 M_\odot$ and $5.048 M_\odot$, with completed geometric mass-equivalent shares of 38.69% and 40.93%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.284 < 9.921 < 15.732$ km, and component 2 gives $6.334 < 7.598 < 12.017$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7779$ and 1.8049; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.246 and 13.715 h. The determinant-averaged frame-dragging rates at those shells are 7947.19 and 10095.78 rad s $^{-1}$, approaching surface-limit values of 10909.58 and 13949.08 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.921/13.657 h at 1 km inward to 0.0381/0.0440 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.14 GW191216_213338

GW191216_213338 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 12.940 M_\odot$ and $m_2 = 8.233 M_\odot$, while the luminosity-distance median is 341.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	13,892
Observed luminosity distance	341.0 Mpc (median)
Detector-frame masses	$m_1 = 12.940 M_\odot$, $m_2 = 8.233 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.636$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0732$; $z_\pi = 0.0787$ (medians)
Post-handoff reorder fraction	12.88%

Table 41: Observational and coordinate summary for GW191216_213338. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.938 [8.531, 16.537]	7.680 [5.777, 10.640]
Physical compact-source mass $M_c (M_\odot)$	7.173 [5.451, 11.290]	4.981 [3.431, 6.557]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.860 [1.418, 6.534]	2.951 [0.586, 5.078]
Geometric share M_{geom}/M_∞ (%)	39.10 [14.81, 49.56]	39.08 [9.24, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.284 [0.047, 0.715]	0.285 [0.023, 0.819]
Polarization factor A_{pol}	1.642 [1.174, 1.982]	1.641 [1.102, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.767 [0.187, 0.985]	0.767 [0.093, 0.995]
GR/CPTG matching radius r_m (km)	17.628 [12.597, 24.419]	11.340 [8.530, 15.711]
Structural Mode N_{BH}	1.7829 [1.5396, 1.9072]	1.7827 [1.5007, 1.9110]
Organization-shell radius r_N (km)	9.291 [6.647, 12.929]	5.981 [4.495, 8.251]
Inner trapping-surface radius r_- (km)	10.829 [8.420, 15.879]	7.369 [5.441, 9.498]
Structural- N lapse $\bar{\alpha}_N$	0.5895 [0.4869, 0.7801]	0.5897 [0.4840, 0.8107]
Structural- N clock $T_{24,N}$ (h)	14.149 [11.686, 18.723]	14.154 [11.615, 19.457]
Structural- N axis ω (rad s $^{-1}$)	6760.64 [1984.23, 9968.95]	9534.56 [1477.87, 14475.20]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7422.48 [2186.38, 11114.94]	11007.67 [1478.41, 15969.92]
Structural- N equatorial ω (rad s $^{-1}$)	7941.18 [2202.70, 12272.46]	12557.19 [1478.68, 17639.10]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8465 [0.5653, 2.8738]	0.5708 [0.3934, 4.2500]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9395.95 [2404.13, 15953.91]	15517.51 [1611.03, 22697.42]
Trapping-surface period P_- (ms)	0.6687 [0.3938, 2.6135]	0.4049 [0.2768, 3.9001]

Table 42: CPTG compact-state, structural, clock, and geometric-rotation results for GW191216_213338. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.5201	13.7436	8503.42	13086.43
500 m inward	8.1973	9.8248	8982.92	14442.17
100 m inward	3.6762	4.4148	9320.34	15330.43
10 m inward	1.1634	1.3970	9388.94	15500.40
1 m inward	0.3679	0.4418	9395.22	15516.23
10 cm inward	0.1163	0.1397	9395.87	15517.36
1 cm inward	0.0368	0.0442	9395.94	15517.50
surface limit	0.0000	0.0000	9395.95	15517.51

Table 43: Near-inner-boundary profile for GW191216_213338. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.173 M_\odot$ and $4.981 M_\odot$, with completed geometric mass-equivalent shares of 39.10% and 39.08%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.291 < 10.829 < 17.628$ km, and component 2 gives $5.981 < 7.369 < 11.340$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7829$ and 1.7827 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.149 and 14.154 h. The determinant-averaged frame-dragging rates at those shells are 7422.48 and 11007.67 rad s $^{-1}$, approaching surface-limit values of 9395.95 and 15517.51 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.520/13.744 h at 1 km inward to 0.0368/0.0442 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.15 GW230904_051013

GW230904_051013 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 12.707 M_\odot$ and $m_2 = 8.573 M_\odot$, while the luminosity-distance median is 1025.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	29,820
Observed luminosity distance	1025.9 Mpc (median)
Detector-frame masses	$m_1 = 12.707 M_\odot$, $m_2 = 8.573 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.675$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2026$; $z_\pi = 0.2179$ (medians)
Post-handoff reorder fraction	14.59%

Table 44: Observational and coordinate summary for GW230904_051013. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	10.347 [7.558, 14.550]	7.081 [5.167, 9.660]
Physical compact-source mass M_c ($M_\odot$)	6.654 [4.772, 10.680]	4.531 [3.137, 6.417]
Geometric mass-equivalent M_{geom} ($M_\odot$)	3.835 [0.615, 5.819]	2.670 [0.436, 4.498]
Geometric share M_{geom}/M_∞ (%)	35.64 [7.17, 49.68]	38.74 [7.18, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.345 [0.040, 0.859]	0.291 [0.026, 0.859]
Polarization factor A_{pol}	1.554 [1.077, 1.987]	1.632 [1.077, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.833 [0.158, 0.997]	0.775 [0.102, 0.997]
GR/CPTG matching radius r_m (km)	15.278 [11.161, 21.485]	10.456 [7.630, 14.265]
Structural Mode N_{BH}	1.7420 [1.4878, 1.9086]	1.7786 [1.4878, 1.9108]
Organization-shell radius r_N (km)	8.050 [5.872, 11.346]	5.509 [4.014, 7.500]
Inner trapping-surface radius r_- (km)	9.765 [7.504, 14.594]	6.754 [4.903, 9.008]
Structural- N lapse $\bar{\alpha}_N$	0.6227 [0.4858, 0.8210]	0.5931 [0.4842, 0.8210]
Structural- N clock $T_{24,N}$ (h)	14.944 [11.659, 19.705]	14.234 [11.620, 19.703]
Structural- N axis ω (rad s $^{-1}$)	6974.40 [2095.08, 10905.82]	9832.68 [1872.21, 16224.93]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8006.64 [2158.90, 12153.90]	11410.14 [1873.35, 17975.00]
Structural- N equatorial ω (rad s $^{-1}$)	9141.38 [2162.59, 13487.88]	13218.98 [1873.90, 19719.80]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.7847 [0.5170, 2.9104]	0.5507 [0.3496, 3.3540]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11316.72 [2358.09, 17583.67]	16544.56 [2041.45, 25216.85]
Trapping-surface period P_- (ms)	0.5552 [0.3573, 2.6645]	0.3798 [0.2492, 3.0778]

Table 45: CPTG compact-state, structural, clock, and geometric-rotation results for GW230904_051013. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.9111	14.3381	10171.09	13867.78
500 m inward	8.4635	10.2770	10797.92	15388.28
100 m inward	3.7881	4.6235	11220.58	16347.49
10 m inward	1.1982	1.4628	11307.60	16524.46
1 m inward	0.3789	0.4626	11315.81	16542.90
10 cm inward	0.1198	0.1463	11316.63	16544.34
1 cm inward	0.0379	0.0463	11316.71	16544.54
surface limit	0.0000	0.0000	11316.72	16544.56

Table 46: Near-inner-boundary profile for GW230904_051013. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.654 M_\odot$ and $4.531 M_\odot$, with completed geometric mass-equivalent shares of 35.64% and 38.74%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.050 < 9.765 < 15.278$ km, and component 2 gives $5.509 < 6.754 < 10.456$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7420$ and 1.7786 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.944 and 14.234 h. The determinant-averaged frame-dragging rates at those shells are 8006.64 and 11410.14 rad s $^{-1}$, approaching surface-limit values of 11316.72 and 16544.56 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.911/14.338 h at 1 km inward to 0.0379/0.0463 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.16 GW240915_105151

GW240915_105151 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 13.076 M_\odot$ and $m_2 = 8.456 M_\odot$, while the luminosity-distance median is 630.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	30,884
Observed luminosity distance	630.2 Mpc (median)
Detector-frame masses	$m_1 = 13.076 M_\odot$, $m_2 = 8.456 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.647$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1300$; $z_\pi = 0.1399$ (medians)
Post-handoff reorder fraction	15.69%

Table 47: Observational and coordinate summary for GW240915_105151. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.479 [8.259, 18.739]	7.484 [4.833, 10.439]
Physical compact-source mass $M_c (M_\odot)$	8.122 [5.524, 14.704]	4.888 [2.970, 7.014]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.554 [0.472, 6.012]	2.522 [0.328, 4.855]
Geometric share M_{geom}/M_∞ (%)	27.86 [4.92, 48.39]	36.27 [5.40, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.480 [0.093, 0.903]	0.334 [0.030, 0.893]
Polarization factor A_{pol}	1.386 [1.052, 1.938]	1.569 [1.057, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.922 [0.347, 0.999]	0.822 [0.119, 0.998]
GR/CPTG matching radius r_m (km)	16.950 [12.196, 27.671]	11.051 [7.137, 15.414]
Structural Mode N_{BH}	1.6561 [1.4746, 1.8938]	1.7493 [1.4773, 1.9102]
Organization-shell radius r_N (km)	8.938 [6.402, 14.617]	5.821 [3.755, 8.114]
Inner trapping-surface radius r_- (km)	11.332 [8.443, 19.718]	7.254 [4.596, 9.755]
Structural- N lapse $\bar{\alpha}_N$	0.6901 [0.4975, 0.8316]	0.6168 [0.4846, 0.8294]
Structural- N clock $T_{24,N}$ (h)	16.562 [11.940, 19.958]	14.803 [11.630, 19.906]
Structural- N axis ω (rad s $^{-1}$)	5871.10 [2530.71, 9786.93]	9216.84 [2039.77, 16994.32]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7071.18 [3471.30, 10948.07]	10820.60 [2040.78, 18915.90]
Structural- N equatorial ω (rad s $^{-1}$)	8442.78 [4217.29, 12248.46]	12661.44 [2041.28, 20909.07]
Structural- N period P_ω (ms)	0.8886 [0.5739, 1.8100]	0.5807 [0.3322, 3.0788]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10719.64 [4772.57, 16075.19]	16105.40 [2224.46, 26865.03]
Trapping-surface period P_- (ms)	0.5861 [0.3909, 1.3165]	0.3901 [0.2339, 2.8246]

Table 48: CPTG compact-state, structural, clock, and geometric-rotation results for GW240915_105151. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.5102	13.7336	9780.90	13755.67
500 m inward	7.3951	9.8012	10302.53	15122.61
100 m inward	3.2857	4.3947	10639.44	15942.09
10 m inward	1.0370	1.3900	10712.35	16092.19
1 m inward	0.3278	0.4396	10718.88	16103.90
10 cm inward	0.1037	0.1390	10719.56	16105.25
1 cm inward	0.0328	0.0440	10719.63	16105.38
surface limit	0.0000	0.0000	10719.64	16105.40

Table 49: Near-inner-boundary profile for GW240915_105151. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.122 M_\odot$ and $4.888 M_\odot$, with completed geometric mass-equivalent shares of 27.86% and 36.27%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.938 < 11.332 < 16.950$ km, and component 2 gives $5.821 < 7.254 < 11.051$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6561$ and 1.7493 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.562 and 14.803 h. The determinant-averaged frame-dragging rates at those shells are 7071.18 and 10820.60 rad s $^{-1}$, approaching surface-limit values of 10719.64 and 16105.40 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.510/13.734 h at 1 km inward to 0.0328/0.0440 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.17 GW190725_174728

GW190725_174728 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.236 M_\odot$ and $m_2 = 7.584 M_\odot$, while the luminosity-distance median is 1026.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	79,172
Observed luminosity distance	1026.0 Mpc (median)
Detector-frame masses	$m_1 = 14.236 M_\odot$, $m_2 = 7.584 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.533$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2029$; $z_\pi = 0.2179$ (medians)
Post-handoff reorder fraction	12.92%

Table 50: Observational and coordinate summary for GW190725_174728. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.652 [7.614, 21.761]	6.221 [3.717, 9.603]
Physical compact-source mass $M_c (M_\odot)$	7.684 [5.540, 17.146]	4.366 [2.401, 6.528]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.879 [0.369, 6.903]	1.948 [0.232, 4.372]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	30.19 [4.04, 49.56]	32.67 [4.36, 49.74]
Exterior spin magnitude $ \hat{a}_\infty $	0.439 [0.048, 0.920]	0.396 [0.036, 0.914]
Polarization factor A_{pol}	1.432 [1.042, 1.982]	1.485 [1.046, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.902 [0.187, 0.999]	0.874 [0.144, 0.999]
GR/CPTG matching radius r_m (km)	17.205 [11.243, 32.133]	9.186 [5.489, 14.181]
Structural Mode N_{BH}	1.6807 [1.4696, 1.9072]	1.7080 [1.4714, 1.9093]
Organization-shell radius r_N (km)	9.062 [5.899, 16.971]	4.836 [2.887, 7.465]
Inner trapping-surface radius r_- (km)	11.161 [8.176, 23.052]	6.247 [3.640, 9.082]
Structural- N lapse $\bar{\alpha}_N$	0.6710 [0.4869, 0.8356]	0.6496 [0.4853, 0.8341]
Structural- N clock $T_{24,N}$ (h)	16.104 [11.686, 20.053]	15.591 [11.647, 20.020]
Structural- N axis ω (rad s $^{-1}$)	5564.34 [1431.01, 9887.80]	10435.57 [2859.82, 20272.58]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6745.42 [1776.79, 11257.99]	12467.10 [2862.55, 22917.76]
Structural- N equatorial ω (rad s $^{-1}$)	8123.85 [2040.95, 12874.91]	14872.53 [2864.31, 25985.37]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9315 [0.5581, 3.5363]	0.5040 [0.2742, 2.1950]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10246.46 [2276.47, 17188.57]	19223.13 [3123.39, 33653.69]
Trapping-surface period P_- (ms)	0.6132 [0.3655, 2.7601]	0.3269 [0.1867, 2.0117]

Table 51: CPTG compact-state, structural, clock, and geometric-rotation results for GW190725_174728. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.8240	14.4614	9367.21	15864.08
500 m inward	7.6332	10.2881	9845.03	17835.58
100 m inward	3.3989	4.5890	10171.12	18992.28
10 m inward	1.0737	1.4491	10238.98	19202.61
1 m inward	0.3395	0.4582	10245.55	19221.18
10 cm inward	0.1073	0.1449	10246.37	19222.87
1 cm inward	0.0339	0.0458	10246.45	19223.11
surface limit	0.0000	0.0000	10246.46	19223.13

Table 52: Near-inner-boundary profile for GW190725_174728. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.684 M_\odot$ and $4.366 M_\odot$, with completed geometric mass-equivalent shares of 30.19% and 32.67%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.062 < 11.161 < 17.205$ km, and component 2 gives $4.836 < 6.247 < 9.186$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6807$ and 1.7080 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.104 and 15.591 h. The determinant-averaged frame-dragging rates at those shells are 6745.42 and 12467.10 rad s $^{-1}$, approaching surface-limit values of 10246.46 and 19223.13 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.824/14.461 h at 1 km inward to 0.0339/0.0458 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.18 GW241114_235258

GW241114_235258 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 13.008 M_\odot$ and $m_2 = 8.884 M_\odot$, while the luminosity-distance median is 794.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,921
Observed luminosity distance	794.3 Mpc (median)
Detector-frame masses	$m_1 = 13.008 M_\odot$, $m_2 = 8.884 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.683$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1608$; $z_\pi = 0.1730$ (medians)
Post-handoff reorder fraction	13.92%

Table 53: Observational and coordinate summary for GW241114_235258. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.077 [8.305, 16.291]	7.640 [5.391, 10.186]
Physical compact-source mass $M_c (M_\odot)$	7.342 [5.173, 12.415]	4.864 [3.333, 6.846]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.847 [0.659, 6.117]	2.848 [0.428, 4.746]
Geometric share M_{geom}/M_∞ (%)	33.09 [6.62, 49.55]	38.57 [6.75, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.389 [0.048, 0.870]	0.294 [0.026, 0.867]
Polarization factor A_{pol}	1.494 [1.071, 1.982]	1.628 [1.072, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.869 [0.188, 0.997]	0.778 [0.103, 0.997]
GR/CPTG matching radius r_m (km)	16.356 [12.263, 24.056]	11.281 [7.961, 15.041]
Structural Mode N_{BH}	1.7127 [1.4844, 1.9071]	1.7766 [1.4852, 1.9107]
Organization-shell radius r_N (km)	8.616 [6.454, 12.718]	5.940 [4.187, 7.920]
Inner trapping-surface radius r_- (km)	10.603 [8.178, 16.837]	7.273 [5.188, 9.511]
Structural- N lapse $\bar{\alpha}_N$	0.6460 [0.4870, 0.8237]	0.5947 [0.4842, 0.8231]
Structural- N clock $T_{24,N}$ (h)	15.503 [11.687, 19.768]	14.273 [11.621, 19.754]
Structural- N axis ω (rad s $^{-1}$)	6323.33 [2139.22, 10133.32]	9139.15 [1749.17, 15183.79]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7382.40 [2332.19, 11259.35]	10581.99 [1750.50, 16902.07]
Structural- N equatorial ω (rad s $^{-1}$)	8492.04 [2357.24, 12444.82]	12310.45 [1751.17, 18570.47]
Structural- N period P_ω (ms)	0.8511 [0.5580, 2.6941]	0.5938 [0.3717, 3.5894]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10650.61 [2573.61, 16153.10]	15478.08 [1909.13, 23952.56]
Trapping-surface period P_- (ms)	0.5899 [0.3890, 2.4414]	0.4059 [0.2623, 3.2911]

Table 54: CPTG compact-state, structural, clock, and geometric-rotation results for GW241114_235258. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.3082	13.8470	9698.93	13166.42
500 m inward	8.0106	9.9103	10222.59	14508.02
100 m inward	3.5782	4.4546	10577.51	15309.97
10 m inward	1.1307	1.4098	10641.93	15462.75
1 m inward	0.3575	0.4459	10649.86	15476.87
10 cm inward	0.1131	0.1410	10650.54	15477.99
1 cm inward	0.0358	0.0446	10650.61	15478.07
surface limit	0.0000	0.0000	10650.61	15478.08

Table 55: Near-inner-boundary profile for GW241114_235258. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.342 M_\odot$ and $4.864 M_\odot$, with completed geometric mass-equivalent shares of 33.09% and 38.57%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.616 < 10.603 < 16.356$ km, and component 2 gives $5.940 < 7.273 < 11.281$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7127$ and 1.7766 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.503 and 14.273 h. The determinant-averaged frame-dragging rates at those shells are 7382.40 and 10581.99 rad s $^{-1}$, approaching surface-limit values of 10650.61 and 15478.08 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.308/13.847 h at 1 km inward to 0.0358/0.0446 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.19 GW241130_110422

GW241130_110422 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 13.796 M_\odot$ and $m_2 = 8.106 M_\odot$, while the luminosity-distance median is 1315.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,253
Observed luminosity distance	1315.1 Mpc (median)
Detector-frame masses	$m_1 = 13.796 M_\odot$, $m_2 = 8.106 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.588$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2525$; $z_\pi = 0.2716$ (medians)
Post-handoff reorder fraction	10.57%

Table 56: Observational and coordinate summary for GW241130_110422. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.914 [7.573, 16.086]	6.421 [4.566, 9.123]
Physical compact-source mass $M_c (M_\odot)$	6.750 [4.805, 11.745]	4.252 [2.801, 6.025]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.254 [0.706, 6.253]	2.299 [0.298, 4.221]
Geometric share M_{geom}/M_∞ (%)	37.42 [8.20, 49.79]	36.87 [5.44, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.314 [0.033, 0.839]	0.324 [0.026, 0.893]
Polarization factor A_{pol}	1.598 [1.089, 1.992]	1.584 [1.057, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.802 [0.129, 0.996]	0.812 [0.104, 0.998]
GR/CPTG matching radius r_m (km)	16.115 [11.182, 23.752]	9.482 [6.743, 13.471]
Structural Mode N_{BH}	1.7629 [1.4941, 1.9098]	1.7564 [1.4775, 1.9107]
Organization-shell radius r_N (km)	8.483 [5.892, 12.550]	4.993 [3.544, 7.081]
Inner trapping-surface radius r_- (km)	10.090 [7.521, 16.079]	6.255 [4.373, 8.503]
Structural- N lapse $\bar{\alpha}_N$	0.6058 [0.4849, 0.8160]	0.6111 [0.4842, 0.8292]
Structural- N clock $T_{24,N}$ (h)	14.540 [11.637, 19.584]	14.666 [11.621, 19.902]
Structural- N axis ω (rad s $^{-1}$)	6536.60 [1665.12, 10752.65]	10400.66 [2017.46, 17943.19]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7447.46 [1679.26, 12020.88]	12240.63 [2018.14, 19849.46]
Structural- N equatorial ω (rad s $^{-1}$)	8293.55 [1679.94, 13435.58]	14536.51 [2018.49, 21920.35]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8437 [0.5227, 3.7416]	0.5133 [0.3165, 3.1133]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10083.33 [1831.31, 17689.14]	18505.64 [2199.06, 28456.82]
Trapping-surface period P_- (ms)	0.6231 [0.3552, 3.4310]	0.3395 [0.2208, 2.8572]

Table 57: CPTG compact-state, structural, clock, and geometric-rotation results for GW241130_110422. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.7995	14.7391	9106.78	15228.54
500 m inward	8.3899	10.5556	9647.74	17159.54
100 m inward	3.7553	4.7393	10002.24	18280.07
10 m inward	1.1876	1.4988	10074.60	18482.08
1 m inward	0.3755	0.4739	10082.57	18503.50
10 cm inward	0.1187	0.1499	10083.25	18505.51
1 cm inward	0.0376	0.0474	10083.32	18505.63
surface limit	0.0000	0.0000	10083.33	18505.64

Table 58: Near-inner-boundary profile for GW241130_110422. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.750 M_\odot$ and $4.252 M_\odot$, with completed geometric mass-equivalent shares of 37.42% and 36.87%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.483 < 10.090 < 16.115$ km, and component 2 gives $4.993 < 6.255 < 9.482$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7629$ and 1.7564 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.540 and 14.666 h. The determinant-averaged frame-dragging rates at those shells are 7447.46 and 12240.63 rad s $^{-1}$, approaching surface-limit values of 10083.33 and 18505.64 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.799/14.739 h at 1 km inward to 0.0376/0.0474 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.20 GW230731_215307

GW230731_215307 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 12.426 M_\odot$ and $m_2 = 9.524 M_\odot$, while the luminosity-distance median is 1098.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	28,176
Observed luminosity distance	1098.8 Mpc (median)
Detector-frame masses	$m_1 = 12.426 M_\odot$, $m_2 = 9.524 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.766$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2154$; $z_\pi = 0.2316$ (medians)
Post-handoff reorder fraction	21.09%

Table 59: Observational and coordinate summary for GW230731_215307. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.018 [7.802, 12.976]	7.898 [6.118, 10.121]
Physical compact-source mass $M_c (M_\odot)$	6.707 [5.079, 9.448]	4.869 [3.529, 6.875]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.272 [0.459, 5.873]	3.144 [0.734, 4.744]
Geometric share M_{geom}/M_∞ (%)	32.83 [5.20, 49.68]	40.71 [10.47, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.394 [0.040, 0.897]	0.256 [0.024, 0.796]
Polarization factor A_{pol}	1.489 [1.055, 1.987]	1.687 [1.117, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.872 [0.159, 0.998]	0.727 [0.095, 0.993]
GR/CPTG matching radius r_m (km)	14.793 [11.521, 19.161]	11.663 [9.034, 14.944]
Structural Mode N_{BH}	1.7098 [1.4761, 1.9086]	1.8022 [1.5087, 1.9109]
Organization-shell radius r_N (km)	7.797 [6.055, 10.083]	6.143 [4.759, 7.864]
Inner trapping-surface radius r_- (km)	9.744 [7.877, 12.791]	7.365 [5.663, 9.481]
Structural- N lapse $\bar{\alpha}_N$	0.6482 [0.4858, 0.8303]	0.5736 [0.4840, 0.8044]
Structural- N clock $T_{24,N}$ (h)	15.556 [11.659, 19.928]	13.768 [11.617, 19.305]
Structural- N axis ω (rad s $^{-1}$)	6958.02 [2057.73, 10746.89]	9259.31 [1616.99, 14202.61]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8261.57 [2069.95, 11978.16]	10457.64 [1617.62, 15571.08]
Structural- N equatorial ω (rad s $^{-1}$)	9801.91 [2071.00, 13209.34]	11746.30 [1617.94, 16947.20]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.7605 [0.5246, 3.0354]	0.6008 [0.4035, 3.8842]
Trapping-surface limit Ω_- (rad s $^{-1}$)	12648.46 [2258.25, 17209.55]	14436.87 [1762.84, 21524.69]
Trapping-surface period P_- (ms)	0.4968 [0.3651, 2.7823]	0.4352 [0.2919, 3.5642]

Table 60: CPTG compact-state, structural, clock, and geometric-rotation results for GW230731_215307. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.8647	13.8875	11405.41	12242.25
500 m inward	8.4205	9.9550	12109.72	13485.18
100 m inward	3.7664	4.4808	12555.93	14273.91
10 m inward	1.1907	1.4183	12640.77	14421.75
1 m inward	0.3765	0.4485	12647.94	14435.27
10 cm inward	0.1191	0.1418	12648.38	14436.68
1 cm inward	0.0377	0.0449	12648.45	14436.85
surface limit	0.0000	0.0000	12648.46	14436.87

Table 61: Near-inner-boundary profile for GW230731_215307. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.707 M_\odot$ and $4.869 M_\odot$, with completed geometric mass-equivalent shares of 32.83% and 40.71%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $7.797 < 9.744 < 14.793$ km, and component 2 gives $6.143 < 7.365 < 11.663$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7098$ and 1.8022 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.556 and 13.768 h. The determinant-averaged frame-dragging rates at those shells are 8261.57 and 10457.64 rad s $^{-1}$, approaching surface-limit values of 12648.46 and 14436.87 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.865/13.888 h at 1 km inward to 0.0377/0.0449 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.21 GW240915_001357

GW240915_001357 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 12.845 M_\odot$ and $m_2 = 9.143 M_\odot$, while the luminosity-distance median is 922.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,419
Observed luminosity distance	922.9 Mpc (median)
Detector-frame masses	$m_1 = 12.845 M_\odot$, $m_2 = 9.143 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.712$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1842$; $z_\pi = 0.1981$ (medians)
Post-handoff reorder fraction	19.96%

Table 62: Observational and coordinate summary for GW240915_001357. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.644 [8.047, 16.256]	7.745 [5.339, 10.216]
Physical compact-source mass $M_c (M_\odot)$	7.632 [5.501, 12.361]	5.035 [3.307, 6.910]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.255 [0.533, 5.455]	2.741 [0.385, 4.826]
Geometric share M_{geom}/M_∞ (%)	28.40 [6.03, 45.70]	36.81 [5.98, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.471 [0.159, 0.881]	0.325 [0.030, 0.882]
Polarization factor A_{pol}	1.397 [1.064, 1.842]	1.583 [1.064, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.918 [0.540, 0.998]	0.813 [0.117, 0.998]
GR/CPTG matching radius r_m (km)	15.717 [11.883, 24.004]	11.436 [7.883, 15.086]
Structural Mode N_{BH}	1.6617 [1.4810, 1.8623]	1.7557 [1.4806, 1.9103]
Organization-shell radius r_N (km)	8.300 [6.243, 12.707]	6.028 [4.148, 7.937]
Inner trapping-surface radius r_- (km)	10.624 [8.201, 16.690]	7.503 [5.105, 9.534]
Structural- N lapse $\bar{\alpha}_N$	0.6858 [0.5236, 0.8265]	0.6117 [0.4845, 0.8267]
Structural- N clock $T_{24,N}$ (h)	16.458 [12.565, 19.835]	14.680 [11.629, 19.841]
Structural- N axis ω (rad s $^{-1}$)	6876.24 [3225.39, 10482.59]	9106.97 [1961.56, 15309.25]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8142.07 [4600.37, 11681.45]	10631.51 [1962.45, 16994.49]
Structural- N equatorial ω (rad s $^{-1}$)	9503.95 [5669.53, 12895.71]	12384.22 [1962.87, 18847.23]
Structural- N period P_ω (ms)	0.7717 [0.5379, 1.3658]	0.5910 [0.3697, 3.2017]
Trapping-surface limit Ω_- (rad s $^{-1}$)	12146.06 [7121.65, 16756.70]	15755.41 [2138.90, 24361.15]
Trapping-surface period P_- (ms)	0.5173 [0.3750, 0.8823]	0.3988 [0.2579, 2.9376]

Table 63: CPTG compact-state, structural, clock, and geometric-rotation results for GW240915_001357. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.9594	13.6080	10981.58	13516.20
500 m inward	7.7279	9.7216	11633.48	14806.85
100 m inward	3.4422	4.3657	12051.86	15579.93
10 m inward	1.0868	1.3810	12135.54	15739.00
1 m inward	0.3437	0.4367	12144.90	15753.28
10 cm inward	0.1087	0.1381	12145.95	15755.20
1 cm inward	0.0344	0.0437	12146.05	15755.39
surface limit	0.0000	0.0000	12146.06	15755.41

Table 64: Near-inner-boundary profile for GW240915_001357. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.632 M_\odot$ and $5.035 M_\odot$, with completed geometric mass-equivalent shares of 28.40% and 36.81%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.300 < 10.624 < 15.717$ km, and component 2 gives $6.028 < 7.503 < 11.436$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6617$ and 1.7557 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.458 and 14.680 h. The determinant-averaged frame-dragging rates at those shells are 8142.07 and 10631.51 rad s $^{-1}$, approaching surface-limit values of 12146.06 and 15755.41 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.959/13.608 h at 1 km inward to 0.0344/0.0437 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.22 GW191105_143521

GW191105_143521 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 12.998 M_\odot$ and $m_2 = 9.358 M_\odot$, while the luminosity-distance median is 1154.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	16,861
Observed luminosity distance	1154.5 Mpc (median)
Detector-frame masses	$m_1 = 12.998 M_\odot$, $m_2 = 9.358 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.720$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2254$; $z_\pi = 0.2420$ (medians)
Post-handoff reorder fraction	17.09%

Table 65: Observational and coordinate summary for GW191105_143521. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.410 [7.879, 14.206]	7.655 [5.717, 10.086]
Physical compact-source mass $M_c (M_\odot)$	6.605 [4.988, 9.892]	4.714 [3.329, 6.699]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.016 [0.654, 6.277]	3.066 [0.658, 4.736]
Geometric share M_{geom}/M_∞ (%)	37.69 [7.25, 49.85]	41.23 [10.03, 49.91]
Exterior spin magnitude $ \hat{a}_\infty $	0.309 [0.027, 0.858]	0.246 [0.021, 0.804]
Polarization factor A_{pol}	1.605 [1.078, 1.994]	1.701 [1.111, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.796 [0.109, 0.997]	0.713 [0.083, 0.994]
GR/CPTG matching radius r_m (km)	15.372 [11.634, 20.976]	11.304 [8.441, 14.893]
Structural Mode N_{BH}	1.7661 [1.4883, 1.9105]	1.8085 [1.5058, 1.9113]
Organization-shell radius r_N (km)	8.099 [6.126, 11.047]	5.955 [4.438, 7.836]
Inner trapping-surface radius r_- (km)	9.816 [7.857, 13.569]	7.130 [5.340, 9.351]
Structural- N lapse $\bar{\alpha}_N$	0.6032 [0.4843, 0.8206]	0.5685 [0.4838, 0.8067]
Structural- N clock $T_{24,N}$ (h)	14.477 [11.624, 19.695]	13.643 [11.611, 19.360]
Structural- N axis ω (rad s $^{-1}$)	6641.65 [1335.25, 10601.21]	9290.63 [1438.32, 14896.23]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7757.73 [1343.96, 11798.98]	10537.55 [1438.86, 16493.29]
Structural- N equatorial ω (rad s $^{-1}$)	9008.58 [1344.15, 13105.34]	11919.44 [1439.13, 18078.25]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8099 [0.5325, 4.6751]	0.5963 [0.3810, 4.3668]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11183.29 [1464.27, 17046.99]	14575.16 [1567.97, 23089.90]
Trapping-surface period P_- (ms)	0.5618 [0.3686, 4.2910]	0.4311 [0.2721, 4.0072]

Table 66: CPTG compact-state, structural, clock, and geometric-rotation results for GW191105_143521. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.9660	14.0700	10037.02	12257.64
500 m inward	8.5179	10.0930	10674.30	13577.06
100 m inward	3.8172	4.5406	11092.52	14407.68
10 m inward	1.2073	1.4374	11176.55	14558.46
1 m inward	0.3818	0.4546	11182.27	14573.53
10 cm inward	0.1207	0.1437	11183.19	14574.99
1 cm inward	0.0382	0.0455	11183.28	14575.14
surface limit	0.0000	0.0000	11183.29	14575.16

Table 67: Near-inner-boundary profile for GW191105_143521. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.605 M_\odot$ and $4.714 M_\odot$, with completed geometric mass-equivalent shares of 37.69% and 41.23%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.099 < 9.816 < 15.372$ km, and component 2 gives $5.955 < 7.130 < 11.304$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7661$ and 1.8085 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.477 and 13.643 h. The determinant-averaged frame-dragging rates at those shells are 7757.73 and 10537.55 rad s $^{-1}$, approaching surface-limit values of 11183.29 and 14575.16 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.966/14.070 h at 1 km inward to 0.0382/0.0455 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.23 GW231223_075055

GW231223_075055 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.261 M_\odot$ and $m_2 = 8.126 M_\odot$, while the luminosity-distance median is 968.8 Mpc. Two retained release labels map to this same numerical stream; their exact transformed identity is preserved in the provenance record rather than counted twice as separate physical realizations. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	34,300
Observed luminosity distance	968.8 Mpc (median)
Detector-frame masses	$m_1 = 14.261 M_\odot$, $m_2 = 8.126 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.570$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1924$; $z_\pi = 0.2070$ (medians)
Post-handoff reorder fraction	10.57%

Table 68: Observational and coordinate summary for GW231223_075055. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.784 [8.091, 18.731]	6.714 [4.572, 9.618]
Physical compact-source mass $M_c (M_\odot)$	8.231 [5.305, 14.620]	4.548 [2.835, 6.621]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.614 [0.500, 6.179]	2.239 [0.290, 4.402]
Geometric share M_{geom}/M_∞ (%)	28.70 [4.93, 48.94]	34.60 [5.00, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.465 [0.074, 0.903]	0.363 [0.032, 0.901]
Polarization factor A_{pol}	1.403 [1.052, 1.959]	1.529 [1.053, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.915 [0.285, 0.999]	0.849 [0.128, 0.999]
GR/CPTG matching radius r_m (km)	17.401 [11.947, 27.659]	9.914 [6.751, 14.202]
Structural Mode N_{BH}	1.6649 [1.4746, 1.9002]	1.7300 [1.4750, 1.9099]
Organization-shell radius r_N (km)	9.166 [6.285, 14.627]	5.220 [3.551, 7.476]
Inner trapping-surface radius r_- (km)	11.596 [8.138, 19.680]	6.636 [4.397, 9.175]
Structural- N lapse $\bar{\alpha}_N$	0.6833 [0.4924, 0.8316]	0.6323 [0.4848, 0.8312]
Structural- N clock $T_{24,N}$ (h)	16.399 [11.818, 19.957]	15.174 [11.636, 19.949]
Structural- N axis ω (rad s $^{-1}$)	5836.58 [2272.83, 9942.22]	9950.96 [2407.01, 17832.33]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6958.29 [3047.58, 11177.71]	11794.57 [2408.77, 19833.23]
Structural- N equatorial ω (rad s $^{-1}$)	8107.19 [3526.71, 12567.36]	13939.85 [2409.66, 21973.23]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9030 [0.5621, 2.0617]	0.5327 [0.3168, 2.6085]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10337.90 [3950.27, 16466.72]	17916.66 [2626.98, 28280.38]
Trapping-surface period P_- (ms)	0.6078 [0.3816, 1.5906]	0.3507 [0.2222, 2.3918]

Table 69: CPTG compact-state, structural, clock, and geometric-rotation results for GW231223_075055. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.5093	14.2415	9485.29	15057.57
500 m inward	7.4005	10.1566	9953.57	16735.62
100 m inward	3.2905	4.5480	10270.55	17718.88
10 m inward	1.0389	1.4377	10332.74	17896.87
1 m inward	0.3285	0.4546	10337.10	17914.60
10 cm inward	0.1039	0.1438	10337.82	17916.63
1 cm inward	0.0328	0.0455	10337.89	17916.66
surface limit	0.0000	0.0000	10337.90	17916.66

Table 70: Near-inner-boundary profile for GW231223_075055. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.231 M_\odot$ and $4.548 M_\odot$, with completed geometric mass-equivalent shares of 28.70% and 34.60%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.166 < 11.596 < 17.401$ km, and component 2 gives $5.220 < 6.636 < 9.914$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6649$ and 1.7300 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.399 and 15.174 h. The determinant-averaged frame-dragging rates at those shells are 6958.29 and 11794.57 rad s $^{-1}$, approaching surface-limit values of 10337.90 and 17916.66 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.509/14.241 h at 1 km inward to 0.0328/0.0455 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.24 GW240601_231004

GW240601_231004 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 13.114 M_\odot$ and $m_2 = 9.539 M_\odot$, while the luminosity-distance median is 1272.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,987
Observed luminosity distance	1272.6 Mpc (median)
Detector-frame masses	$m_1 = 13.114 M_\odot$, $m_2 = 9.539 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.727$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2453$; $z_\pi = 0.2638$ (medians)
Post-handoff reorder fraction	18.62%

Table 71: Observational and coordinate summary for GW240601_231004. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	10.341 [7.872, 15.214]	7.692 [5.193, 10.196]
Physical compact-source mass M_c ($M_\odot$)	7.585 [5.327, 11.911]	5.016 [3.296, 7.085]
Geometric mass-equivalent M_{geom} ($M_\odot$)	2.732 [0.335, 5.700]	2.646 [0.394, 4.741]
Geometric share M_{geom}/M_∞ (%)	25.76 [3.50, 48.26]	35.87 [6.22, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.517 [0.096, 0.931]	0.341 [0.032, 0.878]
Polarization factor A_{pol}	1.347 [1.036, 1.933]	1.559 [1.066, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.938 [0.360, 0.999]	0.829 [0.127, 0.998]
GR/CPTG matching radius r_m (km)	15.270 [11.624, 22.465]	11.358 [7.668, 15.055]
Structural Mode N_{BH}	1.6349 [1.4667, 1.8923]	1.7446 [1.4821, 1.9099]
Organization-shell radius r_N (km)	8.039 [6.102, 11.862]	5.983 [4.038, 7.920]
Inner trapping-surface radius r_- (km)	10.558 [8.080, 16.027]	7.400 [5.032, 9.686]
Structural- N lapse $\bar{\alpha}_N$	0.7065 [0.4987, 0.8379]	0.6205 [0.4848, 0.8256]
Structural- N clock $T_{24,N}$ (h)	16.955 [11.969, 20.111]	14.893 [11.635, 19.814]
Structural- N axis ω (rad s $^{-1}$)	6331.31 [2608.13, 10347.49]	9206.41 [2144.12, 15424.50]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7777.13 [3640.73, 11562.89]	10774.82 [2145.60, 17214.46]
Structural- N equatorial ω (rad s $^{-1}$)	9391.83 [4360.91, 12869.56]	12474.89 [2146.34, 19251.27]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8079 [0.5434, 1.7258]	0.5831 [0.3650, 2.9284]
Trapping-surface limit Ω_- (rad s $^{-1}$)	12355.00 [5018.42, 16897.15]	15911.66 [2339.69, 24778.89]
Trapping-surface period P_- (ms)	0.5086 [0.3718, 1.2520]	0.3949 [0.2536, 2.6855]

Table 72: CPTG compact-state, structural, clock, and geometric-rotation results for GW240601_231004. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.9019	13.6173	11265.45	13704.41
500 m inward	7.6665	9.7119	11897.03	14972.50
100 m inward	3.3995	4.3569	12278.57	15744.86
10 m inward	1.0729	1.3779	12349.69	15896.19
1 m inward	0.3392	0.4357	12354.44	15910.55
10 cm inward	0.1073	0.1378	12354.98	15911.57
1 cm inward	0.0339	0.0436	12355.00	15911.66
surface limit	0.0000	0.0000	12355.00	15911.66

Table 73: Near-inner-boundary profile for GW240601_231004. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.585 M_\odot$ and $5.016 M_\odot$, with completed geometric mass-equivalent shares of 25.76% and 35.87%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.039 < 10.558 < 15.270$ km, and component 2 gives $5.983 < 7.400 < 11.358$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6349$ and 1.7446 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.955 and 14.893 h. The determinant-averaged frame-dragging rates at those shells are 7777.13 and 10774.82 rad s $^{-1}$, approaching surface-limit values of 12355.00 and 15911.66 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.902/13.617 h at 1 km inward to 0.0339/0.0436 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.25 GW191204_171526

GW191204_171526 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 13.247 M_\odot$ and $m_2 = 9.420 M_\odot$, while the luminosity-distance median is 643.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	17,547
Observed luminosity distance	643.6 Mpc (median)
Detector-frame masses	$m_1 = 13.247 M_\odot$, $m_2 = 9.420 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.711$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1326$; $z_\pi = 0.1427$ (medians)
Post-handoff reorder fraction	20.95%

Table 74: Observational and coordinate summary for GW191204_171526. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.435 [8.339, 14.904]	8.459 [6.591, 11.516]
Physical compact-source mass $M_c (M_\odot)$	8.078 [6.184, 11.668]	5.632 [3.877, 7.403]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.029 [0.606, 5.702]	3.053 [0.702, 5.488]
Geometric share M_{geom}/M_∞ (%)	26.76 [6.16, 45.34]	36.94 [9.42, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.499 [0.167, 0.879]	0.322 [0.028, 0.816]
Polarization factor A_{pol}	1.365 [1.066, 1.829]	1.586 [1.104, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.931 [0.559, 0.998]	0.811 [0.113, 0.995]
GR/CPTG matching radius r_m (km)	16.886 [12.313, 22.008]	12.491 [9.733, 17.005]
Structural Mode N_{BH}	1.6449 [1.4817, 1.8579]	1.7572 [1.5018, 1.9104]
Organization-shell radius r_N (km)	8.914 [6.473, 11.626]	6.584 [5.125, 8.934]
Inner trapping-surface radius r_- (km)	11.308 [8.879, 15.661]	8.222 [6.125, 10.359]
Structural- N lapse $\bar{\alpha}_N$	0.6988 [0.5272, 0.8259]	0.6104 [0.4844, 0.8098]
Structural- N clock $T_{24,N}$ (h)	16.771 [12.653, 19.821]	14.651 [11.626, 19.436]
Structural- N axis ω (rad s $^{-1}$)	6537.68 [2953.70, 9452.91]	8528.89 [1642.88, 13216.18]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7834.99 [4365.19, 10755.32]	9976.26 [1643.34, 14604.76]
Structural- N equatorial ω (rad s $^{-1}$)	9074.39 [5850.73, 12131.17]	11508.31 [1643.56, 15850.29]
Structural- N period P_ω (ms)	0.8019 [0.5842, 1.4394]	0.6298 [0.4302, 3.8234]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11687.38 [7159.83, 16068.33]	14599.34 [1790.34, 20144.48]
Trapping-surface period P_- (ms)	0.5376 [0.3910, 0.8776]	0.4304 [0.3119, 3.5095]

Table 75: CPTG compact-state, structural, clock, and geometric-rotation results for GW191204_171526. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.6630	12.9531	10641.53	12700.47
500 m inward	7.5031	9.2338	11217.83	13780.69
100 m inward	3.3331	4.1442	11602.78	14455.03
10 m inward	1.0524	1.3111	11679.76	14584.06
1 m inward	0.3327	0.4146	11686.69	14597.46
10 cm inward	0.1052	0.1311	11687.36	14599.15
1 cm inward	0.0333	0.0415	11687.38	14599.32
surface limit	0.0000	0.0000	11687.38	14599.34

Table 76: Near-inner-boundary profile for GW191204_171526. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.078 M_\odot$ and $5.632 M_\odot$, with completed geometric mass-equivalent shares of 26.76% and 36.94%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.914 < 11.308 < 16.886$ km, and component 2 gives $6.584 < 8.222 < 12.491$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6449$ and 1.7572; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.771 and 14.651 h. The determinant-averaged frame-dragging rates at those shells are 7834.99 and 9976.26 rad s $^{-1}$, approaching surface-limit values of 11687.38 and 14599.34 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.663/12.953 h at 1 km inward to 0.0333/0.0415 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.26 GW231223_202619

GW231223_202619 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 13.080 M_\odot$ and $m_2 = 9.830 M_\odot$, while the luminosity-distance median is 893.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	27,980
Observed luminosity distance	893.9 Mpc (median)
Detector-frame masses	$m_1 = 13.080 M_\odot$, $m_2 = 9.830 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.752$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1789$; $z_\pi = 0.1925$ (medians)
Post-handoff reorder fraction	24.41%

Table 77: Observational and coordinate summary for GW231223_202619. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	10.701 [8.130, 14.888]	8.440 [6.267, 11.259]
Physical compact-source mass $M_c (M_\odot)$	7.568 [5.536, 11.079]	5.459 [3.821, 7.504]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.137 [0.374, 5.986]	3.152 [0.515, 5.276]
Geometric share M_{geom}/M_∞ (%)	28.83 [4.04, 48.55]	38.17 [7.00, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.463 [0.088, 0.920]	0.301 [0.028, 0.863]
Polarization factor A_{pol}	1.405 [1.042, 1.944]	1.617 [1.075, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.914 [0.331, 0.999]	0.787 [0.112, 0.997]
GR/CPTG matching radius r_m (km)	15.802 [12.005, 21.984]	12.463 [9.254, 16.625]
Structural Mode N_{BH}	1.6662 [1.4696, 1.8957]	1.7718 [1.4867, 1.9104]
Organization-shell radius r_N (km)	8.327 [6.302, 11.607]	6.566 [4.871, 8.748]
Inner trapping-surface radius r_- (km)	10.685 [8.420, 15.040]	8.082 [6.010, 10.428]
Structural- N lapse $\bar{\alpha}_N$	0.6823 [0.4961, 0.8356]	0.5986 [0.4844, 0.8219]
Structural- N clock $T_{24,N}$ (h)	16.374 [11.905, 20.054]	14.366 [11.626, 19.725]
Structural- N axis ω (rad s $^{-1}$)	6533.50 [2740.82, 9997.25]	8402.98 [1711.67, 13276.80]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7811.01 [3721.92, 11191.18]	9705.36 [1712.92, 14719.10]
Structural- N equatorial ω (rad s $^{-1}$)	9259.74 [4005.16, 12433.95]	11215.38 [1713.54, 16225.71]
Structural- N period P_ω (ms)	0.8044 [0.5614, 1.6882]	0.6474 [0.4269, 3.6681]
Trapping-surface limit Ω_- (rad s $^{-1}$)	12017.88 [4406.92, 16321.80]	14129.61 [1868.00, 20864.09]
Trapping-surface period P_- (ms)	0.5228 [0.3850, 1.4258]	0.4447 [0.3011, 3.3636]

Table 78: CPTG compact-state, structural, clock, and geometric-rotation results for GW231223_202619. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.0309	13.1366	10940.30	12305.79
500 m inward	7.7828	9.3761	11552.42	13347.35
100 m inward	3.4656	4.2090	11938.94	14001.28
10 m inward	1.0948	1.3314	12010.19	14118.04
1 m inward	0.3462	0.4211	12016.93	14128.31
10 cm inward	0.1095	0.1331	12017.79	14129.48
1 cm inward	0.0346	0.0421	12017.87	14129.60
surface limit	0.0000	0.0000	12017.88	14129.61

Table 79: Near-inner-boundary profile for GW231223_202619. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.568 M_\odot$ and $5.459 M_\odot$, with completed geometric mass-equivalent shares of 28.83% and 38.17%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.327 < 10.685 < 15.802$ km, and component 2 gives $6.566 < 8.082 < 12.463$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6662$ and 1.7718 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.374 and 14.366 h. The determinant-averaged frame-dragging rates at those shells are 7811.01 and 9705.36 rad s $^{-1}$, approaching surface-limit values of 12017.88 and 14129.61 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.031/13.137 h at 1 km inward to 0.0346/0.0421 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.27 GW240916_184352

GW240916_184352 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 13.175 M_\odot$ and $m_2 = 9.740 M_\odot$, while the luminosity-distance median is 1284.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,484
Observed luminosity distance	1284.5 Mpc (median)
Detector-frame masses	$m_1 = 13.175 M_\odot$, $m_2 = 9.740 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.739$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2474$; $z_\pi = 0.2660$ (medians)
Post-handoff reorder fraction	18.31%

Table 80: Observational and coordinate summary for GW240916_184352. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	10.365 [7.874, 14.681]	7.835 [5.572, 10.262]
Physical compact-source mass M_c ($M_\odot$)	7.243 [5.110, 11.134]	4.933 [3.420, 7.071]
Geometric mass-equivalent M_{geom} ($M_\odot$)	3.140 [0.389, 5.881]	2.920 [0.479, 4.795]
Geometric share M_{geom}/M_∞ (%)	29.57 [4.13, 49.28]	38.61 [7.28, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.450 [0.061, 0.918]	0.293 [0.026, 0.857]
Polarization factor A_{pol}	1.420 [1.043, 1.972]	1.629 [1.079, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.908 [0.237, 0.999]	0.778 [0.102, 0.997]
GR/CPTG matching radius r_m (km)	15.306 [11.626, 21.679]	11.570 [8.228, 15.153]
Structural Mode N_{BH}	1.6741 [1.4701, 1.9040]	1.7770 [1.4884, 1.9108]
Organization-shell radius r_N (km)	8.063 [6.110, 11.440]	6.093 [4.329, 7.970]
Inner trapping-surface radius r_- (km)	10.301 [7.920, 15.040]	7.405 [5.350, 9.703]
Structural- N lapse $\bar{\alpha}_N$	0.6762 [0.4894, 0.8352]	0.5944 [0.4842, 0.8205]
Structural- N clock $T_{24,N}$ (h)	16.228 [11.745, 20.044]	14.265 [11.620, 19.692]
Structural- N axis ω (rad s $^{-1}$)	6499.21 [2496.98, 10431.21]	9056.32 [1742.52, 14822.97]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7790.75 [2961.57, 11679.07]	10455.61 [1743.83, 16437.44]
Structural- N equatorial ω (rad s $^{-1}$)	9245.18 [3063.81, 12951.89]	12030.79 [1744.49, 18183.89]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8065 [0.5380, 2.1216]	0.6009 [0.3822, 3.6031]
Trapping-surface limit Ω_- (rad s $^{-1}$)	12056.63 [3360.96, 16949.56]	15144.88 [1901.31, 23285.80]
Trapping-surface period P_- (ms)	0.5211 [0.3707, 1.8695]	0.4149 [0.2698, 3.3047]

Table 81: CPTG compact-state, structural, clock, and geometric-rotation results for GW240916_184352. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.3037	13.7643	10970.30	13002.42
500 m inward	7.9842	9.8436	11576.04	14210.82
100 m inward	3.5584	4.4254	11968.68	14982.95
10 m inward	1.1241	1.4005	12048.21	15129.59
1 m inward	0.3554	0.4429	12055.90	15142.99
10 cm inward	0.1124	0.1401	12056.53	15144.71
1 cm inward	0.0355	0.0443	12056.62	15144.86
surface limit	0.0000	0.0000	12056.63	15144.88

Table 82: Near-inner-boundary profile for GW240916_184352. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.243 M_\odot$ and $4.933 M_\odot$, with completed geometric mass-equivalent shares of 29.57% and 38.61%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.063 < 10.301 < 15.306$ km, and component 2 gives $6.093 < 7.405 < 11.570$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6741$ and 1.7770 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.228 and 14.265 h. The determinant-averaged frame-dragging rates at those shells are 7790.75 and 10455.61 rad s $^{-1}$, approaching surface-limit values of 12056.63 and 15144.88 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.304/13.764 h at 1 km inward to 0.0355/0.0443 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.28 GW241231_054133

GW241231_054133 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.644 M_\odot$ and $m_2 = 8.349 M_\odot$, while the luminosity-distance median is 913.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,987
Observed luminosity distance	913.8 Mpc (median)
Detector-frame masses	$m_1 = 14.644 M_\odot$, $m_2 = 8.349 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.570$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1825$; $z_\pi = 0.1964$ (medians)
Post-handoff reorder fraction	10.31%

Table 83: Observational and coordinate summary for GW241231_054133. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.255 [8.424, 15.531]	7.017 [5.681, 10.120]
Physical compact-source mass $M_c (M_\odot)$	7.165 [5.259, 10.900]	4.876 [3.350, 6.497]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.960 [1.338, 6.693]	2.445 [0.306, 4.738]
Geometric share M_{geom}/M_∞ (%)	40.72 [14.08, 49.85]	35.13 [4.77, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.255 [0.028, 0.728]	0.354 [0.029, 0.906]
Polarization factor A_{pol}	1.687 [1.164, 1.994]	1.541 [1.050, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.727 [0.111, 0.986]	0.841 [0.117, 0.999]
GR/CPTG matching radius r_m (km)	18.097 [12.439, 22.933]	10.362 [8.388, 14.943]
Structural Mode N_{BH}	1.8024 [1.5341, 1.9105]	1.7360 [1.4737, 1.9103]
Organization-shell radius r_N (km)	9.530 [6.561, 12.119]	5.463 [4.411, 7.860]
Inner trapping-surface radius r_- (km)	10.997 [8.293, 15.070]	7.085 [5.302, 9.285]
Structural- N lapse $\bar{\alpha}_N$	0.5735 [0.4844, 0.7844]	0.6275 [0.4845, 0.8323]
Structural- N clock $T_{24,N}$ (h)	13.765 [11.625, 18.826]	15.059 [11.628, 19.975]
Structural- N axis ω (rad s $^{-1}$)	6345.01 [1332.51, 9751.25]	9215.20 [1956.49, 15327.12]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6983.67 [1333.18, 10843.25]	11043.08 [1957.75, 16889.95]
Structural- N equatorial ω (rad s $^{-1}$)	7478.64 [1333.52, 12102.43]	13369.67 [1958.38, 18395.99]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8997 [0.5795, 4.7129]	0.5690 [0.3720, 3.2094]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8939.33 [1453.21, 15918.94]	17545.04 [2134.64, 23479.95]
Trapping-surface period P_- (ms)	0.7029 [0.3947, 4.3237]	0.3581 [0.2676, 2.9434]

Table 84: CPTG compact-state, structural, clock, and geometric-rotation results for GW241231_054133. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.4885	13.8533	8157.61	14876.75
500 m inward	8.1826	9.8888	8573.38	16489.31
100 m inward	3.6731	4.4326	8873.21	17384.85
10 m inward	1.1622	1.4019	8932.54	17526.00
1 m inward	0.3676	0.4433	8938.66	17543.34
10 cm inward	0.1162	0.1402	8939.27	17544.97
1 cm inward	0.0368	0.0443	8939.32	17545.03
surface limit	0.0000	0.0000	8939.33	17545.04

Table 85: Near-inner-boundary profile for GW241231_054133. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.165 M_\odot$ and $4.876 M_\odot$, with completed geometric mass-equivalent shares of 40.72% and 35.13%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.530 < 10.997 < 18.097$ km, and component 2 gives $5.463 < 7.085 < 10.362$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8024$ and 1.7360 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.765 and 15.059 h. The determinant-averaged frame-dragging rates at those shells are 6983.67 and 11043.08 rad s $^{-1}$, approaching surface-limit values of 8939.33 and 17545.04 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.489/13.853 h at 1 km inward to 0.0368/0.0443 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.29 GW250119_190238

GW250119_190238 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 12.373 M_\odot$ and $m_2 = 10.685 M_\odot$, while the luminosity-distance median is 310.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	20,526
Observed luminosity distance	310.5 Mpc (median)
Detector-frame masses	$m_1 = 12.373 M_\odot$, $m_2 = 10.685 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.864$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0668$; $z_\pi = 0.0720$ (medians)
Post-handoff reorder fraction	30.29%

Table 86: Observational and coordinate summary for GW250119_190238. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	11.207 [9.523, 13.387]	10.146 [8.360, 11.893]
Physical compact-source mass M_c ($M_\odot$)	6.881 [5.850, 9.295]	5.727 [4.722, 7.624]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.394 [1.338, 6.198]	4.472 [2.040, 5.722]
Geometric share M_{geom}/M_∞ (%)	39.29 [13.07, 49.40]	44.77 [22.32, 49.94]
Exterior spin magnitude $ \hat{a}_\infty $	0.281 [0.055, 0.747]	0.179 [0.017, 0.578]
Polarization factor A_{pol}	1.647 [1.150, 1.976]	1.811 [1.287, 1.998]
Compact-source spin parameter $\hat{a}_c$	0.762 [0.216, 0.989]	0.586 [0.069, 0.958]
GR/CPTG matching radius r_m (km)	16.549 [14.061, 19.767]	14.981 [12.344, 17.562]
Structural Mode N_{BH}	1.7852 [1.5267, 1.9054]	1.8512 [1.6021, 1.9116]
Organization-shell radius r_N (km)	8.731 [7.429, 10.407]	7.883 [6.510, 9.226]
Inner trapping-surface radius r_- (km)	10.342 [9.163, 12.671]	8.994 [7.659, 10.792]
Structural- N lapse $\bar{\alpha}_N$	0.5877 [0.4883, 0.7902]	0.5328 [0.4835, 0.7317]
Structural- N clock $T_{24,N}$ (h)	14.104 [11.719, 18.965]	12.788 [11.605, 17.561]
Structural- N axis ω (rad s $^{-1}$)	7473.12 [2431.38, 9568.99]	7319.88 [917.70, 10746.00]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	8216.19 [2523.79, 10516.00]	7876.89 [917.89, 11783.44]
Structural- N equatorial ω (rad s $^{-1}$)	8940.80 [2525.83, 11407.95]	8248.21 [917.98, 12716.14]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.7647 [0.5975, 2.4896]	0.7977 [0.5332, 6.8453]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10879.12 [2757.26, 14494.32]	9280.28 [999.88, 15959.21]
Trapping-surface period P_- (ms)	0.5775 [0.4335, 2.2788]	0.6770 [0.3937, 6.2839]

Table 87: CPTG compact-state, structural, clock, and geometric-rotation results for GW250119_190238. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.8433	12.8362	9719.30	8156.92
500 m inward	8.4428	9.1950	10349.23	8761.57
100 m inward	3.7912	4.1422	10780.69	9188.09
10 m inward	1.1997	1.3118	10869.06	9271.19
1 m inward	0.3794	0.4149	10878.12	9279.37
10 cm inward	0.1200	0.1312	10879.02	9280.19
1 cm inward	0.0379	0.0415	10879.11	9280.27
surface limit	0.0000	0.0000	10879.12	9280.28

Table 88: Near-inner-boundary profile for GW250119_190238. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.881 M_\odot$ and $5.727 M_\odot$, with completed geometric mass-equivalent shares of 39.29% and 44.77%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.731 < 10.342 < 16.549$ km, and component 2 gives $7.883 < 8.994 < 14.981$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7852$ and 1.8512 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.104 and 12.788 h. The determinant-averaged frame-dragging rates at those shells are 8216.19 and 7876.89 rad s $^{-1}$, approaching surface-limit values of 10879.12 and 9280.28 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.843/12.836 h at 1 km inward to 0.0379/0.0415 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.30 GW241225_042553

GW241225_042553 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.066 M_\odot$ and $m_2 = 9.146 M_\odot$, while the luminosity-distance median is 663.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,176
Observed luminosity distance	663.6 Mpc (median)
Detector-frame masses	$m_1 = 14.066 M_\odot$, $m_2 = 9.146 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.650$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1364$; $z_\pi = 0.1468$ (medians)
Post-handoff reorder fraction	9.97%

Table 89: Observational and coordinate summary for GW241225_042553. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.278 [8.999, 15.357]	8.055 [6.542, 10.726]
Physical compact-source mass $M_c (M_\odot)$	8.847 [6.187, 12.736]	5.367 [3.783, 7.511]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.261 [0.430, 6.037]	2.855 [0.513, 4.889]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	26.84 [3.78, 46.14]	35.77 [6.78, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.498 [0.150, 0.925]	0.343 [0.034, 0.867]
Polarization factor A_{pol}	1.367 [1.039, 1.857]	1.557 [1.073, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.930 [0.516, 0.999]	0.831 [0.135, 0.997]
GR/CPTG matching radius r_m (km)	18.130 [13.289, 22.677]	11.894 [9.660, 15.838]
Structural Mode N_{BH}	1.6457 [1.4682, 1.8675]	1.7435 [1.4854, 1.9096]
Organization-shell radius r_N (km)	9.561 [6.991, 11.965]	6.266 [5.083, 8.348]
Inner trapping-surface radius r_- (km)	12.354 [9.206, 16.889]	7.819 [6.068, 10.337]
Structural- N lapse $\bar{\alpha}_N$	0.6981 [0.5192, 0.8367]	0.6215 [0.4850, 0.8229]
Structural- N clock $T_{24,N}$ (h)	16.755 [12.460, 20.081]	14.916 [11.641, 19.750]
Structural- N axis ω (rad s $^{-1}$)	6038.11 [2386.23, 9042.31]	8772.50 [2172.39, 13378.21]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7163.72 [3678.80, 10076.58]	10223.47 [2174.69, 14678.55]
Structural- N equatorial ω (rad s $^{-1}$)	8251.40 [5266.53, 11279.68]	11873.21 [2175.69, 15922.94]
Structural- N period P_ω (ms)	0.8771 [0.6235, 1.7079]	0.6146 [0.4281, 2.8892]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10650.16 [6507.99, 14792.40]	15261.49 [2372.06, 20318.98]
Trapping-surface period P_- (ms)	0.5900 [0.4248, 0.9655]	0.4117 [0.3092, 2.6488]

Table 90: CPTG compact-state, structural, clock, and geometric-rotation results for GW241225_042553. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.2418	13.2853	9821.70	13281.52
500 m inward	7.2142	9.4751	10271.96	14416.61
100 m inward	3.2114	4.2475	10577.62	15123.68
10 m inward	1.0142	1.3434	10643.57	15250.04
1 m inward	0.3207	0.4248	10649.36	15260.59
10 cm inward	0.1014	0.1343	10650.08	15261.33
1 cm inward	0.0321	0.0425	10650.15	15261.47
surface limit	0.0000	0.0000	10650.16	15261.49

Table 91: Near-inner-boundary profile for GW241225_042553. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.847 M_\odot$ and $5.367 M_\odot$, with completed geometric mass-equivalent shares of 26.84% and 35.77%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.561 < 12.354 < 18.130$ km, and component 2 gives $6.266 < 7.819 < 11.894$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6457$ and 1.7435 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.755 and 14.916 h. The determinant-averaged frame-dragging rates at those shells are 7163.72 and 10223.47 rad s $^{-1}$, approaching surface-limit values of 10650.16 and 15261.49 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.242/13.285 h at 1 km inward to 0.0321/0.0425 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.31 GW231113_200417

GW231113_200417 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.169 M_\odot$ and $m_2 = 9.058 M_\odot$, while the luminosity-distance median is 1162.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	28,120
Observed luminosity distance	1162.2 Mpc (median)
Detector-frame masses	$m_1 = 14.169 M_\odot$, $m_2 = 9.058 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.639$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2264$; $z_\pi = 0.2435$ (medians)
Post-handoff reorder fraction	16.02%

Table 92: Observational and coordinate summary for GW231113_200417. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.323 [7.892, 16.475]	7.335 [5.283, 10.493]
Physical compact-source mass $M_c (M_\odot)$	7.944 [5.527, 13.657]	5.009 [3.240, 6.937]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.091 [0.424, 5.730]	2.495 [0.312, 4.872]
Geometric share M_{geom}/M_∞ (%)	26.83 [4.29, 47.06]	34.84 [4.87, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.498 [0.128, 0.915]	0.359 [0.032, 0.904]
Polarization factor A_{pol}	1.367 [1.045, 1.889]	1.535 [1.051, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.930 [0.458, 0.999]	0.845 [0.128, 0.999]
GR/CPTG matching radius r_m (km)	16.721 [11.654, 24.327]	10.832 [7.801, 15.494]
Structural Mode N_{BH}	1.6456 [1.4710, 1.8783]	1.7327 [1.4743, 1.9099]
Organization-shell radius r_N (km)	8.819 [6.122, 12.833]	5.702 [4.107, 8.147]
Inner trapping-surface radius r_- (km)	11.161 [8.188, 18.164]	7.285 [5.054, 9.711]
Structural- N lapse $\bar{\alpha}_N$	0.6982 [0.5103, 0.8345]	0.6301 [0.4848, 0.8318]
Structural- N clock $T_{24,N}$ (h)	16.758 [12.246, 20.027]	15.122 [11.635, 19.964]
Structural- N axis ω (rad s $^{-1}$)	6281.98 [2445.10, 10197.81]	9128.99 [2140.84, 15490.35]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7549.43 [3589.18, 11426.14]	10823.02 [2142.09, 17241.52]
Structural- N equatorial ω (rad s $^{-1}$)	8878.86 [4811.18, 12777.85]	12856.73 [2142.72, 19177.41]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8323 [0.5499, 1.7506]	0.5805 [0.3644, 2.9332]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11303.24 [5897.38, 16764.11]	16637.08 [2335.20, 24679.26]
Trapping-surface period P_- (ms)	0.5559 [0.3748, 1.0654]	0.3777 [0.2546, 2.6906]

Table 93: CPTG compact-state, structural, clock, and geometric-rotation results for GW231113_200417. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.6763	13.6209	10286.22	14229.28
500 m inward	7.5110	9.7050	10849.59	15648.38
100 m inward	3.3394	4.3471	11221.23	16459.77
10 m inward	1.0539	1.3745	11295.22	16618.70
1 m inward	0.3332	0.4347	11302.32	16635.05
10 cm inward	0.1054	0.1374	11303.15	16636.88
1 cm inward	0.0333	0.0435	11303.23	16637.06
surface limit	0.0000	0.0000	11303.24	16637.08

Table 94: Near-inner-boundary profile for GW231113_200417. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.944 M_\odot$ and $5.009 M_\odot$, with completed geometric mass-equivalent shares of 26.83% and 34.84%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.819 < 11.161 < 16.721$ km, and component 2 gives $5.702 < 7.285 < 10.832$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6456$ and 1.7327 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.758 and 15.122 h. The determinant-averaged frame-dragging rates at those shells are 7549.43 and 10823.02 rad s $^{-1}$, approaching surface-limit values of 11303.24 and 16637.08 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.676/13.621 h at 1 km inward to 0.0333/0.0435 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.32 GW240629_145256

GW240629_145256 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 13.681 M_\odot$ and $m_2 = 9.664 M_\odot$, while the luminosity-distance median is 1275.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,012
Observed luminosity distance	1275.1 Mpc (median)
Detector-frame masses	$m_1 = 13.681 M_\odot$, $m_2 = 9.664 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.706$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2458$; $z_\pi = 0.2643$ (medians)
Post-handoff reorder fraction	16.56%

Table 95: Observational and coordinate summary for GW240629_145256. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	10.756 [7.923, 13.871]	7.803 [6.131, 10.493]
Physical compact-source mass M_c ($M_\odot$)	6.723 [5.072, 9.803]	4.923 [3.632, 6.840]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.197 [0.633, 6.177]	3.059 [0.559, 4.912]
Geometric share M_{geom}/M_∞ (%)	38.28 [7.22, 49.81]	39.80 [8.17, 49.90]
Exterior spin magnitude $ \hat{a}_\infty $	0.299 [0.031, 0.858]	0.272 [0.022, 0.840]
Polarization factor A_{pol}	1.620 [1.078, 1.992]	1.661 [1.089, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.784 [0.124, 0.997]	0.750 [0.088, 0.996]
GR/CPTG matching radius r_m (km)	15.883 [11.700, 20.482]	11.521 [9.054, 15.493]
Structural Mode N_{BH}	1.7731 [1.4881, 1.9100]	1.7913 [1.4939, 1.9111]
Organization-shell radius r_N (km)	8.367 [6.151, 10.800]	6.068 [4.764, 8.142]
Inner trapping-surface radius r_- (km)	10.043 [8.019, 13.411]	7.363 [5.783, 9.678]
Structural- N lapse $\bar{\alpha}_N$	0.5976 [0.4847, 0.8208]	0.5827 [0.4839, 0.8161]
Structural- N clock $T_{24,N}$ (h)	14.342 [11.633, 19.699]	13.984 [11.613, 19.587]
Structural- N axis ω (rad s $^{-1}$)	6761.83 [1572.61, 10302.31]	8949.19 [1461.66, 14080.69]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7711.77 [1573.90, 11462.93]	10337.09 [1462.19, 15561.82]
Structural- N equatorial ω (rad s $^{-1}$)	8753.53 [1574.55, 12769.66]	11962.57 [1462.45, 16986.93]
Structural- N period P_ω (ms)	0.8148 [0.5481, 3.9921]	0.6078 [0.4038, 4.2971]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10742.52 [1715.91, 16782.49]	14800.24 [1593.34, 21620.79]
Trapping-surface period P_- (ms)	0.5849 [0.3744, 3.6617]	0.4245 [0.2906, 3.9434]

Table 96: CPTG compact-state, structural, clock, and geometric-rotation results for GW240629_145256. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.8658	13.8251	9691.81	12582.87
500 m inward	8.4461	9.8940	10274.98	13841.59
100 m inward	3.7885	4.4508	10655.03	14631.79
10 m inward	1.1986	1.4089	10734.47	14783.62
1 m inward	0.3791	0.4455	10742.08	14799.24
10 cm inward	0.1199	0.1409	10742.48	14800.19
1 cm inward	0.0379	0.0446	10742.52	14800.24
surface limit	0.0000	0.0000	10742.52	14800.24

Table 97: Near-inner-boundary profile for GW240629_145256. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $6.723 M_\odot$ and $4.923 M_\odot$, with completed geometric mass-equivalent shares of 38.28% and 39.80%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.367 < 10.043 < 15.883$ km, and component 2 gives $6.068 < 7.363 < 11.521$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7731$ and 1.7913 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.342 and 13.984 h. The determinant-averaged frame-dragging rates at those shells are 7711.77 and 10337.09 rad s $^{-1}$, approaching surface-limit values of 10742.52 and 14800.24 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.866/13.825 h at 1 km inward to 0.0379/0.0446 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.33 GW190707_093326

GW190707_093326 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.070 M_\odot$ and $m_2 = 9.285 M_\odot$, while the luminosity-distance median is 851.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	19,897
Observed luminosity distance	851.8 Mpc (median)
Detector-frame masses	$m_1 = 14.070 M_\odot$, $m_2 = 9.285 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.660$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1716$; $z_\pi = 0.1843$ (medians)
Post-handoff reorder fraction	14.95%

Table 98: Observational and coordinate summary for GW190707_093326. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.837 [8.249, 14.506]	7.918 [6.516, 11.427]
Physical compact-source mass $M_c (M_\odot)$	7.054 [5.718, 9.951]	5.015 [3.621, 7.056]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.850 [0.710, 6.676]	3.190 [0.917, 5.356]
Geometric share M_{geom}/M_∞ (%)	40.55 [7.98, 49.89]	40.93 [12.39, 49.92]
Exterior spin magnitude $ \hat{a}_\infty $	0.258 [0.023, 0.844]	0.252 [0.020, 0.760]
Polarization factor A_{pol}	1.682 [1.087, 1.996]	1.693 [1.141, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.731 [0.093, 0.996]	0.721 [0.081, 0.990]
GR/CPTG matching radius r_m (km)	17.478 [12.180, 21.420]	11.692 [9.621, 16.874]
Structural Mode N_{BH}	1.8004 [1.4927, 1.9110]	1.8049 [1.5219, 1.9113]
Organization-shell radius r_N (km)	9.205 [6.386, 11.277]	6.162 [5.064, 8.872]
Inner trapping-surface radius r_- (km)	10.763 [8.793, 13.703]	7.448 [5.863, 10.238]
Structural- N lapse $\bar{\alpha}_N$	0.5752 [0.4840, 0.8171]	0.5714 [0.4838, 0.7940]
Structural- N clock $T_{24,N}$ (h)	13.804 [11.615, 19.610]	13.714 [11.610, 19.056]
Structural- N axis ω (rad s $^{-1}$)	6102.68 [1055.01, 9306.52]	9099.96 [1317.32, 13737.77]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6991.71 [1059.30, 10489.07]	10301.37 [1317.83, 14985.79]
Structural- N equatorial ω (rad s $^{-1}$)	7971.05 [1059.43, 11924.17]	11607.75 [1318.10, 16150.28]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8987 [0.5990, 5.9315]	0.6099 [0.4193, 4.7678]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9602.15 [1154.02, 15914.95]	14179.61 [1436.18, 20249.96]
Trapping-surface period P_- (ms)	0.6544 [0.3948, 5.4446]	0.4431 [0.3103, 4.3749]

Table 99: CPTG compact-state, structural, clock, and geometric-rotation results for GW190707_093326. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.5889	13.7490	8673.22	12008.76
500 m inward	8.2553	9.8398	9178.95	13205.38
100 m inward	3.7062	4.4250	9521.84	14009.23
10 m inward	1.1727	1.4004	9596.03	14162.73
1 m inward	0.3709	0.4429	9601.34	14177.86
10 cm inward	0.1173	0.1401	9602.07	14179.43
1 cm inward	0.0371	0.0443	9602.14	14179.59
surface limit	0.0000	0.0000	9602.15	14179.61

Table 100: Near-inner-boundary profile for GW190707_093326. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.054 M_\odot$ and $5.015 M_\odot$, with completed geometric mass-equivalent shares of 40.55% and 40.93%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.205 < 10.763 < 17.478$ km, and component 2 gives $6.162 < 7.448 < 11.692$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8004$ and 1.8049 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.804 and 13.714 h. The determinant-averaged frame-dragging rates at those shells are 6991.71 and 10301.37 rad s $^{-1}$, approaching surface-limit values of 9602.15 and 14179.61 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.589/13.749 h at 1 km inward to 0.0371/0.0443 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.34 GW191103_012549

GW191103_012549 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.029 M_\odot$ and $m_2 = 9.421 M_\odot$, while the luminosity-distance median is 988.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	9,859
Observed luminosity distance	988.1 Mpc (median)
Detector-frame masses	$m_1 = 14.029 M_\odot$, $m_2 = 9.421 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.672$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1962$; $z_\pi = 0.2107$ (medians)
Post-handoff reorder fraction	19.32%

Table 101: Observational and coordinate summary for GW191103_012549. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.442 [8.100, 17.779]	7.904 [5.380, 11.194]
Physical compact-source mass $M_c (M_\odot)$	8.517 [6.171, 14.317]	5.334 [3.362, 7.338]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	2.843 [0.409, 5.555]	2.696 [0.360, 5.249]
Geometric share M_{geom}/M_∞ (%)	23.57 [4.11, 43.08]	35.42 [5.26, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.556 [0.212, 0.919]	0.349 [0.033, 0.896]
Polarization factor A_{pol}	1.308 [1.043, 1.757]	1.549 [1.056, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.951 [0.654, 0.999]	0.836 [0.130, 0.998]
GR/CPTG matching radius r_m (km)	16.895 [11.961, 26.253]	11.671 [7.944, 16.529]
Structural Mode N_{BH}	1.6138 [1.4700, 1.8309]	1.7394 [1.4765, 1.9098]
Organization-shell radius r_N (km)	8.919 [6.282, 13.862]	6.151 [4.172, 8.682]
Inner trapping-surface radius r_- (km)	11.714 [8.804, 19.139]	7.778 [5.161, 10.244]
Structural- N lapse $\bar{\alpha}_N$	0.7227 [0.5498, 0.8353]	0.6247 [0.4849, 0.8300]
Structural- N clock $T_{24,N}$ (h)	17.346 [13.196, 20.046]	14.993 [11.637, 19.921]
Structural- N axis ω (rad s $^{-1}$)	5835.13 [2647.81, 9464.48]	8691.64 [2075.09, 15139.00]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7303.86 [3923.91, 10750.64]	10253.99 [2077.12, 16807.31]
Structural- N equatorial ω (rad s $^{-1}$)	8845.84 [5199.76, 12134.31]	11992.76 [2078.15, 18704.04]
Structural- N period P_ω (ms)	0.8603 [0.5844, 1.6013]	0.6128 [0.3738, 3.0249]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11578.53 [6860.66, 16173.14]	15376.93 [2266.43, 24126.51]
Trapping-surface period P_- (ms)	0.5427 [0.3885, 0.9158]	0.4086 [0.2604, 2.7723]

Table 102: CPTG compact-state, structural, clock, and geometric-rotation results for GW191103_012549. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.1466	13.1763	10593.79	13358.25
500 m inward	7.1066	9.3878	11132.09	14550.39
100 m inward	3.1427	4.2033	11499.40	15237.14
10 m inward	0.9918	1.3291	11570.12	15364.28
1 m inward	0.3135	0.4203	11577.67	15376.68
10 cm inward	0.0992	0.1329	11578.44	15376.95
1 cm inward	0.0314	0.0420	11578.52	15376.93
surface limit	0.0000	0.0000	11578.53	15376.93

Table 103: Near-inner-boundary profile for GW191103_012549. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.517 M_\odot$ and $5.334 M_\odot$, with completed geometric mass-equivalent shares of 23.57% and 35.42%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.919 < 11.714 < 16.895$ km, and component 2 gives $6.151 < 7.778 < 11.671$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6138$ and 1.7394; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.346 and 14.993 h. The determinant-averaged frame-dragging rates at those shells are 7303.86 and 10253.99 rad s $^{-1}$, approaching surface-limit values of 11578.53 and 15376.93 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.147/13.176 h at 1 km inward to 0.0314/0.0420 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.35 GW240922_142106

GW240922_142106 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.083 M_\odot$ and $m_2 = 9.544 M_\odot$, while the luminosity-distance median is 1231.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,650
Observed luminosity distance	1231.0 Mpc (median)
Detector-frame masses	$m_1 = 14.083 M_\odot$, $m_2 = 9.544 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.678$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2382$; $z_\pi = 0.2562$ (medians)
Post-handoff reorder fraction	16.74%

Table 104: Observational and coordinate summary for GW240922_142106. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.128 [7.957, 15.088]	7.738 [5.878, 10.841]
Physical compact-source mass $M_c (M_\odot)$	7.309 [5.295, 10.946]	5.083 [3.509, 7.021]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.926 [0.545, 6.058]	2.859 [0.437, 5.057]
Geometric share M_{geom}/M_∞ (%)	33.85 [6.06, 49.36]	37.76 [6.35, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.376 [0.057, 0.881]	0.308 [0.027, 0.875]
Polarization factor A_{pol}	1.512 [1.065, 1.975]	1.607 [1.068, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.859 [0.224, 0.998]	0.795 [0.109, 0.998]
GR/CPTG matching radius r_m (km)	16.432 [11.749, 22.279]	11.427 [8.680, 16.008]
Structural Mode N_{BH}	1.7213 [1.4812, 1.9049]	1.7669 [1.4828, 1.9105]
Organization-shell radius r_N (km)	8.663 [6.163, 11.775]	6.026 [4.567, 8.423]
Inner trapping-surface radius r_- (km)	10.602 [8.211, 14.982]	7.511 [5.565, 9.912]
Structural- N lapse $\bar{\alpha}_N$	0.6391 [0.4887, 0.8263]	0.6026 [0.4843, 0.8250]
Structural- N clock $T_{24,N}$ (h)	15.339 [11.729, 19.831]	14.462 [11.624, 19.799]
Structural- N axis ω (rad s $^{-1}$)	6707.10 [2514.89, 10035.07]	8903.08 [1780.36, 14336.93]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7682.08 [2800.43, 11246.43]	10363.46 [1781.44, 15826.35]
Structural- N equatorial ω (rad s $^{-1}$)	8803.69 [2811.70, 12538.33]	12144.72 [1781.77, 17311.27]
Structural- N period P_ω (ms)	0.8179 [0.5587, 2.2437]	0.6063 [0.3970, 3.5270]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11012.37 [3070.74, 16524.56]	15473.39 [1941.28, 22127.98]
Trapping-surface period P_- (ms)	0.5706 [0.3802, 2.0462]	0.4061 [0.2839, 3.2366]

Table 105: CPTG compact-state, structural, clock, and geometric-rotation results for GW240922_142106. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.3330	13.5893	9990.85	13292.70
500 m inward	8.0303	9.7045	10545.96	14539.19
100 m inward	3.5891	4.3575	10930.85	15322.41
10 m inward	1.1348	1.3787	11004.25	15458.83
1 m inward	0.3588	0.4360	11011.90	15472.02
10 cm inward	0.1135	0.1379	11012.33	15473.23
1 cm inward	0.0359	0.0436	11012.37	15473.37
surface limit	0.0000	0.0000	11012.37	15473.39

Table 106: Near-inner-boundary profile for GW240922_142106. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.309 M_\odot$ and $5.083 M_\odot$, with completed geometric mass-equivalent shares of 33.85% and 37.76%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.663 < 10.602 < 16.432$ km, and component 2 gives $6.026 < 7.511 < 11.427$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7213$ and 1.7669 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.339 and 14.462 h. The determinant-averaged frame-dragging rates at those shells are 7682.08 and 10363.46 rad s $^{-1}$, approaching surface-limit values of 11012.37 and 15473.39 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.333/13.589 h at 1 km inward to 0.0359/0.0436 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.36 GW151226_033853

GW151226_033853 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 15.511 M_\odot$ and $m_2 = 8.170 M_\odot$, while the luminosity-distance median is 460.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	133,905
Observed luminosity distance	460.6 Mpc (median)
Detector-frame masses	$m_1 = 15.511 M_\odot$, $m_2 = 8.170 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.527$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0973$; $z_\pi = 0.1046$ (medians)
Post-handoff reorder fraction	9.20%

Table 107: Observational and coordinate summary for GW151226_033853. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	14.077 [9.459, 25.093]	7.409 [4.666, 10.873]
Physical compact-source mass $M_c (M_\odot)$	10.950 [6.981, 23.082]	5.074 [2.932, 7.524]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	2.609 [0.342, 5.663]	2.310 [0.316, 4.912]
Geometric share M_{geom}/M_∞ (%)	18.32 [2.28, 40.10]	33.09 [4.95, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.650 [0.267, 0.955]	0.389 [0.037, 0.902]
Polarization factor A_{pol}	1.224 [1.023, 1.669]	1.494 [1.052, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.975 [0.743, 1.000]	0.869 [0.147, 0.999]
GR/CPTG matching radius r_m (km)	20.786 [13.967, 37.053]	10.940 [6.889, 16.055]
Structural Mode N_{BH}	1.5673 [1.4602, 1.7949]	1.7127 [1.4747, 1.9092]
Organization-shell radius r_N (km)	10.955 [7.343, 19.372]	5.759 [3.628, 8.456]
Inner trapping-surface radius r_- (km)	14.824 [9.942, 30.671]	7.352 [4.495, 10.424]
Structural- N lapse $\bar{\alpha}_N$	0.7586 [0.5797, 0.8432]	0.6460 [0.4854, 0.8315]
Structural- N clock $T_{24,N}$ (h)	18.206 [13.913, 20.237]	15.503 [11.649, 19.955]
Structural- N axis ω (rad s $^{-1}$)	4186.79 [1329.52, 8308.78]	9092.08 [2528.93, 17238.46]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	5635.97 [2134.84, 9438.64]	10763.15 [2530.02, 19276.35]
Structural- N equatorial ω (rad s $^{-1}$)	7073.78 [3317.53, 10685.68]	12664.42 [2530.81, 21433.13]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.1148 [0.6657, 2.9432]	0.5838 [0.3260, 2.4835]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9310.40 [4731.83, 14091.54]	16345.34 [2759.80, 27596.88]
Trapping-surface period P_- (ms)	0.6749 [0.4459, 1.3279]	0.3844 [0.2277, 2.2767]

Table 108: CPTG compact-state, structural, clock, and geometric-rotation results for GW151226_033853. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.6932	13.4639	8708.86	14081.33
500 m inward	6.0663	9.5692	9044.25	15396.35
100 m inward	2.6786	4.2765	9261.65	16185.67
10 m inward	0.8445	1.3514	9305.26	16331.78
1 m inward	0.2670	0.4273	9309.93	16344.38
10 cm inward	0.0844	0.1351	9310.35	16345.19
1 cm inward	0.0267	0.0427	9310.40	16345.33
surface limit	0.0000	0.0000	9310.40	16345.34

Table 109: Near-inner-boundary profile for GW151226_033853. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $10.950 M_\odot$ and $5.074 M_\odot$, with completed geometric mass-equivalent shares of 18.32% and 33.09%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $10.955 < 14.824 < 20.786$ km, and component 2 gives $5.759 < 7.352 < 10.940$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5673$ and 1.7127; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.206 and 15.503 h. The determinant-averaged frame-dragging rates at those shells are 5635.97 and 10763.15 rad s $^{-1}$, approaching surface-limit values of 9310.40 and 16345.34 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.693/13.464 h at 1 km inward to 0.0267/0.0427 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.37 GW231020_142947

GW231020_142947 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.670 M_\odot$ and $m_2 = 9.054 M_\odot$, while the luminosity-distance median is 1230.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	33,482
Observed luminosity distance	1230.1 Mpc (median)
Detector-frame masses	$m_1 = 14.670 M_\odot$, $m_2 = 9.054 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.617$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2381$; $z_\pi = 0.2560$ (medians)
Post-handoff reorder fraction	14.18%

Table 110: Observational and coordinate summary for GW231020_142947. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.807 [8.218, 20.166]	7.296 [4.487, 10.447]
Physical compact-source mass $M_c (M_\odot)$	8.190 [5.535, 15.507]	4.829 [2.850, 6.952]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.869 [0.627, 6.129]	2.380 [0.309, 4.876]
Geometric share M_{geom}/M_∞ (%)	29.11 [6.58, 47.96]	35.42 [5.24, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.458 [0.105, 0.871]	0.349 [0.031, 0.897]
Polarization factor A_{pol}	1.411 [1.070, 1.922]	1.548 [1.055, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.912 [0.388, 0.998]	0.836 [0.123, 0.998]
GR/CPTG matching radius r_m (km)	17.434 [12.136, 29.777]	10.774 [6.626, 15.427]
Structural Mode N_{BH}	1.6691 [1.4842, 1.8888]	1.7394 [1.4764, 1.9101]
Organization-shell radius r_N (km)	9.195 [6.390, 15.745]	5.677 [3.485, 8.117]
Inner trapping-surface radius r_- (km)	11.537 [8.378, 20.879]	7.144 [4.366, 9.737]
Structural- N lapse $\bar{\alpha}_N$	0.6800 [0.5016, 0.8239]	0.6248 [0.4847, 0.8301]
Structural- N clock $T_{24,N}$ (h)	16.320 [12.038, 19.773]	14.994 [11.632, 19.924]
Structural- N axis ω (rad s $^{-1}$)	6013.61 [2610.57, 9922.11]	9369.51 [2104.23, 17978.08]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7106.45 [3494.32, 11096.74]	11095.00 [2105.65, 20112.32]
Structural- N equatorial ω (rad s $^{-1}$)	8265.65 [4209.00, 12383.91]	13024.73 [2106.34, 22345.98]
Structural- N period P_ω (ms)	0.8842 [0.5662, 1.7981]	0.5663 [0.3124, 2.9840]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10378.83 [5131.67, 16250.24]	16658.87 [2295.95, 28793.54]
Trapping-surface period P_- (ms)	0.6054 [0.3867, 1.2244]	0.3772 [0.2182, 2.7366]

Table 111: CPTG compact-state, structural, clock, and geometric-rotation results for GW231020_142947. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.5160	13.7858	9478.47	14170.57
500 m inward	7.4075	9.8242	9973.19	15608.77
100 m inward	3.2933	4.4005	10306.18	16489.14
10 m inward	1.0399	1.3911	10371.80	16640.84
1 m inward	0.3288	0.4399	10378.15	16656.99
10 cm inward	0.1040	0.1391	10378.76	16658.70
1 cm inward	0.0329	0.0440	10378.82	16658.85
surface limit	0.0000	0.0000	10378.83	16658.87

Table 112: Near-inner-boundary profile for GW231020_142947. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.190 M_\odot$ and $4.829 M_\odot$, with completed geometric mass-equivalent shares of 29.11% and 35.42%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.195 < 11.537 < 17.434$ km, and component 2 gives $5.677 < 7.144 < 10.774$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6691$ and 1.7394 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.320 and 14.994 h. The determinant-averaged frame-dragging rates at those shells are 7106.45 and 11095.00 rad s $^{-1}$, approaching surface-limit values of 10378.83 and 16658.87 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.516/13.786 h at 1 km inward to 0.0329/0.0440 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.38 GW190728_064510

GW190728_064510 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.557 M_\odot$ and $m_2 = 9.391 M_\odot$, while the luminosity-distance median is 878.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	11,756
Observed luminosity distance	878.0 Mpc (median)
Detector-frame masses	$m_1 = 14.557 M_\odot$, $m_2 = 9.391 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.645$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1764$; $z_\pi = 0.1894$ (medians)
Post-handoff reorder fraction	15.16%

Table 113: Observational and coordinate summary for GW190728_064510. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.214 [8.647, 19.140]	7.995 [5.407, 11.271]
Physical compact-source mass $M_c (M_\odot)$	7.976 [5.801, 14.436]	5.151 [3.255, 7.030]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.295 [1.045, 6.490]	2.986 [0.530, 5.320]
Geometric share M_{geom}/M_∞ (%)	32.86 [10.13, 48.58]	39.01 [8.16, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.393 [0.087, 0.803]	0.286 [0.027, 0.840]
Polarization factor A_{pol}	1.489 [1.113, 1.945]	1.640 [1.089, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.872 [0.328, 0.994]	0.769 [0.109, 0.996]
GR/CPTG matching radius r_m (km)	18.035 [12.768, 28.262]	11.806 [7.984, 16.643]
Structural Mode N_{BH}	1.7101 [1.5065, 1.8960]	1.7818 [1.4938, 1.9105]
Organization-shell radius r_N (km)	9.522 [6.735, 14.957]	6.220 [4.198, 8.747]
Inner trapping-surface radius r_- (km)	11.544 [8.800, 19.550]	7.684 [5.082, 10.149]
Structural- N lapse $\bar{\alpha}_N$	0.6479 [0.4958, 0.8062]	0.5904 [0.4843, 0.8162]
Structural- N clock $T_{24,N}$ (h)	15.551 [11.899, 19.348]	14.170 [11.624, 19.588]
Structural- N axis ω (rad s $^{-1}$)	6393.07 [2473.68, 9627.25]	8917.97 [1719.97, 15211.80]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7263.94 [2997.00, 10684.23]	10321.96 [1721.45, 16862.28]
Structural- N equatorial ω (rad s $^{-1}$)	8123.50 [3174.25, 11939.78]	11874.13 [1722.20, 18546.80]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8650 [0.5881, 2.0965]	0.6087 [0.3726, 3.6499]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9940.31 [3645.91, 15637.20]	14811.14 [1877.84, 23883.32]
Trapping-surface period P_- (ms)	0.6321 [0.4018, 1.7234]	0.4242 [0.2631, 3.3460]

Table 114: CPTG compact-state, structural, clock, and geometric-rotation results for GW190728_064510. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.8735	13.4801	9063.72	12640.76
500 m inward	7.6911	9.6281	9537.13	13880.30
100 m inward	3.4323	4.3245	9870.14	14650.24
10 m inward	1.0842	1.3680	9932.21	14793.21
1 m inward	0.3428	0.4326	9939.57	14809.02
10 cm inward	0.1084	0.1368	9940.25	14810.93
1 cm inward	0.0343	0.0433	9940.31	14811.12
surface limit	0.0000	0.0000	9940.31	14811.14

Table 115: Near-inner-boundary profile for GW190728_064510. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.976 M_\odot$ and $5.151 M_\odot$, with completed geometric mass-equivalent shares of 32.86% and 39.01%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.522 < 11.544 < 18.035$ km, and component 2 gives $6.220 < 7.684 < 11.806$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7101$ and 1.7818 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.551 and 14.170 h. The determinant-averaged frame-dragging rates at those shells are 7263.94 and 10321.96 rad s $^{-1}$, approaching surface-limit values of 9940.31 and 14811.14 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.874/13.480 h at 1 km inward to 0.0343/0.0433 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.39 GW240830_211120

GW240830_211120 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.454 M_\odot$ and $m_2 = 9.548 M_\odot$, while the luminosity-distance median is 1143.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,084
Observed luminosity distance	1143.2 Mpc (median)
Detector-frame masses	$m_1 = 14.454 M_\odot$, $m_2 = 9.548 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.661$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2231$; $z_\pi = 0.2399$ (medians)
Post-handoff reorder fraction	16.03%

Table 116: Observational and coordinate summary for GW240830_211120. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.576 [8.169, 15.268]	7.811 [6.092, 11.002]
Physical compact-source mass $M_c (M_\odot)$	7.285 [5.331, 10.814]	5.154 [3.625, 7.075]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.408 [0.660, 6.586]	2.887 [0.435, 5.118]
Geometric share M_{geom}/M_∞ (%)	37.24 [7.12, 49.75]	37.59 [6.26, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.317 [0.036, 0.860]	0.311 [0.028, 0.877]
Polarization factor A_{pol}	1.593 [1.077, 1.990]	1.602 [1.067, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.805 [0.141, 0.997]	0.798 [0.110, 0.998]
GR/CPTG matching radius r_m (km)	17.093 [12.063, 22.544]	11.534 [8.995, 16.246]
Structural Mode N_{BH}	1.7607 [1.4874, 1.9094]	1.7649 [1.4823, 1.9105]
Organization-shell radius r_N (km)	9.004 [6.349, 11.903]	6.080 [4.732, 8.533]
Inner trapping-surface radius r_- (km)	10.795 [8.338, 14.868]	7.586 [5.765, 10.084]
Structural- N lapse $\bar{\alpha}_N$	0.6076 [0.4852, 0.8213]	0.6042 [0.4844, 0.8254]
Structural- N clock $T_{24,N}$ (h)	14.582 [11.645, 19.711]	14.502 [11.625, 19.810]
Structural- N axis ω (rad s $^{-1}$)	6345.76 [1652.25, 9879.18]	8653.91 [1778.34, 14077.57]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7270.99 [1665.77, 11002.28]	10199.67 [1779.06, 15534.66]
Structural- N equatorial ω (rad s $^{-1}$)	8214.20 [1666.37, 12262.00]	12078.93 [1779.43, 16959.83]
Structural- N period P_ω (ms)	0.8641 [0.5711, 3.7719]	0.6160 [0.4045, 3.5317]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10091.92 [1816.54, 16166.17]	15408.17 [1938.89, 21775.00]
Trapping-surface period P_- (ms)	0.6226 [0.3887, 3.4589]	0.4078 [0.2886, 3.2406]

Table 117: CPTG compact-state, structural, clock, and geometric-rotation results for GW240830_211120. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.3735	13.5087	9167.65	13213.18
500 m inward	8.0779	9.6332	9669.67	14465.95
100 m inward	3.6180	4.3251	10013.74	15243.02
10 m inward	1.1443	1.3680	10083.27	15392.68
1 m inward	0.3619	0.4326	10091.24	15406.92
10 cm inward	0.1144	0.1368	10091.86	15408.06
1 cm inward	0.0362	0.0433	10091.91	15408.16
surface limit	0.0000	0.0000	10091.92	15408.17

Table 118: Near-inner-boundary profile for GW240830_211120. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.285 M_\odot$ and $5.154 M_\odot$, with completed geometric mass-equivalent shares of 37.24% and 37.59%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.004 < 10.795 < 17.093$ km, and component 2 gives $6.080 < 7.586 < 11.534$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7607$ and 1.7649 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.582 and 14.502 h. The determinant-averaged frame-dragging rates at those shells are 7270.99 and 10199.67 rad s $^{-1}$, approaching surface-limit values of 10091.92 and 15408.17 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.373/13.509 h at 1 km inward to 0.0362/0.0433 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.40 GW231018_233037

GW231018_233037 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 14.793 M_\odot$ and $m_2 = 9.349 M_\odot$, while the luminosity-distance median is 1536.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	32,552
Observed luminosity distance	1536.0 Mpc (median)
Detector-frame masses	$m_1 = 14.793 M_\odot$, $m_2 = 9.349 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.632$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2893$; $z_\pi = 0.3111$ (medians)
Post-handoff reorder fraction	14.26%

Table 119: Observational and coordinate summary for GW231018_233037. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.273 [7.871, 16.426]	7.204 [5.096, 10.275]
Physical compact-source mass $M_c (M_\odot)$	7.156 [5.191, 11.394]	4.745 [3.132, 6.787]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.298 [0.581, 6.854]	2.593 [0.374, 4.769]
Geometric share M_{geom}/M_∞ (%)	36.97 [6.37, 49.81]	37.09 [5.95, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.322 [0.031, 0.875]	0.320 [0.026, 0.883]
Polarization factor A_{pol}	1.587 [1.068, 1.992]	1.590 [1.063, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.810 [0.124, 0.998]	0.808 [0.105, 0.998]
GR/CPTG matching radius r_m (km)	16.647 [11.623, 24.255]	10.638 [7.525, 15.172]
Structural Mode N_{BH}	1.7576 [1.4829, 1.9101]	1.7590 [1.4805, 1.9107]
Organization-shell radius r_N (km)	8.765 [6.105, 12.792]	5.603 [3.960, 7.979]
Inner trapping-surface radius r_- (km)	10.613 [8.027, 15.807]	6.990 [4.865, 9.544]
Structural- N lapse $\bar{\alpha}_N$	0.6101 [0.4847, 0.8249]	0.6090 [0.4842, 0.8269]
Structural- N clock $T_{24,N}$ (h)	14.643 [11.633, 19.797]	14.616 [11.621, 19.845]
Structural- N axis ω (rad s $^{-1}$)	6111.12 [1411.73, 10179.13]	9441.19 [1830.20, 16158.96]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7140.11 [1441.84, 11410.47]	11021.34 [1831.03, 17921.67]
Structural- N equatorial ω (rad s $^{-1}$)	8253.10 [1442.20, 12794.22]	12939.26 [1831.30, 19813.80]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8800 [0.5507, 4.3578]	0.5701 [0.3506, 3.4315]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10191.31 [1571.63, 16868.86]	16466.21 [1994.99, 25440.01]
Trapping-surface period P_- (ms)	0.6165 [0.3725, 3.9979]	0.3816 [0.2470, 3.1495]

Table 120: CPTG compact-state, structural, clock, and geometric-rotation results for GW231018_233037. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.4350	13.9914	9240.13	13952.75
500 m inward	8.1157	9.9941	9758.86	15406.83
100 m inward	3.6345	4.4843	10110.70	16284.89
10 m inward	1.1491	1.4178	10184.65	16449.06
1 m inward	0.3634	0.4483	10190.59	16464.90
10 cm inward	0.1149	0.1418	10191.24	16466.05
1 cm inward	0.0363	0.0448	10191.31	16466.20
surface limit	0.0000	0.0000	10191.31	16466.21

Table 121: Near-inner-boundary profile for GW231018_233037. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.156 M_\odot$ and $4.745 M_\odot$, with completed geometric mass-equivalent shares of 36.97% and 37.09%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.765 < 10.613 < 16.647$ km, and component 2 gives $5.603 < 6.990 < 10.638$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7576$ and 1.7590 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.643 and 14.616 h. The determinant-averaged frame-dragging rates at those shells are 7140.11 and 11021.34 rad s $^{-1}$, approaching surface-limit values of 10191.31 and 16466.21 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.435/13.991 h at 1 km inward to 0.0363/0.0448 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.41 GW190930_133541

GW190930_133541 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 16.362 M_\odot$ and $m_2 = 7.983 M_\odot$, while the luminosity-distance median is 768.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C01:Mixed
Posterior samples	34,482
Observed luminosity distance	768.2 Mpc (median)
Detector-frame masses	$m_1 = 16.362 M_\odot$, $m_2 = 7.983 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.488$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1562$; $z_\pi = 0.1678$ (medians)
Post-handoff reorder fraction	8.34%

Table 122: Observational and coordinate summary for GW190930_133541. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	14.053 [9.003, 21.954]	6.843 [4.781, 10.464]
Physical compact-source mass $M_c (M_\odot)$	9.196 [6.197, 17.469]	4.874 [2.858, 6.928]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.654 [0.856, 6.750]	2.118 [0.308, 4.752]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	30.74 [7.85, 49.15]	32.47 [4.98, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.430 [0.066, 0.846]	0.400 [0.037, 0.902]
Polarization factor A_{pol}	1.444 [1.085, 1.967]	1.481 [1.052, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.896 [0.256, 0.996]	0.877 [0.145, 0.999]
GR/CPTG matching radius r_m (km)	20.751 [13.295, 32.418]	10.104 [7.060, 15.451]
Structural Mode N_{BH}	1.6867 [1.4919, 1.9026]	1.7058 [1.4749, 1.9092]
Organization-shell radius r_N (km)	10.950 [6.997, 17.105]	5.317 [3.718, 8.127]
Inner trapping-surface radius r_- (km)	13.179 [9.214, 23.365]	6.970 [4.472, 9.845]
Structural- N lapse $\bar{\alpha}_N$	0.6664 [0.4905, 0.8177]	0.6514 [0.4854, 0.8313]
Structural- N clock $T_{24,N}$ (h)	15.993 [11.773, 19.625]	15.633 [11.649, 19.952]
Structural- N axis ω (rad s $^{-1}$)	5442.05 [1890.65, 8718.67]	9388.15 [2654.77, 17398.07]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6337.17 [2337.78, 9804.10]	11343.50 [2659.58, 19308.05]
Structural- N equatorial ω (rad s $^{-1}$)	7089.61 [2639.52, 11140.57]	13593.27 [2660.17, 21127.31]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9915 [0.6409, 2.6877]	0.5539 [0.3254, 2.3625]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8791.45 [2990.14, 14780.34]	17956.81 [2900.94, 26770.75]
Trapping-surface period P_- (ms)	0.7147 [0.4251, 2.1013]	0.3499 [0.2347, 2.1659]

Table 123: CPTG compact-state, structural, clock, and geometric-rotation results for GW190930_133541. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.0699	13.7745	8148.66	15384.78
500 m inward	7.1098	9.7915	8497.68	16922.45
100 m inward	3.1708	4.3762	8738.29	17787.55
10 m inward	1.0018	1.3826	8785.74	17941.78
1 m inward	0.3168	0.4372	8790.83	17955.03
10 cm inward	0.1002	0.1383	8791.38	17956.55
1 cm inward	0.0317	0.0437	8791.44	17956.78
surface limit	0.0000	0.0000	8791.45	17956.81

Table 124: Near-inner-boundary profile for GW190930_133541. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $9.196 M_\odot$ and $4.874 M_\odot$, with completed geometric mass-equivalent shares of 30.74% and 32.47%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $10.950 < 13.179 < 20.751$ km, and component 2 gives $5.317 < 6.970 < 10.104$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6867$ and 1.7058 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.993 and 15.633 h. The determinant-averaged frame-dragging rates at those shells are 6337.17 and 11343.50 rad s $^{-1}$, approaching surface-limit values of 8791.45 and 17956.81 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.070/13.775 h at 1 km inward to 0.0317/0.0437 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.42 GW240520_213616

GW240520_213616 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 14.453 M_\odot$ and $m_2 = 10.114 M_\odot$, while the luminosity-distance median is 1402.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,432
Observed luminosity distance	1402.0 Mpc (median)
Detector-frame masses	$m_1 = 14.453 M_\odot$, $m_2 = 10.114 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.700$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2672$; $z_\pi = 0.2873$ (medians)
Post-handoff reorder fraction	17.30%

Table 125: Observational and coordinate summary for GW240520_213616. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	11.126 [8.231, 14.806]	7.933 [6.119, 10.702]
Physical compact-source mass M_c ($M_\odot$)	7.041 [5.312, 10.202]	5.053 [3.592, 7.123]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.251 [0.607, 6.667]	3.117 [0.546, 5.019]
Geometric share M_{geom}/M_∞ (%)	37.75 [6.60, 49.84]	39.81 [7.80, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.308 [0.029, 0.870]	0.272 [0.024, 0.847]
Polarization factor A_{pol}	1.607 [1.071, 1.994]	1.662 [1.085, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.795 [0.114, 0.998]	0.750 [0.095, 0.996]
GR/CPTG matching radius r_m (km)	16.428 [12.154, 21.863]	11.715 [9.035, 15.802]
Structural Mode N_{BH}	1.7668 [1.4843, 1.9104]	1.7915 [1.4916, 1.9110]
Organization-shell radius r_N (km)	8.653 [6.394, 11.514]	6.170 [4.757, 8.308]
Inner trapping-surface radius r_- (km)	10.473 [8.308, 14.042]	7.552 [5.736, 9.943]
Structural- N lapse $\bar{\alpha}_N$	0.6026 [0.4844, 0.8238]	0.5825 [0.4840, 0.8179]
Structural- N clock $T_{24,N}$ (h)	14.463 [11.627, 19.771]	13.980 [11.616, 19.631]
Structural- N axis ω (rad s $^{-1}$)	6267.00 [1339.81, 10013.70]	8709.22 [1532.44, 14035.89]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7283.11 [1340.33, 11149.65]	10053.93 [1532.92, 15467.64]
Structural- N equatorial ω (rad s $^{-1}$)	8469.00 [1340.59, 12419.25]	11617.08 [1533.16, 16867.98]
Structural- N period P_ω (ms)	0.8627 [0.5635, 4.6878]	0.6249 [0.4062, 4.0988]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10519.16 [1460.63, 16238.61]	14410.82 [1670.14, 21591.86]
Trapping-surface period P_- (ms)	0.5973 [0.3869, 4.3017]	0.4360 [0.2910, 3.7621]

Table 126: CPTG compact-state, structural, clock, and geometric-rotation results for GW240520_213616. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	11.5737	13.6355	9540.88	12345.54
500 m inward	8.2312	9.7512	10083.07	13508.70
100 m inward	3.6900	4.3846	10441.73	14260.01
10 m inward	1.1671	1.3876	10510.73	14397.08
1 m inward	0.3691	0.4388	10518.08	14409.73
10 cm inward	0.1167	0.1388	10519.06	14410.65
1 cm inward	0.0369	0.0439	10519.15	14410.80
surface limit	0.0000	0.0000	10519.16	14410.82

Table 127: Near-inner-boundary profile for GW240520_213616. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $7.041 M_\odot$ and $5.053 M_\odot$, with completed geometric mass-equivalent shares of 37.75% and 39.81%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $8.653 < 10.473 < 16.428$ km, and component 2 gives $6.170 < 7.552 < 11.715$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7668$ and 1.7915 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.463 and 13.980 h. The determinant-averaged frame-dragging rates at those shells are 7283.11 and 10053.93 rad s $^{-1}$, approaching surface-limit values of 10519.16 and 14410.82 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 11.574/13.636 h at 1 km inward to 0.0369/0.0439 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.43 GW240627_131622

GW240627_131622 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 16.566 M_\odot$ and $m_2 = 8.452 M_\odot$, while the luminosity-distance median is 1352.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,210
Observed luminosity distance	1352.9 Mpc (median)
Detector-frame masses	$m_1 = 16.566 M_\odot$, $m_2 = 8.452 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.510$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2589$; $z_\pi = 0.2784$ (medians)
Post-handoff reorder fraction	8.90%

Table 128: Observational and coordinate summary for GW240627_131622. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.969 [8.518, 18.585]	6.622 [4.796, 10.015]
Physical compact-source mass $M_c (M_\odot)$	8.208 [5.657, 14.081]	4.588 [2.882, 6.698]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.590 [0.708, 7.361]	2.207 [0.279, 4.569]
Geometric share M_{geom}/M_∞ (%)	34.51 [6.83, 49.73]	34.26 [4.72, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.364 [0.037, 0.866]	0.369 [0.032, 0.907]
Polarization factor A_{pol}	1.527 [1.073, 1.989]	1.521 [1.050, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.850 [0.145, 0.997]	0.853 [0.128, 0.999]
GR/CPTG matching radius r_m (km)	19.150 [12.578, 27.443]	9.778 [7.082, 14.789]
Structural Mode N_{BH}	1.7289 [1.4857, 1.9092]	1.7260 [1.4734, 1.9099]
Organization-shell radius r_N (km)	10.095 [6.619, 14.500]	5.145 [3.723, 7.794]
Inner trapping-surface radius r_- (km)	12.154 [8.615, 19.072]	6.615 [4.538, 9.422]
Structural- N lapse $\bar{\alpha}_N$	0.6331 [0.4853, 0.8227]	0.6354 [0.4848, 0.8325]
Structural- N clock $T_{24,N}$ (h)	15.194 [11.648, 19.745]	15.249 [11.636, 19.980]
Structural- N axis ω (rad s $^{-1}$)	5475.74 [1548.86, 9354.45]	9753.70 [2378.97, 17486.28]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6339.95 [1575.47, 10486.54]	11651.91 [2380.91, 19390.03]
Structural- N equatorial ω (rad s $^{-1}$)	7136.77 [1576.53, 11876.28]	14007.13 [2381.48, 21230.31]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9910 [0.5992, 3.9881]	0.5392 [0.3240, 2.6390]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8887.84 [1718.82, 15583.60]	18110.35 [2594.35, 27264.02]
Trapping-surface period P_- (ms)	0.7069 [0.4032, 3.6555]	0.3469 [0.2305, 2.4219]

Table 129: CPTG compact-state, structural, clock, and geometric-rotation results for GW240627_131622. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.6712	14.1682	8218.25	15195.95
500 m inward	7.5587	10.0961	8581.54	16897.36
100 m inward	3.3790	4.5188	8831.55	17894.81
10 m inward	1.0684	1.4280	8882.40	18092.69
1 m inward	0.3378	0.4516	8887.29	18108.15
10 cm inward	0.1068	0.1428	8887.79	18110.17
1 cm inward	0.0338	0.0452	8887.84	18110.33
surface limit	0.0000	0.0000	8887.84	18110.35

Table 130: Near-inner-boundary profile for GW240627_131622. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.208 M_\odot$ and $4.588 M_\odot$, with completed geometric mass-equivalent shares of 34.51% and 34.26%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $10.095 < 12.154 < 19.150$ km, and component 2 gives $5.145 < 6.615 < 9.778$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7289$ and 1.7260 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.194 and 15.249 h. The determinant-averaged frame-dragging rates at those shells are 6339.95 and 11651.91 rad s $^{-1}$, approaching surface-limit values of 8887.84 and 18110.35 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.671/14.168 h at 1 km inward to 0.0338/0.0452 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.44 GW240512_024139

GW240512_024139 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 15.340 M_\odot$ and $m_2 = 9.721 M_\odot$, while the luminosity-distance median is 1166.4 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,999
Observed luminosity distance	1166.4 Mpc (median)
Detector-frame masses	$m_1 = 15.340 M_\odot$, $m_2 = 9.721 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.634$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2271$; $z_\pi = 0.2442$ (medians)
Post-handoff reorder fraction	15.86%

Table 131: Observational and coordinate summary for GW240512_024139. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.339 [8.504, 17.192]	7.936 [5.850, 11.462]
Physical compact-source mass $M_c (M_\odot)$	8.515 [6.047, 13.692]	5.527 [3.581, 7.579]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.551 [0.460, 6.359]	2.579 [0.334, 5.352]
Geometric share M_{geom}/M_∞ (%)	28.44 [4.57, 47.54]	33.40 [4.60, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.470 [0.116, 0.910]	0.384 [0.035, 0.909]
Polarization factor A_{pol}	1.397 [1.048, 1.906]	1.502 [1.048, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.918 [0.423, 0.999]	0.865 [0.140, 0.999]
GR/CPTG matching radius r_m (km)	18.221 [12.558, 25.387]	11.719 [8.638, 16.925]
Structural Mode N_{BH}	1.6621 [1.4726, 1.8839]	1.7163 [1.4727, 1.9094]
Organization-shell radius r_N (km)	9.610 [6.592, 13.400]	6.170 [4.545, 8.907]
Inner trapping-surface radius r_- (km)	12.069 [8.973, 18.342]	7.965 [5.563, 10.619]
Structural- N lapse $\bar{\alpha}_N$	0.6854 [0.5056, 0.8332]	0.6431 [0.4852, 0.8331]
Structural- N clock $T_{24,N}$ (h)	16.450 [12.135, 19.996]	15.435 [11.644, 19.994]
Structural- N axis ω (rad s $^{-1}$)	5941.76 [2390.92, 9292.05]	8254.63 [2148.16, 14229.16]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7043.64 [3450.37, 10435.18]	9893.89 [2149.67, 15790.61]
Structural- N equatorial ω (rad s $^{-1}$)	8180.03 [4368.76, 11707.27]	11940.61 [2150.42, 17376.58]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.8920 [0.6021, 1.8210]	0.6351 [0.3979, 2.9229]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10419.45 [5118.12, 15487.11]	15623.91 [2344.16, 22309.73]
Trapping-surface period P_- (ms)	0.6030 [0.4057, 1.2276]	0.4022 [0.2816, 2.6804]

Table 132: CPTG compact-state, structural, clock, and geometric-rotation results for GW240512_024139. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.3557	12.9833	9576.67	13697.81
500 m inward	7.2950	9.2206	10034.87	14807.59
100 m inward	3.2448	4.1236	10348.79	15480.35
10 m inward	1.0247	1.3034	10413.03	15611.34
1 m inward	0.3240	0.4121	10418.70	15623.44
10 cm inward	0.1025	0.1303	10419.37	15623.82
1 cm inward	0.0324	0.0412	10419.44	15623.90
surface limit	0.0000	0.0000	10419.45	15623.91

Table 133: Near-inner-boundary profile for GW240512_024139. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.515 M_\odot$ and $5.527 M_\odot$, with completed geometric mass-equivalent shares of 28.44% and 33.40%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.610 < 12.069 < 18.221$ km, and component 2 gives $6.170 < 7.965 < 11.719$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6621$ and 1.7163 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.450 and 15.435 h. The determinant-averaged frame-dragging rates at those shells are 7043.64 and 9893.89 rad s $^{-1}$, approaching surface-limit values of 10419.45 and 15623.91 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.356/12.983 h at 1 km inward to 0.0324/0.0412 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.45 GW231118_090602

GW231118_090602 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 16.097 M_\odot$ and $m_2 = 9.208 M_\odot$, while the luminosity-distance median is 1361.7 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	29,856
Observed luminosity distance	1361.7 Mpc (median)
Detector-frame masses	$m_1 = 16.097 M_\odot$, $m_2 = 9.208 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.572$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2604$; $z_\pi = 0.2800$ (medians)
Post-handoff reorder fraction	9.88%

Table 134: Observational and coordinate summary for GW231118_090602. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.695 [8.667, 24.980]	7.259 [4.129, 10.408]
Physical compact-source mass $M_c (M_\odot)$	8.568 [5.573, 20.468]	4.797 [2.709, 6.926]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.278 [0.708, 6.579]	2.327 [0.296, 4.789]
Geometric share M_{geom}/M_∞ (%)	29.96 [6.50, 49.07]	35.06 [5.09, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.443 [0.070, 0.872]	0.355 [0.033, 0.900]
Polarization factor A_{pol}	1.428 [1.070, 1.963]	1.540 [1.054, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.904 [0.268, 0.998]	0.842 [0.130, 0.999]
GR/CPTG matching radius r_m (km)	18.746 [12.798, 36.886]	10.718 [6.098, 15.368]
Structural Mode N_{BH}	1.6782 [1.4837, 1.9016]	1.7352 [1.4755, 1.9098]
Organization-shell radius r_N (km)	9.882 [6.738, 19.442]	5.647 [3.208, 8.084]
Inner trapping-surface radius r_- (km)	12.199 [8.628, 27.201]	7.106 [4.055, 9.748]
Structural- N lapse $\bar{\alpha}_N$	0.6729 [0.4913, 0.8242]	0.6281 [0.4849, 0.8308]
Structural- N clock $T_{24,N}$ (h)	16.151 [11.791, 19.782]	15.074 [11.637, 19.940]
Structural- N axis ω (rad s $^{-1}$)	5511.45 [1752.23, 9388.38]	9430.53 [2182.88, 18805.94]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6472.69 [2357.27, 10505.11]	11206.36 [2184.25, 21260.77]
Structural- N equatorial ω (rad s $^{-1}$)	7394.54 [2914.04, 11768.03]	13124.51 [2184.95, 23831.85]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9707 [0.5981, 2.6655]	0.5607 [0.2955, 2.8766]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9162.00 [3446.29, 15436.54]	16901.56 [2381.54, 30998.30]
Trapping-surface period P_- (ms)	0.6858 [0.4070, 1.8232]	0.3718 [0.2027, 2.6383]

Table 135: CPTG compact-state, structural, clock, and geometric-rotation results for GW231118_090602. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.3661	13.8678	8442.87	14383.09
500 m inward	7.3078	9.8890	8832.56	15843.83
100 m inward	3.2540	4.4292	9098.27	16736.10
10 m inward	1.0278	1.4004	9155.96	16885.56
1 m inward	0.3250	0.4428	9161.46	16901.00
10 cm inward	0.1028	0.1400	9161.95	16901.45
1 cm inward	0.0325	0.0443	9161.99	16901.55
surface limit	0.0000	0.0000	9162.00	16901.56

Table 136: Near-inner-boundary profile for GW231118_090602. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.568 M_\odot$ and $4.797 M_\odot$, with completed geometric mass-equivalent shares of 29.96% and 35.06%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.882 < 12.199 < 18.746$ km, and component 2 gives $5.647 < 7.106 < 10.718$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6782$ and 1.7352 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.151 and 15.074 h. The determinant-averaged frame-dragging rates at those shells are 6472.69 and 11206.36 rad s $^{-1}$, approaching surface-limit values of 9162.00 and 16901.56 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.366/13.868 h at 1 km inward to 0.0325/0.0443 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.46 GW190720_000836

GW190720_000836 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 16.569 M_\odot$ and $m_2 = 8.756 M_\odot$, while the luminosity-distance median is 773.4 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	26,439
Observed luminosity distance	773.4 Mpc (median)
Detector-frame masses	$m_1 = 16.569 M_\odot$, $m_2 = 8.756 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.528$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1572$; $z_\pi = 0.1688$ (medians)
Post-handoff reorder fraction	10.02%

Table 137: Observational and coordinate summary for GW190720_000836. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	14.048 [8.806, 19.603]	7.416 [5.601, 11.692]
Physical compact-source mass $M_c (M_\odot)$	9.132 [6.408, 15.040]	5.321 [3.301, 7.252]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.775 [0.894, 7.023]	2.341 [0.326, 5.444]
Geometric share M_{geom}/M_∞ (%)	32.57 [8.36, 48.89]	32.51 [4.80, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.398 [0.076, 0.836]	0.399 [0.035, 0.905]
Polarization factor A_{pol}	1.483 [1.091, 1.957]	1.482 [1.050, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.876 [0.292, 0.996]	0.876 [0.140, 0.999]
GR/CPTG matching radius r_m (km)	20.744 [13.003, 28.946]	10.951 [8.270, 17.265]
Structural Mode N_{BH}	1.7069 [1.4951, 1.8996]	1.7062 [1.4739, 1.9094]
Organization-shell radius r_N (km)	10.944 [6.840, 15.294]	5.762 [4.350, 9.068]
Inner trapping-surface radius r_- (km)	13.175 [9.501, 20.259]	7.536 [5.232, 10.542]
Structural- N lapse $\bar{\alpha}_N$	0.6505 [0.4929, 0.8152]	0.6511 [0.4852, 0.8322]
Structural- N clock $T_{24,N}$ (h)	15.612 [11.830, 19.564]	15.626 [11.644, 19.972]
Structural- N axis ω (rad s $^{-1}$)	5645.95 [2185.02, 8361.84]	8811.19 [2200.18, 15416.89]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6472.81 [2714.10, 9424.90]	10682.29 [2202.43, 16944.01]
Structural- N equatorial ω (rad s $^{-1}$)	7171.87 [3009.19, 10951.80]	12772.27 [2203.61, 18373.91]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9707 [0.6667, 2.3150]	0.5882 [0.3708, 2.8528]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8833.82 [3301.10, 14792.28]	16785.70 [2403.61, 23450.05]
Trapping-surface period P_- (ms)	0.7113 [0.4248, 1.9034]	0.3743 [0.2679, 2.6141]

Table 138: CPTG compact-state, structural, clock, and geometric-rotation results for GW190720_000836. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.1451	13.2171	8171.24	14548.66
500 m inward	7.1680	9.3742	8529.61	15859.22
100 m inward	3.1979	4.1830	8776.44	16633.94
10 m inward	1.0105	1.3220	8828.60	16772.87
1 m inward	0.3195	0.4180	8833.27	16784.03
10 cm inward	0.1010	0.1322	8833.77	16785.53
1 cm inward	0.0320	0.0418	8833.82	16785.68
surface limit	0.0000	0.0000	8833.82	16785.70

Table 139: Near-inner-boundary profile for GW190720_000836. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $9.132 M_\odot$ and $5.321 M_\odot$, with completed geometric mass-equivalent shares of 32.57% and 32.51%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $10.944 < 13.175 < 20.744$ km, and component 2 gives $5.762 < 7.536 < 10.951$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7069$ and 1.7062 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.612 and 15.626 h. The determinant-averaged frame-dragging rates at those shells are 6472.81 and 10682.29 rad s $^{-1}$, approaching surface-limit values of 8833.82 and 16785.70 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.145/13.217 h at 1 km inward to 0.0320/0.0418 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.47 GW241009_084816

GW241009_084816 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 15.476 M_\odot$ and $m_2 = 9.886 M_\odot$, while the luminosity-distance median is 1132.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,840
Observed luminosity distance	1132.3 Mpc (median)
Detector-frame masses	$m_1 = 15.476 M_\odot$, $m_2 = 9.886 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.639$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2212$; $z_\pi = 0.2379$ (medians)
Post-handoff reorder fraction	10.76%

Table 140: Observational and coordinate summary for GW241009_084816. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	12.518 [9.134, 16.500]	8.023 [6.126, 10.895]
Physical compact-source mass M_c ($M_\odot$)	8.725 [6.113, 13.257]	5.316 [3.609, 7.672]
Geometric mass-equivalent M_{geom} ($M_\odot$)	3.439 [0.446, 6.880]	2.865 [0.434, 4.979]
Geometric share M_{geom}/M_∞ (%)	27.84 [3.89, 49.34]	36.21 [5.99, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.480 [0.058, 0.923]	0.335 [0.031, 0.882]
Polarization factor A_{pol}	1.386 [1.040, 1.974]	1.568 [1.064, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.923 [0.227, 0.999]	0.823 [0.123, 0.998]
GR/CPTG matching radius r_m (km)	18.484 [13.488, 24.365]	11.847 [9.045, 16.088]
Structural Mode N_{BH}	1.6559 [1.4688, 1.9047]	1.7487 [1.4807, 1.9101]
Organization-shell radius r_N (km)	9.734 [7.093, 12.849]	6.239 [4.756, 8.474]
Inner trapping-surface radius r_- (km)	12.374 [9.266, 17.686]	7.791 [5.739, 10.510]
Structural- N lapse $\bar{\alpha}_N$	0.6902 [0.4889, 0.8362]	0.6173 [0.4847, 0.8267]
Structural- N clock $T_{24,N}$ (h)	16.566 [11.733, 20.070]	14.816 [11.632, 19.840]
Structural- N axis ω (rad s $^{-1}$)	5382.99 [1950.75, 8881.23]	8617.37 [2008.58, 13937.44]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6547.45 [2407.36, 9963.26]	10027.70 [2009.84, 15389.45]
Structural- N equatorial ω (rad s $^{-1}$)	7793.46 [2444.99, 11175.39]	11710.49 [2010.64, 16830.19]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9596 [0.6306, 2.6100]	0.6266 [0.4083, 3.1262]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10142.83 [2670.70, 14657.30]	14875.71 [2191.99, 21543.10]
Trapping-surface period P_- (ms)	0.6195 [0.4287, 2.3526]	0.4224 [0.2917, 2.8664]

Table 141: CPTG compact-state, structural, clock, and geometric-rotation results for GW241009_084816. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.3021	13.3038	9420.32	12950.56
500 m inward	7.2608	9.4826	9818.21	14046.73
100 m inward	3.2327	4.2528	10082.23	14728.76
10 m inward	1.0213	1.3452	10137.07	14862.74
1 m inward	0.3229	0.4254	10142.29	14875.05
10 cm inward	0.1021	0.1345	10142.78	14875.64
1 cm inward	0.0323	0.0425	10142.82	14875.70
surface limit	0.0000	0.0000	10142.83	14875.71

Table 142: Near-inner-boundary profile for GW241009_084816. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.725 M_\odot$ and $5.316 M_\odot$, with completed geometric mass-equivalent shares of 27.84% and 36.21%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.734 < 12.374 < 18.484$ km, and component 2 gives $6.239 < 7.791 < 11.847$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6559$ and 1.7487 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.566 and 14.816 h. The determinant-averaged frame-dragging rates at those shells are 6547.45 and 10027.70 rad s $^{-1}$, approaching surface-limit values of 10142.83 and 14875.71 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.302/13.304 h at 1 km inward to 0.0323/0.0425 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.48 GW200316_215756

GW200316_215756 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 15.989 M_\odot$ and $m_2 = 9.516 M_\odot$, while the luminosity-distance median is 1123.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	129,403
Observed luminosity distance	1123.9 Mpc (median)
Detector-frame masses	$m_1 = 15.989 M_\odot$, $m_2 = 9.516 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.595$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2201$; $z_\pi = 0.2363$ (medians)
Post-handoff reorder fraction	12.72%

Table 143: Observational and coordinate summary for GW200316_215756. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.918 [8.767, 23.007]	7.735 [4.874, 11.178]
Physical compact-source mass $M_c (M_\odot)$	8.251 [5.896, 17.255]	5.212 [3.176, 7.162]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.790 [1.024, 7.056]	2.570 [0.407, 5.187]
Geometric share M_{geom}/M_∞ (%)	33.51 [10.19, 49.20]	35.05 [6.31, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.382 [0.064, 0.801]	0.355 [0.035, 0.876]
Polarization factor A_{pol}	1.504 [1.113, 1.968]	1.540 [1.067, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.864 [0.250, 0.994]	0.842 [0.138, 0.998]
GR/CPTG matching radius r_m (km)	19.075 [12.945, 33.973]	11.422 [7.196, 16.506]
Structural Mode N_{BH}	1.7174 [1.5069, 1.9031]	1.7352 [1.4826, 1.9095]
Organization-shell radius r_N (km)	10.059 [6.812, 17.980]	6.020 [3.786, 8.684]
Inner trapping-surface radius r_- (km)	12.028 [8.967, 23.450]	7.612 [4.772, 10.207]
Structural- N lapse $\bar{\alpha}_N$	0.6422 [0.4901, 0.8058]	0.6281 [0.4851, 0.8251]
Structural- N clock $T_{24,N}$ (h)	15.413 [11.763, 19.340]	15.075 [11.643, 19.803]
Structural- N axis ω (rad s $^{-1}$)	5874.49 [2185.50, 9207.28]	9058.34 [2206.43, 16029.66]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6727.30 [2550.69, 10321.35]	10710.11 [2208.24, 18076.36]
Structural- N equatorial ω (rad s $^{-1}$)	7499.57 [2728.80, 11573.96]	12534.88 [2209.06, 20276.61]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9340 [0.6088, 2.4633]	0.5867 [0.3476, 2.8453]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9093.73 [3011.33, 15229.90]	16106.89 [2408.18, 26280.25]
Trapping-surface period P_- (ms)	0.6909 [0.4126, 2.0865]	0.3901 [0.2391, 2.6091]

Table 144: CPTG compact-state, structural, clock, and geometric-rotation results for GW200316_215756. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.6505	13.3769	8337.17	13828.71
500 m inward	7.5341	9.5231	8742.12	15147.75
100 m inward	3.3649	4.2649	9029.14	15947.37
10 m inward	1.0635	1.3483	9087.47	16090.14
1 m inward	0.3363	0.4264	9093.15	16105.55
10 cm inward	0.1063	0.1348	9093.67	16106.88
1 cm inward	0.0336	0.0426	9093.72	16106.89
surface limit	0.0000	0.0000	9093.73	16106.89

Table 145: Near-inner-boundary profile for GW200316_215756. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.251 M_\odot$ and $5.212 M_\odot$, with completed geometric mass-equivalent shares of 33.51% and 35.05%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $10.059 < 12.028 < 19.075$ km, and component 2 gives $6.020 < 7.612 < 11.422$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7174$ and 1.7352 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.413 and 15.075 h. The determinant-averaged frame-dragging rates at those shells are 6727.30 and 10710.11 rad s $^{-1}$, approaching surface-limit values of 9093.73 and 16106.89 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.651/13.377 h at 1 km inward to 0.0336/0.0426 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.49 GW230729_082317

GW230729_082317 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 15.883 M_\odot$ and $m_2 = 9.935 M_\odot$, while the luminosity-distance median is 1635.7 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	25,978
Observed luminosity distance	1635.7 Mpc (median)
Detector-frame masses	$m_1 = 15.883 M_\odot$, $m_2 = 9.935 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.626$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3056$; $z_\pi = 0.3286$ (medians)
Post-handoff reorder fraction	15.46%

Table 146: Observational and coordinate summary for GW230729_082317. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	11.975 [8.265, 19.556]	7.546 [4.879, 10.945]
Physical compact-source mass $M_c (M_\odot)$	8.114 [5.557, 15.093]	5.073 [3.089, 7.243]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.965 [0.575, 6.592]	2.481 [0.309, 5.109]
Geometric share M_{geom}/M_∞ (%)	30.69 [5.76, 49.00]	34.97 [4.90, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.431 [0.072, 0.887]	0.357 [0.032, 0.903]
Polarization factor A_{pol}	1.443 [1.061, 1.961]	1.538 [1.052, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.897 [0.277, 0.998]	0.843 [0.126, 0.999]
GR/CPTG matching radius r_m (km)	17.683 [12.204, 28.876]	11.143 [7.204, 16.161]
Structural Mode N_{BH}	1.6861 [1.4793, 1.9009]	1.7342 [1.4744, 1.9100]
Organization-shell radius r_N (km)	9.324 [6.417, 15.256]	5.869 [3.792, 8.500]
Inner trapping-surface radius r_- (km)	11.587 [8.434, 20.368]	7.429 [4.719, 10.171]
Structural- N lapse $\bar{\alpha}_N$	0.6668 [0.4919, 0.8278]	0.6289 [0.4848, 0.8317]
Structural- N clock $T_{24,N}$ (h)	16.003 [11.805, 19.866]	15.093 [11.634, 19.961]
Structural- N axis ω (rad s $^{-1}$)	5911.48 [2225.42, 9718.44]	8904.52 [2068.80, 16523.74]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6960.27 [2791.84, 10875.38]	10570.45 [2070.27, 18435.93]
Structural- N equatorial ω (rad s $^{-1}$)	8089.46 [3161.10, 12211.72]	12524.66 [2071.01, 20484.54]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9027 [0.5777, 2.2505]	0.5944 [0.3408, 3.0350]
Trapping-surface limit Ω_- (rad s $^{-1}$)	10143.51 [3535.68, 16072.61]	16095.21 [2257.00, 26399.30]
Trapping-surface period P_- (ms)	0.6194 [0.3909, 1.7771]	0.3904 [0.2380, 2.7839]

Table 147: CPTG compact-state, structural, clock, and geometric-rotation results for GW230729_082317. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.6402	13.4795	9275.93	13840.38
500 m inward	7.5083	9.5977	9744.23	15163.67
100 m inward	3.3432	4.2984	10068.04	15940.12
10 m inward	1.0561	1.3591	10135.89	16081.82
1 m inward	0.3339	0.4298	10142.90	16093.62
10 cm inward	0.1056	0.1359	10143.45	16094.99
1 cm inward	0.0334	0.0430	10143.50	16095.19
surface limit	0.0000	0.0000	10143.51	16095.21

Table 148: Near-inner-boundary profile for GW230729_082317. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.114 M_\odot$ and $5.073 M_\odot$, with completed geometric mass-equivalent shares of 30.69% and 34.97%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.324 < 11.587 < 17.683$ km, and component 2 gives $5.869 < 7.429 < 11.143$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6861$ and 1.7342 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.003 and 15.093 h. The determinant-averaged frame-dragging rates at those shells are 6960.27 and 10570.45 rad s $^{-1}$, approaching surface-limit values of 10143.51 and 16095.21 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.640/13.480 h at 1 km inward to 0.0334/0.0430 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.50 GW240825_055146

GW240825_055146 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 16.306 M_\odot$ and $m_2 = 9.722 M_\odot$, while the luminosity-distance median is 1254.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	20,685
Observed luminosity distance	1254.3 Mpc (median)
Detector-frame masses	$m_1 = 16.306 M_\odot$, $m_2 = 9.722 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.596$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2422$; $z_\pi = 0.2605$ (medians)
Post-handoff reorder fraction	9.41%

Table 149: Observational and coordinate summary for GW240825_055146. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	12.988 [9.117, 18.765]	7.777 [5.562, 10.740]
Physical compact-source mass $M_c (M_\odot)$	8.906 [5.816, 14.769]	5.297 [3.428, 7.571]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.080 [0.620, 6.805]	2.518 [0.304, 4.935]
Geometric share M_{geom}/M_∞ (%)	30.25 [5.65, 49.29]	33.55 [4.32, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.438 [0.061, 0.889]	0.381 [0.031, 0.915]
Polarization factor A_{pol}	1.434 [1.060, 1.972]	1.505 [1.045, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.901 [0.236, 0.998]	0.863 [0.125, 0.999]
GR/CPTG matching radius r_m (km)	19.179 [13.462, 27.709]	11.483 [8.213, 15.859]
Structural Mode N_{BH}	1.6814 [1.4787, 1.9041]	1.7179 [1.4712, 1.9100]
Organization-shell radius r_N (km)	10.101 [7.078, 14.645]	6.046 [4.326, 8.340]
Inner trapping-surface radius r_- (km)	12.683 [9.008, 19.834]	7.726 [5.323, 10.368]
Structural- N lapse $\bar{\alpha}_N$	0.6705 [0.4893, 0.8283]	0.6418 [0.4847, 0.8343]
Structural- N clock $T_{24,N}$ (h)	16.091 [11.744, 19.878]	15.404 [11.633, 20.024]
Structural- N axis ω (rad s $^{-1}$)	5431.20 [1983.76, 8975.02]	8443.08 [2026.02, 14667.70]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	6407.97 [2463.97, 10000.38]	10108.17 [2026.60, 16343.31]
Structural- N equatorial ω (rad s $^{-1}$)	7388.31 [2630.32, 11165.34]	12071.67 [2027.18, 18081.10]
Structural- N period $P_{\bar{\omega}}$ (ms)	0.9805 [0.6283, 2.5500]	0.6216 [0.3844, 3.1004]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9416.72 [2876.24, 14607.65]	15721.86 [2211.97, 23333.88]
Trapping-surface period P_- (ms)	0.6672 [0.4301, 2.1845]	0.3996 [0.2693, 2.8405]

Table 150: CPTG compact-state, structural, clock, and geometric-rotation results for GW240825_055146. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.2094	13.2725	8743.94	13724.79
500 m inward	7.2074	9.4454	9109.17	14880.58
100 m inward	3.2124	4.2298	9361.41	15570.28
10 m inward	1.0150	1.3369	9410.93	15704.67
1 m inward	0.3209	0.4228	9416.31	15720.49
10 cm inward	0.1015	0.1337	9416.68	15721.71
1 cm inward	0.0321	0.0423	9416.72	15721.84
surface limit	0.0000	0.0000	9416.72	15721.86

Table 151: Near-inner-boundary profile for GW240825_055146. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.906 M_\odot$ and $5.297 M_\odot$, with completed geometric mass-equivalent shares of 30.25% and 33.55%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $10.101 < 12.683 < 19.179$ km, and component 2 gives $6.046 < 7.726 < 11.483$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6814$ and 1.7179 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.091 and 15.404 h. The determinant-averaged frame-dragging rates at those shells are 6407.97 and 10108.17 rad s $^{-1}$, approaching surface-limit values of 9416.72 and 15721.86 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.209/13.273 h at 1 km inward to 0.0321/0.0423 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.51 GW231104_133418

GW231104_133418 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 15.454 M_\odot$ and $m_2 = 10.977 M_\odot$, while the luminosity-distance median is 1470.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	28,914
Observed luminosity distance	1470.6 Mpc (median)
Detector-frame masses	$m_1 = 15.454 M_\odot$, $m_2 = 10.977 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.710$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2786$; $z_\pi = 0.2995$ (medians)
Post-handoff reorder fraction	22.30%

Table 152: Observational and coordinate summary for GW231104_133418. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	11.717 [8.578, 16.412]	8.669 [6.355, 11.956]
Physical compact-source mass M_c ($M_\odot$)	8.222 [6.119, 12.303]	5.761 [3.865, 7.795]
Geometric mass-equivalent M_{geom} ($M_\odot$)	3.471 [0.475, 6.368]	3.116 [0.467, 5.622]
Geometric share M_{geom}/M_∞ (%)	29.00 [4.76, 47.32]	36.90 [6.33, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.460 [0.122, 0.906]	0.323 [0.030, 0.875]
Polarization factor A_{pol}	1.408 [1.050, 1.898]	1.585 [1.068, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.913 [0.439, 0.999]	0.811 [0.120, 0.998]
GR/CPTG matching radius r_m (km)	17.301 [12.667, 24.234]	12.801 [9.383, 17.654]
Structural Mode N_{BH}	1.6680 [1.4736, 1.8814]	1.7568 [1.4827, 1.9102]
Organization-shell radius r_N (km)	9.133 [6.649, 12.798]	6.748 [4.932, 9.281]
Inner trapping-surface radius r_- (km)	11.634 [9.021, 16.673]	8.456 [6.040, 10.967]
Structural- N lapse $\bar{\alpha}_N$	0.6808 [0.5077, 0.8323]	0.6108 [0.4846, 0.8251]
Structural- N clock $T_{24,N}$ (h)	16.340 [12.184, 19.976]	14.659 [11.631, 19.801]
Structural- N axis ω (rad s $^{-1}$)	6217.02 [2709.37, 9393.56]	8110.25 [1734.67, 13179.73]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	7348.14 [3884.95, 10523.83]	9424.15 [1735.54, 14613.89]
Structural- N equatorial ω (rad s $^{-1}$)	8603.63 [4669.46, 11733.27]	11010.97 [1736.08, 16038.03]
Structural- N period P_ω (ms)	0.8551 [0.5970, 1.6173]	0.6667 [0.4299, 3.6203]
Trapping-surface limit Ω_- (rad s $^{-1}$)	11092.96 [5305.70, 15436.63]	14016.48 [1892.32, 20529.88]
Trapping-surface period P_- (ms)	0.5664 [0.4070, 1.1842]	0.4483 [0.3061, 3.3204]

Table 153: CPTG compact-state, structural, clock, and geometric-rotation results for GW231104_133418. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	10.5741	12.7727	10158.87	12274.50
500 m inward	7.4589	9.0978	10668.36	13264.74
100 m inward	3.3205	4.0786	11020.12	13885.25
10 m inward	1.0487	1.2898	11087.13	14004.06
1 m inward	0.3316	0.4079	11092.08	14014.90
10 cm inward	0.1049	0.1290	11092.87	14016.32
1 cm inward	0.0332	0.0408	11092.95	14016.46
surface limit	0.0000	0.0000	11092.96	14016.48

Table 154: Near-inner-boundary profile for GW231104_133418. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.222 M_\odot$ and $5.761 M_\odot$, with completed geometric mass-equivalent shares of 29.00% and 36.90%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $9.133 < 11.634 < 17.301$ km, and component 2 gives $6.748 < 8.456 < 12.801$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6680$ and 1.7568 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.340 and 14.659 h. The determinant-averaged frame-dragging rates at those shells are 7348.14 and 9424.15 rad s $^{-1}$, approaching surface-limit values of 11092.96 and 14016.48 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 10.574/12.773 h at 1 km inward to 0.0332/0.0408 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.52 GW240530_012417

GW240530_012417 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 16.697 M_\odot$ and $m_2 = 9.867 M_\odot$, while the luminosity-distance median is 1006.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	19,570
Observed luminosity distance	1006.5 Mpc (median)
Detector-frame masses	$m_1 = 16.697 M_\odot$, $m_2 = 9.867 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.591$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1991$; $z_\pi = 0.2142$ (medians)
Post-handoff reorder fraction	20.45%

Table 155: Observational and coordinate summary for GW240530_012417. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	13.670 [8.875, 22.218]	8.262 [5.216, 13.305]
Physical compact-source mass $M_c (M_\odot)$	8.938 [6.408, 17.451]	5.933 [3.357, 8.116]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.491 [0.348, 7.719]	2.448 [0.246, 6.315]
Geometric share M_{geom}/M_∞ (%)	30.03 [3.52, 49.46]	32.15 [3.48, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.442 [0.053, 0.930]	0.405 [0.032, 0.931]
Polarization factor A_{pol}	1.429 [1.036, 1.979]	1.474 [1.036, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.903 [0.206, 0.999]	0.881 [0.127, 0.999]
GR/CPTG matching radius r_m (km)	20.186 [13.104, 32.808]	12.199 [7.702, 19.646]
Structural Mode N_{BH}	1.6790 [1.4668, 1.9061]	1.7021 [1.4666, 1.9099]
Organization-shell radius r_N (km)	10.641 [6.849, 17.328]	6.417 [4.054, 10.316]
Inner trapping-surface radius r_- (km)	12.966 [9.691, 23.472]	8.500 [5.108, 11.813]
Structural- N lapse $\bar{\alpha}_N$	0.6723 [0.4878, 0.8379]	0.6542 [0.4848, 0.8380]
Structural- N clock $T_{24,N}$ (h)	16.136 [11.706, 20.109]	15.702 [11.635, 20.112]
Structural- N axis ω (rad s $^{-1}$)	4821.51 [1672.07, 8176.36]	7369.11 [1748.10, 15173.27]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	5857.15 [1950.58, 9319.48]	9088.40 [1748.99, 17052.87]
Structural- N equatorial ω (rad s $^{-1}$)	7036.49 [2066.52, 10696.78]	11212.12 [1749.44, 19115.21]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.0727 [0.6742, 3.2212]	0.6913 [0.3685, 3.5925]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8726.26 [2273.20, 14418.42]	14904.61 [1906.49, 24756.65]
Trapping-surface period P_- (ms)	0.7200 [0.4358, 2.7640]	0.4216 [0.2538, 3.2957]

Table 156: CPTG compact-state, structural, clock, and geometric-rotation results for GW240530_012417. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	9.9448	12.3700	8071.86	13272.94
500 m inward	7.0062	8.7586	8417.88	14216.53
100 m inward	3.1203	3.9075	8670.63	14787.30
10 m inward	0.9858	1.2349	8720.11	14895.68
1 m inward	0.3117	0.3905	8725.61	14903.47
10 cm inward	0.0986	0.1235	8726.19	14904.48
1 cm inward	0.0312	0.0390	8726.25	14904.60
surface limit	0.0000	0.0000	8726.26	14904.61

Table 157: Near-inner-boundary profile for GW240530_012417. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $8.938 M_\odot$ and $5.933 M_\odot$, with completed geometric mass-equivalent shares of 30.03% and 32.15%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $10.641 < 12.966 < 20.186$ km, and component 2 gives $6.417 < 8.500 < 12.199$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6790$ and 1.7021 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.136 and 15.702 h. The determinant-averaged frame-dragging rates at those shells are 5857.15 and 9088.40 rad s $^{-1}$, approaching surface-limit values of 8726.26 and 14904.61 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 9.945/12.370 h at 1 km inward to 0.0312/0.0390 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.53 GW241011_233834

GW241011_233834 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 20.438 M_\odot$ and $m_2 = 6.237 M_\odot$, while the luminosity-distance median is 212.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,339
Observed luminosity distance	212.0 Mpc (median)
Detector-frame masses	$m_1 = 20.438 M_\odot$, $m_2 = 6.237 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.305$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0463$; $z_\pi = 0.0499$ (medians)
Post-handoff reorder fraction	0.00%

Table 158: Observational and coordinate summary for GW241011_233834. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	19.476 [17.461, 22.015]	5.943 [5.364, 6.526]
Physical compact-source mass $M_c (M_\odot)$	17.104 [15.046, 19.173]	3.869 [2.917, 5.453]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	2.443 [1.683, 3.281]	2.089 [0.363, 3.028]
Geometric share M_{geom}/M_∞ (%)	12.51 [8.93, 16.31]	35.13 [6.35, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.758 [0.687, 0.825]	0.354 [0.034, 0.875]
Polarization factor A_{pol}	1.143 [1.098, 1.195]	1.542 [1.068, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.990 [0.981, 0.995]	0.841 [0.134, 0.998]
GR/CPTG matching radius r_m (km)	28.759 [25.783, 32.508]	8.775 [7.921, 9.637]
Structural Mode N_{BH}	1.5228 [1.4987, 1.5511]	1.7361 [1.4829, 1.9097]
Organization-shell radius r_N (km)	15.085 [13.528, 17.075]	4.625 [4.162, 5.080]
Inner trapping-surface radius r_- (km)	22.438 [19.782, 25.155]	5.624 [4.809, 7.217]
Structural- N lapse $\bar{\alpha}_N$	0.7933 [0.7712, 0.8123]	0.6274 [0.4850, 0.8249]
Structural- N clock $T_{24,N}$ (h)	19.039 [18.508, 19.496]	15.057 [11.640, 19.799]
Structural- N axis ω (rad s $^{-1}$)	2311.64 [1950.07, 2875.55]	12454.35 [3220.53, 17068.21]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3386.54 [2951.19, 4040.61]	14514.62 [3222.75, 18633.46]
Structural- N equatorial ω (rad s $^{-1}$)	4703.15 [4191.49, 5387.94]	16598.00 [3223.87, 19971.44]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.8553 [1.5550, 2.1290]	0.4329 [0.3372, 1.9496]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6675.59 [5955.35, 7561.85]	21293.00 [3514.26, 25277.39]
Trapping-surface period P_- (ms)	0.9412 [0.8309, 1.0550]	0.2951 [0.2486, 1.7879]

Table 159: CPTG compact-state, structural, clock, and geometric-rotation results for GW241011_233834. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.5723	15.6330	6458.15	17188.50
500 m inward	4.5805	11.2460	6579.15	19668.50
100 m inward	2.0225	5.0570	6657.76	21042.97
10 m inward	0.6377	1.5998	6673.67	21266.51
1 m inward	0.2016	0.5059	6675.39	21289.29
10 cm inward	0.0637	0.1600	6675.57	21292.63
1 cm inward	0.0202	0.0506	6675.58	21292.97
surface limit	0.0000	0.0000	6675.59	21293.00

Table 160: Near-inner-boundary profile for GW241011_233834. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $17.104 M_\odot$ and $3.869 M_\odot$, with completed geometric mass-equivalent shares of 12.51% and 35.13%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $15.085 < 22.438 < 28.759$ km, and component 2 gives $4.625 < 5.624 < 8.775$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5228$ and 1.7361 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.039 and 15.057 h. The determinant-averaged frame-dragging rates at those shells are 3386.54 and 14514.62 rad s $^{-1}$, approaching surface-limit values of 6675.59 and 21293.00 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 6.572/15.633 h at 1 km inward to 0.0202/0.0506 h at 1 cm inward for components 1/2. The corresponding interior frame-dragging rate is also in the upper M1 quartile for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.54 GW241110_124123

GW241110_124123 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 18.806 M_\odot$ and $m_2 = 9.225 M_\odot$, while the luminosity-distance median is 738.7 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,356
Observed luminosity distance	738.7 Mpc (median)
Detector-frame masses	$m_1 = 18.806 M_\odot$, $m_2 = 9.225 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.491$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1505$; $z_\pi = 0.1619$ (medians)
Post-handoff reorder fraction	1.96%

Table 161: Observational and coordinate summary for GW241110_124123. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	16.218 [12.005, 21.223]	7.957 [6.343, 10.301]
Physical compact-source mass $M_c (M_\odot)$	12.803 [9.782, 16.280]	5.720 [3.703, 8.378]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	2.999 [0.373, 8.372]	2.238 [0.242, 4.343]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	18.74 [2.70, 42.37]	28.11 [3.15, 49.57]
Exterior spin magnitude $ \hat{a}_\infty $	0.643 [0.225, 0.946]	0.476 [0.047, 0.937]
Polarization factor A_{pol}	1.231 [1.028, 1.735]	1.391 [1.033, 1.983]
Compact-source spin parameter $\hat{a}_c$	0.973 [0.678, 1.000]	0.920 [0.183, 0.999]
GR/CPTG matching radius r_m (km)	23.948 [17.727, 31.339]	11.749 [9.366, 15.211]
Structural Mode N_{BH}	1.5708 [1.4624, 1.8222]	1.6587 [1.4648, 1.9074]
Organization-shell radius r_N (km)	12.617 [9.282, 16.548]	6.180 [4.926, 8.013]
Inner trapping-surface radius r_- (km)	17.501 [13.564, 21.789]	8.046 [5.903, 11.255]
Structural- N lapse $\bar{\alpha}_N$	0.7559 [0.5570, 0.8414]	0.6881 [0.4868, 0.8395]
Structural- N clock $T_{24,N}$ (h)	18.142 [13.368, 20.194]	16.514 [11.683, 20.147]
Structural- N axis ω (rad s $^{-1}$)	3643.55 [1907.69, 5754.00]	7929.93 [2932.35, 13675.76]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4816.89 [3025.09, 6646.15]	9678.47 [3243.06, 15025.47]
Structural- N equatorial ω (rad s $^{-1}$)	6044.12 [4400.39, 7887.32]	11757.03 [3245.50, 16349.45]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.3044 [0.9454, 2.0770]	0.6492 [0.4182, 1.9374]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8158.04 [5462.57, 10727.86]	15508.97 [3542.07, 20820.25]
Trapping-surface period P_- (ms)	0.7702 [0.5857, 1.1502]	0.4051 [0.3018, 1.7739]

Table 162: CPTG compact-state, structural, clock, and geometric-rotation results for GW241110_124123. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.1172	12.7769	7781.98	13731.65
500 m inward	5.6838	9.0227	7992.74	14767.76
100 m inward	2.5180	4.0173	8127.24	15390.27
10 m inward	0.7944	1.2681	8155.42	15501.74
1 m inward	0.2512	0.4009	8157.76	15507.80
10 cm inward	0.0794	0.1268	8158.02	15508.77
1 cm inward	0.0251	0.0401	8158.04	15508.95
surface limit	0.0000	0.0000	8158.04	15508.97

Table 163: Near-inner-boundary profile for GW241110_124123. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $12.803 M_\odot$ and $5.720 M_\odot$, with completed geometric mass-equivalent shares of 18.74% and 28.11%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $12.617 < 17.501 < 23.948$ km, and component 2 gives $6.180 < 8.046 < 11.749$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5708$ and 1.6587 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.142 and 16.514 h. The determinant-averaged frame-dragging rates at those shells are 4816.89 and 9678.47 rad s $^{-1}$, approaching surface-limit values of 8158.04 and 15508.97 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.117/12.777 h at 1 km inward to 0.0251/0.0401 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.55 GW240526_093944

GW240526_093944 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 19.484 M_\odot$ and $m_2 = 11.309 M_\odot$, while the luminosity-distance median is 1939.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,843
Observed luminosity distance	1939.2 Mpc (median)
Detector-frame masses	$m_1 = 19.484 M_\odot$, $m_2 = 11.309 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.580$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3539$; $z_\pi = 0.3806$ (medians)
Post-handoff reorder fraction	10.05%

Table 164: Observational and coordinate summary for GW240526_093944. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	14.191 [9.547, 20.224]	8.226 [5.968, 11.816]
Physical compact-source mass $M_c (M_\odot)$	11.051 [7.159, 17.718]	5.706 [3.650, 8.276]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	2.733 [0.355, 6.085]	2.642 [0.333, 5.294]
Geometric share M_{geom}/M_∞ (%)	19.54 [2.59, 41.47]	32.93 [4.52, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.628 [0.242, 0.949]	0.392 [0.037, 0.911]
Polarization factor A_{pol}	1.243 [1.027, 1.709]	1.491 [1.047, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.706, 1.000]	0.871 [0.146, 0.999]
GR/CPTG matching radius r_m (km)	20.955 [14.097, 29.864]	12.146 [8.813, 17.448]
Structural Mode N_{BH}	1.5776 [1.4618, 1.8114]	1.7110 [1.4723, 1.9092]
Organization-shell radius r_N (km)	11.041 [7.396, 15.677]	6.392 [4.636, 9.191]
Inner trapping-surface radius r_- (km)	14.986 [10.184, 23.504]	8.203 [5.701, 11.408]
Structural- N lapse $\bar{\alpha}_N$	0.7507 [0.5660, 0.8419]	0.6473 [0.4854, 0.8334]
Structural- N clock $T_{24,N}$ (h)	18.016 [13.584, 20.205]	15.536 [11.649, 20.002]
Structural- N axis ω (rad s $^{-1}$)	4288.46 [1775.13, 7961.97]	8093.20 [2305.62, 14039.01]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	5642.84 [2817.72, 9069.41]	9585.15 [2306.86, 15582.12]
Structural- N equatorial ω (rad s $^{-1}$)	6970.95 [4282.85, 10382.83]	11379.00 [2307.66, 17058.59]
Structural- N period P_ω (ms)	1.1135 [0.6928, 2.2299]	0.6555 [0.4032, 2.7237]
Trapping-surface limit Ω_- (rad s $^{-1}$)	9239.97 [5903.43, 13812.06]	14765.08 [2515.54, 21936.07]
Trapping-surface period P_- (ms)	0.6800 [0.4549, 1.0643]	0.4255 [0.2864, 2.4977]

Table 165: CPTG compact-state, structural, clock, and geometric-rotation results for GW240526_093944. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.7588	12.7657	8668.90	13025.04
500 m inward	6.1223	9.0635	8981.37	14030.35
100 m inward	2.7073	4.0522	9193.64	14635.90
10 m inward	0.8537	1.2804	9235.96	14751.28
1 m inward	0.2699	0.4048	9239.31	14764.12
10 cm inward	0.0854	0.1280	9239.90	14764.91
1 cm inward	0.0270	0.0405	9239.96	14765.06
surface limit	0.0000	0.0000	9239.97	14765.08

Table 166: Near-inner-boundary profile for GW240526_093944. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $11.051 M_\odot$ and $5.706 M_\odot$, with completed geometric mass-equivalent shares of 19.54% and 32.93%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $11.041 < 14.986 < 20.955$ km, and component 2 gives $6.392 < 8.203 < 12.146$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5776$ and 1.7110; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.016 and 15.536 h. The determinant-averaged frame-dragging rates at those shells are 5642.84 and 9585.15 rad s $^{-1}$, approaching surface-limit values of 9239.97 and 14765.08 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.759/12.766 h at 1 km inward to 0.0270/0.0405 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.56 GW230605_065343

GW230605_065343 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 20.947 M_\odot$ and $m_2 = 13.298 M_\odot$, while the luminosity-distance median is 1064.2 Mpc. Two retained release labels map to this same numerical stream; their exact transformed identity is preserved in the provenance record rather than counted twice as separate physical realizations. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	30,436
Observed luminosity distance	1064.2 Mpc (median)
Detector-frame masses	$m_1 = 20.947 M_\odot$, $m_2 = 13.298 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.635$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2093$; $z_\pi = 0.2251$ (medians)
Post-handoff reorder fraction	13.58%

Table 167: Observational and coordinate summary for GW230605_065343. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	17.008 [11.902, 23.642]	10.933 [8.143, 15.333]
Physical compact-source mass $M_c (M_\odot)$	10.804 [7.775, 16.758]	7.242 [4.940, 10.117]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	6.416 [0.954, 9.815]	3.931 [0.546, 7.129]
Geometric share M_{geom}/M_∞ (%)	36.43 [6.98, 49.73]	36.84 [5.73, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.331 [0.037, 0.863]	0.324 [0.028, 0.887]
Polarization factor A_{pol}	1.573 [1.075, 1.989]	1.583 [1.061, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.819 [0.145, 0.997]	0.812 [0.111, 0.998]
GR/CPTG matching radius r_m (km)	25.114 [17.575, 34.910]	16.144 [12.024, 22.641]
Structural Mode N_{BH}	1.7512 [1.4866, 1.9092]	1.7560 [1.4792, 1.9105]
Organization-shell radius r_N (km)	13.226 [9.248, 18.451]	8.505 [6.324, 11.907]
Inner trapping-surface radius r_- (km)	15.990 [12.063, 23.067]	10.638 [7.765, 14.249]
Structural- N lapse $\bar{\alpha}_N$	0.6152 [0.4853, 0.8220]	0.6114 [0.4844, 0.8279]
Structural- N clock $T_{24,N}$ (h)	14.766 [11.648, 19.727]	14.674 [11.625, 19.869]
Structural- N axis ω (rad s $^{-1}$)	4273.17 [1151.81, 6790.23]	6201.47 [1260.20, 10261.70]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4887.01 [1163.37, 7577.41]	7309.05 [1260.87, 11375.05]
Structural- N equatorial ω (rad s $^{-1}$)	5555.92 [1163.85, 8474.75]	8624.36 [1261.24, 12502.23]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.2857 [0.8292, 5.4008]	0.8596 [0.5524, 4.9832]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6858.66 [1268.99, 11173.88]	11029.51 [1374.62, 16061.53]
Trapping-surface period P_- (ms)	0.9161 [0.5623, 4.9513]	0.5697 [0.3912, 4.5709]

Table 168: CPTG compact-state, structural, clock, and geometric-rotation results for GW230605_065343. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	9.3543	11.4361	6461.62	10017.55
500 m inward	6.6240	8.1240	6672.93	10582.54
100 m inward	2.9630	3.6382	6823.47	10950.56
10 m inward	0.9370	1.1505	6854.66	11021.74
1 m inward	0.2963	0.3638	6858.27	11028.65
10 cm inward	0.0937	0.1150	6858.62	11029.42
1 cm inward	0.0296	0.0364	6858.66	11029.50
surface limit	0.0000	0.0000	6858.66	11029.51

Table 169: Near-inner-boundary profile for GW230605_065343. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $10.804 M_\odot$ and $7.242 M_\odot$, with completed geometric mass-equivalent shares of 36.43% and 36.84%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $13.226 < 15.990 < 25.114$ km, and component 2 gives $8.505 < 10.638 < 16.144$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7512$ and 1.7560 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.766 and 14.674 h. The determinant-averaged frame-dragging rates at those shells are 4887.01 and 7309.05 rad s $^{-1}$, approaching surface-limit values of 6858.66 and 11029.51 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 9.354/11.436 h at 1 km inward to 0.0296/0.0364 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.57 GW230723_101834

GW230723_101834 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 21.388 M_\odot$ and $m_2 = 13.805 M_\odot$, while the luminosity-distance median is 1614.2 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	32,642
Observed luminosity distance	1614.2 Mpc (median)
Detector-frame masses	$m_1 = 21.388 M_\odot$, $m_2 = 13.805 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.645$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3021$; $z_\pi = 0.3248$ (medians)
Post-handoff reorder fraction	15.61%

Table 170: Observational and coordinate summary for GW230723_101834. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	16.203 [11.272, 21.909]	10.569 [7.753, 15.239]
Physical compact-source mass $M_c (M_\odot)$	11.317 [8.616, 16.392]	7.333 [4.656, 10.675]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.208 [0.477, 9.819]	3.441 [0.480, 6.948]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	26.58 [3.51, 49.17]	33.14 [5.04, 49.74]
Exterior spin magnitude $ \hat{a}_\infty $	0.503 [0.065, 0.930]	0.388 [0.037, 0.900]
Polarization factor A_{pol}	1.362 [1.036, 1.967]	1.496 [1.053, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.932 [0.253, 0.999]	0.868 [0.145, 0.999]
GR/CPTG matching radius r_m (km)	23.925 [16.645, 32.351]	15.607 [11.448, 22.503]
Structural Mode N_{BH}	1.6431 [1.4667, 1.9028]	1.7133 [1.4752, 1.9092]
Organization-shell radius r_N (km)	12.609 [8.738, 17.011]	8.217 [6.025, 11.843]
Inner trapping-surface radius r_- (km)	16.207 [12.378, 22.093]	10.545 [7.308, 14.698]
Structural- N lapse $\bar{\alpha}_N$	0.7002 [0.4904, 0.8379]	0.6454 [0.4853, 0.8311]
Structural- N clock $T_{24,N}$ (h)	16.804 [11.768, 20.110]	15.491 [11.648, 19.945]
Structural- N axis ω (rad s $^{-1}$)	4109.59 [1564.02, 6557.69]	6332.79 [1705.51, 10900.78]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	5082.00 [1855.88, 7501.87]	7512.46 [1707.37, 12084.52]
Structural- N equatorial ω (rad s $^{-1}$)	6162.41 [1860.37, 8612.00]	8909.88 [1708.30, 13255.35]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.2364 [0.8375, 3.3856]	0.8364 [0.5199, 3.6800]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8079.17 [2032.45, 11601.67]	11501.35 [1863.34, 16941.63]
Trapping-surface period P_- (ms)	0.7777 [0.5416, 3.0914]	0.5463 [0.3709, 3.3720]

Table 171: CPTG compact-state, structural, clock, and geometric-rotation results for GW230723_101834. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.9696	11.2990	7660.13	10458.24
500 m inward	6.3223	8.0080	7892.58	11040.14
100 m inward	2.8162	3.5802	8046.75	11420.71
10 m inward	0.8897	1.1317	8075.63	11491.62
1 m inward	0.2813	0.3579	8078.92	11500.36
10 cm inward	0.0890	0.1132	8079.15	11501.24
1 cm inward	0.0281	0.0358	8079.17	11501.34
surface limit	0.0000	0.0000	8079.17	11501.35

Table 172: Near-inner-boundary profile for GW230723_101834. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $11.317 M_\odot$ and $7.333 M_\odot$, with completed geometric mass-equivalent shares of 26.58% and 33.14%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $12.609 < 16.207 < 23.925$ km, and component 2 gives $8.217 < 10.545 < 15.607$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6431$ and 1.7133 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.804 and 15.491 h. The determinant-averaged frame-dragging rates at those shells are 5082.00 and 7512.46 rad s $^{-1}$, approaching surface-limit values of 8079.17 and 11501.35 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.970/11.299 h at 1 km inward to 0.0281/0.0358 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.58 GW190708_232457

GW190708_232457 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 23.407 M_\odot$ and $m_2 = 13.702 M_\odot$, while the luminosity-distance median is 927.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	17,719
Observed luminosity distance	927.8 Mpc (median)
Detector-frame masses	$m_1 = 23.407 M_\odot$, $m_2 = 13.702 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.585$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1854$; $z_\pi = 0.1991$ (medians)
Post-handoff reorder fraction	8.66%

Table 173: Observational and coordinate summary for GW190708_232457. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	19.555 [13.969, 23.810]	11.494 [9.530, 15.872]
Physical compact-source mass $M_c (M_\odot)$	11.567 [8.728, 16.718]	7.500 [5.425, 10.530]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	7.881 [1.657, 10.610]	4.479 [1.018, 7.409]
Geometric share M_{geom}/M_∞ (%)	40.44 [10.55, 49.86]	38.84 [9.28, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.261 [0.026, 0.795]	0.289 [0.027, 0.819]
Polarization factor A_{pol}	1.679 [1.118, 1.995]	1.635 [1.102, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.734 [0.104, 0.993]	0.772 [0.107, 0.995]
GR/CPTG matching radius r_m (km)	28.876 [20.628, 35.158]	16.972 [14.073, 23.436]
Structural Mode N_{BH}	1.7990 [1.5092, 1.9107]	1.7798 [1.5009, 1.9106]
Organization-shell radius r_N (km)	15.210 [10.870, 18.548]	8.945 [7.414, 12.332]
Inner trapping-surface radius r_- (km)	17.685 [13.606, 23.013]	11.037 [8.741, 14.879]
Structural- N lapse $\bar{\alpha}_N$	0.5763 [0.4842, 0.8040]	0.5921 [0.4843, 0.8106]
Structural- N clock $T_{24,N}$ (h)	13.832 [11.621, 19.295]	14.210 [11.623, 19.453]
Structural- N axis ω (rad s $^{-1}$)	3929.58 [775.71, 5922.34]	6147.81 [1138.43, 9275.81]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4385.49 [775.98, 6624.48]	7089.82 [1138.99, 10173.77]
Structural- N equatorial ω (rad s $^{-1}$)	4825.30 [776.12, 7364.84]	8070.99 [1139.27, 11009.73]
Structural- N period P_ω (ms)	1.4327 [0.9485, 8.0971]	0.8862 [0.6176, 5.5164]
Trapping-surface limit Ω_- (rad s $^{-1}$)	5802.86 [845.56, 9584.81]	10069.14 [1241.50, 13992.30]
Trapping-surface period P_- (ms)	1.0828 [0.6555, 7.4308]	0.6240 [0.4490, 5.0610]

Table 174: CPTG compact-state, structural, clock, and geometric-rotation results for GW190708_232457. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	9.1306	11.3073	5489.20	9141.25
500 m inward	6.4798	8.0400	5653.80	9650.39
100 m inward	2.9042	3.6048	5773.44	9991.07
10 m inward	0.9188	1.1406	5800.30	10060.00
1 m inward	0.2906	0.3607	5802.62	10068.60
10 cm inward	0.0919	0.1141	5802.84	10069.05
1 cm inward	0.0291	0.0361	5802.86	10069.13
surface limit	0.0000	0.0000	5802.86	10069.14

Table 175: Near-inner-boundary profile for GW190708_232457. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $11.567 M_\odot$ and $7.500 M_\odot$, with completed geometric mass-equivalent shares of 40.44% and 38.84%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $15.210 < 17.685 < 28.876$ km, and component 2 gives $8.945 < 11.037 < 16.972$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7990$ and 1.7798 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.832 and 14.210 h. The determinant-averaged frame-dragging rates at those shells are 4385.49 and 7089.82 rad s $^{-1}$, approaching surface-limit values of 5802.86 and 10069.14 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 9.131/11.307 h at 1 km inward to 0.0291/0.0361 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.59 GW230706_104333

GW230706_104333 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 21.843 M_\odot$ and $m_2 = 15.502 M_\odot$, while the luminosity-distance median is 1913.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	25,780
Observed luminosity distance	1913.3 Mpc (median)
Detector-frame masses	$m_1 = 21.843 M_\odot$, $m_2 = 15.502 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.710$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3498$; $z_\pi = 0.3762$ (medians)
Post-handoff reorder fraction	20.52%

Table 176: Observational and coordinate summary for GW230706_104333. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	15.685 [11.437, 21.252]	11.536 [8.594, 16.001]
Physical compact-source mass $M_c (M_\odot)$	11.933 [8.454, 17.771]	7.910 [5.189, 11.145]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.254 [0.349, 7.796]	3.820 [0.507, 7.405]
Geometric share M_{geom}/M_∞ (%)	21.01 [2.43, 44.70]	33.84 [4.89, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.602 [0.180, 0.952]	0.376 [0.035, 0.903]
Polarization factor A_{pol}	1.266 [1.025, 1.808]	1.511 [1.051, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.964 [0.589, 1.000]	0.859 [0.140, 0.999]
GR/CPTG matching radius r_m (km)	23.161 [16.888, 31.381]	17.034 [12.690, 23.628]
Structural Mode N_{BH}	1.5903 [1.4609, 1.8503]	1.7212 [1.4744, 1.9094]
Organization-shell radius r_N (km)	12.193 [8.863, 16.542]	8.974 [6.680, 12.425]
Inner trapping-surface radius r_- (km)	16.368 [12.244, 23.690]	11.443 [8.159, 15.262]
Structural- N lapse $\bar{\alpha}_N$	0.7408 [0.5336, 0.8426]	0.6392 [0.4852, 0.8317]
Structural- N clock $T_{24,N}$ (h)	17.779 [12.806, 20.222]	15.341 [11.644, 19.961]
Structural- N axis ω (rad s $^{-1}$)	3990.94 [1745.74, 6731.23]	5858.98 [1547.73, 9840.56]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	5091.66 [2724.02, 7643.52]	6936.68 [1548.87, 10865.22]
Structural- N equatorial ω (rad s $^{-1}$)	6283.10 [3953.40, 8653.96]	8198.21 [1549.37, 11910.77]
Structural- N period P_ω (ms)	1.2340 [0.8220, 2.3066]	0.9058 [0.5783, 4.0566]
Trapping-surface limit Ω_- (rad s $^{-1}$)	8346.12 [5156.96, 11484.68]	10597.50 [1690.20, 15310.63]
Trapping-surface period P_- (ms)	0.7528 [0.5471, 1.2184]	0.5929 [0.4104, 3.7174]

Table 177: CPTG compact-state, structural, clock, and geometric-rotation results for GW230706_104333. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.4664	10.9063	7909.37	9752.52
500 m inward	5.9356	7.7343	8150.39	10228.34
100 m inward	2.6331	3.4599	8310.57	10533.56
10 m inward	0.8308	1.0939	8342.76	10591.61
1 m inward	0.2627	0.3459	8345.78	10597.09
10 cm inward	0.0831	0.1094	8346.08	10597.51
1 cm inward	0.0263	0.0346	8346.12	10597.50
surface limit	0.0000	0.0000	8346.12	10597.50

Table 178: Near-inner-boundary profile for GW230706_104333. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $11.933 M_\odot$ and $7.910 M_\odot$, with completed geometric mass-equivalent shares of 21.01% and 33.84%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $12.193 < 16.368 < 23.161$ km, and component 2 gives $8.974 < 11.443 < 17.034$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5903$ and 1.7212 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.779 and 15.341 h. The determinant-averaged frame-dragging rates at those shells are 5091.66 and 6936.68 rad s $^{-1}$, approaching surface-limit values of 8346.12 and 10597.50 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.466/10.906 h at 1 km inward to 0.0263/0.0346 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.60 GW240622_004008

GW240622_004008 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 23.586 M_\odot$ and $m_2 = 14.432 M_\odot$, while the luminosity-distance median is 1471.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,440
Observed luminosity distance	1471.8 Mpc (median)
Detector-frame masses	$m_1 = 23.586 M_\odot$, $m_2 = 14.432 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.612$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2787$; $z_\pi = 0.2997$ (medians)
Post-handoff reorder fraction	13.50%

Table 179: Observational and coordinate summary for GW240622_004008. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	18.204 [12.636, 25.760]	11.210 [7.958, 16.069]
Physical compact-source mass $M_c (M_\odot)$	14.557 [10.163, 22.255]	8.112 [4.911, 11.410]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.057 [0.368, 7.205]	3.234 [0.347, 7.176]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	16.77 [2.21, 37.00]	29.57 [3.37, 49.61]
Exterior spin magnitude $ \hat{a}_\infty $	0.678 [0.321, 0.956]	0.450 [0.044, 0.933]
Polarization factor A_{pol}	1.202 [1.023, 1.587]	1.420 [1.035, 1.985]
Compact-source spin parameter $\hat{a}_c$	0.979 [0.809, 1.000]	0.908 [0.175, 0.999]
GR/CPTG matching radius r_m (km)	26.881 [18.659, 38.038]	16.553 [11.751, 23.728]
Structural Mode N_{BH}	1.5548 [1.4598, 1.7579]	1.6741 [1.4660, 1.9078]
Organization-shell radius r_N (km)	14.140 [9.781, 20.034]	8.705 [6.186, 12.517]
Inner trapping-surface radius r_- (km)	19.581 [14.102, 29.572]	11.510 [7.655, 15.609]
Structural- N lapse $\bar{\alpha}_N$	0.7683 [0.6099, 0.8435]	0.6762 [0.4864, 0.8385]
Structural- N clock $T_{24,N}$ (h)	18.439 [14.637, 20.244]	16.228 [11.674, 20.124]
Structural- N axis ω (rad s $^{-1}$)	3015.51 [1454.70, 5665.63]	5678.91 [1984.12, 10324.83]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4161.99 [2315.83, 6555.20]	6881.62 [2009.04, 11437.73]
Structural- N equatorial ω (rad s $^{-1}$)	5402.53 [3521.83, 7582.97]	8327.66 [2010.38, 12581.54]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.5097 [0.9585, 2.7131]	0.9130 [0.5493, 3.1275]
Trapping-surface limit Ω_- (rad s $^{-1}$)	7268.98 [4928.43, 10230.45]	11000.18 [2193.20, 16235.43]
Trapping-surface period P_- (ms)	0.8644 [0.6142, 1.2749]	0.5712 [0.3870, 2.8649]

Table 180: CPTG compact-state, structural, clock, and geometric-rotation results for GW240622_004008. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.3512	10.6469	6948.73	10191.13
500 m inward	5.1361	7.5241	7122.56	10642.34
100 m inward	2.2715	3.3574	7241.91	10934.62
10 m inward	0.7165	1.0613	7266.28	10994.22
1 m inward	0.2265	0.3355	7268.79	10999.85
10 cm inward	0.0716	0.1061	7268.96	11000.17
1 cm inward	0.0226	0.0336	7268.98	11000.18
surface limit	0.0000	0.0000	7268.98	11000.18

Table 181: Near-inner-boundary profile for GW240622_004008. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $14.557 M_\odot$ and $8.112 M_\odot$, with completed geometric mass-equivalent shares of 16.77% and 29.57%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $14.140 < 19.581 < 26.881$ km, and component 2 gives $8.705 < 11.510 < 16.553$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5548$ and 1.6741 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.439 and 16.228 h. The determinant-averaged frame-dragging rates at those shells are 4161.99 and 6881.62 rad s $^{-1}$, approaching surface-limit values of 7268.98 and 11000.18 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.351/10.647 h at 1 km inward to 0.0226/0.0336 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.61 GW231114_043211

GW231114_043211 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 28.613 M_\odot$ and $m_2 = 10.343 M_\odot$, while the luminosity-distance median is 1388.1 Mpc. Two retained release labels map to this same numerical stream; their exact transformed identity is preserved in the provenance record rather than counted twice as separate physical realizations. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	32,178
Observed luminosity distance	1388.1 Mpc (median)
Detector-frame masses	$m_1 = 28.613 M_\odot$, $m_2 = 10.343 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.361$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2648$; $z_\pi = 0.2848$ (medians)
Post-handoff reorder fraction	1.45%

Table 182: Observational and coordinate summary for GW231114_043211. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	22.418 [15.974, 31.528]	8.060 [5.949, 10.650]
Physical compact-source mass M_c ($M_\odot$)	12.576 [9.117, 21.294]	5.628 [3.625, 8.314]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.705 [4.893, 12.111]	2.369 [0.292, 4.652]
Geometric share M_{geom}/M_∞ (%)	42.97 [24.82, 49.90]	30.21 [3.87, 49.67]
Exterior spin magnitude $ \hat{a}_\infty $	0.214 [0.023, 0.533]	0.439 [0.041, 0.923]
Polarization factor A_{pol}	1.753 [1.330, 1.996]	1.433 [1.040, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.658 [0.091, 0.944]	0.901 [0.161, 0.999]
GR/CPTG matching radius r_m (km)	33.102 [23.588, 46.555]	11.902 [8.784, 15.726]
Structural Mode N_{BH}	1.8295 [1.6257, 1.9111]	1.6809 [1.4687, 1.9085]
Organization-shell radius r_N (km)	17.416 [12.423, 24.654]	6.263 [4.627, 8.282]
Inner trapping-surface radius r_- (km)	19.758 [14.421, 30.108]	8.022 [5.666, 11.247]
Structural- N lapse $\bar{\alpha}_N$	0.5510 [0.4839, 0.7135]	0.6708 [0.4859, 0.8363]
Structural- N clock $T_{24,N}$ (h)	13.223 [11.614, 17.125]	16.100 [11.661, 20.072]
Structural- N axis ω (rad s $^{-1}$)	3357.64 [615.13, 5282.06]	8116.61 [2609.35, 13973.18]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3594.40 [615.29, 5800.72]	9808.35 [2627.44, 15509.43]
Structural- N equatorial ω (rad s $^{-1}$)	3738.31 [615.38, 6361.59]	11758.31 [2628.30, 17059.31]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.7480 [1.0832, 10.2117]	0.6406 [0.4051, 2.3914]
Trapping-surface limit Ω_- (rad s $^{-1}$)	4349.49 [670.37, 7862.31]	15383.36 [2864.87, 21947.70]
Trapping-surface period P_- (ms)	1.4446 [0.7992, 9.3727]	0.4084 [0.2863, 2.1932]

Table 183: CPTG compact-state, structural, clock, and geometric-rotation results for GW231114_043211. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.7705	12.8805	4161.70	13573.04
500 m inward	6.2265	9.1225	4260.10	14616.31
100 m inward	2.7920	4.0686	4332.21	15252.30
10 m inward	0.8833	1.2850	4347.82	15370.86
1 m inward	0.2794	0.4063	4349.32	15381.75
10 cm inward	0.0883	0.1285	4349.47	15383.18
1 cm inward	0.0279	0.0406	4349.49	15383.34
surface limit	0.0000	0.0000	4349.49	15383.36

Table 184: Near-inner-boundary profile for GW231114_043211. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $12.576 M_\odot$ and $5.628 M_\odot$, with completed geometric mass-equivalent shares of 42.97% and 30.21%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $17.416 < 19.758 < 33.102$ km, and component 2 gives $6.263 < 8.022 < 11.902$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8295$ and 1.6809 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.223 and 16.100 h. The determinant-averaged frame-dragging rates at those shells are 3594.40 and 9808.35 rad s $^{-1}$, approaching surface-limit values of 4349.49 and 15383.36 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.770/12.881 h at 1 km inward to 0.0279/0.0406 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.62 GW240428_225440

GW240428_225440 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 22.996 M_\odot$ and $m_2 = 17.088 M_\odot$, while the luminosity-distance median is 840.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,573
Observed luminosity distance	840.6 Mpc (median)
Detector-frame masses	$m_1 = 22.996 M_\odot$, $m_2 = 17.088 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.743$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1693$; $z_\pi = 0.1821$ (medians)
Post-handoff reorder fraction	17.77%

Table 185: Observational and coordinate summary for GW240428_225440. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	19.314 [14.825, 24.410]	14.715 [11.606, 19.017]
Physical compact-source mass $M_c (M_\odot)$	12.269 [9.314, 17.765]	9.003 [6.710, 12.687]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	7.276 [1.152, 10.961]	5.992 [1.295, 8.951]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	37.03 [6.89, 49.81]	41.46 [9.88, 49.91]
Exterior spin magnitude $ \hat{a}_\infty $	0.321 [0.031, 0.864]	0.242 [0.021, 0.807]
Polarization factor A_{pol}	1.588 [1.074, 1.993]	1.708 [1.110, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.809 [0.122, 0.997]	0.706 [0.084, 0.994]
GR/CPTG matching radius r_m (km)	28.519 [21.891, 36.045]	21.728 [17.138, 28.082]
Structural Mode N_{BH}	1.7583 [1.4861, 1.9101]	1.8113 [1.5048, 1.9112]
Organization-shell radius r_N (km)	15.014 [11.518, 18.978]	11.441 [9.021, 14.768]
Inner trapping-surface radius r_- (km)	18.186 [14.799, 24.027]	13.696 [10.788, 17.670]
Structural- N lapse $\bar{\alpha}_N$	0.6096 [0.4847, 0.8224]	0.5661 [0.4838, 0.8075]
Structural- N clock $T_{24,N}$ (h)	14.630 [11.632, 19.737]	13.587 [11.611, 19.379]
Structural- N axis ω (rad s $^{-1}$)	3727.05 [855.73, 5618.35]	4819.25 [754.78, 7524.15]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4303.44 [856.30, 6256.87]	5454.05 [755.00, 8284.25]
Structural- N equatorial ω (rad s $^{-1}$)	4941.19 [856.58, 6921.78]	6195.41 [755.11, 8987.64]
Structural- N period P_ω (ms)	1.4600 [1.0042, 7.3376]	1.1520 [0.7584, 8.3221]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6175.68 [933.69, 9037.40]	7566.26 [822.60, 11445.54]
Trapping-surface period P_- (ms)	1.0174 [0.6952, 6.7294]	0.8304 [0.5490, 7.6382]

Table 186: CPTG compact-state, structural, clock, and geometric-rotation results for GW240428_225440. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.8594	10.3417	5877.49	7007.09
500 m inward	6.2772	7.3535	6039.70	7305.79
100 m inward	2.8094	3.2997	6152.19	7516.27
10 m inward	0.8886	1.0442	6173.39	7561.03
1 m inward	0.2810	0.3302	6175.51	7565.77
10 cm inward	0.0889	0.1044	6175.65	7566.21
1 cm inward	0.0281	0.0330	6175.67	7566.26
surface limit	0.0000	0.0000	6175.68	7566.26

Table 187: Near-inner-boundary profile for GW240428_225440. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $12.269 M_\odot$ and $9.003 M_\odot$, with completed geometric mass-equivalent shares of 37.03% and 41.46%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $15.014 < 18.186 < 28.519$ km, and component 2 gives $11.441 < 13.696 < 21.728$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7583$ and 1.8113; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.630 and 13.587 h. The determinant-averaged frame-dragging rates at those shells are 4303.44 and 5454.05 rad s $^{-1}$, approaching surface-limit values of 6175.68 and 7566.26 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.859/10.342 h at 1 km inward to 0.0281/0.0330 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.63 GW200225_060421

GW200225_060421 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 23.577 M_\odot$ and $m_2 = 17.199 M_\odot$, while the luminosity-distance median is 1149.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	15,593
Observed luminosity distance	1149.0 Mpc (median)
Detector-frame masses	$m_1 = 23.577 M_\odot$, $m_2 = 17.199 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.729$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2245$; $z_\pi = 0.2410$ (medians)
Post-handoff reorder fraction	14.92%

Table 188: Observational and coordinate summary for GW200225_060421. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	18.901 [14.430, 24.016]	13.989 [10.342, 18.258]
Physical compact-source mass $M_c (M_\odot)$	15.013 [10.141, 20.828]	9.134 [6.110, 13.147]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.406 [0.444, 8.920]	4.912 [0.846, 8.441]
Geometric share M_{geom}/M_∞ (%)	17.98 [2.55, 45.43]	36.39 [6.89, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.656 [0.165, 0.949]	0.332 [0.029, 0.865]
Polarization factor A_{pol}	1.219 [1.026, 1.832]	1.572 [1.074, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.976 [0.554, 1.000]	0.820 [0.115, 0.997]
GR/CPTG matching radius r_m (km)	27.910 [21.308, 35.463]	20.657 [15.271, 26.960]
Structural Mode N_{BH}	1.5645 [1.4616, 1.8590]	1.7508 [1.4860, 1.9103]
Organization-shell radius r_N (km)	14.677 [11.179, 18.655]	10.883 [8.043, 14.196]
Inner trapping-surface radius r_- (km)	20.231 [15.073, 27.693]	13.479 [9.651, 17.787]
Structural- N lapse $\bar{\alpha}_N$	0.7608 [0.5263, 0.8420]	0.6156 [0.4845, 0.8224]
Structural- N clock $T_{24,N}$ (h)	18.258 [12.631, 20.209]	14.775 [11.628, 19.738]
Structural- N axis ω (rad s $^{-1}$)	2936.55 [1422.64, 5481.27]	5025.49 [1073.36, 8129.50]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3982.08 [2269.65, 6194.49]	5858.30 [1073.94, 8972.02]
Structural- N equatorial ω (rad s $^{-1}$)	5086.48 [3373.67, 6932.21]	6770.03 [1074.23, 9809.42]
Structural- N period P_ω (ms)	1.5779 [1.0143, 2.7684]	1.0725 [0.7003, 5.8506]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6852.41 [4318.92, 9141.81]	8627.98 [1170.75, 12530.88]
Trapping-surface period P_- (ms)	0.9169 [0.6873, 1.4548]	0.7282 [0.5014, 5.3668]

Table 189: CPTG compact-state, structural, clock, and geometric-rotation results for GW200225_060421. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.3958	10.2362	6592.72	8061.50
500 m inward	5.1763	7.2636	6735.43	8377.38
100 m inward	2.2922	3.2524	6829.73	8581.36
10 m inward	0.7229	1.0289	6850.33	8624.01
1 m inward	0.2286	0.3254	6852.27	8627.47
10 cm inward	0.0723	0.1029	6852.39	8627.93
1 cm inward	0.0229	0.0325	6852.41	8627.97
surface limit	0.0000	0.0000	6852.41	8627.98

Table 190: Near-inner-boundary profile for GW200225_060421. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $15.013 M_\odot$ and $9.134 M_\odot$, with completed geometric mass-equivalent shares of 17.98% and 36.39%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $14.677 < 20.231 < 27.910$ km, and component 2 gives $10.883 < 13.479 < 20.657$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5645$ and 1.7508 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.258 and 14.775 h. The determinant-averaged frame-dragging rates at those shells are 3982.08 and 5858.30 rad s $^{-1}$, approaching surface-limit values of 6852.41 and 8627.98 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.396/10.236 h at 1 km inward to 0.0229/0.0325 h at 1 cm inward for components 1/2. Both Structural-Mode frame-dragging rates lie in the lower M1 quartile. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.64 GW190412_053044

GW190412_053044 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 31.761 M_\odot$ and $m_2 = 10.342 M_\odot$, while the luminosity-distance median is 721.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C01:Mixed
Posterior samples	7,886
Observed luminosity distance	721.1 Mpc (median)
Detector-frame masses	$m_1 = 31.761 M_\odot$, $m_2 = 10.342 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.326$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1474$; $z_\pi = 0.1584$ (medians)
Post-handoff reorder fraction	0.08%

Table 191: Observational and coordinate summary for GW190412_053044. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	27.483 [21.558, 33.411]	8.935 [7.561, 10.883]
Physical compact-source mass $M_c (M_\odot)$	17.254 [11.891, 23.969]	6.134 [4.296, 8.652]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	10.147 [7.130, 12.089]	2.861 [0.408, 4.984]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	37.37 [24.94, 47.41]	32.19 [4.82, 49.74]
Exterior spin magnitude $ \hat{a}_\infty $	0.315 [0.120, 0.531]	0.405 [0.036, 0.905]
Polarization factor A_{pol}	1.597 [1.332, 1.902]	1.475 [1.051, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.803 [0.433, 0.943]	0.880 [0.144, 0.999]
GR/CPTG matching radius r_m (km)	40.583 [31.832, 49.335]	13.194 [11.165, 16.070]
Structural Mode N_{BH}	1.7622 [1.6269, 1.8825]	1.7026 [1.4740, 1.9093]
Organization-shell radius r_N (km)	21.468 [16.763, 26.150]	6.953 [5.877, 8.468]
Inner trapping-surface radius r_- (km)	25.381 [18.952, 32.947]	8.831 [6.925, 11.576]
Structural- N lapse $\bar{\alpha}_N$	0.6064 [0.5068, 0.7126]	0.6539 [0.4853, 0.8321]
Structural- N clock $T_{24,N}$ (h)	14.553 [12.163, 17.103]	15.693 [11.647, 19.970]
Structural- N axis ω (rad s $^{-1}$)	3299.72 [2148.43, 4263.11]	7496.03 [2201.59, 11720.77]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3577.87 [2498.71, 4539.99]	9070.22 [2203.23, 12916.53]
Structural- N equatorial ω (rad s $^{-1}$)	3767.09 [2653.57, 4773.21]	10723.77 [2204.06, 13967.46]
Structural- N period P_ω (ms)	1.7561 [1.3840, 2.5146]	0.6927 [0.4864, 2.8518]
Trapping-surface limit Ω_- (rad s $^{-1}$)	4516.79 [2939.34, 5704.30]	13979.79 [2402.81, 17810.92]
Trapping-surface period P_- (ms)	1.3911 [1.1015, 2.1376]	0.4494 [0.3528, 2.6149]

Table 192: CPTG compact-state, structural, clock, and geometric-rotation results for GW190412_053044. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.5761	12.4858	4341.86	12500.60
500 m inward	5.3645	8.8566	4433.89	13361.40
100 m inward	2.4005	3.9604	4500.91	13859.69
10 m inward	0.7592	1.2519	4515.18	13967.05
1 m inward	0.2401	0.3959	4516.63	13978.45
10 cm inward	0.0759	0.1252	4516.78	13979.65
1 cm inward	0.0240	0.0396	4516.79	13979.77
surface limit	0.0000	0.0000	4516.79	13979.79

Table 193: Near-inner-boundary profile for GW190412_053044. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $17.254 M_\odot$ and $6.134 M_\odot$, with completed geometric mass-equivalent shares of 37.37% and 32.19%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.468 < 25.381 < 40.583$ km, and component 2 gives $6.953 < 8.831 < 13.194$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7622$ and 1.7026 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.553 and 15.693 h. The determinant-averaged frame-dragging rates at those shells are 3577.87 and 9070.22 rad s $^{-1}$, approaching surface-limit values of 4516.79 and 13979.79 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.576/12.486 h at 1 km inward to 0.0240/0.0396 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.65 GW241113_163507

GW241113_163507 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 24.065 M_\odot$ and $m_2 = 18.084 M_\odot$, while the luminosity-distance median is 1355.4 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	13,408
Observed luminosity distance	1355.4 Mpc (median)
Detector-frame masses	$m_1 = 24.065 M_\odot$, $m_2 = 18.084 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.751$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2593$; $z_\pi = 0.2789$ (medians)
Post-handoff reorder fraction	11.51%

Table 194: Observational and coordinate summary for GW241113_163507. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	18.904 [15.294, 24.373]	14.205 [10.877, 17.703]
Physical compact-source mass M_c ($M_\odot$)	17.023 [13.431, 22.469]	10.448 [6.389, 14.333]
Geometric mass-equivalent M_{geom} ($M_\odot$)	1.591 [0.226, 4.625]	3.826 [0.531, 7.157]
Geometric share M_{geom}/M_∞ (%)	8.34 [1.22, 23.07]	26.93 [3.99, 49.15]
Exterior spin magnitude $ \hat{a}_\infty $	0.837 [0.565, 0.976]	0.496 [0.066, 0.921]
Polarization factor A_{pol}	1.091 [1.012, 1.300]	1.369 [1.042, 1.967]
Compact-source spin parameter $\hat{a}_c$	0.996 [0.954, 1.000]	0.930 [0.256, 0.999]
GR/CPTG matching radius r_m (km)	27.915 [22.584, 35.989]	20.975 [16.061, 26.140]
Structural Mode N_{BH}	1.4950 [1.4547, 1.6091]	1.6466 [1.4693, 1.9026]
Organization-shell radius r_N (km)	14.622 [11.796, 18.843]	11.039 [8.441, 13.791]
Inner trapping-surface radius r_- (km)	22.581 [17.920, 29.925]	14.467 [10.132, 19.074]
Structural- N lapse $\bar{\alpha}_N$	0.8153 [0.7264, 0.8476]	0.6974 [0.4905, 0.8358]
Structural- N clock $T_{24,N}$ (h)	19.566 [17.433, 20.343]	16.738 [11.772, 20.059]
Structural- N axis ω (rad s $^{-1}$)	1993.57 [1302.46, 3778.18]	4701.53 [1934.95, 7835.79]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3094.20 [2143.16, 4853.97]	5696.79 [2655.53, 8680.58]
Structural- N equatorial ω (rad s $^{-1}$)	4628.93 [3459.89, 6004.87]	6729.18 [2675.04, 9517.26]
Structural- N period P_ω (ms)	2.0306 [1.2944, 2.9317]	1.1029 [0.7238, 2.3661]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6607.42 [5001.70, 8274.97]	8833.89 [2922.22, 12103.53]
Trapping-surface period P_- (ms)	0.9509 [0.7593, 1.2562]	0.7113 [0.5191, 2.1501]

Table 195: CPTG compact-state, structural, clock, and geometric-rotation results for GW241113_163507. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0201	9.4028	6464.35	8328.24
500 m inward	4.1647	6.6293	6552.60	8608.26
100 m inward	1.8229	2.9542	6597.11	8791.04
10 m inward	0.5736	0.9330	6605.71	8829.89
1 m inward	0.1813	0.2950	6607.28	8833.53
10 cm inward	0.0573	0.0933	6607.41	8833.86
1 cm inward	0.0181	0.0295	6607.42	8833.88
surface limit	0.0000	0.0000	6607.42	8833.89

Table 196: Near-inner-boundary profile for GW241113_163507. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $17.023 M_\odot$ and $10.448 M_\odot$, with completed geometric mass-equivalent shares of 8.34% and 26.93%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $14.622 < 22.581 < 27.915$ km, and component 2 gives $11.039 < 14.467 < 20.975$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.4950$ and 1.6466; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.566 and 16.738 h. The determinant-averaged frame-dragging rates at those shells are 3094.20 and 5696.79 rad s $^{-1}$, approaching surface-limit values of 6607.42 and 8833.89 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 6.020/9.403 h at 1 km inward to 0.0181/0.0295 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.66 GW231110_040320

GW231110_040320 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 25.758 M_\odot$ and $m_2 = 16.852 M_\odot$, while the luminosity-distance median is 1887.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	30,912
Observed luminosity distance	1887.8 Mpc (median)
Detector-frame masses	$m_1 = 25.758 M_\odot$, $m_2 = 16.852 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.654$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3458$; $z_\pi = 0.3719$ (medians)
Post-handoff reorder fraction	17.27%

Table 197: Observational and coordinate summary for GW231110_040320. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	18.729 [13.086, 24.793]	12.526 [9.469, 18.132]
Physical compact-source mass $M_c (M_\odot)$	13.705 [9.790, 19.969]	8.776 [5.661, 12.187]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	4.541 [0.532, 9.273]	4.060 [0.524, 8.369]
Geometric share M_{geom}/M_∞ (%)	24.18 [3.43, 45.10]	33.11 [4.63, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.545 [0.172, 0.932]	0.389 [0.036, 0.908]
Polarization factor A_{pol}	1.319 [1.036, 1.821]	1.495 [1.049, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.948 [0.570, 0.999]	0.869 [0.141, 0.999]
GR/CPTG matching radius r_m (km)	27.655 [19.323, 36.611]	18.497 [13.983, 26.774]
Structural Mode N_{BH}	1.6196 [1.4663, 1.8551]	1.7129 [1.4729, 1.9094]
Organization-shell radius r_N (km)	14.584 [10.123, 19.317]	9.736 [7.361, 14.083]
Inner trapping-surface radius r_- (km)	18.993 [14.160, 26.570]	12.605 [8.935, 16.964]
Structural- N lapse $\bar{\alpha}_N$	0.7183 [0.5296, 0.8382]	0.6458 [0.4852, 0.8329]
Structural- N clock $T_{24,N}$ (h)	17.238 [12.710, 20.118]	15.499 [11.645, 19.990]
Structural- N axis ω (rad s $^{-1}$)	3660.17 [1656.20, 5790.74]	5226.66 [1397.03, 8964.22]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4509.58 [2539.53, 6577.76]	6266.77 [1398.26, 9912.13]
Structural- N equatorial ω (rad s $^{-1}$)	5399.53 [3530.53, 7485.76]	7510.81 [1398.94, 10865.17]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.3933 [0.9552, 2.4742]	1.0026 [0.6339, 4.4936]
Trapping-surface limit Ω_- (rad s $^{-1}$)	7015.83 [4419.69, 9972.37]	9783.24 [1525.43, 13943.07]
Trapping-surface period P_- (ms)	0.8956 [0.6301, 1.4216]	0.6422 [0.4506, 4.1190]

Table 198: CPTG compact-state, structural, clock, and geometric-rotation results for GW231110_040320. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.0005	10.3412	6690.42	9123.82
500 m inward	5.6242	7.3179	6866.59	9492.61
100 m inward	2.5007	3.2725	6987.70	9731.57
10 m inward	0.7898	1.0347	7013.28	9778.12
1 m inward	0.2497	0.3272	7015.54	9782.54
10 cm inward	0.0790	0.1035	7015.82	9783.17
1 cm inward	0.0250	0.0327	7015.83	9783.24
surface limit	0.0000	0.0000	7015.83	9783.24

Table 199: Near-inner-boundary profile for GW231110_040320. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $13.705 M_\odot$ and $8.776 M_\odot$, with completed geometric mass-equivalent shares of 24.18% and 33.11%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $14.584 < 18.993 < 27.655$ km, and component 2 gives $9.736 < 12.605 < 18.497$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6196$ and 1.7129 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.238 and 15.499 h. The determinant-averaged frame-dragging rates at those shells are 4509.58 and 6266.77 rad s $^{-1}$, approaching surface-limit values of 7015.83 and 9783.24 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.001/10.341 h at 1 km inward to 0.0250/0.0327 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.67 GW231118_005626

GW231118_005626 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 27.790 M_\odot$ and $m_2 = 15.035 M_\odot$, while the luminosity-distance median is 2194.9 Mpc. Two retained release labels map to this same numerical stream; their exact transformed identity is preserved in the provenance record rather than counted twice as separate physical realizations. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	30,726
Observed luminosity distance	2194.9 Mpc (median)
Detector-frame masses	$m_1 = 27.790 M_\odot$, $m_2 = 15.035 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.541$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3933$; $z_\pi = 0.4231$ (medians)
Post-handoff reorder fraction	7.75%

Table 200: Observational and coordinate summary for GW231118_005626. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	19.569 [13.281, 26.997]	10.626 [8.270, 15.465]
Physical compact-source mass $M_c (M_\odot)$	15.863 [11.127, 24.922]	7.969 [4.931, 11.286]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.108 [0.406, 6.542]	2.862 [0.300, 6.536]
Geometric share M_{geom}/M_∞ (%)	15.99 [2.21, 33.22]	26.99 [2.97, 49.45]
Exterior spin magnitude $ \hat{a}_\infty $	0.693 [0.387, 0.956]	0.495 [0.053, 0.941]
Polarization factor A_{pol}	1.190 [1.023, 1.497]	1.370 [1.031, 1.978]
Compact-source spin parameter $\hat{a}_c$	0.982 [0.867, 1.000]	0.929 [0.208, 1.000]
GR/CPTG matching radius r_m (km)	28.896 [19.612, 39.865]	15.691 [12.212, 22.836]
Structural Mode N_{BH}	1.5486 [1.4598, 1.7142]	1.6472 [1.4638, 1.9059]
Organization-shell radius r_N (km)	15.226 [10.260, 20.879]	8.251 [6.420, 12.041]
Inner trapping-surface radius r_- (km)	21.215 [15.249, 33.129]	11.064 [7.800, 15.432]
Structural- N lapse $\bar{\alpha}_N$	0.7731 [0.6448, 0.8435]	0.6970 [0.4879, 0.8402]
Structural- N clock $T_{24,N}$ (h)	18.555 [15.474, 20.244]	16.727 [11.709, 20.166]
Structural- N axis ω (rad s $^{-1}$)	2784.02 [1212.78, 5113.02]	5696.07 [2259.05, 10279.41]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3894.45 [1973.56, 5934.31]	7029.56 [2599.31, 11367.22]
Structural- N equatorial ω (rad s $^{-1}$)	5072.52 [3114.84, 6999.68]	8688.35 [2601.63, 12462.57]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.6134 [1.0588, 3.1837]	0.8938 [0.5527, 2.4173]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6853.60 [4493.57, 9577.80]	11565.36 [2839.67, 15999.18]
Trapping-surface period P_- (ms)	0.9168 [0.6560, 1.3983]	0.5433 [0.3927, 2.2127]

Table 201: CPTG compact-state, structural, clock, and geometric-rotation results for GW231118_005626. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.0750	10.6840	6575.61	10694.37
500 m inward	4.9407	7.5349	6727.70	11192.57
100 m inward	2.1863	3.3541	6831.39	11501.16
10 m inward	0.6896	1.0591	6851.28	11558.59
1 m inward	0.2180	0.3349	6853.32	11565.31
10 cm inward	0.0689	0.1059	6853.57	11565.32
1 cm inward	0.0218	0.0335	6853.59	11565.36
surface limit	0.0000	0.0000	6853.60	11565.36

Table 202: Near-inner-boundary profile for GW231118_005626. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $15.863 M_\odot$ and $7.969 M_\odot$, with completed geometric mass-equivalent shares of 15.99% and 26.99%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $15.226 < 21.215 < 28.896$ km, and component 2 gives $8.251 < 11.064 < 15.691$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5486$ and 1.6472 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.555 and 16.727 h. The determinant-averaged frame-dragging rates at those shells are 3894.45 and 7029.56 rad s $^{-1}$, approaching surface-limit values of 6853.60 and 11565.36 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.075/10.684 h at 1 km inward to 0.0218/0.0335 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.68 GW190925_232845

GW190925_232845 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 24.689 M_\odot$ and $m_2 = 18.465 M_\odot$, while the luminosity-distance median is 926.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	29,450
Observed luminosity distance	926.9 Mpc (median)
Detector-frame masses	$m_1 = 24.689 M_\odot$, $m_2 = 18.465 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.748$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1852$; $z_\pi = 0.1989$ (medians)
Post-handoff reorder fraction	21.45%

Table 203: Observational and coordinate summary for GW190925_232845. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	20.184 [15.420, 27.070]	15.675 [11.788, 20.542]
Physical compact-source mass $M_c (M_\odot)$	14.216 [10.201, 21.320]	9.939 [6.996, 13.902]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	5.749 [0.779, 11.057]	6.016 [1.115, 9.619]
Geometric share M_{geom}/M_∞ (%)	28.06 [4.22, 49.13]	39.23 [8.15, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.477 [0.067, 0.916]	0.282 [0.025, 0.840]
Polarization factor A_{pol}	1.390 [1.044, 1.966]	1.646 [1.089, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.921 [0.259, 0.999]	0.764 [0.101, 0.996]
GR/CPTG matching radius r_m (km)	29.804 [22.769, 39.973]	23.146 [17.407, 30.333]
Structural Mode N_{BH}	1.6582 [1.4707, 1.9024]	1.7845 [1.4937, 1.9108]
Organization-shell radius r_N (km)	15.708 [11.956, 21.094]	12.194 [9.163, 15.944]
Inner trapping-surface radius r_- (km)	20.056 [15.729, 28.577]	14.867 [11.060, 19.198]
Structural- N lapse $\bar{\alpha}_N$	0.6885 [0.4907, 0.8347]	0.5882 [0.4841, 0.8163]
Structural- N clock $T_{24,N}$ (h)	16.523 [11.776, 20.034]	14.117 [11.619, 19.590]
Structural- N axis ω (rad s $^{-1}$)	3415.11 [1277.52, 5321.78]	4542.60 [844.92, 7219.35]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4130.05 [1602.20, 5956.77]	5231.48 [845.27, 7982.40]
Structural- N equatorial ω (rad s $^{-1}$)	4887.75 [1698.45, 6612.41]	5987.74 [845.41, 8707.32]
Structural- N period P_ω (ms)	1.5213 [1.0548, 3.9216]	1.2010 [0.7871, 7.4333]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6340.31 [1881.37, 8644.90]	7479.72 [920.98, 11119.95]
Trapping-surface period P_- (ms)	0.9910 [0.7268, 3.3397]	0.8400 [0.5650, 6.8223]

Table 204: CPTG compact-state, structural, clock, and geometric-rotation results for GW190925_232845. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.0504	9.8408	6074.53	7014.38
500 m inward	5.6776	6.9835	6217.66	7269.59
100 m inward	2.5330	3.1293	6316.59	7441.56
10 m inward	0.8004	0.9900	6338.31	7475.90
1 m inward	0.2531	0.3131	6340.08	7479.31
10 cm inward	0.0800	0.0990	6340.29	7479.68
1 cm inward	0.0253	0.0313	6340.31	7479.72
surface limit	0.0000	0.0000	6340.31	7479.72

Table 205: Near-inner-boundary profile for GW190925_232845. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $14.216 M_\odot$ and $9.939 M_\odot$, with completed geometric mass-equivalent shares of 28.06% and 39.23%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $15.708 < 20.056 < 29.804$ km, and component 2 gives $12.194 < 14.867 < 23.146$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6582$ and 1.7845 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.523 and 14.117 h. The determinant-averaged frame-dragging rates at those shells are 4130.05 and 5231.48 rad s $^{-1}$, approaching surface-limit values of 6340.31 and 7479.72 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.050/9.841 h at 1 km inward to 0.0253/0.0313 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.69 GW240930_035959

GW240930_035959 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 30.451 M_\odot$ and $m_2 = 13.352 M_\odot$, while the luminosity-distance median is 954.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,895
Observed luminosity distance	954.0 Mpc (median)
Detector-frame masses	$m_1 = 30.451 M_\odot$, $m_2 = 13.352 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.438$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1898$; $z_\pi = 0.2041$ (medians)
Post-handoff reorder fraction	1.12%

Table 206: Observational and coordinate summary for GW240930_035959. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	25.221 [20.345, 30.169]	11.073 [9.292, 13.289]
Physical compact-source mass $M_c (M_\odot)$	14.024 [11.689, 17.722]	7.425 [5.266, 10.825]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	11.404 [5.992, 14.282]	3.666 [0.607, 5.944]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	45.36 [27.52, 49.95]	33.35 [5.68, 49.72]
Exterior spin magnitude $ \hat{a}_\infty $	0.166 [0.016, 0.486]	0.385 [0.038, 0.888]
Polarization factor A_{pol}	1.830 [1.380, 1.998]	1.500 [1.060, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.557 [0.064, 0.925]	0.866 [0.149, 0.998]
GR/CPTG matching radius r_m (km)	37.242 [30.041, 44.548]	16.350 [13.721, 19.623]
Structural Mode N_{BH}	1.8582 [1.6527, 1.9117]	1.7157 [1.4789, 1.9091]
Organization-shell radius r_N (km)	19.580 [15.849, 23.403]	8.611 [7.216, 10.339]
Inner trapping-surface radius r_- (km)	22.193 [18.643, 26.428]	10.708 [8.519, 14.482]
Structural- N lapse $\bar{\alpha}_N$	0.5269 [0.4835, 0.6927]	0.6436 [0.4855, 0.8281]
Structural- N clock $T_{24,N}$ (h)	12.647 [11.603, 16.626]	15.446 [11.651, 19.874]
Structural- N axis ω (rad s $^{-1}$)	2925.15 [341.98, 4453.57]	6399.09 [1828.50, 9607.59]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3062.39 [342.05, 4875.71]	7512.95 [1830.85, 10522.60]
Structural- N equatorial ω (rad s $^{-1}$)	3106.95 [342.08, 5311.40]	8675.48 [1831.55, 11371.27]
Structural- N period $P_{\bar{\omega}}$ (ms)	2.0517 [1.2887, 18.3694]	0.8363 [0.5971, 3.4318]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3487.47 [372.58, 6622.67]	11218.51 [1997.69, 14375.56]
Trapping-surface period P_- (ms)	1.8016 [0.9487, 16.8638]	0.5601 [0.4371, 3.1452]

Table 207: CPTG compact-state, structural, clock, and geometric-rotation results for GW240930_035959. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.3051	11.4037	3339.62	10296.57
500 m inward	5.8954	8.0893	3417.74	10805.45
100 m inward	2.6434	3.6183	3474.27	11143.20
10 m inward	0.8364	1.1442	3486.13	11212.32
1 m inward	0.2645	0.3618	3487.30	11218.22
10 cm inward	0.0836	0.1144	3487.45	11218.47
1 cm inward	0.0265	0.0362	3487.47	11218.51
surface limit	0.0000	0.0000	3487.47	11218.51

Table 208: Near-inner-boundary profile for GW240930_035959. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $14.024 M_\odot$ and $7.425 M_\odot$, with completed geometric mass-equivalent shares of 45.36% and 33.35%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $19.580 < 22.193 < 37.242$ km, and component 2 gives $8.611 < 10.708 < 16.350$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8582$ and 1.7157 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 12.647 and 15.446 h. The determinant-averaged frame-dragging rates at those shells are 3062.39 and 7512.95 rad s $^{-1}$, approaching surface-limit values of 3487.47 and 11218.51 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.305/11.404 h at 1 km inward to 0.0265/0.0362 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.70 GW190828_065509

GW190828_065509 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 30.447 M_\odot$ and $m_2 = 13.436 M_\odot$, while the luminosity-distance median is 1545.4 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C01:Mixed
Posterior samples	71,641
Observed luminosity distance	1545.4 Mpc (median)
Detector-frame masses	$m_1 = 30.447 M_\odot$, $m_2 = 13.436 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.441$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2913$; $z_\pi = 0.3128$ (medians)
Post-handoff reorder fraction	3.73%

Table 209: Observational and coordinate summary for GW190828_065509. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	23.278 [15.786, 29.932]	10.270 [8.088, 14.773]
Physical compact-source mass $M_c (M_\odot)$	14.046 [9.990, 20.948]	7.264 [4.755, 10.521]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	9.184 [2.192, 12.695]	3.263 [0.399, 6.536]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	39.33 [11.80, 49.80]	31.91 [4.12, 49.71]
Exterior spin magnitude $ \hat{a}_\infty $	0.280 [0.031, 0.771]	0.410 [0.039, 0.918]
Polarization factor A_{pol}	1.648 [1.134, 1.992]	1.469 [1.043, 1.988]
Compact-source spin parameter $\hat{a}_c$	0.761 [0.125, 0.991]	0.883 [0.153, 0.999]
GR/CPTG matching radius r_m (km)	34.372 [23.309, 44.198]	15.164 [11.943, 21.815]
Structural Mode N_{BH}	1.7857 [1.5178, 1.9100]	1.6995 [1.4701, 1.9089]
Organization-shell radius r_N (km)	18.112 [12.277, 23.333]	7.981 [6.282, 11.495]
Inner trapping-surface radius r_- (km)	21.296 [15.413, 28.856]	10.309 [7.559, 14.626]
Structural- N lapse $\bar{\alpha}_N$	0.5873 [0.4847, 0.7972]	0.6563 [0.4856, 0.8352]
Structural- N clock $T_{24,N}$ (h)	14.095 [11.634, 19.133]	15.751 [11.654, 20.045]
Structural- N axis ω (rad s $^{-1}$)	3332.72 [774.55, 5062.78]	6287.97 [1870.32, 10648.71]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3718.89 [781.78, 5667.76]	7586.27 [1875.11, 11759.96]
Structural- N equatorial ω (rad s $^{-1}$)	4048.05 [782.50, 6402.31]	9086.32 [1875.76, 12811.74]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.6895 [1.1086, 8.0370]	0.8282 [0.5343, 3.3508]
Trapping-surface limit Ω_- (rad s $^{-1}$)	4884.23 [852.91, 8321.44]	11904.98 [2044.95, 16352.12]
Trapping-surface period P_- (ms)	1.2864 [0.7551, 7.3667]	0.5278 [0.3842, 3.0725]

Table 210: CPTG compact-state, structural, clock, and geometric-rotation results for GW190828_065509. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.2830	11.3763	4674.61	10876.60
500 m inward	5.8690	8.0529	4784.42	11461.92
100 m inward	2.6277	3.5951	4864.37	11825.19
10 m inward	0.8311	1.1360	4882.19	11896.38
1 m inward	0.2628	0.3592	4884.05	11904.26
10 cm inward	0.0831	0.1136	4884.21	11904.91
1 cm inward	0.0263	0.0359	4884.23	11904.97
surface limit	0.0000	0.0000	4884.23	11904.98

Table 211: Near-inner-boundary profile for GW190828_065509. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $14.046 M_\odot$ and $7.264 M_\odot$, with completed geometric mass-equivalent shares of 39.33% and 31.91%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.112 < 21.296 < 34.372$ km, and component 2 gives $7.981 < 10.309 < 15.164$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7857$ and 1.6995 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.095 and 15.751 h. The determinant-averaged frame-dragging rates at those shells are 3718.89 and 7586.27 rad s $^{-1}$, approaching surface-limit values of 4884.23 and 11904.98 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.283/11.376 h at 1 km inward to 0.0263/0.0359 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.71 GW190512_180714

GW190512_180714 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 29.341 M_\odot$ and $m_2 = 15.826 M_\odot$, while the luminosity-distance median is 1455.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	19,213
Observed luminosity distance	1455.8 Mpc (median)
Detector-frame masses	$m_1 = 29.341 M_\odot$, $m_2 = 15.826 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.539$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2765$; $z_\pi = 0.2969$ (medians)
Post-handoff reorder fraction	7.93%

Table 212: Observational and coordinate summary for GW190512_180714. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	22.788 [15.109, 28.379]	12.362 [9.814, 18.031]
Physical compact-source mass $M_c (M_\odot)$	13.262 [10.102, 19.712]	8.404 [5.713, 12.021]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	9.488 [1.797, 12.890]	4.331 [0.651, 8.244]
Geometric share M_{geom}/M_∞ (%)	41.82 [10.50, 49.90]	35.41 [5.70, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.235 [0.023, 0.795]	0.349 [0.031, 0.888]
Polarization factor A_{pol}	1.719 [1.117, 1.996]	1.548 [1.060, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.695 [0.090, 0.993]	0.836 [0.122, 0.998]
GR/CPTG matching radius r_m (km)	33.649 [22.311, 41.906]	18.255 [14.492, 26.625]
Structural Mode N_{BH}	1.8156 [1.5089, 1.9111]	1.7393 [1.4790, 1.9101]
Organization-shell radius r_N (km)	17.707 [11.763, 22.082]	9.609 [7.628, 14.024]
Inner trapping-surface radius r_- (km)	20.485 [15.536, 27.247]	12.151 [9.116, 17.051]
Structural- N lapse $\bar{\alpha}_N$	0.5625 [0.4839, 0.8042]	0.6248 [0.4847, 0.8280]
Structural- N clock $T_{24,N}$ (h)	13.501 [11.614, 19.301]	14.996 [11.632, 19.872]
Structural- N axis ω (rad s $^{-1}$)	3199.81 [535.95, 5090.14]	5469.00 [1235.45, 8854.15]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3601.81 [536.14, 5754.06]	6437.60 [1236.22, 9731.38]
Structural- N equatorial ω (rad s $^{-1}$)	3948.29 [536.24, 6556.48]	7593.62 [1236.63, 10551.92]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.7445 [1.0920, 11.7192]	0.9760 [0.6457, 5.0826]
Trapping-surface limit Ω_- (rad s $^{-1}$)	4685.50 [584.23, 8720.23]	9766.02 [1348.92, 13459.46]
Trapping-surface period P_- (ms)	1.3410 [0.7205, 10.7546]	0.6434 [0.4668, 4.6579]

Table 213: CPTG compact-state, structural, clock, and geometric-rotation results for GW190512_180714. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.4956	10.6377	4479.22	9054.11
500 m inward	6.0264	7.5391	4585.91	9448.57
100 m inward	2.6996	3.3729	4666.16	9708.38
10 m inward	0.8540	1.0664	4683.75	9760.23
1 m inward	0.2701	0.3372	4685.32	9765.40
10 cm inward	0.0854	0.1066	4685.48	9765.96
1 cm inward	0.0270	0.0337	4685.50	9766.02
surface limit	0.0000	0.0000	4685.50	9766.02

Table 214: Near-inner-boundary profile for GW190512_180714. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $13.262 M_\odot$ and $8.404 M_\odot$, with completed geometric mass-equivalent shares of 41.82% and 35.41%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $17.707 < 20.485 < 33.649$ km, and component 2 gives $9.609 < 12.151 < 18.255$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8156$ and 1.7393 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.501 and 14.996 h. The determinant-averaged frame-dragging rates at those shells are 3601.81 and 6437.60 rad s $^{-1}$, approaching surface-limit values of 4685.50 and 9766.02 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.496/10.638 h at 1 km inward to 0.0270/0.0337 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.72 GW151012_095443

GW151012_095443 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 29.624 M_\odot$ and $m_2 = 16.300 M_\odot$, while the luminosity-distance median is 999.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	9,566
Observed luminosity distance	999.5 Mpc (median)
Detector-frame masses	$m_1 = 29.624 M_\odot$, $m_2 = 16.300 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.550$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1982$; $z_\pi = 0.2128$ (medians)
Post-handoff reorder fraction	10.50%

Table 215: Observational and coordinate summary for GW151012_095443. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	24.448 [16.314, 38.867]	13.452 [8.548, 20.132]
Physical compact-source mass $M_c (M_\odot)$	16.627 [11.090, 29.798]	9.170 [5.462, 13.874]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	7.532 [0.979, 14.151]	4.292 [0.586, 9.003]
Geometric share M_{geom}/M_∞ (%)	29.57 [4.52, 49.43]	33.63 [5.04, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.450 [0.054, 0.911]	0.380 [0.036, 0.900]
Polarization factor A_{pol}	1.420 [1.047, 1.977]	1.507 [1.053, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.908 [0.211, 0.999]	0.862 [0.142, 0.999]
GR/CPTG matching radius r_m (km)	36.101 [24.089, 57.392]	19.863 [12.622, 29.727]
Structural Mode N_{BH}	1.6741 [1.4723, 1.9057]	1.7188 [1.4752, 1.9094]
Organization-shell radius r_N (km)	19.030 [12.666, 30.301]	10.455 [6.645, 15.656]
Inner trapping-surface radius r_- (km)	23.891 [16.745, 40.286]	13.256 [8.337, 19.178]
Structural- N lapse $\bar{\alpha}_N$	0.6762 [0.4880, 0.8334]	0.6411 [0.4852, 0.8310]
Structural- N clock $T_{24,N}$ (h)	16.228 [11.713, 20.002]	15.386 [11.646, 19.945]
Structural- N axis ω (rad s $^{-1}$)	2748.72 [924.50, 4773.64]	4960.00 [1324.74, 9282.73]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3316.94 [1160.68, 5394.89]	5901.22 [1325.83, 10337.36]
Structural- N equatorial ω (rad s $^{-1}$)	3900.42 [1212.92, 6108.92]	6947.59 [1326.38, 11545.96]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.8943 [1.1647, 5.4134]	1.0647 [0.6078, 4.7391]
Trapping-surface limit Ω_- (rad s $^{-1}$)	4998.51 [1324.81, 8099.03]	8947.54 [1446.15, 14950.31]
Trapping-surface period P_- (ms)	1.2570 [0.7758, 4.7427]	0.7022 [0.4203, 4.3448]

Table 216: CPTG compact-state, structural, clock, and geometric-rotation results for GW151012_095443. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.3764	10.0839	4822.20	8362.30
500 m inward	5.1982	7.1216	4917.15	8695.29
100 m inward	2.3187	3.1817	4983.74	8899.23
10 m inward	0.7328	1.0063	4997.44	8944.35
1 m inward	0.2317	0.3182	4998.35	8947.76
10 cm inward	0.0733	0.1006	4998.49	8947.56
1 cm inward	0.0232	0.0318	4998.51	8947.54
surface limit	0.0000	0.0000	4998.51	8947.54

Table 217: Near-inner-boundary profile for GW151012_095443. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $16.627 M_\odot$ and $9.170 M_\odot$, with completed geometric mass-equivalent shares of 29.57% and 33.63%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $19.030 < 23.891 < 36.101$ km, and component 2 gives $10.455 < 13.256 < 19.863$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6741$ and 1.7188 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.228 and 15.386 h. The determinant-averaged frame-dragging rates at those shells are 3316.94 and 5901.22 rad s $^{-1}$, approaching surface-limit values of 4998.51 and 8947.54 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.376/10.084 h at 1 km inward to 0.0232/0.0318 h at 1 cm inward for components 1/2. Both Structural-Mode frame-dragging rates lie in the lower M1 quartile. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.73 GW240527_230910

GW240527_230910 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 33.938 M_\odot$ and $m_2 = 12.296 M_\odot$, while the luminosity-distance median is 1193.0 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,200
Observed luminosity distance	1193.0 Mpc (median)
Detector-frame masses	$m_1 = 33.938 M_\odot$, $m_2 = 12.296 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.362$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2317$; $z_\pi = 0.2492$ (medians)
Post-handoff reorder fraction	4.61%

Table 218: Observational and coordinate summary for GW240527_230910. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	27.180 [16.329, 35.986]	9.833 [7.796, 15.926]
Physical compact-source mass $M_c (M_\odot)$	16.055 [11.782, 25.751]	7.258 [4.508, 11.101]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	10.457 [1.710, 15.348]	2.967 [0.350, 6.729]
Geometric share M_{geom}/M_∞ (%)	38.31 [9.01, 49.85]	29.94 [3.59, 49.67]
Exterior spin magnitude $ \hat{a}_\infty $	0.298 [0.027, 0.824]	0.444 [0.041, 0.929]
Polarization factor A_{pol}	1.621 [1.099, 1.994]	1.427 [1.037, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.784 [0.109, 0.995]	0.904 [0.162, 0.999]
GR/CPTG matching radius r_m (km)	40.134 [24.112, 53.138]	14.519 [11.511, 23.517]
Structural Mode N_{BH}	1.7734 [1.4992, 1.9105]	1.6780 [1.4671, 1.9085]
Organization-shell radius r_N (km)	21.134 [12.701, 28.095]	7.634 [6.053, 12.409]
Inner trapping-surface radius r_- (km)	24.606 [17.337, 35.395]	10.100 [7.219, 15.652]
Structural- N lapse $\bar{\alpha}_N$	0.5973 [0.4843, 0.8119]	0.6731 [0.4859, 0.8376]
Structural- N clock $T_{24,N}$ (h)	14.335 [11.624, 19.487]	16.154 [11.662, 20.102]
Structural- N axis ω (rad s $^{-1}$)	2761.09 [549.46, 4578.24]	6137.90 [1971.47, 11170.13]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3135.86 [549.69, 5231.87]	7483.09 [2003.27, 12302.25]
Structural- N equatorial ω (rad s $^{-1}$)	3440.52 [549.80, 6068.78]	9211.71 [2003.86, 13338.00]
Structural- N period $P_{\bar{\omega}}$ (ms)	2.0037 [1.2009, 11.4305]	0.8397 [0.5107, 3.1365]
Trapping-surface limit Ω_- (rad s $^{-1}$)	4178.66 [599.06, 8094.72]	12153.71 [2184.05, 16992.44]
Trapping-surface period P_- (ms)	1.5036 [0.7762, 10.4884]	0.5170 [0.3698, 2.8768]

Table 219: CPTG compact-state, structural, clock, and geometric-rotation results for GW240527_230910. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.6835	11.2396	4029.62	11080.09
500 m inward	5.4413	7.9278	4106.34	11699.69
100 m inward	2.4355	3.5298	4165.68	12074.83
10 m inward	0.7702	1.1143	4177.41	12145.55
1 m inward	0.2436	0.3523	4178.54	12152.73
10 cm inward	0.0770	0.1114	4178.65	12153.61
1 cm inward	0.0244	0.0352	4178.66	12153.70
surface limit	0.0000	0.0000	4178.66	12153.71

Table 220: Near-inner-boundary profile for GW240527_230910. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $16.055 M_\odot$ and $7.258 M_\odot$, with completed geometric mass-equivalent shares of 38.31% and 29.94%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.134 < 24.606 < 40.134$ km, and component 2 gives $7.634 < 10.100 < 14.519$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7734$ and 1.6780 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.335 and 16.154 h. The determinant-averaged frame-dragging rates at those shells are 3135.86 and 7483.09 rad s $^{-1}$, approaching surface-limit values of 4178.66 and 12153.71 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.684/11.240 h at 1 km inward to 0.0244/0.0352 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.74 GW230927_153832

GW230927_153832 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 26.510 M_\odot$ and $m_2 = 20.374 M_\odot$, while the luminosity-distance median is 1194.8 Mpc. Two retained release labels map to this same numerical stream; their exact transformed identity is preserved in the provenance record rather than counted twice as separate physical realizations. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	27,030
Observed luminosity distance	1194.8 Mpc (median)
Detector-frame masses	$m_1 = 26.510 M_\odot$, $m_2 = 20.374 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.769$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2320$; $z_\pi = 0.2495$ (medians)
Post-handoff reorder fraction	20.64%

Table 221: Observational and coordinate summary for GW230927_153832. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	21.039 [16.279, 25.157]	16.728 [13.846, 21.511]
Physical compact-source mass $M_c (M_\odot)$	13.701 [10.388, 19.335]	10.299 [7.762, 14.498]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	7.092 [1.236, 11.263]	6.815 [1.737, 10.085]
Geometric share M_{geom}/M_∞ (%)	33.63 [6.82, 49.58]	41.25 [11.33, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.380 [0.046, 0.866]	0.246 [0.023, 0.780]
Polarization factor A_{pol}	1.507 [1.073, 1.983]	1.702 [1.128, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.862 [0.182, 0.997]	0.712 [0.092, 0.992]
GR/CPTG matching radius r_m (km)	31.066 [24.038, 37.148]	24.700 [20.445, 31.764]
Structural Mode N_{BH}	1.7189 [1.4857, 1.9075]	1.8088 [1.5145, 1.9110]
Organization-shell radius r_N (km)	16.372 [12.641, 19.596]	13.009 [10.770, 16.713]
Inner trapping-surface radius r_- (km)	19.945 [16.294, 25.818]	15.571 [12.585, 19.969]
Structural- N lapse $\bar{\alpha}_N$	0.6410 [0.4867, 0.8227]	0.5682 [0.4839, 0.7998]
Structural- N clock $T_{24,N}$ (h)	15.385 [11.681, 19.745]	13.637 [11.615, 19.195]
Structural- N axis ω (rad s $^{-1}$)	3587.29 [1155.44, 5146.25]	4360.07 [738.97, 6472.63]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	4146.24 [1192.25, 5748.56]	4907.30 [739.17, 7083.09]
Structural- N equatorial ω (rad s $^{-1}$)	4739.96 [1192.88, 6340.35]	5486.78 [739.27, 7651.85]
Structural- N period P_ω (ms)	1.5154 [1.0930, 5.2700]	1.2804 [0.8871, 8.5004]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6006.49 [1301.76, 8267.17]	6696.99 [805.32, 9692.30]
Trapping-surface period P_- (ms)	1.0461 [0.7600, 4.8267]	0.9382 [0.6483, 7.8021]

Table 222: CPTG compact-state, structural, clock, and geometric-rotation results for GW230927_153832. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.3800	9.7027	5743.82	6275.35
500 m inward	5.9286	6.8905	5883.94	6503.62
100 m inward	2.6509	3.0895	5982.01	6660.44
10 m inward	0.8382	0.9774	6004.63	6693.78
1 m inward	0.2651	0.3091	6006.25	6696.59
10 cm inward	0.0838	0.0977	6006.46	6696.95
1 cm inward	0.0265	0.0309	6006.49	6696.99
surface limit	0.0000	0.0000	6006.49	6696.99

Table 223: Near-inner-boundary profile for GW230927_153832. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $13.701 M_\odot$ and $10.299 M_\odot$, with completed geometric mass-equivalent shares of 33.63% and 41.25%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $16.372 < 19.945 < 31.066$ km, and component 2 gives $13.009 < 15.571 < 24.700$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7189$ and 1.8088 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.385 and 13.637 h. The determinant-averaged frame-dragging rates at those shells are 4146.24 and 4907.30 rad s $^{-1}$, approaching surface-limit values of 6006.49 and 6696.99 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.380/9.703 h at 1 km inward to 0.0265/0.0309 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.75 GW231231_154016

GW231231_154016 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 27.150 M_\odot$ and $m_2 = 20.879 M_\odot$, while the luminosity-distance median is 1069.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	27,016
Observed luminosity distance	1069.6 Mpc (median)
Detector-frame masses	$m_1 = 27.150 M_\odot$, $m_2 = 20.879 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.769$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2103$; $z_\pi = 0.2261$ (medians)
Post-handoff reorder fraction	21.37%

Table 224: Observational and coordinate summary for GW231231_154016. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	21.773 [16.756, 27.636]	17.366 [13.858, 22.325]
Physical compact-source mass $M_c (M_\odot)$	14.658 [10.910, 21.247]	10.791 [7.944, 15.316]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	6.787 [0.922, 12.492]	6.893 [1.395, 10.436]
Geometric share M_{geom}/M_∞ (%)	31.62 [4.71, 49.58]	40.44 [8.88, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.415 [0.046, 0.907]	0.260 [0.024, 0.826]
Polarization factor A_{pol}	1.462 [1.049, 1.983]	1.679 [1.097, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.887 [0.182, 0.999]	0.734 [0.095, 0.995]
GR/CPTG matching radius r_m (km)	32.150 [24.742, 40.808]	25.642 [20.462, 32.965]
Structural Mode N_{BH}	1.6964 [1.4734, 1.9074]	1.7990 [1.4984, 1.9109]
Organization-shell radius r_N (km)	16.937 [13.001, 21.480]	13.501 [10.766, 17.340]
Inner trapping-surface radius r_- (km)	21.176 [16.936, 28.377]	16.301 [12.769, 21.046]
Structural- N lapse $\bar{\alpha}_N$	0.6588 [0.4867, 0.8325]	0.5763 [0.4840, 0.8126]
Structural- N clock $T_{24,N}$ (h)	15.811 [11.681, 19.981]	13.831 [11.616, 19.502]
Structural- N axis ω (rad s $^{-1}$)	3202.97 [1057.22, 4962.03]	4146.06 [715.05, 6343.89]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3819.26 [1106.54, 5525.43]	4702.39 [715.25, 6953.74]
Structural- N equatorial ω (rad s $^{-1}$)	4497.03 [1107.61, 6121.76]	5331.91 [715.36, 7514.99]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.6451 [1.1371, 5.6783]	1.3362 [0.9036, 8.7846]
Trapping-surface limit Ω_- (rad s $^{-1}$)	5809.61 [1209.59, 7970.59]	6598.11 [779.29, 9553.71]
Trapping-surface period P_- (ms)	1.0815 [0.7883, 5.1945]	0.9523 [0.6577, 8.0627]

Table 225: CPTG compact-state, structural, clock, and geometric-rotation results for GW231231_154016. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	8.0341	9.4657	5584.36	6218.49
500 m inward	5.6809	6.7192	5706.16	6426.35
100 m inward	2.5388	3.0119	5791.22	6563.85
10 m inward	0.8025	0.9528	5807.96	6594.83
1 m inward	0.2538	0.3013	5809.38	6597.74
10 cm inward	0.0803	0.0953	5809.58	6598.08
1 cm inward	0.0254	0.0301	5809.60	6598.11
surface limit	0.0000	0.0000	5809.61	6598.11

Table 226: Near-inner-boundary profile for GW231231_154016. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $14.658 M_\odot$ and $10.791 M_\odot$, with completed geometric mass-equivalent shares of 31.62% and 40.44%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $16.937 < 21.176 < 32.150$ km, and component 2 gives $13.501 < 16.301 < 25.642$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6964$ and 1.7990 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.811 and 13.831 h. The determinant-averaged frame-dragging rates at those shells are 3819.26 and 4702.39 rad s $^{-1}$, approaching surface-limit values of 5809.61 and 6598.11 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 8.034/9.466 h at 1 km inward to 0.0254/0.0301 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.76 GW240920_073424

GW240920_073424 is an M1 lower observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 31.406 M_\odot$ and $m_2 = 17.022 M_\odot$, while the luminosity-distance median is 1441.7 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	20,745
Observed luminosity distance	1441.7 Mpc (median)
Detector-frame masses	$m_1 = 31.406 M_\odot$, $m_2 = 17.022 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.542$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2738$; $z_\pi = 0.2944$ (medians)
Post-handoff reorder fraction	9.34%

Table 227: Observational and coordinate summary for GW240920_073424. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	24.226 [15.331, 31.849]	13.185 [10.030, 20.942]
Physical compact-source mass M_c ($M_\odot$)	16.732 [12.511, 26.693]	10.136 [6.126, 13.669]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.797 [0.507, 13.669]	3.250 [0.350, 9.398]
Geometric share M_{geom}/M_∞ (%)	24.47 [2.49, 49.25]	25.13 [2.78, 49.58]
Exterior spin magnitude $ \hat{a}_\infty $	0.540 [0.062, 0.951]	0.528 [0.046, 0.945]
Polarization factor A_{pol}	1.324 [1.025, 1.970]	1.336 [1.029, 1.983]
Compact-source spin parameter $\hat{a}_c$	0.946 [0.242, 1.000]	0.942 [0.182, 1.000]
GR/CPTG matching radius r_m (km)	35.773 [22.638, 47.029]	19.470 [14.811, 30.923]
Structural Mode N_{BH}	1.6223 [1.4612, 1.9037]	1.6287 [1.4628, 1.9074]
Organization-shell radius r_N (km)	18.840 [11.855, 24.705]	10.238 [7.788, 16.265]
Inner trapping-surface radius r_- (km)	23.925 [18.019, 35.556]	13.912 [9.552, 19.408]
Structural- N lapse $\bar{\alpha}_N$	0.7162 [0.4897, 0.8423]	0.7112 [0.4867, 0.8411]
Structural- N clock $T_{24,N}$ (h)	17.188 [11.752, 20.216]	17.069 [11.681, 20.186]
Structural- N axis ω (rad s $^{-1}$)	2615.89 [937.86, 4324.57]	4377.81 [1633.29, 8334.95]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3346.54 [1223.45, 4913.73]	5592.49 [1649.62, 9208.34]
Structural- N equatorial ω (rad s $^{-1}$)	4077.01 [1246.58, 5834.80]	7058.62 [1650.59, 10103.26]
Structural- N period P_ω (ms)	1.8775 [1.2787, 5.1356]	1.1235 [0.6823, 3.8089]
Trapping-surface limit Ω_- (rad s $^{-1}$)	5332.53 [1362.55, 8132.81]	9477.31 [1801.12, 12977.65]
Trapping-surface period P_- (ms)	1.1783 [0.7726, 4.6113]	0.6630 [0.4842, 3.4885]

Table 228: CPTG compact-state, structural, clock, and geometric-rotation results for GW240920_073424. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.2744	9.3748	5156.79	8940.67
500 m inward	5.1211	6.5934	5249.80	9246.62
100 m inward	2.2817	2.9331	5316.63	9435.82
10 m inward	0.7206	0.9264	5330.74	9472.17
1 m inward	0.2279	0.2929	5332.32	9476.64
10 cm inward	0.0721	0.0926	5332.51	9477.24
1 cm inward	0.0228	0.0293	5332.53	9477.30
surface limit	0.0000	0.0000	5332.53	9477.31

Table 229: Near-inner-boundary profile for GW240920_073424. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $16.732 M_\odot$ and $10.136 M_\odot$, with completed geometric mass-equivalent shares of 24.47% and 25.13%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.840 < 23.925 < 35.773$ km, and component 2 gives $10.238 < 13.912 < 19.470$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6223$ and 1.6287 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.188 and 17.069 h. The determinant-averaged frame-dragging rates at those shells are 3346.54 and 5592.49 rad s $^{-1}$, approaching surface-limit values of 5332.53 and 9477.31 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.274/9.375 h at 1 km inward to 0.0228/0.0293 h at 1 cm inward for components 1/2. Both component geometric shares fall in the lower quartile of the M1 distribution. Both Structural-Mode shell clocks lie in the upper quartile of the M1 time-accumulation distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.77 GW231014_040532

GW231014_040532 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 29.073 M_\odot$ and $m_2 = 20.961 M_\odot$, while the luminosity-distance median is 2339.7 Mpc. One retained release label defines this numerical stream.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	30,564
Observed luminosity distance	2339.7 Mpc (median)
Detector-frame masses	$m_1 = 29.073 M_\odot$, $m_2 = 20.961 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.721$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4152$; $z_\pi = 0.4467$ (medians)
Post-handoff reorder fraction	19.44%

Table 230: Observational and coordinate summary for GW231014_040532. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	19.814 [14.514, 27.988]	14.707 [10.246, 20.156]
Physical compact-source mass $M_c (M_\odot)$	15.407 [10.312, 24.833]	9.852 [6.326, 14.474]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.649 [0.424, 9.641]	4.876 [0.690, 9.075]
Geometric share M_{geom}/M_∞ (%)	18.42 [2.17, 46.09]	34.42 [5.30, 49.74]
Exterior spin magnitude $ \hat{a}_\infty $	0.648 [0.151, 0.957]	0.366 [0.036, 0.895]
Polarization factor A_{pol}	1.226 [1.022, 1.855]	1.525 [1.056, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.974 [0.519, 1.000]	0.851 [0.143, 0.998]
GR/CPTG matching radius r_m (km)	29.258 [21.432, 41.327]	21.716 [15.130, 29.762]
Structural Mode N_{BH}	1.5682 [1.4596, 1.8668]	1.7279 [1.4767, 1.9093]
Organization-shell radius r_N (km)	15.398 [11.242, 21.683]	11.434 [7.953, 15.676]
Inner trapping-surface radius r_- (km)	20.955 [15.206, 33.019]	14.379 [9.844, 19.664]
Structural- N lapse $\bar{\alpha}_N$	0.7579 [0.5198, 0.8437]	0.6339 [0.4853, 0.8299]
Structural- N clock $T_{24,N}$ (h)	18.190 [12.474, 20.248]	15.214 [11.647, 19.916]
Structural- N axis ω (rad s $^{-1}$)	2871.08 [1170.53, 5344.94]	4669.79 [1314.14, 8024.98]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3831.84 [1817.47, 6044.62]	5505.76 [1315.66, 8901.70]
Structural- N equatorial ω (rad s $^{-1}$)	4863.50 [2681.22, 6846.24]	6428.86 [1316.32, 9791.40]
Structural- N period P_ω (ms)	1.6397 [1.0395, 3.4571]	1.1412 [0.7058, 4.7757]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6505.88 [3501.98, 9104.72]	8262.58 [1435.43, 12584.55]
Trapping-surface period P_- (ms)	0.9658 [0.6901, 1.7942]	0.7604 [0.4993, 4.3772]

Table 231: CPTG compact-state, structural, clock, and geometric-rotation results for GW231014_040532. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.3099	9.7894	6267.94	7759.60
500 m inward	5.1218	6.9345	6398.99	8033.17
100 m inward	2.2701	3.1005	6485.83	8220.54
10 m inward	0.7164	0.9803	6503.53	8258.66
1 m inward	0.2265	0.3100	6505.66	8262.06
10 cm inward	0.0716	0.0980	6505.85	8262.53
1 cm inward	0.0226	0.0310	6505.87	8262.57
surface limit	0.0000	0.0000	6505.88	8262.58

Table 232: Near-inner-boundary profile for GW231014_040532. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $15.407 M_\odot$ and $9.852 M_\odot$, with completed geometric mass-equivalent shares of 18.42% and 34.42%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $15.398 < 20.955 < 29.258$ km, and component 2 gives $11.434 < 14.379 < 21.716$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5682$ and 1.7279 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.190 and 15.214 h. The determinant-averaged frame-dragging rates at those shells are 3831.84 and 5505.76 rad s $^{-1}$, approaching surface-limit values of 6505.88 and 8262.58 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.310/9.789 h at 1 km inward to 0.0226/0.0310 h at 1 cm inward for components 1/2. Both Structural-Mode frame-dragging rates lie in the lower M1 quartile. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.78 GW240507_041632

GW240507_041632 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 40.169 M_\odot$ and $m_2 = 10.907 M_\odot$, while the luminosity-distance median is 1454.5 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	21,033
Observed luminosity distance	1454.5 Mpc (median)
Detector-frame masses	$m_1 = 40.169 M_\odot$, $m_2 = 10.907 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.272$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2759$; $z_\pi = 0.2966$ (medians)
Post-handoff reorder fraction	2.13%

Table 233: Observational and coordinate summary for GW240507_041632. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	30.762 [18.925, 39.888]	8.393 [6.558, 14.911]
Physical compact-source mass M_c ($M_\odot$)	19.573 [12.542, 29.836]	6.196 [3.834, 10.525]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.324 [1.679, 16.858]	2.517 [0.283, 6.047]
Geometric share M_{geom}/M_∞ (%)	34.24 [6.95, 49.62]	29.62 [3.40, 49.62]
Exterior spin magnitude $ \hat{a}_\infty $	0.369 [0.044, 0.863]	0.449 [0.044, 0.933]
Polarization factor A_{pol}	1.521 [1.075, 1.985]	1.421 [1.035, 1.985]
Compact-source spin parameter $\hat{a}_c$	0.854 [0.172, 0.997]	0.907 [0.173, 0.999]
GR/CPTG matching radius r_m (km)	45.424 [27.946, 58.900]	12.394 [9.684, 22.018]
Structural Mode N_{BH}	1.7257 [1.4864, 1.9080]	1.6746 [1.4661, 1.9079]
Organization-shell radius r_N (km)	23.927 [14.723, 31.118]	6.527 [5.098, 11.613]
Inner trapping-surface radius r_- (km)	28.968 [18.678, 40.319]	8.617 [6.106, 14.783]
Structural- N lapse $\bar{\alpha}_N$	0.6356 [0.4863, 0.8221]	0.6758 [0.4864, 0.8384]
Structural- N clock $T_{24,N}$ (h)	15.255 [11.672, 19.730]	16.219 [11.672, 20.122]
Structural- N axis ω (rad s $^{-1}$)	2451.98 [768.20, 4115.06]	7029.13 [2284.29, 13179.27]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	2811.04 [791.07, 4659.32]	8685.06 [2493.13, 14521.68]
Structural- N equatorial ω (rad s $^{-1}$)	3165.06 [791.55, 5321.39]	10741.94 [2494.85, 15802.95]
Structural- N period P_ω (ms)	2.2352 [1.3485, 7.9426]	0.7234 [0.4327, 2.5202]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3976.59 [863.60, 6926.42]	14153.55 [2722.91, 20157.48]
Trapping-surface period P_- (ms)	1.5800 [0.9071, 7.2756]	0.4439 [0.3117, 2.3075]

Table 234: CPTG compact-state, structural, clock, and geometric-rotation results for GW240507_041632. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.9726	12.1348	3863.46	12617.09
500 m inward	4.9310	8.5504	3921.77	13522.88
100 m inward	2.2040	3.7995	3966.81	14037.04
10 m inward	0.6969	1.1990	3975.65	14146.53
1 m inward	0.2204	0.3790	3976.51	14153.34
10 cm inward	0.0697	0.1199	3976.58	14153.53
1 cm inward	0.0220	0.0379	3976.59	14153.55
surface limit	0.0000	0.0000	3976.59	14153.55

Table 235: Near-inner-boundary profile for GW240507_041632. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $19.573 M_\odot$ and $6.196 M_\odot$, with completed geometric mass-equivalent shares of 34.24% and 29.62%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $23.927 < 28.968 < 45.424$ km, and component 2 gives $6.527 < 8.617 < 12.394$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7257$ and 1.6746 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.255 and 16.219 h. The determinant-averaged frame-dragging rates at those shells are 2811.04 and 8685.06 rad s $^{-1}$, approaching surface-limit values of 3976.59 and 14153.55 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 6.973/12.135 h at 1 km inward to 0.0220/0.0379 h at 1 cm inward for components 1/2. Its downstream compact-state readouts remain near the central M1 ranges rather than defining an extreme of the class. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.79 GW240531_040326

GW240531_040326 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 29.435 M_\odot$ and $m_2 = 22.005 M_\odot$, while the luminosity-distance median is 3043.2 Mpc. One retained release label defines this numerical stream.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,768
Observed luminosity distance	3043.2 Mpc (median)
Detector-frame masses	$m_1 = 29.435 M_\odot$, $m_2 = 22.005 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.748$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5178$; $z_\pi = 0.5576$ (medians)
Post-handoff reorder fraction	18.17%

Table 236: Observational and coordinate summary for GW240531_040326. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	18.711 [14.045, 26.354]	14.228 [9.758, 18.939]
Physical compact-source mass $M_c (M_\odot)$	14.875 [9.863, 23.315]	9.351 [5.987, 13.771]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	3.237 [0.375, 9.094]	4.840 [0.722, 8.542]
Geometric share M_{geom}/M_∞ (%)	17.45 [2.01, 45.36]	35.47 [5.63, 49.78]
Exterior spin magnitude $ \hat{a}_\infty $	0.666 [0.167, 0.960]	0.348 [0.033, 0.889]
Polarization factor A_{pol}	1.211 [1.021, 1.830]	1.550 [1.060, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.977 [0.558, 1.000]	0.835 [0.132, 0.998]
GR/CPTG matching radius r_m (km)	27.629 [20.740, 38.915]	21.010 [14.409, 27.966]
Structural Mode N_{BH}	1.5602 [1.4588, 1.8582]	1.7400 [1.4786, 1.9097]
Organization-shell radius r_N (km)	14.521 [10.887, 20.451]	11.058 [7.584, 14.721]
Inner trapping-surface radius r_- (km)	20.116 [14.527, 31.098]	13.744 [9.319, 18.645]
Structural- N lapse $\bar{\alpha}_N$	0.7641 [0.5270, 0.8443]	0.6243 [0.4849, 0.8283]
Structural- N clock $T_{24,N}$ (h)	18.338 [12.648, 20.264]	14.982 [11.639, 19.880]
Structural- N axis ω (rad s $^{-1}$)	2898.92 [1251.76, 5644.21]	4935.30 [1253.93, 8341.40]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3940.27 [1966.74, 6379.65]	5749.36 [1254.36, 9296.72]
Structural- N equatorial ω (rad s $^{-1}$)	5077.53 [2997.43, 7183.41]	6650.92 [1254.57, 10258.42]
Structural- N period P_ω (ms)	1.5946 [0.9849, 3.1947]	1.0929 [0.6758, 5.0091]
Trapping-surface limit Ω_- (rad s $^{-1}$)	6841.29 [3976.46, 9440.71]	8504.76 [1367.79, 13136.00]
Trapping-surface period P_- (ms)	0.9184 [0.6655, 1.5801]	0.7388 [0.4783, 4.5937]

Table 237: CPTG compact-state, structural, clock, and geometric-rotation results for GW240531_040326. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.3790	10.0805	6595.94	7950.53
500 m inward	5.1638	7.1454	6733.44	8249.32
100 m inward	2.2874	3.1983	6820.26	8460.48
10 m inward	0.7215	1.0112	6839.46	8500.65
1 m inward	0.2281	0.3198	6841.25	8504.42
10 cm inward	0.0721	0.1011	6841.28	8504.72
1 cm inward	0.0228	0.0320	6841.29	8504.75
surface limit	0.0000	0.0000	6841.29	8504.76

Table 238: Near-inner-boundary profile for GW240531_040326. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $14.875 M_\odot$ and $9.351 M_\odot$, with completed geometric mass-equivalent shares of 17.45% and 35.47%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $14.521 < 20.116 < 27.629$ km, and component 2 gives $11.058 < 13.744 < 21.010$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5602$ and 1.7400 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.338 and 14.982 h. The determinant-averaged frame-dragging rates at those shells are 3940.27 and 5749.36 rad s $^{-1}$, approaching surface-limit values of 6841.29 and 8504.76 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.379/10.080 h at 1 km inward to 0.0228/0.0320 h at 1 cm inward for components 1/2. Both Structural-Mode frame-dragging rates lie in the lower M1 quartile. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.80 GW240921_201835

GW240921_201835 is an M1 lower observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 41.480 M_\odot$ and $m_2 = 11.699 M_\odot$, while the luminosity-distance median is 917.7 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,020
Observed luminosity distance	917.7 Mpc (median)
Detector-frame masses	$m_1 = 41.480 M_\odot$, $m_2 = 11.699 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.282$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1833$; $z_\pi = 0.1971$ (medians)
Post-handoff reorder fraction	0.55%

Table 239: Observational and coordinate summary for GW240921_201835. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	34.454 [25.076, 43.307]	9.719 [8.018, 12.639]
Physical compact-source mass $M_c (M_\odot)$	19.323 [14.958, 27.977]	7.001 [4.636, 10.312]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	15.119 [6.295, 18.842]	2.794 [0.314, 5.266]
Geometric share M_{geom}/M_∞ (%)	43.70 [22.57, 49.93]	28.56 [3.31, 49.59]
Exterior spin magnitude $ \hat{a}_\infty $	0.200 [0.019, 0.574]	0.468 [0.046, 0.934]
Polarization factor A_{pol}	1.776 [1.291, 1.997]	1.400 [1.034, 1.984]
Compact-source spin parameter $\hat{a}_c$	0.631 [0.076, 0.957]	0.917 [0.179, 0.999]
GR/CPTG matching radius r_m (km)	50.875 [37.028, 63.948]	14.351 [11.840, 18.663]
Structural Mode N_{BH}	1.8383 [1.6044, 1.9114]	1.6634 [1.4657, 1.9076]
Organization-shell radius r_N (km)	26.754 [19.542, 33.781]	7.556 [6.228, 9.838]
Inner trapping-surface radius r_- (km)	30.410 [23.546, 40.060]	9.788 [7.447, 13.895]
Structural- N lapse $\bar{\alpha}_N$	0.5436 [0.4837, 0.7299]	0.6845 [0.4866, 0.8388]
Structural- N clock $T_{24,N}$ (h)	13.047 [11.608, 17.518]	16.427 [11.678, 20.130]
Structural- N axis ω (rad s $^{-1}$)	2200.46 [323.29, 3396.97]	6463.36 [2286.80, 10866.13]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	2370.66 [323.35, 3771.31]	7961.99 [2565.99, 11927.35]
Structural- N equatorial ω (rad s $^{-1}$)	2466.67 [323.39, 4175.39]	9662.36 [2568.10, 12969.17]
Structural- N period $P_{\bar{\omega}}$ (ms)	2.6504 [1.6660, 19.4313]	0.7891 [0.5268, 2.4486]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2831.69 [352.23, 5287.47]	12737.76 [2803.69, 16541.74]
Trapping-surface period P_- (ms)	2.2189 [1.1883, 17.8382]	0.4933 [0.3798, 2.2410]

Table 240: CPTG compact-state, structural, clock, and geometric-rotation results for GW240921_201835. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.0906	11.5930	2747.60	11605.46
500 m inward	5.0246	8.1896	2791.28	12253.95
100 m inward	2.2507	3.6483	2823.38	12655.06
10 m inward	0.7120	1.1523	2830.87	12730.57
1 m inward	0.2251	0.3643	2831.61	12736.91
10 cm inward	0.0712	0.1152	2831.68	12737.67
1 cm inward	0.0225	0.0364	2831.69	12737.75
surface limit	0.0000	0.0000	2831.69	12737.76

Table 241: Near-inner-boundary profile for GW240921_201835. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $19.323 M_\odot$ and $7.001 M_\odot$, with completed geometric mass-equivalent shares of 43.70% and 28.56%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.754 < 30.410 < 50.875$ km, and component 2 gives $7.556 < 9.788 < 14.351$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8383$ and 1.6634 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 13.047 and 16.427 h. The determinant-averaged frame-dragging rates at those shells are 2370.66 and 7961.99 rad s $^{-1}$, approaching surface-limit values of 2831.69 and 12737.76 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.091/11.593 h at 1 km inward to 0.0225/0.0364 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.81 GW231108_125142

GW231108_125142 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 31.530 M_\odot$ and $m_2 = 23.802 M_\odot$, while the luminosity-distance median is 2046.7 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	26,956
Observed luminosity distance	2046.7 Mpc (median)
Detector-frame masses	$m_1 = 31.530 M_\odot$, $m_2 = 23.802 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.755$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3706$; $z_\pi = 0.3986$ (medians)
Post-handoff reorder fraction	23.41%

Table 242: Observational and coordinate summary for GW231108_125142. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	22.175 [16.676, 28.235]	17.531 [13.961, 23.317]
Physical compact-source mass $M_c (M_\odot)$	15.261 [11.413, 21.406]	11.220 [8.009, 15.654]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	6.355 [0.745, 12.806]	6.801 [1.334, 10.903]
Geometric share M_{geom}/M_∞ (%)	29.16 [3.84, 49.50]	39.25 [8.46, 49.88]
Exterior spin magnitude $ \hat{a}_\infty $	0.457 [0.051, 0.924]	0.282 [0.025, 0.834]
Polarization factor A_{pol}	1.412 [1.040, 1.980]	1.646 [1.092, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.911 [0.199, 0.999]	0.763 [0.100, 0.996]
GR/CPTG matching radius r_m (km)	32.745 [24.625, 41.693]	25.886 [20.615, 34.431]
Structural Mode N_{BH}	1.6698 [1.4685, 1.9065]	1.7847 [1.4957, 1.9108]
Organization-shell radius r_N (km)	17.243 [12.931, 21.951]	13.636 [10.853, 18.107]
Inner trapping-surface radius r_- (km)	21.906 [17.410, 28.666]	16.713 [12.882, 21.667]
Structural- N lapse $\bar{\alpha}_N$	0.6795 [0.4874, 0.8365]	0.5880 [0.4841, 0.8147]
Structural- N clock $T_{24,N}$ (h)	16.308 [11.698, 20.075]	14.113 [11.619, 19.553]
Structural- N axis ω (rad s $^{-1}$)	3051.41 [1089.26, 4806.12]	4050.04 [747.48, 6266.26]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3706.82 [1162.59, 5402.08]	4636.21 [747.71, 6871.17]
Structural- N equatorial ω (rad s $^{-1}$)	4448.53 [1163.74, 6031.73]	5317.67 [747.83, 7469.61]
Structural- N period $P_{\bar{\omega}}$ (ms)	1.6950 [1.1631, 5.4045]	1.3552 [0.9144, 8.4032]
Trapping-surface limit Ω_- (rad s $^{-1}$)	5801.68 [1270.42, 7943.29]	6652.07 [814.68, 9529.34]
Trapping-surface period P_- (ms)	1.0830 [0.7910, 4.9458]	0.9445 [0.6594, 7.7124]

Table 243: CPTG compact-state, structural, clock, and geometric-rotation results for GW231108_125142. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.8064	9.2770	5589.54	6284.39
500 m inward	5.5111	6.5801	5703.28	6479.62
100 m inward	2.4602	2.9474	5784.60	6618.26
10 m inward	0.7775	0.9323	5799.94	6648.05
1 m inward	0.2459	0.2948	5801.56	6651.68
10 cm inward	0.0777	0.0932	5801.66	6652.04
1 cm inward	0.0246	0.0295	5801.68	6652.07
surface limit	0.0000	0.0000	5801.68	6652.07

Table 244: Near-inner-boundary profile for GW231108_125142. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $15.261 M_\odot$ and $11.220 M_\odot$, with completed geometric mass-equivalent shares of 29.16% and 39.25%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $17.243 < 21.906 < 32.745$ km, and component 2 gives $13.636 < 16.713 < 25.886$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6698$ and 1.7847 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.308 and 14.113 h. The determinant-averaged frame-dragging rates at those shells are 3706.82 and 4636.21 rad s $^{-1}$, approaching surface-limit values of 5801.68 and 6652.07 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.806/9.277 h at 1 km inward to 0.0246/0.0295 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

3.82 GW190408_181802

GW190408_181802 is an M1 lower observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 31.696 M_\odot$ and $m_2 = 23.831 M_\odot$, while the luminosity-distance median is 1539.6 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M1: Lower observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	10,527
Observed luminosity distance	1539.6 Mpc (median)
Detector-frame masses	$m_1 = 31.696 M_\odot$, $m_2 = 23.831 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.752$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2904$; $z_\pi = 0.3117$ (medians)
Post-handoff reorder fraction	19.98%

Table 245: Observational and coordinate summary for GW190408_181802. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	23.933 [18.250, 29.700]	18.607 [14.315, 24.528]
Physical compact-source mass $M_c (M_\odot)$	16.063 [12.067, 22.758]	11.586 [8.221, 16.101]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	7.404 [1.174, 13.506]	7.329 [1.640, 11.497]
Geometric share M_{geom}/M_∞ (%)	30.99 [5.61, 49.60]	40.52 [9.97, 49.90]
Exterior spin magnitude $ \hat{a}_\infty $	0.425 [0.045, 0.889]	0.259 [0.023, 0.806]
Polarization factor A_{pol}	1.449 [1.059, 1.984]	1.681 [1.111, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.893 [0.178, 0.998]	0.732 [0.090, 0.994]
GR/CPTG matching radius r_m (km)	35.340 [26.949, 43.856]	27.476 [21.139, 36.218]
Structural Mode N_{BH}	1.6894 [1.4785, 1.9077]	1.8000 [1.5054, 1.9111]
Organization-shell radius r_N (km)	18.633 [14.163, 23.111]	14.485 [11.130, 19.051]
Inner trapping-surface radius r_- (km)	23.277 [18.709, 30.307]	17.486 [13.131, 22.547]
Structural- N lapse $\bar{\alpha}_N$	0.6642 [0.4865, 0.8284]	0.5755 [0.4839, 0.8070]
Structural- N clock $T_{24,N}$ (h)	15.941 [11.677, 19.882]	13.811 [11.614, 19.367]
Structural- N axis ω (rad s $^{-1}$)	2975.64 [956.13, 4465.96]	3851.16 [615.50, 6087.92]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	3545.66 [973.48, 4997.86]	4385.84 [615.71, 6684.73]
Structural- N equatorial ω (rad s $^{-1}$)	4151.86 [974.07, 5562.99]	4983.36 [615.82, 7265.38]
Structural- N period P_ω (ms)	1.7721 [1.2572, 6.4543]	1.4326 [0.9399, 10.2047]
Trapping-surface limit Ω_- (rad s $^{-1}$)	5345.37 [1062.25, 7297.82]	6123.79 [670.91, 9188.54]
Trapping-surface period P_- (ms)	1.1754 [0.8610, 5.9150]	1.0260 [0.6838, 9.3652]

Table 246: CPTG compact-state, structural, clock, and geometric-rotation results for GW190408_181802. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.6777	9.1254	5159.07	5788.12
500 m inward	5.4254	6.4746	5260.47	5967.50
100 m inward	2.4242	2.9018	5328.47	6093.60
10 m inward	0.7664	0.9179	5343.81	6120.45
1 m inward	0.2423	0.2903	5345.16	6123.45
10 cm inward	0.0766	0.0918	5345.35	6123.75
1 cm inward	0.0242	0.0290	5345.37	6123.78
surface limit	0.0000	0.0000	5345.37	6123.79

Table 247: Near-inner-boundary profile for GW190408_181802. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $16.063 M_\odot$ and $11.586 M_\odot$, with completed geometric mass-equivalent shares of 30.99% and 40.52%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.633 < 23.277 < 35.340$ km, and component 2 gives $14.485 < 17.486 < 27.476$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6894$ and 1.8000 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.941 and 13.811 h. The determinant-averaged frame-dragging rates at those shells are 3545.66 and 4385.84 rad s $^{-1}$, approaching surface-limit values of 5345.37 and 6123.79 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 7.678/9.125 h at 1 km inward to 0.0242/0.0290 h at 1 cm inward for components 1/2. Its geometric mass-equivalent share is comparatively large within M1 for at least one component. At least one Structural-Mode shell is among the more strongly time-dilated M1 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4 M2: Intermediate Observed-Mass Black-Hole Binaries

The second candidate class contains 94 unique event-level numerical streams with detector-frame total masses constructed from released component marginal medians spanning $57.388\text{--}105.052 M_\odot$. The class boundary was fixed from the observational distribution before any CPTG output was examined. Within M2, 4 systems fall in the strongly asymmetric mass-ratio cross-cut, 21 are moderately asymmetric, and 69 are near-equal in the same observational screen. The distance cross-cuts contain 52 nearer, 37 intermediate-distance, and 5 farther-distance systems; 5 candidates carry the predeclared broad-posterior flag. Candidate subsections are ordered by detector-frame total mass from lower to higher values. Neither Structural Mode nor any compact-state, clock, or rotation quantity controls class membership or ordering.

As in M1, each subsection begins with the observational state and then gives the two post-handoff compact components separately. Five M2 events have two retained release labels that are numerically identical under the declared transformation; their labels are preserved together for provenance but their identical numerical stream is reported once. The tables expose the physical compact-source mass, completed geometric mass-equivalent contribution, GR/CPTG matching radius, Structural-Mode organization shell, inner trapping surface, proper-time accumulation, and geometric frame dragging for every M2 binary. The descriptive comparisons in the closing paragraph of each subsection are made only after the fixed calculation has been completed and do not feed back into the classification or solve.

4.1 GW240513_183302

GW240513_183302 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 33.776 M_\odot$ and $m_2 = 23.613 M_\odot$, while the luminosity-distance median is 2246.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,735
Observed luminosity distance	2246.8 Mpc (median)
Detector-frame masses	$m_1 = 33.776 M_\odot$, $m_2 = 23.613 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.699$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4012$; $z_\pi = 0.4316$ (medians)
Post-handoff reorder fraction	16.84%

Table 248: Observational and coordinate summary for GW240513_183302. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	23.575 [17.467, 29.746]	16.681 [13.043, 22.770]
Physical compact-source mass M_c ($M_\odot$)	18.570 [13.169, 26.030]	11.583 [7.692, 15.995]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.114 [0.505, 10.162]	5.612 [0.757, 10.305]
Geometric share M_{geom}/M_∞ (%)	17.59 [2.31, 40.56]	33.59 [4.95, 49.68]
Exterior spin magnitude $ \hat{a}_\infty $	0.663 [0.258, 0.954]	0.380 [0.040, 0.902]
Polarization factor A_{pol}	1.213 [1.024, 1.682]	1.506 [1.052, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.977 [0.731, 1.000]	0.863 [0.160, 0.999]
GR/CPTG matching radius r_m (km)	34.812 [25.792, 43.923]	24.631 [19.259, 33.623]
Structural Mode N_{BH}	1.5614 [1.4603, 1.8004]	1.7184 [1.4747, 1.9086]
Organization-shell radius r_N (km)	18.326 [13.517, 23.113]	12.965 [10.124, 17.733]
Inner trapping-surface radius r_- (km)	25.132 [18.906, 34.596]	16.695 [12.236, 21.828]
Structural- N lapse $\bar{\alpha}_N$	0.7632 [0.5751, 0.8431]	0.6414 [0.4858, 0.8315]
Structural- N clock $T_{24,N}$ (h)	18.317 [13.803, 20.234]	15.394 [11.660, 19.956]
Structural- N axis ω (rad s^{-1})	2430.13 [1164.31, 4359.06]	4095.28 [1261.14, 6580.52]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3299.57 [1871.34, 4946.14]	4813.45 [1262.60, 7263.64]
Structural- N equatorial ω (rad s^{-1})	4179.18 [2924.37, 5597.70]	5663.33 [1263.36, 7925.70]
Structural- N period P_ω (ms)	1.9042 [1.2703, 3.3576]	1.3053 [0.8650, 4.9764]
Trapping-surface limit Ω_- (rad s^{-1})	5592.03 [4068.65, 7480.12]	7336.72 [1378.27, 10134.22]
Trapping-surface period P_- (ms)	1.1236 [0.8400, 1.5443]	0.8564 [0.6200, 4.5587]

Table 249: CPTG compact-state, structural, clock, and geometric-rotation results for GW240513_183302. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.5951	9.0610	5410.63	6975.40
500 m inward	4.6244	6.4141	5505.60	7176.42
100 m inward	2.0537	2.8667	5575.87	7307.19
10 m inward	0.6481	0.9062	5590.48	7334.26
1 m inward	0.2049	0.2865	5591.92	7336.25
10 cm inward	0.0648	0.0906	5592.02	7336.67
1 cm inward	0.0205	0.0287	5592.03	7336.71
surface limit	0.0000	0.0000	5592.03	7336.72

Table 250: Near-inner-boundary profile for GW240513_183302. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $18.570 M_\odot$ and $11.583 M_\odot$, with completed geometric mass-equivalent shares of 17.59% and 33.59%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.326 < 25.132 < 34.812 \text{ km}$, and component 2 gives $12.965 < 16.695 < 24.631 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5614$ and 1.7184 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.317 and 15.394 h. The determinant-averaged frame-dragging rates at those shells are 3299.57 and $4813.45 \text{ rad s}^{-1}$, approaching surface-limit values of 5592.03 and $7336.72 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.595/9.061 h at 1 km inward to 0.0205/0.0287 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.2 GW191215_223052

GW191215_223052 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 33.483 M_\odot$ and $m_2 = 24.458 M_\odot$, while the luminosity-distance median is 1935.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	8,884
Observed luminosity distance	1935.0 Mpc (median)
Detector-frame masses	$m_1 = 33.483 M_\odot$, $m_2 = 24.458 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.730$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3537$; $z_\pi = 0.3799$ (medians)
Post-handoff reorder fraction	19.27%

Table 251: Observational and coordinate summary for GW191215_223052. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	24.047 [17.875, 31.417]	18.156 [13.769, 24.280]
Physical compact-source mass M_c ($M_\odot$)	18.003 [12.341, 26.830]	11.894 [8.096, 16.768]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.057 [0.597, 12.528]	6.567 [1.090, 11.306]
Geometric share M_{geom}/M_∞ (%)	21.20 [2.69, 48.57]	37.00 [6.90, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.598 [0.087, 0.947]	0.321 [0.029, 0.864]
Polarization factor A_{pol}	1.269 [1.028, 1.944]	1.587 [1.074, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.963 [0.329, 1.000]	0.809 [0.116, 0.997]
GR/CPTG matching radius r_m (km)	35.508 [26.394, 46.392]	26.809 [20.331, 35.853]
Structural Mode N_{BH}	1.5920 [1.4623, 1.8958]	1.7579 [1.4861, 1.9103]
Organization-shell radius r_N (km)	18.679 [13.821, 24.430]	14.123 [10.702, 18.873]
Inner trapping-surface radius r_- (km)	24.755 [18.782, 35.541]	17.556 [12.916, 23.090]
Structural- N lapse $\bar{\alpha}_N$	0.7395 [0.4959, 0.8414]	0.6098 [0.4845, 0.8224]
Structural- N clock $T_{24,N}$ (h)	17.747 [11.902, 20.195]	14.636 [11.628, 19.737]
Structural- N axis ω (rad s^{-1})	2493.35 [1063.49, 4370.96]	3877.52 [825.52, 6272.33]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3230.57 [1560.33, 4902.77]	4504.39 [826.20, 6875.80]
Structural- N equatorial ω (rad s^{-1})	4058.58 [1898.22, 5513.95]	5190.30 [826.49, 7512.22]
Structural- N period P_ω (ms)	1.9449 [1.2816, 4.0268]	1.3949 [0.9138, 7.6049]
Trapping-surface limit Ω_- (rad s^{-1})	5365.64 [2089.81, 7312.94]	6619.18 [900.43, 9588.73]
Trapping-surface period P_- (ms)	1.1710 [0.8592, 3.0066]	0.9492 [0.6553, 6.9780]

Table 252: CPTG compact-state, structural, clock, and geometric-rotation results for GW191215_223052. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.8893	8.9725	5202.30	6306.49
500 m inward	4.8409	6.3602	5291.56	6474.40
100 m inward	2.1540	2.8455	5352.50	6596.26
10 m inward	0.6801	0.9002	5364.67	6617.52
1 m inward	0.2150	0.2847	5365.73	6619.00
10 cm inward	0.0680	0.0900	5365.65	6619.16
1 cm inward	0.0215	0.0285	5365.64	6619.18
surface limit	0.0000	0.0000	5365.64	6619.18

Table 253: Near-inner-boundary profile for GW191215_223052. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $18.003 M_\odot$ and $11.894 M_\odot$, with completed geometric mass-equivalent shares of 21.20% and 37.00%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.679 < 24.755 < 35.508 \text{ km}$, and component 2 gives $14.123 < 17.556 < 26.809 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5920$ and 1.7579 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.747 and 14.636 h. The determinant-averaged frame-dragging rates at those shells are 3230.57 and $4504.39 \text{ rad s}^{-1}$, approaching surface-limit values of 5365.64 and $6619.18 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.889/8.972 \text{ h}$ at 1 km inward to $0.0215/0.0285 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.3 GW241111_111552

GW241111_111552 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 31.983 M_\odot$ and $m_2 = 26.394 M_\odot$, while the luminosity-distance median is 1545.2 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	12,522
Observed luminosity distance	1545.2 Mpc (median)
Detector-frame masses	$m_1 = 31.983 M_\odot$, $m_2 = 26.394 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.825$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2908$; $z_\pi = 0.3127$ (medians)
Post-handoff reorder fraction	29.78%

Table 254: Observational and coordinate summary for GW241111_111552. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	24.036 [19.186, 28.998]	20.918 [15.961, 26.548]
Physical compact-source mass M_c ($M_\odot$)	17.211 [12.322, 24.039]	12.806 [9.185, 18.672]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.350 [0.827, 12.579]	8.271 [1.941, 12.373]
Geometric share M_{geom}/M_∞ (%)	26.69 [3.65, 48.91]	40.67 [10.28, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.501 [0.075, 0.928]	0.256 [0.024, 0.800]
Polarization factor A_{pol}	1.364 [1.038, 1.957]	1.686 [1.115, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.931 [0.289, 0.999]	0.728 [0.094, 0.993]
GR/CPTG matching radius r_m (km)	35.492 [28.331, 42.820]	30.888 [23.569, 39.201]
Structural Mode N_{BH}	1.6442 [1.4675, 1.8998]	1.8018 [1.5075, 1.9110]
Organization-shell radius r_N (km)	18.698 [14.918, 22.584]	16.275 [12.421, 20.648]
Inner trapping-surface radius r_- (km)	24.042 [19.183, 31.919]	19.449 [14.722, 25.483]
Structural- N lapse $\bar{\alpha}_N$	0.6993 [0.4927, 0.8373]	0.5740 [0.4840, 0.8054]
Structural- N clock $T_{24,N}$ (h)	16.783 [11.825, 20.095]	13.775 [11.616, 19.329]
Structural- N axis ω (rad s^{-1})	2838.39 [1136.92, 4401.66]	3476.86 [617.58, 5449.71]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3468.01 [1589.98, 4884.10]	3918.43 [617.79, 5979.75]
Structural- N equatorial ω (rad s^{-1})	4102.01 [1631.47, 5388.11]	4389.76 [617.89, 6481.90]
Structural- N period P_ω (ms)	1.8118 [1.2865, 3.9518]	1.6035 [1.0507, 10.1705]
Trapping-surface limit Ω_- (rad s^{-1})	5374.11 [1789.45, 7016.20]	5396.22 [673.15, 8240.72]
Trapping-surface period P_- (ms)	1.1692 [0.8955, 3.5112]	1.1644 [0.7625, 9.3340]

Table 255: CPTG compact-state, structural, clock, and geometric-rotation results for GW241111_111552. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.3277	8.6873	5194.12	5149.13
500 m inward	5.1660	6.1633	5287.96	5281.69
100 m inward	2.3036	2.7612	5356.04	5372.46
10 m inward	0.7280	0.8736	5372.39	5393.53
1 m inward	0.2302	0.2763	5374.00	5395.96
10 cm inward	0.0728	0.0874	5374.10	5396.19
1 cm inward	0.0230	0.0276	5374.11	5396.22
surface limit	0.0000	0.0000	5374.11	5396.22

Table 256: Near-inner-boundary profile for GW241111_111552. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $17.211 M_\odot$ and $12.806 M_\odot$, with completed geometric mass-equivalent shares of 26.69% and 40.67%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.698 < 24.042 < 35.492 \text{ km}$, and component 2 gives $16.275 < 19.449 < 30.888 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6442$ and 1.8018 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.783 and 13.775 h. The determinant-averaged frame-dragging rates at those shells are 3468.01 and $3918.43 \text{ rad s}^{-1}$, approaching surface-limit values of 5374.11 and $5396.22 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $7.328/8.687 \text{ h}$ at 1 km inward to $0.0230/0.0276 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.4 GW230624_113103

GW230624_113103 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 36.573 M_\odot$ and $m_2 = 21.894 M_\odot$, while the luminosity-distance median is 1929.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	27,328
Observed luminosity distance	1929.0 Mpc (median)
Detector-frame masses	$m_1 = 36.573 M_\odot$, $m_2 = 21.894 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.599$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3523$; $z_\pi = 0.3789$ (medians)
Post-handoff reorder fraction	13.32%

Table 257: Observational and coordinate summary for GW230624_113103. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	26.356 [17.520, 40.469]	15.911 [11.372, 23.710]
Physical compact-source mass M_c ($M_\odot$)	18.355 [12.182, 33.824]	10.995 [6.920, 16.333]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.920 [0.849, 13.926]	5.150 [0.623, 10.737]
Geometric share M_{geom}/M_∞ (%)	25.74 [3.64, 49.06]	33.41 [4.25, 49.78]
Exterior spin magnitude $ \hat{a}_\infty $	0.517 [0.070, 0.928]	0.384 [0.034, 0.916]
Polarization factor A_{pol}	1.347 [1.038, 1.963]	1.502 [1.044, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.938 [0.269, 0.999]	0.865 [0.133, 0.999]
GR/CPTG matching radius r_m (km)	38.918 [25.870, 59.757]	23.494 [16.792, 35.011]
Structural Mode N_{BH}	1.6347 [1.4674, 1.9015]	1.7164 [1.4708, 1.9097]
Organization-shell radius r_N (km)	20.506 [13.584, 31.524]	12.371 [8.837, 18.413]
Inner trapping-surface radius r_- (km)	25.933 [18.276, 44.941]	15.834 [10.827, 22.700]
Structural- N lapse $\bar{\alpha}_N$	0.7066 [0.4914, 0.8373]	0.6430 [0.4850, 0.8346]
Structural- N clock $T_{24,N}$ (h)	16.959 [11.793, 20.096]	15.433 [11.639, 20.031]
Structural- N axis ω (rad s^{-1})	2468.78 [852.50, 4390.60]	4086.82 [1049.51, 7297.79]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3056.15 [1181.03, 4960.40]	4890.85 [1050.36, 8093.41]
Structural- N equatorial ω (rad s^{-1})	3664.08 [1403.30, 5638.48]	5834.87 [1050.79, 8900.33]
Structural- N period P_ω (ms)	2.0559 [1.2667, 5.3201]	1.2847 [0.7763, 5.9819]
Trapping-surface limit Ω_- (rad s^{-1})	4705.25 [1591.38, 7491.36]	7564.44 [1145.66, 11456.25]
Trapping-surface period P_- (ms)	1.3354 [0.8387, 3.9483]	0.8306 [0.5485, 5.4843]

Table 258: CPTG compact-state, structural, clock, and geometric-rotation results for GW230624_113103. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.9241	9.2306	4554.10	7175.37
500 m inward	4.8756	6.5275	4632.97	7387.04
100 m inward	2.1718	2.9155	4691.36	7530.21
10 m inward	0.6860	0.9217	4703.84	7561.41
1 m inward	0.2169	0.2914	4705.07	7563.92
10 cm inward	0.0686	0.0922	4705.23	7564.39
1 cm inward	0.0217	0.0291	4705.25	7564.43
surface limit	0.0000	0.0000	4705.25	7564.44

Table 259: Near-inner-boundary profile for GW230624_113103. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $18.355 M_\odot$ and $10.995 M_\odot$, with completed geometric mass-equivalent shares of 25.74% and 33.41%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $20.506 < 25.933 < 38.918 \text{ km}$, and component 2 gives $12.371 < 15.834 < 23.494 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6347$ and 1.7164 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.959 and 15.433 h. The determinant-averaged frame-dragging rates at those shells are 3056.15 and $4890.85 \text{ rad s}^{-1}$, approaching surface-limit values of 4705.25 and $7564.44 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.924/9.231 \text{ h}$ at 1 km inward to $0.0217/0.0291 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.5 GW170104_101158

GW170104_101158 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 34.784 M_\odot$ and $m_2 = 25.252 M_\odot$, while the luminosity-distance median is 1109.8 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	28,078
Observed luminosity distance	1109.8 Mpc (median)
Detector-frame masses	$m_1 = 34.784 M_\odot$, $m_2 = 25.252 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.726$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2177$; $z_\pi = 0.2337$ (medians)
Post-handoff reorder fraction	18.42%

Table 260: Observational and coordinate summary for GW170104_101158. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.921 [20.871, 34.912]	20.908 [15.926, 27.948]
Physical compact-source mass M_c ($M_\odot$)	19.054 [14.016, 27.901]	13.272 [9.279, 18.507]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.081 [1.190, 15.248]	8.070 [1.616, 13.112]
Geometric share M_{geom}/M_∞ (%)	29.24 [4.98, 49.42]	39.58 [8.85, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.456 [0.054, 0.902]	0.276 [0.024, 0.827]
Polarization factor A_{pol}	1.413 [1.052, 1.977]	1.655 [1.097, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.911 [0.212, 0.999]	0.756 [0.095, 0.995]
GR/CPTG matching radius r_m (km)	41.229 [30.819, 51.552]	30.873 [23.516, 41.269]
Structural Mode N_{BH}	1.6706 [1.4749, 1.9057]	1.7886 [1.4982, 1.9109]
Organization-shell radius r_N (km)	21.729 [16.186, 27.163]	16.260 [12.375, 21.707]
Inner trapping-surface radius r_- (km)	27.219 [21.541, 37.071]	19.851 [14.754, 25.758]
Structural- N lapse $\bar{\alpha}_N$	0.6789 [0.4881, 0.8313]	0.5849 [0.4840, 0.8127]
Structural- N clock $T_{24,N}$ (h)	16.293 [11.714, 19.952]	14.037 [11.616, 19.506]
Structural- N axis ω (rad s^{-1})	2514.03 [903.76, 3869.77]	3420.29 [586.46, 5423.61]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3036.15 [1006.10, 4320.85]	3910.39 [586.64, 5960.87]
Structural- N equatorial ω (rad s^{-1})	3563.88 [1007.04, 4825.96]	4471.39 [586.73, 6476.26]
Structural- N period P_ω (ms)	2.0695 [1.4542, 6.2451]	1.6068 [1.0541, 10.7105]
Trapping-surface limit Ω_- (rad s^{-1})	4603.96 [1100.19, 6359.01]	5568.62 [639.18, 8242.35]
Trapping-surface period P_- (ms)	1.3647 [0.9881, 5.7110]	1.1283 [0.7623, 9.8301]

Table 261: CPTG compact-state, structural, clock, and geometric-rotation results for GW170104_101158. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.9883	8.5286	4463.69	5311.93
500 m inward	4.9340	6.0467	4537.01	5448.79
100 m inward	2.2024	2.7087	4591.62	5544.21
10 m inward	0.6962	0.8569	4602.83	5566.07
1 m inward	0.2201	0.2710	4603.88	5568.35
10 cm inward	0.0696	0.0857	4603.95	5568.60
1 cm inward	0.0220	0.0271	4603.96	5568.62
surface limit	0.0000	0.0000	4603.96	5568.62

Table 262: Near-inner-boundary profile for GW170104_101158. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $19.054 M_\odot$ and $13.272 M_\odot$, with completed geometric mass-equivalent shares of 29.24% and 39.58%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.729 < 27.219 < 41.229 \text{ km}$, and component 2 gives $16.260 < 19.851 < 30.873 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6706$ and 1.7886 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.293 and 14.037 h. The determinant-averaged frame-dragging rates at those shells are 3036.15 and $3910.39 \text{ rad s}^{-1}$, approaching surface-limit values of 4603.96 and $5568.62 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.988/8.529 \text{ h}$ at 1 km inward to $0.0220/0.0271 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.6 GW241006_015333

GW241006_015333 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 32.856 M_\odot$ and $m_2 = 27.238 M_\odot$, while the luminosity-distance median is 1848.5 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,107
Observed luminosity distance	1848.5 Mpc (median)
Detector-frame masses	$m_1 = 32.856 M_\odot$, $m_2 = 27.238 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.829$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3396$; $z_\pi = 0.3652$ (medians)
Post-handoff reorder fraction	30.99%

Table 263: Observational and coordinate summary for GW241006_015333. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	23.319 [18.909, 28.287]	20.645 [16.850, 25.585]
Physical compact-source mass M_c ($M_\odot$)	16.640 [12.292, 22.719]	12.493 [9.487, 17.350]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.422 [0.931, 12.289]	8.513 [2.513, 12.066]
Geometric share M_{geom}/M_∞ (%)	27.65 [4.30, 48.59]	41.76 [13.35, 49.91]
Exterior spin magnitude $ \hat{a}_\infty $	0.484 [0.086, 0.915]	0.236 [0.021, 0.742]
Polarization factor A_{pol}	1.382 [1.045, 1.945]	1.717 [1.154, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.924 [0.326, 0.999]	0.697 [0.084, 0.988]
GR/CPTG matching radius r_m (km)	34.434 [27.922, 41.770]	30.485 [24.881, 37.779]
Structural Mode N_{BH}	1.6539 [1.4711, 1.8962]	1.8150 [1.5287, 1.9112]
Organization-shell radius r_N (km)	18.159 [14.661, 22.008]	16.053 [13.099, 19.878]
Inner trapping-surface radius r_- (km)	23.274 [19.052, 30.220]	19.076 [15.375, 23.809]
Structural- N lapse $\bar{\alpha}_N$	0.6918 [0.4957, 0.8344]	0.5631 [0.4838, 0.7886]
Structural- N clock $T_{24,N}$ (h)	16.602 [11.896, 20.026]	13.514 [11.611, 18.927]
Structural- N axis ω (rad s^{-1})	3065.45 [1230.33, 4461.86]	3601.79 [536.89, 5296.27]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3676.01 [1760.51, 4966.16]	4030.20 [536.98, 5804.35]
Structural- N equatorial ω (rad s^{-1})	4296.07 [1882.45, 5483.25]	4462.79 [537.02, 6272.77]
Structural- N period P_ω (ms)	1.7092 [1.2652, 3.5690]	1.5590 [1.0825, 11.7010]
Trapping-surface limit Ω_- (rad s^{-1})	5596.93 [2066.71, 7121.55]	5361.32 [584.90, 7912.88]
Trapping-surface period P_- (ms)	1.1226 [0.8823, 3.0402]	1.1719 [0.7940, 10.7423]

Table 264: CPTG compact-state, structural, clock, and geometric-rotation results for GW241006_015333. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.4967	8.8203	5396.80	5074.10
500 m inward	5.2897	6.2590	5503.32	5229.08
100 m inward	2.3601	2.8054	5580.72	5335.20
10 m inward	0.7458	0.8875	5595.13	5359.03
1 m inward	0.2358	0.2807	5596.69	5361.05
10 cm inward	0.0746	0.0888	5596.91	5361.29
1 cm inward	0.0236	0.0281	5596.93	5361.31
surface limit	0.0000	0.0000	5596.93	5361.32

Table 265: Near-inner-boundary profile for GW241006_015333. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $16.640 M_\odot$ and $12.493 M_\odot$, with completed geometric mass-equivalent shares of 27.65% and 41.76%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.159 < 23.274 < 34.434 \text{ km}$, and component 2 gives $16.053 < 19.076 < 30.485 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6539$ and 1.8150 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.602 and 13.514 h. The determinant-averaged frame-dragging rates at those shells are 3676.01 and $4030.20 \text{ rad s}^{-1}$, approaching surface-limit values of 5596.93 and $5361.32 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $7.497/8.820 \text{ h}$ at 1 km inward to $0.0236/0.0281 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.7 GW240109_050431

GW240109_050431 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 36.859 M_\odot$ and $m_2 = 23.283 M_\odot$, while the luminosity-distance median is 1499.1 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	29,374
Observed luminosity distance	1499.1 Mpc (median)
Detector-frame masses	$m_1 = 36.859 M_\odot$, $m_2 = 23.283 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.632$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2833$; $z_\pi = 0.3046$ (medians)
Post-handoff reorder fraction	13.82%

Table 266: Observational and coordinate summary for GW240109_050431. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.998 [19.292, 35.718]	18.016 [13.749, 26.105]
Physical compact-source mass M_c ($M_\odot$)	18.331 [13.629, 26.585]	12.189 [8.164, 17.294]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.977 [1.093, 16.132]	6.373 [0.958, 12.109]
Geometric share M_{geom}/M_∞ (%)	32.42 [4.88, 49.64]	35.98 [5.83, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.401 [0.043, 0.904]	0.339 [0.030, 0.885]
Polarization factor A_{pol}	1.480 [1.051, 1.986]	1.562 [1.062, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.877 [0.168, 0.999]	0.827 [0.118, 0.998]
GR/CPTG matching radius r_m (km)	41.342 [28.487, 52.743]	26.602 [20.302, 38.547]
Structural Mode N_{BH}	1.7052 [1.4743, 1.9082]	1.7460 [1.4798, 1.9102]
Organization-shell radius r_N (km)	21.798 [14.957, 27.783]	14.016 [10.682, 20.279]
Inner trapping-surface radius r_- (km)	26.861 [20.537, 35.692]	17.677 [12.924, 24.343]
Structural- N lapse $\bar{\alpha}_N$	0.6518 [0.4862, 0.8318]	0.6195 [0.4845, 0.8274]
Structural- N clock $T_{24,N}$ (h)	15.643 [11.668, 19.963]	14.867 [11.629, 19.857]
Structural- N axis ω (rad s^{-1})	2536.53 [772.86, 3943.05]	3751.54 [819.78, 6203.95]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3025.13 [788.63, 4455.89]	4419.93 [820.31, 6838.62]
Structural- N equatorial ω (rad s^{-1})	3537.78 [788.98, 5081.98]	5193.68 [820.58, 7483.80]
Structural- N period P_ω (ms)	2.0770 [1.4101, 7.9672]	1.4216 [0.9188, 7.6595]
Trapping-surface limit Ω_- (rad s^{-1})	4517.38 [860.31, 6814.60]	6666.97 [894.21, 9599.26]
Trapping-surface period P_- (ms)	1.3909 [0.9220, 7.3034]	0.9424 [0.6545, 7.0265]

Table 267: CPTG compact-state, structural, clock, and geometric-rotation results for GW240109_050431. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.1504	8.8317	4378.81	6348.27
500 m inward	5.0567	6.2513	4451.87	6520.73
100 m inward	2.2595	2.7952	4505.46	6641.41
10 m inward	0.7145	0.8841	4515.91	6664.47
1 m inward	0.2259	0.2796	4517.22	6666.78
10 cm inward	0.0714	0.0884	4517.37	6666.94
1 cm inward	0.0226	0.0280	4517.38	6666.97
surface limit	0.0000	0.0000	4517.38	6666.97

Table 268: Near-inner-boundary profile for GW240109_050431. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $18.331 M_\odot$ and $12.189 M_\odot$, with completed geometric mass-equivalent shares of 32.42% and 35.98%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.798 < 26.861 < 41.342 \text{ km}$, and component 2 gives $14.016 < 17.677 < 26.602 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7052$ and 1.7460 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.643 and 14.867 h. The determinant-averaged frame-dragging rates at those shells are 3025.13 and $4419.93 \text{ rad s}^{-1}$, approaching surface-limit values of 4517.38 and $6666.97 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $7.150/8.832 \text{ h}$ at 1 km inward to $0.0226/0.0280 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.8 GW230919_215712

GW230919_215712 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 34.276 M_\odot$ and $m_2 = 27.082 M_\odot$, while the luminosity-distance median is 1329.3 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	28,540
Observed luminosity distance	1329.3 Mpc (median)
Detector-frame masses	$m_1 = 34.276 M_\odot$, $m_2 = 27.082 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.790$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2549$; $z_\pi = 0.2741$ (medians)
Post-handoff reorder fraction	27.07%

Table 269: Observational and coordinate summary for GW230919_215712. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	26.134 [20.329, 32.396]	21.790 [16.944, 27.934]
Physical compact-source mass M_c ($M_\odot$)	19.981 [14.542, 27.271]	14.008 [9.853, 19.390]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.523 [0.747, 12.466]	8.119 [1.699, 12.996]
Geometric share M_{geom}/M_∞ (%)	21.23 [3.09, 44.29]	37.79 [8.70, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.597 [0.188, 0.939]	0.307 [0.030, 0.830]
Polarization factor A_{pol}	1.270 [1.032, 1.795]	1.607 [1.095, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.963 [0.607, 0.999]	0.794 [0.117, 0.995]
GR/CPTG matching radius r_m (km)	38.590 [30.019, 47.836]	32.176 [25.020, 41.248]
Structural Mode N_{BH}	1.5924 [1.4644, 1.8454]	1.7672 [1.4972, 1.9103]
Organization-shell radius r_N (km)	20.338 [15.776, 25.217]	16.961 [13.174, 21.711]
Inner trapping-surface radius r_- (km)	27.251 [21.385, 36.195]	20.825 [15.731, 26.374]
Structural- N lapse $\bar{\alpha}_N$	0.7392 [0.5376, 0.8398]	0.6023 [0.4845, 0.8135]
Structural- N clock $T_{24,N}$ (h)	17.742 [12.904, 20.154]	14.455 [11.629, 19.524]
Structural- N axis ω (rad s^{-1})	2450.06 [1123.64, 3928.35]	3363.19 [720.03, 5140.51]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3139.11 [1779.95, 4428.36]	3848.94 [720.32, 5648.37]
Structural- N equatorial ω (rad s^{-1})	3805.92 [2626.92, 4968.29]	4379.06 [720.46, 6135.04]
Structural- N period P_ω (ms)	2.0016 [1.4189, 3.5300]	1.6324 [1.1124, 8.7228]
Trapping-surface limit Ω_- (rad s^{-1})	5046.87 [3422.72, 6532.95]	5519.49 [784.99, 7809.16]
Trapping-surface period P_- (ms)	1.2450 [0.9618, 1.8357]	1.1384 [0.8046, 8.0041]

Table 270: CPTG compact-state, structural, clock, and geometric-rotation results for GW230919_215712. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.5792	8.3099	4904.13	5287.26
500 m inward	4.6271	5.8875	4980.99	5410.84
100 m inward	2.0583	2.6356	5034.03	5498.35
10 m inward	0.6500	0.8336	5045.57	5516.79
1 m inward	0.2055	0.2636	5046.73	5519.23
10 cm inward	0.0650	0.0834	5046.86	5519.48
1 cm inward	0.0206	0.0264	5046.87	5519.49
surface limit	0.0000	0.0000	5046.87	5519.49

Table 271: Near-inner-boundary profile for GW230919_215712. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $19.981 M_\odot$ and $14.008 M_\odot$, with completed geometric mass-equivalent shares of 21.23% and 37.79%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $20.338 < 27.251 < 38.590 \text{ km}$, and component 2 gives $16.961 < 20.825 < 32.176 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5924$ and 1.7672 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.742 and 14.455 h. The determinant-averaged frame-dragging rates at those shells are 3139.11 and $3848.94 \text{ rad s}^{-1}$, approaching surface-limit values of 5046.87 and $5519.49 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.579/8.310 \text{ h}$ at 1 km inward to $0.0206/0.0264 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.9 GW230712_090405

GW230712_090405 is an M2 intermediate observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 45.170 M_\odot$ and $m_2 = 17.052 M_\odot$, while the luminosity-distance median is 2033.1 Mpc . 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the nearer-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M2: Intermediate observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	49,376
Observed luminosity distance	2033.1 Mpc (median)
Detector-frame masses	$m_1 = 45.170 M_\odot$, $m_2 = 17.052 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.378$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3685$; $z_\pi = 0.3963$ (medians)
Post-handoff reorder fraction	5.76%

Table 272: Observational and coordinate summary for GW230712_090405. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	31.357 [20.765, 47.831]	12.272 [7.212, 28.261]
Physical compact-source mass M_c ($M_\odot$)	25.120 [14.657, 42.390]	9.084 [4.508, 21.116]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.183 [0.575, 14.559]	3.368 [0.383, 11.194]
Geometric share M_{geom}/M_∞ (%)	16.53 [1.84, 45.54]	27.99 [3.29, 49.56]
Exterior spin magnitude $ \hat{a}_\infty $	0.683 [0.163, 0.963]	0.478 [0.047, 0.935]
Polarization factor A_{pol}	1.198 [1.019, 1.836]	1.389 [1.034, 1.983]
Compact-source spin parameter $\hat{a}_c$	0.980 [0.549, 1.000]	0.921 [0.185, 0.999]
GR/CPTG matching radius r_m (km)	46.302 [30.662, 70.629]	18.121 [10.650, 41.731]
Structural Mode N_{BH}	1.5528 [1.4579, 1.8603]	1.6575 [1.4655, 1.9073]
Organization-shell radius r_N (km)	24.354 [16.107, 37.095]	9.541 [5.603, 21.974]
Inner trapping-surface radius r_- (km)	33.964 [21.477, 56.410]	12.719 [6.935, 28.875]
Structural- N lapse $\bar{\alpha}_N$	0.7698 [0.5252, 0.8450]	0.6890 [0.4868, 0.8389]
Structural- N clock $T_{24,N}$ (h)	18.476 [12.605, 20.281]	16.537 [11.684, 20.133]
Structural- N axis ω (rad s^{-1})	1684.33 [699.16, 3617.57]	4418.39 [1281.37, 11162.35]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2303.92 [1102.44, 4144.59]	5646.30 [1645.94, 12504.24]
Structural- N equatorial ω (rad s^{-1})	2990.48 [1656.17, 4759.17]	7009.79 [1705.82, 13871.47]
Structural- N period P_ω (ms)	2.7272 [1.5160, 5.6993]	1.1128 [0.5025, 3.8174]
Trapping-surface limit Ω_- (rad s^{-1})	4033.48 [2224.74, 6284.50]	9032.25 [1863.84, 17887.34]
Trapping-surface period P_- (ms)	1.5578 [0.9998, 2.8242]	0.6956 [0.3513, 3.3711]

Table 273: CPTG compact-state, structural, clock, and geometric-rotation results for GW230712_090405. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5483	9.6138	3950.59	8365.15
500 m inward	3.8935	6.7374	3994.42	8735.20
100 m inward	1.7299	2.9857	4026.86	8978.52
10 m inward	0.5463	0.9422	4032.93	9027.51
1 m inward	0.1727	0.2979	4033.41	9031.83
10 cm inward	0.0546	0.0942	4033.47	9032.21
1 cm inward	0.0173	0.0298	4033.47	9032.24
surface limit	0.0000	0.0000	4033.48	9032.25

Table 274: Near-inner-boundary profile for GW230712_090405. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $25.120 M_\odot$ and $9.084 M_\odot$, with completed geometric mass-equivalent shares of 16.53% and 27.99%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.354 < 33.964 < 46.302 \text{ km}$, and component 2 gives $9.541 < 12.719 < 18.121 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5528$ and 1.6575 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.476 and 16.537 h. The determinant-averaged frame-dragging rates at those shells are 2303.92 and $5646.30 \text{ rad s}^{-1}$, approaching surface-limit values of 4033.48 and $9032.25 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.548/9.614 \text{ h}$ at 1 km inward to $0.0173/0.0298 \text{ h}$ at 1 cm inward for components 1/2. Its strong observational mass asymmetry makes the contrast between the two solved compact interiors particularly useful for component-by-component comparison. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.10 GW170814_103043

GW170814_103043 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 34.696 M_\odot$ and $m_2 = 28.030 M_\odot$, while the luminosity-distance median is 605.6 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	28,098
Observed luminosity distance	605.6 Mpc (median)
Detector-frame masses	$m_1 = 34.696 M_\odot$, $m_2 = 28.030 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.808$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1256$; $z_\pi = 0.1349$ (medians)
Post-handoff reorder fraction	24.59%

Table 275: Observational and coordinate summary for GW170814_103043. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	30.024 [24.019, 36.008]	25.332 [20.760, 31.335]
Physical compact-source mass M_c ($M_\odot$)	22.397 [15.777, 30.488]	15.929 [11.690, 22.418]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.715 [0.824, 15.673]	9.821 [2.368, 14.551]
Geometric share M_{geom}/M_∞ (%)	22.79 [2.95, 48.45]	39.12 [10.14, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.570 [0.091, 0.942]	0.284 [0.026, 0.802]
Polarization factor A_{pol}	1.295 [1.030, 1.940]	1.643 [1.113, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.955 [0.341, 1.000]	0.766 [0.104, 0.994]
GR/CPTG matching radius r_m (km)	44.334 [35.467, 53.171]	37.407 [30.654, 46.271]
Structural Mode N_{BH}	1.6065 [1.4637, 1.8945]	1.7831 [1.5065, 1.9107]
Organization-shell radius r_N (km)	23.336 [18.610, 27.996]	19.701 [16.144, 24.382]
Inner trapping-surface radius r_- (km)	30.821 [24.387, 40.504]	23.850 [18.941, 30.169]
Structural- N lapse $\bar{\alpha}_N$	0.7283 [0.4970, 0.8404]	0.5893 [0.4842, 0.8061]
Structural- N clock $T_{24,N}$ (h)	17.480 [11.927, 20.169]	14.144 [11.621, 19.346]
Structural- N axis ω (rad s^{-1})	2093.38 [907.07, 3441.03]	2911.78 [564.52, 4288.71]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2656.78 [1354.78, 3831.08]	3311.04 [564.81, 4681.66]
Structural- N equatorial ω (rad s^{-1})	3262.37 [1514.54, 4252.24]	3734.14 [564.96, 5057.45]
Structural- N period P_ω (ms)	2.3650 [1.6401, 4.6378]	1.8976 [1.3421, 11.1244]
Trapping-surface limit Ω_- (rad s^{-1})	4336.73 [1662.99, 5559.27]	4690.26 [615.68, 6406.40]
Trapping-surface period P_- (ms)	1.4488 [1.1302, 3.7783]	1.3396 [0.9808, 10.2052]

Table 276: CPTG compact-state, structural, clock, and geometric-rotation results for GW170814_103043. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.2908	7.8260	4231.87	4521.35
500 m inward	4.4288	5.5456	4286.67	4611.06
100 m inward	1.9738	2.4832	4327.75	4675.29
10 m inward	0.6237	0.7854	4335.66	4688.68
1 m inward	0.1972	0.2484	4336.65	4690.06
10 cm inward	0.0624	0.0785	4336.72	4690.24
1 cm inward	0.0197	0.0248	4336.73	4690.25
surface limit	0.0000	0.0000	4336.73	4690.26

Table 277: Near-inner-boundary profile for GW170814_103043. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.397 M_\odot$ and $15.929 M_\odot$, with completed geometric mass-equivalent shares of 22.79% and 39.12%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $23.336 < 30.821 < 44.334 \text{ km}$, and component 2 gives $19.701 < 23.850 < 37.407 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6065$ and 1.7831 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.480 and 14.144 h. The determinant-averaged frame-dragging rates at those shells are 2656.78 and $3311.04 \text{ rad s}^{-1}$, approaching surface-limit values of 4336.73 and $4690.26 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.291/7.826 \text{ h}$ at 1 km inward to $0.0197/0.0248 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.11 GW240902_143306

GW240902_143306 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 36.515 M_\odot$ and $m_2 = 27.437 M_\odot$, while the luminosity-distance median is 2864.1 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,797
Observed luminosity distance	2864.1 Mpc (median)
Detector-frame masses	$m_1 = 36.515 M_\odot$, $m_2 = 27.437 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.751$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4922$; $z_\pi = 0.5300$ (medians)
Post-handoff reorder fraction	21.16%

Table 278: Observational and coordinate summary for GW240902_143306. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	23.295 [17.389, 30.470]	18.429 [12.831, 24.711]
Physical compact-source mass M_c ($M_\odot$)	18.198 [12.401, 26.457]	12.049 [7.906, 17.779]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.206 [0.483, 11.663]	6.353 [0.928, 11.246]
Geometric share M_{geom}/M_∞ (%)	18.44 [2.16, 46.81]	35.61 [5.76, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.648 [0.134, 0.957]	0.345 [0.032, 0.887]
Polarization factor A_{pol}	1.226 [1.022, 1.880]	1.553 [1.061, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.974 [0.475, 1.000]	0.833 [0.125, 0.998]
GR/CPTG matching radius r_m (km)	34.398 [25.677, 44.993]	27.213 [18.946, 36.489]
Structural Mode N_{BH}	1.5683 [1.4595, 1.8754]	1.7416 [1.4794, 1.9100]
Organization-shell radius r_N (km)	18.089 [13.476, 23.661]	14.339 [9.967, 19.234]
Inner trapping-surface radius r_- (km)	24.791 [18.372, 35.257]	17.744 [12.283, 24.199]
Structural- N lapse $\bar{\alpha}_N$	0.7578 [0.5126, 0.8437]	0.6230 [0.4847, 0.8278]
Structural- N clock $T_{24,N}$ (h)	18.187 [12.302, 20.249]	14.951 [11.634, 19.866]
Structural- N axis ω (rad s^{-1})	2413.54 [1083.74, 4451.71]	3781.99 [886.21, 6346.68]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3240.46 [1697.90, 5031.79]	4418.05 [886.82, 7035.20]
Structural- N equatorial ω (rad s^{-1})	4118.76 [2435.48, 5680.22]	5129.92 [887.13, 7810.80]
Structural- N period P_ω (ms)	1.9390 [1.2487, 3.7006]	1.4222 [0.8931, 7.0851]
Trapping-surface limit Ω_- (rad s^{-1})	5546.59 [3000.20, 7541.71]	6513.47 [967.04, 10134.30]
Trapping-surface period P_- (ms)	1.1328 [0.8331, 2.0943]	0.9646 [0.6200, 6.4974]

Table 279: CPTG compact-state, structural, clock, and geometric-rotation results for GW240902_143306. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.7341	8.8834	5381.68	6206.15
500 m inward	4.7209	6.2905	5471.20	6370.78
100 m inward	2.0964	2.8138	5532.12	6487.96
10 m inward	0.6618	0.8899	5545.73	6512.34
1 m inward	0.2092	0.2814	5546.50	6513.31
10 cm inward	0.0662	0.0890	5546.59	6513.46
1 cm inward	0.0209	0.0281	5546.59	6513.47
surface limit	0.0000	0.0000	5546.59	6513.47

Table 280: Near-inner-boundary profile for GW240902_143306. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $18.198 M_\odot$ and $12.049 M_\odot$, with completed geometric mass-equivalent shares of 18.44% and 35.61%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $18.089 < 24.791 < 34.398 \text{ km}$, and component 2 gives $14.339 < 17.744 < 27.213 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5683$ and 1.7416 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.187 and 14.951 h. The determinant-averaged frame-dragging rates at those shells are 3240.46 and $4418.05 \text{ rad s}^{-1}$, approaching surface-limit values of 5546.59 and $6513.47 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.734/8.883 \text{ h}$ at 1 km inward to $0.0209/0.0281 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.12 GW230814_230901

GW230814_230901 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 35.805 M_\odot$ and $m_2 = 29.919 M_\odot$, while the luminosity-distance median is 289.5 Mpc . 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	42,693
Observed luminosity distance	289.5 Mpc (median)
Detector-frame masses	$m_1 = 35.805 M_\odot$, $m_2 = 29.919 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.836$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0625$; $z_\pi = 0.0673$ (medians)
Post-handoff reorder fraction	14.42%

Table 281: Observational and coordinate summary for GW230814_230901. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.388 [28.211, 36.366]	28.258 [24.962, 33.517]
Physical compact-source mass M_c ($M_\odot$)	18.472 [16.413, 24.645]	15.461 [13.398, 20.733]
Geometric mass-equivalent M_{geom} ($M_\odot$)	14.997 [6.396, 17.481]	13.091 [7.287, 15.955]
Geometric share M_{geom}/M_∞ (%)	44.81 [21.47, 49.95]	46.64 [26.95, 49.97]
Exterior spin magnitude $ \hat{a}_\infty $	0.178 [0.016, 0.593]	0.138 [0.013, 0.496]
Polarization factor A_{pol}	1.812 [1.273, 1.998]	1.874 [1.369, 1.999]
Compact-source spin parameter $\hat{a}_c$	0.584 [0.063, 0.962]	0.486 [0.051, 0.929]
GR/CPTG matching radius r_m (km)	49.301 [41.657, 53.699]	41.727 [36.859, 49.493]
Structural Mode N_{BH}	1.8517 [1.5945, 1.9117]	1.8734 [1.6468, 1.9119]
Organization-shell radius r_N (km)	25.928 [22.014, 28.241]	21.934 [19.409, 26.010]
Inner trapping-surface radius r_- (km)	29.316 [26.686, 34.034]	24.756 [22.095, 29.882]
Structural- N lapse $\bar{\alpha}_N$	0.5324 [0.4835, 0.7376]	0.5143 [0.4833, 0.6973]
Structural- N clock $T_{24,N}$ (h)	12.778 [11.603, 17.702]	12.343 [11.599, 16.734]
Structural- N axis ω (rad s^{-1})	2219.11 [263.71, 3217.80]	2303.27 [251.60, 3648.61]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2389.81 [263.75, 3562.14]	2401.09 [251.62, 3958.56]
Structural- N equatorial ω (rad s^{-1})	2507.09 [263.77, 3866.18]	2451.82 [251.64, 4227.72]
Structural- N period P_ω (ms)	2.6292 [1.7639, 23.8227]	2.6168 [1.5872, 24.9707]
Trapping-surface limit Ω_- (rad s^{-1})	2820.15 [287.27, 4885.83]	2719.50 [274.04, 5252.00]
Trapping-surface period P_- (ms)	2.2280 [1.2860, 21.8721]	2.3104 [1.1963, 22.9278]

Table 282: CPTG compact-state, structural, clock, and geometric-rotation results for GW230814_230901. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.2727	7.9274	2728.51	2614.33
500 m inward	5.1580	5.6268	2775.94	2669.28
100 m inward	2.3115	2.5233	2811.61	2710.04
10 m inward	0.7313	0.7984	2819.28	2718.68
1 m inward	0.2313	0.2525	2820.06	2719.41
10 cm inward	0.0731	0.0798	2820.14	2719.49
1 cm inward	0.0231	0.0252	2820.15	2719.50
surface limit	0.0000	0.0000	2820.15	2719.50

Table 283: Near-inner-boundary profile for GW230814_230901. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $18.472 M_\odot$ and $15.461 M_\odot$, with completed geometric mass-equivalent shares of 44.81% and 46.64%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.928 < 29.316 < 49.301$ km, and component 2 gives $21.934 < 24.756 < 41.727$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8517$ and 1.8734 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 12.778 and 12.343 h. The determinant-averaged frame-dragging rates at those shells are 2389.81 and $2401.09 \text{ rad s}^{-1}$, approaching surface-limit values of 2820.15 and $2719.50 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $7.273/7.927$ h at 1 km inward to $0.0231/0.0252$ h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.13 GW190630_185205

GW190630_185205 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 41.396 M_\odot$ and $m_2 = 28.236 M_\odot$, while the luminosity-distance median is 873.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	16,770
Observed luminosity distance	873.3 Mpc (median)
Detector-frame masses	$m_1 = 41.396 M_\odot$, $m_2 = 28.236 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.682$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1755$; $z_\pi = 0.1885$ (medians)
Post-handoff reorder fraction	21.20%

Table 284: Observational and coordinate summary for GW190630_185205. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	34.035 [23.956, 41.249]	24.280 [18.530, 34.911]
Physical compact-source mass M_c ($M_\odot$)	21.917 [16.899, 30.361]	16.017 [10.741, 21.844]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.547 [1.689, 18.405]	9.216 [1.837, 16.569]
Geometric share M_{geom}/M_∞ (%)	33.64 [6.56, 49.49]	38.76 [8.60, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.380 [0.051, 0.871]	0.290 [0.025, 0.832]
Polarization factor A_{pol}	1.507 [1.070, 1.980]	1.633 [1.094, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.862 [0.200, 0.998]	0.774 [0.101, 0.996]
GR/CPTG matching radius r_m (km)	50.257 [35.374, 60.909]	35.853 [27.362, 51.550]
Structural Mode N_{BH}	1.7189 [1.4841, 1.9064]	1.7789 [1.4966, 1.9108]
Organization-shell radius r_N (km)	26.503 [18.533, 32.137]	18.880 [14.410, 27.076]
Inner trapping-surface radius r_- (km)	32.073 [25.659, 41.041]	23.583 [17.166, 31.220]
Structural- N lapse $\bar{\alpha}_N$	0.6410 [0.4875, 0.8240]	0.5928 [0.4841, 0.8140]
Structural- N clock $T_{24,N}$ (h)	15.384 [11.700, 19.775]	14.228 [11.619, 19.536]
Structural- N axis ω (rad s^{-1})	2248.09 [776.80, 3208.92]	2871.09 [515.72, 4682.49]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2610.74 [800.82, 3610.38]	3332.24 [515.87, 5154.86]
Structural- N equatorial ω (rad s^{-1})	2988.60 [801.34, 4100.22]	3833.23 [515.95, 5615.97]
Structural- N period P_ω (ms)	2.4067 [1.7403, 7.8459]	1.8856 [1.2189, 12.1797]
Trapping-surface limit Ω_- (rad s^{-1})	3750.08 [874.37, 5467.38]	4805.04 [562.07, 7138.72]
Trapping-surface period P_- (ms)	1.6755 [1.1492, 7.1860]	1.3076 [0.8802, 11.1786]

Table 285: CPTG compact-state, structural, clock, and geometric-rotation results for GW190630_185205. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.5800	7.7327	3649.80	4630.40
500 m inward	4.6527	5.4722	3701.63	4720.76
100 m inward	2.0794	2.4486	3740.86	4788.90
10 m inward	0.6574	0.7744	3748.92	4803.55
1 m inward	0.2079	0.2449	3749.97	4804.85
10 cm inward	0.0657	0.0774	3750.07	4805.02
1 cm inward	0.0208	0.0245	3750.08	4805.04
surface limit	0.0000	0.0000	3750.08	4805.04

Table 286: Near-inner-boundary profile for GW190630_185205. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $21.917 M_\odot$ and $16.017 M_\odot$, with completed geometric mass-equivalent shares of 33.64% and 38.76%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.503 < 32.073 < 50.257 \text{ km}$, and component 2 gives $18.880 < 23.583 < 35.853 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7189$ and 1.7789 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.384 and 14.228 h. The determinant-averaged frame-dragging rates at those shells are 2610.74 and $3332.24 \text{ rad s}^{-1}$, approaching surface-limit values of 3750.08 and $4805.04 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.580/7.733 \text{ h}$ at 1 km inward to $0.0208/0.0245 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.14 GW230911_195324

GW230911_195324 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 42.781 M_\odot$ and $m_2 = 27.429 M_\odot$, while the luminosity-distance median is 1345.3 Mpc . 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the nearer-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	30,300
Observed luminosity distance	1345.3 Mpc (median)
Detector-frame masses	$m_1 = 42.781 M_\odot$, $m_2 = 27.429 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.641$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2576$; $z_\pi = 0.2770$ (medians)
Post-handoff reorder fraction	13.07%

Table 287: Observational and coordinate summary for GW230911_195324. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.127 [23.013, 41.645]	21.432 [15.722, 30.626]
Physical compact-source mass M_c ($M_\odot$)	23.237 [15.922, 35.671]	14.540 [9.544, 20.840]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.151 [0.975, 17.259]	7.225 [0.984, 14.079]
Geometric share M_{geom}/M_∞ (%)	25.17 [3.25, 49.12]	34.58 [5.13, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.527 [0.068, 0.935]	0.363 [0.032, 0.899]
Polarization factor A_{pol}	1.336 [1.034, 1.965]	1.529 [1.054, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.942 [0.261, 0.999]	0.849 [0.125, 0.999]
GR/CPTG matching radius r_m (km)	48.917 [33.981, 61.493]	31.648 [23.216, 45.223]
Structural Mode N_{BH}	1.6291 [1.4653, 1.9022]	1.7297 [1.4757, 1.9100]
Organization-shell radius r_N (km)	25.738 [17.860, 32.375]	16.660 [12.212, 23.803]
Inner trapping-surface radius r_- (km)	32.794 [24.041, 47.306]	21.067 [15.015, 28.952]
Structural- N lapse $\bar{\alpha}_N$	0.7109 [0.4909, 0.8390]	0.6325 [0.4848, 0.8307]
Structural- N clock $T_{24,N}$ (h)	17.061 [11.781, 20.136]	15.179 [11.634, 19.936]
Structural- N axis ω (rad s^{-1})	1977.86 [756.33, 3351.60]	3135.31 [744.29, 5293.79]
Structural- N determinant-averaged $\bar{\omega}$ (rad s^{-1})	2467.04 [1057.67, 3789.62]	3716.12 [745.03, 5888.46]
Structural- N equatorial ω (rad s^{-1})	2948.64 [1104.51, 4306.40]	4393.35 [745.42, 6469.51]
Structural- N period P_ω (ms)	2.5469 [1.6580, 5.9406]	1.6908 [1.0670, 8.4334]
Trapping-surface limit Ω_- (rad s^{-1})	3835.70 [1207.55, 5727.97]	5672.61 [813.07, 8323.21]
Trapping-surface period P_- (ms)	1.6381 [1.0969, 5.2033]	1.1076 [0.7549, 7.7277]

Table 288: CPTG compact-state, structural, clock, and geometric-rotation results for GW230911_195324. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1924	8.0713	3745.77	5456.53
500 m inward	4.3653	5.7105	3792.91	5575.38
100 m inward	1.9464	2.5537	3828.12	5655.50
10 m inward	0.6150	0.8075	3835.08	5670.57
1 m inward	0.1945	0.2553	3835.64	5672.41
10 cm inward	0.0615	0.0807	3835.69	5672.59
1 cm inward	0.0194	0.0255	3835.69	5672.61
surface limit	0.0000	0.0000	3835.70	5672.61

Table 289: Near-inner-boundary profile for GW230911_195324. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.237 M_\odot$ and $14.540 M_\odot$, with completed geometric mass-equivalent shares of 25.17% and 34.58%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.738 < 32.794 < 48.917 \text{ km}$, and component 2 gives $16.660 < 21.067 < 31.648 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6291$ and 1.7297 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.061 and 15.179 h. The determinant-averaged frame-dragging rates at those shells are 2467.04 and $3716.12 \text{ rad s}^{-1}$, approaching surface-limit values of 3835.70 and $5672.61 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.192/8.071 \text{ h}$ at 1 km inward to $0.0194/0.0255 \text{ h}$ at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.15 GW170809_082821

GW170809_082821 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 41.187 M_\odot$ and $m_2 = 29.248 M_\odot$, while the luminosity-distance median is 1065.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	32,872
Observed luminosity distance	1065.0 Mpc (median)
Detector-frame masses	$m_1 = 41.187 M_\odot$, $m_2 = 29.248 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.710$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2098$; $z_\pi = 0.2253$ (medians)
Post-handoff reorder fraction	18.66%

Table 290: Observational and coordinate summary for GW170809_082821. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.276 [24.913, 41.609]	24.268 [18.671, 32.664]
Physical compact-source mass M_c ($M_\odot$)	22.681 [16.621, 34.356]	15.653 [10.717, 22.147]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.434 [1.207, 18.087]	9.158 [1.737, 15.245]
Geometric share M_{geom}/M_∞ (%)	28.78 [4.05, 49.37]	38.69 [7.95, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.464 [0.057, 0.920]	0.292 [0.027, 0.844]
Polarization factor A_{pol}	1.404 [1.042, 1.975]	1.631 [1.086, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.915 [0.222, 0.999]	0.776 [0.106, 0.996]
GR/CPTG matching radius r_m (km)	49.137 [36.788, 61.441]	35.835 [27.571, 48.233]
Structural Mode N_{BH}	1.6657 [1.4697, 1.9051]	1.7781 [1.4925, 1.9106]
Organization-shell radius r_N (km)	25.892 [19.355, 32.375]	18.875 [14.510, 25.394]
Inner trapping-surface radius r_- (km)	32.457 [25.377, 45.621]	23.252 [17.177, 30.603]
Structural- N lapse $\bar{\alpha}_N$	0.6827 [0.4886, 0.8355]	0.5935 [0.4843, 0.8173]
Structural- N clock $T_{24,N}$ (h)	16.384 [11.726, 20.053]	14.244 [11.622, 19.614]
Structural- N axis ω (rad s^{-1})	2109.58 [733.44, 3277.12]	2929.09 [567.73, 4643.45]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2543.73 [895.34, 3666.94]	3356.77 [567.97, 5093.81]
Structural- N equatorial ω (rad s^{-1})	2984.66 [898.93, 4089.08]	3846.33 [568.12, 5539.34]
Structural- N period P_ω (ms)	2.4701 [1.7135, 7.0177]	1.8718 [1.2335, 11.0626]
Trapping-surface limit Ω_- (rad s^{-1})	3844.45 [981.79, 5363.53]	4820.47 [619.14, 7035.00]
Trapping-surface period P_- (ms)	1.6344 [1.1715, 6.3998]	1.3034 [0.8931, 10.1483]

Table 291: CPTG compact-state, structural, clock, and geometric-rotation results for GW170809_082821. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.4095	7.8614	3747.73	4638.14
500 m inward	4.5263	5.5682	3798.65	4731.96
100 m inward	2.0213	2.4924	3835.50	4804.74
10 m inward	0.6389	0.7883	3843.55	4819.23
1 m inward	0.2020	0.2493	3844.39	4820.44
10 cm inward	0.0639	0.0788	3844.45	4820.47
1 cm inward	0.0202	0.0249	3844.45	4820.47
surface limit	0.0000	0.0000	3844.45	4820.47

Table 292: Near-inner-boundary profile for GW170809_082821. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.681 M_\odot$ and $15.653 M_\odot$, with completed geometric mass-equivalent shares of 28.78% and 38.69%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.892 < 32.457 < 49.137 \text{ km}$, and component 2 gives $18.875 < 23.252 < 35.835 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6657$ and 1.7781 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.384 and 14.244 h. The determinant-averaged frame-dragging rates at those shells are 2543.73 and $3356.77 \text{ rad s}^{-1}$, approaching surface-limit values of 3844.45 and $4820.47 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.410/7.861 \text{ h}$ at 1 km inward to $0.0202/0.0249 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.16 GW150914_095045

GW150914_095045 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 37.990 M_\odot$ and $m_2 = 32.956 M_\odot$, while the luminosity-distance median is 473.3 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	3,337
Observed luminosity distance	473.3 Mpc (median)
Detector-frame masses	$m_1 = 37.990 M_\odot$, $m_2 = 32.956 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.867$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0998$; $z_\pi = 0.1073$ (medians)
Post-handoff reorder fraction	28.86%

Table 293: Observational and coordinate summary for GW150914_095045. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.607 [28.495, 38.669]	30.641 [25.457, 35.347]
Physical compact-source mass M_c ($M_\odot$)	25.758 [18.447, 33.533]	18.839 [14.496, 26.101]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.370 [1.044, 16.271]	12.000 [3.116, 16.625]
Geometric share M_{geom}/M_∞ (%)	21.85 [3.35, 46.58]	38.87 [11.34, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.586 [0.140, 0.934]	0.288 [0.028, 0.780]
Polarization factor A_{pol}	1.280 [1.035, 1.872]	1.636 [1.128, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.960 [0.490, 0.999]	0.772 [0.112, 0.992]
GR/CPTG matching radius r_m (km)	49.624 [42.077, 57.099]	45.246 [37.591, 52.194]
Structural Mode N_{BH}	1.5979 [1.4658, 1.8726]	1.7802 [1.5146, 1.9104]
Organization-shell radius r_N (km)	26.158 [22.033, 30.084]	23.831 [19.750, 27.582]
Inner trapping-surface radius r_- (km)	34.791 [28.653, 44.528]	28.498 [23.542, 34.834]
Structural- N lapse $\bar{\alpha}_N$	0.7350 [0.5149, 0.8386]	0.5918 [0.4844, 0.7997]
Structural- N clock $T_{24,N}$ (h)	17.639 [12.358, 20.127]	14.202 [11.626, 19.194]
Structural- N axis ω (rad s^{-1})	1906.41 [853.67, 2998.92]	2488.93 [505.49, 3480.39]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2458.22 [1356.93, 3315.88]	2824.61 [505.74, 3824.28]
Structural- N equatorial ω (rad s^{-1})	2976.35 [1940.33, 3618.61]	3183.25 [505.87, 4142.12]
Structural- N period P_ω (ms)	2.5560 [1.8949, 4.6304]	2.2244 [1.6430, 12.4237]
Trapping-surface limit Ω_- (rad s^{-1})	3945.63 [2230.50, 4685.89]	3946.58 [551.27, 5267.24]
Trapping-surface period P_- (ms)	1.5924 [1.3409, 2.8170]	1.5921 [1.1929, 11.3978]

Table 294: CPTG compact-state, structural, clock, and geometric-rotation results for GW150914_095045. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8341	7.2281	3849.65	3824.64
500 m inward	4.1067	5.1239	3901.58	3889.87
100 m inward	1.8297	2.2928	3937.18	3936.55
10 m inward	0.5782	0.7252	3944.46	3946.22
1 m inward	0.1828	0.2293	3945.50	3946.55
10 cm inward	0.0578	0.0725	3945.62	3946.57
1 cm inward	0.0183	0.0229	3945.63	3946.58
surface limit	0.0000	0.0000	3945.63	3946.58

Table 295: Near-inner-boundary profile for GW150914_095045. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $25.758 M_\odot$ and $18.839 M_\odot$, with completed geometric mass-equivalent shares of 21.85% and 38.87%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.158 < 34.791 < 49.624 \text{ km}$, and component 2 gives $23.831 < 28.498 < 45.246 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5979$ and 1.7802 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.639 and 14.202 h. The determinant-averaged frame-dragging rates at those shells are 2458.22 and $2824.61 \text{ rad s}^{-1}$, approaching surface-limit values of 3945.63 and $3946.58 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.834/7.228 \text{ h}$ at 1 km inward to $0.0183/0.0229 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.17 GW250114_082203

GW250114_082203 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 36.636 M_\odot$ and $m_2 = 35.035 M_\odot$, while the luminosity-distance median is 404.7 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	13,969
Observed luminosity distance	404.7 Mpc (median)
Detector-frame masses	$m_1 = 36.636 M_\odot$, $m_2 = 35.035 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.956$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.0860$; $z_\pi = 0.0926$ (medians)
Post-handoff reorder fraction	30.58%

Table 296: Observational and coordinate summary for GW250114_082203. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.317 [31.465, 34.781]	32.397 [30.724, 33.939]
Physical compact-source mass M_c ($M_\odot$)	18.040 [16.755, 23.033]	16.700 [15.759, 20.049]
Geometric mass-equivalent M_{geom} ($M_\odot$)	15.153 [9.585, 17.092]	15.682 [12.177, 16.694]
Geometric share M_{geom}/M_∞ (%)	45.69 [29.54, 49.95]	48.79 [38.07, 49.99]
Exterior spin magnitude $ \hat{a}_\infty $	0.159 [0.016, 0.451]	0.080 [0.008, 0.303]
Polarization factor A_{pol}	1.841 [1.419, 1.998]	1.953 [1.615, 2.000]
Compact-source spin parameter $\hat{a}_c$	0.540 [0.064, 0.908]	0.304 [0.031, 0.789]
GR/CPTG matching radius r_m (km)	49.197 [46.462, 51.359]	47.838 [45.368, 50.115]
Structural Mode N_{BH}	1.8622 [1.6738, 1.9117]	1.8984 [1.7706, 1.9121]
Organization-shell radius r_N (km)	25.860 [24.544, 26.973]	25.083 [23.821, 26.310]
Inner trapping-surface radius r_- (km)	28.990 [27.800, 32.285]	27.721 [26.365, 29.919]
Structural- N lapse $\bar{\alpha}_N$	0.5236 [0.4835, 0.6764]	0.4939 [0.4831, 0.5996]
Structural- N clock $T_{24,N}$ (h)	12.567 [11.603, 16.233]	11.853 [11.595, 14.391]
Structural- N axis ω (rad s^{-1})	2240.87 [266.24, 3062.83]	1339.01 [135.35, 2999.41]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2324.82 [266.28, 3314.18]	1345.26 [135.36, 3205.56]
Structural- N equatorial ω (rad s^{-1})	2361.40 [266.30, 3521.69]	1347.64 [135.36, 3367.17]
Structural- N period P_ω (ms)	2.7027 [1.8958, 23.5961]	4.6706 [1.9601, 46.4191]
Trapping-surface limit Ω_- (rad s^{-1})	2632.67 [290.03, 4347.52]	1476.17 [147.40, 4013.42]
Trapping-surface period P_- (ms)	2.3866 [1.4452, 21.6636]	4.2564 [1.5655, 42.6276]

Table 297: CPTG compact-state, structural, clock, and geometric-rotation results for GW250114_082203. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	7.3536	7.5680	2545.05	1428.97
500 m inward	5.2171	5.3725	2589.28	1452.77
100 m inward	2.3388	2.4095	2624.58	1471.50
10 m inward	0.7400	0.7624	2631.87	1475.70
1 m inward	0.2340	0.2411	2632.59	1476.12
10 cm inward	0.0740	0.0762	2632.67	1476.16
1 cm inward	0.0234	0.0241	2632.67	1476.17
surface limit	0.0000	0.0000	2632.67	1476.17

Table 298: Near-inner-boundary profile for GW250114_082203. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $18.040 M_\odot$ and $16.700 M_\odot$, with completed geometric mass-equivalent shares of 45.69% and 48.79%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.860 < 28.990 < 49.197 \text{ km}$, and component 2 gives $25.083 < 27.721 < 47.838 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.8622$ and 1.8984 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 12.567 and 11.853 h. The determinant-averaged frame-dragging rates at those shells are 2324.82 and $1345.26 \text{ rad s}^{-1}$, approaching surface-limit values of 2632.67 and $1476.17 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $7.354/7.568 \text{ h}$ at 1 km inward to $0.0234/0.0241 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.18 GW241210_060606

GW241210_060606 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 41.761 M_\odot$ and $m_2 = 31.212 M_\odot$, while the luminosity-distance median is 2595.8 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,229
Observed luminosity distance	2595.8 Mpc (median)
Detector-frame masses	$m_1 = 41.761 M_\odot$, $m_2 = 31.212 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.747$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4532$; $z_\pi = 0.4878$ (medians)
Post-handoff reorder fraction	19.30%

Table 299: Observational and coordinate summary for GW241210_060606. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.829 [20.797, 36.446]	21.369 [15.490, 28.887]
Physical compact-source mass M_c ($M_\odot$)	20.264 [13.965, 29.972]	13.760 [9.296, 20.313]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.755 [0.746, 15.048]	7.784 [1.267, 13.187]
Geometric share M_{geom}/M_∞ (%)	24.64 [2.87, 48.83]	37.64 [6.58, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.537 [0.078, 0.943]	0.310 [0.029, 0.871]
Polarization factor A_{pol}	1.327 [1.030, 1.954]	1.604 [1.070, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.945 [0.298, 1.000]	0.797 [0.114, 0.998]
GR/CPTG matching radius r_m (km)	41.092 [30.710, 53.818]	31.554 [22.873, 42.656]
Structural Mode N_{BH}	1.6240 [1.4633, 1.8989]	1.7655 [1.4842, 1.9104]
Organization-shell radius r_N (km)	21.622 [16.157, 28.339]	16.623 [12.035, 22.468]
Inner trapping-surface radius r_- (km)	28.282 [21.275, 39.878]	20.437 [14.627, 27.874]
Structural- N lapse $\bar{\alpha}_N$	0.7149 [0.4934, 0.8407]	0.6038 [0.4844, 0.8239]
Structural- N clock $T_{24,N}$ (h)	17.157 [11.842, 20.177]	14.490 [11.627, 19.773]
Structural- N axis ω (rad s^{-1})	2294.76 [914.53, 3879.20]	3253.06 [701.50, 5391.16]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2863.13 [1310.26, 4348.81]	3780.21 [701.95, 5968.15]
Structural- N equatorial ω (rad s^{-1})	3469.57 [1419.63, 4865.33]	4345.83 [702.21, 6563.10]
Structural- N period P_ω (ms)	2.1945 [1.4448, 4.7954]	1.6621 [1.0528, 8.9510]
Trapping-surface limit Ω_- (rad s^{-1})	4570.47 [1553.66, 6381.15]	5501.25 [765.40, 8401.02]
Trapping-surface period P_- (ms)	1.3747 [0.9846, 4.0441]	1.1421 [0.7479, 8.2090]

Table 300: CPTG compact-state, structural, clock, and geometric-rotation results for GW241210_060606. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.6300	8.3493	4450.52	5272.07
500 m inward	4.6722	5.9115	4515.85	5395.05
100 m inward	2.0827	2.6458	4560.44	5479.30
10 m inward	0.6582	0.8367	4569.74	5500.18
1 m inward	0.2081	0.2646	4570.38	5501.04
10 cm inward	0.0658	0.0837	4570.45	5501.23
1 cm inward	0.0208	0.0265	4570.47	5501.25
surface limit	0.0000	0.0000	4570.47	5501.25

Table 301: Near-inner-boundary profile for GW241210_060606. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $20.264 M_\odot$ and $13.760 M_\odot$, with completed geometric mass-equivalent shares of 24.64% and 37.64%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.622 < 28.282 < 41.092 \text{ km}$, and component 2 gives $16.623 < 20.437 < 31.554 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6240$ and 1.7655 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.157 and 14.490 h. The determinant-averaged frame-dragging rates at those shells are 2863.13 and $3780.21 \text{ rad s}^{-1}$, approaching surface-limit values of 4570.47 and $5501.25 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.630/8.349 \text{ h}$ at 1 km inward to $0.0208/0.0265 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.19 GW230924_124453

GW230924_124453 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 40.541 M_\odot$ and $m_2 = 32.992 M_\odot$, while the luminosity-distance median is 2379.5 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	41,241
Observed luminosity distance	2379.5 Mpc (median)
Detector-frame masses	$m_1 = 40.541 M_\odot$, $m_2 = 32.992 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.814$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4212$; $z_\pi = 0.4532$ (medians)
Post-handoff reorder fraction	26.99%

Table 302: Observational and coordinate summary for GW230924_124453. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.332 [21.507, 33.703]	23.454 [18.458, 29.720]
Physical compact-source mass M_c ($M_\odot$)	19.188 [13.974, 27.574]	14.369 [10.565, 20.573]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.503 [0.995, 14.631]	9.433 [2.134, 13.865]
Geometric share M_{geom}/M_∞ (%)	27.90 [3.90, 49.19]	41.06 [10.11, 49.90]
Exterior spin magnitude $ \hat{a}_\infty $	0.479 [0.065, 0.923]	0.249 [0.023, 0.803]
Polarization factor A_{pol}	1.387 [1.041, 1.968]	1.697 [1.112, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.922 [0.251, 0.999]	0.718 [0.090, 0.994]
GR/CPTG matching radius r_m (km)	40.359 [31.757, 49.767]	34.633 [27.256, 43.885]
Structural Mode N_{BH}	1.6566 [1.4688, 1.9030]	1.8064 [1.5063, 1.9111]
Organization-shell radius r_N (km)	21.265 [16.687, 26.204]	18.237 [14.346, 23.095]
Inner trapping-surface radius r_- (km)	27.062 [21.656, 36.599]	21.807 [16.992, 28.130]
Structural- N lapse $\bar{\alpha}_N$	0.6897 [0.4902, 0.8362]	0.5702 [0.4839, 0.8063]
Structural- N clock $T_{24,N}$ (h)	16.554 [11.765, 20.069]	13.684 [11.614, 19.350]
Structural- N axis ω (rad s^{-1})	2536.38 [947.83, 3882.71]	3075.44 [523.21, 4754.18]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	3066.64 [1220.18, 4332.97]	3478.75 [523.33, 5213.96]
Structural- N equatorial ω (rad s^{-1})	3617.53 [1237.01, 4793.30]	3908.39 [523.38, 5641.20]
Structural- N period P_ω (ms)	2.0489 [1.4501, 5.1494]	1.8062 [1.2051, 12.0063]
Trapping-surface limit Ω_- (rad s^{-1})	4708.81 [1351.84, 6241.66]	4777.89 [570.18, 7175.56]
Trapping-surface period P_- (ms)	1.3343 [1.0067, 4.6479]	1.3151 [0.8756, 11.0196]

Table 303: CPTG compact-state, structural, clock, and geometric-rotation results for GW230924_124453. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.9615	8.2194	4572.69	4576.67
500 m inward	4.9124	5.8287	4644.13	4682.60
100 m inward	2.1920	2.6114	4696.90	4760.76
10 m inward	0.6928	0.8260	4707.62	4775.88
1 m inward	0.2191	0.2612	4708.69	4777.64
10 cm inward	0.0693	0.0826	4708.80	4777.86
1 cm inward	0.0219	0.0261	4708.81	4777.89
surface limit	0.0000	0.0000	4708.81	4777.89

Table 304: Near-inner-boundary profile for GW230924_124453. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $19.188 M_\odot$ and $14.369 M_\odot$, with completed geometric mass-equivalent shares of 27.90% and 41.06%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.265 < 27.062 < 40.359 \text{ km}$, and component 2 gives $18.237 < 21.807 < 34.633 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6566$ and 1.8064 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.554 and 13.684 h. The determinant-averaged frame-dragging rates at those shells are 3066.64 and $3478.75 \text{ rad s}^{-1}$, approaching surface-limit values of 4708.81 and $4777.89 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.962/8.219 \text{ h}$ at 1 km inward to $0.0219/0.0261 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.20 GW241130_034908

GW241130_034908 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 40.448 M_\odot$ and $m_2 = 33.133 M_\odot$, while the luminosity-distance median is 1987.4 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,275
Observed luminosity distance	1987.4 Mpc (median)
Detector-frame masses	$m_1 = 40.448 M_\odot$, $m_2 = 33.133 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.819$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3614$; $z_\pi = 0.3887$ (medians)
Post-handoff reorder fraction	27.66%

Table 305: Observational and coordinate summary for GW241130_034908. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	28.364 [22.693, 34.981]	24.582 [19.792, 31.267]
Physical compact-source mass M_c ($M_\odot$)	20.617 [14.895, 28.218]	15.194 [11.228, 21.427]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.280 [0.889, 15.138]	9.899 [2.353, 14.590]
Geometric share M_{geom}/M_∞ (%)	25.76 [3.46, 48.63]	40.82 [10.46, 49.91]
Exterior spin magnitude $ \hat{a}_\infty $	0.517 [0.085, 0.931]	0.254 [0.022, 0.796]
Polarization factor A_{pol}	1.347 [1.036, 1.947]	1.690 [1.117, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.938 [0.322, 0.999]	0.724 [0.087, 0.993]
GR/CPTG matching radius r_m (km)	41.883 [33.510, 51.653]	36.298 [29.225, 46.170]
Structural Mode N_{BH}	1.6349 [1.4664, 1.8966]	1.8036 [1.5087, 1.9112]
Organization-shell radius r_N (km)	22.067 [17.609, 27.227]	19.121 [15.395, 24.281]
Inner trapping-surface radius r_- (km)	28.765 [22.947, 37.495]	22.993 [18.090, 29.363]
Structural- N lapse $\bar{\alpha}_N$	0.7064 [0.4953, 0.8381]	0.5725 [0.4839, 0.8044]
Structural- N clock $T_{24,N}$ (h)	16.954 [11.888, 20.115]	13.741 [11.613, 19.306]
Structural- N axis ω (rad s^{-1})	2368.94 [991.52, 3680.74]	2952.20 [481.91, 4448.21]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2907.79 [1417.17, 4103.89]	3327.37 [482.10, 4876.45]
Structural- N equatorial ω (rad s^{-1})	3469.18 [1487.11, 4525.08]	3755.07 [482.19, 5290.22]
Structural- N period P_ω (ms)	2.1608 [1.5310, 4.4336]	1.8883 [1.2885, 13.0330]
Trapping-surface limit Ω_- (rad s^{-1})	4567.16 [1632.60, 5906.65]	4610.24 [525.35, 6708.30]
Trapping-surface period P_- (ms)	1.3757 [1.0637, 3.8486]	1.3629 [0.9366, 11.9600]

Table 306: CPTG compact-state, structural, clock, and geometric-rotation results for GW241130_034908. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.6459	8.0040	4447.09	4425.89
500 m inward	4.6867	5.6736	4510.70	4525.05
100 m inward	2.0899	2.5415	4555.45	4595.20
10 m inward	0.6604	0.8039	4565.64	4608.82
1 m inward	0.2088	0.2542	4567.05	4610.09
10 cm inward	0.0660	0.0804	4567.15	4610.22
1 cm inward	0.0209	0.0254	4567.16	4610.24
surface limit	0.0000	0.0000	4567.16	4610.24

Table 307: Near-inner-boundary profile for GW241130_034908. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $20.617 M_\odot$ and $15.194 M_\odot$, with completed geometric mass-equivalent shares of 25.76% and 40.82%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $22.067 < 28.765 < 41.883 \text{ km}$, and component 2 gives $19.121 < 22.993 < 36.298 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6349$ and 1.8036 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.954 and 13.741 h. The determinant-averaged frame-dragging rates at those shells are 2907.79 and $3327.37 \text{ rad s}^{-1}$, approaching surface-limit values of 4567.16 and $4610.24 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.646/8.004 \text{ h}$ at 1 km inward to $0.0209/0.0254 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.21 GW200302_015811

GW200302_015811 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 48.445 M_\odot$ and $m_2 = 25.500 M_\odot$, while the luminosity-distance median is 1483.3 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	18,183
Observed luminosity distance	1483.3 Mpc (median)
Detector-frame masses	$m_1 = 48.445 M_\odot$, $m_2 = 25.500 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.526$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2811$; $z_\pi = 0.3018$ (medians)
Post-handoff reorder fraction	7.50%

Table 308: Observational and coordinate summary for GW200302_015811. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	37.070 [25.650, 45.871]	19.685 [14.060, 30.857]
Physical compact-source mass M_c ($M_\odot$)	24.401 [17.414, 37.434]	13.683 [8.465, 21.298]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.385 [1.475, 20.052]	6.365 [0.800, 13.594]
Geometric share M_{geom}/M_∞ (%)	30.96 [4.54, 49.60]	33.02 [4.50, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.426 [0.045, 0.910]	0.390 [0.035, 0.911]
Polarization factor A_{pol}	1.449 [1.048, 1.984]	1.493 [1.047, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.894 [0.178, 0.999]	0.870 [0.138, 0.999]
GR/CPTG matching radius r_m (km)	54.738 [37.875, 67.735]	29.067 [20.761, 45.564]
Structural Mode N_{BH}	1.6891 [1.4724, 1.9077]	1.7120 [1.4722, 1.9095]
Organization-shell radius r_N (km)	28.824 [19.873, 35.706]	15.315 [10.927, 24.003]
Inner trapping-surface radius r_- (km)	35.563 [26.519, 49.798]	19.578 [13.307, 29.757]
Structural- N lapse $\bar{\alpha}_N$	0.6644 [0.4865, 0.8333]	0.6465 [0.4851, 0.8335]
Structural- N clock $T_{24,N}$ (h)	15.947 [11.677, 20.000]	15.516 [11.643, 20.004]
Structural- N axis ω (rad s^{-1})	1905.93 [613.89, 2975.14]	3325.81 [891.08, 5971.97]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2276.35 [666.16, 3332.81]	3967.08 [895.51, 6629.29]
Structural- N equatorial ω (rad s^{-1})	2643.83 [666.63, 3833.48]	4716.80 [896.04, 7260.83]
Structural- N period P_ω (ms)	2.7602 [1.8853, 9.4320]	1.5838 [0.9478, 7.0163]
Trapping-surface limit Ω_- (rad s^{-1})	3373.66 [727.49, 5110.36]	6077.23 [977.49, 9308.10]
Trapping-surface period P_- (ms)	1.8624 [1.2295, 8.6368]	1.0339 [0.6750, 6.4279]

Table 309: CPTG compact-state, structural, clock, and geometric-rotation results for GW200302_015811. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1984	8.2489	3299.78	5820.27
500 m inward	4.3791	5.8320	3337.37	5955.99
100 m inward	1.9561	2.6049	3367.05	6053.49
10 m inward	0.6185	0.8236	3373.07	6075.17
1 m inward	0.1956	0.2604	3373.59	6077.07
10 cm inward	0.0619	0.0824	3373.65	6077.21
1 cm inward	0.0196	0.0260	3373.66	6077.23
surface limit	0.0000	0.0000	3373.66	6077.23

Table 310: Near-inner-boundary profile for GW200302_015811. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.401 M_\odot$ and $13.683 M_\odot$, with completed geometric mass-equivalent shares of 30.96% and 33.02%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.824 < 35.563 < 54.738 \text{ km}$, and component 2 gives $15.315 < 19.578 < 29.067 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6891$ and 1.7120 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.947 and 15.516 h. The determinant-averaged frame-dragging rates at those shells are 2276.35 and $3967.08 \text{ rad s}^{-1}$, approaching surface-limit values of 3373.66 and $6077.23 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.198/8.249 \text{ h}$ at 1 km inward to $0.0196/0.0260 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.22 GW200129_065458

GW200129_065458 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 40.207 M_\odot$ and $m_2 = 34.094 M_\odot$, while the luminosity-distance median is 893.4 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	8,775
Observed luminosity distance	893.4 Mpc (median)
Detector-frame masses	$m_1 = 40.207 M_\odot$, $m_2 = 34.094 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.848$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.1792$; $z_\pi = 0.1924$ (medians)
Post-handoff reorder fraction	28.46%

Table 311: Observational and coordinate summary for GW200129_065458. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.338 [28.147, 43.868]	29.411 [19.429, 34.630]
Physical compact-source mass M_c ($M_\odot$)	26.107 [18.407, 41.278]	18.517 [12.868, 26.068]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.360 [0.792, 13.961]	10.074 [1.298, 16.184]
Geometric share M_{geom}/M_∞ (%)	19.31 [2.14, 42.31]	35.51 [5.32, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.632 [0.226, 0.957]	0.347 [0.030, 0.895]
Polarization factor A_{pol}	1.239 [1.022, 1.733]	1.551 [1.056, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.971 [0.680, 1.000]	0.835 [0.120, 0.998]
GR/CPTG matching radius r_m (km)	49.227 [41.563, 64.776]	43.429 [28.690, 51.136]
Structural Mode N_{BH}	1.5756 [1.4594, 1.8215]	1.7404 [1.4769, 1.9102]
Organization-shell radius r_N (km)	25.952 [21.857, 33.805]	22.880 [15.080, 26.952]
Inner trapping-surface radius r_- (km)	35.102 [27.672, 55.069]	27.875 [19.304, 34.626]
Structural- N lapse $\bar{\alpha}_N$	0.7522 [0.5576, 0.8438]	0.6239 [0.4846, 0.8298]
Structural- N clock $T_{24,N}$ (h)	18.052 [13.383, 20.251]	14.974 [11.631, 19.914]
Structural- N axis ω (rad s^{-1})	1822.99 [694.28, 3130.45]	2508.52 [565.96, 4033.18]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2411.28 [1136.65, 3464.20]	2914.16 [566.23, 4564.97]
Structural- N equatorial ω (rad s^{-1})	2985.88 [1832.72, 3780.44]	3348.87 [566.37, 5137.68]
Structural- N period P_ω (ms)	2.6057 [1.8137, 5.5278]	2.1561 [1.3764, 11.0964]
Trapping-surface limit Ω_- (rad s^{-1})	3947.02 [2611.17, 4859.97]	4261.77 [617.19, 6709.95]
Trapping-surface period P_- (ms)	1.5919 [1.2928, 2.4063]	1.4743 [0.9364, 10.1804]

Table 312: CPTG compact-state, structural, clock, and geometric-rotation results for GW200129_065458. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6843	7.1963	3855.48	4144.31
500 m inward	3.9960	5.0930	3903.22	4203.58
100 m inward	1.7788	2.2784	3939.46	4250.29
10 m inward	0.5618	0.7205	3945.97	4260.80
1 m inward	0.1776	0.2279	3946.89	4261.65
10 cm inward	0.0562	0.0721	3947.01	4261.75
1 cm inward	0.0178	0.0228	3947.02	4261.77
surface limit	0.0000	0.0000	3947.02	4261.77

Table 313: Near-inner-boundary profile for GW200129_065458. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.107 M_\odot$ and $18.517 M_\odot$, with completed geometric mass-equivalent shares of 19.31% and 35.51%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.952 < 35.102 < 49.227 \text{ km}$, and component 2 gives $22.880 < 27.875 < 43.429 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5756$ and 1.7404 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.052 and 14.974 h. The determinant-averaged frame-dragging rates at those shells are 2411.28 and $2914.16 \text{ rad s}^{-1}$, approaching surface-limit values of 3947.02 and $4261.77 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 5.684/7.196 h at 1 km inward to 0.0178/0.0228 h at 1 cm inward for components 1/2. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.23 GW190915_235702

GW190915_235702 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 43.047 M_\odot$ and $m_2 = 32.496 M_\odot$, while the luminosity-distance median is 1747.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	21,302
Observed luminosity distance	1747.0 Mpc (median)
Detector-frame masses	$m_1 = 43.047 M_\odot$, $m_2 = 32.496 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.755$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3240$; $z_\pi = 0.3479$ (medians)
Post-handoff reorder fraction	19.75%

Table 314: Observational and coordinate summary for GW190915_235702. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	31.430 [24.004, 40.624]	24.557 [18.278, 31.966]
Physical compact-source mass M_c ($M_\odot$)	24.745 [16.874, 35.705]	16.031 [10.699, 22.729]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.663 [0.654, 15.167]	8.728 [1.558, 14.607]
Geometric share M_{geom}/M_∞ (%)	17.99 [2.17, 46.01]	36.46 [7.14, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.656 [0.152, 0.957]	0.331 [0.030, 0.860]
Polarization factor A_{pol}	1.219 [1.022, 1.852]	1.574 [1.077, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.976 [0.523, 1.000]	0.819 [0.120, 0.997]
GR/CPTG matching radius r_m (km)	46.410 [35.445, 59.987]	36.261 [26.990, 47.202]
Structural Mode N_{BH}	1.5646 [1.4596, 1.8660]	1.7515 [1.4875, 1.9102]
Organization-shell radius r_N (km)	24.421 [18.614, 31.519]	19.104 [14.191, 24.870]
Inner trapping-surface radius r_- (km)	33.469 [25.261, 47.523]	23.714 [16.913, 30.835]
Structural- N lapse $\bar{\alpha}_N$	0.7607 [0.5205, 0.8437]	0.6150 [0.4846, 0.8212]
Structural- N clock $T_{24,N}$ (h)	18.256 [12.492, 20.248]	14.760 [11.630, 19.709]
Structural- N axis ω (rad s^{-1})	1760.07 [820.49, 3259.68]	2900.31 [660.83, 4654.48]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2385.36 [1287.26, 3678.35]	3368.74 [661.32, 5123.73]
Structural- N equatorial ω (rad s^{-1})	3058.43 [1901.02, 4129.68]	3871.56 [661.47, 5590.28]
Structural- N period P_ω (ms)	2.6341 [1.7082, 4.8811]	1.8651 [1.2263, 9.5010]
Trapping-surface limit Ω_- (rad s^{-1})	4122.09 [2461.24, 5469.48]	4928.24 [720.87, 7109.88]
Trapping-surface period P_- (ms)	1.5243 [1.1488, 2.5529]	1.2749 [0.8837, 8.7161]

Table 315: CPTG compact-state, structural, clock, and geometric-rotation results for GW190915_235702. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.7055	7.7577	4033.79	4761.28
500 m inward	4.0075	5.4909	4080.03	4848.71
100 m inward	1.7819	2.4573	4114.11	4912.50
10 m inward	0.5627	0.7771	4121.30	4926.74
1 m inward	0.1779	0.2457	4122.02	4928.11
10 cm inward	0.0563	0.0777	4122.08	4928.23
1 cm inward	0.0178	0.0246	4122.09	4928.24
surface limit	0.0000	0.0000	4122.09	4928.24

Table 316: Near-inner-boundary profile for GW190915_235702. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.745 M_\odot$ and $16.031 M_\odot$, with completed geometric mass-equivalent shares of 17.99% and 36.46%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.421 < 33.469 < 46.410 \text{ km}$, and component 2 gives $19.104 < 23.714 < 36.261 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5646$ and 1.7515 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.256 and 14.760 h. The determinant-averaged frame-dragging rates at those shells are 2385.36 and $3368.74 \text{ rad s}^{-1}$, approaching surface-limit values of 4122.09 and $4928.24 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.706/7.758 \text{ h}$ at 1 km inward to $0.0178/0.0246 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.24 GW170818_022509

GW170818_022509 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 42.143 M_\odot$ and $m_2 = 33.416 M_\odot$, while the luminosity-distance median is 1081.8 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	33,331
Observed luminosity distance	1081.8 Mpc (median)
Detector-frame masses	$m_1 = 42.143 M_\odot$, $m_2 = 33.416 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.793$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2128$; $z_\pi = 0.2284$ (medians)
Post-handoff reorder fraction	23.93%

Table 317: Observational and coordinate summary for GW170818_022509. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.598 [26.436, 40.779]	27.979 [22.240, 35.122]
Physical compact-source mass M_c ($M_\odot$)	26.497 [18.436, 35.797]	18.177 [12.913, 25.429]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.850 [0.658, 16.715]	10.239 [1.847, 16.230]
Geometric share M_{geom}/M_∞ (%)	17.70 [2.06, 46.77]	36.79 [7.20, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.662 [0.135, 0.959]	0.325 [0.033, 0.859]
Polarization factor A_{pol}	1.215 [1.021, 1.878]	1.582 [1.078, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.977 [0.478, 1.000]	0.813 [0.130, 0.997]
GR/CPTG matching radius r_m (km)	49.612 [39.036, 60.216]	41.315 [32.839, 51.863]
Structural Mode N_{BH}	1.5622 [1.4590, 1.8749]	1.7555 [1.4879, 1.9098]
Organization-shell radius r_N (km)	26.096 [20.451, 31.680]	21.760 [17.283, 27.330]
Inner trapping-surface radius r_- (km)	35.745 [27.774, 47.586]	26.886 [20.670, 34.062]
Structural- N lapse $\bar{\alpha}_N$	0.7625 [0.5131, 0.8441]	0.6118 [0.4849, 0.8209]
Structural- N clock $T_{24,N}$ (h)	18.301 [12.314, 20.259]	14.684 [11.637, 19.702]
Structural- N axis ω (rad s^{-1})	1632.94 [789.34, 2994.86]	2573.31 [625.91, 3881.95]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2215.93 [1250.62, 3360.89]	2967.31 [626.35, 4283.54]
Structural- N equatorial ω (rad s^{-1})	2864.24 [1824.97, 3760.32]	3408.15 [626.57, 4670.43]
Structural- N period P_ω (ms)	2.8355 [1.8695, 5.0241]	2.1175 [1.4668, 10.0314]
Trapping-surface limit Ω_- (rad s^{-1})	3869.83 [2137.59, 4976.29]	4341.79 [682.98, 5964.08]
Trapping-surface period P_- (ms)	1.6236 [1.2626, 2.9394]	1.4471 [1.0535, 9.1997]

Table 318: CPTG compact-state, structural, clock, and geometric-rotation results for GW170818_022509. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5230	7.3049	3793.64	4210.32
500 m inward	3.8820	5.1717	3834.69	4278.64
100 m inward	1.7269	2.3145	3863.42	4330.05
10 m inward	0.5452	0.7320	3869.14	4340.68
1 m inward	0.1724	0.2315	3869.76	4341.74
10 cm inward	0.0545	0.0732	3869.83	4341.78
1 cm inward	0.0172	0.0231	3869.83	4341.79
surface limit	0.0000	0.0000	3869.83	4341.79

Table 319: Near-inner-boundary profile for GW170818_022509. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.497 M_\odot$ and $18.177 M_\odot$, with completed geometric mass-equivalent shares of 17.70% and 36.79%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.096 < 35.745 < 49.612 \text{ km}$, and component 2 gives $21.760 < 26.886 < 41.315 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5622$ and 1.7555 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.301 and 14.684 h. The determinant-averaged frame-dragging rates at those shells are 2215.93 and $2967.31 \text{ rad s}^{-1}$, approaching surface-limit values of 3869.83 and $4341.79 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 5.523/7.305 h at 1 km inward to 0.0172/0.0231 h at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.25 GW190513_205428

GW190513_205428 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 49.985 M_\odot$ and $m_2 = 25.603 M_\odot$, while the luminosity-distance median is 2209.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	13,281
Observed luminosity distance	2209.8 Mpc (median)
Detector-frame masses	$m_1 = 49.985 M_\odot$, $m_2 = 25.603 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.512$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3961$; $z_\pi = 0.4256$ (medians)
Post-handoff reorder fraction	8.17%

Table 320: Observational and coordinate summary for GW190513_205428. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.203 [23.255, 45.603]	17.960 [13.404, 27.701]
Physical compact-source mass M_c ($M_\odot$)	23.618 [16.065, 38.275]	12.602 [7.852, 19.414]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.846 [1.294, 18.521]	5.807 [0.780, 12.191]
Geometric share M_{geom}/M_∞ (%)	28.46 [4.48, 49.39]	32.68 [4.64, 49.74]
Exterior spin magnitude $ \hat{a}_\infty $	0.470 [0.056, 0.911]	0.396 [0.036, 0.908]
Polarization factor A_{pol}	1.398 [1.047, 1.976]	1.485 [1.049, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.917 [0.218, 0.999]	0.874 [0.143, 0.999]
GR/CPTG matching radius r_m (km)	51.982 [34.339, 67.339]	26.520 [19.792, 40.904]
Structural Mode N_{BH}	1.6624 [1.4721, 1.9053]	1.7081 [1.4730, 1.9093]
Organization-shell radius r_N (km)	27.383 [18.051, 35.496]	13.959 [10.415, 21.539]
Inner trapping-surface radius r_- (km)	33.971 [24.008, 50.617]	18.023 [12.465, 26.783]
Structural- N lapse $\bar{\alpha}_N$	0.6852 [0.4884, 0.8336]	0.6495 [0.4853, 0.8329]
Structural- N clock $T_{24,N}$ (h)	16.446 [11.721, 20.007]	15.589 [11.647, 19.989]
Structural- N axis ω (rad s^{-1})	1988.24 [693.63, 3310.56]	3567.31 [1011.15, 6412.36]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2386.97 [863.94, 3732.39]	4272.93 [1017.02, 7069.83]
Structural- N equatorial ω (rad s^{-1})	2776.63 [867.37, 4263.98]	5109.03 [1017.22, 7696.69]
Structural- N period P_ω (ms)	2.6323 [1.6834, 7.2727]	1.4705 [0.8887, 6.1780]
Trapping-surface limit Ω_- (rad s^{-1})	3541.75 [947.41, 5659.23]	6619.26 [1108.31, 9823.58]
Trapping-surface period P_- (ms)	1.7740 [1.1103, 6.6320]	0.9492 [0.6396, 5.6692]

Table 321: CPTG compact-state, structural, clock, and geometric-rotation results for GW190513_205428. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.2039	8.6093	3458.85	6311.65
500 m inward	4.3778	6.0838	3501.13	6476.70
100 m inward	1.9546	2.7183	3534.22	6593.68
10 m inward	0.6178	0.8592	3541.15	6617.43
1 m inward	0.1954	0.2717	3541.70	6618.98
10 cm inward	0.0618	0.0859	3541.73	6619.23
1 cm inward	0.0195	0.0272	3541.74	6619.26
surface limit	0.0000	0.0000	3541.75	6619.26

Table 322: Near-inner-boundary profile for GW190513_205428. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.618 M_\odot$ and $12.602 M_\odot$, with completed geometric mass-equivalent shares of 28.46% and 32.68%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.383 < 33.971 < 51.982 \text{ km}$, and component 2 gives $13.959 < 18.023 < 26.520 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6624$ and 1.7081 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.446 and 15.589 h. The determinant-averaged frame-dragging rates at those shells are 2386.97 and $4272.93 \text{ rad s}^{-1}$, approaching surface-limit values of 3541.75 and $6619.26 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.204/8.609 \text{ h}$ at 1 km inward to $0.0195/0.0272 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.26 GW200311_115853

GW200311_115853 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 41.832 M_\odot$ and $m_2 = 33.997 M_\odot$, while the luminosity-distance median is 1172.1 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	20,276
Observed luminosity distance	1172.1 Mpc (median)
Detector-frame masses	$m_1 = 41.832 M_\odot$, $m_2 = 33.997 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.813$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2285$; $z_\pi = 0.2453$ (medians)
Post-handoff reorder fraction	25.39%

Table 323: Observational and coordinate summary for GW200311_115853. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.022 [26.483, 40.093]	28.174 [21.548, 34.599]
Physical compact-source mass M_c ($M_\odot$)	24.076 [17.312, 33.893]	17.306 [12.807, 24.198]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.066 [1.128, 16.990]	11.005 [2.330, 16.225]
Geometric share M_{geom}/M_∞ (%)	24.95 [3.61, 48.48]	39.75 [9.49, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.531 [0.090, 0.928]	0.273 [0.024, 0.815]
Polarization factor A_{pol}	1.332 [1.037, 1.941]	1.660 [1.105, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.943 [0.338, 0.999]	0.751 [0.094, 0.994]
GR/CPTG matching radius r_m (km)	48.761 [39.105, 59.202]	41.603 [31.818, 51.089]
Structural Mode N_{BH}	1.6270 [1.4673, 1.8949]	1.7907 [1.5023, 1.9110]
Organization-shell radius r_N (km)	25.695 [20.544, 31.154]	21.916 [16.746, 26.889]
Inner trapping-surface radius r_- (km)	33.278 [26.935, 44.817]	26.258 [20.359, 32.853]
Structural- N lapse $\bar{\alpha}_N$	0.7126 [0.4967, 0.8375]	0.5831 [0.4840, 0.8095]
Structural- N clock $T_{24,N}$ (h)	17.101 [11.920, 20.099]	13.995 [11.616, 19.427]
Structural- N axis ω (rad s^{-1})	2027.98 [821.30, 3137.38]	2621.30 [449.49, 3969.38]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2519.23 [1207.54, 3497.01]	2983.43 [449.64, 4371.09]
Structural- N equatorial ω (rad s^{-1})	3018.31 [1382.08, 3870.98]	3376.18 [449.71, 4788.11]
Structural- N period P_ω (ms)	2.4941 [1.7967, 5.2033]	2.1060 [1.4374, 13.9739]
Trapping-surface limit Ω_- (rad s^{-1})	3963.35 [1518.94, 5036.00]	4171.95 [489.94, 6118.72]
Trapping-surface period P_- (ms)	1.5853 [1.2477, 4.1366]	1.5061 [1.0269, 12.8243]

Table 324: CPTG compact-state, structural, clock, and geometric-rotation results for GW200311_115853. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1256	7.4930	3870.84	4030.43
500 m inward	4.3180	5.3083	3920.24	4104.71
100 m inward	1.9252	2.3772	3955.19	4158.03
10 m inward	0.6084	0.7519	3962.57	4171.02
1 m inward	0.1924	0.2378	3963.26	4171.87
10 cm inward	0.0608	0.0752	3963.34	4171.95
1 cm inward	0.0192	0.0238	3963.35	4171.95
surface limit	0.0000	0.0000	3963.35	4171.95

Table 325: Near-inner-boundary profile for GW200311_115853. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.076 M_\odot$ and $17.306 M_\odot$, with completed geometric mass-equivalent shares of 24.95% and 39.75%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.695 < 33.278 < 48.761$ km, and component 2 gives $21.916 < 26.258 < 41.603$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6270$ and 1.7907 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.101 and 13.995 h. The determinant-averaged frame-dragging rates at those shells are 2519.23 and $2983.43 \text{ rad s}^{-1}$, approaching surface-limit values of 3963.35 and $4171.95 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.126/7.493$ h at 1 km inward to $0.0192/0.0238$ h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.27 GW241129_021832

GW241129_021832 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 43.539 M_\odot$ and $m_2 = 33.264 M_\odot$, while the luminosity-distance median is 2552.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,142
Observed luminosity distance	2552.3 Mpc (median)
Detector-frame masses	$m_1 = 43.539 M_\odot$, $m_2 = 33.264 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.764$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4468$; $z_\pi = 0.4809$ (medians)
Post-handoff reorder fraction	21.59%

Table 326: Observational and coordinate summary for GW241129_021832. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	29.114 [22.354, 36.329]	22.977 [17.710, 29.907]
Physical compact-source mass M_c ($M_\odot$)	22.480 [15.789, 31.818]	15.060 [10.316, 21.514]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.495 [0.624, 14.295]	8.176 [1.329, 13.762]
Geometric share M_{geom}/M_∞ (%)	19.29 [2.26, 46.03]	36.25 [6.27, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.633 [0.152, 0.955]	0.334 [0.031, 0.877]
Polarization factor A_{pol}	1.239 [1.023, 1.853]	1.569 [1.067, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.971 [0.522, 1.000]	0.823 [0.121, 0.998]
GR/CPTG matching radius r_m (km)	42.991 [33.009, 53.645]	33.928 [26.150, 44.162]
Structural Mode N_{BH}	1.5754 [1.4600, 1.8662]	1.7491 [1.4823, 1.9101]
Organization-shell radius r_N (km)	22.624 [17.324, 28.245]	17.862 [13.750, 23.272]
Inner trapping-surface radius r_- (km)	30.619 [23.419, 42.248]	22.195 [16.425, 28.975]
Structural- N lapse $\bar{\alpha}_N$	0.7523 [0.5203, 0.8433]	0.6169 [0.4846, 0.8254]
Structural- N clock $T_{24,N}$ (h)	18.056 [12.486, 20.239]	14.806 [11.631, 19.809]
Structural- N axis ω (rad s^{-1})	2018.93 [903.99, 3566.10]	3079.77 [701.77, 4898.82]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2671.64 [1447.95, 4012.27]	3578.45 [702.09, 5414.07]
Structural- N equatorial ω (rad s^{-1})	3347.70 [2153.76, 4498.25]	4126.89 [702.24, 5868.70]
Structural- N period P_ω (ms)	2.3518 [1.5660, 4.3394]	1.7558 [1.1605, 8.9493]
Trapping-surface limit Ω_- (rad s^{-1})	4471.07 [2686.01, 5920.71]	5263.19 [765.20, 7458.31]
Trapping-surface period P_- (ms)	1.4053 [1.0612, 2.3392]	1.1938 [0.8424, 8.2112]

Table 327: CPTG compact-state, structural, clock, and geometric-rotation results for GW241129_021832. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0848	7.9986	4359.92	5068.97
500 m inward	4.2786	5.6618	4419.56	5170.04
100 m inward	1.9030	2.5341	4462.97	5243.78
10 m inward	0.6011	0.8013	4470.33	5260.66
1 m inward	0.1900	0.2534	4470.98	5262.95
10 cm inward	0.0601	0.0801	4471.06	5263.17
1 cm inward	0.0190	0.0253	4471.07	5263.19
surface limit	0.0000	0.0000	4471.07	5263.19

Table 328: Near-inner-boundary profile for GW241129_021832. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.480 M_\odot$ and $15.060 M_\odot$, with completed geometric mass-equivalent shares of 19.29% and 36.25%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $22.624 < 30.619 < 42.991 \text{ km}$, and component 2 gives $17.862 < 22.195 < 33.928 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5754$ and 1.7491 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.056 and 14.806 h. The determinant-averaged frame-dragging rates at those shells are 2671.64 and $3578.45 \text{ rad s}^{-1}$, approaching surface-limit values of 4471.07 and $5263.19 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.085/7.999 \text{ h}$ at 1 km inward to $0.0190/0.0253 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.28 GW230704_021211

GW230704_021211 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 47.941 M_\odot$ and $m_2 = 29.501 M_\odot$, while the luminosity-distance median is 2716.2 Mpc . 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	41,796
Observed luminosity distance	2716.2 Mpc (median)
Detector-frame masses	$m_1 = 47.941 M_\odot$, $m_2 = 29.501 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.615$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4708$; $z_\pi = 0.5068$ (medians)
Post-handoff reorder fraction	13.41%

Table 329: Observational and coordinate summary for GW230704_021211. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	31.539 [21.332, 43.652]	19.683 [13.676, 29.211]
Physical compact-source mass M_c ($M_\odot$)	20.396 [14.562, 30.610]	13.162 [8.353, 19.329]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.973 [1.441, 19.432]	6.924 [0.961, 13.429]
Geometric share M_{geom}/M_∞ (%)	34.92 [5.74, 49.75]	36.43 [5.60, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.357 [0.036, 0.887]	0.331 [0.029, 0.890]
Polarization factor A_{pol}	1.537 [1.061, 1.990]	1.573 [1.059, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.844 [0.142, 0.998]	0.820 [0.115, 0.998]
GR/CPTG matching radius r_m (km)	46.572 [31.499, 64.457]	29.064 [20.194, 43.133]
Structural Mode N_{BH}	1.7337 [1.4793, 1.9094]	1.7512 [1.4784, 1.9103]
Organization-shell radius r_N (km)	24.538 [16.549, 33.951]	15.314 [10.631, 22.706]
Inner trapping-surface radius r_- (km)	30.110 [21.945, 42.218]	19.212 [13.073, 27.304]
Structural- N lapse $\bar{\alpha}_N$	0.6293 [0.4852, 0.8278]	0.6153 [0.4845, 0.8285]
Structural- N clock $T_{24,N}$ (h)	15.104 [11.646, 19.868]	14.766 [11.628, 19.883]
Structural- N axis ω (rad s^{-1})	2246.82 [568.50, 3672.38]	3413.69 [735.67, 6005.43]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2639.92 [571.95, 4136.46]	4021.83 [736.10, 6665.36]
Structural- N equatorial ω (rad s^{-1})	3066.44 [572.21, 4697.71]	4717.13 [736.27, 7365.90]
Structural- N period P_ω (ms)	2.3801 [1.5190, 10.9855]	1.5623 [0.9427, 8.5358]
Trapping-surface limit Ω_- (rad s^{-1})	3857.21 [623.89, 6202.13]	5999.61 [802.30, 9467.89]
Trapping-surface period P_- (ms)	1.6289 [1.0131, 10.0710]	1.0473 [0.6636, 7.8315]

Table 330: CPTG compact-state, structural, clock, and geometric-rotation results for GW230704_021211. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.7818	8.4668	3749.91	5733.99
500 m inward	4.7973	5.9925	3805.52	5875.19
100 m inward	2.1449	2.6800	3847.02	5976.22
10 m inward	0.6782	0.8475	3856.24	5997.05
1 m inward	0.2145	0.2680	3857.15	5999.33
10 cm inward	0.0678	0.0848	3857.20	5999.59
1 cm inward	0.0214	0.0268	3857.21	5999.61
surface limit	0.0000	0.0000	3857.21	5999.61

Table 331: Near-inner-boundary profile for GW230704_021211. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $20.396 M_\odot$ and $13.162 M_\odot$, with completed geometric mass-equivalent shares of 34.92% and 36.43%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.538 < 30.110 < 46.572 \text{ km}$, and component 2 gives $15.314 < 19.212 < 29.064 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7337$ and 1.7512 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.104 and 14.766 h. The determinant-averaged frame-dragging rates at those shells are 2639.92 and $4021.83 \text{ rad s}^{-1}$, approaching surface-limit values of 3857.21 and $5999.61 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.782/8.467 \text{ h}$ at 1 km inward to $0.0214/0.0268 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.29 GW240630_101703

GW240630_101703 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 43.921 M_\odot$ and $m_2 = 33.542 M_\odot$, while the luminosity-distance median is 3195.9 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,167
Observed luminosity distance	3195.9 Mpc (median)
Detector-frame masses	$m_1 = 43.921 M_\odot$, $m_2 = 33.542 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.764$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5393$; $z_\pi = 0.5810$ (medians)
Post-handoff reorder fraction	19.11%

Table 332: Observational and coordinate summary for GW240630_101703. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.384 [20.972, 35.358]	21.398 [15.891, 28.105]
Physical compact-source mass M_c ($M_\odot$)	22.576 [14.891, 32.062]	14.339 [9.589, 20.678]
Geometric mass-equivalent M_{geom} ($M_\odot$)	3.867 [0.451, 12.107]	7.173 [0.988, 12.593]
Geometric share M_{geom}/M_∞ (%)	14.36 [1.69, 43.70]	33.92 [5.11, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.723 [0.200, 0.966]	0.375 [0.034, 0.899]
Polarization factor A_{pol}	1.168 [1.017, 1.776]	1.513 [1.054, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.986 [0.631, 1.000]	0.858 [0.135, 0.999]
GR/CPTG matching radius r_m (km)	40.436 [30.967, 52.211]	31.597 [23.466, 41.501]
Structural Mode N_{BH}	1.5362 [1.4571, 1.8383]	1.7221 [1.4756, 1.9096]
Organization-shell radius r_N (km)	21.243 [16.261, 27.413]	16.637 [12.346, 21.854]
Inner trapping-surface radius r_- (km)	30.220 [21.862, 42.649]	20.936 [15.113, 27.864]
Structural- N lapse $\bar{\alpha}_N$	0.7828 [0.5436, 0.8457]	0.6385 [0.4850, 0.8307]
Structural- N clock $T_{24,N}$ (h)	18.787 [13.047, 20.297]	15.323 [11.640, 19.938]
Structural- N axis ω (rad s^{-1})	1787.63 [899.32, 3708.33]	3233.17 [870.30, 5241.60]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2540.07 [1458.59, 4212.80]	3793.32 [870.85, 5792.41]
Structural- N equatorial ω (rad s^{-1})	3419.90 [2269.50, 4752.62]	4426.83 [871.13, 6396.50]
Structural- N period P_ω (ms)	2.4736 [1.4915, 4.3077]	1.6564 [1.0847, 7.2150]
Trapping-surface limit Ω_- (rad s^{-1})	4672.80 [3114.12, 6281.05]	5699.52 [949.51, 8194.32]
Trapping-surface period P_- (ms)	1.3446 [1.0003, 2.0176]	1.1024 [0.7668, 6.6173]

Table 333: CPTG compact-state, structural, clock, and geometric-rotation results for GW240630_101703. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.7638	8.1509	4570.80	5480.64
500 m inward	4.0388	5.7684	4627.13	5597.61
100 m inward	1.7912	2.5784	4663.90	5681.73
10 m inward	0.5652	0.8153	4672.25	5697.90
1 m inward	0.1787	0.2578	4672.72	5699.29
10 cm inward	0.0565	0.0815	4672.80	5699.50
1 cm inward	0.0179	0.0258	4672.80	5699.52
surface limit	0.0000	0.0000	4672.80	5699.52

Table 334: Near-inner-boundary profile for GW240630_101703. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.576 M_\odot$ and $14.339 M_\odot$, with completed geometric mass-equivalent shares of 14.36% and 33.92%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.243 < 30.220 < 40.436$ km, and component 2 gives $16.637 < 20.936 < 31.597$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5362$ and 1.7221 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.787 and 15.323 h. The determinant-averaged frame-dragging rates at those shells are 2540.07 and $3793.32 \text{ rad s}^{-1}$, approaching surface-limit values of 4672.80 and $5699.52 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 5.764/8.151 h at 1 km inward to 0.0179/0.0258 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.30 GW240615_113620

GW240615_113620 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 43.796 M_\odot$ and $m_2 = 34.148 M_\odot$, while the luminosity-distance median is 1564.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,327
Observed luminosity distance	1564.8 Mpc (median)
Detector-frame masses	$m_1 = 43.796 M_\odot$, $m_2 = 34.148 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.780$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2940$; $z_\pi = 0.3162$ (medians)
Post-handoff reorder fraction	19.70%

Table 335: Observational and coordinate summary for GW240615_113620. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.047 [26.065, 38.741]	26.443 [21.279, 33.266]
Physical compact-source mass M_c ($M_\odot$)	23.749 [16.980, 33.672]	17.144 [12.331, 24.894]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.269 [1.102, 16.738]	9.532 [1.444, 15.507]
Geometric share M_{geom}/M_∞ (%)	25.40 [3.53, 48.71]	36.79 [5.83, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.523 [0.082, 0.930]	0.325 [0.028, 0.885]
Polarization factor A_{pol}	1.340 [1.037, 1.950]	1.582 [1.062, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.940 [0.313, 0.999]	0.813 [0.111, 0.998]
GR/CPTG matching radius r_m (km)	48.798 [38.488, 57.205]	39.046 [31.422, 49.122]
Structural Mode N_{BH}	1.6313 [1.4668, 1.8975]	1.7555 [1.4798, 1.9105]
Organization-shell radius r_N (km)	25.703 [20.271, 30.127]	20.558 [16.549, 25.879]
Inner trapping-surface radius r_- (km)	32.982 [26.297, 44.547]	25.448 [19.714, 33.157]
Structural- N lapse $\bar{\alpha}_N$	0.7092 [0.4946, 0.8378]	0.6118 [0.4844, 0.8274]
Structural- N clock $T_{24,N}$ (h)	17.021 [11.870, 20.108]	14.684 [11.625, 19.858]
Structural- N axis ω (rad s^{-1})	2039.77 [821.30, 3214.23]	2595.29 [571.20, 4129.07]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2531.72 [1210.29, 3566.39]	3031.19 [571.46, 4529.92]
Structural- N equatorial ω (rad s^{-1})	3003.45 [1311.09, 3940.91]	3523.94 [571.59, 4913.94]
Structural- N period P_ω (ms)	2.4818 [1.7618, 5.1915]	2.0728 [1.3870, 10.9950]
Trapping-surface limit Ω_- (rad s^{-1})	3926.56 [1437.02, 5125.65]	4537.55 [622.85, 6277.61]
Trapping-surface period P_- (ms)	1.6002 [1.2258, 4.3724]	1.3847 [1.0009, 10.0878]

Table 336: CPTG compact-state, structural, clock, and geometric-rotation results for GW240615_113620. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1863	7.5178	3833.93	4397.44
500 m inward	4.3626	5.3233	3881.07	4474.37
100 m inward	1.9463	2.3822	3917.32	4524.31
10 m inward	0.6152	0.7533	3925.83	4536.67
1 m inward	0.1945	0.2382	3926.52	4537.60
10 cm inward	0.0615	0.0753	3926.55	4537.55
1 cm inward	0.0195	0.0238	3926.55	4537.55
surface limit	0.0000	0.0000	3926.56	4537.55

Table 337: Near-inner-boundary profile for GW240615_113620. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.749 M_\odot$ and $17.144 M_\odot$, with completed geometric mass-equivalent shares of 25.40% and 36.79%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.703 < 32.982 < 48.798 \text{ km}$, and component 2 gives $20.558 < 25.448 < 39.046 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6313$ and 1.7555 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.021 and 14.684 h. The determinant-averaged frame-dragging rates at those shells are 2531.72 and $3031.19 \text{ rad s}^{-1}$, approaching surface-limit values of 3926.56 and $4537.55 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.186/7.518 h at 1 km inward to 0.0195/0.0238 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.31 GW190828_063405

GW190828_063405 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 43.266 M_\odot$ and $m_2 = 35.398 M_\odot$, while the luminosity-distance median is 2074.3 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	42,966
Observed luminosity distance	2074.3 Mpc (median)
Detector-frame masses	$m_1 = 43.266 M_\odot$, $m_2 = 35.398 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.818$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3754$; $z_\pi = 0.4032$ (medians)
Post-handoff reorder fraction	24.35%

Table 338: Observational and coordinate summary for GW190828_063405. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	30.676 [24.424, 36.618]	26.099 [20.130, 32.420]
Physical compact-source mass M_c ($M_\odot$)	22.905 [16.468, 31.483]	16.115 [11.659, 22.980]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.900 [0.903, 15.153]	10.093 [2.478, 15.052]
Geometric share M_{geom}/M_∞ (%)	23.09 [3.11, 46.76]	39.60 [10.51, 49.88]
Exterior spin magnitude $ \hat{a}_\infty $	0.564 [0.135, 0.938]	0.275 [0.024, 0.795]
Polarization factor A_{pol}	1.300 [1.032, 1.878]	1.656 [1.117, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.954 [0.478, 0.999]	0.755 [0.097, 0.993]
GR/CPTG matching radius r_m (km)	45.297 [36.065, 54.072]	38.538 [29.725, 47.872]
Structural Mode N_{BH}	1.6093 [1.4646, 1.8748]	1.7889 [1.5090, 1.9109]
Organization-shell radius r_N (km)	23.864 [18.954, 28.500]	20.299 [15.666, 25.197]
Inner trapping-surface radius r_- (km)	31.437 [24.845, 41.784]	24.374 [18.651, 31.117]
Structural- N lapse $\bar{\alpha}_N$	0.7262 [0.5131, 0.8397]	0.5846 [0.4841, 0.8042]
Structural- N clock $T_{24,N}$ (h)	17.428 [12.315, 20.152]	14.030 [11.617, 19.300]
Structural- N axis ω (rad s^{-1})	2164.36 [915.39, 3398.24]	2856.28 [504.25, 4327.63]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2723.70 [1430.75, 3809.54]	3239.53 [504.43, 4749.58]
Structural- N equatorial ω (rad s^{-1})	3251.42 [2053.49, 4226.82]	3628.15 [504.51, 5152.76]
Structural- N period P_ω (ms)	2.3069 [1.6493, 4.3915]	1.9395 [1.3229, 12.4561]
Trapping-surface limit Ω_- (rad s^{-1})	4282.82 [2366.82, 5510.92]	4481.00 [549.66, 6550.30]
Trapping-surface period P_- (ms)	1.4671 [1.1401, 2.6547]	1.4022 [0.9592, 11.4311]

Table 339: CPTG compact-state, structural, clock, and geometric-rotation results for GW190828_063405. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.2285	7.7694	4174.75	4316.13
500 m inward	4.3876	5.5052	4231.24	4401.62
100 m inward	1.9549	2.4654	4273.06	4466.21
10 m inward	0.6177	0.7798	4282.18	4479.79
1 m inward	0.1953	0.2466	4282.73	4480.83
10 cm inward	0.0618	0.0780	4282.81	4480.98
1 cm inward	0.0195	0.0247	4282.82	4481.00
surface limit	0.0000	0.0000	4282.82	4481.00

Table 340: Near-inner-boundary profile for GW190828_063405. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.905 M_\odot$ and $16.115 M_\odot$, with completed geometric mass-equivalent shares of 23.09% and 39.60%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $23.864 < 31.437 < 45.297 \text{ km}$, and component 2 gives $20.299 < 24.374 < 38.538 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6093$ and 1.7889 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.428 and 14.030 h. The determinant-averaged frame-dragging rates at those shells are 2723.70 and $3239.53 \text{ rad s}^{-1}$, approaching surface-limit values of 4282.82 and $4481.00 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.229/7.769 \text{ h}$ at 1 km inward to $0.0195/0.0247 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.32 GW250101_011205

GW250101_011205 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 46.040 M_\odot$ and $m_2 = 32.943 M_\odot$, while the luminosity-distance median is 3226.0 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	19,073
Observed luminosity distance	3226.0 Mpc (median)
Detector-frame masses	$m_1 = 46.040 M_\odot$, $m_2 = 32.943 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.716$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5435$; $z_\pi = 0.5856$ (medians)
Post-handoff reorder fraction	18.38%

Table 341: Observational and coordinate summary for GW250101_011205. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	28.629 [21.037, 37.846]	20.888 [13.514, 29.263]
Physical compact-source mass M_c ($M_\odot$)	21.298 [14.593, 31.425]	13.739 [8.539, 20.238]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.254 [0.675, 15.511]	7.151 [0.992, 13.365]
Geometric share M_{geom}/M_∞ (%)	22.10 [2.56, 48.51]	35.69 [5.67, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.582 [0.089, 0.949]	0.344 [0.031, 0.888]
Polarization factor A_{pol}	1.284 [1.026, 1.942]	1.555 [1.060, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.959 [0.335, 1.000]	0.832 [0.122, 0.998]
GR/CPTG matching radius r_m (km)	42.274 [31.064, 55.884]	30.844 [19.955, 43.210]
Structural Mode N_{BH}	1.6002 [1.4616, 1.8952]	1.7426 [1.4789, 1.9101]
Organization-shell radius r_N (km)	22.229 [16.307, 29.452]	16.242 [10.491, 22.736]
Inner trapping-surface radius r_- (km)	29.611 [21.823, 41.773]	20.221 [13.141, 27.819]
Structural- N lapse $\bar{\alpha}_N$	0.7332 [0.4964, 0.8420]	0.6222 [0.4847, 0.8281]
Structural- N clock $T_{24,N}$ (h)	17.597 [11.914, 20.208]	14.932 [11.632, 19.875]
Structural- N axis ω (rad s^{-1})	2118.30 [890.13, 3715.23]	3297.94 [755.07, 5827.93]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2720.78 [1318.24, 4201.95]	3877.53 [755.47, 6513.76]
Structural- N equatorial ω (rad s^{-1})	3361.62 [1542.76, 4757.98]	4515.19 [755.66, 7276.31]
Structural- N period P_ω (ms)	2.3093 [1.4953, 4.7664]	1.6204 [0.9646, 8.3170]
Trapping-surface limit Ω_- (rad s^{-1})	4464.11 [1695.48, 6291.96]	5755.12 [823.51, 9408.84]
Trapping-surface period P_- (ms)	1.4075 [0.9986, 3.7059]	1.0918 [0.6678, 7.6297]

Table 342: CPTG compact-state, structural, clock, and geometric-rotation results for GW250101_011205. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.3587	8.2871	4353.44	5510.04
500 m inward	4.4754	5.8649	4413.95	5637.42
100 m inward	1.9927	2.6240	4453.96	5731.66
10 m inward	0.6296	0.8297	4463.34	5753.06
1 m inward	0.1991	0.2624	4464.04	5755.06
10 cm inward	0.0630	0.0830	4464.10	5755.14
1 cm inward	0.0199	0.0262	4464.10	5755.12
surface limit	0.0000	0.0000	4464.11	5755.12

Table 343: Near-inner-boundary profile for GW250101_011205. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $21.298 M_\odot$ and $13.739 M_\odot$, with completed geometric mass-equivalent shares of 22.10% and 35.69%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $22.229 < 29.611 < 42.274 \text{ km}$, and component 2 gives $16.242 < 20.221 < 30.844 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6002$ and 1.7426 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.597 and 14.932 h. The determinant-averaged frame-dragging rates at those shells are 2720.78 and $3877.53 \text{ rad s}^{-1}$, approaching surface-limit values of 4464.11 and $5755.12 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.359/8.287 h at 1 km inward to 0.0199/0.0262 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.33 GW240615_160735

GW240615_160735 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 46.133 M_\odot$ and $m_2 = 32.927 M_\odot$, while the luminosity-distance median is 3971.2 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,859
Observed luminosity distance	3971.2 Mpc (median)
Detector-frame masses	$m_1 = 46.133 M_\odot$, $m_2 = 32.927 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.714$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6454$; $z_\pi = 0.6963$ (medians)
Post-handoff reorder fraction	19.38%

Table 344: Observational and coordinate summary for GW240615_160735. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.061 [19.962, 35.401]	19.811 [13.947, 28.126]
Physical compact-source mass M_c ($M_\odot$)	21.842 [15.255, 30.282]	13.714 [8.498, 20.154]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.132 [0.464, 13.131]	6.207 [0.802, 12.582]
Geometric share M_{geom}/M_∞ (%)	15.69 [1.84, 43.58]	32.34 [4.51, 49.65]
Exterior spin magnitude $ \hat{a}_\infty $	0.698 [0.202, 0.963]	0.402 [0.042, 0.911]
Polarization factor A_{pol}	1.186 [1.019, 1.772]	1.478 [1.047, 1.986]
Compact-source spin parameter $\hat{a}_c$	0.983 [0.635, 1.000]	0.878 [0.165, 0.999]
GR/CPTG matching radius r_m (km)	39.958 [29.477, 52.273]	29.253 [20.594, 41.531]
Structural Mode N_{BH}	1.5463 [1.4579, 1.8369]	1.7043 [1.4723, 1.9083]
Organization-shell radius r_N (km)	20.994 [15.436, 27.533]	15.401 [10.839, 21.865]
Inner trapping-surface radius r_- (km)	29.539 [21.931, 40.384]	19.778 [13.282, 27.545]
Structural- N lapse $\bar{\alpha}_N$	0.7749 [0.5448, 0.8451]	0.6525 [0.4860, 0.8334]
Structural- N clock $T_{24,N}$ (h)	18.597 [13.076, 20.281]	15.661 [11.665, 20.003]
Structural- N axis ω (rad s^{-1})	1912.36 [965.88, 3637.21]	3365.24 [1061.05, 5937.73]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2662.41 [1548.78, 4177.86]	4018.08 [1072.73, 6597.27]
Structural- N equatorial ω (rad s^{-1})	3499.47 [2364.08, 4829.23]	4768.21 [1073.23, 7251.65]
Structural- N period P_ω (ms)	2.3600 [1.5039, 4.0569]	1.5637 [0.9524, 5.8572]
Trapping-surface limit Ω_- (rad s^{-1})	4762.24 [3170.71, 6502.41]	6164.28 [1170.62, 9297.09]
Trapping-surface period P_- (ms)	1.3194 [0.9663, 1.9816]	1.0193 [0.6758, 5.3674]

Table 345: CPTG compact-state, structural, clock, and geometric-rotation results for GW240615_160735. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.9429	8.2570	4654.20	5906.35
500 m inward	4.1631	5.8386	4713.20	6043.09
100 m inward	1.8465	2.6092	4753.50	6142.37
10 m inward	0.5829	0.8249	4761.39	6161.92
1 m inward	0.1843	0.2608	4762.16	6164.13
10 cm inward	0.0583	0.0825	4762.23	6164.27
1 cm inward	0.0184	0.0261	4762.24	6164.28
surface limit	0.0000	0.0000	4762.24	6164.28

Table 346: Near-inner-boundary profile for GW240615_160735. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $21.842 M_\odot$ and $13.714 M_\odot$, with completed geometric mass-equivalent shares of 15.69% and 32.34%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $20.994 < 29.539 < 39.958 \text{ km}$, and component 2 gives $15.401 < 19.778 < 29.253 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5463$ and 1.7043 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.597 and 15.661 h. The determinant-averaged frame-dragging rates at those shells are 2662.41 and $4018.08 \text{ rad s}^{-1}$, approaching surface-limit values of 4762.24 and $6164.28 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.943/8.257 \text{ h}$ at 1 km inward to $0.0184/0.0261 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.34 GW200112_155838

GW200112_155838 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 44.008 M_\odot$ and $m_2 = 35.178 M_\odot$, while the luminosity-distance median is 1252.1 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	4,323
Observed luminosity distance	1252.1 Mpc (median)
Detector-frame masses	$m_1 = 44.008 M_\odot$, $m_2 = 35.178 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.799$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2422$; $z_\pi = 0.2601$ (medians)
Post-handoff reorder fraction	23.99%

Table 347: Observational and coordinate summary for GW200112_155838. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	34.176 [27.199, 41.742]	28.706 [22.103, 36.306]
Physical compact-source mass M_c ($M_\odot$)	23.969 [17.666, 33.460]	17.626 [12.775, 24.813]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.600 [1.702, 18.477]	11.275 [2.831, 16.905]
Geometric share M_{geom}/M_∞ (%)	28.29 [5.35, 49.18]	40.28 [11.02, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.473 [0.065, 0.894]	0.263 [0.026, 0.786]
Polarization factor A_{pol}	1.394 [1.057, 1.968]	1.674 [1.124, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.919 [0.252, 0.998]	0.738 [0.104, 0.992]
GR/CPTG matching radius r_m (km)	50.465 [40.162, 61.637]	42.388 [32.638, 53.610]
Structural Mode N_{BH}	1.6605 [1.4770, 1.9029]	1.7970 [1.5125, 1.9107]
Organization-shell radius r_N (km)	26.639 [21.087, 32.530]	22.321 [17.204, 28.211]
Inner trapping-surface radius r_- (km)	33.826 [27.285, 44.353]	26.643 [20.409, 33.999]
Structural- N lapse $\bar{\alpha}_N$	0.6867 [0.4903, 0.8296]	0.5779 [0.4842, 0.8014]
Structural- N clock $T_{24,N}$ (h)	16.480 [11.766, 19.911]	13.871 [11.621, 19.234]
Structural- N axis ω (rad s^{-1})	2088.83 [804.57, 3119.96]	2597.31 [489.26, 3948.87]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2508.80 [963.59, 3468.35]	2912.94 [489.49, 4337.08]
Structural- N equatorial ω (rad s^{-1})	2941.89 [975.57, 3813.16]	3267.72 [489.60, 4714.61]
Structural- N period P_ω (ms)	2.5045 [1.8116, 6.5207]	2.1570 [1.4487, 12.8362]
Trapping-surface limit Ω_- (rad s^{-1})	3824.16 [1065.93, 4929.17]	4040.30 [533.51, 5931.96]
Trapping-surface period P_- (ms)	1.6430 [1.2747, 5.8945]	1.5551 [1.0592, 11.7770]

Table 348: CPTG compact-state, structural, clock, and geometric-rotation results for GW200112_155838. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.2397	7.4330	3729.61	3891.70
500 m inward	4.4064	5.2621	3778.88	3969.50
100 m inward	1.9662	2.3564	3815.22	4025.00
10 m inward	0.6216	0.7453	3823.22	4038.78
1 m inward	0.1966	0.2357	3824.06	4040.16
10 cm inward	0.0622	0.0745	3824.15	4040.29
1 cm inward	0.0197	0.0236	3824.16	4040.30
surface limit	0.0000	0.0000	3824.16	4040.30

Table 349: Near-inner-boundary profile for GW200112_155838. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.969 M_\odot$ and $17.626 M_\odot$, with completed geometric mass-equivalent shares of 28.29% and 40.28%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.639 < 33.826 < 50.465 \text{ km}$, and component 2 gives $22.321 < 26.643 < 42.388 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6605$ and 1.7970 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.480 and 13.871 h. The determinant-averaged frame-dragging rates at those shells are 2508.80 and $2912.94 \text{ rad s}^{-1}$, approaching surface-limit values of 3824.16 and $4040.30 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.240/7.433 \text{ h}$ at 1 km inward to $0.0197/0.0236 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.35 GW230811_032116

GW230811_032116 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 48.449 M_\odot$ and $m_2 = 30.738 M_\odot$, while the luminosity-distance median is 2095.3 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	49,503
Observed luminosity distance	2095.3 Mpc (median)
Detector-frame masses	$m_1 = 48.449 M_\odot$, $m_2 = 30.738 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.634$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3781$; $z_\pi = 0.4067$ (medians)
Post-handoff reorder fraction	13.01%

Table 350: Observational and coordinate summary for GW230811_032116. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	34.480 [24.016, 43.322]	22.066 [16.761, 32.199]
Physical compact-source mass M_c ($M_\odot$)	24.028 [16.594, 37.102]	14.988 [9.953, 21.570]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.797 [1.124, 18.017]	7.730 [1.104, 14.737]
Geometric share M_{geom}/M_∞ (%)	26.08 [3.60, 49.09]	35.51 [5.61, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.511 [0.069, 0.929]	0.347 [0.031, 0.889]
Polarization factor A_{pol}	1.353 [1.037, 1.964]	1.551 [1.059, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.936 [0.265, 0.999]	0.835 [0.124, 0.998]
GR/CPTG matching radius r_m (km)	50.913 [35.463, 63.971]	32.583 [24.749, 47.546]
Structural Mode N_{BH}	1.6381 [1.4672, 1.9019]	1.7404 [1.4785, 1.9100]
Organization-shell radius r_N (km)	26.831 [18.626, 33.632]	17.158 [13.014, 25.010]
Inner trapping-surface radius r_- (km)	33.927 [25.023, 49.203]	21.711 [15.744, 30.181]
Structural- N lapse $\bar{\alpha}_N$	0.7040 [0.4911, 0.8375]	0.6239 [0.4847, 0.8284]
Structural- N clock $T_{24,N}$ (h)	16.896 [11.786, 20.100]	14.974 [11.633, 19.883]
Structural- N axis ω (rad s^{-1})	1971.61 [726.26, 3206.20]	3043.22 [705.26, 5120.53]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2424.98 [1015.04, 3612.68]	3591.21 [705.64, 5643.12]
Structural- N equatorial ω (rad s^{-1})	2871.23 [1078.62, 4118.80]	4235.12 [705.79, 6156.20]
Structural- N period P_ω (ms)	2.5910 [1.7392, 6.1901]	1.7496 [1.1134, 8.9043]
Trapping-surface limit Ω_- (rad s^{-1})	3716.41 [1178.79, 5475.60]	5434.47 [769.00, 7877.68]
Trapping-surface period P_- (ms)	1.6907 [1.1475, 5.3302]	1.1562 [0.7976, 8.1706]

Table 351: CPTG compact-state, structural, clock, and geometric-rotation results for GW230811_032116. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1386	7.9531	3629.46	5228.35
500 m inward	4.3297	5.6273	3674.41	5339.11
100 m inward	1.9316	2.5155	3708.70	5416.73
10 m inward	0.6105	0.7954	3715.54	5432.83
1 m inward	0.1930	0.2515	3716.33	5434.25
10 cm inward	0.0610	0.0795	3716.41	5434.45
1 cm inward	0.0193	0.0252	3716.41	5434.47
surface limit	0.0000	0.0000	3716.41	5434.47

Table 352: Near-inner-boundary profile for GW230811_032116. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.028 M_\odot$ and $14.988 M_\odot$, with completed geometric mass-equivalent shares of 26.08% and 35.51%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.831 < 33.927 < 50.913 \text{ km}$, and component 2 gives $17.158 < 21.711 < 32.583 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6381$ and 1.7404 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.896 and 14.974 h. The determinant-averaged frame-dragging rates at those shells are 2424.98 and $3591.21 \text{ rad s}^{-1}$, approaching surface-limit values of 3716.41 and $5434.47 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.139/7.953 \text{ h}$ at 1 km inward to $0.0193/0.0252 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.36 GW230708_053705

GW230708_053705 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 44.734 M_\odot$ and $m_2 = 35.343 M_\odot$, while the luminosity-distance median is 3254.9 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	44,415
Observed luminosity distance	3254.9 Mpc (median)
Detector-frame masses	$m_1 = 44.734 M_\odot$, $m_2 = 35.343 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.790$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5476$; $z_\pi = 0.5899$ (medians)
Post-handoff reorder fraction	22.88%

Table 353: Observational and coordinate summary for GW230708_053705. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.613 [20.971, 36.423]	22.734 [16.845, 29.942]
Physical compact-source mass M_c ($M_\odot$)	21.861 [14.475, 31.407]	14.906 [10.065, 22.035]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.925 [0.543, 13.842]	7.929 [1.174, 13.520]
Geometric share M_{geom}/M_∞ (%)	18.05 [2.03, 46.73]	35.66 [5.65, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.655 [0.136, 0.960]	0.345 [0.032, 0.889]
Polarization factor A_{pol}	1.220 [1.021, 1.877]	1.554 [1.060, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.975 [0.480, 1.000]	0.832 [0.129, 0.998]
GR/CPTG matching radius r_m (km)	40.774 [30.966, 53.782]	33.570 [24.874, 44.214]
Structural Mode N_{BH}	1.5651 [1.4588, 1.8745]	1.7422 [1.4787, 1.9099]
Organization-shell radius r_N (km)	21.445 [16.261, 28.288]	17.678 [13.102, 23.309]
Inner trapping-surface radius r_- (km)	29.674 [21.604, 41.797]	21.951 [15.918, 29.737]
Structural- N lapse $\bar{\alpha}_N$	0.7603 [0.5134, 0.8443]	0.6225 [0.4849, 0.8283]
Structural- N clock $T_{24,N}$ (h)	18.247 [12.322, 20.263]	14.940 [11.636, 19.878]
Structural- N axis ω (rad s^{-1})	1987.92 [910.21, 3809.31]	3066.38 [766.82, 5012.73]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2676.04 [1434.65, 4295.26]	3567.62 [767.39, 5541.78]
Structural- N equatorial ω (rad s^{-1})	3428.96 [2088.99, 4804.72]	4126.95 [767.62, 6072.57]
Structural- N period P_ω (ms)	2.3479 [1.4628, 4.3796]	1.7612 [1.1338, 8.1877]
Trapping-surface limit Ω_- (rad s^{-1})	4631.27 [2563.91, 6326.05]	5280.08 [836.71, 7778.06]
Trapping-surface period P_- (ms)	1.3567 [0.9932, 2.4506]	1.1900 [0.8078, 7.5094]

Table 354: CPTG compact-state, structural, clock, and geometric-rotation results for GW230708_053705. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0981	8.0230	4524.44	5083.91
500 m inward	4.2826	5.6793	4581.62	5188.94
100 m inward	1.9039	2.5406	4622.23	5263.85
10 m inward	0.6012	0.8034	4630.47	5278.26
1 m inward	0.1901	0.2540	4631.22	5279.86
10 cm inward	0.0601	0.0803	4631.26	5280.06
1 cm inward	0.0190	0.0254	4631.27	5280.08
surface limit	0.0000	0.0000	4631.27	5280.08

Table 355: Near-inner-boundary profile for GW230708_053705. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $21.861 M_\odot$ and $14.906 M_\odot$, with completed geometric mass-equivalent shares of 18.05% and 35.66%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.445 < 29.674 < 40.774 \text{ km}$, and component 2 gives $17.678 < 21.951 < 33.570 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5651$ and 1.7422 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.247 and 14.940 h. The determinant-averaged frame-dragging rates at those shells are 2676.04 and $3567.62 \text{ rad s}^{-1}$, approaching surface-limit values of 4631.27 and $5280.08 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.098/8.023 h at 1 km inward to 0.0190/0.0254 h at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.37 GW240420_175625

GW240420_175625 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 47.239 M_\odot$ and $m_2 = 32.930 M_\odot$, while the luminosity-distance median is 2031.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	11,908
Observed luminosity distance	2031.1 Mpc (median)
Detector-frame masses	$m_1 = 47.239 M_\odot$, $m_2 = 32.930 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.697$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3682$; $z_\pi = 0.3960$ (medians)
Post-handoff reorder fraction	18.86%

Table 356: Observational and coordinate summary for GW240420_175625. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.201 [23.963, 43.831]	23.860 [16.394, 33.993]
Physical compact-source mass M_c ($M_\odot$)	22.949 [16.351, 34.067]	15.526 [9.984, 22.879]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.363 [1.050, 19.084]	8.590 [1.408, 15.601]
Geometric share M_{geom}/M_∞ (%)	28.57 [3.60, 49.45]	37.57 [6.80, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.468 [0.053, 0.929]	0.311 [0.030, 0.866]
Polarization factor A_{pol}	1.400 [1.037, 1.978]	1.602 [1.073, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.917 [0.207, 0.999]	0.799 [0.118, 0.997]
GR/CPTG matching radius r_m (km)	49.026 [35.384, 64.721]	35.233 [24.207, 50.195]
Structural Mode N_{BH}	1.6635 [1.4672, 1.9060]	1.7647 [1.4855, 1.9102]
Organization-shell radius r_N (km)	25.824 [18.546, 34.127]	18.553 [12.747, 26.385]
Inner trapping-surface radius r_- (km)	32.855 [24.730, 45.472]	22.970 [15.587, 31.851]
Structural- N lapse $\bar{\alpha}_N$	0.6844 [0.4878, 0.8375]	0.6044 [0.4846, 0.8228]
Structural- N clock $T_{24,N}$ (h)	16.424 [11.708, 20.100]	14.506 [11.630, 19.748]
Structural- N axis ω (rad s^{-1})	2010.93 [710.46, 3303.61]	2894.39 [631.37, 5050.07]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2448.19 [812.97, 3714.09]	3386.78 [631.68, 5616.09]
Structural- N equatorial ω (rad s^{-1})	2934.10 [816.38, 4177.79]	3903.89 [631.84, 6191.46]
Structural- N period P_ω (ms)	2.5665 [1.6917, 7.7287]	1.8552 [1.1188, 9.9467]
Trapping-surface limit Ω_- (rad s^{-1})	3819.31 [891.47, 5530.09]	4907.51 [688.55, 7948.83]
Trapping-surface period P_- (ms)	1.6451 [1.1362, 7.0481]	1.2803 [0.7905, 9.1252]

Table 357: CPTG compact-state, structural, clock, and geometric-rotation results for GW240420_175625. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.3176	7.8252	3733.23	4731.45
500 m inward	4.4624	5.5357	3778.15	4829.06
100 m inward	1.9926	2.4761	3811.19	4894.57
10 m inward	0.6298	0.7832	3818.45	4905.70
1 m inward	0.1991	0.2477	3819.22	4907.35
10 cm inward	0.0630	0.0783	3819.30	4907.49
1 cm inward	0.0199	0.0248	3819.31	4907.51
surface limit	0.0000	0.0000	3819.31	4907.51

Table 358: Near-inner-boundary profile for GW240420_175625. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.949 M_\odot$ and $15.526 M_\odot$, with completed geometric mass-equivalent shares of 28.57% and 37.57%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.824 < 32.855 < 49.026 \text{ km}$, and component 2 gives $18.553 < 22.970 < 35.233 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6635$ and 1.7647 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.424 and 14.506 h. The determinant-averaged frame-dragging rates at those shells are 2448.19 and $3386.78 \text{ rad s}^{-1}$, approaching surface-limit values of 3819.31 and $4907.51 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.318/7.825 \text{ h}$ at 1 km inward to $0.0199/0.0248 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.38 GW231005_091549

GW231005_091549 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 46.120 M_\odot$ and $m_2 = 34.882 M_\odot$, while the luminosity-distance median is 3766.5 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	40,647
Observed luminosity distance	3766.5 Mpc (median)
Detector-frame masses	$m_1 = 46.120 M_\odot$, $m_2 = 34.882 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.756$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6179$; $z_\pi = 0.6663$ (medians)
Post-handoff reorder fraction	21.50%

Table 359: Observational and coordinate summary for GW231005_091549. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	27.206 [19.833, 37.891]	21.258 [14.595, 29.428]
Physical compact-source mass M_c ($M_\odot$)	20.034 [13.663, 30.578]	13.693 [8.885, 20.581]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.301 [0.687, 15.509]	7.661 [1.140, 13.389]
Geometric share M_{geom}/M_∞ (%)	23.56 [2.75, 48.87]	37.40 [6.24, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.556 [0.077, 0.945]	0.314 [0.028, 0.877]
Polarization factor A_{pol}	1.308 [1.028, 1.956]	1.598 [1.067, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.951 [0.294, 1.000]	0.802 [0.112, 0.998]
GR/CPTG matching radius r_m (km)	40.173 [29.285, 55.951]	31.390 [21.552, 43.455]
Structural Mode N_{BH}	1.6137 [1.4626, 1.8993]	1.7627 [1.4822, 1.9104]
Organization-shell radius r_N (km)	21.144 [15.378, 29.451]	16.534 [11.344, 22.882]
Inner trapping-surface radius r_- (km)	27.996 [20.512, 41.041]	20.319 [13.883, 28.307]
Structural- N lapse $\bar{\alpha}_N$	0.7228 [0.4931, 0.8412]	0.6060 [0.4844, 0.8255]
Structural- N clock $T_{24,N}$ (h)	17.348 [11.834, 20.189]	14.544 [11.626, 19.812]
Structural- N axis ω (rad s^{-1})	2253.49 [886.64, 3993.98]	3285.99 [691.47, 5630.04]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2865.13 [1223.29, 4500.33]	3804.61 [691.89, 6251.85]
Structural- N equatorial ω (rad s^{-1})	3532.64 [1336.20, 5047.18]	4371.25 [692.09, 6906.57]
Structural- N period P_ω (ms)	2.1930 [1.3962, 5.1363]	1.6515 [1.0050, 9.0812]
Trapping-surface limit Ω_- (rad s^{-1})	4680.93 [1479.70, 6661.79]	5505.12 [754.26, 8879.36]
Trapping-surface period P_- (ms)	1.3423 [0.9432, 4.2463]	1.1413 [0.7076, 8.3302]

Table 360: CPTG compact-state, structural, clock, and geometric-rotation results for GW231005_091549. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.6032	8.3419	4560.01	5270.23
500 m inward	4.6496	5.9072	4624.24	5395.48
100 m inward	2.0714	2.6433	4670.40	5484.48
10 m inward	0.6545	0.8359	4679.80	5503.05
1 m inward	0.2069	0.2643	4680.91	5504.87
10 cm inward	0.0654	0.0836	4680.93	5505.09
1 cm inward	0.0207	0.0264	4680.93	5505.12
surface limit	0.0000	0.0000	4680.93	5505.12

Table 361: Near-inner-boundary profile for GW231005_091549. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $20.034 M_\odot$ and $13.693 M_\odot$, with completed geometric mass-equivalent shares of 23.56% and 37.40%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $21.144 < 27.996 < 40.173 \text{ km}$, and component 2 gives $16.534 < 20.319 < 31.390 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6137$ and 1.7627 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.348 and 14.544 h. The determinant-averaged frame-dragging rates at those shells are 2865.13 and $3804.61 \text{ rad s}^{-1}$, approaching surface-limit values of 4680.93 and $5505.12 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.603/8.342 \text{ h}$ at 1 km inward to $0.0207/0.0264 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.39 GW240621_195059

GW240621_195059 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 44.965 M_\odot$ and $m_2 = 36.382 M_\odot$, while the luminosity-distance median is 1208.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,813
Observed luminosity distance	1208.0 Mpc (median)
Detector-frame masses	$m_1 = 44.965 M_\odot$, $m_2 = 36.382 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.809$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2343$; $z_\pi = 0.2519$ (medians)
Post-handoff reorder fraction	20.94%

Table 362: Observational and coordinate summary for GW240621_195059. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.742 [28.438, 40.605]	29.950 [24.104, 36.752]
Physical compact-source mass M_c ($M_\odot$)	21.618 [17.642, 30.638]	17.406 [13.470, 25.161]
Geometric mass-equivalent M_{geom} ($M_\odot$)	13.947 [2.873, 19.158]	12.718 [4.018, 17.399]
Geometric share M_{geom}/M_∞ (%)	38.92 [9.01, 49.86]	43.70 [14.49, 49.93]
Exterior spin magnitude $ \hat{a}_\infty $	0.288 [0.026, 0.824]	0.200 [0.018, 0.721]
Polarization factor A_{pol}	1.637 [1.099, 1.994]	1.776 [1.169, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.771 [0.105, 0.995]	0.631 [0.072, 0.986]
GR/CPTG matching radius r_m (km)	52.777 [41.993, 59.958]	44.224 [35.592, 54.268]
Structural Mode N_{BH}	1.7808 [1.4992, 1.9107]	1.8383 [1.5372, 1.9115]
Organization-shell radius r_N (km)	27.793 [22.138, 31.596]	23.289 [18.755, 28.538]
Inner trapping-surface radius r_- (km)	32.754 [28.175, 40.819]	27.082 [21.869, 34.460]
Structural- N lapse $\bar{\alpha}_N$	0.5913 [0.4842, 0.8119]	0.5436 [0.4836, 0.7820]
Structural- N clock $T_{24,N}$ (h)	14.191 [11.622, 19.486]	13.047 [11.606, 18.768]
Structural- N axis ω (rad s^{-1})	2119.39 [400.30, 3000.45]	2380.38 [323.69, 3700.59]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2424.40 [400.46, 3332.70]	2639.78 [323.74, 4057.93]
Structural- N equatorial ω (rad s^{-1})	2734.94 [400.53, 3669.64]	2884.07 [323.77, 4380.92]
Structural- N period P_ω (ms)	2.5916 [1.8853, 15.6901]	2.3802 [1.5484, 19.4081]
Trapping-surface limit Ω_- (rad s^{-1})	3389.85 [436.40, 4761.15]	3376.10 [352.64, 5531.21]
Trapping-surface period P_- (ms)	1.8535 [1.3197, 14.3979]	1.8611 [1.1360, 17.8177]

Table 363: CPTG compact-state, structural, clock, and geometric-rotation results for GW240621_195059. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.7353	7.4783	3304.30	3254.08
500 m inward	4.7707	5.3029	3348.52	3316.70
100 m inward	2.1356	2.3756	3382.87	3364.91
10 m inward	0.6755	0.7515	3389.05	3375.05
1 m inward	0.2136	0.2377	3389.74	3375.99
10 cm inward	0.0675	0.0752	3389.84	3376.09
1 cm inward	0.0214	0.0238	3389.85	3376.10
surface limit	0.0000	0.0000	3389.85	3376.10

Table 364: Near-inner-boundary profile for GW240621_195059. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $21.618 M_\odot$ and $17.406 M_\odot$, with completed geometric mass-equivalent shares of 38.92% and 43.70%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.793 < 32.754 < 52.777 \text{ km}$, and component 2 gives $23.289 < 27.082 < 44.224 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7808$ and 1.8383 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 14.191 and 13.047 h. The determinant-averaged frame-dragging rates at those shells are 2424.40 and $2639.78 \text{ rad s}^{-1}$, approaching surface-limit values of 3389.85 and $3376.10 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.735/7.478 \text{ h}$ at 1 km inward to $0.0214/0.0238 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.40 GW240703_191355

GW240703_191355 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 46.248 M_\odot$ and $m_2 = 36.036 M_\odot$, while the luminosity-distance median is 1881.7 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,705
Observed luminosity distance	1881.7 Mpc (median)
Detector-frame masses	$m_1 = 46.248 M_\odot$, $m_2 = 36.036 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.779$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3448$; $z_\pi = 0.3708$ (medians)
Post-handoff reorder fraction	22.23%

Table 365: Observational and coordinate summary for GW240703_191355. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	32.948 [24.822, 42.196]	26.619 [19.263, 34.771]
Physical compact-source mass M_c ($M_\odot$)	23.783 [16.457, 34.813]	16.633 [11.486, 24.508]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.217 [1.017, 17.734]	10.095 [1.878, 16.020]
Geometric share M_{geom}/M_∞ (%)	25.33 [3.30, 49.03]	39.25 [8.10, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.525 [0.071, 0.935]	0.282 [0.026, 0.841]
Polarization factor A_{pol}	1.339 [1.034, 1.962]	1.646 [1.088, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.941 [0.273, 0.999]	0.763 [0.102, 0.996]
GR/CPTG matching radius r_m (km)	48.653 [36.653, 62.307]	39.307 [28.445, 51.344]
Structural Mode N_{BH}	1.6307 [1.4656, 1.9012]	1.7847 [1.4935, 1.9108]
Organization-shell radius r_N (km)	25.601 [19.263, 32.803]	20.710 [14.966, 27.037]
Inner trapping-surface radius r_- (km)	33.295 [25.191, 46.211]	25.043 [18.107, 33.448]
Structural- N lapse $\bar{\alpha}_N$	0.7097 [0.4916, 0.8388]	0.5881 [0.4842, 0.8165]
Structural- N clock $T_{24,N}$ (h)	17.033 [11.799, 20.132]	14.113 [11.620, 19.595]
Structural- N axis ω (rad s^{-1})	1975.56 [778.98, 3285.42]	2685.29 [520.97, 4368.62]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2451.91 [1072.80, 3682.56]	3077.93 [521.23, 4836.28]
Structural- N equatorial ω (rad s^{-1})	2955.02 [1123.07, 4092.30]	3486.25 [521.36, 5286.67]
Structural- N period P_ω (ms)	2.5626 [1.7062, 5.8568]	2.0414 [1.2992, 12.0546]
Trapping-surface limit Ω_- (rad s^{-1})	3882.27 [1228.18, 5363.03]	4340.04 [568.16, 6791.15]
Trapping-surface period P_- (ms)	1.6184 [1.1716, 5.1159]	1.4477 [0.9252, 11.0588]

Table 366: CPTG compact-state, structural, clock, and geometric-rotation results for GW240703_191355. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1586	7.6187	3797.55	4195.54
500 m inward	4.3420	5.3991	3842.96	4271.69
100 m inward	1.9370	2.4165	3874.41	4326.26
10 m inward	0.6121	0.7643	3881.56	4338.47
1 m inward	0.1936	0.2417	3882.23	4340.06
10 cm inward	0.0612	0.0764	3882.27	4340.04
1 cm inward	0.0194	0.0242	3882.27	4340.04
surface limit	0.0000	0.0000	3882.27	4340.04

Table 367: Near-inner-boundary profile for GW240703_191355. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.783 M_\odot$ and $16.633 M_\odot$, with completed geometric mass-equivalent shares of 25.33% and 39.25%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.601 < 33.295 < 48.653 \text{ km}$, and component 2 gives $20.710 < 25.043 < 39.307 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6307$ and 1.7847 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.033 and 14.113 h. The determinant-averaged frame-dragging rates at those shells are 2451.91 and $3077.93 \text{ rad s}^{-1}$, approaching surface-limit values of 3882.27 and $4340.04 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.159/7.619 h at 1 km inward to 0.0194/0.0242 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.41 GW230628_231200

GW230628_231200 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 45.057 M_\odot$ and $m_2 = 37.895 M_\odot$, while the luminosity-distance median is 2261.9 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	38,628
Observed luminosity distance	2261.9 Mpc (median)
Detector-frame masses	$m_1 = 45.057 M_\odot$, $m_2 = 37.895 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.841$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4035$; $z_\pi = 0.4341$ (medians)
Post-handoff reorder fraction	28.27%

Table 368: Observational and coordinate summary for GW230628_231200. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	30.950 [25.142, 37.681]	27.408 [21.468, 33.801]
Physical compact-source mass M_c ($M_\odot$)	22.856 [16.056, 32.690]	16.798 [12.513, 24.394]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.244 [0.856, 15.892]	10.810 [2.196, 15.733]
Geometric share M_{geom}/M_∞ (%)	23.82 [2.81, 48.59]	40.15 [8.80, 49.88]
Exterior spin magnitude $ \hat{a}_\infty $	0.551 [0.086, 0.944]	0.266 [0.024, 0.828]
Polarization factor A_{pol}	1.313 [1.029, 1.945]	1.671 [1.096, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.950 [0.326, 1.000]	0.742 [0.096, 0.995]
GR/CPTG matching radius r_m (km)	45.702 [37.125, 55.641]	40.472 [31.700, 49.911]
Structural Mode N_{BH}	1.6161 [1.4630, 1.8961]	1.7955 [1.4979, 1.9109]
Organization-shell radius r_N (km)	24.058 [19.513, 29.261]	21.310 [16.687, 26.275]
Inner trapping-surface radius r_- (km)	31.504 [24.998, 43.410]	25.505 [20.025, 32.864]
Structural- N lapse $\bar{\alpha}_N$	0.7210 [0.4957, 0.8409]	0.5792 [0.4840, 0.8130]
Structural- N clock $T_{24,N}$ (h)	17.303 [11.896, 20.182]	13.900 [11.617, 19.512]
Structural- N axis ω (rad s^{-1})	2059.23 [837.14, 3375.58]	2634.58 [484.79, 4032.53]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2590.44 [1247.00, 3753.16]	2992.67 [485.03, 4429.75]
Structural- N equatorial ω (rad s^{-1})	3143.27 [1429.14, 4135.22]	3387.95 [485.14, 4839.25]
Structural- N period P_ω (ms)	2.4255 [1.6741, 5.0386]	2.0995 [1.4184, 12.9543]
Trapping-surface limit Ω_- (rad s^{-1})	4148.61 [1566.15, 5366.59]	4206.65 [528.67, 6161.66]
Trapping-surface period P_- (ms)	1.5145 [1.1708, 4.0119]	1.4936 [1.0197, 11.8850]

Table 369: CPTG compact-state, structural, clock, and geometric-rotation results for GW230628_231200. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.2604	7.6110	4051.81	4063.79
500 m inward	4.4106	5.3937	4102.05	4138.60
100 m inward	1.9660	2.4155	4139.37	4194.14
10 m inward	0.6213	0.7641	4147.58	4205.43
1 m inward	0.1964	0.2416	4148.50	4206.54
10 cm inward	0.0621	0.0764	4148.60	4206.64
1 cm inward	0.0196	0.0242	4148.61	4206.64
surface limit	0.0000	0.0000	4148.61	4206.65

Table 370: Near-inner-boundary profile for GW230628_231200. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.856 M_\odot$ and $16.798 M_\odot$, with completed geometric mass-equivalent shares of 23.82% and 40.15%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.058 < 31.504 < 45.702 \text{ km}$, and component 2 gives $21.310 < 25.505 < 40.472 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6161$ and 1.7955 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.303 and 13.900 h. The determinant-averaged frame-dragging rates at those shells are 2590.44 and $2992.67 \text{ rad s}^{-1}$, approaching surface-limit values of 4148.61 and $4206.65 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.260/7.611 \text{ h}$ at 1 km inward to $0.0196/0.0242 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.42 GW190527_092055

GW190527_092055 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 51.054 M_\odot$ and $m_2 = 32.364 M_\odot$, while the luminosity-distance median is 2516.2 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	13,819
Observed luminosity distance	2516.2 Mpc (median)
Detector-frame masses	$m_1 = 51.054 M_\odot$, $m_2 = 32.364 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.634$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4421$; $z_\pi = 0.4751$ (medians)
Post-handoff reorder fraction	12.31%

Table 371: Observational and coordinate summary for GW190527_092055. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	34.470 [24.268, 52.992]	21.861 [13.129, 33.846]
Physical compact-source mass M_c ($M_\odot$)	23.489 [16.360, 42.065]	14.201 [8.300, 22.664]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.053 [1.245, 19.952]	7.717 [1.202, 14.955]
Geometric share M_{geom}/M_∞ (%)	29.39 [3.68, 49.46]	36.71 [6.83, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.453 [0.053, 0.927]	0.326 [0.029, 0.866]
Polarization factor A_{pol}	1.416 [1.038, 1.979]	1.580 [1.073, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.909 [0.206, 0.999]	0.815 [0.115, 0.997]
GR/CPTG matching radius r_m (km)	50.899 [35.834, 78.250]	32.281 [19.387, 49.978]
Structural Mode N_{BH}	1.6721 [1.4677, 1.9061]	1.7545 [1.4857, 1.9103]
Organization-shell radius r_N (km)	26.806 [18.863, 41.219]	16.998 [10.196, 26.272]
Inner trapping-surface radius r_- (km)	33.632 [24.826, 56.551]	21.020 [12.680, 32.017]
Structural- N lapse $\bar{\alpha}_N$	0.6777 [0.4878, 0.8372]	0.6126 [0.4845, 0.8227]
Structural- N clock $T_{24,N}$ (h)	16.264 [11.706, 20.092]	14.703 [11.627, 19.745]
Structural- N axis ω (rad s^{-1})	1967.89 [538.17, 3276.90]	3167.88 [653.63, 5953.03]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2383.37 [685.53, 3687.64]	3682.02 [655.13, 6641.38]
Structural- N equatorial ω (rad s^{-1})	2812.37 [767.17, 4147.07]	4238.14 [655.27, 7418.74]
Structural- N period P_ω (ms)	2.6363 [1.7039, 9.1655]	1.7064 [0.9461, 9.5907]
Trapping-surface limit Ω_- (rad s^{-1})	3621.61 [849.44, 5458.97]	5297.40 [714.25, 9469.94]
Trapping-surface period P_- (ms)	1.7349 [1.1510, 7.3968]	1.1861 [0.6635, 8.7969]

Table 372: CPTG compact-state, structural, clock, and geometric-rotation results for GW190527_092055. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.2734	8.1653	3539.46	5070.38
500 m inward	4.4307	5.7766	3582.42	5187.16
100 m inward	1.9792	2.5811	3614.75	5276.61
10 m inward	0.6257	0.8160	3621.14	5295.46
1 m inward	0.1979	0.2580	3621.57	5297.16
10 cm inward	0.0626	0.0816	3621.61	5297.37
1 cm inward	0.0198	0.0258	3621.61	5297.39
surface limit	0.0000	0.0000	3621.61	5297.40

Table 373: Near-inner-boundary profile for GW190527_092055. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.489 M_\odot$ and $14.201 M_\odot$, with completed geometric mass-equivalent shares of 29.39% and 36.71%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.806 < 33.632 < 50.899 \text{ km}$, and component 2 gives $16.998 < 21.020 < 32.281 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6721$ and 1.7545 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.264 and 14.703 h. The determinant-averaged frame-dragging rates at those shells are 2383.37 and $3682.02 \text{ rad s}^{-1}$, approaching surface-limit values of 3621.61 and $5297.40 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.273/8.165 \text{ h}$ at 1 km inward to $0.0198/0.0258 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.43 GW240920_124024

GW240920_124024 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 44.807 M_\odot$ and $m_2 = 38.659 M_\odot$, while the luminosity-distance median is 1090.7 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,926
Observed luminosity distance	1090.7 Mpc (median)
Detector-frame masses	$m_1 = 44.807 M_\odot$, $m_2 = 38.659 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.863$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2140$; $z_\pi = 0.2301$ (medians)
Post-handoff reorder fraction	33.66%

Table 374: Observational and coordinate summary for GW240920_124024. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.528 [29.718, 40.330]	32.670 [27.849, 38.288]
Physical compact-source mass M_c ($M_\odot$)	26.259 [19.666, 34.127]	19.560 [15.379, 27.209]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.626 [1.328, 17.093]	13.502 [4.201, 18.132]
Geometric share M_{geom}/M_∞ (%)	24.55 [4.04, 45.93]	41.65 [13.80, 49.90]
Exterior spin magnitude $ \hat{a}_\infty $	0.538 [0.154, 0.920]	0.239 [0.023, 0.733]
Polarization factor A_{pol}	1.325 [1.042, 1.850]	1.714 [1.160, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.946 [0.528, 0.999]	0.700 [0.091, 0.987]
GR/CPTG matching radius r_m (km)	52.462 [43.882, 59.552]	48.242 [41.122, 56.537]
Structural Mode N_{BH}	1.6231 [1.4696, 1.8650]	1.8136 [1.5320, 1.9111]
Organization-shell radius r_N (km)	27.670 [23.065, 31.424]	25.411 [21.656, 29.751]
Inner trapping-surface radius r_- (km)	35.946 [30.196, 45.156]	30.085 [25.068, 36.538]
Structural- N lapse $\bar{\alpha}_N$	0.7156 [0.5213, 0.8356]	0.5642 [0.4839, 0.7860]
Structural- N clock $T_{24,N}$ (h)	17.174 [12.510, 20.054]	13.542 [11.614, 18.865]
Structural- N axis ω (rad s^{-1})	1993.64 [866.99, 2869.03]	2320.32 [381.39, 3263.98]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2464.16 [1368.51, 3193.82]	2595.71 [381.51, 3566.32]
Structural- N equatorial ω (rad s^{-1})	2879.25 [1992.37, 3503.36]	2868.21 [381.56, 3835.93]
Structural- N period P_ω (ms)	2.5498 [1.9673, 4.5913]	2.4206 [1.7618, 16.4694]
Trapping-surface limit Ω_- (rad s^{-1})	3758.84 [2284.87, 4540.00]	3432.81 [415.67, 4840.02]
Trapping-surface period P_- (ms)	1.6716 [1.3840, 2.7499]	1.8303 [1.2982, 15.1158]

Table 375: CPTG compact-state, structural, clock, and geometric-rotation results for GW240920_124024. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8803	7.0780	3674.37	3322.30
500 m inward	4.1465	5.0167	3718.64	3379.98
100 m inward	1.8500	2.2469	3751.04	3422.39
10 m inward	0.5847	0.7108	3758.02	3431.78
1 m inward	0.1849	0.2248	3758.76	3432.71
10 cm inward	0.0585	0.0711	3758.84	3432.80
1 cm inward	0.0185	0.0225	3758.84	3432.81
surface limit	0.0000	0.0000	3758.84	3432.81

Table 376: Near-inner-boundary profile for GW240920_124024. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.259 M_\odot$ and $19.560 M_\odot$, with completed geometric mass-equivalent shares of 24.55% and 41.65%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.670 < 35.946 < 52.462 \text{ km}$, and component 2 gives $25.411 < 30.085 < 48.242 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6231$ and 1.8136 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.174 and 13.542 h. The determinant-averaged frame-dragging rates at those shells are 2464.16 and $2595.71 \text{ rad s}^{-1}$, approaching surface-limit values of 3758.84 and $3432.81 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.880/7.078 \text{ h}$ at 1 km inward to $0.0185/0.0225 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.44 GW240414_054515

GW240414_054515 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 49.977 M_\odot$ and $m_2 = 33.531 M_\odot$, while the luminosity-distance median is 1525.5 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,167
Observed luminosity distance	1525.5 Mpc (median)
Detector-frame masses	$m_1 = 49.977 M_\odot$, $m_2 = 33.531 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.671$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2876$; $z_\pi = 0.3092$ (medians)
Post-handoff reorder fraction	16.44%

Table 377: Observational and coordinate summary for GW240414_054515. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	37.786 [27.495, 50.152]	25.766 [17.540, 36.187]
Physical compact-source mass M_c ($M_\odot$)	25.977 [18.576, 40.811]	17.065 [10.719, 25.126]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.347 [1.076, 21.593]	8.824 [1.293, 16.668]
Geometric share M_{geom}/M_∞ (%)	28.24 [3.13, 49.51]	35.93 [5.84, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.473 [0.050, 0.938]	0.340 [0.030, 0.885]
Polarization factor A_{pol}	1.394 [1.032, 1.981]	1.561 [1.062, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.919 [0.195, 0.999]	0.828 [0.121, 0.998]
GR/CPTG matching radius r_m (km)	55.796 [40.599, 74.055]	38.047 [25.900, 53.434]
Structural Mode N_{BH}	1.6601 [1.4647, 1.9067]	1.7454 [1.4798, 1.9101]
Organization-shell radius r_N (km)	29.371 [21.298, 38.967]	20.047 [13.631, 28.139]
Inner trapping-surface radius r_- (km)	37.295 [28.223, 54.450]	25.094 [16.663, 34.459]
Structural- N lapse $\bar{\alpha}_N$	0.6870 [0.4873, 0.8395]	0.6199 [0.4846, 0.8274]
Structural- N clock $T_{24,N}$ (h)	16.488 [11.695, 20.149]	14.879 [11.631, 19.857]
Structural- N axis ω (rad s^{-1})	1742.54 [569.38, 2916.18]	2654.34 [614.94, 4696.45]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2120.39 [658.27, 3266.89]	3100.67 [615.54, 5235.31]
Structural- N equatorial ω (rad s^{-1})	2561.13 [683.15, 3669.30]	3623.05 [615.84, 5783.93]
Structural- N period P_ω (ms)	2.9632 [1.9233, 9.5449]	2.0264 [1.2002, 10.2077]
Trapping-surface limit Ω_- (rad s^{-1})	3330.64 [745.70, 4839.70]	4620.86 [671.45, 7418.64]
Trapping-surface period P_- (ms)	1.8865 [1.2983, 8.4259]	1.3597 [0.8469, 9.3577]

Table 378: CPTG compact-state, structural, clock, and geometric-rotation results for GW240414_054515. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.9412	7.4582	3265.08	4467.25
500 m inward	4.1963	5.2759	3299.75	4548.92
100 m inward	1.8737	2.3599	3324.42	4607.53
10 m inward	0.5923	0.7462	3330.29	4619.12
1 m inward	0.1873	0.2360	3330.57	4620.68
10 cm inward	0.0592	0.0746	3330.64	4620.84
1 cm inward	0.0187	0.0236	3330.64	4620.86
surface limit	0.0000	0.0000	3330.64	4620.86

Table 379: Near-inner-boundary profile for GW240414_054515. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $25.977 M_\odot$ and $17.065 M_\odot$, with completed geometric mass-equivalent shares of 28.24% and 35.93%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $29.371 < 37.295 < 55.796$ km, and component 2 gives $20.047 < 25.094 < 38.047$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6601$ and 1.7454 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.488 and 14.879 h. The determinant-averaged frame-dragging rates at those shells are 2120.39 and $3100.67 \text{ rad s}^{-1}$, approaching surface-limit values of 3330.64 and $4620.86 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.941/7.458$ h at 1 km inward to $0.0187/0.0236$ h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.45 GW241101_220523

GW241101_220523 is an M2 intermediate observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 56.513 M_\odot$ and $m_2 = 27.030 M_\odot$, while the luminosity-distance median is 2178.9 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,545
Observed luminosity distance	2178.9 Mpc (median)
Detector-frame masses	$m_1 = 56.513 M_\odot$, $m_2 = 27.030 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.478$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3909$; $z_\pi = 0.4205$ (medians)
Post-handoff reorder fraction	7.89%

Table 380: Observational and coordinate summary for GW241101_220523. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	39.601 [25.500, 52.486]	18.898 [13.290, 31.532]
Physical compact-source mass M_c ($M_\odot$)	26.761 [17.937, 45.116]	13.655 [8.010, 22.126]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.542 [1.163, 21.276]	5.582 [0.664, 13.496]
Geometric share M_{geom}/M_∞ (%)	27.42 [3.36, 49.46]	30.22 [3.67, 49.68]
Exterior spin magnitude $ \hat{a}_\infty $	0.488 [0.053, 0.933]	0.439 [0.040, 0.927]
Polarization factor A_{pol}	1.378 [1.035, 1.978]	1.433 [1.038, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.926 [0.207, 0.999]	0.901 [0.159, 0.999]
GR/CPTG matching radius r_m (km)	58.477 [37.655, 77.502]	27.906 [19.624, 46.560]
Structural Mode N_{BH}	1.6516 [1.4659, 1.9060]	1.6810 [1.4676, 1.9086]
Organization-shell radius r_N (km)	30.788 [19.767, 40.742]	14.684 [10.325, 24.501]
Inner trapping-surface radius r_- (km)	38.485 [26.617, 59.887]	19.275 [12.561, 30.864]
Structural- N lapse $\bar{\alpha}_N$	0.6936 [0.4878, 0.8386]	0.6708 [0.4858, 0.8372]
Structural- N clock $T_{24,N}$ (h)	16.646 [11.707, 20.126]	16.098 [11.659, 20.093]
Structural- N axis ω (rad s^{-1})	1657.41 [568.84, 2936.94]	3224.76 [1012.46, 6290.24]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2050.67 [724.51, 3324.00]	3936.48 [1042.79, 6999.46]
Structural- N equatorial ω (rad s^{-1})	2428.29 [738.10, 3835.75]	4808.59 [1043.32, 7682.40]
Structural- N period P_ω (ms)	3.0640 [1.8902, 8.6723]	1.5961 [0.8977, 6.0254]
Trapping-surface limit Ω_- (rad s^{-1})	3127.01 [805.92, 5113.98]	6287.41 [1137.63, 9855.03]
Trapping-surface period P_- (ms)	2.0093 [1.2286, 7.7963]	0.9993 [0.6376, 5.5230]

Table 381: CPTG compact-state, structural, clock, and geometric-rotation results for GW241101_220523. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8058	8.1544	3065.30	6032.20
500 m inward	4.0972	5.7532	3097.03	6162.69
100 m inward	1.8292	2.5659	3121.17	6263.43
10 m inward	0.5782	0.8108	3126.43	6285.17
1 m inward	0.1828	0.2564	3126.93	6287.16
10 cm inward	0.0578	0.0811	3127.01	6287.38
1 cm inward	0.0183	0.0256	3127.01	6287.40
surface limit	0.0000	0.0000	3127.01	6287.41

Table 382: Near-inner-boundary profile for GW241101_220523. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.761 M_\odot$ and $13.655 M_\odot$, with completed geometric mass-equivalent shares of 27.42% and 30.22%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $30.788 < 38.485 < 58.477 \text{ km}$, and component 2 gives $14.684 < 19.275 < 27.906 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6516$ and 1.6810 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.646 and 16.098 h. The determinant-averaged frame-dragging rates at those shells are 2050.67 and $3936.48 \text{ rad s}^{-1}$, approaching surface-limit values of 3127.01 and $6287.41 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.806/8.154 \text{ h}$ at 1 km inward to $0.0183/0.0256 \text{ h}$ at 1 cm inward for components 1/2. Its strong observational mass asymmetry makes the contrast between the two solved compact interiors particularly useful for component-by-component comparison. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.46 GW230920_071124

GW230920_071124 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 48.113 M_\odot$ and $m_2 = 35.908 M_\odot$, while the luminosity-distance median is 2876.4 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	37,770
Observed luminosity distance	2876.4 Mpc (median)
Detector-frame masses	$m_1 = 48.113 M_\odot$, $m_2 = 35.908 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.746$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4940$; $z_\pi = 0.5319$ (medians)
Post-handoff reorder fraction	21.77%

Table 383: Observational and coordinate summary for GW230920_071124. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	30.886 [22.827, 40.562]	23.922 [16.552, 32.879]
Physical compact-source mass M_c ($M_\odot$)	22.604 [15.668, 33.788]	15.467 [10.044, 22.873]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.030 [0.752, 16.771]	8.530 [1.347, 15.055]
Geometric share M_{geom}/M_∞ (%)	23.26 [2.63, 48.81]	37.16 [6.66, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.561 [0.079, 0.948]	0.318 [0.028, 0.869]
Polarization factor A_{pol}	1.303 [1.027, 1.954]	1.591 [1.071, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.953 [0.301, 1.000]	0.806 [0.112, 0.997]
GR/CPTG matching radius r_m (km)	45.606 [33.707, 59.894]	35.324 [24.441, 48.550]
Structural Mode N_{BH}	1.6108 [1.4620, 1.8987]	1.7599 [1.4847, 1.9104]
Organization-shell radius r_N (km)	24.008 [17.708, 31.521]	18.607 [12.867, 25.546]
Inner trapping-surface radius r_- (km)	31.510 [23.659, 44.905]	22.944 [15.740, 31.454]
Structural- N lapse $\bar{\alpha}_N$	0.7250 [0.4936, 0.8417]	0.6083 [0.4844, 0.8235]
Structural- N clock $T_{24,N}$ (h)	17.400 [11.847, 20.201]	14.599 [11.626, 19.764]
Structural- N axis ω (rad s^{-1})	2011.30 [804.56, 3475.84]	2920.36 [607.73, 5014.29]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2553.11 [1147.89, 3895.90]	3398.43 [608.08, 5549.22]
Structural- N equatorial ω (rad s^{-1})	3132.82 [1276.09, 4384.64]	3912.41 [608.22, 6116.53]
Structural- N period P_ω (ms)	2.4610 [1.6128, 5.4737]	1.8489 [1.1323, 10.3329]
Trapping-surface limit Ω_- (rad s^{-1})	4145.61 [1405.20, 5788.70]	4928.46 [662.77, 7845.47]
Trapping-surface period P_- (ms)	1.5156 [1.0854, 4.4714]	1.2749 [0.8009, 9.4802]

Table 384: CPTG compact-state, structural, clock, and geometric-rotation results for GW230920_071124. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.2274	7.8486	4048.17	4745.63
500 m inward	4.3856	5.5564	4099.09	4840.99
100 m inward	1.9549	2.4861	4136.16	4912.62
10 m inward	0.6177	0.7862	4144.58	4926.63
1 m inward	0.1953	0.2486	4145.55	4928.28
10 cm inward	0.0618	0.0786	4145.61	4928.44
1 cm inward	0.0195	0.0249	4145.61	4928.46
surface limit	0.0000	0.0000	4145.61	4928.46

Table 385: Near-inner-boundary profile for GW230920_071124. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.604 M_\odot$ and $15.467 M_\odot$, with completed geometric mass-equivalent shares of 23.26% and 37.16%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.008 < 31.510 < 45.606 \text{ km}$, and component 2 gives $18.607 < 22.944 < 35.324 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6108$ and 1.7599 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.400 and 14.599 h. The determinant-averaged frame-dragging rates at those shells are 2553.11 and $3398.43 \text{ rad s}^{-1}$, approaching surface-limit values of 4145.61 and $4928.46 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.227/7.849 \text{ h}$ at 1 km inward to $0.0195/0.0249 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.47 GW230702_185453

GW230702_185453 is an M2 intermediate observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 58.140 M_\odot$ and $m_2 = 26.017 M_\odot$, while the luminosity-distance median is 2472.8 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:Mixed
Posterior samples	49,131
Observed luminosity distance	2472.8 Mpc (median)
Detector-frame masses	$m_1 = 58.140 M_\odot$, $m_2 = 26.017 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.447$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4351$; $z_\pi = 0.4682$ (medians)
Post-handoff reorder fraction	6.89%

Table 386: Observational and coordinate summary for GW230702_185453. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	39.559 [24.490, 60.269]	17.551 [11.081, 28.499]
Physical compact-source mass M_c ($M_\odot$)	25.540 [17.017, 43.728]	12.329 [6.901, 19.923]
Geometric mass-equivalent M_{geom} ($M_\odot$)	13.888 [1.602, 24.166]	5.098 [0.601, 12.532]
Geometric share M_{geom}/M_∞ (%)	34.18 [5.34, 49.70]	30.90 [3.83, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.370 [0.039, 0.895]	0.427 [0.037, 0.924]
Polarization factor A_{pol}	1.519 [1.056, 1.988]	1.447 [1.040, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.855 [0.154, 0.998]	0.894 [0.146, 0.999]
GR/CPTG matching radius r_m (km)	58.415 [36.163, 88.995]	25.917 [16.362, 42.083]
Structural Mode N_{BH}	1.7251 [1.4769, 1.9088]	1.6884 [1.4685, 1.9092]
Organization-shell radius r_N (km)	30.757 [18.994, 46.982]	13.640 [8.609, 22.186]
Inner trapping-surface radius r_- (km)	37.629 [25.432, 59.820]	17.705 [10.662, 27.826]
Structural- N lapse $\bar{\alpha}_N$	0.6362 [0.4856, 0.8297]	0.6650 [0.4854, 0.8365]
Structural- N clock $T_{24,N}$ (h)	15.268 [11.655, 19.913]	15.961 [11.649, 20.076]
Structural- N axis ω (rad s^{-1})	1739.17 [500.47, 3087.73]	3572.52 [1010.49, 7330.93]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2040.14 [517.02, 3515.71]	4353.16 [1014.34, 8195.01]
Structural- N equatorial ω (rad s^{-1})	2349.49 [517.37, 4037.66]	5246.30 [1015.54, 9073.85]
Structural- N period P_ω (ms)	3.0798 [1.7872, 12.1528]	1.4434 [0.7667, 6.1943]
Trapping-surface limit Ω_- (rad s^{-1})	2933.46 [564.10, 5361.23]	6804.71 [1108.33, 11723.62]
Trapping-surface period P_- (ms)	2.1419 [1.1720, 11.1383]	0.9234 [0.5359, 5.6691]

Table 387: CPTG compact-state, structural, clock, and geometric-rotation results for GW230702_185453. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0081	8.5622	2868.00	6484.55
500 m inward	4.2450	6.0422	2902.15	6662.92
100 m inward	1.8970	2.6961	2927.36	6779.14
10 m inward	0.5999	0.8519	2932.72	6802.17
1 m inward	0.1897	0.2694	2933.38	6804.52
10 cm inward	0.0600	0.0852	2933.45	6804.69
1 cm inward	0.0190	0.0269	2933.45	6804.71
surface limit	0.0000	0.0000	2933.46	6804.71

Table 388: Near-inner-boundary profile for GW230702_185453. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $25.540 M_\odot$ and $12.329 M_\odot$, with completed geometric mass-equivalent shares of 34.18% and 30.90%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $30.757 < 37.629 < 58.415 \text{ km}$, and component 2 gives $13.640 < 17.705 < 25.917 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7251$ and 1.6884 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.268 and 15.961 h. The determinant-averaged frame-dragging rates at those shells are 2040.14 and $4353.16 \text{ rad s}^{-1}$, approaching surface-limit values of 2933.46 and $6804.71 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.008/8.562 h at 1 km inward to 0.0190/0.0269 h at 1 cm inward for components 1/2. Its strong observational mass asymmetry makes the contrast between the two solved compact interiors particularly useful for component-by-component comparison. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.48 GW231206_233901

GW231206_233901 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 47.863 M_\odot$ and $m_2 = 36.479 M_\odot$, while the luminosity-distance median is 1494.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	41,133
Observed luminosity distance	1494.0 Mpc (median)
Detector-frame masses	$m_1 = 47.863 M_\odot$, $m_2 = 36.479 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.762$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2824$; $z_\pi = 0.3037$ (medians)
Post-handoff reorder fraction	22.13%

Table 389: Observational and coordinate summary for GW231206_233901. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	36.380 [28.127, 43.529]	28.638 [22.021, 37.196]
Physical compact-source mass M_c ($M_\odot$)	26.568 [19.234, 36.748]	18.435 [12.786, 25.812]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.575 [1.118, 18.857]	10.748 [2.081, 17.381]
Geometric share M_{geom}/M_∞ (%)	23.96 [3.34, 48.42]	38.22 [8.10, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.549 [0.092, 0.934]	0.300 [0.028, 0.841]
Polarization factor A_{pol}	1.315 [1.035, 1.939]	1.619 [1.088, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.949 [0.345, 0.999]	0.786 [0.109, 0.996]
GR/CPTG matching radius r_m (km)	53.720 [41.534, 64.276]	42.288 [32.517, 54.924]
Structural Mode N_{BH}	1.6175 [1.4658, 1.8941]	1.7723 [1.4935, 1.9105]
Organization-shell radius r_N (km)	28.309 [21.814, 33.860]	22.281 [17.113, 28.926]
Inner trapping-surface radius r_- (km)	36.763 [29.338, 48.756]	27.412 [20.460, 35.147]
Structural- N lapse $\bar{\alpha}_N$	0.7199 [0.4973, 0.8386]	0.5982 [0.4843, 0.8165]
Structural- N clock $T_{24,N}$ (h)	17.277 [11.935, 20.127]	14.356 [11.624, 19.596]
Structural- N axis ω (rad s^{-1})	1806.35 [762.45, 2837.06]	2514.34 [513.52, 3941.33]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2272.10 [1125.77, 3185.22]	2881.29 [513.69, 4330.26]
Structural- N equatorial ω (rad s^{-1})	2735.89 [1270.01, 3565.59]	3287.32 [513.78, 4709.13]
Structural- N period P_ω (ms)	2.7654 [1.9726, 5.5812]	2.1807 [1.4510, 12.2314]
Trapping-surface limit Ω_- (rad s^{-1})	3598.56 [1393.54, 4701.56]	4125.75 [559.75, 6008.77]
Trapping-surface period P_- (ms)	1.7460 [1.3364, 4.5088]	1.5229 [1.0457, 11.2250]

Table 390: CPTG compact-state, structural, clock, and geometric-rotation results for GW231206_233901. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.7920	7.2367	3524.49	3996.39
500 m inward	4.0830	5.1250	3563.21	4065.50
100 m inward	1.8206	2.2937	3591.93	4114.00
10 m inward	0.5753	0.7254	3597.93	4124.55
1 m inward	0.1819	0.2294	3598.52	4125.66
10 cm inward	0.0575	0.0725	3598.56	4125.74
1 cm inward	0.0182	0.0229	3598.56	4125.74
surface limit	0.0000	0.0000	3598.56	4125.75

Table 391: Near-inner-boundary profile for GW231206_233901. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.568 M_\odot$ and $18.435 M_\odot$, with completed geometric mass-equivalent shares of 23.96% and 38.22%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.309 < 36.763 < 53.720 \text{ km}$, and component 2 gives $22.281 < 27.412 < 42.288 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6175$ and 1.7723 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.277 and 14.356 h. The determinant-averaged frame-dragging rates at those shells are 2272.10 and $2881.29 \text{ rad s}^{-1}$, approaching surface-limit values of 3598.56 and $4125.75 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.792/7.237 \text{ h}$ at 1 km inward to $0.0182/0.0229 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.49 GW190517_055101

GW190517_055101 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 52.919 M_\odot$ and $m_2 = 32.747 M_\odot$, while the luminosity-distance median is 1791.6 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	15,688
Observed luminosity distance	1791.6 Mpc (median)
Detector-frame masses	$m_1 = 52.919 M_\odot$, $m_2 = 32.747 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.619$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3311$; $z_\pi = 0.3555$ (medians)
Post-handoff reorder fraction	4.09%

Table 392: Observational and coordinate summary for GW190517_055101. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	38.499 [28.806, 52.466]	23.580 [15.853, 31.446]
Physical compact-source mass M_c ($M_\odot$)	35.918 [25.855, 50.005]	18.017 [10.038, 27.441]
Geometric mass-equivalent M_{geom} ($M_\odot$)	1.931 [0.279, 7.162]	4.926 [0.537, 11.589]
Geometric share M_{geom}/M_∞ (%)	4.99 [0.70, 18.57]	21.26 [2.39, 48.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.901 [0.646, 0.986]	0.597 [0.077, 0.953]
Polarization factor A_{pol}	1.052 [1.007, 1.228]	1.270 [1.024, 1.955]
Compact-source spin parameter $\hat{a}_c$	0.999 [0.974, 1.000]	0.963 [0.295, 1.000]
GR/CPTG matching radius r_m (km)	56.848 [42.535, 77.473]	34.819 [23.410, 46.434]
Structural Mode N_{BH}	1.4749 [1.4521, 1.5694]	1.5926 [1.4607, 1.8992]
Organization-shell radius r_N (km)	29.688 [22.211, 40.406]	18.315 [12.295, 24.390]
Inner trapping-surface radius r_- (km)	47.883 [34.398, 67.370]	24.676 [15.444, 36.600]
Structural- N lapse $\bar{\alpha}_N$	0.8313 [0.7570, 0.8497]	0.7391 [0.4932, 0.8428]
Structural- N clock $T_{24,N}$ (h)	19.951 [18.167, 20.394]	17.737 [11.836, 20.226]
Structural- N axis ω (rad s^{-1})	852.70 [560.98, 1698.25]	2442.54 [1006.54, 5006.26]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1370.69 [932.66, 2322.52]	3193.52 [1547.64, 5601.59]
Structural- N equatorial ω (rad s^{-1})	2167.28 [1536.90, 3079.26]	3952.78 [1903.35, 6301.44]
Structural- N period P_ω (ms)	4.5839 [2.7053, 6.7369]	1.9675 [1.1217, 4.0599]
Trapping-surface limit Ω_- (rad s^{-1})	3123.11 [2220.70, 4326.67]	5249.91 [2086.61, 8155.46]
Trapping-surface period P_- (ms)	2.0118 [1.4522, 2.8294]	1.1968 [0.7704, 3.0112]

Table 393: CPTG compact-state, structural, clock, and geometric-rotation results for GW190517_055101. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.6081	6.8937	3108.88	5088.42
500 m inward	2.5096	4.8484	3117.88	5174.38
100 m inward	1.1059	2.1553	3122.14	5235.52
10 m inward	0.3486	0.6805	3123.15	5248.58
1 m inward	0.1102	0.2152	3123.13	5249.75
10 cm inward	0.0349	0.0680	3123.11	5249.90
1 cm inward	0.0110	0.0215	3123.11	5249.91
surface limit	0.0000	0.0000	3123.11	5249.91

Table 394: Near-inner-boundary profile for GW190517_055101. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $35.918 M_\odot$ and $18.017 M_\odot$, with completed geometric mass-equivalent shares of 4.99% and 21.26%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $29.688 < 47.883 < 56.848 \text{ km}$, and component 2 gives $18.315 < 24.676 < 34.819 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.4749$ and 1.5926 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.951 and 17.737 h. The determinant-averaged frame-dragging rates at those shells are 1370.69 and $3193.52 \text{ rad s}^{-1}$, approaching surface-limit values of 3123.11 and $5249.91 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $3.608/6.894 \text{ h}$ at 1 km inward to $0.0110/0.0215 \text{ h}$ at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.50 GW241002_030559

GW241002_030559 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 47.482 M_\odot$ and $m_2 = 38.284 M_\odot$, while the luminosity-distance median is 1456.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,435
Observed luminosity distance	1456.0 Mpc (median)
Detector-frame masses	$m_1 = 47.482 M_\odot$, $m_2 = 38.284 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.806$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2761$; $z_\pi = 0.2969$ (medians)
Post-handoff reorder fraction	27.15%

Table 395: Observational and coordinate summary for GW241002_030559. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.548 [27.814, 44.982]	30.459 [22.390, 38.518]
Physical compact-source mass M_c ($M_\odot$)	26.451 [19.348, 36.644]	19.009 [13.453, 26.958]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.007 [0.912, 19.216]	11.573 [2.101, 18.035]
Geometric share M_{geom}/M_∞ (%)	23.05 [2.76, 48.27]	39.20 [8.03, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.565 [0.096, 0.945]	0.283 [0.026, 0.843]
Polarization factor A_{pol}	1.300 [1.028, 1.933]	1.645 [1.087, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.954 [0.360, 1.000]	0.764 [0.104, 0.996]
GR/CPTG matching radius r_m (km)	52.491 [41.071, 66.421]	44.976 [33.062, 56.876]
Structural Mode N_{BH}	1.6089 [1.4627, 1.8924]	1.7841 [1.4930, 1.9107]
Organization-shell radius r_N (km)	27.623 [21.592, 34.956]	23.698 [17.404, 29.909]
Inner trapping-surface radius r_- (km)	36.573 [29.011, 48.905]	28.684 [21.221, 36.625]
Structural- N lapse $\bar{\alpha}_N$	0.7265 [0.4987, 0.8411]	0.5886 [0.4842, 0.8168]
Structural- N clock $T_{24,N}$ (h)	17.435 [11.969, 20.187]	14.126 [11.621, 19.604]
Structural- N axis ω (rad s^{-1})	1777.09 [752.52, 2892.38]	2384.45 [459.33, 3765.32]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2254.09 [1128.97, 3252.31]	2727.73 [459.52, 4166.20]
Structural- N equatorial ω (rad s^{-1})	2757.45 [1269.00, 3641.31]	3104.33 [459.62, 4558.22]
Structural- N period P_ω (ms)	2.7875 [1.9319, 5.5654]	2.3035 [1.5081, 13.6733]
Trapping-surface limit Ω_- (rad s^{-1})	3656.99 [1420.27, 4781.74]	3874.31 [500.81, 5832.15]
Trapping-surface period P_- (ms)	1.7181 [1.3140, 4.4239]	1.6218 [1.0773, 12.5461]

Table 396: CPTG compact-state, structural, clock, and geometric-rotation results for GW241002_030559. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.7695	7.1398	3584.85	3759.36
500 m inward	4.0647	5.0572	3623.19	3823.01
100 m inward	1.8129	2.2641	3650.17	3864.74
10 m inward	0.5729	0.7161	3656.57	3873.38
1 m inward	0.1812	0.2265	3656.96	3874.22
10 cm inward	0.0573	0.0716	3656.99	3874.31
1 cm inward	0.0181	0.0226	3656.99	3874.31
surface limit	0.0000	0.0000	3656.99	3874.31

Table 397: Near-inner-boundary profile for GW241002_030559. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.451 M_\odot$ and $19.009 M_\odot$, with completed geometric mass-equivalent shares of 23.05% and 39.20%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.623 < 36.573 < 52.491$ km, and component 2 gives $23.698 < 28.684 < 44.976$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6089$ and 1.7841 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.435 and 14.126 h. The determinant-averaged frame-dragging rates at those shells are 2254.09 and $2727.73 \text{ rad s}^{-1}$, approaching surface-limit values of 3656.99 and $3874.31 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.770/7.140$ h at 1 km inward to $0.0181/0.0226$ h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.51 GW230805_034249

GW230805_034249 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 51.198 M_\odot$ and $m_2 = 35.912 M_\odot$, while the luminosity-distance median is 3486.2 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	41,526
Observed luminosity distance	3486.2 Mpc (median)
Detector-frame masses	$m_1 = 51.198 M_\odot$, $m_2 = 35.912 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.701$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5797$; $z_\pi = 0.6247$ (medians)
Post-handoff reorder fraction	18.27%

Table 398: Observational and coordinate summary for GW230805_034249. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	30.662 [21.916, 43.405]	22.510 [14.460, 31.895]
Physical compact-source mass M_c ($M_\odot$)	23.471 [15.082, 38.383]	14.746 [9.042, 22.960]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.854 [0.659, 15.670]	7.516 [1.027, 14.223]
Geometric share M_{geom}/M_∞ (%)	19.10 [2.19, 48.00]	35.21 [5.34, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.636 [0.104, 0.956]	0.352 [0.032, 0.895]
Polarization factor A_{pol}	1.236 [1.022, 1.923]	1.543 [1.056, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.972 [0.385, 1.000]	0.839 [0.127, 0.998]
GR/CPTG matching radius r_m (km)	45.277 [32.361, 64.093]	33.239 [21.352, 47.097]
Structural Mode N_{BH}	1.5738 [1.4597, 1.8893]	1.7370 [1.4770, 1.9099]
Organization-shell radius r_N (km)	23.819 [17.005, 33.645]	17.500 [11.236, 24.820]
Inner trapping-surface radius r_- (km)	32.130 [22.665, 50.997]	21.755 [13.900, 31.410]
Structural- N lapse $\bar{\alpha}_N$	0.7535 [0.5012, 0.8436]	0.6267 [0.4848, 0.8297]
Structural- N clock $T_{24,N}$ (h)	18.085 [12.029, 20.246]	15.040 [11.635, 19.912]
Structural- N axis ω (rad s^{-1})	1821.94 [743.84, 3571.30]	3051.23 [743.64, 5570.15]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2425.66 [1139.55, 4033.66]	3574.54 [744.90, 6219.84]
Structural- N equatorial ω (rad s^{-1})	3083.45 [1582.12, 4566.72]	4158.12 [745.19, 6910.16]
Structural- N period P_ω (ms)	2.5903 [1.5577, 5.5137]	1.7578 [1.0102, 8.4349]
Trapping-surface limit Ω_- (rad s^{-1})	4118.72 [1862.96, 6022.78]	5279.93 [812.03, 8888.23]
Trapping-surface period P_- (ms)	1.5255 [1.0432, 3.3727]	1.1900 [0.7069, 7.7376]

Table 399: CPTG compact-state, structural, clock, and geometric-rotation results for GW230805_034249. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.9018	7.9867	4030.71	5081.65
500 m inward	4.1480	5.6517	4078.42	5187.64
100 m inward	1.8453	2.5270	4111.57	5260.87
10 m inward	0.5828	0.7990	4117.98	5278.12
1 m inward	0.1843	0.2526	4118.66	5279.70
10 cm inward	0.0583	0.0799	4118.72	5279.91
1 cm inward	0.0184	0.0253	4118.72	5279.93
surface limit	0.0000	0.0000	4118.72	5279.93

Table 400: Near-inner-boundary profile for GW230805_034249. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.471 M_\odot$ and $14.746 M_\odot$, with completed geometric mass-equivalent shares of 19.10% and 35.21%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $23.819 < 32.130 < 45.277 \text{ km}$, and component 2 gives $17.500 < 21.755 < 33.239 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5738$ and 1.7370 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.085 and 15.040 h. The determinant-averaged frame-dragging rates at those shells are 2425.66 and $3574.54 \text{ rad s}^{-1}$, approaching surface-limit values of 4118.72 and $5279.93 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.902/7.987 \text{ h}$ at 1 km inward to $0.0184/0.0253 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.52 GW230726_002940

GW230726_002940 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 48.651 M_\odot$ and $m_2 = 38.559 M_\odot$, while the luminosity-distance median is 2061.1 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	34,980
Observed luminosity distance	2061.1 Mpc (median)
Detector-frame masses	$m_1 = 48.651 M_\odot$, $m_2 = 38.559 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.793$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3728$; $z_\pi = 0.4010$ (medians)
Post-handoff reorder fraction	24.86%

Table 401: Observational and coordinate summary for GW230726_002940. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.839 [25.998, 43.639]	28.196 [21.526, 36.691]
Physical compact-source mass M_c ($M_\odot$)	24.921 [17.550, 35.755]	17.882 [12.719, 25.783]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.972 [0.918, 18.421]	10.674 [1.914, 16.920]
Geometric share M_{geom}/M_∞ (%)	23.88 [2.89, 48.76]	38.49 [7.59, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.550 [0.081, 0.943]	0.295 [0.027, 0.851]
Polarization factor A_{pol}	1.314 [1.030, 1.952]	1.626 [1.082, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.950 [0.307, 1.000]	0.780 [0.106, 0.997]
GR/CPTG matching radius r_m (km)	49.967 [38.389, 64.438]	41.635 [31.786, 54.179]
Structural Mode N_{BH}	1.6167 [1.4634, 1.8981]	1.7757 [1.4903, 1.9106]
Organization-shell radius r_N (km)	26.303 [20.165, 33.916]	21.926 [16.747, 28.505]
Inner trapping-surface radius r_- (km)	34.674 [26.632, 47.578]	26.768 [20.182, 35.216]
Structural- N lapse $\bar{\alpha}_N$	0.7205 [0.4941, 0.8406]	0.5955 [0.4843, 0.8190]
Structural- N clock $T_{24,N}$ (h)	17.291 [11.859, 20.174]	14.291 [11.622, 19.656]
Structural- N axis ω (rad s^{-1})	1868.13 [762.70, 3126.24]	2523.26 [504.13, 3979.56]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2354.37 [1084.63, 3503.16]	2906.84 [504.31, 4387.43]
Structural- N equatorial ω (rad s^{-1})	2873.84 [1179.18, 3908.76]	3324.02 [504.41, 4788.29]
Structural- N period P_ω (ms)	2.6687 [1.7936, 5.7929]	2.1615 [1.4321, 12.4589]
Trapping-surface limit Ω_- (rad s^{-1})	3807.40 [1295.04, 5109.58]	4168.50 [549.55, 6112.44]
Trapping-surface period P_- (ms)	1.6503 [1.2297, 4.8517]	1.5073 [1.0279, 11.4332]

Table 402: CPTG compact-state, structural, clock, and geometric-rotation results for GW230726_002940. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.9659	7.3552	3726.50	4039.39
500 m inward	4.2041	5.2087	3769.88	4107.37
100 m inward	1.8750	2.3310	3800.39	4156.67
10 m inward	0.5925	0.7373	3806.63	4167.23
1 m inward	0.1874	0.2332	3807.35	4168.36
10 cm inward	0.0592	0.0737	3807.39	4168.49
1 cm inward	0.0187	0.0233	3807.39	4168.50
surface limit	0.0000	0.0000	3807.40	4168.50

Table 403: Near-inner-boundary profile for GW230726_002940. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.921 M_\odot$ and $17.882 M_\odot$, with completed geometric mass-equivalent shares of 23.88% and 38.49%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.303 < 34.674 < 49.967 \text{ km}$, and component 2 gives $21.926 < 26.768 < 41.635 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6167$ and 1.7757 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.291 and 14.291 h. The determinant-averaged frame-dragging rates at those shells are 2354.37 and $2906.84 \text{ rad s}^{-1}$, approaching surface-limit values of 3807.40 and $4168.50 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.966/7.355 \text{ h}$ at 1 km inward to $0.0187/0.0233 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.53 GW241009_220455

GW241009_220455 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 49.745 M_\odot$ and $m_2 = 38.837 M_\odot$, while the luminosity-distance median is 2931.7 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,784
Observed luminosity distance	2931.7 Mpc (median)
Detector-frame masses	$m_1 = 49.745 M_\odot$, $m_2 = 38.837 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.781$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5019$; $z_\pi = 0.5405$ (medians)
Post-handoff reorder fraction	21.50%

Table 404: Observational and coordinate summary for GW241009_220455. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	31.625 [23.399, 40.227]	25.541 [17.024, 33.360]
Physical compact-source mass M_c ($M_\odot$)	24.561 [16.111, 35.085]	16.463 [10.675, 24.385]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.880 [0.681, 15.667]	8.789 [1.362, 15.174]
Geometric share M_{geom}/M_∞ (%)	18.99 [2.24, 47.37]	35.89 [6.14, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.638 [0.121, 0.955]	0.341 [0.031, 0.879]
Polarization factor A_{pol}	1.234 [1.023, 1.900]	1.560 [1.065, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.972 [0.436, 1.000]	0.829 [0.124, 0.998]
GR/CPTG matching radius r_m (km)	46.698 [34.552, 59.400]	37.715 [25.138, 49.261]
Structural Mode N_{BH}	1.5729 [1.4600, 1.8820]	1.7449 [1.4816, 1.9100]
Organization-shell radius r_N (km)	24.567 [18.151, 31.249]	19.859 [13.226, 25.927]
Inner trapping-surface radius r_- (km)	33.411 [24.125, 46.611]	24.463 [16.556, 32.814]
Structural- N lapse $\bar{\alpha}_N$	0.7542 [0.5072, 0.8434]	0.6203 [0.4847, 0.8260]
Structural- N clock $T_{24,N}$ (h)	18.101 [12.173, 20.241]	14.887 [11.633, 19.823]
Structural- N axis ω (rad s^{-1})	1804.55 [804.00, 3368.67]	2761.39 [667.51, 4676.62]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2400.75 [1248.79, 3775.32]	3217.82 [667.94, 5212.70]
Structural- N equatorial ω (rad s^{-1})	3028.78 [1791.02, 4279.91]	3703.51 [668.16, 5817.24]
Structural- N period P_ω (ms)	2.6172 [1.6643, 5.0314]	1.9526 [1.2054, 9.4068]
Trapping-surface limit Ω_- (rad s^{-1})	4055.72 [2105.49, 5626.29]	4720.36 [728.29, 7486.92]
Trapping-surface period P_- (ms)	1.5492 [1.1168, 2.9842]	1.3311 [0.8392, 8.6274]

Table 405: CPTG compact-state, structural, clock, and geometric-rotation results for GW241009_220455. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8073	7.6109	3966.70	4560.16
500 m inward	4.0808	5.3902	4012.99	4644.78
100 m inward	1.8157	2.4115	4047.88	4705.17
10 m inward	0.5734	0.7626	4054.78	4718.79
1 m inward	0.1813	0.2411	4055.58	4720.25
10 cm inward	0.0573	0.0763	4055.70	4720.34
1 cm inward	0.0181	0.0241	4055.72	4720.35
surface limit	0.0000	0.0000	4055.72	4720.36

Table 406: Near-inner-boundary profile for GW241009_220455. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.561 M_\odot$ and $16.463 M_\odot$, with completed geometric mass-equivalent shares of 18.99% and 35.89%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.567 < 33.411 < 46.698 \text{ km}$, and component 2 gives $19.859 < 24.463 < 37.715 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5729$ and 1.7449 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.101 and 14.887 h. The determinant-averaged frame-dragging rates at those shells are 2400.75 and $3217.82 \text{ rad s}^{-1}$, approaching surface-limit values of 4055.72 and $4720.36 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.807/7.611 \text{ h}$ at 1 km inward to $0.0181/0.0241 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.54 GW230922_020344

GW230922_020344 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 51.146 M_\odot$ and $m_2 = 38.145 M_\odot$, while the luminosity-distance median is 1639.9 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	43,575
Observed luminosity distance	1639.9 Mpc (median)
Detector-frame masses	$m_1 = 51.146 M_\odot$, $m_2 = 38.145 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.746$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3063$; $z_\pi = 0.3293$ (medians)
Post-handoff reorder fraction	21.24%

Table 407: Observational and coordinate summary for GW230922_020344. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	37.824 [28.507, 48.792]	29.348 [22.591, 38.885]
Physical compact-source mass M_c ($M_\odot$)	27.221 [19.149, 40.819]	18.920 [13.199, 26.926]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.424 [1.133, 20.030]	11.015 [1.894, 18.011]
Geometric share M_{geom}/M_∞ (%)	25.11 [3.26, 48.85]	38.24 [7.08, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.528 [0.078, 0.935]	0.300 [0.027, 0.861]
Polarization factor A_{pol}	1.335 [1.034, 1.955]	1.619 [1.076, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.942 [0.297, 0.999]	0.785 [0.106, 0.997]
GR/CPTG matching radius r_m (km)	55.851 [42.095, 72.048]	43.335 [33.359, 57.418]
Structural Mode N_{BH}	1.6286 [1.4654, 1.8991]	1.7726 [1.4872, 1.9106]
Organization-shell radius r_N (km)	29.412 [22.096, 37.941]	22.821 [17.552, 30.215]
Inner trapping-surface radius r_- (km)	38.019 [29.270, 54.186]	28.084 [21.084, 36.798]
Structural- N lapse $\bar{\alpha}_N$	0.7113 [0.4933, 0.8390]	0.5979 [0.4843, 0.8215]
Structural- N clock $T_{24,N}$ (h)	17.072 [11.839, 20.136]	14.351 [11.622, 19.715]
Structural- N axis ω (rad s^{-1})	1733.69 [685.82, 2841.70]	2385.72 [472.42, 3798.59]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2148.81 [976.14, 3180.77]	2754.79 [472.67, 4184.52]
Structural- N equatorial ω (rad s^{-1})	2587.23 [1065.87, 3544.93]	3181.92 [472.80, 4562.34]
Structural- N period P_ω (ms)	2.9240 [1.9754, 6.4367]	2.2808 [1.5015, 13.2931]
Trapping-surface limit Ω_- (rad s^{-1})	3384.80 [1167.38, 4661.04]	4026.05 [515.23, 5825.89]
Trapping-surface period P_- (ms)	1.8563 [1.3480, 5.3823]	1.5606 [1.0785, 12.1949]

Table 408: CPTG compact-state, structural, clock, and geometric-rotation results for GW230922_020344. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.7226	7.1508	3316.72	3909.39
500 m inward	4.0360	5.0618	3352.42	3969.78
100 m inward	1.8003	2.2650	3378.55	4015.05
10 m inward	0.5689	0.7164	3384.29	4025.05
1 m inward	0.1799	0.2266	3384.75	4025.99
10 cm inward	0.0569	0.0716	3384.79	4026.04
1 cm inward	0.0180	0.0227	3384.80	4026.05
surface limit	0.0000	0.0000	3384.80	4026.05

Table 409: Near-inner-boundary profile for GW230922_020344. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.221 M_\odot$ and $18.920 M_\odot$, with completed geometric mass-equivalent shares of 25.11% and 38.24%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $29.412 < 38.019 < 55.851 \text{ km}$, and component 2 gives $22.821 < 28.084 < 43.335 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6286$ and 1.7726 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.072 and 14.351 h. The determinant-averaged frame-dragging rates at those shells are 2148.81 and $2754.79 \text{ rad s}^{-1}$, approaching surface-limit values of 3384.80 and $4026.05 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.723/7.151 \text{ h}$ at 1 km inward to $0.0180/0.0227 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.55 GW240716_034900

GW240716_034900 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 53.945 M_\odot$ and $m_2 = 35.672 M_\odot$, while the luminosity-distance median is 1861.7 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,295
Observed luminosity distance	1861.7 Mpc (median)
Detector-frame masses	$m_1 = 53.945 M_\odot$, $m_2 = 35.672 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.661$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3417$; $z_\pi = 0.3675$ (medians)
Post-handoff reorder fraction	13.95%

Table 410: Observational and coordinate summary for GW240716_034900. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	39.091 [27.690, 48.301]	26.276 [19.684, 37.414]
Physical compact-source mass M_c ($M_\odot$)	26.615 [19.181, 37.632]	17.546 [11.728, 25.478]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.252 [1.280, 21.639]	9.317 [1.412, 17.090]
Geometric share M_{geom}/M_∞ (%)	29.35 [3.85, 49.44]	36.12 [5.97, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.454 [0.053, 0.924]	0.337 [0.030, 0.882]
Polarization factor A_{pol}	1.416 [1.040, 1.978]	1.566 [1.064, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.910 [0.209, 0.999]	0.825 [0.120, 0.998]
GR/CPTG matching radius r_m (km)	57.722 [40.888, 71.322]	38.799 [29.066, 55.247]
Structural Mode N_{BH}	1.6718 [1.4686, 1.9059]	1.7476 [1.4806, 1.9102]
Organization-shell radius r_N (km)	30.417 [21.472, 37.552]	20.433 [15.286, 29.072]
Inner trapping-surface radius r_- (km)	38.339 [29.000, 50.092]	25.508 [18.533, 35.387]
Structural- N lapse $\bar{\alpha}_N$	0.6779 [0.4879, 0.8364]	0.6182 [0.4846, 0.8267]
Structural- N clock $T_{24,N}$ (h)	16.271 [11.710, 20.073]	14.836 [11.631, 19.842]
Structural- N axis ω (rad s^{-1})	1782.60 [635.64, 2777.25]	2618.79 [585.18, 4284.74]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2151.54 [716.67, 3125.48]	3056.53 [585.71, 4741.37]
Structural- N equatorial ω (rad s^{-1})	2533.34 [717.57, 3574.24]	3571.11 [585.97, 5178.00]
Structural- N period P_ω (ms)	2.9203 [2.0103, 8.7672]	2.0557 [1.3252, 10.7275]
Trapping-surface limit Ω_- (rad s^{-1})	3279.13 [784.20, 4754.52]	4552.23 [638.98, 6651.55]
Trapping-surface period P_- (ms)	1.9161 [1.3215, 8.0123]	1.3802 [0.9446, 9.8331]

Table 411: CPTG compact-state, structural, clock, and geometric-rotation results for GW240716_034900. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.9001	7.3741	3213.10	4404.74
500 m inward	4.1654	5.2150	3248.40	4482.69
100 m inward	1.8608	2.3321	3273.12	4538.78
10 m inward	0.5883	0.7375	3278.55	4550.66
1 m inward	0.1860	0.2332	3279.14	4552.09
10 cm inward	0.0588	0.0738	3279.13	4552.21
1 cm inward	0.0186	0.0233	3279.13	4552.23
surface limit	0.0000	0.0000	3279.13	4552.23

Table 412: Near-inner-boundary profile for GW240716_034900. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.615 M_\odot$ and $17.546 M_\odot$, with completed geometric mass-equivalent shares of 29.35% and 36.12%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $30.417 < 38.339 < 57.722 \text{ km}$, and component 2 gives $20.433 < 25.508 < 38.799 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6718$ and 1.7476 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.271 and 14.836 h. The determinant-averaged frame-dragging rates at those shells are 2151.54 and $3056.53 \text{ rad s}^{-1}$, approaching surface-limit values of 3279.13 and $4552.23 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.900/7.374 \text{ h}$ at 1 km inward to $0.0186/0.0233 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.56 GW190503_185404

GW190503_185404 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 53.322 M_\odot$ and $m_2 = 36.451 M_\odot$, while the luminosity-distance median is 1522.4 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	24,159
Observed luminosity distance	1522.4 Mpc (median)
Detector-frame masses	$m_1 = 53.322 M_\odot$, $m_2 = 36.451 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.684$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2875$; $z_\pi = 0.3087$ (medians)
Post-handoff reorder fraction	16.30%

Table 413: Observational and coordinate summary for GW190503_185404. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.052 [29.359, 50.881]	28.256 [18.841, 39.605]
Physical compact-source mass M_c ($M_\odot$)	28.559 [20.422, 42.144]	18.536 [11.680, 26.968]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.860 [1.118, 21.425]	9.799 [1.389, 18.279]
Geometric share M_{geom}/M_∞ (%)	25.05 [3.14, 49.03]	36.39 [5.86, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.529 [0.071, 0.938]	0.332 [0.029, 0.884]
Polarization factor A_{pol}	1.334 [1.032, 1.962]	1.572 [1.062, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.942 [0.274, 0.999]	0.820 [0.117, 0.998]
GR/CPTG matching radius r_m (km)	59.142 [43.352, 75.132]	41.724 [27.820, 58.482]
Structural Mode N_{BH}	1.6280 [1.4647, 1.9011]	1.7507 [1.4800, 1.9103]
Organization-shell radius r_N (km)	31.153 [22.752, 39.552]	21.976 [14.621, 30.774]
Inner trapping-surface radius r_- (km)	40.072 [30.804, 56.040]	27.336 [18.045, 37.219]
Structural- N lapse $\bar{\alpha}_N$	0.7118 [0.4917, 0.8395]	0.6157 [0.4845, 0.8273]
Structural- N clock $T_{24,N}$ (h)	17.083 [11.800, 20.148]	14.776 [11.629, 19.854]
Structural- N axis ω (rad s^{-1})	1649.74 [641.60, 2672.58]	2440.18 [545.98, 4295.95]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2048.23 [889.59, 3015.16]	2864.73 [546.22, 4780.10]
Structural- N equatorial ω (rad s^{-1})	2462.35 [931.87, 3399.02]	3331.45 [546.35, 5320.51]
Structural- N period P_ω (ms)	3.0676 [2.0839, 7.0630]	2.1933 [1.3144, 11.5030]
Trapping-surface limit Ω_- (rad s^{-1})	3225.14 [1019.18, 4491.90]	4232.23 [595.30, 6831.90]
Trapping-surface period P_- (ms)	1.9482 [1.3988, 6.1650]	1.4846 [0.9197, 10.5546]

Table 414: CPTG compact-state, structural, clock, and geometric-rotation results for GW190503_185404. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5943	7.1533	3165.82	4108.40
500 m inward	3.9434	5.0592	3197.92	4174.26
100 m inward	1.7591	2.2632	3219.90	4221.24
10 m inward	0.5559	0.7157	3224.51	4231.17
1 m inward	0.1758	0.2263	3225.06	4232.20
10 cm inward	0.0556	0.0716	3225.13	4232.22
1 cm inward	0.0176	0.0226	3225.14	4232.23
surface limit	0.0000	0.0000	3225.14	4232.23

Table 415: Near-inner-boundary profile for GW190503_185404. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.559 M_\odot$ and $18.536 M_\odot$, with completed geometric mass-equivalent shares of 25.05% and 36.39%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.153 < 40.072 < 59.142 \text{ km}$, and component 2 gives $21.976 < 27.336 < 41.724 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6280$ and 1.7507 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.083 and 14.776 h. The determinant-averaged frame-dragging rates at those shells are 2048.23 and $2864.73 \text{ rad s}^{-1}$, approaching surface-limit values of 3225.14 and $4232.23 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.594/7.153 \text{ h}$ at 1 km inward to $0.0176/0.0226 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.57 GW240515_005301

GW240515_005301 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 59.875 M_\odot$ and $m_2 = 30.246 M_\odot$, while the luminosity-distance median is 3830.6 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,489
Observed luminosity distance	3830.6 Mpc (median)
Detector-frame masses	$m_1 = 59.875 M_\odot$, $m_2 = 30.246 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.505$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6266$; $z_\pi = 0.6757$ (medians)
Post-handoff reorder fraction	5.36%

Table 416: Observational and coordinate summary for GW240515_005301. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.940 [25.077, 47.176]	18.010 [13.640, 27.943]
Physical compact-source mass M_c ($M_\odot$)	30.617 [21.072, 41.901]	13.709 [8.022, 22.687]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.540 [0.576, 11.443]	4.593 [0.526, 10.341]
Geometric share M_{geom}/M_∞ (%)	12.64 [1.81, 30.28]	24.99 [2.83, 49.24]
Exterior spin magnitude $ \hat{a}_\infty $	0.755 [0.438, 0.964]	0.530 [0.063, 0.944]
Polarization factor A_{pol}	1.145 [1.018, 1.434]	1.333 [1.029, 1.970]
Compact-source spin parameter $\hat{a}_c$	0.989 [0.901, 1.000]	0.943 [0.244, 1.000]
GR/CPTG matching radius r_m (km)	53.070 [37.030, 69.661]	26.594 [20.141, 41.262]
Structural Mode N_{BH}	1.5236 [1.4577, 1.6817]	1.6274 [1.4631, 1.9035]
Organization-shell radius r_N (km)	27.875 [19.357, 36.564]	13.995 [10.609, 21.713]
Inner trapping-surface radius r_- (km)	40.728 [28.810, 55.564]	18.913 [12.712, 30.332]
Structural- N lapse $\bar{\alpha}_N$	0.7926 [0.6702, 0.8452]	0.7122 [0.4898, 0.8409]
Structural- N clock $T_{24,N}$ (h)	19.023 [16.085, 20.284]	17.094 [11.755, 20.181]
Structural- N axis ω (rad s^{-1})	1288.33 [738.84, 2572.59]	3223.11 [1234.15, 6343.68]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1886.98 [1195.79, 3080.25]	4021.75 [1743.33, 7000.90]
Structural- N equatorial ω (rad s^{-1})	2603.16 [1865.81, 3699.83]	5013.31 [1883.26, 7624.99]
Structural- N period P_ω (ms)	3.3298 [2.0398, 5.2544]	1.5623 [0.8975, 3.6041]
Trapping-surface limit Ω_- (rad s^{-1})	3612.78 [2670.55, 5066.18]	6619.89 [2061.03, 9723.85]
Trapping-surface period P_- (ms)	1.7392 [1.2402, 2.3528]	0.9491 [0.6462, 3.0486]

Table 417: CPTG compact-state, structural, clock, and geometric-rotation results for GW240515_005301. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.7829	8.0295	3556.70	6353.85
500 m inward	3.3535	5.6480	3586.95	6502.62
100 m inward	1.4886	2.5157	3608.27	6594.97
10 m inward	0.4700	0.7944	3612.23	6617.25
1 m inward	0.1486	0.2512	3612.73	6619.62
10 cm inward	0.0470	0.0794	3612.77	6619.86
1 cm inward	0.0149	0.0251	3612.78	6619.89
surface limit	0.0000	0.0000	3612.78	6619.89

Table 418: Near-inner-boundary profile for GW240515_005301. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.617 M_\odot$ and $13.709 M_\odot$, with completed geometric mass-equivalent shares of 12.64% and 24.99%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.875 < 40.728 < 53.070 \text{ km}$, and component 2 gives $13.995 < 18.913 < 26.594 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5236$ and 1.6274 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.023 and 17.094 h. The determinant-averaged frame-dragging rates at those shells are 1886.98 and $4021.75 \text{ rad s}^{-1}$, approaching surface-limit values of 3612.78 and $6619.89 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $4.783/8.030 \text{ h}$ at 1 km inward to $0.0149/0.0251 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.58 GW240525_031210

GW240525_031210 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 53.211 M_\odot$ and $m_2 = 37.362 M_\odot$, while the luminosity-distance median is 4541.2 Mpc . One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	19,070
Observed luminosity distance	4541.2 Mpc (median)
Detector-frame masses	$m_1 = 53.211 M_\odot$, $m_2 = 37.362 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.702$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7205$; $z_\pi = 0.7781$ (medians)
Post-handoff reorder fraction	16.38%

Table 419: Observational and coordinate summary for GW240525_031210. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	29.529 [20.924, 41.987]	21.041 [14.087, 29.922]
Physical compact-source mass M_c ($M_\odot$)	22.809 [14.920, 34.804]	14.100 [8.699, 21.595]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.659 [0.614, 16.178]	6.901 [0.959, 13.264]
Geometric share M_{geom}/M_∞ (%)	19.58 [2.24, 47.64]	34.11 [5.11, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.627 [0.114, 0.956]	0.371 [0.035, 0.899]
Polarization factor A_{pol}	1.243 [1.023, 1.910]	1.518 [1.054, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.415, 1.000]	0.856 [0.140, 0.999]
GR/CPTG matching radius r_m (km)	43.603 [30.898, 61.999]	31.070 [20.801, 44.183]
Structural Mode N_{BH}	1.5779 [1.4599, 1.8851]	1.7243 [1.4756, 1.9094]
Organization-shell radius r_N (km)	22.926 [16.249, 32.655]	16.362 [10.942, 23.253]
Inner trapping-surface radius r_- (km)	31.400 [21.897, 46.415]	20.583 [13.501, 29.529]
Structural- N lapse $\bar{\alpha}_N$	0.7504 [0.5046, 0.8434]	0.6368 [0.4852, 0.8307]
Structural- N clock $T_{24,N}$ (h)	18.009 [12.111, 20.241]	15.282 [11.645, 19.938]
Structural- N axis ω (rad s^{-1})	1915.15 [847.76, 3640.19]	3203.08 [893.20, 5789.10]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2523.78 [1286.56, 4159.59]	3784.26 [894.84, 6424.21]
Structural- N equatorial ω (rad s^{-1})	3207.50 [1695.51, 4726.65]	4425.14 [895.13, 7095.22]
Structural- N period P_ω (ms)	2.4896 [1.5105, 4.8837]	1.6603 [0.9780, 7.0216]
Trapping-surface limit Ω_- (rad s^{-1})	4315.98 [1989.51, 6272.12]	5677.64 [975.71, 9130.23]
Trapping-surface period P_- (ms)	1.4558 [1.0018, 3.1582]	1.1067 [0.6882, 6.4396]

Table 420: CPTG compact-state, structural, clock, and geometric-rotation results for GW240525_031210. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.9997	8.1789	4217.14	5450.80
500 m inward	4.2181	5.7831	4273.21	5575.18
100 m inward	1.8767	2.5860	4308.24	5658.22
10 m inward	0.5927	0.8178	4315.04	5676.05
1 m inward	0.1874	0.2586	4315.91	5677.54
10 cm inward	0.0593	0.0818	4315.97	5677.63
1 cm inward	0.0187	0.0259	4315.98	5677.64
surface limit	0.0000	0.0000	4315.98	5677.64

Table 421: Near-inner-boundary profile for GW240525_031210. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.809 M_\odot$ and $14.100 M_\odot$, with completed geometric mass-equivalent shares of 19.58% and 34.11%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $22.926 < 31.400 < 43.603 \text{ km}$, and component 2 gives $16.362 < 20.583 < 31.070 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5779$ and 1.7243 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.009 and 15.282 h. The determinant-averaged frame-dragging rates at those shells are 2523.78 and $3784.26 \text{ rad s}^{-1}$, approaching surface-limit values of 4315.98 and $5677.64 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.000/8.179 \text{ h}$ at 1 km inward to $0.0187/0.0259 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. Its Structural-Mode frame dragging is in the upper part of the M2 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.59 GW240919_061559

GW240919_061559 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 51.059 M_\odot$ and $m_2 = 39.593 M_\odot$, while the luminosity-distance median is 1868.6 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,355
Observed luminosity distance	1868.6 Mpc (median)
Detector-frame masses	$m_1 = 51.059 M_\odot$, $m_2 = 39.593 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.775$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3428$; $z_\pi = 0.3686$ (medians)
Post-handoff reorder fraction	25.88%

Table 422: Observational and coordinate summary for GW240919_061559. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	36.534 [27.891, 44.052]	29.990 [23.195, 39.174]
Physical compact-source mass M_c ($M_\odot$)	25.387 [18.747, 35.186]	18.894 [13.556, 26.435]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.115 [1.277, 19.587]	11.734 [2.321, 18.361]
Geometric share M_{geom}/M_∞ (%)	27.99 [3.99, 49.25]	39.71 [8.80, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.478 [0.062, 0.921]	0.273 [0.026, 0.828]
Polarization factor A_{pol}	1.389 [1.042, 1.971]	1.659 [1.096, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.921 [0.241, 0.999]	0.752 [0.102, 0.995]
GR/CPTG matching radius r_m (km)	53.947 [41.184, 65.048]	44.285 [34.251, 57.846]
Structural Mode N_{BH}	1.6575 [1.4694, 1.9037]	1.7903 [1.4978, 1.9108]
Organization-shell radius r_N (km)	28.416 [21.598, 34.284]	23.323 [18.028, 30.408]
Inner trapping-surface radius r_- (km)	35.948 [28.853, 46.717]	28.376 [21.675, 36.373]
Structural- N lapse $\bar{\alpha}_N$	0.6890 [0.4896, 0.8358]	0.5835 [0.4842, 0.8130]
Structural- N clock $T_{24,N}$ (h)	16.537 [11.751, 20.059]	14.004 [11.620, 19.512]
Structural- N axis ω (rad s^{-1})	1914.99 [739.69, 2886.24]	2402.09 [447.40, 3737.55]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2313.82 [885.05, 3239.77]	2727.26 [447.64, 4098.92]
Structural- N equatorial ω (rad s^{-1})	2733.88 [885.98, 3619.17]	3103.96 [447.77, 4465.49]
Structural- N period P_ω (ms)	2.7155 [1.9394, 7.0993]	2.3038 [1.5329, 14.0362]
Trapping-surface limit Ω_- (rad s^{-1})	3560.33 [968.05, 4770.35]	3846.42 [488.03, 5715.50]
Trapping-surface period P_- (ms)	1.7648 [1.3171, 6.4906]	1.6335 [1.0993, 12.8746]

Table 423: CPTG compact-state, structural, clock, and geometric-rotation results for GW240919_061559. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0311	7.1590	3482.47	3730.87
500 m inward	4.2565	5.0702	3522.27	3791.56
100 m inward	1.9004	2.2700	3553.10	3835.83
10 m inward	0.6008	0.7180	3559.72	3845.11
1 m inward	0.1900	0.2270	3560.22	3846.29
10 cm inward	0.0601	0.0718	3560.32	3846.41
1 cm inward	0.0190	0.0227	3560.33	3846.42
surface limit	0.0000	0.0000	3560.33	3846.42

Table 424: Near-inner-boundary profile for GW240919_061559. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $25.387 M_\odot$ and $18.894 M_\odot$, with completed geometric mass-equivalent shares of 27.99% and 39.71%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.416 < 35.948 < 53.947 \text{ km}$, and component 2 gives $23.323 < 28.376 < 44.285 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6575$ and 1.7903 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.537 and 14.004 h. The determinant-averaged frame-dragging rates at those shells are 2313.82 and $2727.26 \text{ rad s}^{-1}$, approaching surface-limit values of 3560.33 and $3846.42 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.031/7.159 h at 1 km inward to 0.0190/0.0227 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.60 GW190413_052954

GW190413_052954 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 52.707 M_\odot$ and $m_2 = 37.960 M_\odot$, while the luminosity-distance median is 3317.8 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	27,700
Observed luminosity distance	3317.8 Mpc (median)
Detector-frame masses	$m_1 = 52.707 M_\odot$, $m_2 = 37.960 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.720$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5570$; $z_\pi = 0.5995$ (medians)
Post-handoff reorder fraction	19.83%

Table 425: Observational and coordinate summary for GW190413_052954. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	32.122 [23.674, 43.185]	24.099 [16.717, 33.144]
Physical compact-source mass M_c ($M_\odot$)	23.346 [16.425, 35.370]	15.751 [10.295, 22.777]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.623 [0.865, 17.795]	8.559 [1.278, 15.274]
Geometric share M_{geom}/M_∞ (%)	24.09 [2.89, 48.84]	36.78 [6.28, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.546 [0.078, 0.943]	0.325 [0.029, 0.876]
Polarization factor A_{pol}	1.317 [1.030, 1.955]	1.582 [1.067, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.948 [0.298, 1.000]	0.813 [0.117, 0.998]
GR/CPTG matching radius r_m (km)	47.432 [34.958, 63.768]	35.584 [24.684, 48.942]
Structural Mode N_{BH}	1.6187 [1.4634, 1.8990]	1.7553 [1.4824, 1.9103]
Organization-shell radius r_N (km)	24.966 [18.345, 33.560]	18.745 [12.992, 25.766]
Inner trapping-surface radius r_- (km)	32.627 [24.708, 47.173]	23.251 [15.978, 31.414]
Structural- N lapse $\bar{\alpha}_N$	0.7189 [0.4934, 0.8406]	0.6120 [0.4845, 0.8253]
Structural- N clock $T_{24,N}$ (h)	17.254 [11.841, 20.175]	14.687 [11.628, 19.807]
Structural- N axis ω (rad s^{-1})	1960.28 [764.26, 3315.82]	2876.85 [626.54, 4911.39]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2470.16 [1083.07, 3755.29]	3362.57 [627.14, 5451.14]
Structural- N equatorial ω (rad s^{-1})	3028.55 [1211.18, 4225.65]	3906.42 [627.40, 6021.37]
Structural- N period P_ω (ms)	2.5436 [1.6732, 5.8013]	1.8686 [1.1526, 10.0188]
Trapping-surface limit Ω_- (rad s^{-1})	3989.49 [1325.33, 5591.75]	4954.44 [683.73, 7710.04]
Trapping-surface period P_- (ms)	1.5749 [1.1237, 4.7408]	1.2682 [0.8149, 9.1896]

Table 426: CPTG compact-state, structural, clock, and geometric-rotation results for GW190413_052954. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1415	7.7757	3902.08	4771.50
500 m inward	4.3275	5.5039	3947.07	4868.99
100 m inward	1.9300	2.4620	3981.48	4937.36
10 m inward	0.6099	0.7786	3988.60	4952.97
1 m inward	0.1928	0.2462	3989.39	4954.31
10 cm inward	0.0610	0.0779	3989.48	4954.42
1 cm inward	0.0193	0.0246	3989.49	4954.44
surface limit	0.0000	0.0000	3989.49	4954.44

Table 427: Near-inner-boundary profile for GW190413_052954. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.346 M_\odot$ and $15.751 M_\odot$, with completed geometric mass-equivalent shares of 24.09% and 36.78%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.966 < 32.627 < 47.432 \text{ km}$, and component 2 gives $18.745 < 23.251 < 35.584 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6187$ and 1.7553 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.254 and 14.687 h. The determinant-averaged frame-dragging rates at those shells are 2470.16 and $3362.57 \text{ rad s}^{-1}$, approaching surface-limit values of 3989.49 and $4954.44 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.141/7.776 \text{ h}$ at 1 km inward to $0.0193/0.0246 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.61 GW170823_131358

GW170823_131358 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 51.671 M_\odot$ and $m_2 = 39.420 M_\odot$, while the luminosity-distance median is 1969.0 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	12,443
Observed luminosity distance	1969.0 Mpc (median)
Detector-frame masses	$m_1 = 51.671 M_\odot$, $m_2 = 39.420 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.763$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3590$; $z_\pi = 0.3856$ (medians)
Post-handoff reorder fraction	21.14%

Table 428: Observational and coordinate summary for GW170823_131358. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	36.741 [28.273, 46.914]	29.120 [20.786, 38.427]
Physical compact-source mass M_c ($M_\odot$)	27.191 [18.914, 39.532]	18.564 [12.438, 26.936]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.523 [0.946, 19.418]	10.621 [1.908, 17.631]
Geometric share M_{geom}/M_∞ (%)	23.30 [2.79, 48.54]	37.70 [7.54, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.560 [0.088, 0.945]	0.309 [0.029, 0.852]
Polarization factor A_{pol}	1.304 [1.029, 1.943]	1.605 [1.082, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.953 [0.333, 1.000]	0.796 [0.116, 0.997]
GR/CPTG matching radius r_m (km)	54.252 [41.748, 69.275]	42.999 [30.693, 56.742]
Structural Mode N_{BH}	1.6112 [1.4629, 1.8955]	1.7662 [1.4900, 1.9103]
Organization-shell radius r_N (km)	28.556 [21.945, 36.487]	22.657 [16.168, 29.877]
Inner trapping-surface radius r_- (km)	37.625 [28.711, 52.625]	27.740 [19.536, 36.866]
Structural- N lapse $\bar{\alpha}_N$	0.7247 [0.4962, 0.8410]	0.6031 [0.4845, 0.8193]
Structural- N clock $T_{24,N}$ (h)	17.392 [11.908, 20.185]	14.475 [11.628, 19.662]
Structural- N axis ω (rad s^{-1})	1722.08 [705.16, 2928.15]	2452.92 [545.30, 4063.91]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2178.77 [1038.35, 3258.59]	2831.66 [545.52, 4505.57]
Structural- N equatorial ω (rad s^{-1})	2653.77 [1223.94, 3599.52]	3234.07 [545.63, 4920.77]
Structural- N period P_ω (ms)	2.8838 [1.9282, 6.0511]	2.2189 [1.3945, 11.5179]
Trapping-surface limit Ω_- (rad s^{-1})	3506.05 [1344.50, 4692.22]	4075.66 [594.49, 6270.19]
Trapping-surface period P_- (ms)	1.7921 [1.3391, 4.6732]	1.5416 [1.0021, 10.5690]

Table 429: CPTG compact-state, structural, clock, and geometric-rotation results for GW170823_131358. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6936	7.2073	3437.91	3958.22
500 m inward	4.0125	5.1018	3471.85	4017.35
100 m inward	1.7889	2.2830	3500.54	4064.41
10 m inward	0.5653	0.7219	3505.51	4074.38
1 m inward	0.1787	0.2283	3506.03	4075.53
10 cm inward	0.0565	0.0722	3506.05	4075.65
1 cm inward	0.0179	0.0228	3506.05	4075.66
surface limit	0.0000	0.0000	3506.05	4075.66

Table 430: Near-inner-boundary profile for GW170823_131358. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.191 M_\odot$ and $18.564 M_\odot$, with completed geometric mass-equivalent shares of 23.30% and 37.70%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.556 < 37.625 < 54.252 \text{ km}$, and component 2 gives $22.657 < 27.740 < 42.999 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6112$ and 1.7662 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.392 and 14.475 h. The determinant-averaged frame-dragging rates at those shells are 2178.77 and $2831.66 \text{ rad s}^{-1}$, approaching surface-limit values of 3506.05 and $4075.66 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.694/7.207 \text{ h}$ at 1 km inward to $0.0179/0.0228 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.62 GW250118_170523

GW250118_170523 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 51.226 M_\odot$ and $m_2 = 39.996 M_\odot$, while the luminosity-distance median is 1908.1 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,711
Observed luminosity distance	1908.1 Mpc (median)
Detector-frame masses	$m_1 = 51.226 M_\odot$, $m_2 = 39.996 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.781$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3490$; $z_\pi = 0.3753$ (medians)
Post-handoff reorder fraction	27.19%

Table 431: Observational and coordinate summary for GW250118_170523. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	36.152 [27.953, 45.341]	30.038 [22.866, 39.291]
Physical compact-source mass M_c ($M_\odot$)	27.034 [19.092, 37.448]	19.137 [13.449, 27.047]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.269 [1.003, 18.660]	11.355 [2.109, 18.325]
Geometric share M_{geom}/M_∞ (%)	23.08 [3.06, 47.81]	38.58 [7.99, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.564 [0.109, 0.939]	0.294 [0.027, 0.843]
Polarization factor A_{pol}	1.300 [1.032, 1.916]	1.628 [1.087, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.954 [0.401, 1.000]	0.778 [0.106, 0.996]
GR/CPTG matching radius r_m (km)	53.383 [41.276, 66.952]	44.355 [33.764, 58.018]
Structural Mode N_{BH}	1.6092 [1.4643, 1.8872]	1.7767 [1.4927, 1.9106]
Organization-shell radius r_N (km)	28.097 [21.678, 35.284]	23.370 [17.774, 30.525]
Inner trapping-surface radius r_- (km)	37.239 [28.934, 49.720]	28.596 [21.389, 36.923]
Structural- N lapse $\bar{\alpha}_N$	0.7263 [0.5030, 0.8399]	0.5946 [0.4842, 0.8170]
Structural- N clock $T_{24,N}$ (h)	17.430 [12.071, 20.157]	14.271 [11.622, 19.609]
Structural- N axis ω (rad s^{-1})	1761.82 [775.90, 2878.60]	2382.06 [460.94, 3768.41]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2233.63 [1188.02, 3235.10]	2735.52 [461.18, 4148.20]
Structural- N equatorial ω (rad s^{-1})	2714.27 [1508.54, 3625.02]	3129.35 [461.29, 4515.77]
Structural- N period P_ω (ms)	2.8130 [1.9422, 5.2888]	2.2969 [1.5147, 13.6243]
Trapping-surface limit Ω_- (rad s^{-1})	3587.48 [1664.32, 4752.38]	3932.41 [502.71, 5774.02]
Trapping-surface period P_- (ms)	1.7514 [1.3221, 3.7752]	1.5978 [1.0882, 12.4987]

Table 432: CPTG compact-state, structural, clock, and geometric-rotation results for GW250118_170523. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6894	7.1061	3516.21	3815.65
500 m inward	4.0083	5.0316	3555.81	3877.57
100 m inward	1.7868	2.2518	3581.19	3921.86
10 m inward	0.5648	0.7121	3586.70	3931.33
1 m inward	0.1786	0.2252	3587.39	3932.28
10 cm inward	0.0565	0.0712	3587.47	3932.40
1 cm inward	0.0179	0.0225	3587.48	3932.41
surface limit	0.0000	0.0000	3587.48	3932.41

Table 433: Near-inner-boundary profile for GW250118_170523. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.034 M_\odot$ and $19.137 M_\odot$, with completed geometric mass-equivalent shares of 23.08% and 38.58%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.097 < 37.239 < 53.383 \text{ km}$, and component 2 gives $23.370 < 28.596 < 44.355 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6092$ and 1.7767 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.430 and 14.271 h. The determinant-averaged frame-dragging rates at those shells are 2233.63 and $2735.52 \text{ rad s}^{-1}$, approaching surface-limit values of 3587.48 and $3932.41 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.689/7.106 \text{ h}$ at 1 km inward to $0.0179/0.0225 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.63 GW200208_130117

GW200208_130117 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 52.822 M_\odot$ and $m_2 = 38.515 M_\odot$, while the luminosity-distance median is 2229.9 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	31,299
Observed luminosity distance	2229.9 Mpc (median)
Detector-frame masses	$m_1 = 52.822 M_\odot$, $m_2 = 38.515 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.729$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3992$; $z_\pi = 0.4288$ (medians)
Post-handoff reorder fraction	20.50%

Table 434: Observational and coordinate summary for GW200208_130117. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	36.262 [27.048, 46.108]	27.443 [19.694, 37.037]
Physical compact-source mass M_c ($M_\odot$)	25.393 [18.643, 36.093]	17.583 [11.902, 24.751]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.813 [1.210, 20.464]	10.265 [1.853, 17.313]
Geometric share M_{geom}/M_∞ (%)	27.47 [3.75, 49.26]	38.43 [7.95, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.487 [0.062, 0.926]	0.296 [0.026, 0.844]
Polarization factor A_{pol}	1.379 [1.039, 1.971]	1.624 [1.086, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.926 [0.239, 0.999]	0.781 [0.104, 0.996]
GR/CPTG matching radius r_m (km)	53.545 [39.940, 68.084]	40.523 [29.081, 54.690]
Structural Mode N_{BH}	1.6521 [1.4681, 1.9038]	1.7749 [1.4925, 1.9107]
Organization-shell radius r_N (km)	28.205 [20.992, 35.861]	21.353 [15.299, 28.768]
Inner trapping-surface radius r_- (km)	36.077 [28.312, 48.117]	26.173 [18.676, 34.380]
Structural- N lapse $\bar{\alpha}_N$	0.6932 [0.4895, 0.8368]	0.5961 [0.4842, 0.8172]
Structural- N clock $T_{24,N}$ (h)	16.637 [11.749, 20.084]	14.306 [11.621, 19.613]
Structural- N axis ω (rad s^{-1})	1864.09 [713.17, 2927.37]	2590.26 [491.73, 4239.80]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2287.05 [843.41, 3290.03]	2986.45 [491.97, 4702.36]
Structural- N equatorial ω (rad s^{-1})	2730.41 [846.48, 3697.00]	3427.22 [492.09, 5173.38]
Structural- N period P_ω (ms)	2.7473 [1.9098, 7.4498]	2.1039 [1.3362, 12.7714]
Trapping-surface limit Ω_- (rad s^{-1})	3565.28 [924.99, 4894.52]	4282.67 [536.31, 6644.03]
Trapping-surface period P_- (ms)	1.7623 [1.2837, 6.7927]	1.4671 [0.9457, 11.7157]

Table 435: CPTG compact-state, structural, clock, and geometric-rotation results for GW200208_130117. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0011	7.3872	3489.41	4143.04
500 m inward	4.2366	5.2307	3529.43	4215.48
100 m inward	1.8912	2.3405	3558.52	4270.08
10 m inward	0.5978	0.7402	3564.67	4281.29
1 m inward	0.1890	0.2341	3565.21	4282.54
10 cm inward	0.0598	0.0740	3565.27	4282.66
1 cm inward	0.0189	0.0234	3565.28	4282.67
surface limit	0.0000	0.0000	3565.28	4282.67

Table 436: Near-inner-boundary profile for GW200208_130117. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $25.393 M_\odot$ and $17.583 M_\odot$, with completed geometric mass-equivalent shares of 27.47% and 38.43%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.205 < 36.077 < 53.545 \text{ km}$, and component 2 gives $21.353 < 26.173 < 40.523 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6521$ and 1.7749 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.637 and 14.306 h. The determinant-averaged frame-dragging rates at those shells are 2287.05 and $2986.45 \text{ rad s}^{-1}$, approaching surface-limit values of 3565.28 and $4282.67 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.001/7.387 \text{ h}$ at 1 km inward to $0.0189/0.0234 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.64 GW250119_025138

GW250119_025138 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 50.804 M_\odot$ and $m_2 = 40.726 M_\odot$, while the luminosity-distance median is 3078.1 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,097
Observed luminosity distance	3078.1 Mpc (median)
Detector-frame masses	$m_1 = 50.804 M_\odot$, $m_2 = 40.726 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.802$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5227$; $z_\pi = 0.5630$ (medians)
Post-handoff reorder fraction	23.10%

Table 437: Observational and coordinate summary for GW250119_025138. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	32.205 [25.364, 39.084]	26.728 [19.853, 34.063]
Physical compact-source mass M_c ($M_\odot$)	24.181 [16.720, 33.910]	16.784 [11.813, 24.338]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.985 [0.800, 16.503]	9.984 [1.901, 15.742]
Geometric share M_{geom}/M_∞ (%)	22.17 [2.58, 48.31]	38.30 [7.99, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.581 [0.095, 0.949]	0.298 [0.026, 0.843]
Polarization factor A_{pol}	1.285 [1.027, 1.935]	1.621 [1.087, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.959 [0.355, 1.000]	0.784 [0.105, 0.996]
GR/CPTG matching radius r_m (km)	47.555 [37.453, 57.713]	39.467 [29.315, 50.298]
Structural Mode N_{BH}	1.6007 [1.4618, 1.8929]	1.7733 [1.4928, 1.9107]
Organization-shell radius r_N (km)	25.013 [19.676, 30.376]	20.778 [15.438, 26.483]
Inner trapping-surface radius r_- (km)	33.155 [25.676, 44.975]	25.251 [18.712, 32.991]
Structural- N lapse $\bar{\alpha}_N$	0.7328 [0.4983, 0.8419]	0.5973 [0.4842, 0.8170]
Structural- N clock $T_{24,N}$ (h)	17.586 [11.958, 20.206]	14.336 [11.621, 19.609]
Structural- N axis ω (rad s^{-1})	1913.95 [817.87, 3245.03]	2683.91 [534.76, 4270.82]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2462.78 [1243.40, 3627.83]	3074.24 [535.00, 4702.05]
Structural- N equatorial ω (rad s^{-1})	3025.52 [1500.31, 4035.02]	3527.14 [535.12, 5168.59]
Structural- N period P_ω (ms)	2.5513 [1.7319, 5.0532]	2.0438 [1.3363, 11.7443]
Trapping-surface limit Ω_- (rad s^{-1})	4006.98 [1646.50, 5268.85]	4422.00 [583.12, 6590.90]
Trapping-surface period P_- (ms)	1.5681 [1.1925, 3.8161]	1.4209 [0.9533, 10.7752]

Table 438: CPTG compact-state, structural, clock, and geometric-rotation results for GW250119_025138. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0081	7.5858	3919.76	4277.59
500 m inward	4.2295	5.3746	3966.94	4353.63
100 m inward	1.8848	2.4060	3999.50	4409.68
10 m inward	0.5955	0.7610	4006.46	4420.28
1 m inward	0.1883	0.2407	4006.99	4421.83
10 cm inward	0.0595	0.0761	4006.98	4421.99
1 cm inward	0.0188	0.0241	4006.98	4422.00
surface limit	0.0000	0.0000	4006.98	4422.00

Table 439: Near-inner-boundary profile for GW250119_025138. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.181 M_\odot$ and $16.784 M_\odot$, with completed geometric mass-equivalent shares of 22.17% and 38.30%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.013 < 33.155 < 47.555 \text{ km}$, and component 2 gives $20.778 < 25.251 < 39.467 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6007$ and 1.7733 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.586 and 14.336 h. The determinant-averaged frame-dragging rates at those shells are 2462.78 and $3074.24 \text{ rad s}^{-1}$, approaching surface-limit values of 4006.98 and $4422.00 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.008/7.586 \text{ h}$ at 1 km inward to $0.0188/0.0241 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.65 GW190719_215514

GW190719_215514 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 59.066 M_\odot$ and $m_2 = 32.552 M_\odot$, while the luminosity-distance median is 3697.3 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	20,337
Observed luminosity distance	3697.3 Mpc (median)
Detector-frame masses	$m_1 = 59.066 M_\odot$, $m_2 = 32.552 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.551$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6092$; $z_\pi = 0.6561$ (medians)
Post-handoff reorder fraction	9.76%

Table 440: Observational and coordinate summary for GW190719_215514. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.600 [22.887, 76.841]	19.342 [10.254, 31.154]
Physical compact-source mass M_c ($M_\odot$)	27.469 [16.581, 67.551]	13.531 [6.891, 23.104]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.518 [0.705, 18.429]	5.312 [0.605, 12.944]
Geometric share M_{geom}/M_∞ (%)	17.85 [1.95, 47.06]	29.69 [3.61, 49.62]
Exterior spin magnitude $ \hat{a}_\infty $	0.659 [0.128, 0.961]	0.448 [0.044, 0.928]
Polarization factor A_{pol}	1.217 [1.020, 1.889]	1.422 [1.037, 1.985]
Compact-source spin parameter $\hat{a}_c$	0.976 [0.458, 1.000]	0.906 [0.173, 0.999]
GR/CPTG matching radius r_m (km)	52.569 [33.795, 113.466]	28.562 [15.142, 46.003]
Structural Mode N_{BH}	1.5635 [1.4584, 1.8783]	1.6754 [1.4673, 1.9079]
Organization-shell radius r_N (km)	27.643 [17.726, 59.482]	15.038 [7.976, 24.240]
Inner trapping-surface radius r_- (km)	37.627 [24.182, 89.799]	19.393 [10.212, 31.571]
Structural- N lapse $\bar{\alpha}_N$	0.7616 [0.5103, 0.8446]	0.6751 [0.4864, 0.8375]
Structural- N clock $T_{24,N}$ (h)	18.278 [12.246, 20.270]	16.203 [11.673, 20.099]
Structural- N axis ω (rad s^{-1})	1521.09 [454.57, 3236.81]	3298.76 [1011.69, 7220.53]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2068.45 [675.66, 3704.78]	4008.67 [1148.59, 8162.61]
Structural- N equatorial ω (rad s^{-1})	2652.27 [965.51, 4278.30]	4793.44 [1160.92, 9416.51]
Structural- N period P_ω (ms)	3.0376 [1.6960, 9.2994]	1.5674 [0.7698, 5.4703]
Trapping-surface limit Ω_- (rad s^{-1})	3547.33 [1309.77, 5713.75]	6225.07 [1268.88, 12240.53]
Trapping-surface period P_- (ms)	1.7712 [1.0997, 4.7972]	1.0093 [0.5133, 4.9517]

Table 441: CPTG compact-state, structural, clock, and geometric-rotation results for GW190719_215514. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3175	8.1866	3482.79	5963.38
500 m inward	3.7343	5.7722	3517.99	6103.37
100 m inward	1.6596	2.5733	3540.46	6199.53
10 m inward	0.5242	0.8131	3546.45	6223.50
1 m inward	0.1657	0.2571	3547.20	6224.90
10 cm inward	0.0524	0.0813	3547.31	6225.06
1 cm inward	0.0166	0.0257	3547.32	6225.07
surface limit	0.0000	0.0000	3547.33	6225.07

Table 442: Near-inner-boundary profile for GW190719_215514. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.469 M_\odot$ and $13.531 M_\odot$, with completed geometric mass-equivalent shares of 17.85% and 29.69%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.643 < 37.627 < 52.569 \text{ km}$, and component 2 gives $15.038 < 19.393 < 28.562 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5635$ and 1.6754 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.278 and 16.203 h. The determinant-averaged frame-dragging rates at those shells are 2068.45 and $4008.67 \text{ rad s}^{-1}$, approaching surface-limit values of 3547.33 and $6225.07 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.317/8.187 \text{ h}$ at 1 km inward to $0.0166/0.0257 \text{ h}$ at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.66 GW231226_101520

GW231226_101520 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 49.092 M_\odot$ and $m_2 = 42.939 M_\odot$, while the luminosity-distance median is 1180.2 Mpc . 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	42,525
Observed luminosity distance	1180.2 Mpc (median)
Detector-frame masses	$m_1 = 49.092 M_\odot$, $m_2 = 42.939 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.875$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2295$; $z_\pi = 0.2468$ (medians)
Post-handoff reorder fraction	41.36%

Table 443: Observational and coordinate summary for GW231226_101520. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	37.984 [31.843, 43.593]	36.439 [30.195, 42.016]
Physical compact-source mass M_c ($M_\odot$)	27.339 [20.928, 36.376]	21.066 [17.166, 28.377]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.009 [1.678, 19.039]	15.567 [5.468, 20.068]
Geometric share M_{geom}/M_∞ (%)	26.63 [4.76, 47.10]	43.19 [16.86, 49.92]
Exterior spin magnitude $ \hat{a}_\infty $	0.502 [0.127, 0.906]	0.210 [0.020, 0.677]
Polarization factor A_{pol}	1.363 [1.050, 1.890]	1.760 [1.203, 1.997]
Compact-source spin parameter $\hat{a}_c$	0.932 [0.455, 0.999]	0.649 [0.080, 0.979]
GR/CPTG matching radius r_m (km)	56.089 [47.021, 64.370]	53.806 [44.586, 62.042]
Structural Mode N_{BH}	1.6436 [1.4736, 1.8788]	1.8322 [1.5554, 1.9113]
Organization-shell radius r_N (km)	29.596 [24.713, 33.947]	28.327 [23.495, 32.625]
Inner trapping-surface radius r_- (km)	37.798 [32.246, 48.148]	32.888 [27.738, 38.840]
Structural- N lapse $\bar{\alpha}_N$	0.6997 [0.5098, 0.8324]	0.5487 [0.4837, 0.7678]
Structural- N clock $T_{24,N}$ (h)	16.794 [12.236, 19.976]	13.168 [11.609, 18.427]
Structural- N axis ω (rad s^{-1})	1945.50 [819.78, 2697.88]	2092.25 [303.18, 2989.10]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2345.01 [1249.40, 2991.46]	2298.91 [303.25, 3268.37]
Structural- N equatorial ω (rad s^{-1})	2705.72 [1628.15, 3281.06]	2500.55 [303.28, 3527.06]
Structural- N period P_ω (ms)	2.6794 [2.1004, 5.0290]	2.7331 [1.9224, 20.7195]
Trapping-surface limit Ω_- (rad s^{-1})	3512.68 [1805.89, 4245.69]	2876.51 [330.37, 4446.77]
Trapping-surface period P_- (ms)	1.7887 [1.4799, 3.4793]	2.1843 [1.4130, 19.0186]

Table 444: CPTG compact-state, structural, clock, and geometric-rotation results for GW231226_101520. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8271	6.8143	3436.56	2790.27
500 m inward	4.1120	4.8294	3476.35	2834.69
100 m inward	1.8355	2.1633	3505.61	2868.34
10 m inward	0.5802	0.6844	3511.98	2875.71
1 m inward	0.1835	0.2164	3512.62	2876.43
10 cm inward	0.0580	0.0684	3512.68	2876.50
1 cm inward	0.0183	0.0216	3512.68	2876.51
surface limit	0.0000	0.0000	3512.68	2876.51

Table 445: Near-inner-boundary profile for GW231226_101520. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.339 M_\odot$ and $21.066 M_\odot$, with completed geometric mass-equivalent shares of 26.63% and 43.19%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $29.596 < 37.798 < 56.089 \text{ km}$, and component 2 gives $28.327 < 32.888 < 53.806 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6436$ and 1.8322 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.794 and 13.168 h. The determinant-averaged frame-dragging rates at those shells are 2345.01 and $2298.91 \text{ rad s}^{-1}$, approaching surface-limit values of 3512.68 and $2876.51 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.827/6.814 \text{ h}$ at 1 km inward to $0.0183/0.0216 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.67 GW240531_075248

GW240531_075248 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 54.401 M_\odot$ and $m_2 = 37.779 M_\odot$, while the luminosity-distance median is 4137.2 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,444
Observed luminosity distance	4137.2 Mpc (median)
Detector-frame masses	$m_1 = 54.401 M_\odot$, $m_2 = 37.779 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.694$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6675$; $z_\pi = 0.7203$ (medians)
Post-handoff reorder fraction	17.39%

Table 446: Observational and coordinate summary for GW240531_075248. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	31.339 [22.160, 42.325]	22.340 [15.642, 32.545]
Physical compact-source mass M_c ($M_\odot$)	23.156 [15.532, 34.299]	14.918 [9.480, 22.993]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.841 [0.684, 17.786]	7.531 [1.073, 14.495]
Geometric share M_{geom}/M_∞ (%)	22.55 [2.37, 48.89]	35.04 [5.27, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.574 [0.076, 0.953]	0.355 [0.031, 0.896]
Polarization factor A_{pol}	1.291 [1.024, 1.957]	1.539 [1.056, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.957 [0.291, 1.000]	0.842 [0.123, 0.998]
GR/CPTG matching radius r_m (km)	46.275 [32.722, 62.498]	32.988 [23.097, 48.057]
Structural Mode N_{BH}	1.6043 [1.4607, 1.8996]	1.7350 [1.4766, 1.9101]
Organization-shell radius r_N (km)	24.327 [17.178, 32.924]	17.372 [12.160, 25.306]
Inner trapping-surface radius r_- (km)	32.396 [23.057, 45.762]	21.789 [14.840, 31.693]
Structural- N lapse $\bar{\alpha}_N$	0.7300 [0.4929, 0.8428]	0.6282 [0.4847, 0.8300]
Structural- N clock $T_{24,N}$ (h)	17.520 [11.830, 20.227]	15.077 [11.632, 19.920]
Structural- N axis ω (rad s^{-1})	1914.05 [779.34, 3506.99]	2997.51 [725.05, 5317.25]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2449.75 [1098.77, 3955.96]	3549.50 [726.15, 5883.35]
Structural- N equatorial ω (rad s^{-1})	3045.47 [1194.61, 4483.30]	4154.41 [726.31, 6447.28]
Structural- N period P_ω (ms)	2.5648 [1.5883, 5.7184]	1.7702 [1.0680, 8.6528]
Trapping-surface limit Ω_- (rad s^{-1})	4048.42 [1306.64, 5933.19]	5320.69 [791.40, 8245.73]
Trapping-surface period P_- (ms)	1.5520 [1.0590, 4.8086]	1.1809 [0.7620, 7.9393]

Table 447: CPTG compact-state, structural, clock, and geometric-rotation results for GW240531_075248. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1045	7.9551	3962.98	5127.34
500 m inward	4.2972	5.6256	4008.36	5234.01
100 m inward	1.9138	2.5157	4040.79	5304.73
10 m inward	0.6046	0.7954	4047.44	5319.37
1 m inward	0.1912	0.2515	4048.34	5320.61
10 cm inward	0.0604	0.0795	4048.41	5320.67
1 cm inward	0.0191	0.0252	4048.42	5320.69
surface limit	0.0000	0.0000	4048.42	5320.69

Table 448: Near-inner-boundary profile for GW240531_075248. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.156 M_\odot$ and $14.918 M_\odot$, with completed geometric mass-equivalent shares of 22.55% and 35.04%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $24.327 < 32.396 < 46.275 \text{ km}$, and component 2 gives $17.372 < 21.789 < 32.988 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6043$ and 1.7350 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.520 and 15.077 h. The determinant-averaged frame-dragging rates at those shells are 2449.75 and $3549.50 \text{ rad s}^{-1}$, approaching surface-limit values of 4048.42 and $5320.69 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.105/7.955 \text{ h}$ at 1 km inward to $0.0191/0.0252 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.68 GW190521_074359

GW190521_074359 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 52.188 M_\odot$ and $m_2 = 40.358 M_\odot$, while the luminosity-distance median is 1079.7 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	8,111
Observed luminosity distance	1079.7 Mpc (median)
Detector-frame masses	$m_1 = 52.188 M_\odot$, $m_2 = 40.358 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.773$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2124$; $z_\pi = 0.2280$ (medians)
Post-handoff reorder fraction	25.74%

Table 449: Observational and coordinate summary for GW190521_074359. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	41.556 [31.792, 48.509]	33.946 [26.386, 44.648]
Physical compact-source mass M_c ($M_\odot$)	28.600 [21.601, 39.408]	21.740 [15.236, 29.586]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.006 [1.846, 22.134]	13.058 [3.434, 21.089]
Geometric share M_{geom}/M_∞ (%)	27.22 [5.06, 49.47]	39.40 [11.23, 49.88]
Exterior spin magnitude $ \hat{a}_\infty $	0.491 [0.052, 0.900]	0.279 [0.024, 0.782]
Polarization factor A_{pol}	1.374 [1.053, 1.979]	1.650 [1.126, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.927 [0.203, 0.999]	0.760 [0.097, 0.992]
GR/CPTG matching radius r_m (km)	61.363 [46.945, 71.630]	50.126 [38.962, 65.929]
Structural Mode N_{BH}	1.6496 [1.4754, 1.9063]	1.7865 [1.5138, 1.9109]
Organization-shell radius r_N (km)	32.349 [24.592, 37.750]	26.404 [20.488, 34.692]
Inner trapping-surface radius r_- (km)	40.375 [33.177, 52.023]	32.205 [24.182, 40.909]
Structural- N lapse $\bar{\alpha}_N$	0.6951 [0.4876, 0.8310]	0.5866 [0.4841, 0.8003]
Structural- N clock $T_{24,N}$ (h)	16.683 [11.703, 19.943]	14.077 [11.617, 19.208]
Structural- N axis ω (rad s^{-1})	1673.47 [615.34, 2505.48]	2158.85 [372.53, 3324.38]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2065.49 [669.01, 2825.15]	2460.29 [372.61, 3631.15]
Structural- N equatorial ω (rad s^{-1})	2441.53 [669.50, 3168.28]	2791.57 [372.65, 3923.68]
Structural- N period P_ω (ms)	3.0420 [2.2240, 9.3917]	2.5538 [1.7304, 16.8628]
Trapping-surface limit Ω_- (rad s^{-1})	3186.73 [730.71, 4179.70]	3473.66 [405.91, 4974.83]
Trapping-surface period P_- (ms)	1.9717 [1.5033, 8.5988]	1.8088 [1.2630, 15.4794]

Table 450: CPTG compact-state, structural, clock, and geometric-rotation results for GW190521_074359. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6798	6.6824	3121.67	3374.51
500 m inward	4.0081	4.7307	3156.12	3424.81
100 m inward	1.7900	2.1162	3181.66	3465.17
10 m inward	0.5658	0.6692	3186.36	3472.84
1 m inward	0.1789	0.2116	3186.70	3473.58
10 cm inward	0.0566	0.0669	3186.73	3473.65
1 cm inward	0.0179	0.0212	3186.73	3473.66
surface limit	0.0000	0.0000	3186.73	3473.66

Table 451: Near-inner-boundary profile for GW190521_074359. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.600 M_\odot$ and $21.740 M_\odot$, with completed geometric mass-equivalent shares of 27.22% and 39.40%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.349 < 40.375 < 61.363 \text{ km}$, and component 2 gives $26.404 < 32.205 < 50.126 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6496$ and 1.7865 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.683 and 14.077 h. The determinant-averaged frame-dragging rates at those shells are 2065.49 and $2460.29 \text{ rad s}^{-1}$, approaching surface-limit values of 3186.73 and $3473.66 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.680/6.682 \text{ h}$ at 1 km inward to $0.0179/0.0212 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.69 GW240930_234614

GW240930_234614 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 57.212 M_\odot$ and $m_2 = 35.528 M_\odot$, while the luminosity-distance median is 3612.7 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	19,143
Observed luminosity distance	3612.7 Mpc (median)
Detector-frame masses	$m_1 = 57.212 M_\odot$, $m_2 = 35.528 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.621$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5970$; $z_\pi = 0.6436$ (medians)
Post-handoff reorder fraction	11.27%

Table 452: Observational and coordinate summary for GW240930_234614. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	34.566 [23.733, 48.789]	21.393 [14.312, 30.432]
Physical compact-source mass M_c ($M_\odot$)	27.388 [16.367, 44.271]	14.610 [8.942, 22.497]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.526 [0.699, 15.322]	6.614 [0.791, 13.376]
Geometric share M_{geom}/M_∞ (%)	15.75 [2.03, 46.69]	32.44 [4.10, 49.71]
Exterior spin magnitude $ \hat{a}_\infty $	0.697 [0.137, 0.960]	0.400 [0.038, 0.919]
Polarization factor A_{pol}	1.187 [1.021, 1.876]	1.480 [1.043, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.982 [0.482, 1.000]	0.877 [0.150, 0.999]
GR/CPTG matching radius r_m (km)	51.041 [35.044, 72.044]	31.589 [21.134, 44.937]
Structural Mode N_{BH}	1.5467 [1.4588, 1.8740]	1.7054 [1.4700, 1.9090]
Organization-shell radius r_N (km)	26.843 [18.426, 37.777]	16.627 [11.108, 23.666]
Inner trapping-surface radius r_- (km)	36.892 [24.395, 58.835]	21.154 [13.787, 30.502]
Structural- N lapse $\bar{\alpha}_N$	0.7746 [0.5138, 0.8443]	0.6517 [0.4855, 0.8353]
Structural- N clock $T_{24,N}$ (h)	18.589 [12.331, 20.262]	15.640 [11.652, 20.047]
Structural- N axis ω (rad s^{-1})	1457.85 [657.25, 3291.53]	3078.58 [907.69, 5653.72]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2036.32 [1049.16, 3736.40]	3674.37 [917.86, 6275.18]
Structural- N equatorial ω (rad s^{-1})	2687.65 [1607.47, 4225.13]	4358.20 [918.33, 6934.12]
Structural- N period P_ω (ms)	3.0856 [1.6816, 5.9888]	1.7100 [1.0013, 6.8455]
Trapping-surface limit Ω_- (rad s^{-1})	3609.01 [2174.43, 5579.38]	5631.60 [1001.60, 8951.04]
Trapping-surface period P_- (ms)	1.7410 [1.1261, 2.8896]	1.1157 [0.7020, 6.2731]

Table 453: CPTG compact-state, structural, clock, and geometric-rotation results for GW240930_234614. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.2252	7.9867	3539.62	5429.19
500 m inward	3.6664	5.6444	3575.05	5538.10
100 m inward	1.6289	2.5224	3602.21	5614.42
10 m inward	0.5141	0.7975	3608.17	5631.06
1 m inward	0.1625	0.2522	3608.93	5631.46
10 cm inward	0.0514	0.0798	3609.00	5631.57
1 cm inward	0.0163	0.0252	3609.01	5631.59
surface limit	0.0000	0.0000	3609.01	5631.60

Table 454: Near-inner-boundary profile for GW240930_234614. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.388 M_\odot$ and $14.610 M_\odot$, with completed geometric mass-equivalent shares of 15.75% and 32.44%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.843 < 36.892 < 51.041$ km, and component 2 gives $16.627 < 21.154 < 31.589$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5467$ and 1.7054 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.589 and 15.640 h. The determinant-averaged frame-dragging rates at those shells are 2036.32 and $3674.37 \text{ rad s}^{-1}$, approaching surface-limit values of 3609.01 and $5631.60 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.225/7.987$ h at 1 km inward to $0.0163/0.0252$ h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.70 GW230606_004305

GW230606_004305 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 55.228 M_\odot$ and $m_2 = 38.144 M_\odot$, while the luminosity-distance median is 2757.5 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	41,919
Observed luminosity distance	2757.5 Mpc (median)
Detector-frame masses	$m_1 = 55.228 M_\odot$, $m_2 = 38.144 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.691$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4768$; $z_\pi = 0.5133$ (medians)
Post-handoff reorder fraction	17.22%

Table 455: Observational and coordinate summary for GW230606_004305. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	36.106 [26.264, 49.861]	25.518 [16.999, 36.736]
Physical compact-source mass M_c ($M_\odot$)	27.080 [18.761, 41.108]	17.174 [10.599, 25.619]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.587 [0.821, 19.866]	8.418 [1.120, 16.658]
Geometric share M_{geom}/M_∞ (%)	21.36 [2.47, 48.41]	34.58 [5.10, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.595 [0.092, 0.951]	0.363 [0.033, 0.899]
Polarization factor A_{pol}	1.272 [1.025, 1.938]	1.529 [1.054, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.962 [0.345, 1.000]	0.849 [0.129, 0.999]
GR/CPTG matching radius r_m (km)	53.315 [38.782, 73.627]	37.681 [25.101, 54.245]
Structural Mode N_{BH}	1.5935 [1.4612, 1.8941]	1.7297 [1.4756, 1.9098]
Organization-shell radius r_N (km)	28.052 [20.340, 38.731]	19.837 [13.207, 28.555]
Inner trapping-surface radius r_- (km)	37.458 [27.863, 54.756]	25.051 [16.405, 35.308]
Structural- N lapse $\bar{\alpha}_N$	0.7384 [0.4973, 0.8424]	0.6325 [0.4849, 0.8308]
Structural- N clock $T_{24,N}$ (h)	17.721 [11.936, 20.218]	15.180 [11.637, 19.939]
Structural- N axis ω (rad s^{-1})	1643.09 [683.11, 2909.01]	2642.66 [657.75, 4754.44]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2131.03 [975.34, 3295.38]	3130.85 [658.15, 5284.09]
Structural- N equatorial ω (rad s^{-1})	2670.98 [1194.28, 3739.30]	3664.38 [658.36, 5863.95]
Structural- N period $P_{\bar{\omega}}$ (ms)	2.9484 [1.9067, 6.4420]	2.0069 [1.1891, 9.5467]
Trapping-surface limit Ω_- (rad s^{-1})	3561.60 [1359.44, 4990.52]	4703.65 [717.57, 7578.75]
Trapping-surface period P_- (ms)	1.7641 [1.2590, 4.6219]	1.3358 [0.8291, 8.7561]

Table 456: CPTG compact-state, structural, clock, and geometric-rotation results for GW230606_004305. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5918	7.3940	3494.34	4558.86
500 m inward	3.9383	5.2276	3530.50	4633.97
100 m inward	1.7549	2.3372	3555.43	4690.38
10 m inward	0.5544	0.7390	3560.99	4702.20
1 m inward	0.1753	0.2337	3561.50	4703.52
10 cm inward	0.0554	0.0739	3561.59	4703.63
1 cm inward	0.0175	0.0234	3561.60	4703.65
surface limit	0.0000	0.0000	3561.60	4703.65

Table 457: Near-inner-boundary profile for GW230606_004305. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.080 M_\odot$ and $17.174 M_\odot$, with completed geometric mass-equivalent shares of 21.36% and 34.58%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.052 < 37.458 < 53.315 \text{ km}$, and component 2 gives $19.837 < 25.051 < 37.681 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5935$ and 1.7297 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.721 and 15.180 h. The determinant-averaged frame-dragging rates at those shells are 2131.03 and $3130.85 \text{ rad s}^{-1}$, approaching surface-limit values of 3561.60 and $4703.65 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.592/7.394 \text{ h}$ at 1 km inward to $0.0175/0.0234 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.71 GW240505_133552

GW240505_133552 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 53.824 M_\odot$ and $m_2 = 40.732 M_\odot$, while the luminosity-distance median is 4693.1 Mpc . One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	13,399
Observed luminosity distance	4693.1 Mpc (median)
Detector-frame masses	$m_1 = 53.824 M_\odot$, $m_2 = 40.732 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.757$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7402$; $z_\pi = 0.7996$ (medians)
Post-handoff reorder fraction	22.25%

Table 458: Observational and coordinate summary for GW240505_133552. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	29.186 [21.735, 38.892]	23.048 [15.927, 31.865]
Physical compact-source mass M_c ($M_\odot$)	22.589 [15.575, 32.855]	15.137 [9.754, 22.181]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.473 [0.584, 15.306]	8.079 [1.190, 14.527]
Geometric share M_{geom}/M_∞ (%)	19.13 [2.12, 47.41]	36.29 [6.10, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.636 [0.120, 0.958]	0.334 [0.031, 0.880]
Polarization factor A_{pol}	1.236 [1.022, 1.901]	1.570 [1.065, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.972 [0.433, 1.000]	0.822 [0.122, 0.998]
GR/CPTG matching radius r_m (km)	43.097 [32.094, 57.429]	34.033 [23.518, 47.052]
Structural Mode N_{BH}	1.5741 [1.4593, 1.8824]	1.7496 [1.4814, 1.9101]
Organization-shell radius r_N (km)	22.670 [16.836, 30.274]	17.924 [12.349, 24.767]
Inner trapping-surface radius r_- (km)	30.898 [22.998, 43.832]	22.283 [15.223, 30.526]
Structural- N lapse $\bar{\alpha}_N$	0.7534 [0.5069, 0.8439]	0.6166 [0.4847, 0.8262]
Structural- N clock $T_{24,N}$ (h)	18.081 [12.165, 20.253]	14.798 [11.632, 19.828]
Structural- N axis ω (rad s^{-1})	1938.02 [864.22, 3550.94]	2980.84 [697.16, 5129.97]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2576.76 [1331.36, 4010.93]	3492.45 [697.38, 5689.04]
Structural- N equatorial ω (rad s^{-1})	3281.16 [1810.25, 4535.75]	4070.41 [697.49, 6291.87]
Structural- N period P_ω (ms)	2.4384 [1.5665, 4.7194]	1.7991 [1.1044, 9.0097]
Trapping-surface limit Ω_- (rad s^{-1})	4399.21 [2129.97, 6029.34]	5148.45 [759.87, 8129.78]
Trapping-surface period P_- (ms)	1.4283 [1.0421, 2.9499]	1.2204 [0.7729, 8.2688]

Table 459: CPTG compact-state, structural, clock, and geometric-rotation results for GW240505_133552. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.0263	7.9282	4294.95	4956.81
500 m inward	4.2337	5.6116	4351.89	5059.09
100 m inward	1.8833	2.5106	4390.44	5129.25
10 m inward	0.5947	0.7939	4398.06	5145.99
1 m inward	0.1880	0.2510	4399.08	5148.29
10 cm inward	0.0595	0.0794	4399.20	5148.44
1 cm inward	0.0188	0.0251	4399.21	5148.45
surface limit	0.0000	0.0000	4399.21	5148.45

Table 460: Near-inner-boundary profile for GW240505_133552. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $22.589 M_\odot$ and $15.137 M_\odot$, with completed geometric mass-equivalent shares of 19.13% and 36.29%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $22.670 < 30.898 < 43.097 \text{ km}$, and component 2 gives $17.924 < 22.283 < 34.033 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5741$ and 1.7496 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.081 and 14.798 h. The determinant-averaged frame-dragging rates at those shells are 2576.76 and $3492.45 \text{ rad s}^{-1}$, approaching surface-limit values of 4399.21 and $5148.45 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 6.026/7.928 h at 1 km inward to 0.0188/0.0251 h at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.72 GW230609_064958

GW230609_064958 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 55.076 M_\odot$ and $m_2 = 39.739 M_\odot$, while the luminosity-distance median is 3357.8 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	40,845
Observed luminosity distance	3357.8 Mpc (median)
Detector-frame masses	$m_1 = 55.076 M_\odot$, $m_2 = 39.739 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.722$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5619$; $z_\pi = 0.6055$ (medians)
Post-handoff reorder fraction	19.60%

Table 461: Observational and coordinate summary for GW230609_064958. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.771 [24.515, 45.091]	25.118 [17.247, 35.355]
Physical compact-source mass M_c ($M_\odot$)	24.940 [17.450, 37.084]	16.577 [10.691, 24.093]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.483 [0.855, 18.524]	8.762 [1.217, 16.171]
Geometric share M_{geom}/M_∞ (%)	22.59 [2.78, 48.49]	36.06 [5.70, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.573 [0.089, 0.945]	0.338 [0.030, 0.888]
Polarization factor A_{pol}	1.292 [1.029, 1.942]	1.564 [1.060, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.956 [0.337, 1.000]	0.826 [0.120, 0.998]
GR/CPTG matching radius r_m (km)	49.868 [36.199, 66.583]	37.090 [25.468, 52.207]
Structural Mode N_{BH}	1.6047 [1.4628, 1.8950]	1.7469 [1.4790, 1.9102]
Organization-shell radius r_N (km)	26.248 [18.991, 35.049]	19.540 [13.392, 27.468]
Inner trapping-surface radius r_- (km)	34.712 [25.883, 49.378]	24.344 [16.610, 33.418]
Structural- N lapse $\bar{\alpha}_N$	0.7298 [0.4966, 0.8411]	0.6187 [0.4846, 0.8280]
Structural- N clock $T_{24,N}$ (h)	17.514 [11.917, 20.186]	14.849 [11.630, 19.872]
Structural- N axis ω (rad s^{-1})	1829.47 [757.59, 3151.81]	2738.88 [610.19, 4728.29]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2336.43 [1099.98, 3564.01]	3210.35 [610.79, 5266.29]
Structural- N equatorial ω (rad s^{-1})	2882.83 [1275.27, 4037.22]	3749.75 [611.18, 5838.27]
Structural- N period P_ω (ms)	2.6892 [1.7630, 5.7121]	1.9572 [1.1931, 10.2871]
Trapping-surface limit Ω_- (rad s^{-1})	3817.43 [1406.64, 5368.80]	4765.11 [666.87, 7537.83]
Trapping-surface period P_- (ms)	1.6459 [1.1703, 4.4668]	1.3186 [0.8336, 9.4219]

Table 462: CPTG compact-state, structural, clock, and geometric-rotation results for GW230609_064958. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8887	7.5578	3738.87	4602.88
500 m inward	4.1464	5.3482	3780.01	4689.53
100 m inward	1.8476	2.3924	3810.62	4751.67
10 m inward	0.5838	0.7565	3816.76	4763.65
1 m inward	0.1846	0.2392	3817.33	4764.94
10 cm inward	0.0584	0.0757	3817.42	4765.10
1 cm inward	0.0185	0.0239	3817.43	4765.11
surface limit	0.0000	0.0000	3817.43	4765.11

Table 463: Near-inner-boundary profile for GW230609_064958. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.940 M_\odot$ and $16.577 M_\odot$, with completed geometric mass-equivalent shares of 22.59% and 36.06%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.248 < 34.712 < 49.868 \text{ km}$, and component 2 gives $19.540 < 24.344 < 37.090 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6047$ and 1.7469 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.514 and 14.849 h. The determinant-averaged frame-dragging rates at those shells are 2336.43 and $3210.35 \text{ rad s}^{-1}$, approaching surface-limit values of 3817.43 and $4765.11 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 5.889/7.558 h at 1 km inward to 0.0185/0.0239 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.73 GW231026_130704

GW231026_130704 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 59.066 M_\odot$ and $m_2 = 36.124 M_\odot$, while the luminosity-distance median is 4433.9 Mpc . One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,361
Observed luminosity distance	4433.9 Mpc (median)
Detector-frame masses	$m_1 = 59.066 M_\odot$, $m_2 = 36.124 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.612$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7066$; $z_\pi = 0.7629$ (medians)
Post-handoff reorder fraction	12.99%

Table 464: Observational and coordinate summary for GW231026_130704. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	32.560 [20.774, 59.152]	20.457 [12.191, 31.669]
Physical compact-source mass M_c ($M_\odot$)	24.868 [15.029, 49.125]	13.925 [7.795, 23.032]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.151 [0.627, 18.131]	6.259 [0.785, 13.711]
Geometric share M_{geom}/M_∞ (%)	18.89 [2.03, 47.91]	32.51 [4.34, 49.71]
Exterior spin magnitude $ \hat{a}_\infty $	0.640 [0.106, 0.960]	0.399 [0.038, 0.914]
Polarization factor A_{pol}	1.233 [1.021, 1.920]	1.482 [1.045, 1.988]
Compact-source spin parameter $\hat{a}_c$	0.972 [0.392, 1.000]	0.876 [0.152, 0.999]
GR/CPTG matching radius r_m (km)	48.079 [30.675, 87.345]	30.207 [18.002, 46.763]
Structural Mode N_{BH}	1.5721 [1.4589, 1.8883]	1.7062 [1.4713, 1.9089]
Organization-shell radius r_N (km)	25.291 [16.137, 45.820]	15.904 [9.474, 24.622]
Inner trapping-surface radius r_- (km)	34.187 [21.895, 65.954]	20.169 [11.815, 31.707]
Structural- N lapse $\bar{\alpha}_N$	0.7549 [0.5020, 0.8443]	0.6511 [0.4856, 0.8342]
Structural- N clock $T_{24,N}$ (h)	18.117 [12.049, 20.262]	15.626 [11.653, 20.021]
Structural- N axis ω (rad s^{-1})	1670.14 [614.56, 3561.01]	3205.40 [904.21, 6435.89]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2247.55 [913.68, 4097.33]	3824.30 [940.50, 7238.70]
Structural- N equatorial ω (rad s^{-1})	2886.12 [1206.63, 4728.21]	4526.76 [940.98, 8089.27]
Structural- N period P_ω (ms)	2.7956 [1.5335, 6.8768]	1.6430 [0.8680, 6.6807]
Trapping-surface limit Ω_- (rad s^{-1})	3841.00 [1522.57, 6309.18]	5797.21 [1026.28, 10389.20]
Trapping-surface period P_- (ms)	1.6358 [0.9959, 4.1267]	1.0838 [0.6048, 6.1223]

Table 465: CPTG compact-state, structural, clock, and geometric-rotation results for GW231026_130704. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6337	8.1387	3767.66	5563.68
500 m inward	3.9553	5.7493	3804.44	5693.98
100 m inward	1.7585	2.5689	3834.29	5776.44
10 m inward	0.5553	0.8118	3840.10	5794.81
1 m inward	0.1756	0.2567	3840.91	5796.91
10 cm inward	0.0555	0.0812	3840.99	5797.18
1 cm inward	0.0176	0.0257	3841.00	5797.21
surface limit	0.0000	0.0000	3841.00	5797.21

Table 466: Near-inner-boundary profile for GW231026_130704. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.868 M_\odot$ and $13.925 M_\odot$, with completed geometric mass-equivalent shares of 18.89% and 32.51%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.291 < 34.187 < 48.079 \text{ km}$, and component 2 gives $15.904 < 20.169 < 30.207 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5721$ and 1.7062 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.117 and 15.626 h. The determinant-averaged frame-dragging rates at those shells are 2247.55 and $3824.30 \text{ rad s}^{-1}$, approaching surface-limit values of 3841.00 and $5797.21 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.634/8.139 \text{ h}$ at 1 km inward to $0.0176/0.0257 \text{ h}$ at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.74 GW200224_222234

GW200224_222234 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 52.279 M_\odot$ and $m_2 = 42.995 M_\odot$, while the luminosity-distance median is 1705.7 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	6,348
Observed luminosity distance	1705.7 Mpc (median)
Detector-frame masses	$m_1 = 52.279 M_\odot$, $m_2 = 42.995 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.822$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3173$; $z_\pi = 0.3407$ (medians)
Post-handoff reorder fraction	25.55%

Table 467: Observational and coordinate summary for GW200224_222234. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	38.515 [30.826, 45.926]	33.180 [25.070, 40.180]
Physical compact-source mass M_c ($M_\odot$)	29.009 [20.675, 39.942]	20.686 [14.688, 28.975]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.365 [1.219, 18.971]	12.416 [2.879, 18.705]
Geometric share M_{geom}/M_∞ (%)	22.01 [3.24, 47.35]	38.29 [9.80, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.584 [0.121, 0.936]	0.299 [0.029, 0.809]
Polarization factor A_{pol}	1.282 [1.033, 1.900]	1.621 [1.109, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.959 [0.437, 0.999]	0.784 [0.117, 0.994]
GR/CPTG matching radius r_m (km)	56.872 [45.519, 67.815]	48.994 [37.019, 59.331]
Structural Mode N_{BH}	1.5993 [1.4653, 1.8818]	1.7733 [1.5043, 1.9103]
Organization-shell radius r_N (km)	29.976 [23.883, 35.671]	25.801 [19.509, 31.259]
Inner trapping-surface radius r_- (km)	39.487 [31.487, 53.027]	31.119 [23.323, 38.913]
Structural- N lapse $\bar{\alpha}_N$	0.7339 [0.5074, 0.8391]	0.5974 [0.4845, 0.8079]
Structural- N clock $T_{24,N}$ (h)	17.613 [12.177, 20.138]	14.338 [11.629, 19.389]
Structural- N axis ω (rad s^{-1})	1654.60 [715.99, 2670.37]	2288.15 [478.65, 3400.13]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2134.78 [1100.39, 3001.38]	2601.21 [478.86, 3748.28]
Structural- N equatorial ω (rad s^{-1})	2590.68 [1516.50, 3325.50]	2905.75 [478.97, 4077.16]
Structural- N period P_ω (ms)	2.9432 [2.0934, 5.7100]	2.4155 [1.6763, 13.1211]
Trapping-surface limit Ω_- (rad s^{-1})	3424.33 [1720.23, 4343.11]	3658.26 [521.92, 5215.52]
Trapping-surface period P_- (ms)	1.8349 [1.4467, 3.6525]	1.7175 [1.2047, 12.0387]

Table 468: CPTG compact-state, structural, clock, and geometric-rotation results for GW200224_222234. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.4880	6.8718	3356.74	3555.30
500 m inward	3.8685	4.8657	3393.99	3609.86
100 m inward	1.7248	2.1784	3418.04	3648.57
10 m inward	0.5448	0.6890	3423.72	3657.34
1 m inward	0.1723	0.2179	3424.27	3658.18
10 cm inward	0.0545	0.0689	3424.32	3658.25
1 cm inward	0.0172	0.0218	3424.33	3658.26
surface limit	0.0000	0.0000	3424.33	3658.26

Table 469: Near-inner-boundary profile for GW200224_222234. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $29.009 M_\odot$ and $20.686 M_\odot$, with completed geometric mass-equivalent shares of 22.01% and 38.29%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $29.976 < 39.487 < 56.872 \text{ km}$, and component 2 gives $25.801 < 31.119 < 48.994 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5993$ and 1.7733 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.613 and 14.338 h. The determinant-averaged frame-dragging rates at those shells are 2134.78 and $2601.21 \text{ rad s}^{-1}$, approaching surface-limit values of 3424.33 and $3658.26 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.488/6.872 \text{ h}$ at 1 km inward to $0.0172/0.0218 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.75 GW230927_043729

GW230927_043729 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 53.443 M_\odot$ and $m_2 = 42.107 M_\odot$, while the luminosity-distance median is 3237.1 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	39,195
Observed luminosity distance	3237.1 Mpc (median)
Detector-frame masses	$m_1 = 53.443 M_\odot$, $m_2 = 42.107 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.788$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5451$; $z_\pi = 0.5872$ (medians)
Post-handoff reorder fraction	25.64%

Table 470: Observational and coordinate summary for GW230927_043729. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.076 [24.942, 42.472]	27.471 [20.033, 36.812]
Physical compact-source mass M_c ($M_\odot$)	23.899 [16.720, 35.079]	17.193 [11.842, 25.237]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.202 [0.954, 17.882]	10.546 [2.120, 16.972]
Geometric share M_{geom}/M_∞ (%)	25.28 [3.15, 48.87]	39.61 [8.76, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.525 [0.077, 0.937]	0.275 [0.026, 0.829]
Polarization factor A_{pol}	1.338 [1.033, 1.956]	1.656 [1.096, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.941 [0.294, 0.999]	0.755 [0.102, 0.995]
GR/CPTG matching radius r_m (km)	48.841 [36.830, 62.715]	40.564 [29.582, 54.358]
Structural Mode N_{BH}	1.6302 [1.4648, 1.8994]	1.7891 [1.4976, 1.9108]
Organization-shell radius r_N (km)	25.719 [19.340, 33.028]	21.361 [15.572, 28.629]
Inner trapping-surface radius r_- (km)	33.413 [25.462, 46.677]	25.829 [18.780, 34.925]
Structural- N lapse $\bar{\alpha}_N$	0.7101 [0.4931, 0.8395]	0.5845 [0.4842, 0.8132]
Structural- N clock $T_{24,N}$ (h)	17.042 [11.834, 20.147]	14.027 [11.620, 19.516]
Structural- N axis ω (rad s^{-1})	1977.53 [778.94, 3255.73]	2596.78 [489.33, 4255.58]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2455.44 [1098.79, 3650.29]	2958.67 [489.56, 4677.95]
Structural- N equatorial ω (rad s^{-1})	2953.52 [1190.14, 4075.04]	3357.15 [489.68, 5115.83]
Structural- N period $P_{\bar{\omega}}$ (ms)	2.5589 [1.7213, 5.7183]	2.1237 [1.3432, 12.8343]
Trapping-surface limit Ω_- (rad s^{-1})	3873.12 [1304.66, 5341.19]	4154.24 [533.60, 6526.22]
Trapping-surface period P_- (ms)	1.6223 [1.1764, 4.8159]	1.5125 [0.9628, 11.7750]

Table 471: CPTG compact-state, structural, clock, and geometric-rotation results for GW230927_043729. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1370	7.4921	3788.39	4015.79
500 m inward	4.3259	5.3060	3833.39	4087.77
100 m inward	1.9294	2.3750	3865.78	4141.92
10 m inward	0.6098	0.7512	3872.53	4152.75
1 m inward	0.1928	0.2375	3873.07	4154.08
10 cm inward	0.0610	0.0751	3873.11	4154.22
1 cm inward	0.0193	0.0238	3873.12	4154.24
surface limit	0.0000	0.0000	3873.12	4154.24

Table 472: Near-inner-boundary profile for GW230927_043729. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.899 M_\odot$ and $17.193 M_\odot$, with completed geometric mass-equivalent shares of 25.28% and 39.61%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.719 < 33.413 < 48.841$ km, and component 2 gives $21.361 < 25.829 < 40.564$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6302$ and 1.7891 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.042 and 14.027 h. The determinant-averaged frame-dragging rates at those shells are 2455.44 and $2958.67 \text{ rad s}^{-1}$, approaching surface-limit values of 3873.12 and $4154.24 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.137/7.492$ h at 1 km inward to $0.0193/0.0238$ h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.76 GW250109_074552

GW250109_074552 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 54.733 M_\odot$ and $m_2 = 43.000 M_\odot$, while the luminosity-distance median is 2311.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,817
Observed luminosity distance	2311.3 Mpc (median)
Detector-frame masses	$m_1 = 54.733 M_\odot$, $m_2 = 43.000 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.786$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4109$; $z_\pi = 0.4421$ (medians)
Post-handoff reorder fraction	24.40%

Table 473: Observational and coordinate summary for GW250109_074552. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	37.115 [28.328, 46.311]	30.558 [22.567, 39.970]
Physical compact-source mass M_c ($M_\odot$)	27.367 [18.960, 39.170]	19.268 [13.237, 28.013]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.476 [1.012, 19.494]	11.576 [2.275, 18.376]
Geometric share M_{geom}/M_∞ (%)	23.38 [2.92, 48.52]	38.86 [8.42, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.559 [0.088, 0.942]	0.289 [0.026, 0.835]
Polarization factor A_{pol}	1.305 [1.030, 1.943]	1.636 [1.092, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.952 [0.334, 1.000]	0.772 [0.102, 0.996]
GR/CPTG matching radius r_m (km)	54.805 [41.830, 68.383]	45.122 [33.322, 59.020]
Structural Mode N_{BH}	1.6120 [1.4636, 1.8954]	1.7800 [1.4954, 1.9107]
Organization-shell radius r_N (km)	28.865 [21.944, 36.013]	23.781 [17.537, 31.058]
Inner trapping-surface radius r_- (km)	37.941 [28.916, 52.001]	28.888 [21.077, 38.279]
Structural- N lapse $\bar{\alpha}_N$	0.7241 [0.4963, 0.8405]	0.5919 [0.4842, 0.8149]
Structural- N clock $T_{24,N}$ (h)	17.379 [11.911, 20.171]	14.206 [11.620, 19.558]
Structural- N axis ω (rad s^{-1})	1709.99 [709.68, 2853.46]	2339.63 [449.69, 3765.82]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2166.83 [1048.21, 3206.27]	2672.46 [449.84, 4163.57]
Structural- N equatorial ω (rad s^{-1})	2632.84 [1209.00, 3574.52]	3041.48 [449.92, 4539.71]
Structural- N period P_ω (ms)	2.8997 [1.9597, 5.9942]	2.3511 [1.5091, 13.9676]
Trapping-surface limit Ω_- (rad s^{-1})	3475.33 [1330.67, 4689.20]	3780.47 [490.17, 5787.51]
Trapping-surface period P_- (ms)	1.8079 [1.3399, 4.7218]	1.6620 [1.0856, 12.8183]

Table 474: CPTG compact-state, structural, clock, and geometric-rotation results for GW250109_074552. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6813	7.0814	3405.37	3665.67
500 m inward	4.0058	5.0156	3441.50	3727.66
100 m inward	1.7865	2.2447	3468.52	3770.94
10 m inward	0.5647	0.7099	3474.66	3779.79
1 m inward	0.1786	0.2245	3475.29	3780.40
10 cm inward	0.0565	0.0710	3475.33	3780.46
1 cm inward	0.0179	0.0225	3475.33	3780.47
surface limit	0.0000	0.0000	3475.33	3780.47

Table 475: Near-inner-boundary profile for GW250109_074552. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.367 M_\odot$ and $19.268 M_\odot$, with completed geometric mass-equivalent shares of 23.38% and 38.86%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.865 < 37.941 < 54.805 \text{ km}$, and component 2 gives $23.781 < 28.888 < 45.122 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6120$ and 1.7800 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.379 and 14.206 h. The determinant-averaged frame-dragging rates at those shells are 2166.83 and $2672.46 \text{ rad s}^{-1}$, approaching surface-limit values of 3475.33 and $3780.47 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.681/7.081 \text{ h}$ at 1 km inward to $0.0179/0.0225 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.77 GW231206_233134

GW231206_233134 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 54.227 M_\odot$ and $m_2 = 43.682 M_\odot$, while the luminosity-distance median is 3182.3 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	39,501
Observed luminosity distance	3182.3 Mpc (median)
Detector-frame masses	$m_1 = 54.227 M_\odot$, $m_2 = 43.682 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.806$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5374$; $z_\pi = 0.5789$ (medians)
Post-handoff reorder fraction	25.17%

Table 476: Observational and coordinate summary for GW231206_233134. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.746 [25.637, 43.583]	28.514 [20.537, 37.650]
Physical compact-source mass M_c ($M_\odot$)	26.334 [17.513, 38.236]	18.220 [12.362, 27.716]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.196 [0.674, 16.693]	10.200 [1.702, 16.953]
Geometric share M_{geom}/M_∞ (%)	18.62 [2.04, 47.02]	37.04 [6.59, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.645 [0.129, 0.959]	0.321 [0.029, 0.870]
Polarization factor A_{pol}	1.229 [1.021, 1.888]	1.588 [1.071, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.973 [0.461, 1.000]	0.809 [0.116, 0.998]
GR/CPTG matching radius r_m (km)	49.830 [37.856, 64.356]	42.105 [30.325, 55.595]
Structural Mode N_{BH}	1.5698 [1.4589, 1.8779]	1.7584 [1.4842, 1.9103]
Organization-shell radius r_N (km)	26.207 [19.892, 33.829]	22.174 [15.962, 29.288]
Inner trapping-surface radius r_- (km)	35.785 [26.333, 50.860]	27.114 [19.458, 37.451]
Structural- N lapse $\bar{\alpha}_N$	0.7567 [0.5106, 0.8442]	0.6095 [0.4845, 0.8238]
Structural- N clock $T_{24,N}$ (h)	18.160 [12.253, 20.262]	14.627 [11.628, 19.772]
Structural- N axis ω (rad s^{-1})	1666.35 [741.33, 3137.47]	2471.38 [544.47, 4073.84]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2225.41 [1169.83, 3513.80]	2853.27 [544.73, 4505.91]
Structural- N equatorial ω (rad s^{-1})	2826.68 [1713.68, 3936.50]	3277.92 [544.86, 4949.09]
Structural- N period P_ω (ms)	2.8234 [1.7881, 5.3710]	2.2021 [1.3944, 11.5345]
Trapping-surface limit Ω_- (rad s^{-1})	3789.18 [2070.41, 5178.76]	4142.77 [593.74, 6348.72]
Trapping-surface period P_- (ms)	1.6582 [1.2133, 3.0348]	1.5167 [0.9897, 10.5824]

Table 477: CPTG compact-state, structural, clock, and geometric-rotation results for GW231206_233134. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5732	7.2628	3715.10	4015.84
500 m inward	3.9173	5.1394	3754.55	4082.25
100 m inward	1.7427	2.2999	3781.96	4131.13
10 m inward	0.5504	0.7273	3788.54	4141.45
1 m inward	0.1740	0.2300	3789.13	4142.65
10 cm inward	0.0550	0.0727	3789.17	4142.76
1 cm inward	0.0174	0.0230	3789.18	4142.77
surface limit	0.0000	0.0000	3789.18	4142.77

Table 478: Near-inner-boundary profile for GW231206_233134. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.334 M_\odot$ and $18.220 M_\odot$, with completed geometric mass-equivalent shares of 18.62% and 37.04%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.207 < 35.785 < 49.830 \text{ km}$, and component 2 gives $22.174 < 27.114 < 42.105 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5698$ and 1.7584 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.160 and 14.627 h. The determinant-averaged frame-dragging rates at those shells are 2225.41 and $2853.27 \text{ rad s}^{-1}$, approaching surface-limit values of 3789.18 and $4142.77 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.573/7.263 \text{ h}$ at 1 km inward to $0.0174/0.0230 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.78 GW200209_085452

GW200209_085452 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 55.702 M_\odot$ and $m_2 = 42.950 M_\odot$, while the luminosity-distance median is 3418.9 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	9,015
Observed luminosity distance	3418.9 Mpc (median)
Detector-frame masses	$m_1 = 55.702 M_\odot$, $m_2 = 42.950 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.771$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5710$; $z_\pi = 0.6147$ (medians)
Post-handoff reorder fraction	23.48%

Table 479: Observational and coordinate summary for GW200209_085452. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.700 [25.248, 44.846]	27.241 [18.860, 37.022]
Physical compact-source mass M_c ($M_\odot$)	26.527 [18.120, 38.650]	17.938 [11.656, 26.155]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.887 [0.544, 17.376]	9.477 [1.311, 16.921]
Geometric share M_{geom}/M_∞ (%)	17.72 [1.67, 46.63]	35.73 [5.50, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.661 [0.139, 0.967]	0.343 [0.031, 0.892]
Polarization factor A_{pol}	1.215 [1.017, 1.874]	1.556 [1.058, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.977 [0.487, 1.000]	0.831 [0.122, 0.998]
GR/CPTG matching radius r_m (km)	49.762 [37.282, 66.221]	40.225 [27.849, 54.668]
Structural Mode N_{BH}	1.5624 [1.4570, 1.8732]	1.7430 [1.4778, 1.9101]
Organization-shell radius r_N (km)	26.159 [19.548, 34.799]	21.192 [14.654, 28.802]
Inner trapping-surface radius r_- (km)	36.075 [26.749, 51.671]	26.404 [18.207, 35.857]
Structural- N lapse $\bar{\alpha}_N$	0.7624 [0.5144, 0.8458]	0.6218 [0.4847, 0.8290]
Structural- N clock $T_{24,N}$ (h)	18.298 [12.347, 20.298]	14.924 [11.632, 19.895]
Structural- N axis ω (rad s^{-1})	1608.84 [728.68, 3066.32]	2552.98 [572.28, 4307.71]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2183.66 [1143.52, 3460.34]	3001.73 [572.86, 4777.54]
Structural- N equatorial ω (rad s^{-1})	2817.35 [1664.86, 3906.09]	3480.88 [573.17, 5289.19]
Structural- N period P_ω (ms)	2.8774 [1.8158, 5.4946]	2.0932 [1.3151, 10.9681]
Trapping-surface limit Ω_- (rad s^{-1})	3799.68 [2047.22, 5198.16]	4419.27 [625.22, 6806.96]
Trapping-surface period P_- (ms)	1.6536 [1.2087, 3.0691]	1.4218 [0.9231, 10.0496]

Table 480: CPTG compact-state, structural, clock, and geometric-rotation results for GW200209_085452. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.4922	7.2769	3738.08	4278.74
500 m inward	3.8602	5.1492	3768.98	4354.56
100 m inward	1.7165	2.3039	3793.27	4407.86
10 m inward	0.5419	0.7287	3799.07	4418.78
1 m inward	0.1713	0.2304	3799.70	4419.16
10 cm inward	0.0542	0.0729	3799.68	4419.26
1 cm inward	0.0171	0.0230	3799.68	4419.27
surface limit	0.0000	0.0000	3799.68	4419.27

Table 481: Near-inner-boundary profile for GW200209_085452. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.527 M_\odot$ and $17.938 M_\odot$, with completed geometric mass-equivalent shares of 17.72% and 35.73%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.159 < 36.075 < 49.762 \text{ km}$, and component 2 gives $21.192 < 26.404 < 40.225 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5624$ and 1.7430 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.298 and 14.924 h. The determinant-averaged frame-dragging rates at those shells are 2183.66 and $3001.73 \text{ rad s}^{-1}$, approaching surface-limit values of 3799.68 and $4419.27 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.492/7.277 \text{ h}$ at 1 km inward to $0.0171/0.0230 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.79 GW240511_031507

GW240511_031507 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 54.883 M_\odot$ and $m_2 = 43.844 M_\odot$, while the luminosity-distance median is 1990.6 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,800
Observed luminosity distance	1990.6 Mpc (median)
Detector-frame masses	$m_1 = 54.883 M_\odot$, $m_2 = 43.844 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.799$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3619$; $z_\pi = 0.3892$ (medians)
Post-handoff reorder fraction	24.75%

Table 482: Observational and coordinate summary for GW240511_031507. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	38.754 [30.594, 46.940]	32.479 [25.052, 41.073]
Physical compact-source mass M_c ($M_\odot$)	27.644 [19.992, 39.132]	20.179 [14.583, 28.610]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.236 [1.338, 20.366]	12.766 [2.636, 19.230]
Geometric share M_{geom}/M_∞ (%)	26.76 [3.72, 48.89]	40.04 [9.17, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.499 [0.076, 0.926]	0.268 [0.024, 0.821]
Polarization factor A_{pol}	1.365 [1.039, 1.957]	1.668 [1.101, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.931 [0.291, 0.999]	0.744 [0.095, 0.995]
GR/CPTG matching radius r_m (km)	57.224 [45.175, 69.313]	47.960 [36.993, 60.650]
Structural Mode N_{BH}	1.6449 [1.4679, 1.8996]	1.7942 [1.5002, 1.9110]
Organization-shell radius r_N (km)	30.160 [23.724, 36.540]	25.251 [19.482, 31.891]
Inner trapping-surface radius r_- (km)	38.729 [30.869, 51.896]	30.459 [23.212, 38.896]
Structural- N lapse $\bar{\alpha}_N$	0.6987 [0.4929, 0.8370]	0.5803 [0.4840, 0.8111]
Structural- N clock $T_{24,N}$ (h)	16.770 [11.829, 20.088]	13.926 [11.616, 19.467]
Structural- N axis ω (rad s^{-1})	1763.29 [697.73, 2725.32]	2231.95 [381.65, 3451.93]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2150.03 [974.98, 3034.83]	2541.36 [381.75, 3790.47]
Structural- N equatorial ω (rad s^{-1})	2547.41 [1011.20, 3361.39]	2862.01 [381.80, 4136.15]
Structural- N period P_ω (ms)	2.9224 [2.0704, 6.4445]	2.4724 [1.6576, 16.4588]
Trapping-surface limit Ω_- (rad s^{-1})	3333.25 [1107.47, 4377.01]	3549.54 [415.91, 5297.06]
Trapping-surface period P_- (ms)	1.8850 [1.4355, 5.6734]	1.7701 [1.1862, 15.1071]

Table 483: CPTG compact-state, structural, clock, and geometric-rotation results for GW240511_031507. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.7548	6.9387	3267.90	3450.92
500 m inward	4.0609	4.9151	3301.86	3503.54
100 m inward	1.8129	2.2001	3327.28	3541.39
10 m inward	0.5730	0.6959	3332.64	3548.72
1 m inward	0.1812	0.2201	3333.18	3549.44
10 cm inward	0.0573	0.0696	3333.24	3549.53
1 cm inward	0.0181	0.0220	3333.25	3549.54
surface limit	0.0000	0.0000	3333.25	3549.54

Table 484: Near-inner-boundary profile for GW240511_031507. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.644 M_\odot$ and $20.179 M_\odot$, with completed geometric mass-equivalent shares of 26.76% and 40.04%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $30.160 < 38.729 < 57.224 \text{ km}$, and component 2 gives $25.251 < 30.459 < 47.960 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6449$ and 1.7942 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.770 and 13.926 h. The determinant-averaged frame-dragging rates at those shells are 2150.03 and $2541.36 \text{ rad s}^{-1}$, approaching surface-limit values of 3333.25 and $3549.54 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.755/6.939 \text{ h}$ at 1 km inward to $0.0181/0.0220 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.80 GW240104_164932

GW240104_164932 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 56.909 M_\odot$ and $m_2 = 43.687 M_\odot$, while the luminosity-distance median is 1907.1 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	41,535
Observed luminosity distance	1907.1 Mpc (median)
Detector-frame masses	$m_1 = 56.909 M_\odot$, $m_2 = 43.687 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.768$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3488$; $z_\pi = 0.3752$ (medians)
Post-handoff reorder fraction	21.74%

Table 485: Observational and coordinate summary for GW240104_164932. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.656 [31.109, 50.832]	32.416 [23.481, 42.660]
Physical compact-source mass M_c ($M_\odot$)	31.818 [21.716, 44.403]	21.115 [14.023, 30.944]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.516 [0.912, 19.427]	11.405 [1.923, 19.293]
Geometric share M_{geom}/M_∞ (%)	18.78 [2.34, 45.94]	36.13 [6.70, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.642 [0.154, 0.953]	0.336 [0.032, 0.868]
Polarization factor A_{pol}	1.231 [1.024, 1.850]	1.566 [1.072, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.973 [0.527, 1.000]	0.825 [0.129, 0.997]
GR/CPTG matching radius r_m (km)	60.033 [45.936, 75.060]	47.867 [34.672, 62.993]
Structural Mode N_{BH}	1.5711 [1.4605, 1.8652]	1.7477 [1.4849, 1.9099]
Organization-shell radius r_N (km)	31.593 [24.108, 39.457]	25.216 [18.258, 33.171]
Inner trapping-surface radius r_- (km)	43.089 [32.445, 58.975]	31.184 [22.059, 41.777]
Structural- N lapse $\bar{\alpha}_N$	0.7556 [0.5211, 0.8429]	0.6181 [0.4848, 0.8233]
Structural- N clock $T_{24,N}$ (h)	18.135 [12.508, 20.231]	14.834 [11.636, 19.759]
Structural- N axis ω (rad s^{-1})	1411.18 [651.92, 2544.54]	2206.54 [537.98, 3593.25]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1887.15 [1030.40, 2869.73]	2546.82 [538.42, 3967.77]
Structural- N equatorial ω (rad s^{-1})	2387.09 [1541.97, 3231.55]	2918.52 [538.67, 4341.69]
Structural- N period P_ω (ms)	3.3295 [2.1895, 6.0978]	2.4671 [1.5836, 11.6697]
Trapping-surface limit Ω_- (rad s^{-1})	3195.30 [1973.99, 4253.87]	3698.91 [587.28, 5551.82]
Trapping-surface period P_- (ms)	1.9664 [1.4771, 3.1830]	1.6987 [1.1317, 10.6988]

Table 486: CPTG compact-state, structural, clock, and geometric-rotation results for GW240104_164932. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0752	6.7434	3141.84	3606.07
500 m inward	3.5715	4.7719	3170.52	3654.74
100 m inward	1.5902	2.1346	3190.59	3690.85
10 m inward	0.5024	0.6750	3194.75	3698.11
1 m inward	0.1589	0.2135	3195.27	3698.81
10 cm inward	0.0502	0.0675	3195.30	3698.90
1 cm inward	0.0159	0.0213	3195.30	3698.91
surface limit	0.0000	0.0000	3195.30	3698.91

Table 487: Near-inner-boundary profile for GW240104_164932. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.818 M_\odot$ and $21.115 M_\odot$, with completed geometric mass-equivalent shares of 18.78% and 36.13%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.593 < 43.089 < 60.033 \text{ km}$, and component 2 gives $25.216 < 31.184 < 47.867 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5711$ and 1.7477 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.135 and 14.834 h. The determinant-averaged frame-dragging rates at those shells are 1887.15 and $2546.82 \text{ rad s}^{-1}$, approaching surface-limit values of 3195.30 and $3698.91 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.075/6.743 \text{ h}$ at 1 km inward to $0.0159/0.0213 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.81 GW190803_022701

GW190803_022701 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 57.599 M_\odot$ and $m_2 = 43.089 M_\odot$, while the luminosity-distance median is 3188.6 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	20,702
Observed luminosity distance	3188.6 Mpc (median)
Detector-frame masses	$m_1 = 57.599 M_\odot$, $m_2 = 43.089 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.748$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5389$; $z_\pi = 0.5799$ (medians)
Post-handoff reorder fraction	20.59%

Table 488: Observational and coordinate summary for GW190803_022701. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.852 [26.756, 46.455]	27.687 [18.709, 37.728]
Physical compact-source mass M_c ($M_\odot$)	26.154 [18.177, 38.957]	17.714 [11.577, 26.163]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.455 [0.995, 19.330]	10.015 [1.538, 17.395]
Geometric share M_{geom}/M_∞ (%)	23.95 [2.92, 49.00]	37.64 [6.51, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.549 [0.072, 0.942]	0.310 [0.028, 0.872]
Polarization factor A_{pol}	1.315 [1.030, 1.961]	1.604 [1.070, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.949 [0.278, 1.000]	0.797 [0.110, 0.998]
GR/CPTG matching radius r_m (km)	52.941 [39.509, 68.597]	40.883 [27.626, 55.710]
Structural Mode N_{BH}	1.6174 [1.4636, 1.9008]	1.7655 [1.4838, 1.9105]
Organization-shell radius r_N (km)	27.868 [20.765, 36.074]	21.530 [14.542, 29.299]
Inner trapping-surface radius r_- (km)	36.401 [27.598, 51.757]	26.406 [18.012, 35.978]
Structural- N lapse $\bar{\alpha}_N$	0.7200 [0.4919, 0.8405]	0.6038 [0.4844, 0.8242]
Structural- N clock $T_{24,N}$ (h)	17.279 [11.807, 20.171]	14.490 [11.625, 19.781]
Structural- N axis ω (rad s^{-1})	1764.65 [687.97, 2989.76]	2532.49 [535.49, 4313.32]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2221.33 [965.83, 3355.01]	2931.90 [535.90, 4807.65]
Structural- N equatorial ω (rad s^{-1})	2710.67 [1022.89, 3774.05]	3372.66 [536.10, 5325.73]
Structural- N period P_ω (ms)	2.8286 [1.8728, 6.5054]	2.1430 [1.3069, 11.7246]
Trapping-surface limit Ω_- (rad s^{-1})	3573.07 [1123.22, 4965.28]	4238.99 [584.46, 6894.06]
Trapping-surface period P_- (ms)	1.7585 [1.2654, 5.5939]	1.4822 [0.9114, 10.7504]

Table 489: CPTG compact-state, structural, clock, and geometric-rotation results for GW190803_022701. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8248	7.3358	3500.14	4105.11
500 m inward	4.1039	5.1946	3537.90	4178.45
100 m inward	1.8300	2.3243	3566.48	4226.42
10 m inward	0.5781	0.7350	3572.45	4237.45
1 m inward	0.1828	0.2324	3572.99	4238.82
10 cm inward	0.0578	0.0735	3573.06	4238.97
1 cm inward	0.0183	0.0232	3573.07	4238.98
surface limit	0.0000	0.0000	3573.07	4238.99

Table 490: Near-inner-boundary profile for GW190803_022701. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.154 M_\odot$ and $17.714 M_\odot$, with completed geometric mass-equivalent shares of 23.95% and 37.64%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.868 < 36.401 < 52.941$ km, and component 2 gives $21.530 < 26.406 < 40.883$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6174$ and 1.7655 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.279 and 14.490 h. The determinant-averaged frame-dragging rates at those shells are 2221.33 and $2931.90 \text{ rad s}^{-1}$, approaching surface-limit values of 3573.07 and $4238.99 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.825/7.336$ h at 1 km inward to $0.0183/0.0232$ h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.82 GW190910_112807

GW190910_112807 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 56.543 M_\odot$ and $m_2 = 44.497 M_\odot$, while the luminosity-distance median is 1517.1 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	17,857
Observed luminosity distance	1517.1 Mpc (median)
Detector-frame masses	$m_1 = 56.543 M_\odot$, $m_2 = 44.497 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.787$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2867$; $z_\pi = 0.3078$ (medians)
Post-handoff reorder fraction	23.39%

Table 491: Observational and coordinate summary for GW190910_112807. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	42.007 [32.252, 50.627]	34.601 [26.435, 44.635]
Physical compact-source mass M_c ($M_\odot$)	28.587 [21.324, 40.415]	21.209 [15.518, 29.825]
Geometric mass-equivalent M_{geom} ($M_\odot$)	12.186 [1.627, 22.774]	13.961 [3.587, 20.837]
Geometric share M_{geom}/M_∞ (%)	29.66 [4.42, 49.44]	41.01 [11.84, 49.90]
Exterior spin magnitude $ \hat{a}_\infty $	0.449 [0.054, 0.912]	0.250 [0.023, 0.770]
Polarization factor A_{pol}	1.422 [1.046, 1.978]	1.695 [1.134, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.907 [0.210, 0.999]	0.719 [0.090, 0.991]
GR/CPTG matching radius r_m (km)	62.029 [47.624, 74.757]	51.093 [39.035, 65.909]
Structural Mode N_{BH}	1.6750 [1.4718, 1.9058]	1.8059 [1.5180, 1.9111]
Organization-shell radius r_N (km)	32.691 [25.006, 39.381]	26.905 [20.569, 34.685]
Inner trapping-surface radius r_- (km)	40.926 [32.887, 53.695]	32.133 [24.834, 41.407]
Structural- N lapse $\bar{\alpha}_N$	0.6754 [0.4880, 0.8338]	0.5706 [0.4839, 0.7970]
Structural- N clock $T_{24,N}$ (h)	16.211 [11.712, 20.012]	13.695 [11.614, 19.129]
Structural- N axis ω (rad s^{-1})	1693.68 [593.86, 2558.76]	2111.11 [354.77, 3236.03]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2033.98 [663.16, 2849.66]	2391.65 [354.84, 3578.23]
Structural- N equatorial ω (rad s^{-1})	2379.18 [663.76, 3168.86]	2665.67 [354.88, 3909.51]
Structural- N period P_ω (ms)	3.0891 [2.2049, 9.4746]	2.6271 [1.7559, 17.7069]
Trapping-surface limit Ω_- (rad s^{-1})	3077.16 [724.55, 4143.08]	3235.66 [386.55, 4989.64]
Trapping-surface period P_- (ms)	2.0419 [1.5166, 8.6719]	1.9419 [1.2592, 16.2545]

Table 492: CPTG compact-state, structural, clock, and geometric-rotation results for GW190910_112807. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.7352	6.7655	3018.70	3143.06
500 m inward	4.0514	4.7923	3049.41	3190.13
100 m inward	1.8096	2.1453	3071.85	3226.79
10 m inward	0.5720	0.6786	3076.45	3235.03
1 m inward	0.1809	0.2146	3077.09	3235.60
10 cm inward	0.0572	0.0679	3077.16	3235.65
1 cm inward	0.0181	0.0215	3077.16	3235.66
surface limit	0.0000	0.0000	3077.16	3235.66

Table 493: Near-inner-boundary profile for GW190910_112807. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.587 M_\odot$ and $21.209 M_\odot$, with completed geometric mass-equivalent shares of 29.66% and 41.01%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.691 < 40.926 < 62.029 \text{ km}$, and component 2 gives $26.905 < 32.133 < 51.093 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6750$ and 1.8059 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.211 and 13.695 h. The determinant-averaged frame-dragging rates at those shells are 2033.98 and $2391.65 \text{ rad s}^{-1}$, approaching surface-limit values of 3077.16 and $3235.66 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.735/6.765 \text{ h}$ at 1 km inward to $0.0181/0.0215 \text{ h}$ at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M2 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M2 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.83 GW250116_015318

GW250116_015318 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 61.710 M_\odot$ and $m_2 = 39.582 M_\odot$, while the luminosity-distance median is 5159.0 Mpc . One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	10,825
Observed luminosity distance	5159.0 Mpc (median)
Detector-frame masses	$m_1 = 61.710 M_\odot$, $m_2 = 39.582 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.641$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7997$; $z_\pi = 0.8647$ (medians)
Post-handoff reorder fraction	13.46%

Table 494: Observational and coordinate summary for GW250116_015318. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	32.574 [21.744, 47.372]	20.875 [12.581, 31.615]
Physical compact-source mass M_c ($M_\odot$)	24.248 [15.393, 40.857]	14.122 [8.103, 22.650]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.819 [0.759, 17.322]	6.632 [0.827, 13.959]
Geometric share M_{geom}/M_∞ (%)	21.13 [2.50, 48.35]	33.63 [4.51, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.599 [0.094, 0.950]	0.380 [0.036, 0.911]
Polarization factor A_{pol}	1.268 [1.026, 1.936]	1.507 [1.047, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.963 [0.352, 1.000]	0.862 [0.142, 0.999]
GR/CPTG matching radius r_m (km)	48.100 [32.107, 69.951]	30.825 [18.577, 46.684]
Structural Mode N_{BH}	1.5915 [1.4613, 1.8933]	1.7188 [1.4722, 1.9094]
Organization-shell radius r_N (km)	25.305 [16.881, 36.779]	16.237 [9.789, 24.556]
Inner trapping-surface radius r_- (km)	33.584 [22.712, 54.351]	20.599 [12.257, 31.027]
Structural- N lapse $\bar{\alpha}_N$	0.7399 [0.4979, 0.8423]	0.6411 [0.4852, 0.8335]
Structural- N clock $T_{24,N}$ (h)	17.758 [11.950, 20.215]	15.387 [11.646, 20.003]
Structural- N axis ω (rad s^{-1})	1823.03 [693.05, 3500.61]	3168.34 [849.32, 6134.19]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2364.89 [1029.74, 3976.08]	3777.96 [865.88, 6972.19]
Structural- N equatorial ω (rad s^{-1})	2932.22 [1317.00, 4550.05]	4435.93 [869.91, 7833.10]
Structural- N period P_ω (ms)	2.6569 [1.5802, 6.1017]	1.6631 [0.9012, 7.2565]
Trapping-surface limit Ω_- (rad s^{-1})	3872.24 [1562.36, 6055.56]	5686.56 [948.38, 10159.10]
Trapping-surface period P_- (ms)	1.6226 [1.0376, 4.0216]	1.1049 [0.6185, 6.6251]

Table 495: CPTG compact-state, structural, clock, and geometric-rotation results for GW250116_015318. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8813	8.1181	3783.38	5454.50
500 m inward	4.1367	5.7370	3828.49	5580.05
100 m inward	1.8417	2.5646	3863.23	5666.75
10 m inward	0.5819	0.8110	3871.11	5684.61
1 m inward	0.1840	0.2565	3872.12	5686.37
10 cm inward	0.0582	0.0811	3872.22	5686.54
1 cm inward	0.0184	0.0256	3872.23	5686.56
surface limit	0.0000	0.0000	3872.24	5686.56

Table 496: Near-inner-boundary profile for GW250116_015318. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.248 M_\odot$ and $14.122 M_\odot$, with completed geometric mass-equivalent shares of 21.13% and 33.63%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.305 < 33.584 < 48.100 \text{ km}$, and component 2 gives $16.237 < 20.599 < 30.825 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5915$ and 1.7188 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.758 and 15.387 h. The determinant-averaged frame-dragging rates at those shells are 2364.89 and $3777.96 \text{ rad s}^{-1}$, approaching surface-limit values of 3872.24 and $5686.56 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.881/8.118 \text{ h}$ at 1 km inward to $0.0184/0.0256 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.84 GW250109_010541

GW250109_010541 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 58.256 M_\odot$ and $m_2 = 43.366 M_\odot$, while the luminosity-distance median is 3377.3 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,800
Observed luminosity distance	3377.3 Mpc (median)
Detector-frame masses	$m_1 = 58.256 M_\odot$, $m_2 = 43.366 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.744$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5646$; $z_\pi = 0.6084$ (medians)
Post-handoff reorder fraction	20.02%

Table 497: Observational and coordinate summary for GW250109_010541. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.707 [26.536, 50.991]	27.342 [17.973, 37.334]
Physical compact-source mass M_c ($M_\odot$)	28.814 [19.128, 45.763]	18.313 [11.329, 27.396]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.479 [0.608, 17.084]	8.791 [1.151, 16.666]
Geometric share M_{geom}/M_∞ (%)	15.52 [1.71, 45.08]	33.29 [4.83, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.702 [0.172, 0.966]	0.386 [0.037, 0.905]
Polarization factor A_{pol}	1.184 [1.017, 1.821]	1.499 [1.051, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.983 [0.571, 1.000]	0.867 [0.147, 0.999]
GR/CPTG matching radius r_m (km)	52.726 [39.184, 75.295]	40.373 [26.540, 55.128]
Structural Mode N_{BH}	1.5449 [1.4572, 1.8548]	1.7150 [1.4741, 1.9091]
Organization-shell radius r_N (km)	27.706 [20.548, 39.498]	21.248 [13.948, 29.035]
Inner trapping-surface radius r_- (km)	38.874 [28.137, 60.949]	26.776 [17.368, 37.182]
Structural- N lapse $\bar{\alpha}_N$	0.7760 [0.5298, 0.8456]	0.6441 [0.4854, 0.8320]
Structural- N clock $T_{24,N}$ (h)	18.623 [12.715, 20.294]	15.458 [11.649, 19.968]
Structural- N axis ω (rad s^{-1})	1418.63 [635.55, 2881.12]	2494.65 [702.10, 4416.63]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1985.02 [1010.58, 3271.77]	2959.75 [703.63, 4950.38]
Structural- N equatorial ω (rad s^{-1})	2633.22 [1531.12, 3712.25]	3450.77 [703.94, 5556.02]
Structural- N period P_ω (ms)	3.1653 [1.9204, 6.2174]	2.1229 [1.2692, 8.9297]
Trapping-surface limit Ω_- (rad s^{-1})	3572.93 [2093.43, 4942.19]	4439.59 [767.61, 7186.44]
Trapping-surface period P_- (ms)	1.7586 [1.2713, 3.0014]	1.4153 [0.8743, 8.1854]

Table 498: CPTG compact-state, structural, clock, and geometric-rotation results for GW250109_010541. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.1322	7.1700	3514.58	4309.36
500 m inward	3.6023	5.0682	3544.05	4382.78
100 m inward	1.6009	2.2650	3567.07	4431.25
10 m inward	0.5055	0.7161	3572.91	4438.65
1 m inward	0.1598	0.2264	3572.95	4439.48
10 cm inward	0.0505	0.0716	3572.94	4439.58
1 cm inward	0.0160	0.0226	3572.94	4439.59
surface limit	0.0000	0.0000	3572.93	4439.59

Table 499: Near-inner-boundary profile for GW250109_010541. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.814 M_\odot$ and $18.313 M_\odot$, with completed geometric mass-equivalent shares of 15.52% and 33.29%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.706 < 38.874 < 52.726 \text{ km}$, and component 2 gives $21.248 < 26.776 < 40.373 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5449$ and 1.7150 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.623 and 15.458 h. The determinant-averaged frame-dragging rates at those shells are 1985.02 and $2959.75 \text{ rad s}^{-1}$, approaching surface-limit values of 3572.93 and $4439.59 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.132/7.170 \text{ h}$ at 1 km inward to $0.0160/0.0226 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.85 GW241210_120900

GW241210_120900 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 64.432 M_\odot$ and $m_2 = 38.191 M_\odot$, while the luminosity-distance median is 3585.1 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,608
Observed luminosity distance	3585.1 Mpc (median)
Detector-frame masses	$m_1 = 64.432 M_\odot$, $m_2 = 38.191 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.593$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5932$; $z_\pi = 0.6395$ (medians)
Post-handoff reorder fraction	12.38%

Table 500: Observational and coordinate summary for GW241210_120900. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	38.632 [25.566, 54.372]	23.093 [15.132, 35.423]
Physical compact-source mass M_c ($M_\odot$)	26.691 [17.906, 41.732]	15.728 [9.413, 24.068]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.799 [1.294, 22.751]	7.551 [0.959, 16.048]
Geometric share M_{geom}/M_∞ (%)	28.25 [3.97, 49.35]	34.32 [4.78, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.473 [0.058, 0.921]	0.368 [0.033, 0.906]
Polarization factor A_{pol}	1.394 [1.041, 1.974]	1.522 [1.050, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.919 [0.226, 0.999]	0.853 [0.129, 0.999]
GR/CPTG matching radius r_m (km)	57.045 [37.751, 80.287]	34.100 [22.345, 52.306]
Structural Mode N_{BH}	1.6602 [1.4692, 1.9048]	1.7267 [1.4737, 1.9099]
Organization-shell radius r_N (km)	30.029 [19.811, 42.266]	17.958 [11.748, 27.504]
Inner trapping-surface radius r_- (km)	38.309 [26.613, 56.068]	22.739 [14.526, 33.734]
Structural- N lapse $\bar{\alpha}_N$	0.6869 [0.4888, 0.8359]	0.6349 [0.4849, 0.8323]
Structural- N clock $T_{24,N}$ (h)	16.486 [11.731, 20.061]	15.237 [11.636, 19.974]
Structural- N axis ω (rad s^{-1})	1718.75 [636.72, 2985.95]	2836.65 [705.03, 5323.43]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2101.71 [754.54, 3388.36]	3392.23 [705.81, 5948.18]
Structural- N equatorial ω (rad s^{-1})	2505.53 [761.57, 3878.28]	4017.49 [706.11, 6598.88]
Structural- N period P_ω (ms)	2.9896 [1.8543, 8.3272]	1.8522 [1.0563, 8.9021]
Trapping-surface limit Ω_- (rad s^{-1})	3237.75 [833.79, 5136.91]	5141.04 [769.75, 8500.48]
Trapping-surface period P_- (ms)	1.9406 [1.2231, 7.5357]	1.2222 [0.7392, 8.1626]

Table 501: CPTG compact-state, structural, clock, and geometric-rotation results for GW241210_120900. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.8249	7.6880	3176.41	4965.22
500 m inward	4.1121	5.4347	3205.54	5059.76
100 m inward	1.8351	2.4285	3232.34	5128.33
10 m inward	0.5800	0.7679	3237.08	5139.52
1 m inward	0.1834	0.2428	3237.69	5140.85
10 cm inward	0.0580	0.0768	3237.74	5141.02
1 cm inward	0.0183	0.0243	3237.75	5141.03
surface limit	0.0000	0.0000	3237.75	5141.04

Table 502: Near-inner-boundary profile for GW241210_120900. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.691 M_\odot$ and $15.728 M_\odot$, with completed geometric mass-equivalent shares of 28.25% and 34.32%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $30.029 < 38.309 < 57.045 \text{ km}$, and component 2 gives $17.958 < 22.739 < 34.100 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6602$ and 1.7267 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.486 and 15.237 h. The determinant-averaged frame-dragging rates at those shells are 2101.71 and $3392.23 \text{ rad s}^{-1}$, approaching surface-limit values of 3237.75 and $5141.04 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.825/7.688 \text{ h}$ at 1 km inward to $0.0183/0.0243 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.86 GW241127_061008

GW241127_061008 is an M2 intermediate observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 77.558 M_\odot$ and $m_2 = 25.173 M_\odot$, while the luminosity-distance median is 1077.2 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,755
Observed luminosity distance	1077.2 Mpc (median)
Detector-frame masses	$m_1 = 77.558 M_\odot$, $m_2 = 25.173 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.325$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2116$; $z_\pi = 0.2276$ (medians)
Post-handoff reorder fraction	0.43%

Table 503: Observational and coordinate summary for GW241127_061008. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	62.996 [55.330, 71.670]	20.503 [17.182, 26.920]
Physical compact-source mass M_c ($M_\odot$)	49.933 [39.576, 59.421]	15.734 [9.934, 23.271]
Geometric mass-equivalent M_{geom} ($M_\odot$)	13.182 [7.910, 19.432]	4.814 [0.512, 10.628]
Geometric share M_{geom}/M_∞ (%)	20.62 [12.73, 31.96]	23.40 [2.39, 49.30]
Exterior spin magnitude $ \hat{a}_\infty $	0.608 [0.409, 0.754]	0.559 [0.060, 0.953]
Polarization factor A_{pol}	1.260 [1.146, 1.470]	1.305 [1.024, 1.972]
Compact-source spin parameter $\hat{a}_c$	0.966 [0.883, 0.989]	0.952 [0.233, 1.000]
GR/CPTG matching radius r_m (km)	93.022 [81.701, 105.829]	30.275 [25.371, 39.750]
Structural Mode N_{BH}	1.5869 [1.5243, 1.7000]	1.6121 [1.4607, 1.9043]
Organization-shell radius r_N (km)	49.118 [43.113, 55.808]	15.926 [13.338, 20.893]
Inner trapping-surface radius r_- (km)	66.728 [55.283, 78.396]	21.412 [15.984, 31.222]
Structural- N lapse $\bar{\alpha}_N$	0.7434 [0.6559, 0.7921]	0.7240 [0.4892, 0.8427]
Structural- N clock $T_{24,N}$ (h)	17.842 [15.742, 19.011]	17.376 [11.741, 20.226]
Structural- N axis ω (rad s^{-1})	1018.31 [707.32, 1551.87]	2790.87 [1126.52, 5088.52]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1337.08 [1018.27, 1770.67]	3605.57 [1519.31, 5611.29]
Structural- N equatorial ω (rad s^{-1})	1638.95 [1367.21, 1958.93]	4520.12 [1578.79, 6102.52]
Structural- N period P_ω (ms)	4.6992 [3.5485, 6.1705]	1.7426 [1.1197, 4.1356]
Trapping-surface limit Ω_- (rad s^{-1})	2195.17 [1896.17, 2514.03]	6059.83 [1728.71, 7781.86]
Trapping-surface period P_- (ms)	2.8623 [2.4992, 3.3136]	1.0369 [0.8074, 3.6346]

Table 504: CPTG compact-state, structural, clock, and geometric-rotation results for GW241127_061008. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.1492	7.5249	2166.20	5861.46
500 m inward	2.9265	5.2926	2181.11	5969.22
100 m inward	1.3059	2.3558	2192.41	6041.51
10 m inward	0.4127	0.7440	2194.86	6057.85
1 m inward	0.1305	0.2353	2195.14	6059.66
10 cm inward	0.0413	0.0744	2195.17	6059.80
1 cm inward	0.0131	0.0235	2195.17	6059.83
surface limit	0.0000	0.0000	2195.17	6059.83

Table 505: Near-inner-boundary profile for GW241127_061008. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $49.933 M_\odot$ and $15.734 M_\odot$, with completed geometric mass-equivalent shares of 20.62% and 23.40%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $49.118 < 66.728 < 93.022 \text{ km}$, and component 2 gives $15.926 < 21.412 < 30.275 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5869$ and 1.6121 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.842 and 17.376 h. The determinant-averaged frame-dragging rates at those shells are 1337.08 and $3605.57 \text{ rad s}^{-1}$, approaching surface-limit values of 2195.17 and $6059.83 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $4.149/7.525 \text{ h}$ at 1 km inward to $0.0131/0.0235 \text{ h}$ at 1 cm inward for components 1/2. Its strong observational mass asymmetry makes the contrast between the two solved compact interiors particularly useful for component-by-component comparison. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.87 GW200219_094415

GW200219_094415 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 58.573 M_\odot$ and $m_2 = 44.406 M_\odot$, while the luminosity-distance median is 3417.7 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	9,991
Observed luminosity distance	3417.7 Mpc (median)
Detector-frame masses	$m_1 = 58.573 M_\odot$, $m_2 = 44.406 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.758$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5709$; $z_\pi = 0.6145$ (medians)
Post-handoff reorder fraction	21.81%

Table 506: Observational and coordinate summary for GW200219_094415. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.622 [26.379, 46.460]	28.107 [19.053, 37.695]
Physical compact-source mass M_c ($M_\odot$)	27.010 [18.734, 39.313]	18.225 [11.844, 26.888]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.173 [0.745, 18.943]	9.768 [1.404, 17.300]
Geometric share M_{geom}/M_∞ (%)	20.62 [2.24, 47.87]	36.19 [6.00, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.608 [0.108, 0.955]	0.335 [0.031, 0.882]
Polarization factor A_{pol}	1.260 [1.023, 1.918]	1.567 [1.064, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.966 [0.396, 1.000]	0.824 [0.123, 0.998]
GR/CPTG matching radius r_m (km)	52.600 [38.952, 68.604]	41.503 [28.134, 55.662]
Structural Mode N_{BH}	1.5869 [1.4600, 1.8878]	1.7485 [1.4808, 1.9101]
Organization-shell radius r_N (km)	27.691 [20.462, 36.166]	21.839 [14.812, 29.287]
Inner trapping-surface radius r_- (km)	37.193 [27.734, 52.378]	26.952 [18.402, 36.650]
Structural- N lapse $\bar{\alpha}_N$	0.7434 [0.5025, 0.8434]	0.6175 [0.4847, 0.8266]
Structural- N clock $T_{24,N}$ (h)	17.842 [12.059, 20.241]	14.819 [11.633, 19.838]
Structural- N axis ω (rad s^{-1})	1671.72 [713.64, 2936.34]	2480.35 [577.87, 4229.11]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2180.42 [1094.30, 3320.65]	2901.14 [578.27, 4710.53]
Structural- N equatorial ω (rad s^{-1})	2721.48 [1419.30, 3768.30]	3364.84 [578.47, 5197.72]
Structural- N period P_ω (ms)	2.8816 [1.8922, 5.7418]	2.1658 [1.3339, 10.8655]
Trapping-surface limit Ω_- (rad s^{-1})	3623.30 [1574.63, 4966.97]	4257.79 [630.59, 6757.87]
Trapping-surface period P_- (ms)	1.7341 [1.2650, 3.9903]	1.4757 [0.9298, 9.9640]

Table 507: CPTG compact-state, structural, clock, and geometric-rotation results for GW200219_094415. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5830	7.2260	3553.05	4135.95
500 m inward	3.9285	5.1165	3590.11	4204.25
100 m inward	1.7501	2.2888	3617.58	4247.47
10 m inward	0.5527	0.7239	3622.93	4256.77
1 m inward	0.1748	0.2289	3623.23	4257.79
10 cm inward	0.0553	0.0724	3623.29	4257.79
1 cm inward	0.0175	0.0229	3623.30	4257.79
surface limit	0.0000	0.0000	3623.30	4257.79

Table 508: Near-inner-boundary profile for GW200219_094415. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.010 M_\odot$ and $18.225 M_\odot$, with completed geometric mass-equivalent shares of 20.62% and 36.19%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.691 < 37.193 < 52.600 \text{ km}$, and component 2 gives $21.839 < 26.952 < 41.503 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5869$ and 1.7485 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.842 and 14.819 h. The determinant-averaged frame-dragging rates at those shells are 2180.42 and $2901.14 \text{ rad s}^{-1}$, approaching surface-limit values of 3623.30 and $4257.79 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.583/7.226 \text{ h}$ at 1 km inward to $0.0175/0.0229 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.88 GW231213_111417

GW231213_111417 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 58.510 M_\odot$ and $m_2 = 45.299 M_\odot$, while the luminosity-distance median is 4039.4 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	42,438
Observed luminosity distance	4039.4 Mpc (median)
Detector-frame masses	$m_1 = 58.510 M_\odot$, $m_2 = 45.299 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.774$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6545$; $z_\pi = 0.7062$ (medians)
Post-handoff reorder fraction	23.18%

Table 509: Observational and coordinate summary for GW231213_111417. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	33.608 [25.153, 44.771]	27.261 [18.947, 37.088]
Physical compact-source mass M_c ($M_\odot$)	24.986 [17.056, 37.138]	17.345 [11.541, 25.804]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.652 [0.879, 18.040]	10.008 [1.657, 16.855]
Geometric share M_{geom}/M_∞ (%)	23.13 [2.81, 48.47]	37.98 [7.08, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.563 [0.090, 0.944]	0.304 [0.028, 0.861]
Polarization factor A_{pol}	1.301 [1.029, 1.941]	1.612 [1.076, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.954 [0.339, 1.000]	0.791 [0.109, 0.997]
GR/CPTG matching radius r_m (km)	49.627 [37.142, 66.110]	40.255 [27.978, 54.765]
Structural Mode N_{BH}	1.6097 [1.4629, 1.8948]	1.7696 [1.4872, 1.9105]
Organization-shell radius r_N (km)	26.121 [19.507, 34.818]	21.205 [14.729, 28.837]
Inner trapping-surface radius r_- (km)	34.608 [25.726, 49.498]	25.902 [18.100, 35.528]
Structural- N lapse $\bar{\alpha}_N$	0.7259 [0.4967, 0.8410]	0.6004 [0.4843, 0.8215]
Structural- N clock $T_{24,N}$ (h)	17.421 [11.922, 20.183]	14.410 [11.624, 19.715]
Structural- N axis ω (rad s^{-1})	1849.40 [757.35, 3200.47]	2588.44 [541.24, 4358.45]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2348.65 [1109.92, 3598.45]	2987.63 [541.63, 4815.87]
Structural- N equatorial ω (rad s^{-1})	2879.71 [1325.83, 4024.72]	3418.88 [541.76, 5318.83]
Structural- N period P_ω (ms)	2.6752 [1.7461, 5.6609]	2.1031 [1.3047, 11.6006]
Trapping-surface limit Ω_- (rad s^{-1})	3813.51 [1463.24, 5281.07]	4289.48 [590.37, 6823.56]
Trapping-surface period P_- (ms)	1.6476 [1.1898, 4.2940]	1.4648 [0.9208, 10.6427]

Table 510: CPTG compact-state, structural, clock, and geometric-rotation results for GW231213_111417. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.9166	7.4324	3733.79	4150.41
500 m inward	4.1675	5.2613	3776.39	4222.37
100 m inward	1.8582	2.3540	3806.71	4276.59
10 m inward	0.5871	0.7445	3812.78	4288.32
1 m inward	0.1857	0.2354	3813.42	4289.34
10 cm inward	0.0587	0.0744	3813.51	4289.47
1 cm inward	0.0186	0.0235	3813.51	4289.48
surface limit	0.0000	0.0000	3813.51	4289.48

Table 511: Near-inner-boundary profile for GW231213_111417. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $24.986 M_\odot$ and $17.345 M_\odot$, with completed geometric mass-equivalent shares of 23.13% and 37.98%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $26.121 < 34.608 < 49.627 \text{ km}$, and component 2 gives $21.205 < 25.902 < 40.255 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6097$ and 1.7696 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.421 and 14.410 h. The determinant-averaged frame-dragging rates at those shells are 2348.65 and $2987.63 \text{ rad s}^{-1}$, approaching surface-limit values of 3813.51 and $4289.48 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.917/7.432 \text{ h}$ at 1 km inward to $0.0186/0.0235 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.89 GW230930_110730

GW230930_110730 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 60.295 M_\odot$ and $m_2 = 43.606 M_\odot$, while the luminosity-distance median is 4866.0 Mpc . One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	34,812
Observed luminosity distance	4866.0 Mpc (median)
Detector-frame masses	$m_1 = 60.295 M_\odot$, $m_2 = 43.606 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.723$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7624$; $z_\pi = 0.8239$ (medians)
Post-handoff reorder fraction	19.87%

Table 512: Observational and coordinate summary for GW230930_110730. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	32.549 [23.076, 46.312]	24.218 [16.270, 34.786]
Physical compact-source mass M_c ($M_\odot$)	23.362 [15.859, 36.350]	15.684 [9.972, 23.787]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.224 [0.944, 19.058]	8.661 [1.340, 15.756]
Geometric share M_{geom}/M_∞ (%)	25.68 [3.22, 49.11]	37.32 [6.47, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.518 [0.068, 0.936]	0.316 [0.028, 0.873]
Polarization factor A_{pol}	1.345 [1.033, 1.965]	1.595 [1.069, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.938 [0.262, 0.999]	0.803 [0.110, 0.998]
GR/CPTG matching radius r_m (km)	48.063 [34.075, 68.385]	35.761 [24.025, 51.365]
Structural Mode N_{BH}	1.6341 [1.4652, 1.9021]	1.7617 [1.4835, 1.9105]
Organization-shell radius r_N (km)	25.301 [17.893, 36.006]	18.838 [12.643, 27.036]
Inner trapping-surface radius r_- (km)	32.928 [23.785, 48.826]	23.226 [15.504, 33.138]
Structural- N lapse $\bar{\alpha}_N$	0.7071 [0.4909, 0.8392]	0.6068 [0.4844, 0.8244]
Structural- N clock $T_{24,N}$ (h)	16.970 [11.782, 20.140]	14.563 [11.625, 19.786]
Structural- N axis ω (rad s^{-1})	1962.22 [742.93, 3412.43]	2836.50 [591.67, 5041.77]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2444.08 [976.13, 3846.40]	3309.61 [591.90, 5602.74]
Structural- N equatorial ω (rad s^{-1})	2970.58 [1026.16, 4338.19]	3822.77 [592.01, 6166.97]
Structural- N period P_ω (ms)	2.5708 [1.6335, 6.4368]	1.8985 [1.1214, 10.6153]
Trapping-surface limit Ω_- (rad s^{-1})	3895.23 [1125.62, 5706.39]	4811.74 [645.02, 7924.53]
Trapping-surface period P_- (ms)	1.6130 [1.1011, 5.5820]	1.3058 [0.7929, 9.7410]

Table 513: CPTG compact-state, structural, clock, and geometric-rotation results for GW230930_110730. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	6.1666	7.7814	3802.30	4638.19
500 m inward	4.3489	5.5070	3851.69	4732.38
100 m inward	1.9399	2.4638	3886.81	4798.50
10 m inward	0.6131	0.7792	3894.48	4810.24
1 m inward	0.1939	0.2464	3895.21	4811.55
10 cm inward	0.0613	0.0779	3895.22	4811.72
1 cm inward	0.0194	0.0246	3895.23	4811.74
surface limit	0.0000	0.0000	3895.23	4811.74

Table 514: Near-inner-boundary profile for GW230930_110730. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $23.362 M_\odot$ and $15.684 M_\odot$, with completed geometric mass-equivalent shares of 25.68% and 37.32%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $25.301 < 32.928 < 48.063 \text{ km}$, and component 2 gives $18.838 < 23.226 < 35.761 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6341$ and 1.7617 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.970 and 14.563 h. The determinant-averaged frame-dragging rates at those shells are 2444.08 and $3309.61 \text{ rad s}^{-1}$, approaching surface-limit values of 3895.23 and $4811.74 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $6.167/7.781 \text{ h}$ at 1 km inward to $0.0194/0.0246 \text{ h}$ at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M2 descriptive distribution, making it a useful representative intermediate-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.90 GW240908_082628

GW240908_082628 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 60.993 M_\odot$ and $m_2 = 43.214 M_\odot$, while the luminosity-distance median is 3109.3 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,245
Observed luminosity distance	3109.3 Mpc (median)
Detector-frame masses	$m_1 = 60.993 M_\odot$, $m_2 = 43.214 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.709$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5271$; $z_\pi = 0.5678$ (medians)
Post-handoff reorder fraction	17.91%

Table 515: Observational and coordinate summary for GW240908_082628. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	38.029 [26.906, 50.383]	27.519 [17.993, 38.712]
Physical compact-source mass M_c ($M_\odot$)	28.131 [18.658, 43.352]	18.150 [11.145, 27.481]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.118 [0.883, 20.261]	9.205 [1.302, 17.549]
Geometric share M_{geom}/M_∞ (%)	21.84 [2.46, 48.65]	35.17 [5.52, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.587 [0.085, 0.951]	0.353 [0.034, 0.891]
Polarization factor A_{pol}	1.279 [1.025, 1.947]	1.542 [1.058, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.960 [0.320, 1.000]	0.840 [0.134, 0.998]
GR/CPTG matching radius r_m (km)	56.154 [39.730, 74.397]	40.635 [26.569, 57.163]
Structural Mode N_{BH}	1.5978 [1.4611, 1.8968]	1.7365 [1.4780, 1.9096]
Organization-shell radius r_N (km)	29.556 [20.867, 39.141]	21.397 [13.979, 30.106]
Inner trapping-surface radius r_- (km)	39.128 [27.758, 57.513]	26.685 [17.342, 37.445]
Structural- N lapse $\bar{\alpha}_N$	0.7350 [0.4952, 0.8424]	0.6271 [0.4850, 0.8288]
Structural- N clock $T_{24,N}$ (h)	17.641 [11.884, 20.218]	15.049 [11.640, 19.892]
Structural- N axis ω (rad s^{-1})	1593.09 [635.07, 2882.27]	2472.15 [637.64, 4450.06]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2056.74 [937.78, 3260.86]	2909.14 [638.43, 4977.78]
Structural- N equatorial ω (rad s^{-1})	2537.68 [1127.64, 3714.69]	3397.93 [638.81, 5545.45]
Structural- N period P_ω (ms)	3.0549 [1.9268, 6.7001]	2.1598 [1.2622, 9.8416]
Trapping-surface limit Ω_- (rad s^{-1})	3361.71 [1235.86, 4910.83]	4313.53 [696.55, 7180.29]
Trapping-surface period P_- (ms)	1.8690 [1.2795, 5.0840]	1.4566 [0.8751, 9.0205]

Table 516: CPTG compact-state, structural, clock, and geometric-rotation results for GW240908_082628. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5236	7.2055	3301.45	4187.16
500 m inward	3.8917	5.0967	3334.21	4251.04
100 m inward	1.7345	2.2794	3356.26	4299.77
10 m inward	0.5481	0.7205	3361.32	4311.95
1 m inward	0.1733	0.2279	3361.62	4313.37
10 cm inward	0.0548	0.0721	3361.70	4313.52
1 cm inward	0.0173	0.0228	3361.71	4313.53
surface limit	0.0000	0.0000	3361.71	4313.53

Table 517: Near-inner-boundary profile for GW240908_082628. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.131 M_\odot$ and $18.150 M_\odot$, with completed geometric mass-equivalent shares of 21.84% and 35.17%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $29.556 < 39.128 < 56.154 \text{ km}$, and component 2 gives $21.397 < 26.685 < 40.635 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5978$ and 1.7365 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.641 and 15.049 h. The determinant-averaged frame-dragging rates at those shells are 2056.74 and $2909.14 \text{ rad s}^{-1}$, approaching surface-limit values of 3361.71 and $4313.53 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.524/7.205 \text{ h}$ at 1 km inward to $0.0173/0.0228 \text{ h}$ at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.91 GW231113_122623

GW231113_122623 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 62.163 M_\odot$ and $m_2 = 42.195 M_\odot$, while the luminosity-distance median is 3378.8 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	31,978
Observed luminosity distance	3378.8 Mpc (median)
Detector-frame masses	$m_1 = 62.163 M_\odot$, $m_2 = 42.195 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.679$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5648$; $z_\pi = 0.6087$ (medians)
Post-handoff reorder fraction	12.35%

Table 518: Observational and coordinate summary for GW231113_122623. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	38.389 [27.068, 55.633]	26.001 [16.477, 36.644]
Physical compact-source mass M_c ($M_\odot$)	32.179 [20.623, 49.302]	18.150 [10.418, 28.454]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.682 [0.545, 17.321]	7.291 [0.930, 15.417]
Geometric share M_{geom}/M_∞ (%)	12.46 [1.46, 42.03]	29.30 [3.93, 49.57]
Exterior spin magnitude $ \hat{a}_\infty $	0.758 [0.231, 0.971]	0.455 [0.047, 0.922]
Polarization factor A_{pol}	1.142 [1.015, 1.725]	1.414 [1.041, 1.983]
Compact-source spin parameter $\hat{a}_c$	0.990 [0.689, 1.000]	0.910 [0.183, 0.999]
GR/CPTG matching radius r_m (km)	56.687 [39.969, 82.150]	38.394 [24.331, 54.109]
Structural Mode N_{BH}	1.5224 [1.4559, 1.8182]	1.6712 [1.4690, 1.9074]
Organization-shell radius r_N (km)	29.735 [20.940, 43.132]	20.205 [12.801, 28.515]
Inner trapping-surface radius r_- (km)	43.156 [29.430, 65.786]	26.007 [15.988, 38.165]
Structural- N lapse $\bar{\alpha}_N$	0.7936 [0.5604, 0.8466]	0.6784 [0.4868, 0.8361]
Structural- N clock $T_{24,N}$ (h)	19.046 [13.449, 20.319]	16.282 [11.682, 20.066]
Structural- N axis ω (rad s^{-1})	1192.09 [599.49, 2623.97]	2515.20 [874.64, 4778.97]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1735.34 [960.21, 3037.30]	3035.14 [969.49, 5366.21]
Structural- N equatorial ω (rad s^{-1})	2392.22 [1467.60, 3547.44]	3596.58 [969.89, 6003.07]
Structural- N period P_ω (ms)	3.6207 [2.0687, 6.5436]	2.0701 [1.1709, 6.4809]
Trapping-surface limit Ω_- (rad s^{-1})	3300.38 [2015.00, 4779.54]	4678.17 [1059.80, 7760.33]
Trapping-surface period P_- (ms)	1.9038 [1.3146, 3.1182]	1.3431 [0.8097, 5.9286]

Table 519: CPTG compact-state, structural, clock, and geometric-rotation results for GW231113_122623. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.6486	7.1271	3256.90	4543.51
500 m inward	3.2569	5.0306	3280.13	4612.12
100 m inward	1.4457	2.2460	3296.92	4665.32
10 m inward	0.4565	0.7098	3300.10	4677.39
1 m inward	0.1443	0.2245	3300.36	4678.16
10 cm inward	0.0456	0.0710	3300.38	4678.16
1 cm inward	0.0144	0.0224	3300.38	4678.17
surface limit	0.0000	0.0000	3300.38	4678.17

Table 520: Near-inner-boundary profile for GW231113_122623. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $32.179 M_\odot$ and $18.150 M_\odot$, with completed geometric mass-equivalent shares of 12.46% and 29.30%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $29.735 < 43.156 < 56.687 \text{ km}$, and component 2 gives $20.205 < 26.007 < 38.394 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5224$ and 1.6712 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.046 and 16.282 h. The determinant-averaged frame-dragging rates at those shells are 1735.34 and $3035.14 \text{ rad s}^{-1}$, approaching surface-limit values of 3300.38 and $4678.17 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $4.649/7.127 \text{ h}$ at 1 km inward to $0.0144/0.0224 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.92 GW241007_082943

GW241007_082943 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 60.092 M_\odot$ and $m_2 = 44.356 M_\odot$, while the luminosity-distance median is 4034.6 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,180
Observed luminosity distance	4034.6 Mpc (median)
Detector-frame masses	$m_1 = 60.092 M_\odot$, $m_2 = 44.356 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.738$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6539$; $z_\pi = 0.7055$ (medians)
Post-handoff reorder fraction	18.52%

Table 521: Observational and coordinate summary for GW241007_082943. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	34.854 [25.360, 48.804]	26.375 [18.114, 36.533]
Physical compact-source mass M_c ($M_\odot$)	27.673 [17.833, 42.502]	17.634 [11.126, 27.284]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.892 [0.619, 17.637]	8.542 [1.201, 16.016]
Geometric share M_{geom}/M_∞ (%)	17.16 [1.79, 46.77]	33.56 [4.98, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.671 [0.135, 0.964]	0.381 [0.035, 0.902]
Polarization factor A_{pol}	1.207 [1.018, 1.879]	1.505 [1.052, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.978 [0.477, 1.000]	0.863 [0.137, 0.999]
GR/CPTG matching radius r_m (km)	51.466 [37.447, 72.065]	38.946 [26.748, 53.945]
Structural Mode N_{BH}	1.5579 [1.4576, 1.8750]	1.7181 [1.4749, 1.9095]
Organization-shell radius r_N (km)	27.034 [19.680, 37.874]	20.509 [14.058, 28.389]
Inner trapping-surface radius r_- (km)	37.639 [26.407, 56.597]	25.698 [17.356, 36.857]
Structural- N lapse $\bar{\alpha}_N$	0.7659 [0.5130, 0.8453]	0.6417 [0.4851, 0.8313]
Structural- N clock $T_{24,N}$ (h)	18.382 [12.312, 20.286]	15.401 [11.642, 19.952]
Structural- N axis ω (rad s^{-1})	1518.01 [674.43, 3083.14]	2549.06 [683.54, 4490.42]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2065.12 [1058.70, 3480.94]	3023.60 [685.91, 5007.70]
Structural- N equatorial ω (rad s^{-1})	2691.99 [1565.33, 3929.12]	3560.48 [686.18, 5556.25]
Structural- N period P_ω (ms)	3.0425 [1.8050, 5.9348]	2.0780 [1.2547, 9.1603]
Trapping-surface limit Ω_- (rad s^{-1})	3646.64 [1970.07, 5183.91]	4571.91 [748.12, 7167.98]
Trapping-surface period P_- (ms)	1.7230 [1.2121, 3.1893]	1.3743 [0.8766, 8.3986]

Table 522: CPTG compact-state, structural, clock, and geometric-rotation results for GW241007_082943. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3311	7.3215	3582.04	4429.00
500 m inward	3.7454	5.1773	3615.76	4504.61
100 m inward	1.6650	2.3142	3640.79	4558.33
10 m inward	0.5260	0.7317	3646.19	4570.74
1 m inward	0.1663	0.2314	3646.62	4571.80
10 cm inward	0.0526	0.0732	3646.64	4571.89
1 cm inward	0.0166	0.0231	3646.64	4571.91
surface limit	0.0000	0.0000	3646.64	4571.91

Table 523: Near-inner-boundary profile for GW241007_082943. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.673 M_\odot$ and $17.634 M_\odot$, with completed geometric mass-equivalent shares of 17.16% and 33.56%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.034 < 37.639 < 51.466$ km, and component 2 gives $20.509 < 25.698 < 38.946$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5579$ and 1.7181; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.382 and 15.401 h. The determinant-averaged frame-dragging rates at those shells are 2065.12 and 3023.60 rad s^{-1} , approaching surface-limit values of 3646.64 and 4571.91 rad s^{-1} . Across the registered radial sequence, the median clock falls from 5.331/7.322 h at 1 km inward to 0.0166/0.0231 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.93 GW241114_024711

GW241114_024711 is an M2 intermediate observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 68.075 M_\odot$ and $m_2 = 36.573 M_\odot$, while the luminosity-distance median is 2582.7 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,867
Observed luminosity distance	2582.7 Mpc (median)
Detector-frame masses	$m_1 = 68.075 M_\odot$, $m_2 = 36.573 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.537$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4513$; $z_\pi = 0.4857$ (medians)
Post-handoff reorder fraction	5.08%

Table 524: Observational and coordinate summary for GW241114_024711. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	45.873 [32.347, 55.668]	24.473 [17.353, 34.547]
Physical compact-source mass M_c ($M_\odot$)	36.488 [22.860, 50.156]	17.420 [10.718, 26.414]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.639 [0.878, 21.079]	6.885 [0.749, 14.911]
Geometric share M_{geom}/M_∞ (%)	16.93 [2.04, 46.32]	28.90 [3.32, 49.61]
Exterior spin magnitude $ \hat{a}_\infty $	0.676 [0.146, 0.959]	0.462 [0.044, 0.934]
Polarization factor A_{pol}	1.204 [1.021, 1.863]	1.407 [1.034, 1.985]
Compact-source spin parameter $\hat{a}_c$	0.979 [0.505, 1.000]	0.914 [0.175, 0.999]
GR/CPTG matching radius r_m (km)	67.738 [47.764, 82.200]	36.137 [25.624, 51.013]
Structural Mode N_{BH}	1.5560 [1.4589, 1.8696]	1.6670 [1.4657, 1.9078]
Organization-shell radius r_N (km)	35.614 [25.089, 43.181]	19.008 [13.485, 26.892]
Inner trapping-surface radius r_- (km)	49.079 [33.960, 66.890]	24.691 [16.683, 35.982]
Structural- N lapse $\bar{\alpha}_N$	0.7673 [0.5174, 0.8442]	0.6816 [0.4864, 0.8387]
Structural- N clock $T_{24,N}$ (h)	18.416 [12.418, 20.262]	16.359 [11.674, 20.130]
Structural- N axis ω (rad s^{-1})	1179.14 [562.79, 2323.42]	2577.12 [885.34, 4704.50]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1617.74 [905.30, 2629.88]	3147.06 [922.91, 5230.13]
Structural- N equatorial ω (rad s^{-1})	2078.64 [1405.62, 3031.53]	3817.54 [923.54, 5794.20]
Structural- N period P_ω (ms)	3.8839 [2.3892, 6.9405]	1.9965 [1.2013, 6.8081]
Trapping-surface limit Ω_- (rad s^{-1})	2796.09 [1789.27, 4035.23]	5009.57 [1007.71, 7508.79]
Trapping-surface period P_- (ms)	2.2471 [1.5571, 3.5116]	1.2542 [0.8368, 6.2351]

Table 525: CPTG compact-state, structural, clock, and geometric-rotation results for GW241114_024711. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.6539	7.2651	2757.48	4856.97
500 m inward	3.2752	5.1253	2776.68	4937.53
100 m inward	1.4584	2.2875	2791.76	4997.72
10 m inward	0.4605	0.7231	2795.69	5008.52
1 m inward	0.1456	0.2286	2796.05	5009.55
10 cm inward	0.0460	0.0723	2796.08	5009.58
1 cm inward	0.0146	0.0229	2796.09	5009.57
surface limit	0.0000	0.0000	2796.09	5009.57

Table 526: Near-inner-boundary profile for GW241114_024711. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $36.488 M_\odot$ and $17.420 M_\odot$, with completed geometric mass-equivalent shares of 16.93% and 28.90%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $35.614 < 49.079 < 67.738 \text{ km}$, and component 2 gives $19.008 < 24.691 < 36.137 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5560$ and 1.6670 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.416 and 16.359 h. The determinant-averaged frame-dragging rates at those shells are 1617.74 and $3147.06 \text{ rad s}^{-1}$, approaching surface-limit values of 2796.09 and $5009.57 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $4.654/7.265 \text{ h}$ at 1 km inward to $0.0146/0.0229 \text{ h}$ at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M2 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

4.94 GW190727_060333

GW190727_060333 is an M2 intermediate observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 58.873 M_\odot$ and $m_2 = 46.179 M_\odot$, while the luminosity-distance median is 3071.9 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M2: Intermediate observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	11,530
Observed luminosity distance	3071.9 Mpc (median)
Detector-frame masses	$m_1 = 58.873 M_\odot$, $m_2 = 46.179 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.784$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5225$; $z_\pi = 0.5621$ (medians)
Post-handoff reorder fraction	22.18%

Table 527: Observational and coordinate summary for GW190727_060333. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	37.127 [28.654, 46.681]	30.230 [21.321, 39.032]
Physical compact-source mass M_c ($M_\odot$)	28.727 [19.705, 40.879]	19.438 [13.001, 28.218]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.028 [0.817, 18.480]	10.685 [1.783, 17.775]
Geometric share M_{geom}/M_∞ (%)	19.24 [2.30, 47.23]	36.69 [6.86, 49.78]
Exterior spin magnitude $ \hat{a}_\infty $	0.633 [0.124, 0.954]	0.327 [0.033, 0.865]
Polarization factor A_{pol}	1.238 [1.024, 1.895]	1.580 [1.074, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.971 [0.446, 1.000]	0.815 [0.131, 0.997]
GR/CPTG matching radius r_m (km)	54.823 [42.311, 68.930]	44.639 [31.484, 57.635]
Structural Mode N_{BH}	1.5750 [1.4603, 1.8804]	1.7543 [1.4859, 1.9098]
Organization-shell radius r_N (km)	28.815 [22.229, 36.203]	23.507 [16.571, 30.357]
Inner trapping-surface radius r_- (km)	39.033 [29.652, 54.408]	28.893 [20.221, 38.149]
Structural- N lapse $\bar{\alpha}_N$	0.7526 [0.5085, 0.8431]	0.6128 [0.4849, 0.8225]
Structural- N clock $T_{24,N}$ (h)	18.063 [12.205, 20.234]	14.706 [11.638, 19.741]
Structural- N axis ω (rad s^{-1})	1559.63 [691.55, 2787.92]	2380.92 [567.94, 3895.10]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	2062.94 [1090.77, 3143.74]	2748.87 [568.26, 4338.03]
Structural- N equatorial ω (rad s^{-1})	2600.97 [1557.91, 3525.23]	3142.48 [568.47, 4786.08]
Structural- N period P_ω (ms)	3.0457 [1.9986, 5.7603]	2.2857 [1.4484, 11.0569]
Trapping-surface limit Ω_- (rad s^{-1})	3486.11 [1819.14, 4635.54]	3994.79 [619.69, 6075.91]
Trapping-surface period P_- (ms)	1.8024 [1.3554, 3.4539]	1.5728 [1.0341, 10.1392]

Table 528: CPTG compact-state, structural, clock, and geometric-rotation results for GW190727_060333. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3793	7.0320	3423.23	3877.35
500 m inward	3.7868	4.9765	3457.51	3936.06
100 m inward	1.6857	2.2268	3479.88	3982.74
10 m inward	0.5325	0.7042	3485.63	3993.55
1 m inward	0.1684	0.2227	3486.05	3994.66
10 cm inward	0.0533	0.0704	3486.10	3994.78
1 cm inward	0.0168	0.0223	3486.11	3994.79
surface limit	0.0000	0.0000	3486.11	3994.79

Table 529: Near-inner-boundary profile for GW190727_060333. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.727 M_\odot$ and $19.438 M_\odot$, with completed geometric mass-equivalent shares of 19.24% and 36.69%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.815 < 39.033 < 54.823 \text{ km}$, and component 2 gives $23.507 < 28.893 < 44.639 \text{ km}$. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5750$ and 1.7543 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.063 and 14.706 h. The determinant-averaged frame-dragging rates at those shells are 2062.94 and $2748.87 \text{ rad s}^{-1}$, approaching surface-limit values of 3486.11 and $3994.79 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $5.379/7.032 \text{ h}$ at 1 km inward to $0.0168/0.0223 \text{ h}$ at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M2 systems. Its Structural-Mode frame dragging is comparatively slow within M2. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5 M3: Upper Observed-Mass Black-Hole Binaries

The third candidate class contains 47 unique event-level numerical streams with detector-frame total masses constructed from released component marginal medians spanning $107.782\text{--}148.961 M_\odot$. The class boundary was fixed from the observational distribution before any CPTG output was examined. Within M3, 2 systems fall in the strongly asymmetric mass-ratio cross-cut, 18 are moderately asymmetric, and 27 are near-equal in the same observational screen. The distance cross-cuts contain 4 nearer, 25 intermediate-distance, and 18 farther-distance systems; 8 candidates carry the predeclared broad-posterior flag. Candidate subsections are ordered by detector-frame total mass from lower to higher values. Neither Structural Mode nor any compact-state, clock, or rotation quantity controls class membership or ordering.

As in M1, each subsection begins with the observational state and then gives the two post-handoff compact components separately. 7 M3 events have two retained release labels that are numerically identical under the declared transformation; their labels are preserved together for provenance but their identical numerical stream is reported once. The tables expose the physical compact-source mass, completed geometric mass-equivalent contribution, GR/CPTG matching radius, Structural-Mode organization shell, inner trapping surface, proper-time accumulation, and geometric frame dragging for every M3 binary. The descriptive comparisons in the closing paragraph of each subsection are made only after the fixed calculation has been completed and do not feed back into the classification or solve.

5.1 GW190421_213856

GW190421_213856 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 60.739 M_\odot$ and $m_2 = 47.043 M_\odot$, while the luminosity-distance median is 2595.5 Mpc . One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	30,870
Observed luminosity distance	2595.5 Mpc (median)
Detector-frame masses	$m_1 = 60.739 M_\odot$, $m_2 = 47.043 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.775$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4538$; $z_\pi = 0.4878$ (medians)
Post-handoff reorder fraction	23.51%

Table 530: Observational and coordinate summary for GW190421_213856. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.016 [30.267, 50.961]	32.418 [21.587, 43.202]
Physical compact-source mass M_c ($M_\odot$)	29.835 [20.878, 42.848]	20.652 [13.518, 30.162]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.669 [0.979, 21.230]	11.712 [1.791, 19.828]
Geometric share M_{geom}/M_∞ (%)	22.13 [2.61, 48.37]	37.66 [6.72, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.581 [0.093, 0.948]	0.310 [0.027, 0.868]
Polarization factor A_{pol}	1.284 [1.027, 1.937]	1.604 [1.072, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.959 [0.349, 1.000]	0.797 [0.109, 0.997]
GR/CPTG matching radius r_m (km)	59.089 [44.693, 75.250]	47.870 [31.876, 63.794]
Structural Mode N_{BH}	1.6005 [1.4619, 1.8936]	1.7657 [1.4850, 1.9105]
Organization-shell radius r_N (km)	31.097 [23.464, 39.574]	25.208 [16.774, 33.571]
Inner trapping-surface radius r_- (km)	41.231 [31.356, 56.939]	30.819 [20.877, 41.280]
Structural- N lapse $\bar{\alpha}_N$	0.7330 [0.4977, 0.8418]	0.6035 [0.4843, 0.8232]
Structural- N clock $T_{24,N}$ (h)	17.592 [11.944, 20.203]	14.485 [11.624, 19.757]
Structural- N axis ω (rad s $^{-1}$)	1539.66 [647.06, 2631.42]	2181.13 [441.79, 3752.99]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1970.34 [966.83, 2959.85]	2522.77 [442.02, 4175.53]
Structural- N equatorial ω (rad s $^{-1}$)	2429.09 [1144.25, 3327.40]	2909.50 [442.15, 4625.94]
Structural- N period P_ω (ms)	3.1889 [2.1228, 6.4987]	2.4906 [1.5048, 14.2146]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3227.33 [1271.35, 4386.20]	3652.73 [481.87, 5970.09]
Trapping-surface period P_- (ms)	1.9469 [1.4325, 4.9422]	1.7201 [1.0524, 13.0392]

Table 531: CPTG compact-state, structural, clock, and geometric-rotation results for GW190421_213856. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3870	6.8014	3170.90	3554.26
500 m inward	3.7954	4.8142	3199.32	3604.69
100 m inward	1.6928	2.1538	3221.81	3643.84
10 m inward	0.5349	0.6811	3226.88	3651.81
1 m inward	0.1691	0.2154	3227.28	3652.64
10 cm inward	0.0535	0.0681	3227.33	3652.72
1 cm inward	0.0169	0.0215	3227.33	3652.73
surface limit	0.0000	0.0000	3227.33	3652.73

Table 532: Near-inner-boundary profile for GW190421_213856. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $29.835 M_\odot$ and $20.652 M_\odot$, with completed geometric mass-equivalent shares of 22.13% and 37.66%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.097 < 41.231 < 59.089$ km, and component 2 gives $25.208 < 30.819 < 47.870$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6005$ and 1.7657 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.592 and 14.485 h. The determinant-averaged frame-dragging rates at those shells are 1970.34 and 2522.77 rad s $^{-1}$, approaching surface-limit values of 3227.33 and 3652.73 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.387/6.801 h at 1 km inward to 0.0169/0.0215 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.2 GW231008_142521

GW231008_142521 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 68.580 M_\odot$ and $m_2 = 39.543 M_\odot$, while the luminosity-distance median is 2993.1 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	49,473
Observed luminosity distance	2993.1 Mpc (median)
Detector-frame masses	$m_1 = 68.580 M_\odot$, $m_2 = 39.543 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.577$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5107$; $z_\pi = 0.5499$ (medians)
Post-handoff reorder fraction	11.40%

Table 533: Observational and coordinate summary for GW231008_142521. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.556 [29.175, 60.520]	25.118 [15.536, 39.227]
Physical compact-source mass M_c ($M_\odot$)	30.446 [20.311, 49.325]	17.231 [9.798, 27.087]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.352 [1.230, 24.738]	7.835 [0.955, 17.522]
Geometric share M_{geom}/M_∞ (%)	26.56 [3.18, 49.35]	33.15 [4.45, 49.74]
Exterior spin magnitude $ \hat{a}_\infty $	0.503 [0.058, 0.937]	0.388 [0.036, 0.912]
Polarization factor A_{pol}	1.362 [1.033, 1.974]	1.496 [1.047, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.932 [0.226, 0.999]	0.868 [0.142, 0.999]
GR/CPTG matching radius r_m (km)	64.316 [43.081, 89.366]	37.090 [22.940, 57.924]
Structural Mode N_{BH}	1.6429 [1.4649, 1.9048]	1.7134 [1.4719, 1.9093]
Organization-shell radius r_N (km)	33.856 [22.625, 47.043]	19.525 [12.062, 30.495]
Inner trapping-surface radius r_- (km)	43.458 [30.305, 65.705]	24.920 [15.037, 37.900]
Structural- N lapse $\bar{\alpha}_N$	0.7003 [0.4888, 0.8394]	0.6454 [0.4853, 0.8337]
Structural- N clock $T_{24,N}$ (h)	16.807 [11.731, 20.145]	15.490 [11.646, 20.009]
Structural- N axis ω (rad s $^{-1}$)	1490.55 [520.48, 2627.23]	2577.79 [700.73, 5093.36]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1839.41 [658.13, 2969.79]	3093.29 [705.23, 5699.05]
Structural- N equatorial ω (rad s $^{-1}$)	2212.45 [683.52, 3395.13]	3680.17 [705.83, 6363.61]
Structural- N period P_ω (ms)	3.4159 [2.1157, 9.5470]	2.0312 [1.1025, 8.9094]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2877.34 [746.66, 4498.34]	4727.46 [769.74, 8236.30]
Trapping-surface period P_- (ms)	2.1837 [1.3968, 8.4151]	1.3291 [0.7629, 8.1628]

Table 534: CPTG compact-state, structural, clock, and geometric-rotation results for GW231008_142521. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.4176	7.3015	2827.25	4579.80
500 m inward	3.8225	5.1577	2853.52	4656.25
100 m inward	1.7063	2.3043	2872.30	4715.42
10 m inward	0.5394	0.7285	2876.84	4726.32
1 m inward	0.1706	0.2304	2877.28	4727.32
10 cm inward	0.0539	0.0728	2877.33	4727.44
1 cm inward	0.0171	0.0230	2877.34	4727.46
surface limit	0.0000	0.0000	2877.34	4727.46

Table 535: Near-inner-boundary profile for GW231008_142521. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.446 M_\odot$ and $17.231 M_\odot$, with completed geometric mass-equivalent shares of 26.56% and 33.15%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.856 < 43.458 < 64.316$ km, and component 2 gives $19.525 < 24.920 < 37.090$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6429$ and 1.7134 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.807 and 15.490 h. The determinant-averaged frame-dragging rates at those shells are 1839.41 and 3093.29 rad s $^{-1}$, approaching surface-limit values of 2877.34 and 4727.46 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.418/7.301 h at 1 km inward to 0.0171/0.0230 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M3 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.3 GW241109_033317

GW241109_033317 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 61.519 M_\odot$ and $m_2 = 47.113 M_\odot$, while the luminosity-distance median is 2652.1 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,821
Observed luminosity distance	2652.1 Mpc (median)
Detector-frame masses	$m_1 = 61.519 M_\odot$, $m_2 = 47.113 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.766$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4615$; $z_\pi = 0.4967$ (medians)
Post-handoff reorder fraction	20.47%

Table 536: Observational and coordinate summary for GW241109_033317. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.315 [30.033, 52.692]	31.796 [21.689, 43.728]
Physical compact-source mass M_c ($M_\odot$)	30.148 [20.343, 43.922]	20.511 [13.250, 30.805]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.857 [0.984, 21.673]	11.293 [1.819, 19.804]
Geometric share M_{geom}/M_∞ (%)	22.32 [2.58, 48.57]	36.95 [6.65, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.578 [0.087, 0.949]	0.322 [0.028, 0.869]
Polarization factor A_{pol}	1.287 [1.026, 1.945]	1.586 [1.071, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.958 [0.328, 1.000]	0.810 [0.113, 0.997]
GR/CPTG matching radius r_m (km)	59.530 [44.347, 77.807]	46.950 [32.026, 64.570]
Structural Mode N_{BH}	1.6022 [1.4617, 1.8959]	1.7574 [1.4846, 1.9104]
Organization-shell radius r_N (km)	31.325 [23.313, 40.938]	24.739 [16.837, 34.034]
Inner trapping-surface radius r_- (km)	41.770 [30.746, 58.341]	30.413 [20.715, 42.195]
Structural- N lapse $\bar{\alpha}_N$	0.7317 [0.4958, 0.8419]	0.6103 [0.4844, 0.8235]
Structural- N clock $T_{24,N}$ (h)	17.560 [11.900, 20.206]	14.647 [11.626, 19.765]
Structural- N axis ω (rad s $^{-1}$)	1514.15 [632.14, 2654.76]	2197.91 [475.07, 3766.10]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1931.42 [922.62, 2979.14]	2544.36 [475.19, 4193.62]
Structural- N equatorial ω (rad s $^{-1}$)	2386.30 [1065.29, 3347.88]	2943.59 [475.25, 4612.86]
Structural- N period P_ω (ms)	3.2531 [2.1091, 6.8102]	2.4695 [1.4983, 13.2224]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3173.54 [1170.58, 4406.34]	3713.55 [517.70, 5956.24]
Trapping-surface period P_- (ms)	1.9799 [1.4259, 5.3676]	1.6920 [1.0549, 12.1368]

Table 537: CPTG compact-state, structural, clock, and geometric-rotation results for GW241109_033317. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3591	6.8168	3122.91	3612.39
500 m inward	3.7767	4.8236	3149.44	3663.99
100 m inward	1.6835	2.1579	3169.32	3705.54
10 m inward	0.5322	0.6824	3173.20	3712.99
1 m inward	0.1683	0.2158	3173.46	3713.56
10 cm inward	0.0532	0.0682	3173.53	3713.54
1 cm inward	0.0168	0.0216	3173.54	3713.54
surface limit	0.0000	0.0000	3173.54	3713.55

Table 538: Near-inner-boundary profile for GW241109_033317. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.148 M_\odot$ and $20.511 M_\odot$, with completed geometric mass-equivalent shares of 22.32% and 36.95%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.325 < 41.770 < 59.530$ km, and component 2 gives $24.739 < 30.413 < 46.950$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6022$ and 1.7574 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.560 and 14.647 h. The determinant-averaged frame-dragging rates at those shells are 1931.42 and 2544.36 rad s $^{-1}$, approaching surface-limit values of 3173.54 and 3713.55 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.359/6.817 h at 1 km inward to 0.0168/0.0216 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.4 GW190731_140936

GW190731_140936 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 64.908 M_\odot$ and $m_2 = 46.274 M_\odot$, while the luminosity-distance median is 3327.7 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	5,529
Observed luminosity distance	3327.7 Mpc (median)
Detector-frame masses	$m_1 = 64.908 M_\odot$, $m_2 = 46.274 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.713$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5584$; $z_\pi = 0.6010$ (medians)
Post-handoff reorder fraction	17.96%

Table 539: Observational and coordinate summary for GW190731_140936. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	39.950 [28.662, 53.254]	28.815 [18.656, 41.559]
Physical compact-source mass M_c ($M_\odot$)	28.290 [19.402, 43.480]	18.857 [11.491, 29.428]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.089 [1.107, 22.353]	9.874 [1.549, 18.739]
Geometric share M_{geom}/M_∞ (%)	25.99 [2.96, 49.41]	36.15 [5.95, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.513 [0.055, 0.941]	0.336 [0.031, 0.883]
Polarization factor A_{pol}	1.351 [1.031, 1.977]	1.566 [1.063, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.936 [0.215, 1.000]	0.824 [0.124, 0.998]
GR/CPTG matching radius r_m (km)	58.991 [42.322, 78.636]	42.548 [27.549, 61.367]
Structural Mode N_{BH}	1.6372 [1.4638, 1.9055]	1.7480 [1.4805, 1.9100]
Organization-shell radius r_N (km)	31.023 [22.235, 41.390]	22.402 [14.496, 32.318]
Inner trapping-surface radius r_- (km)	40.075 [29.368, 57.856]	27.896 [17.826, 40.227]
Structural- N lapse $\bar{\alpha}_N$	0.7047 [0.4882, 0.8403]	0.6179 [0.4847, 0.8268]
Structural- N clock $T_{24,N}$ (h)	16.913 [11.717, 20.167]	14.829 [11.633, 19.844]
Structural- N axis ω (rad s $^{-1}$)	1623.07 [544.72, 2759.99]	2379.76 [550.11, 4345.95]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	2009.17 [686.30, 3102.48]	2779.41 [550.32, 4795.46]
Structural- N equatorial ω (rad s $^{-1}$)	2426.03 [709.02, 3485.54]	3223.06 [550.43, 5337.79]
Structural- N period P_ω (ms)	3.1273 [2.0252, 9.1552]	2.2606 [1.3102, 11.4173]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3177.40 [773.81, 4598.33]	4079.00 [599.72, 6854.40]
Trapping-surface period P_- (ms)	1.9775 [1.3664, 8.1199]	1.5404 [0.9167, 10.4769]

Table 540: CPTG compact-state, structural, clock, and geometric-rotation results for GW190731_140936. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6647	7.0724	3117.09	3954.35
500 m inward	3.9977	5.0012	3150.24	4021.44
100 m inward	1.7846	2.2359	3171.04	4067.99
10 m inward	0.5641	0.7069	3176.69	4077.95
1 m inward	0.1784	0.2236	3177.33	4078.89
10 cm inward	0.0564	0.0707	3177.39	4078.99
1 cm inward	0.0178	0.0224	3177.40	4079.00
surface limit	0.0000	0.0000	3177.40	4079.00

Table 541: Near-inner-boundary profile for GW190731_140936. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.290 M_\odot$ and $18.857 M_\odot$, with completed geometric mass-equivalent shares of 25.99% and 36.15%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.023 < 40.075 < 58.991$ km, and component 2 gives $22.402 < 27.896 < 42.548$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6372$ and 1.7480 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.913 and 14.829 h. The determinant-averaged frame-dragging rates at those shells are 2009.17 and 2779.41 rad s $^{-1}$, approaching surface-limit values of 3177.40 and 4079.00 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.665/7.072 h at 1 km inward to 0.0178/0.0224 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M3 descriptive distribution, making it a useful representative upper-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.5 GW231129_081745

GW231129_081745 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 73.043 M_\odot$ and $m_2 = 39.295 M_\odot$, while the luminosity-distance median is 3783.4 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	46,884
Observed luminosity distance	3783.4 Mpc (median)
Detector-frame masses	$m_1 = 73.043 M_\odot$, $m_2 = 39.295 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.538$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6202$; $z_\pi = 0.6688$ (medians)
Post-handoff reorder fraction	7.83%

Table 542: Observational and coordinate summary for GW231129_081745. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.564 [27.671, 58.761]	23.158 [15.083, 36.252]
Physical compact-source mass M_c ($M_\odot$)	28.835 [19.198, 46.725]	16.067 [9.439, 25.423]
Geometric mass-equivalent M_{geom} ($M_\odot$)	13.192 [1.489, 25.084]	7.103 [0.857, 15.921]
Geometric share M_{geom}/M_∞ (%)	31.09 [3.95, 49.65]	32.05 [4.19, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.424 [0.042, 0.922]	0.407 [0.037, 0.917]
Polarization factor A_{pol}	1.451 [1.041, 1.986]	1.472 [1.044, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.892 [0.165, 0.999]	0.882 [0.146, 0.999]
GR/CPTG matching radius r_m (km)	64.328 [40.859, 86.768]	34.196 [22.272, 53.530]
Structural Mode N_{BH}	1.6905 [1.4692, 1.9083]	1.7011 [1.4704, 1.9092]
Organization-shell radius r_N (km)	33.867 [21.487, 45.683]	18.002 [11.724, 28.166]
Inner trapping-surface radius r_- (km)	42.122 [28.651, 62.260]	23.099 [14.614, 35.461]
Structural- N lapse $\bar{\alpha}_N$	0.6633 [0.4860, 0.8359]	0.6551 [0.4854, 0.8349]
Structural- N clock $T_{24,N}$ (h)	15.920 [11.665, 20.063]	15.722 [11.649, 20.038]
Structural- N axis ω (rad s $^{-1}$)	1573.43 [467.97, 2712.10]	2782.88 [778.35, 5304.68]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1883.88 [511.33, 3081.90]	3354.37 [786.25, 5928.93]
Structural- N equatorial ω (rad s $^{-1}$)	2198.62 [516.03, 3540.37]	4003.41 [786.71, 6602.92]
Structural- N period P_ω (ms)	3.3352 [2.0387, 12.2880]	1.8731 [1.0598, 7.9913]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2809.30 [562.89, 4691.92]	5178.84 [858.09, 8509.32]
Trapping-surface period P_- (ms)	2.2366 [1.3391, 11.1623]	1.2132 [0.7384, 7.3223]

Table 543: CPTG compact-state, structural, clock, and geometric-rotation results for GW231129_081745. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6746	7.5818	2759.25	5001.35
500 m inward	4.0082	5.3533	2785.44	5096.54
100 m inward	1.7911	2.3907	2804.49	5163.73
10 m inward	0.5663	0.7557	2808.92	5177.37
1 m inward	0.1791	0.2390	2809.26	5178.64
10 cm inward	0.0566	0.0756	2809.30	5178.82
1 cm inward	0.0179	0.0239	2809.30	5178.84
surface limit	0.0000	0.0000	2809.30	5178.84

Table 544: Near-inner-boundary profile for GW231129_081745. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.835 M_\odot$ and $16.067 M_\odot$, with completed geometric mass-equivalent shares of 31.09% and 32.05%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.867 < 42.122 < 64.328$ km, and component 2 gives $18.002 < 23.099 < 34.196$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6905$ and 1.7011 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.920 and 15.722 h. The determinant-averaged frame-dragging rates at those shells are 1883.88 and 3354.37 rad s $^{-1}$, approaching surface-limit values of 2809.30 and 5178.84 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.675/7.582 h at 1 km inward to 0.0179/0.0239 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.6 GW241230_084504

GW241230_084504 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 63.074 M_\odot$ and $m_2 = 50.529 M_\odot$, while the luminosity-distance median is 2926.2 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,400
Observed luminosity distance	2926.2 Mpc (median)
Detector-frame masses	$m_1 = 63.074 M_\odot$, $m_2 = 50.529 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.801$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5011$; $z_\pi = 0.5396$ (medians)
Post-handoff reorder fraction	27.12%

Table 545: Observational and coordinate summary for GW241230_084504. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.062 [30.941, 50.719]	33.923 [24.611, 44.128]
Physical compact-source mass M_c ($M_\odot$)	31.176 [22.186, 43.823]	21.766 [15.087, 31.077]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.474 [0.863, 19.979]	12.394 [2.087, 20.378]
Geometric share M_{geom}/M_∞ (%)	19.00 [2.24, 45.91]	37.35 [6.96, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.638 [0.155, 0.956]	0.315 [0.028, 0.863]
Polarization factor A_{pol}	1.235 [1.023, 1.849]	1.596 [1.075, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.972 [0.529, 1.000]	0.803 [0.110, 0.997]
GR/CPTG matching radius r_m (km)	59.156 [45.688, 74.892]	50.091 [36.342, 65.160]
Structural Mode N_{BH}	1.5730 [1.4599, 1.8648]	1.7621 [1.4865, 1.9105]
Organization-shell radius r_N (km)	31.113 [23.982, 39.404]	26.396 [19.122, 34.333]
Inner trapping-surface radius r_- (km)	42.452 [32.802, 58.289]	32.404 [23.659, 42.352]
Structural- N lapse $\bar{\alpha}_N$	0.7542 [0.5215, 0.8434]	0.6065 [0.4843, 0.8220]
Structural- N clock $T_{24,N}$ (h)	18.101 [12.516, 20.241]	14.555 [11.624, 19.729]
Structural- N axis ω (rad s $^{-1}$)	1444.57 [661.97, 2544.53]	2118.32 [446.59, 3398.55]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1922.17 [1046.71, 2866.65]	2448.62 [446.77, 3757.53]
Structural- N equatorial ω (rad s $^{-1}$)	2424.06 [1535.85, 3218.54]	2806.45 [446.86, 4135.93]
Structural- N period P_ω (ms)	3.2688 [2.1918, 6.0028]	2.5660 [1.6722, 14.0635]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3249.79 [1927.68, 4262.18]	3539.15 [486.88, 5312.56]
Trapping-surface period P_- (ms)	1.9334 [1.4742, 3.2595]	1.7753 [1.1827, 12.9049]

Table 546: CPTG compact-state, structural, clock, and geometric-rotation results for GW241230_084504. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.1402	6.6540	3196.20	3450.02
500 m inward	3.6163	4.7091	3224.62	3497.56
100 m inward	1.6107	2.1068	3244.60	3531.31
10 m inward	0.5089	0.6663	3249.32	3538.38
1 m inward	0.1609	0.2107	3249.75	3539.07
10 cm inward	0.0509	0.0666	3249.79	3539.14
1 cm inward	0.0161	0.0211	3249.79	3539.15
surface limit	0.0000	0.0000	3249.79	3539.15

Table 547: Near-inner-boundary profile for GW241230_084504. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.176 M_\odot$ and $21.766 M_\odot$, with completed geometric mass-equivalent shares of 19.00% and 37.35%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.113 < 42.452 < 59.156$ km, and component 2 gives $26.396 < 32.404 < 50.091$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5730$ and 1.7621 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.101 and 14.555 h. The determinant-averaged frame-dragging rates at those shells are 1922.17 and 2448.62 rad s $^{-1}$, approaching surface-limit values of 3249.79 and 3539.15 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.140/6.654 h at 1 km inward to 0.0161/0.0211 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.7 GW240907_153833

GW240907_153833 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 64.807 M_\odot$ and $m_2 = 49.210 M_\odot$, while the luminosity-distance median is 4482.7 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,343
Observed luminosity distance	4482.7 Mpc (median)
Detector-frame masses	$m_1 = 64.807 M_\odot$, $m_2 = 49.210 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.759$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7129$; $z_\pi = 0.7698$ (medians)
Post-handoff reorder fraction	21.34%

Table 548: Observational and coordinate summary for GW240907_153833. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.899 [26.163, 47.862]	28.246 [19.140, 39.030]
Physical compact-source mass M_c ($M_\odot$)	26.830 [17.967, 40.513]	18.097 [11.785, 27.264]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.727 [0.843, 19.109]	10.122 [1.521, 17.635]
Geometric share M_{geom}/M_∞ (%)	22.13 [2.44, 48.34]	37.20 [6.46, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.581 [0.094, 0.951]	0.318 [0.028, 0.873]
Polarization factor A_{pol}	1.284 [1.025, 1.936]	1.592 [1.069, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.959 [0.353, 1.000]	0.806 [0.110, 0.998]
GR/CPTG matching radius r_m (km)	53.009 [38.633, 70.674]	41.708 [28.262, 57.633]
Structural Mode N_{BH}	1.6004 [1.4610, 1.8932]	1.7603 [1.4835, 1.9105]
Organization-shell radius r_N (km)	27.889 [20.324, 37.172]	21.963 [14.889, 30.398]
Inner trapping-surface radius r_- (km)	37.088 [27.007, 53.927]	26.967 [18.295, 37.421]
Structural- N lapse $\bar{\alpha}_N$	0.7330 [0.4980, 0.8425]	0.6079 [0.4844, 0.8244]
Structural- N clock $T_{24,N}$ (h)	17.592 [11.953, 20.220]	14.590 [11.625, 19.786]
Structural- N axis ω (rad s $^{-1}$)	1695.14 [688.42, 3032.18]	2452.91 [523.45, 4293.26]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	2167.73 [1023.16, 3405.36]	2862.64 [523.63, 4768.19]
Structural- N equatorial ω (rad s $^{-1}$)	2682.46 [1275.70, 3828.54]	3297.01 [523.72, 5257.55]
Structural- N period P_ω (ms)	2.8985 [1.8451, 6.1410]	2.1949 [1.3177, 11.9993]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3552.10 [1436.33, 5025.31]	4152.30 [570.55, 6780.56]
Trapping-surface period P_- (ms)	1.7689 [1.2503, 4.3745]	1.5132 [0.9266, 11.0125]

Table 549: CPTG compact-state, structural, clock, and geometric-rotation results for GW240907_153833. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6703	7.2450	3485.45	4026.82
500 m inward	3.9940	5.1270	3520.48	4092.67
100 m inward	1.7797	2.2934	3546.80	4140.52
10 m inward	0.5624	0.7252	3551.45	4150.73
1 m inward	0.1778	0.2293	3552.03	4152.11
10 cm inward	0.0562	0.0725	3552.10	4152.28
1 cm inward	0.0178	0.0229	3552.10	4152.30
surface limit	0.0000	0.0000	3552.10	4152.30

Table 550: Near-inner-boundary profile for GW240907_153833. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.830 M_\odot$ and $18.097 M_\odot$, with completed geometric mass-equivalent shares of 22.13% and 37.20%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.889 < 37.088 < 53.009$ km, and component 2 gives $21.963 < 26.967 < 41.708$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6004$ and 1.7603 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.592 and 14.590 h. The determinant-averaged frame-dragging rates at those shells are 2167.73 and 2862.64 rad s $^{-1}$, approaching surface-limit values of 3552.10 and 4152.30 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.670/7.245 h at 1 km inward to 0.0178/0.0229 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.8 GW240501_033534

GW240501_033534 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 67.088 M_\odot$ and $m_2 = 48.864 M_\odot$, while the luminosity-distance median is 4695.8 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,007
Observed luminosity distance	4695.8 Mpc (median)
Detector-frame masses	$m_1 = 67.088 M_\odot$, $m_2 = 48.864 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.728$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7405$; $z_\pi = 0.8000$ (medians)
Post-handoff reorder fraction	19.73%

Table 551: Observational and coordinate summary for GW240501_033534. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	36.438 [26.618, 48.750]	27.420 [18.845, 38.854]
Physical compact-source mass M_c ($M_\odot$)	27.090 [18.323, 40.205]	17.976 [11.508, 27.099]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.984 [0.881, 19.754]	9.586 [1.405, 17.476]
Geometric share M_{geom}/M_∞ (%)	22.42 [2.60, 48.47]	36.25 [6.01, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.576 [0.090, 0.948]	0.334 [0.029, 0.882]
Polarization factor A_{pol}	1.289 [1.027, 1.940]	1.569 [1.064, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.957 [0.340, 1.000]	0.823 [0.116, 0.998]
GR/CPTG matching radius r_m (km)	53.805 [39.304, 71.985]	40.489 [27.827, 57.372]
Structural Mode N_{BH}	1.6030 [1.4619, 1.8947]	1.7491 [1.4808, 1.9103]
Organization-shell radius r_N (km)	28.336 [20.632, 37.890]	21.331 [14.657, 30.182]
Inner trapping-surface radius r_- (km)	37.644 [27.535, 53.469]	26.468 [17.967, 37.420]
Structural- N lapse $\bar{\alpha}_N$	0.7310 [0.4968, 0.8418]	0.6170 [0.4845, 0.8266]
Structural- N clock $T_{24,N}$ (h)	17.544 [11.924, 20.204]	14.807 [11.628, 19.838]
Structural- N axis ω (rad s $^{-1}$)	1669.53 [693.16, 2946.92]	2518.09 [569.49, 4387.92]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	2129.45 [1024.94, 3320.92]	2943.73 [569.71, 4872.92]
Structural- N equatorial ω (rad s $^{-1}$)	2636.24 [1212.20, 3748.55]	3400.69 [569.82, 5375.91]
Structural- N period P_ω (ms)	2.9506 [1.8920, 6.1303]	2.1344 [1.2894, 11.0287]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3500.21 [1333.15, 4943.28]	4317.06 [620.80, 6889.37]
Trapping-surface period P_- (ms)	1.7951 [1.2711, 4.7130]	1.4554 [0.9120, 10.1211]

Table 552: CPTG compact-state, structural, clock, and geometric-rotation results for GW240501_033534. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.6379	7.2762	3431.38	4181.48
500 m inward	3.9751	5.1472	3468.45	4248.64
100 m inward	1.7720	2.3025	3494.42	4304.87
10 m inward	0.5599	0.7283	3499.71	4315.89
1 m inward	0.1771	0.2303	3500.18	4316.94
10 cm inward	0.0560	0.0728	3500.22	4317.05
1 cm inward	0.0177	0.0230	3500.21	4317.06
surface limit	0.0000	0.0000	3500.21	4317.06

Table 553: Near-inner-boundary profile for GW240501_033534. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $27.090 M_\odot$ and $17.976 M_\odot$, with completed geometric mass-equivalent shares of 22.42% and 36.25%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $28.336 < 37.644 < 53.805$ km, and component 2 gives $21.331 < 26.468 < 40.489$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6030$ and 1.7491 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.544 and 14.807 h. The determinant-averaged frame-dragging rates at those shells are 2129.45 and 2943.73 rad s $^{-1}$, approaching surface-limit values of 3500.21 and 4317.06 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.638/7.276 h at 1 km inward to 0.0177/0.0230 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M3 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.9 GW200128_022011

GW200128_022011 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 65.053 M_\odot$ and $m_2 = 51.158 M_\odot$, while the luminosity-distance median is 3352.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	6,638
Observed luminosity distance	3352.3 Mpc (median)
Detector-frame masses	$m_1 = 65.053 M_\odot$, $m_2 = 51.158 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.786$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5618$; $z_\pi = 0.6047$ (medians)
Post-handoff reorder fraction	20.88%

Table 554: Observational and coordinate summary for GW200128_022011. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.181 [30.214, 52.538]	32.686 [22.812, 43.862]
Physical compact-source mass M_c ($M_\odot$)	32.694 [20.982, 47.006]	21.560 [13.795, 32.766]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.292 [0.653, 18.605]	10.876 [1.764, 19.186]
Geometric share M_{geom}/M_∞ (%)	15.77 [1.65, 44.96]	34.13 [5.86, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.697 [0.175, 0.967]	0.371 [0.034, 0.885]
Polarization factor A_{pol}	1.187 [1.017, 1.817]	1.518 [1.062, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.982 [0.577, 1.000]	0.855 [0.134, 0.998]
GR/CPTG matching radius r_m (km)	59.333 [44.615, 77.579]	48.265 [33.685, 64.767]
Structural Mode N_{BH}	1.5469 [1.4569, 1.8535]	1.7246 [1.4799, 1.9096]
Organization-shell radius r_N (km)	31.208 [23.447, 40.785]	25.435 [17.729, 34.159]
Inner trapping-surface radius r_- (km)	43.943 [31.062, 62.660]	31.740 [21.611, 44.238]
Structural- N lapse $\bar{\alpha}_N$	0.7744 [0.5309, 0.8459]	0.6365 [0.4850, 0.8273]
Structural- N clock $T_{24,N}$ (h)	18.586 [12.742, 20.301]	15.277 [11.640, 19.855]
Structural- N axis ω (rad s $^{-1}$)	1274.89 [612.82, 2635.93]	2123.64 [547.26, 3601.24]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1779.21 [984.20, 2962.31]	2484.47 [548.99, 4002.09]
Structural- N equatorial ω (rad s $^{-1}$)	2343.68 [1507.05, 3320.32]	2884.64 [549.18, 4398.81]
Structural- N period P_ω (ms)	3.5315 [2.1210, 6.3841]	2.5290 [1.5700, 11.4450]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3184.27 [2015.34, 4375.50]	3689.26 [598.63, 5670.66]
Trapping-surface period P_- (ms)	1.9732 [1.4360, 3.1177]	1.7031 [1.1080, 10.4960]

Table 555: CPTG compact-state, structural, clock, and geometric-rotation results for GW200128_022011. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.8394	6.6388	3137.58	3593.04
500 m inward	3.4040	4.6941	3162.22	3645.34
100 m inward	1.5142	2.0988	3180.27	3680.51
10 m inward	0.4783	0.6637	3184.01	3688.39
1 m inward	0.1512	0.2099	3184.22	3689.19
10 cm inward	0.0478	0.0664	3184.26	3689.25
1 cm inward	0.0151	0.0210	3184.27	3689.26
surface limit	0.0000	0.0000	3184.27	3689.26

Table 556: Near-inner-boundary profile for GW200128_022011. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $32.694 M_\odot$ and $21.560 M_\odot$, with completed geometric mass-equivalent shares of 15.77% and 34.13%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.208 < 43.943 < 59.333$ km, and component 2 gives $25.435 < 31.740 < 48.265$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5469$ and 1.7246 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.586 and 15.277 h. The determinant-averaged frame-dragging rates at those shells are 1779.21 and 2484.47 rad s $^{-1}$, approaching surface-limit values of 3184.27 and 3689.26 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.839/6.639 h at 1 km inward to 0.0151/0.0210 h at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.10 GW241124_024914

GW241124_024914 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 73.068 M_\odot$ and $m_2 = 45.760 M_\odot$, while the luminosity-distance median is 4538.5 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	17,634
Observed luminosity distance	4538.5 Mpc (median)
Detector-frame masses	$m_1 = 73.068 M_\odot$, $m_2 = 45.760 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.626$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7202$; $z_\pi = 0.7778$ (medians)
Post-handoff reorder fraction	14.55%

Table 557: Observational and coordinate summary for GW241124_024914. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.314 [26.716, 71.740]	25.316 [14.322, 39.258]
Physical compact-source mass M_c ($M_\odot$)	28.465 [18.414, 56.619]	16.916 [9.251, 27.079]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.846 [1.172, 26.704]	8.203 [0.998, 17.487]
Geometric share M_{geom}/M_∞ (%)	27.44 [2.97, 49.48]	34.67 [4.83, 49.78]
Exterior spin magnitude $ \hat{a}_\infty $	0.487 [0.052, 0.941]	0.362 [0.033, 0.905]
Polarization factor A_{pol}	1.378 [1.031, 1.979]	1.531 [1.051, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.926 [0.203, 1.000]	0.848 [0.133, 0.999]
GR/CPTG matching radius r_m (km)	59.529 [39.450, 105.934]	37.382 [21.148, 57.969]
Structural Mode N_{BH}	1.6518 [1.4638, 1.9063]	1.7308 [1.4740, 1.9097]
Organization-shell radius r_N (km)	31.346 [20.744, 55.687]	19.681 [11.129, 30.476]
Inner trapping-surface radius r_- (km)	40.416 [27.457, 75.999]	24.757 [13.997, 37.765]
Structural- N lapse $\bar{\alpha}_N$	0.6934 [0.4876, 0.8403]	0.6316 [0.4850, 0.8320]
Structural- N clock $T_{24,N}$ (h)	16.641 [11.703, 20.166]	15.159 [11.639, 19.969]
Structural- N axis ω (rad s $^{-1}$)	1542.00 [411.81, 2907.64]	2614.82 [659.17, 5360.99]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1919.88 [547.62, 3301.29]	3101.96 [660.24, 6065.39]
Structural- N equatorial ω (rad s $^{-1}$)	2350.27 [593.45, 3745.32]	3648.34 [660.34, 6844.97]
Structural- N period P_ω (ms)	3.2727 [1.9032, 11.4736]	2.0256 [1.0359, 9.5166]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3041.62 [649.25, 4944.16]	4644.06 [719.40, 8796.73]
Trapping-surface period P_- (ms)	2.0657 [1.2708, 9.6776]	1.3530 [0.7143, 8.7340]

Table 558: CPTG compact-state, structural, clock, and geometric-rotation results for GW241124_024914. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5754	7.4109	2989.10	4490.30
500 m inward	3.9333	5.2383	3016.73	4571.94
100 m inward	1.7553	2.3398	3036.48	4629.80
10 m inward	0.5549	0.7398	3041.12	4643.29
1 m inward	0.1754	0.2339	3041.57	4643.90
10 cm inward	0.0555	0.0740	3041.62	4644.05
1 cm inward	0.0175	0.0234	3041.62	4644.06
surface limit	0.0000	0.0000	3041.62	4644.06

Table 559: Near-inner-boundary profile for GW241124_024914. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $28.465 M_\odot$ and $16.916 M_\odot$, with completed geometric mass-equivalent shares of 27.44% and 34.67%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.346 < 40.416 < 59.529$ km, and component 2 gives $19.681 < 24.757 < 37.382$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6518$ and 1.7308 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.641 and 15.159 h. The determinant-averaged frame-dragging rates at those shells are 1919.88 and 3101.96 rad s $^{-1}$, approaching surface-limit values of 3041.62 and 4644.06 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.575/7.411 h at 1 km inward to 0.0175/0.0234 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Its Structural-Mode frame dragging is in the upper part of the M3 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.11 GW241009_022835

GW241009_022835 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 68.326 M_\odot$ and $m_2 = 50.761 M_\odot$, while the luminosity-distance median is 5589.1 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,066
Observed luminosity distance	5589.1 Mpc (median)
Detector-frame masses	$m_1 = 68.326 M_\odot$, $m_2 = 50.761 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.743$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8536$; $z_\pi = 0.9237$ (medians)
Post-handoff reorder fraction	19.26%

Table 560: Observational and coordinate summary for GW241009_022835. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	35.020 [24.799, 49.600]	26.592 [16.677, 38.350]
Physical compact-source mass M_c ($M_\odot$)	26.750 [17.275, 42.567]	17.406 [10.450, 27.967]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.806 [0.736, 18.666]	8.975 [1.296, 16.944]
Geometric share M_{geom}/M_∞ (%)	19.62 [2.10, 48.12]	35.57 [5.57, 49.78]
Exterior spin magnitude $ \hat{a}_\infty $	0.627 [0.101, 0.958]	0.346 [0.033, 0.890]
Polarization factor A_{pol}	1.244 [1.021, 1.927]	1.552 [1.059, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.374, 1.000]	0.834 [0.131, 0.998]
GR/CPTG matching radius r_m (km)	51.712 [36.619, 73.240]	39.267 [24.625, 56.628]
Structural Mode N_{BH}	1.5782 [1.4592, 1.8907]	1.7411 [1.4783, 1.9098]
Organization-shell radius r_N (km)	27.193 [19.241, 38.472]	20.671 [12.938, 29.830]
Inner trapping-surface radius r_- (km)	36.774 [25.577, 56.803]	25.677 [16.089, 38.171]
Structural- N lapse $\bar{\alpha}_N$	0.7501 [0.5001, 0.8439]	0.6234 [0.4849, 0.8286]
Structural- N clock $T_{24,N}$ (h)	18.003 [12.002, 20.255]	14.961 [11.638, 19.887]
Structural- N axis ω (rad s $^{-1}$)	1613.57 [651.25, 3156.85]	2554.66 [660.29, 4752.42]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	2136.85 [980.73, 3587.24]	2991.77 [660.69, 5330.04]
Structural- N equatorial ω (rad s $^{-1}$)	2709.28 [1307.74, 4056.34]	3470.88 [660.88, 5920.72]
Structural- N period P_ω (ms)	2.9404 [1.7515, 6.4066]	2.1002 [1.1788, 9.5101]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3619.44 [1485.34, 5323.26]	4404.32 [720.31, 7621.59]
Trapping-surface period P_- (ms)	1.7360 [1.1803, 4.2301]	1.4266 [0.8244, 8.7229]

Table 561: CPTG compact-state, structural, clock, and geometric-rotation results for GW241009_022835. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.5417	7.3694	3550.43	4257.26
500 m inward	3.8981	5.2110	3586.79	4335.56
100 m inward	1.7349	2.3295	3614.66	4392.92
10 m inward	0.5481	0.7367	3618.85	4403.16
1 m inward	0.1733	0.2330	3619.38	4404.11
10 cm inward	0.0548	0.0737	3619.43	4404.30
1 cm inward	0.0173	0.0233	3619.44	4404.32
surface limit	0.0000	0.0000	3619.44	4404.32

Table 562: Near-inner-boundary profile for GW241009_022835. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $26.750 M_\odot$ and $17.406 M_\odot$, with completed geometric mass-equivalent shares of 19.62% and 35.57%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $27.193 < 36.774 < 51.712$ km, and component 2 gives $20.671 < 25.677 < 39.267$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5782$ and 1.7411 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.003 and 14.961 h. The determinant-averaged frame-dragging rates at those shells are 2136.85 and 2991.77 rad s $^{-1}$, approaching surface-limit values of 3619.44 and 4404.32 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.542/7.369 h at 1 km inward to 0.0173/0.0233 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M3 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.12 GW241102_144729

GW241102_144729 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 69.768 M_\odot$ and $m_2 = 49.705 M_\odot$, while the luminosity-distance median is 3446.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,659
Observed luminosity distance	3446.3 Mpc (median)
Detector-frame masses	$m_1 = 69.768 M_\odot$, $m_2 = 49.705 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.712$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5741$; $z_\pi = 0.6188$ (medians)
Post-handoff reorder fraction	19.13%

Table 563: Observational and coordinate summary for GW241102_144729. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	42.434 [30.862, 55.367]	30.814 [19.338, 43.588]
Physical compact-source mass M_c ($M_\odot$)	30.587 [21.358, 47.110]	20.049 [12.259, 29.558]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.063 [1.163, 22.401]	10.759 [1.552, 20.248]
Geometric share M_{geom}/M_∞ (%)	24.14 [2.98, 48.70]	36.91 [6.10, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.545 [0.083, 0.941]	0.323 [0.028, 0.880]
Polarization factor A_{pol}	1.318 [1.031, 1.949]	1.585 [1.065, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.948 [0.315, 1.000]	0.811 [0.111, 0.998]
GR/CPTG matching radius r_m (km)	62.659 [45.572, 81.756]	45.501 [28.555, 64.363]
Structural Mode N_{BH}	1.6192 [1.4639, 1.8973]	1.7569 [1.4814, 1.9105]
Organization-shell radius r_N (km)	32.992 [23.938, 42.936]	23.953 [15.007, 33.846]
Inner trapping-surface radius r_- (km)	42.732 [32.088, 62.613]	29.661 [18.769, 40.926]
Structural- N lapse $\bar{\alpha}_N$	0.7185 [0.4947, 0.8402]	0.6107 [0.4844, 0.8262]
Structural- N clock $T_{24,N}$ (h)	17.245 [11.873, 20.165]	14.657 [11.625, 19.828]
Structural- N axis ω (rad s $^{-1}$)	1531.90 [587.51, 2552.44]	2236.03 [458.13, 4114.63]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1915.10 [845.26, 2862.49]	2615.81 [458.35, 4597.88]
Structural- N equatorial ω (rad s $^{-1}$)	2317.65 [1006.62, 3239.23]	3028.69 [458.46, 5119.44]
Structural- N period P_ω (ms)	3.2809 [2.1950, 7.4334]	2.4020 [1.3665, 13.7083]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3030.32 [1104.00, 4289.14]	3824.37 [499.59, 6655.68]
Trapping-surface period P_- (ms)	2.0734 [1.4649, 5.6913]	1.6429 [0.9440, 12.5768]

Table 564: CPTG compact-state, structural, clock, and geometric-rotation results for GW241102_144729. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3671	6.8650	2980.24	3720.16
500 m inward	3.7831	4.8556	3005.91	3773.17
100 m inward	1.6875	2.1721	3025.95	3814.52
10 m inward	0.5334	0.6870	3030.05	3823.21
1 m inward	0.1687	0.2172	3030.27	3824.23
10 cm inward	0.0533	0.0687	3030.31	3824.36
1 cm inward	0.0169	0.0217	3030.32	3824.37
surface limit	0.0000	0.0000	3030.32	3824.37

Table 565: Near-inner-boundary profile for GW241102_144729. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.587 M_\odot$ and $20.049 M_\odot$, with completed geometric mass-equivalent shares of 24.14% and 36.91%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.992 < 42.732 < 62.659$ km, and component 2 gives $23.953 < 29.661 < 45.501$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6192$ and 1.7569 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.245 and 14.657 h. The determinant-averaged frame-dragging rates at those shells are 1915.10 and 2615.81 rad s $^{-1}$, approaching surface-limit values of 3030.32 and 3824.37 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.367/6.865 h at 1 km inward to 0.0169/0.0217 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.13 GW191222_033537

GW191222_033537 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 67.152 M_\odot$ and $m_2 = 52.412 M_\odot$, while the luminosity-distance median is 2986.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	8,528
Observed luminosity distance	2986.3 Mpc (median)
Detector-frame masses	$m_1 = 67.152 M_\odot$, $m_2 = 52.412 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.781$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5103$; $z_\pi = 0.5489$ (medians)
Post-handoff reorder fraction	23.67%

Table 566: Observational and coordinate summary for GW191222_033537. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	42.781 [32.365, 54.645]	35.109 [23.645, 46.929]
Physical compact-source mass M_c ($M_\odot$)	30.642 [21.668, 43.820]	21.860 [14.541, 31.572]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.916 [1.377, 23.173]	13.344 [2.544, 21.713]
Geometric share M_{geom}/M_∞ (%)	26.00 [3.53, 49.00]	39.48 [9.04, 49.89]
Exterior spin magnitude $ \hat{a}_\infty $	0.513 [0.072, 0.930]	0.278 [0.023, 0.823]
Polarization factor A_{pol}	1.351 [1.037, 1.961]	1.652 [1.099, 1.996]
Compact-source spin parameter $\hat{a}_c$	0.936 [0.278, 0.999]	0.758 [0.093, 0.995]
GR/CPTG matching radius r_m (km)	63.171 [47.791, 80.690]	51.842 [34.915, 69.297]
Structural Mode N_{BH}	1.6373 [1.4668, 1.9008]	1.7875 [1.4994, 1.9110]
Organization-shell radius r_N (km)	33.279 [25.136, 42.472]	27.291 [18.344, 36.455]
Inner trapping-surface radius r_- (km)	43.074 [32.994, 58.439]	32.878 [22.709, 43.818]
Structural- N lapse $\bar{\alpha}_N$	0.7046 [0.4919, 0.8378]	0.5858 [0.4840, 0.8118]
Structural- N clock $T_{24,N}$ (h)	16.911 [11.807, 20.108]	14.059 [11.616, 19.482]
Structural- N axis ω (rad s $^{-1}$)	1571.98 [610.75, 2494.92]	2031.06 [363.35, 3461.79]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1926.31 [828.46, 2795.06]	2340.83 [363.51, 3851.66]
Structural- N equatorial ω (rad s $^{-1}$)	2302.36 [860.34, 3146.48]	2648.00 [363.59, 4265.80]
Structural- N period P_ω (ms)	3.2618 [2.2480, 7.5842]	2.6842 [1.6313, 17.2847]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3016.38 [943.60, 4138.14]	3273.48 [396.10, 5447.95]
Trapping-surface period P_- (ms)	2.0830 [1.5184, 6.6587]	1.9194 [1.1533, 15.8625]

Table 567: CPTG compact-state, structural, clock, and geometric-rotation results for GW191222_033537. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.4319	6.6283	2965.38	3193.44
500 m inward	3.8313	4.6935	2993.63	3237.17
100 m inward	1.7105	2.1000	3012.72	3266.84
10 m inward	0.5406	0.6642	3015.99	3273.04
1 m inward	0.1710	0.2100	3016.32	3273.44
10 cm inward	0.0541	0.0664	3016.37	3273.48
1 cm inward	0.0171	0.0210	3016.37	3273.48
surface limit	0.0000	0.0000	3016.38	3273.48

Table 568: Near-inner-boundary profile for GW191222_033537. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.642 M_\odot$ and $21.860 M_\odot$, with completed geometric mass-equivalent shares of 26.00% and 39.48%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.279 < 43.074 < 63.171$ km, and component 2 gives $27.291 < 32.878 < 51.842$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6373$ and 1.7875 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.911 and 14.059 h. The determinant-averaged frame-dragging rates at those shells are 1926.31 and 2340.83 rad s $^{-1}$, approaching surface-limit values of 3016.38 and 3273.48 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.432/6.628 h at 1 km inward to 0.0171/0.0210 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.14 GW240514_121713

GW240514_121713 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 67.354 M_\odot$ and $m_2 = 53.333 M_\odot$, while the luminosity-distance median is 2590.1 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,943
Observed luminosity distance	2590.1 Mpc (median)
Detector-frame masses	$m_1 = 67.354 M_\odot$, $m_2 = 53.333 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.792$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4524$; $z_\pi = 0.4869$ (medians)
Post-handoff reorder fraction	23.85%

Table 569: Observational and coordinate summary for GW240514_121713. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	44.829 [35.048, 54.907]	37.007 [25.488, 46.975]
Physical compact-source mass M_c ($M_\odot$)	31.857 [22.844, 46.392]	22.642 [15.524, 33.235]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.620 [1.484, 23.756]	14.125 [2.787, 21.901]
Geometric share M_{geom}/M_∞ (%)	26.31 [3.47, 49.20]	39.76 [8.83, 49.87]
Exterior spin magnitude $ \hat{a}_\infty $	0.507 [0.064, 0.931]	0.273 [0.026, 0.827]
Polarization factor A_{pol}	1.357 [1.036, 1.969]	1.660 [1.097, 1.995]
Compact-source spin parameter $\hat{a}_c$	0.934 [0.249, 0.999]	0.751 [0.102, 0.995]
GR/CPTG matching radius r_m (km)	66.196 [51.753, 81.077]	54.645 [37.637, 69.364]
Structural Mode N_{BH}	1.6403 [1.4665, 1.9031]	1.7908 [1.4980, 1.9108]
Organization-shell radius r_N (km)	34.866 [27.204, 42.629]	28.781 [19.811, 36.548]
Inner trapping-surface radius r_- (km)	44.631 [35.328, 61.486]	34.478 [24.204, 45.103]
Structural- N lapse $\bar{\alpha}_N$	0.7023 [0.4901, 0.8381]	0.5831 [0.4842, 0.8128]
Structural- N clock $T_{24,N}$ (h)	16.855 [11.762, 20.114]	13.993 [11.620, 19.508]
Structural- N axis ω (rad s $^{-1}$)	1487.39 [562.38, 2364.66]	1961.32 [367.51, 3231.53]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1838.07 [735.54, 2641.76]	2225.93 [367.67, 3599.66]
Structural- N equatorial ω (rad s $^{-1}$)	2191.71 [740.53, 2934.18]	2520.34 [367.79, 3960.61]
Structural- N period P_ω (ms)	3.4184 [2.3784, 8.5423]	2.8227 [1.7455, 17.0891]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2865.55 [809.32, 3836.75]	3130.23 [400.88, 5062.37]
Trapping-surface period P_- (ms)	2.1927 [1.6376, 7.7636]	2.0073 [1.2412, 15.6735]

Table 570: CPTG compact-state, structural, clock, and geometric-rotation results for GW240514_121713. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3615	6.5328	2817.65	3052.35
500 m inward	3.7834	4.6259	2842.18	3092.05
100 m inward	1.6889	2.0706	2860.54	3123.01
10 m inward	0.5338	0.6549	2865.27	3129.41
1 m inward	0.1688	0.2071	2865.58	3130.15
10 cm inward	0.0534	0.0655	2865.55	3130.22
1 cm inward	0.0169	0.0207	2865.55	3130.23
surface limit	0.0000	0.0000	2865.55	3130.23

Table 571: Near-inner-boundary profile for GW240514_121713. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.857 M_\odot$ and $22.642 M_\odot$, with completed geometric mass-equivalent shares of 26.31% and 39.76%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $34.866 < 44.631 < 66.196$ km, and component 2 gives $28.781 < 34.478 < 54.645$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6403$ and 1.7908 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.855 and 13.993 h. The determinant-averaged frame-dragging rates at those shells are 1838.07 and 2225.93 rad s $^{-1}$, approaching surface-limit values of 2865.55 and 3130.23 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.362/6.533 h at 1 km inward to 0.0169/0.0207 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.15 GW170729_185629

GW170729_185629 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 77.883 M_\odot$ and $m_2 = 44.135 M_\odot$, while the luminosity-distance median is 2492.5 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	14,844
Observed luminosity distance	2492.5 Mpc (median)
Detector-frame masses	$m_1 = 77.883 M_\odot$, $m_2 = 44.135 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.567$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4386$; $z_\pi = 0.4713$ (medians)
Post-handoff reorder fraction	6.24%

Table 572: Observational and coordinate summary for GW170729_185629. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	53.419 [38.225, 66.159]	29.578 [19.685, 43.483]
Physical compact-source mass M_c ($M_\odot$)	40.602 [27.138, 58.240]	20.749 [12.076, 33.198]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.707 [0.973, 27.440]	8.648 [1.039, 18.199]
Geometric share M_{geom}/M_∞ (%)	18.70 [2.01, 48.71]	30.34 [3.75, 49.66]
Exterior spin magnitude $ \hat{a}_\infty $	0.643 [0.082, 0.960]	0.437 [0.041, 0.926]
Polarization factor A_{pol}	1.230 [1.021, 1.950]	1.435 [1.039, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.973 [0.313, 1.000]	0.900 [0.163, 0.999]
GR/CPTG matching radius r_m (km)	78.880 [56.444, 97.692]	43.676 [29.067, 64.208]
Structural Mode N_{BH}	1.5705 [1.4588, 1.8975]	1.6823 [1.4681, 1.9084]
Organization-shell radius r_N (km)	41.481 [29.588, 51.308]	22.997 [15.288, 33.805]
Inner trapping-surface radius r_- (km)	55.460 [41.132, 77.535]	29.666 [18.743, 44.738]
Structural- N lapse $\bar{\alpha}_N$	0.7561 [0.4946, 0.8443]	0.6698 [0.4860, 0.8368]
Structural- N clock $T_{24,N}$ (h)	18.147 [11.870, 20.264]	16.075 [11.664, 20.084]
Structural- N axis ω (rad s $^{-1}$)	1032.19 [459.77, 1899.71]	2186.96 [701.82, 4162.83]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1396.02 [686.77, 2150.06]	2624.31 [754.17, 4633.88]
Structural- N equatorial ω (rad s $^{-1}$)	1790.66 [812.06, 2508.86]	3132.79 [754.49, 5084.58]
Structural- N period P_ω (ms)	4.5008 [2.9223, 9.1489]	2.3942 [1.3559, 8.3312]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2398.34 [889.18, 3392.18]	4072.75 [822.43, 6522.40]
Trapping-surface period P_- (ms)	2.6198 [1.8523, 7.0663]	1.5427 [0.9633, 7.6398]

Table 573: CPTG compact-state, structural, clock, and geometric-rotation results for GW170729_185629. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.4582	6.6739	2369.12	3969.26
500 m inward	3.1393	4.7133	2384.05	4024.38
100 m inward	1.3996	2.1045	2395.64	4063.76
10 m inward	0.4422	0.6653	2398.06	4071.93
1 m inward	0.1398	0.2104	2398.31	4072.64
10 cm inward	0.0442	0.0665	2398.34	4072.74
1 cm inward	0.0140	0.0210	2398.34	4072.75
surface limit	0.0000	0.0000	2398.34	4072.75

Table 574: Near-inner-boundary profile for GW170729_185629. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $40.602 M_\odot$ and $20.749 M_\odot$, with completed geometric mass-equivalent shares of 18.70% and 30.34%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $41.481 < 55.460 < 78.880$ km, and component 2 gives $22.997 < 29.666 < 43.676$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5705$ and 1.6823 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.147 and 16.075 h. The determinant-averaged frame-dragging rates at those shells are 1396.02 and 2624.31 rad s $^{-1}$, approaching surface-limit values of 2398.34 and 4072.75 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.458/6.674 h at 1 km inward to 0.0140/0.0210 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.16 GW231118_071402

GW231118_071402 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 72.267 M_\odot$ and $m_2 = 50.728 M_\odot$, while the luminosity-distance median is 4311.2 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	46,647
Observed luminosity distance	4311.2 Mpc (median)
Detector-frame masses	$m_1 = 72.267 M_\odot$, $m_2 = 50.728 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.702$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6905$; $z_\pi = 0.7454$ (medians)
Post-handoff reorder fraction	17.79%

Table 575: Observational and coordinate summary for GW231118_071402. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	40.558 [28.416, 56.157]	29.047 [18.444, 42.591]
Physical compact-source mass M_c ($M_\odot$)	30.467 [20.229, 46.580]	19.304 [11.471, 29.999]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.429 [0.907, 22.262]	9.621 [1.323, 18.945]
Geometric share M_{geom}/M_∞ (%)	21.20 [2.41, 48.33]	34.92 [5.32, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.598 [0.094, 0.952]	0.357 [0.033, 0.895]
Polarization factor A_{pol}	1.269 [1.025, 1.935]	1.536 [1.056, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.963 [0.354, 1.000]	0.844 [0.130, 0.998]
GR/CPTG matching radius r_m (km)	59.888 [41.960, 82.922]	42.892 [27.235, 62.890]
Structural Mode N_{BH}	1.5920 [1.4609, 1.8931]	1.7336 [1.4768, 1.9098]
Organization-shell radius r_N (km)	31.499 [22.031, 43.614]	22.592 [14.329, 33.112]
Inner trapping-surface radius r_- (km)	42.260 [29.851, 62.038]	28.294 [17.691, 41.327]
Structural- N lapse $\bar{\alpha}_N$	0.7395 [0.4981, 0.8426]	0.6294 [0.4849, 0.8298]
Structural- N clock $T_{24,N}$ (h)	17.747 [11.955, 20.223]	15.105 [11.638, 19.914]
Structural- N axis ω (rad s $^{-1}$)	1463.48 [608.57, 2686.03]	2327.90 [586.71, 4328.56]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1891.60 [897.66, 3044.63]	2742.69 [587.35, 4847.57]
Structural- N equatorial ω (rad s $^{-1}$)	2365.47 [1114.60, 3469.72]	3204.35 [587.65, 5364.87]
Structural- N period P_ω (ms)	3.3216 [2.0637, 6.9995]	2.2909 [1.2962, 10.6976]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3150.47 [1247.10, 4625.81]	4090.77 [640.71, 6915.44]
Trapping-surface period P_- (ms)	1.9944 [1.3583, 5.0382]	1.5359 [0.9086, 9.8066]

Table 576: CPTG compact-state, structural, clock, and geometric-rotation results for GW231118_071402. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.2563	6.9672	3099.21	3974.35
500 m inward	3.7011	4.9270	3125.94	4036.29
100 m inward	1.6492	2.2022	3145.87	4080.05
10 m inward	0.5210	0.6963	3149.90	4089.76
1 m inward	0.1648	0.2202	3150.40	4090.65
10 cm inward	0.0521	0.0696	3150.46	4090.76
1 cm inward	0.0165	0.0220	3150.47	4090.77
surface limit	0.0000	0.0000	3150.47	4090.77

Table 577: Near-inner-boundary profile for GW231118_071402. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.467 M_\odot$ and $19.304 M_\odot$, with completed geometric mass-equivalent shares of 21.20% and 34.92%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $31.499 < 42.260 < 59.888$ km, and component 2 gives $22.592 < 28.294 < 42.892$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5920$ and 1.7336 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.747 and 15.105 h. The determinant-averaged frame-dragging rates at those shells are 1891.60 and 2742.69 rad s $^{-1}$, approaching surface-limit values of 3150.47 and 4090.77 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.256/6.967 h at 1 km inward to 0.0165/0.0220 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M3 descriptive distribution, making it a useful representative upper-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.17 GW191127_050227

GW191127_050227 is an M3 upper observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 86.410 M_\odot$ and $m_2 = 38.451 M_\odot$, while the luminosity-distance median is 3380.8 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C01:Mixed
Posterior samples	7,811
Observed luminosity distance	3380.8 Mpc (median)
Detector-frame masses	$m_1 = 86.410 M_\odot$, $m_2 = 38.451 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.445$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5658$; $z_\pi = 0.6090$ (medians)
Post-handoff reorder fraction	8.26%

Table 578: Observational and coordinate summary for GW191127_050227. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	51.471 [30.215, 98.335]	23.376 [9.666, 42.100]
Physical compact-source mass M_c ($M_\odot$)	39.770 [23.197, 92.202]	16.510 [6.453, 31.600]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.695 [0.738, 27.719]	5.758 [0.577, 16.931]
Geometric share M_{geom}/M_∞ (%)	15.48 [1.38, 47.28]	28.16 [3.18, 49.66]
Exterior spin magnitude $ \hat{a}_\infty $	0.702 [0.123, 0.973]	0.475 [0.042, 0.937]
Polarization factor A_{pol}	1.183 [1.014, 1.897]	1.392 [1.033, 1.986]
Compact-source spin parameter $\hat{a}_c$	0.983 [0.443, 1.000]	0.920 [0.165, 0.999]
GR/CPTG matching radius r_m (km)	76.003 [44.617, 145.205]	34.518 [14.273, 62.165]
Structural Mode N_{BH}	1.5447 [1.4555, 1.8809]	1.6592 [1.4650, 1.9083]
Organization-shell radius r_N (km)	39.960 [23.448, 75.741]	18.166 [7.495, 32.720]
Inner trapping-surface radius r_- (km)	55.113 [32.851, 122.924]	23.795 [9.457, 42.913]
Structural- N lapse $\bar{\alpha}_N$	0.7762 [0.5081, 0.8470]	0.6877 [0.4860, 0.8393]
Structural- N clock $T_{24,N}$ (h)	18.628 [12.195, 20.327]	16.504 [11.665, 20.144]
Structural- N axis ω (rad s $^{-1}$)	1002.00 [301.74, 2288.34]	2621.93 [766.44, 7390.31]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1384.68 [476.50, 2695.71]	3199.54 [889.06, 8536.05]
Structural- N equatorial ω (rad s $^{-1}$)	1806.86 [727.69, 3172.95]	3858.32 [896.75, 10007.19]
Structural- N period P_ω (ms)	4.5376 [2.3308, 13.1860]	1.9638 [0.7361, 7.0673]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2419.44 [986.69, 4279.94]	5005.78 [978.36, 12983.44]
Trapping-surface period P_- (ms)	2.5970 [1.4681, 6.3680]	1.2552 [0.4839, 6.4223]

Table 579: CPTG compact-state, structural, clock, and geometric-rotation results for GW191127_050227. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.3169	7.3519	2393.88	4841.66
500 m inward	3.0341	5.1803	2407.81	4927.26
100 m inward	1.3502	2.3070	2417.92	4991.30
10 m inward	0.4262	0.7286	2419.14	5004.14
1 m inward	0.1348	0.2304	2419.41	5005.71
10 cm inward	0.0426	0.0728	2419.44	5005.78
1 cm inward	0.0135	0.0230	2419.44	5005.78
surface limit	0.0000	0.0000	2419.44	5005.78

Table 580: Near-inner-boundary profile for GW191127_050227. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $39.770 M_\odot$ and $16.510 M_\odot$, with completed geometric mass-equivalent shares of 15.48% and 28.16%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $39.960 < 55.113 < 76.003$ km, and component 2 gives $18.166 < 23.795 < 34.518$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5447$ and 1.6592 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.628 and 16.504 h. The determinant-averaged frame-dragging rates at those shells are 1384.68 and 3199.54 rad s $^{-1}$, approaching surface-limit values of 2419.44 and 5005.78 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.317/7.352 h at 1 km inward to 0.0135/0.0230 h at 1 cm inward for components 1/2. Its strong observational mass asymmetry makes the contrast between the two solved compact interiors particularly useful for component-by-component comparison. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.18 GW230608_205047

GW230608_205047 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 76.160 M_\odot$ and $m_2 = 48.861 M_\odot$, while the luminosity-distance median is 3478.4 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	43,614
Observed luminosity distance	3478.4 Mpc (median)
Detector-frame masses	$m_1 = 76.160 M_\odot$, $m_2 = 48.861 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.642$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5786$; $z_\pi = 0.6236$ (medians)
Post-handoff reorder fraction	13.69%

Table 581: Observational and coordinate summary for GW230608_205047. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	46.493 [32.657, 60.426]	29.956 [19.561, 44.520]
Physical compact-source mass M_c ($M_\odot$)	32.468 [22.509, 49.700]	20.158 [12.078, 30.957]
Geometric mass-equivalent M_{geom} ($M_\odot$)	12.036 [1.317, 25.703]	9.903 [1.346, 20.087]
Geometric share M_{geom}/M_∞ (%)	26.35 [3.18, 49.36]	34.78 [5.14, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.507 [0.057, 0.937]	0.360 [0.032, 0.899]
Polarization factor A_{pol}	1.358 [1.033, 1.975]	1.533 [1.054, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.934 [0.223, 0.999]	0.846 [0.127, 0.999]
GR/CPTG matching radius r_m (km)	68.652 [48.223, 89.226]	44.234 [28.884, 65.739]
Structural Mode N_{BH}	1.6408 [1.4649, 1.9050]	1.7320 [1.4758, 1.9099]
Organization-shell radius r_N (km)	36.141 [25.319, 46.978]	23.294 [15.192, 34.589]
Inner trapping-surface radius r_- (km)	46.212 [33.846, 66.125]	29.398 [18.703, 42.811]
Structural- N lapse $\bar{\alpha}_N$	0.7019 [0.4886, 0.8394]	0.6307 [0.4848, 0.8306]
Structural- N clock $T_{24,N}$ (h)	16.846 [11.727, 20.145]	15.136 [11.635, 19.935]
Structural- N axis ω (rad s $^{-1}$)	1402.92 [511.17, 2378.42]	2222.49 [540.04, 4156.11]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1735.96 [634.35, 2677.16]	2636.30 [541.55, 4617.46]
Structural- N equatorial ω (rad s $^{-1}$)	2085.35 [639.84, 3045.77]	3090.42 [541.70, 5116.53]
Structural- N period P_ω (ms)	3.6194 [2.3470, 9.9048]	2.3833 [1.3607, 11.6022]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2717.57 [698.99, 4030.26]	3947.57 [590.53, 6572.60]
Trapping-surface period P_- (ms)	2.3121 [1.5590, 8.9890]	1.5917 [0.9560, 10.6398]

Table 582: CPTG compact-state, structural, clock, and geometric-rotation results for GW230608_205047. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.2654	6.8158	2675.59	3839.25
500 m inward	3.7165	4.8191	2696.87	3895.01
100 m inward	1.6594	2.1544	2713.51	3937.02
10 m inward	0.5245	0.6812	2717.10	3946.54
1 m inward	0.1659	0.2154	2717.51	3947.45
10 cm inward	0.0524	0.0681	2717.56	3947.55
1 cm inward	0.0166	0.0215	2717.57	3947.56
surface limit	0.0000	0.0000	2717.57	3947.57

Table 583: Near-inner-boundary profile for GW230608_205047. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $32.468 M_\odot$ and $20.158 M_\odot$, with completed geometric mass-equivalent shares of 26.35% and 34.78%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $36.141 < 46.212 < 68.652$ km, and component 2 gives $23.294 < 29.398 < 44.234$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6408$ and 1.7320 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.846 and 15.136 h. The determinant-averaged frame-dragging rates at those shells are 1735.96 and 2636.30 rad s $^{-1}$, approaching surface-limit values of 2717.57 and 3947.57 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.265/6.816 h at 1 km inward to 0.0166/0.0215 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M3 descriptive distribution, making it a useful representative upper-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.19 GW230825_041334

GW230825_041334 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 76.489 M_\odot$ and $m_2 = 49.177 M_\odot$, while the luminosity-distance median is 4875.1 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	47,475
Observed luminosity distance	4875.1 Mpc (median)
Detector-frame masses	$m_1 = 76.489 M_\odot$, $m_2 = 49.177 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.643$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7636$; $z_\pi = 0.8252$ (medians)
Post-handoff reorder fraction	9.93%

Table 584: Observational and coordinate summary for GW230825_041334. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	41.713 [28.112, 59.282]	26.635 [17.779, 40.087]
Physical compact-source mass M_c ($M_\odot$)	34.615 [21.206, 53.405]	18.754 [10.958, 30.893]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.249 [0.615, 18.507]	7.558 [0.933, 16.473]
Geometric share M_{geom}/M_∞ (%)	12.74 [1.52, 43.17]	29.39 [3.79, 49.59]
Exterior spin magnitude $ \hat{a}_\infty $	0.753 [0.210, 0.970]	0.453 [0.046, 0.925]
Polarization factor A_{pol}	1.146 [1.015, 1.760]	1.416 [1.039, 1.984]
Compact-source spin parameter $\hat{a}_c$	0.989 [0.650, 1.000]	0.909 [0.179, 0.999]
GR/CPTG matching radius r_m (km)	61.594 [41.511, 87.537]	39.330 [26.253, 59.194]
Structural Mode N_{BH}	1.5244 [1.4562, 1.8320]	1.6721 [1.4682, 1.9076]
Organization-shell radius r_N (km)	32.333 [21.749, 45.932]	20.698 [13.823, 31.163]
Inner trapping-surface radius r_- (km)	46.498 [30.389, 71.122]	26.770 [16.992, 41.642]
Structural- N lapse $\bar{\alpha}_N$	0.7920 [0.5489, 0.8464]	0.6777 [0.4866, 0.8367]
Structural- N clock $T_{24,N}$ (h)	19.009 [13.174, 20.313]	16.264 [11.679, 20.080]
Structural- N axis ω (rad s $^{-1}$)	1116.19 [541.86, 2522.53]	2397.90 [791.48, 4603.55]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1618.09 [870.49, 2921.09]	2907.02 [899.93, 5119.77]
Structural- N equatorial ω (rad s $^{-1}$)	2206.93 [1349.00, 3427.75]	3482.39 [900.69, 5661.42]
Structural- N period P_ω (ms)	3.8831 [2.1510, 7.2180]	2.1614 [1.2272, 6.9818]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3034.61 [1869.61, 4616.46]	4527.19 [982.78, 7287.48]
Trapping-surface period P_- (ms)	2.0705 [1.3610, 3.3607]	1.3879 [0.8622, 6.3933]

Table 585: CPTG compact-state, structural, clock, and geometric-rotation results for GW230825_041334. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.5006	6.9696	2995.72	4397.54
500 m inward	3.1567	4.9198	3016.99	4466.55
100 m inward	1.4020	2.1959	3031.09	4515.67
10 m inward	0.4428	0.6941	3034.17	4525.81
1 m inward	0.1400	0.2195	3034.57	4527.08
10 cm inward	0.0443	0.0694	3034.61	4527.18
1 cm inward	0.0140	0.0219	3034.61	4527.18
surface limit	0.0000	0.0000	3034.61	4527.19

Table 586: Near-inner-boundary profile for GW230825_041334. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $34.615 M_\odot$ and $18.754 M_\odot$, with completed geometric mass-equivalent shares of 12.74% and 29.39%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.333 < 46.498 < 61.594$ km, and component 2 gives $20.698 < 26.770 < 39.330$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5244$ and 1.6721 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.009 and 16.264 h. The determinant-averaged frame-dragging rates at those shells are 1618.09 and 2907.02 rad s $^{-1}$, approaching surface-limit values of 3034.61 and 4527.19 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.501/6.970 h at 1 km inward to 0.0140/0.0219 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.20 GW231127_165300

GW231127_165300 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 77.938 M_\odot$ and $m_2 = 49.866 M_\odot$, while the luminosity-distance median is 4461.2 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	38,538
Observed luminosity distance	4461.2 Mpc (median)
Detector-frame masses	$m_1 = 77.938 M_\odot$, $m_2 = 49.866 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.640$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7101$; $z_\pi = 0.7668$ (medians)
Post-handoff reorder fraction	14.60%

Table 587: Observational and coordinate summary for GW231127_165300. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.205 [29.100, 62.068]	27.927 [16.190, 43.209]
Physical compact-source mass M_c ($M_\odot$)	31.634 [20.568, 51.226]	18.859 [10.385, 30.513]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.719 [1.005, 24.650]	8.753 [1.084, 18.963]
Geometric share M_{geom}/M_∞ (%)	22.88 [2.54, 48.91]	33.41 [4.62, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.568 [0.075, 0.950]	0.383 [0.034, 0.909]
Polarization factor A_{pol}	1.297 [1.026, 1.957]	1.502 [1.048, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.955 [0.289, 1.000]	0.865 [0.134, 0.999]
GR/CPTG matching radius r_m (km)	63.797 [42.969, 91.651]	41.238 [23.906, 63.803]
Structural Mode N_{BH}	1.6073 [1.4615, 1.8998]	1.7164 [1.4729, 1.9096]
Organization-shell radius r_N (km)	33.572 [22.560, 48.291]	21.713 [12.574, 33.561]
Inner trapping-surface radius r_- (km)	44.320 [30.435, 68.350]	27.545 [15.761, 42.170]
Structural- N lapse $\bar{\alpha}_N$	0.7277 [0.4927, 0.8421]	0.6430 [0.4850, 0.8330]
Structural- N clock $T_{24,N}$ (h)	17.465 [11.825, 20.211]	15.433 [11.640, 19.991]
Structural- N axis ω (rad s $^{-1}$)	1400.27 [536.07, 2604.98]	2363.23 [623.47, 4776.81]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1786.38 [751.96, 2955.92]	2813.65 [627.29, 5395.19]
Structural- N equatorial ω (rad s $^{-1}$)	2208.40 [862.85, 3384.62]	3312.84 [627.45, 6084.96]
Structural- N period P_ω (ms)	3.5173 [2.1256, 8.3558]	2.2331 [1.1646, 10.0164]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2921.23 [947.39, 4496.61]	4243.07 [684.00, 7853.47]
Trapping-surface period P_- (ms)	2.1509 [1.3973, 6.6321]	1.4808 [0.8001, 9.1859]

Table 588: CPTG compact-state, structural, clock, and geometric-rotation results for GW231127_165300. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.1991	6.9979	2875.36	4120.18
500 m inward	3.6653	4.9435	2899.01	4186.08
100 m inward	1.6346	2.2085	2917.24	4232.93
10 m inward	0.5166	0.6982	2920.79	4242.08
1 m inward	0.1633	0.2208	2921.20	4242.98
10 cm inward	0.0517	0.0698	2921.23	4243.06
1 cm inward	0.0163	0.0221	2921.23	4243.07
surface limit	0.0000	0.0000	2921.23	4243.07

Table 589: Near-inner-boundary profile for GW231127_165300. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.634 M_\odot$ and $18.859 M_\odot$, with completed geometric mass-equivalent shares of 22.88% and 33.41%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.572 < 44.320 < 63.797$ km, and component 2 gives $21.713 < 27.545 < 41.238$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6073$ and 1.7164 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.465 and 15.433 h. The determinant-averaged frame-dragging rates at those shells are 1786.38 and 2813.65 rad s $^{-1}$, approaching surface-limit values of 2921.23 and 4243.07 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.199/6.998 h at 1 km inward to 0.0163/0.0221 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M3 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.21 GW240621_200935

GW240621_200935 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 72.381 M_\odot$ and $m_2 = 56.045 M_\odot$, while the luminosity-distance median is 4788.4 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,047
Observed luminosity distance	4788.4 Mpc (median)
Detector-frame masses	$m_1 = 72.381 M_\odot$, $m_2 = 56.045 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.774$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7525$; $z_\pi = 0.8130$ (medians)
Post-handoff reorder fraction	23.33%

Table 590: Observational and coordinate summary for GW240621_200935. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	39.122 [28.888, 51.558]	31.632 [21.310, 43.495]
Physical compact-source mass M_c ($M_\odot$)	30.060 [20.116, 44.480]	20.455 [13.015, 31.167]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.471 [0.840, 20.126]	11.121 [1.712, 19.653]
Geometric share M_{geom}/M_∞ (%)	19.57 [2.24, 47.52]	36.86 [6.16, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.627 [0.117, 0.955]	0.324 [0.027, 0.879]
Polarization factor A_{pol}	1.243 [1.023, 1.906]	1.584 [1.066, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.424, 1.000]	0.812 [0.108, 0.998]
GR/CPTG matching radius r_m (km)	57.768 [42.656, 76.132]	46.709 [31.467, 64.226]
Structural Mode N_{BH}	1.5779 [1.4600, 1.8837]	1.7562 [1.4817, 1.9106]
Organization-shell radius r_N (km)	30.376 [22.413, 40.033]	24.613 [16.564, 33.842]
Inner trapping-surface radius r_- (km)	41.133 [30.151, 59.170]	30.329 [20.310, 42.603]
Structural- N lapse $\bar{\alpha}_N$	0.7504 [0.5057, 0.8434]	0.6112 [0.4843, 0.8259]
Structural- N clock $T_{24,N}$ (h)	18.010 [12.138, 20.241]	14.669 [11.623, 19.821]
Structural- N axis ω (rad s $^{-1}$)	1467.81 [639.95, 2709.96]	2184.45 [470.26, 3837.15]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1939.95 [984.16, 3062.83]	2542.49 [470.50, 4263.98]
Structural- N equatorial ω (rad s $^{-1}$)	2452.46 [1360.53, 3447.34]	2936.16 [470.63, 4699.51]
Structural- N period P_ω (ms)	3.2388 [2.0514, 6.3843]	2.4713 [1.4735, 13.3541]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3275.98 [1577.66, 4557.07]	3706.19 [512.88, 6025.66]
Trapping-surface period P_- (ms)	1.9180 [1.3788, 3.9826]	1.6953 [1.0427, 12.2508]

Table 591: CPTG compact-state, structural, clock, and geometric-rotation results for GW240621_200935. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.2425	6.8235	3220.09	3605.93
500 m inward	3.6886	4.8266	3248.30	3656.92
100 m inward	1.6429	2.1592	3271.68	3698.30
10 m inward	0.5192	0.6827	3275.47	3705.58
1 m inward	0.1642	0.2159	3275.94	3706.13
10 cm inward	0.0519	0.0683	3275.97	3706.19
1 cm inward	0.0164	0.0216	3275.98	3706.19
surface limit	0.0000	0.0000	3275.98	3706.19

Table 592: Near-inner-boundary profile for GW240621_200935. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.060 M_\odot$ and $20.455 M_\odot$, with completed geometric mass-equivalent shares of 19.57% and 36.86%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $30.376 < 41.133 < 57.768$ km, and component 2 gives $24.613 < 30.329 < 46.709$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5779$ and 1.7562 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.010 and 14.669 h. The determinant-averaged frame-dragging rates at those shells are 1939.95 and 2542.49 rad s $^{-1}$, approaching surface-limit values of 3275.98 and 3706.19 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.243/6.824 h at 1 km inward to 0.0164/0.0216 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.22 GW230831_015414

GW230831_015414 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 74.167 M_\odot$ and $m_2 = 55.071 M_\odot$, while the luminosity-distance median is 4948.8 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	30,332
Observed luminosity distance	4948.8 Mpc (median)
Detector-frame masses	$m_1 = 74.167 M_\odot$, $m_2 = 55.071 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.743$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7730$; $z_\pi = 0.8355$ (medians)
Post-handoff reorder fraction	19.35%

Table 593: Observational and coordinate summary for GW230831_015414. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	39.897 [28.074, 58.858]	30.166 [19.193, 44.423]
Physical compact-source mass M_c ($M_\odot$)	30.821 [19.564, 50.469]	19.994 [12.054, 32.081]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.400 [0.770, 21.434]	9.994 [1.299, 19.583]
Geometric share M_{geom}/M_∞ (%)	18.89 [1.94, 47.97]	34.79 [4.95, 49.80]
Exterior spin magnitude $ \hat{a}_\infty $	0.640 [0.105, 0.961]	0.360 [0.032, 0.902]
Polarization factor A_{pol}	1.233 [1.020, 1.922]	1.534 [1.052, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.973 [0.387, 1.000]	0.846 [0.125, 0.999]
GR/CPTG matching radius r_m (km)	58.913 [41.455, 86.912]	44.543 [28.341, 65.597]
Structural Mode N_{BH}	1.5720 [1.4584, 1.8889]	1.7321 [1.4747, 1.9100]
Organization-shell radius r_N (km)	30.968 [21.770, 45.678]	23.449 [14.898, 34.534]
Inner trapping-surface radius r_- (km)	42.290 [28.984, 67.603]	29.292 [18.574, 43.967]
Structural- N lapse $\bar{\alpha}_N$	0.7549 [0.5015, 0.8446]	0.6305 [0.4847, 0.8315]
Structural- N clock $T_{24,N}$ (h)	18.118 [12.036, 20.271]	15.133 [11.634, 19.955]
Structural- N axis ω (rad s $^{-1}$)	1379.98 [562.17, 2781.77]	2221.18 [547.17, 4143.89]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1840.56 [843.50, 3146.24]	2616.76 [548.83, 4646.49]
Structural- N equatorial ω (rad s $^{-1}$)	2361.42 [1119.21, 3567.68]	3057.91 [548.96, 5162.66]
Structural- N period P_ω (ms)	3.4137 [1.9970, 7.4489]	2.4011 [1.3522, 11.4484]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3168.69 [1319.77, 4705.24]	3897.50 [598.25, 6659.93]
Trapping-surface period P_- (ms)	1.9829 [1.3354, 4.7608]	1.6121 [0.9434, 10.5026]

Table 594: CPTG compact-state, structural, clock, and geometric-rotation results for GW230831_015414. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.1088	6.8512	3118.99	3791.43
500 m inward	3.5942	4.8445	3144.32	3846.67
100 m inward	1.5996	2.1658	3164.29	3887.99
10 m inward	0.5053	0.6847	3168.43	3896.90
1 m inward	0.1598	0.2165	3168.66	3897.45
10 cm inward	0.0505	0.0685	3168.69	3897.50
1 cm inward	0.0160	0.0217	3168.69	3897.50
surface limit	0.0000	0.0000	3168.69	3897.50

Table 595: Near-inner-boundary profile for GW230831_015414. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.821 M_\odot$ and $19.994 M_\odot$, with completed geometric mass-equivalent shares of 18.89% and 34.79%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $30.968 < 42.290 < 58.913$ km, and component 2 gives $23.449 < 29.292 < 44.543$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5720$ and 1.7321 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.118 and 15.133 h. The determinant-averaged frame-dragging rates at those shells are 1840.56 and 2616.76 rad s $^{-1}$, approaching surface-limit values of 3168.69 and 3897.50 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.109/6.851 h at 1 km inward to 0.0160/0.0217 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.23 GW240924_000316

GW240924_000316 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 73.928 M_\odot$ and $m_2 = 56.547 M_\odot$, while the luminosity-distance median is 3734.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,026
Observed luminosity distance	3734.3 Mpc (median)
Detector-frame masses	$m_1 = 73.928 M_\odot$, $m_2 = 56.547 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.765$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6135$; $z_\pi = 0.6616$ (medians)
Post-handoff reorder fraction	20.54%

Table 596: Observational and coordinate summary for GW240924_000316. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.670 [31.815, 58.882]	34.556 [23.430, 47.101]
Physical compact-source mass M_c ($M_\odot$)	33.197 [21.549, 52.858]	22.152 [14.120, 34.663]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.560 [0.948, 21.875]	12.010 [1.953, 20.832]
Geometric share M_{geom}/M_∞ (%)	19.91 [2.21, 48.03]	36.58 [6.53, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.621 [0.103, 0.956]	0.329 [0.030, 0.872]
Polarization factor A_{pol}	1.249 [1.023, 1.924]	1.577 [1.070, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.969 [0.382, 1.000]	0.817 [0.119, 0.998]
GR/CPTG matching radius r_m (km)	64.484 [46.979, 86.947]	51.026 [34.598, 69.551]
Structural Mode N_{BH}	1.5807 [1.4598, 1.8897]	1.7530 [1.4839, 1.9102]
Organization-shell radius r_N (km)	33.924 [24.702, 45.592]	26.877 [18.198, 36.617]
Inner trapping-surface radius r_- (km)	45.512 [32.484, 70.304]	32.919 [22.072, 46.868]
Structural- N lapse $\bar{\alpha}_N$	0.7482 [0.5009, 0.8435]	0.6139 [0.4846, 0.8241]
Structural- N clock $T_{24,N}$ (h)	17.957 [12.021, 20.244]	14.733 [11.630, 19.779]
Structural- N axis ω (rad s^{-1})	1319.23 [535.25, 2509.97]	2033.21 [474.92, 3517.64]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1741.63 [828.44, 2828.21]	2350.76 [475.09, 3902.03]
Structural- N equatorial ω (rad s^{-1})	2183.20 [1176.01, 3170.97]	2695.22 [475.17, 4296.99]
Structural- N period P_ω (ms)	3.6076 [2.2216, 7.5844]	2.6728 [1.6102, 13.2253]
Trapping-surface limit Ω_- (rad s^{-1})	2901.96 [1352.03, 4168.74]	3390.68 [517.70, 5506.27]
Trapping-surface period P_- (ms)	2.1652 [1.5072, 4.6472]	1.8531 [1.1411, 12.1368]

Table 597: CPTG compact-state, structural, clock, and geometric-rotation results for GW240924_000316. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0002	6.5685	2856.94	3308.48
500 m inward	3.5196	4.6482	2880.01	3352.66
100 m inward	1.5687	2.0788	2897.40	3382.18
10 m inward	0.4957	0.6574	2901.54	3389.70
1 m inward	0.1567	0.2079	2901.92	3390.58
10 cm inward	0.0496	0.0657	2901.95	3390.67
1 cm inward	0.0157	0.0208	2901.96	3390.68
surface limit	0.0000	0.0000	2901.96	3390.68

Table 598: Near-inner-boundary profile for GW240924_000316. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $33.197 M_\odot$ and $22.152 M_\odot$, with completed geometric mass-equivalent shares of 19.91% and 36.58%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.924 < 45.512 < 64.484$ km, and component 2 gives $26.877 < 32.919 < 51.026$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5807$ and 1.7530 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.957 and 14.733 h. The determinant-averaged frame-dragging rates at those shells are 1741.63 and $2350.76 \text{ rad s}^{-1}$, approaching surface-limit values of 2901.96 and $3390.68 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 5.000/6.568 h at 1 km inward to 0.0157/0.0208 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M3 descriptive distribution, making it a useful representative upper-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.24 GW190701_203306

GW190701_203306 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 74.559 M_\odot$ and $m_2 = 56.078 M_\odot$, while the luminosity-distance median is 2090.8 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	22,577
Observed luminosity distance	2090.8 Mpc (median)
Detector-frame masses	$m_1 = 74.559 M_\odot$, $m_2 = 56.078 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.752$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3779$; $z_\pi = 0.4059$ (medians)
Post-handoff reorder fraction	21.46%

Table 599: Observational and coordinate summary for GW190701_203306. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	51.973 [39.608, 65.415]	40.732 [27.921, 53.483]
Physical compact-source mass M_c ($M_\odot$)	38.507 [27.248, 54.959]	26.065 [17.137, 37.543]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.567 [1.185, 27.630]	14.605 [2.463, 24.819]
Geometric share M_{geom}/M_∞ (%)	22.56 [2.52, 48.57]	37.39 [7.16, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.574 [0.087, 0.950]	0.314 [0.029, 0.859]
Polarization factor A_{pol}	1.291 [1.026, 1.944]	1.597 [1.077, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.957 [0.329, 1.000]	0.802 [0.115, 0.997]
GR/CPTG matching radius r_m (km)	76.745 [58.486, 96.593]	60.145 [41.228, 78.974]
Structural Mode N_{BH}	1.6043 [1.4614, 1.8959]	1.7625 [1.4877, 1.9103]
Organization-shell radius r_N (km)	40.372 [30.661, 50.897]	31.682 [21.683, 41.568]
Inner trapping-surface radius r_- (km)	53.255 [41.424, 73.167]	38.900 [26.679, 51.055]
Structural- N lapse $\bar{\alpha}_N$	0.7300 [0.4959, 0.8422]	0.6061 [0.4845, 0.8211]
Structural- N clock $T_{24,N}$ (h)	17.520 [11.901, 20.212]	14.547 [11.627, 19.706]
Structural- N axis ω (rad s $^{-1}$)	1182.90 [503.47, 1986.37]	1760.99 [367.71, 2933.80]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1517.49 [738.44, 2242.75]	2030.85 [367.93, 3253.94]
Structural- N equatorial ω (rad s $^{-1}$)	1873.73 [834.58, 2514.52]	2323.11 [368.01, 3597.38]
Structural- N period P_ω (ms)	4.1405 [2.8016, 8.5087]	3.0939 [1.9309, 17.0773]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2483.18 [916.45, 3323.38]	2936.72 [400.99, 4615.75]
Trapping-surface period P_- (ms)	2.5303 [1.8906, 6.8560]	2.1395 [1.3612, 15.6690]

Table 600: CPTG compact-state, structural, clock, and geometric-rotation results for GW190701_203306. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.7410	6.0674	2451.66	2876.97
500 m inward	3.3427	4.2928	2468.12	2908.04
100 m inward	1.4921	1.9206	2480.09	2931.58
10 m inward	0.4715	0.6074	2482.98	2936.25
1 m inward	0.1491	0.1921	2483.16	2936.67
10 cm inward	0.0472	0.0607	2483.18	2936.72
1 cm inward	0.0149	0.0192	2483.18	2936.72
surface limit	0.0000	0.0000	2483.18	2936.72

Table 601: Near-inner-boundary profile for GW190701_203306. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $38.507 M_\odot$ and $26.065 M_\odot$, with completed geometric mass-equivalent shares of 22.56% and 37.39%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $40.372 < 53.255 < 76.745$ km, and component 2 gives $31.682 < 38.900 < 60.145$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6043$ and 1.7625 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.520 and 14.547 h. The determinant-averaged frame-dragging rates at those shells are 1517.49 and 2030.85 rad s $^{-1}$, approaching surface-limit values of 2483.18 and 2936.72 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.741/6.067 h at 1 km inward to 0.0149/0.0192 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.25 GW231223_032836

GW231223_032836 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 76.699 M_\odot$ and $m_2 = 53.975 M_\odot$, while the luminosity-distance median is 4183.6 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	42,558
Observed luminosity distance	4183.6 Mpc (median)
Detector-frame masses	$m_1 = 76.699 M_\odot$, $m_2 = 53.975 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.704$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6737$; $z_\pi = 0.7270$ (medians)
Post-handoff reorder fraction	16.91%

Table 602: Observational and coordinate summary for GW231223_032836. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.577 [31.184, 60.007]	30.759 [16.511, 45.230]
Physical compact-source mass M_c ($M_\odot$)	33.995 [22.623, 50.178]	20.850 [10.882, 32.741]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.622 [0.778, 23.633]	8.964 [1.096, 19.863]
Geometric share M_{geom}/M_∞ (%)	17.90 [1.93, 47.80]	31.86 [4.34, 49.71]
Exterior spin magnitude $ \hat{a}_\infty $	0.658 [0.109, 0.962]	0.410 [0.038, 0.914]
Polarization factor A_{pol}	1.218 [1.020, 1.916]	1.468 [1.045, 1.988]
Compact-source spin parameter $\hat{a}_c$	0.976 [0.401, 1.000]	0.884 [0.152, 0.999]
GR/CPTG matching radius r_m (km)	64.347 [46.048, 88.608]	45.419 [24.381, 66.788]
Structural Mode N_{BH}	1.5639 [1.4583, 1.8870]	1.6990 [1.4713, 1.9089]
Organization-shell radius r_N (km)	33.827 [24.141, 46.601]	23.903 [12.829, 35.188]
Inner trapping-surface radius r_- (km)	46.602 [33.163, 66.852]	30.451 [16.178, 44.631]
Structural- N lapse $\bar{\alpha}_N$	0.7613 [0.5031, 0.8447]	0.6567 [0.4856, 0.8342]
Structural- N clock $T_{24,N}$ (h)	18.270 [12.073, 20.273]	15.761 [11.654, 20.022]
Structural- N axis ω (rad s $^{-1}$)	1237.19 [564.97, 2404.26]	2173.14 [646.00, 4578.69]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1674.92 [856.68, 2741.39]	2592.35 [655.73, 5192.76]
Structural- N equatorial ω (rad s $^{-1}$)	2167.47 [1127.21, 3143.64]	3049.41 [656.07, 5909.77]
Structural- N period P_ω (ms)	3.7513 [2.2920, 7.3343]	2.4237 [1.2100, 9.5819]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2926.19 [1288.51, 4188.71]	3939.53 [715.54, 7633.88]
Trapping-surface period P_- (ms)	2.1472 [1.5000, 4.8763]	1.5949 [0.8231, 8.7810]

Table 603: CPTG compact-state, structural, clock, and geometric-rotation results for GW231223_032836. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.8360	6.6596	2886.05	3838.73
500 m inward	3.4022	4.7038	2908.13	3891.57
100 m inward	1.5148	2.1016	2922.66	3930.26
10 m inward	0.4786	0.6644	2925.82	3939.21
1 m inward	0.1513	0.2101	2926.15	3939.42
10 cm inward	0.0479	0.0664	2926.19	3939.52
1 cm inward	0.0151	0.0210	2926.19	3939.52
surface limit	0.0000	0.0000	2926.19	3939.53

Table 604: Near-inner-boundary profile for GW231223_032836. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $33.995 M_\odot$ and $20.850 M_\odot$, with completed geometric mass-equivalent shares of 17.90% and 31.86%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.827 < 46.602 < 64.347$ km, and component 2 gives $23.903 < 30.451 < 45.419$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5639$ and 1.6990 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.270 and 15.761 h. The determinant-averaged frame-dragging rates at those shells are 1674.92 and 2592.35 rad s $^{-1}$, approaching surface-limit values of 2926.19 and 3939.53 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.836/6.660 h at 1 km inward to 0.0151/0.0210 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.26 GW230709_122727

GW230709_122727 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 77.987 M_\odot$ and $m_2 = 53.370 M_\odot$, while the luminosity-distance median is 4661.1 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	40,194
Observed luminosity distance	4661.1 Mpc (median)
Detector-frame masses	$m_1 = 77.987 M_\odot$, $m_2 = 53.370 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.684$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7361$; $z_\pi = 0.7951$ (medians)
Post-handoff reorder fraction	17.11%

Table 605: Observational and coordinate summary for GW230709_122727. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	42.502 [29.585, 59.034]	29.523 [16.643, 44.471]
Physical compact-source mass M_c ($M_\odot$)	31.555 [20.790, 48.697]	19.612 [10.707, 31.365]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.278 [0.973, 23.319]	9.350 [1.248, 19.679]
Geometric share M_{geom}/M_∞ (%)	22.19 [2.51, 48.57]	34.16 [5.05, 49.78]
Exterior spin magnitude $ \hat{a}_\infty $	0.580 [0.087, 0.950]	0.371 [0.033, 0.900]
Polarization factor A_{pol}	1.285 [1.026, 1.945]	1.519 [1.053, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.958 [0.328, 1.000]	0.855 [0.132, 0.999]
GR/CPTG matching radius r_m (km)	62.760 [43.686, 87.172]	43.595 [24.576, 65.667]
Structural Mode N_{BH}	1.6010 [1.4614, 1.8959]	1.7248 [1.4753, 1.9097]
Organization-shell radius r_N (km)	33.033 [22.925, 45.878]	22.951 [12.922, 34.600]
Inner trapping-surface radius r_- (km)	43.902 [30.892, 64.940]	28.795 [16.257, 43.303]
Structural- N lapse $\bar{\alpha}_N$	0.7326 [0.4958, 0.8422]	0.6363 [0.4849, 0.8310]
Structural- N clock $T_{24,N}$ (h)	17.581 [11.900, 20.214]	15.272 [11.639, 19.944]
Structural- N axis ω (rad s^{-1})	1416.76 [576.02, 2587.16]	2276.23 [584.02, 4614.36]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1815.12 [846.37, 2929.37]	2694.32 [585.05, 5191.19]
Structural- N equatorial ω (rad s^{-1})	2255.13 [1017.81, 3339.57]	3146.17 [585.25, 5852.83]
Structural- N period P_ω (ms)	3.4616 [2.1449, 7.4237]	2.3320 [1.2104, 10.7396]
Trapping-surface limit Ω_- (rad s^{-1})	2985.08 [1118.93, 4432.19]	4015.72 [638.14, 7568.40]
Trapping-surface period P_- (ms)	2.1049 [1.4176, 5.6154]	1.5646 [0.8302, 9.8461]

Table 606: CPTG compact-state, structural, clock, and geometric-rotation results for GW230709_122727. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.1916	6.8773	2937.87	3902.66
500 m inward	3.6579	4.8608	2962.92	3963.05
100 m inward	1.6310	2.1721	2981.02	4005.36
10 m inward	0.5154	0.6867	2984.57	4014.81
1 m inward	0.1630	0.2172	2985.02	4015.60
10 cm inward	0.0515	0.0687	2985.07	4015.71
1 cm inward	0.0163	0.0217	2985.08	4015.72
surface limit	0.0000	0.0000	2985.08	4015.72

Table 607: Near-inner-boundary profile for GW230709_122727. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.555 M_\odot$ and $19.612 M_\odot$, with completed geometric mass-equivalent shares of 22.19% and 34.16%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.033 < 43.902 < 62.760$ km, and component 2 gives $22.951 < 28.795 < 43.595$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6010$ and 1.7248 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.581 and 15.272 h. The determinant-averaged frame-dragging rates at those shells are 1815.12 and $2694.32 \text{ rad s}^{-1}$, approaching surface-limit values of 2985.08 and $4015.72 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from 5.192/6.877 h at 1 km inward to 0.0163/0.0217 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M3 descriptive distribution, making it a useful representative upper-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.27 GW241225_082815

GW241225_082815 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 74.730 M_\odot$ and $m_2 = 56.907 M_\odot$, while the luminosity-distance median is 1875.8 Mpc . One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	20,151
Observed luminosity distance	1875.8 Mpc (median)
Detector-frame masses	$m_1 = 74.730 M_\odot$, $m_2 = 56.907 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.762$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3439$; $z_\pi = 0.3698$ (medians)
Post-handoff reorder fraction	14.03%

Table 608: Observational and coordinate summary for GW241225_082815. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	54.206 [42.043, 65.645]	42.025 [32.241, 53.511]
Physical compact-source mass M_c ($M_\odot$)	47.022 [32.358, 61.512]	31.149 [19.306, 44.364]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.329 [0.653, 19.764]	10.169 [1.136, 22.734]
Geometric share M_{geom}/M_∞ (%)	10.01 [1.23, 36.22]	24.76 [2.72, 49.42]
Exterior spin magnitude $ \hat{a}_\infty $	0.805 [0.335, 0.976]	0.535 [0.054, 0.946]
Polarization factor A_{pol}	1.111 [1.012, 1.568]	1.329 [1.028, 1.977]
Compact-source spin parameter $\hat{a}_c$	0.994 [0.823, 1.000]	0.944 [0.212, 1.000]
GR/CPTG matching radius r_m (km)	80.042 [62.082, 96.933]	62.054 [47.608, 79.016]
Structural Mode N_{BH}	1.5057 [1.4547, 1.7488]	1.6251 [1.4625, 1.9057]
Organization-shell radius r_N (km)	41.947 [32.568, 50.692]	32.609 [25.027, 41.607]
Inner trapping-surface radius r_- (km)	62.347 [45.972, 82.475]	43.408 [30.485, 59.001]
Structural- N lapse $\bar{\alpha}_N$	0.8068 [0.6172, 0.8476]	0.7140 [0.4881, 0.8413]
Structural- N clock $T_{24,N}$ (h)	19.363 [14.813, 20.343]	17.136 [11.714, 20.192]
Structural- N axis ω (rad s $^{-1}$)	764.48 [454.65, 1732.99]	1438.59 [581.37, 2622.31]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1158.40 [752.79, 1983.44]	1812.12 [713.15, 2902.05]
Structural- N equatorial ω (rad s $^{-1}$)	1671.44 [1228.71, 2277.68]	2220.51 [713.73, 3182.16]
Structural- N period P_ω (ms)	5.4240 [3.1678, 8.3466]	3.4673 [2.1651, 8.8105]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2345.71 [1757.00, 3054.05]	2951.27 [779.42, 4081.02]
Trapping-surface period P_- (ms)	2.6786 [2.0573, 3.5761]	2.1290 [1.5396, 8.0613]

Table 609: CPTG compact-state, structural, clock, and geometric-rotation results for GW241225_082815. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.6818	5.3814	2324.00	2906.96
500 m inward	2.5836	3.7965	2334.63	2929.66
100 m inward	1.1486	1.6942	2343.47	2946.74
10 m inward	0.3627	0.5356	2345.59	2950.72
1 m inward	0.1147	0.1694	2345.70	2951.21
10 cm inward	0.0363	0.0536	2345.71	2951.26
1 cm inward	0.0115	0.0169	2345.71	2951.27
surface limit	0.0000	0.0000	2345.71	2951.27

Table 610: Near-inner-boundary profile for GW241225_082815. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $47.022 M_\odot$ and $31.149 M_\odot$, with completed geometric mass-equivalent shares of 10.01% and 24.76%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $41.947 < 62.347 < 80.042$ km, and component 2 gives $32.609 < 43.408 < 62.054$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5057$ and 1.6251 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.363 and 17.136 h. The determinant-averaged frame-dragging rates at those shells are 1158.40 and 1812.12 rad s $^{-1}$, approaching surface-limit values of 2345.71 and 2951.27 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.682/5.381 h at 1 km inward to 0.0115/0.0169 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.28 GW230803_033412

GW230803_033412 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 79.719 M_\odot$ and $m_2 = 52.017 M_\odot$, while the luminosity-distance median is 5023.7 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	28,700
Observed luminosity distance	5023.7 Mpc (median)
Detector-frame masses	$m_1 = 79.719 M_\odot$, $m_2 = 52.017 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.652$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7825$; $z_\pi = 0.8459$ (medians)
Post-handoff reorder fraction	13.81%

Table 611: Observational and coordinate summary for GW230803_033412. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	42.329 [28.264, 61.986]	28.009 [17.683, 41.681]
Physical compact-source mass M_c ($M_\odot$)	31.726 [19.627, 54.041]	18.777 [11.148, 29.829]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.577 [0.998, 21.639]	9.000 [1.162, 18.341]
Geometric share M_{geom}/M_∞ (%)	20.36 [2.50, 48.27]	33.86 [4.79, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.613 [0.096, 0.950]	0.376 [0.034, 0.905]
Polarization factor A_{pol}	1.256 [1.026, 1.933]	1.512 [1.050, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.967 [0.360, 1.000]	0.859 [0.134, 0.999]
GR/CPTG matching radius r_m (km)	62.504 [41.735, 91.530]	41.359 [26.111, 61.548]
Structural Mode N_{BH}	1.5847 [1.4613, 1.8924]	1.7215 [1.4738, 1.9097]
Organization-shell radius r_N (km)	32.888 [21.919, 48.041]	21.772 [13.747, 32.404]
Inner trapping-surface radius r_- (km)	43.732 [29.192, 71.732]	27.412 [17.087, 40.906]
Structural- N lapse $\bar{\alpha}_N$	0.7452 [0.4987, 0.8423]	0.6390 [0.4850, 0.8322]
Structural- N clock $T_{24,N}$ (h)	17.884 [11.969, 20.215]	15.336 [11.640, 19.972]
Structural- N axis ω (rad s $^{-1}$)	1368.76 [535.54, 2717.86]	2374.20 [623.51, 4529.11]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1792.61 [813.25, 3088.34]	2822.53 [625.50, 5058.50]
Structural- N equatorial ω (rad s $^{-1}$)	2239.04 [1117.15, 3516.86]	3315.64 [625.66, 5607.89]
Structural- N period P_ω (ms)	3.5050 [2.0345, 7.7261]	2.2261 [1.2421, 10.0450]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2951.64 [1316.77, 4674.35]	4243.30 [681.83, 7273.58]
Trapping-surface period P_- (ms)	2.1287 [1.3442, 4.7717]	1.4807 [0.8638, 9.2152]

Table 612: CPTG compact-state, structural, clock, and geometric-rotation results for GW230803_033412. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0997	7.0458	2904.54	4117.45
500 m inward	3.5891	4.9804	2929.13	4184.52
100 m inward	1.5990	2.2261	2946.85	4231.68
10 m inward	0.5052	0.7038	2951.19	4242.46
1 m inward	0.1597	0.2225	2951.56	4243.21
10 cm inward	0.0505	0.0704	2951.63	4243.29
1 cm inward	0.0160	0.0223	2951.63	4243.30
surface limit	0.0000	0.0000	2951.64	4243.30

Table 613: Near-inner-boundary profile for GW230803_033412. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.726 M_\odot$ and $18.777 M_\odot$, with completed geometric mass-equivalent shares of 20.36% and 33.86%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.888 < 43.732 < 62.504$ km, and component 2 gives $21.772 < 27.412 < 41.359$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5847$ and 1.7215 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.884 and 15.336 h. The determinant-averaged frame-dragging rates at those shells are 1792.61 and 2822.53 rad s $^{-1}$, approaching surface-limit values of 2951.64 and 4243.30 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.100/7.046 h at 1 km inward to 0.0160/0.0223 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M3 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.29 GW240705_053215

GW240705_053215 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 75.068 M_\odot$ and $m_2 = 56.982 M_\odot$, while the luminosity-distance median is 3733.5 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,248
Observed luminosity distance	3733.5 Mpc (median)
Detector-frame masses	$m_1 = 75.068 M_\odot$, $m_2 = 56.982 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.759$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6134$; $z_\pi = 0.6614$ (medians)
Post-handoff reorder fraction	21.16%

Table 614: Observational and coordinate summary for GW240705_053215. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	44.708 [33.777, 55.325]	35.212 [26.330, 47.397]
Physical compact-source mass M_c ($M_\odot$)	33.254 [23.118, 47.538]	23.146 [15.871, 33.360]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.525 [0.943, 23.420]	12.518 [1.778, 21.841]
Geometric share M_{geom}/M_∞ (%)	21.98 [2.23, 48.57]	36.26 [5.66, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.584 [0.087, 0.956]	0.334 [0.030, 0.888]
Polarization factor A_{pol}	1.282 [1.023, 1.944]	1.569 [1.060, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.959 [0.329, 1.000]	0.822 [0.121, 0.998]
GR/CPTG matching radius r_m (km)	66.017 [49.876, 81.695]	51.994 [38.880, 69.988]
Structural Mode N_{BH}	1.5991 [1.4599, 1.8959]	1.7492 [1.4788, 1.9101]
Organization-shell radius r_N (km)	34.731 [26.166, 43.002]	27.376 [20.426, 36.816]
Inner trapping-surface radius r_- (km)	46.030 [35.109, 63.191]	34.075 [25.141, 45.383]
Structural- N lapse $\bar{\alpha}_N$	0.7341 [0.4959, 0.8434]	0.6169 [0.4846, 0.8282]
Structural- N clock $T_{24,N}$ (h)	17.618 [11.901, 20.242]	14.806 [11.631, 19.877]
Structural- N axis ω (rad s $^{-1}$)	1359.41 [569.03, 2336.88]	1957.11 [440.34, 3187.32]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1755.84 [841.28, 2637.47]	2287.00 [440.65, 3530.50]
Structural- N equatorial ω (rad s $^{-1}$)	2164.09 [996.59, 2964.65]	2669.63 [440.81, 3869.72]
Structural- N period P_ω (ms)	3.5785 [2.3823, 7.4686]	2.7473 [1.7797, 14.2588]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2870.20 [1094.60, 3913.88]	3414.07 [480.54, 4980.99]
Trapping-surface period P_- (ms)	2.1891 [1.6054, 5.7402]	1.8404 [1.2614, 13.0754]

Table 615: CPTG compact-state, structural, clock, and geometric-rotation results for GW240705_053215. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0922	6.4374	2827.61	3337.40
500 m inward	3.5884	4.5546	2849.41	3377.74
100 m inward	1.6000	2.0373	2866.32	3407.62
10 m inward	0.5056	0.6443	2869.83	3413.38
1 m inward	0.1599	0.2037	2870.18	3413.99
10 cm inward	0.0506	0.0644	2870.20	3414.07
1 cm inward	0.0160	0.0204	2870.20	3414.07
surface limit	0.0000	0.0000	2870.20	3414.07

Table 616: Near-inner-boundary profile for GW240705_053215. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $33.254 M_\odot$ and $23.146 M_\odot$, with completed geometric mass-equivalent shares of 21.98% and 36.26%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $34.731 < 46.030 < 66.017$ km, and component 2 gives $27.376 < 34.075 < 51.994$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5991$ and 1.7492 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.618 and 14.806 h. The determinant-averaged frame-dragging rates at those shells are 1755.84 and 2287.00 rad s $^{-1}$, approaching surface-limit values of 2870.20 and 3414.07 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.092/6.437 h at 1 km inward to 0.0160/0.0204 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M3. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.30 GW231221_135041

GW231221_135041 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 80.324 M_\odot$ and $m_2 = 52.015 M_\odot$, while the luminosity-distance median is 4569.5 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	30,978
Observed luminosity distance	4569.5 Mpc (median)
Detector-frame masses	$m_1 = 80.324 M_\odot$, $m_2 = 52.015 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.648$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7242$; $z_\pi = 0.7821$ (medians)
Post-handoff reorder fraction	13.55%

Table 617: Observational and coordinate summary for GW231221_135041. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	44.725 [29.695, 67.480]	28.448 [15.111, 43.285]
Physical compact-source mass M_c ($M_\odot$)	35.286 [21.180, 59.706]	19.150 [9.940, 31.441]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.446 [0.817, 21.984]	8.393 [0.940, 18.754]
Geometric share M_{geom}/M_∞ (%)	16.75 [1.87, 46.60]	32.41 [4.03, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.679 [0.139, 0.963]	0.401 [0.037, 0.920]
Polarization factor A_{pol}	1.201 [1.019, 1.873]	1.479 [1.042, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.980 [0.488, 1.000]	0.878 [0.147, 0.999]
GR/CPTG matching radius r_m (km)	66.042 [43.849, 99.643]	42.007 [22.313, 63.915]
Structural Mode N_{BH}	1.5546 [1.4580, 1.8730]	1.7050 [1.4696, 1.9091]
Organization-shell radius r_N (km)	34.729 [23.014, 52.295]	22.120 [11.731, 33.657]
Inner trapping-surface radius r_- (km)	48.065 [31.049, 79.544]	27.946 [14.796, 43.124]
Structural- N lapse $\bar{\alpha}_N$	0.7685 [0.5146, 0.8449]	0.6520 [0.4854, 0.8356]
Structural- N clock $T_{24,N}$ (h)	18.443 [12.352, 20.278]	15.647 [11.650, 20.055]
Structural- N axis ω (rad s $^{-1}$)	1184.37 [488.62, 2555.22]	2324.81 [662.77, 5018.69]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1615.59 [762.02, 2909.84]	2784.49 [673.84, 5691.39]
Structural- N equatorial ω (rad s $^{-1}$)	2092.85 [1132.91, 3323.97]	3279.92 [674.33, 6453.05]
Structural- N period P_ω (ms)	3.8891 [2.1593, 8.2454]	2.2565 [1.1040, 9.3245]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2814.76 [1485.58, 4399.83]	4215.83 [735.31, 8380.90]
Trapping-surface period P_- (ms)	2.2322 [1.4281, 4.2294]	1.4904 [0.7497, 8.5450]

Table 618: CPTG compact-state, structural, clock, and geometric-rotation results for GW231221_135041. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.6637	6.9425	2777.23	4097.15
500 m inward	3.2791	4.9046	2797.81	4159.36
100 m inward	1.4593	2.1902	2811.42	4204.70
10 m inward	0.4609	0.6925	2814.41	4214.67
1 m inward	0.1457	0.2190	2814.75	4215.71
10 cm inward	0.0461	0.0692	2814.76	4215.82
1 cm inward	0.0146	0.0219	2814.76	4215.83
surface limit	0.0000	0.0000	2814.76	4215.83

Table 619: Near-inner-boundary profile for GW231221_135041. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $35.286 M_\odot$ and $19.150 M_\odot$, with completed geometric mass-equivalent shares of 16.75% and 32.41%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $34.729 < 48.065 < 66.042$ km, and component 2 gives $22.120 < 27.946 < 42.007$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5546$ and 1.7050 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.443 and 15.647 h. The determinant-averaged frame-dragging rates at those shells are 1615.59 and 2784.49 rad s $^{-1}$, approaching surface-limit values of 2814.76 and 4215.83 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.664/6.942 h at 1 km inward to 0.0146/0.0219 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.31 GW240923_204006

GW240923_204006 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 75.269 M_\odot$ and $m_2 = 58.141 M_\odot$, while the luminosity-distance median is 4024.7 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	11,152
Observed luminosity distance	4024.7 Mpc (median)
Detector-frame masses	$m_1 = 75.269 M_\odot$, $m_2 = 58.141 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.772$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6526$; $z_\pi = 0.7040$ (medians)
Post-handoff reorder fraction	23.79%

Table 620: Observational and coordinate summary for GW240923_204006. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.540 [32.835, 54.693]	35.062 [23.564, 47.583]
Physical compact-source mass M_c ($M_\odot$)	31.511 [21.851, 45.900]	22.183 [14.469, 33.520]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.711 [1.242, 22.841]	12.860 [1.919, 21.802]
Geometric share M_{geom}/M_∞ (%)	24.81 [3.06, 48.97]	38.36 [6.51, 49.86]
Exterior spin magnitude $ \hat{a}_\infty $	0.534 [0.073, 0.939]	0.297 [0.027, 0.872]
Polarization factor A_{pol}	1.330 [1.032, 1.960]	1.622 [1.070, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.944 [0.281, 1.000]	0.783 [0.105, 0.998]
GR/CPTG matching radius r_m (km)	64.292 [48.485, 80.761]	51.773 [34.795, 70.262]
Structural Mode N_{BH}	1.6256 [1.4643, 1.9005]	1.7741 [1.4838, 1.9106]
Organization-shell radius r_N (km)	33.859 [25.508, 42.543]	27.244 [18.292, 37.063]
Inner trapping-surface radius r_- (km)	43.873 [33.550, 61.050]	33.160 [22.428, 45.847]
Structural- N lapse $\bar{\alpha}_N$	0.7136 [0.4922, 0.8399]	0.5968 [0.4842, 0.8242]
Structural- N clock $T_{24,N}$ (h)	17.127 [11.812, 20.157]	14.322 [11.622, 19.781]
Structural- N axis ω (rad s $^{-1}$)	1493.45 [578.22, 2466.67]	1982.36 [401.44, 3514.88]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1864.29 [822.20, 2755.39]	2292.43 [401.69, 3897.59]
Structural- N equatorial ω (rad s $^{-1}$)	2244.74 [879.97, 3094.35]	2620.91 [401.81, 4284.49]
Structural- N period P_ω (ms)	3.3703 [2.2803, 7.6419]	2.7408 [1.6121, 15.6419]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2943.07 [963.82, 4056.29]	3301.42 [437.95, 5473.57]
Trapping-surface period P_- (ms)	2.1349 [1.5490, 6.5190]	1.9032 [1.1479, 14.3467]

Table 621: CPTG compact-state, structural, clock, and geometric-rotation results for GW240923_204006. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.3237	6.5711	2893.75	3219.97
500 m inward	3.7550	4.6527	2919.41	3261.24
100 m inward	1.6754	2.0809	2938.59	3293.18
10 m inward	0.5296	0.6580	2942.59	3300.64
1 m inward	0.1675	0.2081	2943.05	3301.35
10 cm inward	0.0530	0.0658	2943.07	3301.41
1 cm inward	0.0167	0.0208	2943.07	3301.42
surface limit	0.0000	0.0000	2943.07	3301.42

Table 622: Near-inner-boundary profile for GW240923_204006. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.511 M_\odot$ and $22.183 M_\odot$, with completed geometric mass-equivalent shares of 24.81% and 38.36%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.859 < 43.873 < 64.292$ km, and component 2 gives $27.244 < 33.160 < 51.773$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6256$ and 1.7741 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.127 and 14.322 h. The determinant-averaged frame-dragging rates at those shells are 1864.29 and 2292.43 rad s $^{-1}$, approaching surface-limit values of 2943.07 and 3301.42 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.324/6.571 h at 1 km inward to 0.0167/0.0208 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.32 GW190413_134308

GW190413_134308 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 83.077 M_\odot$ and $m_2 = 50.536 M_\odot$, while the luminosity-distance median is 3830.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	23,732
Observed luminosity distance	3830.3 Mpc (median)
Detector-frame masses	$m_1 = 83.077 M_\odot$, $m_2 = 50.536 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.608$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6272$; $z_\pi = 0.6757$ (medians)
Post-handoff reorder fraction	11.92%

Table 623: Observational and coordinate summary for GW190413_134308. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	49.383 [33.806, 66.564]	29.805 [17.320, 44.227]
Physical compact-source mass M_c ($M_\odot$)	38.273 [23.407, 59.862]	20.325 [11.040, 32.341]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.465 [0.893, 23.622]	8.889 [1.052, 19.591]
Geometric share M_{geom}/M_∞ (%)	17.40 [1.82, 47.79]	32.15 [4.09, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.667 [0.110, 0.964]	0.405 [0.037, 0.919]
Polarization factor A_{pol}	1.211 [1.019, 1.915]	1.474 [1.043, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.978 [0.403, 1.000]	0.881 [0.147, 0.999]
GR/CPTG matching radius r_m (km)	72.921 [49.919, 98.290]	44.011 [25.575, 65.307]
Structural Mode N_{BH}	1.5598 [1.4578, 1.8869]	1.7022 [1.4699, 1.9091]
Organization-shell radius r_N (km)	38.321 [26.208, 51.590]	23.161 [13.443, 34.386]
Inner trapping-surface radius r_- (km)	52.005 [34.981, 79.723]	29.560 [16.803, 44.052]
Structural- N lapse $\bar{\alpha}_N$	0.7644 [0.5032, 0.8451]	0.6542 [0.4854, 0.8353]
Structural- N clock $T_{24,N}$ (h)	18.345 [12.076, 20.283]	15.701 [11.650, 20.048]
Structural- N axis ω (rad s $^{-1}$)	1075.53 [465.50, 2290.47]	2205.19 [629.29, 4478.66]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1471.33 [732.31, 2598.83]	2633.51 [636.42, 5023.97]
Structural- N equatorial ω (rad s $^{-1}$)	1894.25 [1044.85, 2943.37]	3122.53 [636.63, 5656.57]
Structural- N period P_ω (ms)	4.2704 [2.4177, 8.5800]	2.3859 [1.2506, 9.8727]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2532.34 [1281.19, 3880.81]	4029.43 [694.23, 7274.81]
Trapping-surface period P_- (ms)	2.4812 [1.6190, 4.9042]	1.5593 [0.8637, 9.0505]

Table 624: CPTG compact-state, structural, clock, and geometric-rotation results for GW190413_134308. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.5101	6.7435	2501.80	3920.68
500 m inward	3.1750	4.7654	2517.07	3979.78
100 m inward	1.4145	2.1286	2529.32	4019.68
10 m inward	0.4468	0.6729	2531.98	4028.33
1 m inward	0.1413	0.02128	2532.33	4029.33
10 cm inward	0.0447	0.0673	2532.34	4029.42
1 cm inward	0.0141	0.0213	2532.34	4029.42
surface limit	0.0000	0.0000	2532.34	4029.43

Table 625: Near-inner-boundary profile for GW190413_134308. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $38.273 M_\odot$ and $20.325 M_\odot$, with completed geometric mass-equivalent shares of 17.40% and 32.15%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $38.321 < 52.005 < 72.921$ km, and component 2 gives $23.161 < 29.560 < 44.011$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5598$ and 1.7022 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.345 and 15.701 h. The determinant-averaged frame-dragging rates at those shells are 1471.33 and 2633.51 rad s $^{-1}$, approaching surface-limit values of 2532.34 and 4029.43 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.510/6.744 h at 1 km inward to 0.0141/0.0213 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.33 GW240908_125134

GW240908_125134 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 77.362 M_\odot$ and $m_2 = 57.001 M_\odot$, while the luminosity-distance median is 4655.0 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,022
Observed luminosity distance	4655.0 Mpc (median)
Detector-frame masses	$m_1 = 77.362 M_\odot$, $m_2 = 57.001 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.737$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7353$; $z_\pi = 0.7942$ (medians)
Post-handoff reorder fraction	20.55%

Table 626: Observational and coordinate summary for GW240908_125134. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	42.023 [29.919, 57.500]	31.972 [20.390, 46.008]
Physical compact-source mass M_c ($M_\odot$)	31.120 [20.904, 46.913]	20.764 [12.781, 31.378]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.464 [1.050, 23.351]	11.271 [1.627, 20.915]
Geometric share M_{geom}/M_∞ (%)	22.97 [2.76, 48.54]	36.95 [6.11, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.566 [0.088, 0.945]	0.322 [0.029, 0.880]
Polarization factor A_{pol}	1.298 [1.028, 1.943]	1.586 [1.065, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.954 [0.332, 1.000]	0.810 [0.114, 0.998]
GR/CPTG matching radius r_m (km)	62.052 [44.179, 84.906]	47.210 [30.109, 67.937]
Structural Mode N_{BH}	1.6082 [1.4627, 1.8955]	1.7573 [1.4814, 1.9104]
Organization-shell radius r_N (km)	32.649 [23.218, 44.756]	24.851 [15.860, 35.800]
Inner trapping-surface radius r_- (km)	43.368 [31.136, 62.570]	30.727 [19.734, 43.555]
Structural- N lapse $\bar{\alpha}_N$	0.7270 [0.4962, 0.8412]	0.6103 [0.4844, 0.8261]
Structural- N clock $T_{24,N}$ (h)	17.449 [11.908, 20.188]	14.648 [11.627, 19.826]
Structural- N axis ω (rad s $^{-1}$)	1466.10 [595.95, 2596.61]	2160.45 [471.89, 3925.78]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1862.36 [866.24, 2950.95]	2517.80 [472.18, 4362.26]
Structural- N equatorial ω (rad s $^{-1}$)	2300.39 [1014.66, 3347.78]	2910.34 [472.33, 4878.59]
Structural- N period P_ω (ms)	3.3738 [2.1292, 7.2534]	2.4955 [1.4403, 13.3067]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3046.56 [1121.89, 4407.48]	3665.80 [514.83, 6290.20]
Trapping-surface period P_- (ms)	2.0624 [1.4256, 5.6005]	1.7140 [0.9989, 12.2044]

Table 627: CPTG compact-state, structural, clock, and geometric-rotation results for GW240908_125134. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.2677	6.7536	2994.81	3567.39
500 m inward	3.7144	4.7784	3021.32	3620.85
100 m inward	1.6564	2.1374	3041.54	3657.59
10 m inward	0.5234	0.6758	3046.03	3665.09
1 m inward	0.1655	0.2137	3046.50	3665.77
10 cm inward	0.0523	0.0676	3046.55	3665.80
1 cm inward	0.0166	0.0214	3046.56	3665.80
surface limit	0.0000	0.0000	3046.56	3665.80

Table 628: Near-inner-boundary profile for GW240908_125134. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.120 M_\odot$ and $20.764 M_\odot$, with completed geometric mass-equivalent shares of 22.97% and 36.95%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.649 < 43.368 < 62.052$ km, and component 2 gives $24.851 < 30.727 < 47.210$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6082$ and 1.7573 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.449 and 14.648 h. The determinant-averaged frame-dragging rates at those shells are 1862.36 and 2517.80 rad s $^{-1}$, approaching surface-limit values of 3046.56 and 3665.80 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.268/6.754 h at 1 km inward to 0.0166/0.0214 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.34 GW200216_220804

GW200216_220804 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 84.362 M_\odot$ and $m_2 = 50.739 M_\odot$, while the luminosity-distance median is 3818.1 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	6,038
Observed luminosity distance	3818.1 Mpc (median)
Detector-frame masses	$m_1 = 84.362 M_\odot$, $m_2 = 50.739 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.601$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6255$; $z_\pi = 0.6739$ (medians)
Post-handoff reorder fraction	14.06%

Table 629: Observational and coordinate summary for GW200216_220804. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	49.154 [33.433, 71.452]	29.375 [13.647, 46.768]
Physical compact-source mass M_c ($M_\odot$)	36.137 [23.790, 59.878]	20.013 [9.102, 32.729]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.543 [1.038, 27.804]	8.367 [0.893, 20.594]
Geometric share M_{geom}/M_∞ (%)	21.45 [2.20, 48.81]	32.16 [4.06, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.594 [0.079, 0.956]	0.405 [0.035, 0.920]
Polarization factor A_{pol}	1.273 [1.023, 1.953]	1.474 [1.042, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.962 [0.302, 1.000]	0.880 [0.140, 0.999]
GR/CPTG matching radius r_m (km)	72.582 [49.369, 105.507]	43.375 [20.151, 69.058]
Structural Mode N_{BH}	1.5943 [1.4598, 1.8986]	1.7023 [1.4697, 1.9094]
Organization-shell radius r_N (km)	38.197 [25.952, 55.408]	22.853 [10.617, 36.337]
Inner trapping-surface radius r_- (km)	50.486 [35.132, 79.831]	29.313 [13.557, 45.400]
Structural- N lapse $\bar{\alpha}_N$	0.7378 [0.4937, 0.8435]	0.6541 [0.4852, 0.8355]
Structural- N clock $T_{24,N}$ (h)	17.706 [11.848, 20.245]	15.698 [11.644, 20.051]
Structural- N axis ω (rad s $^{-1}$)	1198.17 [448.07, 2247.48]	2212.27 [633.95, 5358.23]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1556.06 [628.75, 2545.51]	2657.60 [642.86, 6168.13]
Structural- N equatorial ω (rad s $^{-1}$)	1936.91 [710.67, 2921.77]	3133.18 [643.17, 7051.63]
Structural- N period P_ω (ms)	4.0379 [2.4683, 9.9931]	2.3642 [1.0187, 9.7738]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2564.31 [796.11, 3906.27]	4030.22 [701.39, 9268.66]
Trapping-surface period P_- (ms)	2.4502 [1.6085, 7.8924]	1.5590 [0.6779, 8.9583]

Table 630: CPTG compact-state, structural, clock, and geometric-rotation results for GW200216_220804. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.8249	6.7430	2530.61	3923.84
500 m inward	3.4002	4.7602	2546.87	3982.82
100 m inward	1.5167	2.1258	2560.18	4020.79
10 m inward	0.4792	0.6717	2563.91	4029.10
1 m inward	0.1515	0.2124	2564.28	4030.11
10 cm inward	0.0479	0.0672	2564.31	4030.21
1 cm inward	0.0152	0.0212	2564.31	4030.22
surface limit	0.0000	0.0000	2564.31	4030.22

Table 631: Near-inner-boundary profile for GW200216_220804. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $36.137 M_\odot$ and $20.013 M_\odot$, with completed geometric mass-equivalent shares of 21.45% and 32.16%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $38.197 < 50.486 < 72.582$ km, and component 2 gives $22.853 < 29.313 < 43.375$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5943$ and 1.7023 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.706 and 15.698 h. The determinant-averaged frame-dragging rates at those shells are 1556.06 and 2657.60 rad s $^{-1}$, approaching surface-limit values of 2564.31 and 4030.22 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.825/6.743 h at 1 km inward to 0.0152/0.0212 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.35 GW240519_012815

GW240519_012815 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 86.469 M_\odot$ and $m_2 = 53.588 M_\odot$, while the luminosity-distance median is 1742.8 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,123
Observed luminosity distance	1742.8 Mpc (median)
Detector-frame masses	$m_1 = 86.469 M_\odot$, $m_2 = 53.588 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.620$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.3228$; $z_\pi = 0.3471$ (medians)
Post-handoff reorder fraction	13.31%

Table 632: Observational and coordinate summary for GW240519_012815. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	62.954 [42.407, 82.849]	39.003 [23.068, 59.311]
Physical compact-source mass M_c ($M_\odot$)	44.247 [30.335, 65.871]	26.482 [15.029, 40.077]
Geometric mass-equivalent M_{geom} ($M_\odot$)	16.449 [1.840, 34.128]	12.142 [1.448, 26.904]
Geometric share M_{geom}/M_∞ (%)	26.24 [3.49, 49.06]	33.55 [4.49, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.509 [0.070, 0.931]	0.381 [0.035, 0.911]
Polarization factor A_{pol}	1.356 [1.036, 1.963]	1.505 [1.047, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.935 [0.269, 0.999]	0.863 [0.137, 0.999]
GR/CPTG matching radius r_m (km)	92.959 [62.619, 122.337]	57.593 [34.063, 87.580]
Structural Mode N_{BH}	1.6396 [1.4666, 1.9015]	1.7180 [1.4721, 1.9095]
Organization-shell radius r_N (km)	48.996 [32.883, 64.575]	30.305 [17.938, 46.065]
Inner trapping-surface radius r_- (km)	62.585 [45.082, 88.041]	38.630 [22.634, 56.062]
Structural- N lapse $\bar{\alpha}_N$	0.7028 [0.4913, 0.8380]	0.6418 [0.4851, 0.8336]
Structural- N clock $T_{24,N}$ (h)	16.867 [11.792, 20.111]	15.402 [11.642, 20.006]
Structural- N axis ω (rad s $^{-1}$)	1063.11 [426.81, 1767.23]	1686.49 [432.75, 3349.89]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1306.48 [568.18, 2008.95]	2021.63 [433.28, 3788.14]
Structural- N equatorial ω (rad s $^{-1}$)	1563.26 [580.93, 2305.46]	2394.79 [433.48, 4271.11]
Structural- N period P_ω (ms)	4.8093 [3.1276, 11.0584]	3.1080 [1.6586, 14.5014]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2028.65 [636.68, 3074.37]	3090.30 [472.58, 5530.35]
Trapping-surface period P_- (ms)	3.0972 [2.0437, 9.8687]	2.0332 [1.1361, 13.2955]

Table 633: CPTG compact-state, structural, clock, and geometric-rotation results for GW240519_012815. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.4694	5.9098	2005.09	3027.16
500 m inward	3.1539	4.1756	2017.18	3058.29
100 m inward	1.4085	1.8658	2026.52	3083.49
10 m inward	0.4452	0.5898	2028.37	3089.52
1 m inward	0.1408	0.1865	2028.62	3090.21
10 cm inward	0.0445	0.0590	2028.65	3090.29
1 cm inward	0.0141	0.0187	2028.65	3090.30
surface limit	0.0000	0.0000	2028.65	3090.30

Table 634: Near-inner-boundary profile for GW240519_012815. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $44.247 M_\odot$ and $26.482 M_\odot$, with completed geometric mass-equivalent shares of 26.24% and 33.55%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $48.996 < 62.585 < 92.959$ km, and component 2 gives $30.305 < 38.630 < 57.593$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6396$ and 1.7180 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.867 and 15.402 h. The determinant-averaged frame-dragging rates at those shells are 1306.48 and 2021.63 rad s $^{-1}$, approaching surface-limit values of 2028.65 and 3090.30 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.469/5.910 h at 1 km inward to 0.0141/0.0187 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Its Structural-Mode frame dragging is comparatively slow within M3. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.36 GW190620_030421

GW190620_030421 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 87.131 M_\odot$ and $m_2 = 53.348 M_\odot$, while the luminosity-distance median is 2907.2 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	8,895
Observed luminosity distance	2907.2 Mpc (median)
Detector-frame masses	$m_1 = 87.131 M_\odot$, $m_2 = 53.348 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.612$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4990$; $z_\pi = 0.5367$ (medians)
Post-handoff reorder fraction	10.94%

Table 635: Observational and coordinate summary for GW190620_030421. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	56.368 [38.980, 75.708]	34.423 [20.031, 50.261]
Physical compact-source mass M_c ($M_\odot$)	46.650 [31.072, 69.025]	24.498 [12.757, 38.383]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.131 [0.784, 22.859]	9.003 [1.003, 21.406]
Geometric share M_{geom}/M_∞ (%)	12.77 [1.47, 39.22]	27.66 [3.24, 49.49]
Exterior spin magnitude $ \hat{a}_\infty $	0.753 [0.282, 0.971]	0.484 [0.051, 0.936]
Polarization factor A_{pol}	1.146 [1.015, 1.645]	1.382 [1.033, 1.980]
Compact-source spin parameter $\hat{a}_c$	0.989 [0.764, 1.000]	0.924 [0.201, 0.999]
GR/CPTG matching radius r_m (km)	83.235 [57.559, 111.792]	50.830 [29.579, 74.217]
Structural Mode N_{BH}	1.5246 [1.4560, 1.7843]	1.6541 [1.4653, 1.9064]
Organization-shell radius r_N (km)	43.711 [30.124, 58.521]	26.748 [15.570, 39.066]
Inner trapping-surface radius r_- (km)	62.317 [44.336, 91.688]	35.148 [19.352, 51.585]
Structural- N lapse $\bar{\alpha}_N$	0.7919 [0.5884, 0.8466]	0.6916 [0.4875, 0.8391]
Structural- N clock $T_{24,N}$ (h)	19.004 [14.121, 20.318]	16.600 [11.701, 20.138]
Structural- N axis ω (rad s $^{-1}$)	846.89 [425.44, 1733.41]	1821.44 [652.91, 3914.61]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1232.06 [692.57, 2005.85]	2234.24 [769.93, 4409.07]
Structural- N equatorial ω (rad s $^{-1}$)	1672.69 [1087.35, 2380.11]	2691.33 [770.52, 4930.73]
Structural- N period P_ω (ms)	5.0997 [3.1324, 9.0722]	2.8122 [1.4251, 8.1608]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2288.38 [1533.86, 3244.35]	3523.09 [840.67, 6404.67]
Trapping-surface period P_- (ms)	2.7457 [1.9367, 4.0963]	1.7834 [0.9810, 7.4740]

Table 636: CPTG compact-state, structural, clock, and geometric-rotation results for GW190620_030421. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.8789	6.0503	2264.60	3448.54
500 m inward	2.7271	4.2696	2275.96	3487.54
100 m inward	1.2133	1.9060	2286.17	3514.98
10 m inward	0.3832	0.6024	2288.10	3522.33
1 m inward	0.1212	0.1905	2288.35	3523.05
10 cm inward	0.0383	0.0602	2288.37	3523.09
1 cm inward	0.0121	0.0190	2288.38	3523.09
surface limit	0.0000	0.0000	2288.38	3523.09

Table 637: Near-inner-boundary profile for GW190620_030421. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $46.650 M_\odot$ and $24.498 M_\odot$, with completed geometric mass-equivalent shares of 12.77% and 27.66%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $43.711 < 62.317 < 83.235$ km, and component 2 gives $26.748 < 35.148 < 50.830$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5246$ and 1.6541 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.004 and 16.600 h. The determinant-averaged frame-dragging rates at those shells are 1232.06 and 2234.24 rad s $^{-1}$, approaching surface-limit values of 2288.38 and 3523.09 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.879/6.050 h at 1 km inward to 0.0121/0.0190 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.37 GW250104_015122

GW250104_015122 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 79.328 M_\odot$ and $m_2 = 61.308 M_\odot$, while the luminosity-distance median is 4828.0 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,861
Observed luminosity distance	4828.0 Mpc (median)
Detector-frame masses	$m_1 = 79.328 M_\odot$, $m_2 = 61.308 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.773$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7575$; $z_\pi = 0.8186$ (medians)
Post-handoff reorder fraction	21.92%

Table 638: Observational and coordinate summary for GW250104_015122. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.174 [32.305, 54.979]	34.445 [23.669, 47.189]
Physical compact-source mass M_c ($M_\odot$)	33.036 [22.272, 48.327]	22.303 [14.338, 33.623]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.279 [0.912, 21.934]	12.146 [1.935, 21.493]
Geometric share M_{geom}/M_∞ (%)	19.55 [2.20, 47.76]	36.75 [6.45, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.628 [0.111, 0.956]	0.326 [0.031, 0.873]
Polarization factor A_{pol}	1.243 [1.023, 1.914]	1.581 [1.069, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.405, 1.000]	0.814 [0.121, 0.998]
GR/CPTG matching radius r_m (km)	63.752 [47.702, 81.183]	50.863 [34.950, 69.681]
Structural Mode N_{BH}	1.5776 [1.4598, 1.8865]	1.7550 [1.4834, 1.9101]
Organization-shell radius r_N (km)	33.559 [25.047, 42.679]	26.791 [18.376, 36.678]
Inner trapping-surface radius r_- (km)	45.095 [33.349, 64.320]	33.132 [22.428, 45.743]
Structural- N lapse $\bar{\alpha}_N$	0.7506 [0.5035, 0.8435]	0.6122 [0.4846, 0.8245]
Structural- N clock $T_{24,N}$ (h)	18.014 [12.083, 20.245]	14.694 [11.631, 19.788]
Structural- N axis ω (rad s $^{-1}$)	1335.74 [578.78, 2446.57]	2025.73 [459.41, 3515.94]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1774.31 [897.76, 2764.82]	2353.48 [459.68, 3892.79]
Structural- N equatorial ω (rad s $^{-1}$)	2231.37 [1239.59, 3104.40]	2712.65 [459.81, 4291.28]
Structural- N period P_ω (ms)	3.5412 [2.2725, 6.9987]	2.6697 [1.6141, 13.6687]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2980.93 [1410.65, 4094.01]	3425.33 [501.14, 5501.89]
Trapping-surface period P_- (ms)	2.1078 [1.5347, 4.4541]	1.8343 [1.1420, 12.5379]

Table 639: CPTG compact-state, structural, clock, and geometric-rotation results for GW250104_015122. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0075	6.5365	2932.33	3338.34
500 m inward	3.5268	4.6250	2958.25	3384.49
100 m inward	1.5718	2.0691	2976.86	3417.13
10 m inward	0.4966	0.6544	2980.80	3424.16
1 m inward	0.1570	0.2069	2980.92	3425.21
10 cm inward	0.0497	0.0654	2980.93	3425.31
1 cm inward	0.0157	0.0207	2980.93	3425.32
surface limit	0.0000	0.0000	2980.93	3425.33

Table 640: Near-inner-boundary profile for GW250104_015122. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $33.036 M_\odot$ and $22.303 M_\odot$, with completed geometric mass-equivalent shares of 19.55% and 36.75%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.559 < 45.095 < 63.752$ km, and component 2 gives $26.791 < 33.132 < 50.863$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5776$ and 1.7550 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.014 and 14.694 h. The determinant-averaged frame-dragging rates at those shells are 1774.31 and 2353.48 rad s $^{-1}$, approaching surface-limit values of 2980.93 and 3425.33 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.008/6.536 h at 1 km inward to 0.0157/0.0207 h at 1 cm inward for components 1/2. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.38 GW230707_124047

GW230707_124047 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 78.093 M_\odot$ and $m_2 = 62.699 M_\odot$, while the luminosity-distance median is 4448.0 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	39,528
Observed luminosity distance	4448.0 Mpc (median)
Detector-frame masses	$m_1 = 78.093 M_\odot$, $m_2 = 62.699 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.803$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7084$; $z_\pi = 0.7649$ (medians)
Post-handoff reorder fraction	25.33%

Table 641: Observational and coordinate summary for GW230707_124047. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.382 [32.877, 56.555]	36.542 [25.817, 49.266]
Physical compact-source mass M_c ($M_\odot$)	33.262 [22.498, 48.412]	23.325 [15.598, 35.175]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.758 [0.987, 22.173]	13.248 [2.210, 22.245]
Geometric share M_{geom}/M_∞ (%)	20.47 [2.37, 47.67]	37.51 [6.86, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.611 [0.113, 0.953]	0.312 [0.029, 0.865]
Polarization factor A_{pol}	1.257 [1.024, 1.911]	1.600 [1.074, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.966 [0.412, 1.000]	0.800 [0.114, 0.997]
GR/CPTG matching radius r_m (km)	64.059 [48.547, 83.511]	53.959 [38.122, 72.748]
Structural Mode N_{BH}	1.5856 [1.4606, 1.8855]	1.7640 [1.4859, 1.9104]
Organization-shell radius r_N (km)	33.698 [25.493, 43.927]	28.410 [20.062, 38.329]
Inner trapping-surface radius r_- (km)	45.482 [33.904, 64.470]	34.768 [24.457, 48.018]
Structural- N lapse $\bar{\alpha}_N$	0.7444 [0.5043, 0.8428]	0.6049 [0.4845, 0.8225]
Structural- N clock $T_{24,N}$ (h)	17.866 [12.104, 20.228]	14.519 [11.627, 19.741]
Structural- N axis ω (rad s $^{-1}$)	1350.80 [585.30, 2431.01]	1927.67 [415.71, 3207.02]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1771.28 [905.69, 2735.41]	2224.69 [415.92, 3563.44]
Structural- N equatorial ω (rad s $^{-1}$)	2217.30 [1233.60, 3053.67]	2553.08 [416.04, 3925.84]
Structural- N period P_ω (ms)	3.5473 [2.2970, 6.9374]	2.8243 [1.7632, 15.1066]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2954.91 [1415.20, 4017.57]	3207.65 [453.54, 5034.98]
Trapping-surface period P_- (ms)	2.1264 [1.5639, 4.4398]	1.9588 [1.2479, 13.8535]

Table 642: CPTG compact-state, structural, clock, and geometric-rotation results for GW230707_124047. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0131	6.4157	2910.59	3132.82
500 m inward	3.5304	4.5398	2933.82	3171.20
100 m inward	1.5732	2.0310	2950.84	3200.89
10 m inward	0.4971	0.6423	2954.49	3206.96
1 m inward	0.1572	0.2031	2954.88	3207.63
10 cm inward	0.0497	0.0642	2954.90	3207.65
1 cm inward	0.0157	0.0203	2954.91	3207.65
surface limit	0.0000	0.0000	2954.91	3207.65

Table 643: Near-inner-boundary profile for GW230707_124047. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $33.262 M_\odot$ and $23.325 M_\odot$, with completed geometric mass-equivalent shares of 20.47% and 37.51%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $33.698 < 45.482 < 64.059$ km, and component 2 gives $28.410 < 34.768 < 53.959$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5856$ and 1.7640 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.866 and 14.519 h. The determinant-averaged frame-dragging rates at those shells are 1771.28 and 2224.69 rad s $^{-1}$, approaching surface-limit values of 2954.91 and 3207.65 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.013/6.416 h at 1 km inward to 0.0157/0.0203 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M3 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M3 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.39 GW191109_010717

GW191109_010717 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 81.156 M_\odot$ and $m_2 = 59.722 M_\odot$, while the luminosity-distance median is 1294.3 Mpc. One retained release label defines this numerical stream. It lies in the nearer-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	32,415
Observed luminosity distance	1294.3 Mpc (median)
Detector-frame masses	$m_1 = 81.156 M_\odot$, $m_2 = 59.722 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.736$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.2494$; $z_\pi = 0.2678$ (medians)
Post-handoff reorder fraction	11.84%

Table 644: Observational and coordinate summary for GW191109_010717. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	63.317 [48.497, 74.926]	46.886 [33.360, 65.945]
Physical compact-source mass M_c ($M_\odot$)	57.250 [40.897, 70.765]	35.444 [20.076, 56.218]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.600 [0.570, 17.408]	10.712 [1.016, 25.612]
Geometric share M_{geom}/M_∞ (%)	7.39 [0.93, 27.74]	23.03 [2.08, 49.23]
Exterior spin magnitude $ \hat{a}_\infty $	0.855 [0.482, 0.981]	0.565 [0.063, 0.959]
Polarization factor A_{pol}	1.080 [1.009, 1.384]	1.299 [1.021, 1.970]
Compact-source spin parameter $\hat{a}_c$	0.997 [0.923, 1.000]	0.954 [0.245, 1.000]
GR/CPTG matching radius r_m (km)	93.495 [71.612, 110.637]	69.233 [49.260, 97.376]
Structural Mode N_{BH}	1.4891 [1.4532, 1.6549]	1.6087 [1.4591, 1.9034]
Organization-shell radius r_N (km)	48.902 [37.472, 57.845]	36.390 [25.923, 51.201]
Inner trapping-surface radius r_- (km)	75.903 [55.919, 95.264]	48.722 [31.601, 74.830]
Structural- N lapse $\bar{\alpha}_N$	0.8200 [0.6910, 0.8488]	0.7266 [0.4898, 0.8441]
Structural- N clock $T_{24,N}$ (h)	19.679 [16.585, 20.372]	17.438 [11.756, 20.257]
Structural- N axis ω (rad s $^{-1}$)	584.06 [392.47, 1286.89]	1231.09 [475.63, 2499.43]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	911.02 [654.57, 1544.13]	1557.28 [719.70, 2764.18]
Structural- N equatorial ω (rad s $^{-1}$)	1368.64 [1081.79, 1891.92]	1946.74 [783.25, 3027.35]
Structural- N period P_ω (ms)	6.8969 [4.0691, 9.5989]	4.0347 [2.2731, 8.7303]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1955.22 [1550.74, 2605.75]	2587.44 [857.36, 3889.23]
Trapping-surface period P_- (ms)	3.2136 [2.4113, 4.0517]	2.4283 [1.6155, 7.3285]

Table 645: CPTG compact-state, structural, clock, and geometric-rotation results for GW191109_010717. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.1351	4.9795	1946.09	2551.57
500 m inward	2.1986	3.5123	1951.12	2570.51
100 m inward	0.9764	1.5676	1954.47	2584.85
10 m inward	0.3083	0.4954	1955.14	2587.07
1 m inward	0.0975	0.1567	1955.19	2587.39
10 cm inward	0.0308	0.0495	1955.21	2587.44
1 cm inward	0.0097	0.0157	1955.21	2587.44
surface limit	0.0000	0.0000	1955.22	2587.44

Table 646: Near-inner-boundary profile for GW191109_010717. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $57.250 M_\odot$ and $35.444 M_\odot$, with completed geometric mass-equivalent shares of 7.39% and 23.03%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $48.902 < 75.903 < 93.495$ km, and component 2 gives $36.390 < 48.722 < 69.233$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.4891$ and 1.6087 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.679 and 17.438 h. The determinant-averaged frame-dragging rates at those shells are 911.02 and 1557.28 rad s $^{-1}$, approaching surface-limit values of 1955.22 and 2587.44 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.135/4.979 h at 1 km inward to 0.0097/0.0157 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.40 GW250118_023225

GW250118_023225 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 82.315 M_\odot$ and $m_2 = 58.735 M_\odot$, while the luminosity-distance median is 5812.6 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,696
Observed luminosity distance	5812.6 Mpc (median)
Detector-frame masses	$m_1 = 82.315 M_\odot$, $m_2 = 58.735 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.714$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8812$; $z_\pi = 0.9541$ (medians)
Post-handoff reorder fraction	17.87%

Table 647: Observational and coordinate summary for GW250118_023225. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	41.440 [29.416, 57.173]	30.085 [18.360, 44.381]
Physical compact-source mass M_c ($M_\odot$)	30.876 [20.494, 46.875]	19.889 [11.688, 31.062]
Geometric mass-equivalent M_{geom} ($M_\odot$)	9.052 [0.980, 23.117]	9.936 [1.400, 19.736]
Geometric share M_{geom}/M_∞ (%)	22.13 [2.53, 48.80]	34.92 [5.55, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.581 [0.079, 0.950]	0.357 [0.033, 0.891]
Polarization factor A_{pol}	1.284 [1.026, 1.953]	1.537 [1.059, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.959 [0.303, 1.000]	0.844 [0.130, 0.998]
GR/CPTG matching radius r_m (km)	61.191 [43.436, 84.423]	44.425 [27.110, 65.535]
Structural Mode N_{BH}	1.6004 [1.4615, 1.8985]	1.7336 [1.4781, 1.9098]
Organization-shell radius r_N (km)	32.207 [22.789, 44.440]	23.381 [14.270, 34.498]
Inner trapping-surface radius r_- (km)	42.839 [30.623, 62.562]	29.198 [17.723, 42.850]
Structural- N lapse $\bar{\alpha}_N$	0.7330 [0.4938, 0.8422]	0.6294 [0.4849, 0.8287]
Structural- N clock $T_{24,N}$ (h)	17.592 [11.850, 20.212]	15.105 [11.637, 19.889]
Structural- N axis ω (rad s $^{-1}$)	1444.05 [583.27, 2626.50]	2279.47 [571.76, 4318.12]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1852.24 [845.07, 2954.14]	2678.60 [572.77, 4827.12]
Structural- N equatorial ω (rad s $^{-1}$)	2302.87 [961.92, 3371.22]	3105.81 [572.99, 5408.05]
Structural- N period P_ω (ms)	3.3922 [2.1269, 7.4351]	2.3457 [1.3016, 10.9698]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3067.68 [1056.76, 4465.64]	3953.54 [624.50, 6992.85]
Trapping-surface period P_- (ms)	2.0482 [1.4070, 5.9457]	1.5893 [0.8985, 10.0612]

Table 648: CPTG compact-state, structural, clock, and geometric-rotation results for GW250118_023225. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.2474	6.8692	3019.81	3843.60
500 m inward	3.6981	4.8551	3044.55	3901.46
100 m inward	1.6495	2.1704	3063.17	3944.01
10 m inward	0.5212	0.6863	3067.41	3952.81
1 m inward	0.1648	0.2170	3067.65	3953.47
10 cm inward	0.0521	0.0686	3067.68	3953.53
1 cm inward	0.0165	0.0217	3067.68	3953.54
surface limit	0.0000	0.0000	3067.68	3953.54

Table 649: Near-inner-boundary profile for GW250118_023225. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $30.876 M_\odot$ and $19.889 M_\odot$, with completed geometric mass-equivalent shares of 22.13% and 34.92%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.207 < 42.839 < 61.191$ km, and component 2 gives $23.381 < 29.198 < 44.425$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6004$ and 1.7336 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.592 and 15.105 h. The determinant-averaged frame-dragging rates at those shells are 1852.24 and 2678.60 rad s $^{-1}$, approaching surface-limit values of 3067.68 and 3953.54 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.247/6.869 h at 1 km inward to 0.0165/0.0217 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M3 descriptive distribution, making it a useful representative upper-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.41 GW230914_111401

GW230914_111401 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 87.043 M_\odot$ and $m_2 = 54.194 M_\odot$, while the luminosity-distance median is 2702.1 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	48,738
Observed luminosity distance	2702.1 Mpc (median)
Detector-frame masses	$m_1 = 87.043 M_\odot$, $m_2 = 54.194 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.623$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4688$; $z_\pi = 0.5046$ (medians)
Post-handoff reorder fraction	17.78%

Table 650: Observational and coordinate summary for GW230914_111401. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	57.410 [38.925, 69.670]	36.273 [23.844, 57.124]
Physical compact-source mass M_c ($M_\odot$)	38.225 [29.044, 55.913]	25.521 [14.831, 36.892]
Geometric mass-equivalent M_{geom} ($M_\odot$)	16.653 [1.753, 31.065]	11.696 [1.566, 26.588]
Geometric share M_{geom}/M_∞ (%)	29.19 [3.74, 49.45]	33.64 [5.05, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.457 [0.053, 0.926]	0.380 [0.034, 0.900]
Polarization factor A_{pol}	1.412 [1.039, 1.978]	1.507 [1.053, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.911 [0.207, 0.999]	0.862 [0.135, 0.999]
GR/CPTG matching radius r_m (km)	84.773 [57.478, 102.876]	53.562 [35.209, 84.351]
Structural Mode N_{BH}	1.6701 [1.4680, 1.9060]	1.7190 [1.4753, 1.9096]
Organization-shell radius r_N (km)	44.678 [30.095, 54.219]	28.217 [18.510, 44.375]
Inner trapping-surface radius r_- (km)	55.083 [43.277, 74.143]	36.536 [22.925, 52.253]
Structural- N lapse $\bar{\alpha}_N$	0.6793 [0.4878, 0.8369]	0.6410 [0.4850, 0.8310]
Structural- N clock $T_{24,N}$ (h)	16.302 [11.708, 20.085]	15.384 [11.641, 19.945]
Structural- N axis ω (rad s^{-1})	1231.35 [423.92, 1829.59]	1797.35 [440.83, 3409.45]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	1494.92 [481.79, 2082.81]	2156.90 [441.05, 3800.92]
Structural- N equatorial ω (rad s^{-1})	1759.63 [482.68, 2438.70]	2576.27 [441.27, 4216.58]
Structural- N period P_ω (ms)	4.2030 [3.0167, 13.0413]	2.9131 [1.6531, 14.2460]
Trapping-surface limit Ω_- (rad s^{-1})	2257.38 [526.90, 3319.60]	3320.20 [481.30, 5425.62]
Trapping-surface period P_- (ms)	2.7834 [1.8928, 11.9247]	1.8924 [1.1581, 13.0547]

Table 651: CPTG compact-state, structural, clock, and geometric-rotation results for GW230914_111401. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.9047	6.0018	2225.57	3249.92
500 m inward	3.4646	4.2428	2242.06	3286.52
100 m inward	1.5478	1.8963	2254.37	3313.67
10 m inward	0.4893	0.5996	2257.06	3319.66
1 m inward	0.1547	0.1896	2257.34	3320.12
10 cm inward	0.0489	0.0600	2257.38	3320.19
1 cm inward	0.0155	0.0190	2257.38	3320.19
surface limit	0.0000	0.0000	2257.38	3320.20

Table 652: Near-inner-boundary profile for GW230914_111401. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $38.225 M_\odot$ and $25.521 M_\odot$, with completed geometric mass-equivalent shares of 29.19% and 33.64%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $44.678 < 55.083 < 84.773$ km, and component 2 gives $28.217 < 36.536 < 53.562$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6701$ and 1.7190 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.302 and 15.384 h. The determinant-averaged frame-dragging rates at those shells are 1494.92 and $2156.90 \text{ rad s}^{-1}$, approaching surface-limit values of 2257.38 and $3320.20 \text{ rad s}^{-1}$. Across the registered radial sequence, the median clock falls from $4.905/6.002$ h at 1 km inward to $0.0155/0.0190$ h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M3. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.42 GW190929_012149

GW190929_012149 is an M3 upper observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 101.800 M_\odot$ and $m_2 = 41.690 M_\odot$, while the luminosity-distance median is 3125.4 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C01:Mixed
Posterior samples	30,673
Observed luminosity distance	3125.4 Mpc (median)
Detector-frame masses	$m_1 = 101.800 M_\odot$, $m_2 = 41.690 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.410$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5301$; $z_\pi = 0.5702$ (medians)
Post-handoff reorder fraction	3.18%

Table 653: Observational and coordinate summary for GW190929_012149. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	64.549 [45.662, 86.214]	26.063 [15.742, 42.278]
Physical compact-source mass M_c ($M_\odot$)	40.979 [28.232, 71.070]	18.381 [9.991, 31.229]
Geometric mass-equivalent M_{geom} ($M_\odot$)	22.238 [3.174, 35.207]	7.319 [0.879, 17.868]
Geometric share M_{geom}/M_∞ (%)	34.86 [5.30, 49.75]	29.79 [3.62, 49.68]
Exterior spin magnitude $ \hat{a}_\infty $	0.358 [0.036, 0.895]	0.446 [0.041, 0.928]
Polarization factor A_{pol}	1.535 [1.056, 1.990]	1.424 [1.038, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.845 [0.141, 0.998]	0.906 [0.160, 0.999]
GR/CPTG matching radius r_m (km)	95.315 [67.426, 127.306]	38.485 [23.244, 62.429]
Structural Mode N_{BH}	1.7330 [1.4767, 1.9094]	1.6764 [1.4673, 1.9085]
Organization-shell radius r_N (km)	50.205 [35.480, 67.067]	20.247 [12.227, 32.833]
Inner trapping-surface radius r_- (km)	60.845 [44.105, 94.805]	26.195 [15.229, 43.449]
Structural- N lapse $\bar{\alpha}_N$	0.6299 [0.4852, 0.8299]	0.6744 [0.4859, 0.8374]
Structural- N clock $T_{24,N}$ (h)	15.117 [11.645, 19.917]	16.185 [11.661, 20.098]
Structural- N axis ω (rad s $^{-1}$)	1107.10 [272.56, 1762.29]	2391.62 [727.94, 4995.48]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1287.22 [303.02, 1963.97]	2927.80 [772.81, 5589.72]
Structural- N equatorial ω (rad s $^{-1}$)	1463.88 [304.48, 2216.70]	3517.97 [773.25, 6277.51]
Structural- N period P_ω (ms)	4.8812 [3.1992, 20.7351]	2.1460 [1.1241, 8.1304]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1842.27 [332.03, 2866.28]	4569.36 [842.62, 8088.35]
Trapping-surface period P_- (ms)	3.4106 [2.1921, 18.9236]	1.3751 [0.7768, 7.4567]

Table 654: CPTG compact-state, structural, clock, and geometric-rotation results for GW190929_012149. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.8363	7.0063	1819.05	4435.17
500 m inward	3.4199	4.9383	1830.55	4504.09
100 m inward	1.5290	2.2020	1840.14	4557.16
10 m inward	0.4835	0.6959	1842.02	4568.53
1 m inward	0.1529	0.2201	1842.26	4569.32
10 cm inward	0.0484	0.0696	1842.27	4569.36
1 cm inward	0.0153	0.0220	1842.27	4569.36
surface limit	0.0000	0.0000	1842.27	4569.36

Table 655: Near-inner-boundary profile for GW190929_012149. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $40.979 M_\odot$ and $18.381 M_\odot$, with completed geometric mass-equivalent shares of 34.86% and 29.79%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $50.205 < 60.845 < 95.315$ km, and component 2 gives $20.247 < 26.195 < 38.485$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.7330$ and 1.6764 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 15.117 and 16.185 h. The determinant-averaged frame-dragging rates at those shells are 1287.22 and 2927.80 rad s $^{-1}$, approaching surface-limit values of 1842.27 and 4569.36 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.836/7.006 h at 1 km inward to 0.0153/0.0220 h at 1 cm inward for components 1/2. Its strong observational mass asymmetry makes the contrast between the two solved compact interiors particularly useful for component-by-component comparison. Its Structural-Mode frame dragging is in the upper part of the M3 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.43 GW191230_180458

GW191230_180458 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 82.698 M_\odot$ and $m_2 = 62.848 M_\odot$, while the luminosity-distance median is 4283.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C01:Mixed
Posterior samples	7,026
Observed luminosity distance	4283.3 Mpc (median)
Detector-frame masses	$m_1 = 82.698 M_\odot$, $m_2 = 62.848 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.760$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6875$; $z_\pi = 0.7414$ (medians)
Post-handoff reorder fraction	22.08%

Table 656: Observational and coordinate summary for GW191230_180458. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	46.722 [34.651, 61.566]	36.650 [24.262, 50.404]
Physical compact-source mass M_c ($M_\odot$)	36.083 [24.409, 54.641]	23.962 [14.955, 36.578]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.392 [0.871, 23.864]	12.399 [1.913, 22.717]
Geometric share M_{geom}/M_∞ (%)	18.17 [1.88, 47.01]	35.51 [5.94, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.653 [0.130, 0.962]	0.347 [0.033, 0.883]
Polarization factor A_{pol}	1.222 [1.019, 1.887]	1.551 [1.063, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.975 [0.462, 1.000]	0.835 [0.129, 0.998]
GR/CPTG matching radius r_m (km)	68.991 [51.167, 90.909]	54.118 [35.826, 74.428]
Structural Mode N_{BH}	1.5661 [1.4581, 1.8777]	1.7404 [1.4804, 1.9098]
Organization-shell radius r_N (km)	36.275 [26.864, 47.839]	28.497 [18.888, 39.227]
Inner trapping-surface radius r_- (km)	49.297 [36.302, 73.299]	35.561 [23.055, 49.593]
Structural- N lapse $\bar{\alpha}_N$	0.7595 [0.5107, 0.8449]	0.6239 [0.4849, 0.8269]
Structural- N clock $T_{24,N}$ (h)	18.229 [12.257, 20.277]	14.974 [11.637, 19.846]
Structural- N axis ω (rad s $^{-1}$)	1184.69 [503.30, 2254.54]	1891.92 [455.92, 3336.32]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1604.37 [793.92, 2543.94]	2195.09 [456.29, 3704.67]
Structural- N equatorial ω (rad s $^{-1}$)	2054.77 [1138.69, 2867.12]	2554.52 [456.48, 4112.82]
Structural- N period P_ω (ms)	3.9163 [2.4699, 7.9141]	2.8624 [1.6960, 13.7702]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2751.95 [1346.30, 3780.48]	3245.24 [497.64, 5254.08]
Trapping-surface period P_- (ms)	2.2832 [1.6620, 4.6671]	1.9361 [1.1959, 12.6260]

Table 657: CPTG compact-state, structural, clock, and geometric-rotation results for GW191230_180458. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.7093	6.2941	2714.50	3169.25
500 m inward	3.3146	4.4522	2734.04	3209.41
100 m inward	1.4764	1.9914	2748.07	3236.91
10 m inward	0.4664	0.6296	2751.58	3244.58
1 m inward	0.1475	0.1991	2751.91	3245.19
10 cm inward	0.0466	0.0630	2751.94	3245.24
1 cm inward	0.0147	0.0199	2751.95	3245.24
surface limit	0.0000	0.0000	2751.95	3245.24

Table 658: Near-inner-boundary profile for GW191230_180458. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $36.083 M_\odot$ and $23.962 M_\odot$, with completed geometric mass-equivalent shares of 18.17% and 35.51%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $36.275 < 49.297 < 68.991$ km, and component 2 gives $28.497 < 35.561 < 54.118$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5661$ and 1.7404 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.229 and 14.974 h. The determinant-averaged frame-dragging rates at those shells are 1604.37 and 2195.09 rad s $^{-1}$, approaching surface-limit values of 2751.95 and 3245.24 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.709/6.294 h at 1 km inward to 0.0147/0.0199 h at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. Its Structural-Mode frame dragging is comparatively slow within M3. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.44 GW250108_152221

GW250108_152221 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 87.798 M_\odot$ and $m_2 = 58.338 M_\odot$, while the luminosity-distance median is 3838.6 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,752
Observed luminosity distance	3838.6 Mpc (median)
Detector-frame masses	$m_1 = 87.798 M_\odot$, $m_2 = 58.338 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.664$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6276$; $z_\pi = 0.6769$ (medians)
Post-handoff reorder fraction	13.15%

Table 659: Observational and coordinate summary for GW250108_152221. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	52.163 [36.819, 67.850]	34.949 [24.581, 49.585]
Physical compact-source mass M_c ($M_\odot$)	38.549 [26.146, 56.397]	23.729 [15.067, 35.970]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.094 [1.037, 28.493]	11.344 [1.522, 21.957]
Geometric share M_{geom}/M_∞ (%)	21.84 [2.14, 48.64]	33.68 [4.88, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.587 [0.085, 0.957]	0.379 [0.035, 0.904]
Polarization factor A_{pol}	1.279 [1.022, 1.947]	1.508 [1.051, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.960 [0.321, 1.000]	0.861 [0.139, 0.999]
GR/CPTG matching radius r_m (km)	77.026 [54.368, 100.188]	51.607 [36.297, 73.219]
Structural Mode N_{BH}	1.5978 [1.4594, 1.8967]	1.7194 [1.4743, 1.9095]
Organization-shell radius r_N (km)	40.525 [28.512, 52.748]	27.181 [19.088, 38.553]
Inner trapping-surface radius r_- (km)	53.832 [38.872, 75.374]	34.371 [23.529, 49.238]
Structural- N lapse $\bar{\alpha}_N$	0.7350 [0.4952, 0.8438]	0.6406 [0.4852, 0.8318]
Structural- N clock $T_{24,N}$ (h)	17.641 [11.885, 20.251]	15.375 [11.644, 19.963]
Structural- N axis ω (rad s $^{-1}$)	1157.07 [473.69, 2066.25]	1918.17 [528.99, 3330.03]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1492.95 [703.16, 2342.12]	2276.59 [530.35, 3699.76]
Structural- N equatorial ω (rad s $^{-1}$)	1847.82 [817.68, 2669.51]	2675.20 [530.83, 4089.35]
Structural- N period P_ω (ms)	4.2086 [2.6827, 8.9357]	2.7599 [1.6983, 11.8471]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2443.79 [900.97, 3552.89]	3443.59 [579.47, 5276.13]
Trapping-surface period P_- (ms)	2.5711 [1.7685, 6.9738]	1.8246 [1.1909, 10.8430]

Table 660: CPTG compact-state, structural, clock, and geometric-rotation results for GW250108_152221. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.6886	6.3051	2414.23	3370.17
500 m inward	3.3039	4.4561	2430.11	3407.11
100 m inward	1.4742	1.9913	2441.51	3435.91
10 m inward	0.4659	0.6296	2443.57	3442.99
1 m inward	0.1473	0.1991	2443.75	3443.57
10 cm inward	0.0466	0.0630	2443.78	3443.59
1 cm inward	0.0147	0.0199	2443.78	3443.59
surface limit	0.0000	0.0000	2443.79	3443.59

Table 661: Near-inner-boundary profile for GW250108_152221. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $38.549 M_\odot$ and $23.729 M_\odot$, with completed geometric mass-equivalent shares of 21.84% and 33.68%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $40.525 < 53.832 < 77.026$ km, and component 2 gives $27.181 < 34.371 < 51.607$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5978$ and 1.7194 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.641 and 15.375 h. The determinant-averaged frame-dragging rates at those shells are 1492.95 and 2276.59 rad s $^{-1}$, approaching surface-limit values of 2443.79 and 3443.59 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.689/6.305 h at 1 km inward to 0.0147/0.0199 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M3. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.45 GW240426_031451

GW240426_031451 is an M3 upper observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 85.566 M_\odot$ and $m_2 = 61.106 M_\odot$, while the luminosity-distance median is 4000.3 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,745
Observed luminosity distance	4000.3 Mpc (median)
Detector-frame masses	$m_1 = 85.566 M_\odot$, $m_2 = 61.106 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.714$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6493$; $z_\pi = 0.7005$ (medians)
Post-handoff reorder fraction	18.09%

Table 662: Observational and coordinate summary for GW240426_031451. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	49.514 [35.727, 64.727]	35.997 [23.539, 50.885]
Physical compact-source mass M_c ($M_\odot$)	36.880 [24.855, 54.386]	23.610 [14.594, 35.702]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.460 [1.141, 26.370]	12.176 [1.678, 23.191]
Geometric share M_{geom}/M_∞ (%)	21.66 [2.45, 48.48]	35.85 [5.50, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.590 [0.090, 0.951]	0.341 [0.032, 0.892]
Polarization factor A_{pol}	1.277 [1.025, 1.941]	1.559 [1.058, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.961 [0.338, 1.000]	0.829 [0.128, 0.998]
GR/CPTG matching radius r_m (km)	73.113 [52.756, 95.578]	53.154 [34.759, 75.138]
Structural Mode N_{BH}	1.5962 [1.4610, 1.8949]	1.7445 [1.4779, 1.9099]
Organization-shell radius r_N (km)	38.448 [27.710, 50.288]	27.990 [18.274, 39.583]
Inner trapping-surface radius r_- (km)	51.097 [37.129, 72.414]	34.872 [22.593, 48.998]
Structural- N lapse $\bar{\alpha}_N$	0.7363 [0.4967, 0.8425]	0.6207 [0.4848, 0.8290]
Structural- N clock $T_{24,N}$ (h)	17.671 [11.920, 20.220]	14.896 [11.636, 19.895]
Structural- N axis ω (rad s $^{-1}$)	1215.37 [512.23, 2187.52]	1895.50 [467.11, 3439.04]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1565.96 [764.66, 2458.47]	2227.72 [467.72, 3832.21]
Structural- N equatorial ω (rad s $^{-1}$)	1938.19 [911.47, 2783.01]	2591.71 [468.03, 4259.56]
Structural- N period P_ω (ms)	4.0124 [2.5557, 8.2170]	2.8205 [1.6396, 13.4337]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2575.44 [1001.18, 3685.66]	3306.98 [510.65, 5496.18]
Trapping-surface period P_- (ms)	2.4397 [1.7048, 6.2757]	1.9000 [1.1432, 12.3044]

Table 663: CPTG compact-state, structural, clock, and geometric-rotation results for GW240426_031451. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.8001	6.3213	2541.92	3229.77
500 m inward	3.3836	4.4714	2558.81	3270.49
100 m inward	1.5088	1.9997	2572.96	3299.85
10 m inward	0.4769	0.6323	2575.26	3306.35
1 m inward	0.1508	0.1999	2575.43	3306.91
10 cm inward	0.0477	0.0632	2575.44	3306.98
1 cm inward	0.0151	0.0200	2575.44	3306.98
surface limit	0.0000	0.0000	2575.44	3306.98

Table 664: Near-inner-boundary profile for GW240426_031451. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $36.880 M_\odot$ and $23.610 M_\odot$, with completed geometric mass-equivalent shares of 21.66% and 35.85%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $38.448 < 51.097 < 73.113$ km, and component 2 gives $27.990 < 34.872 < 53.154$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5962$ and 1.7445 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.671 and 14.896 h. The determinant-averaged frame-dragging rates at those shells are 1565.96 and 2227.72 rad s $^{-1}$, approaching surface-limit values of 2575.44 and 3306.98 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.800/6.321 h at 1 km inward to 0.0151/0.0200 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M3. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.46 GW190805_211137

GW190805_211137 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 87.187 M_\odot$ and $m_2 = 59.601 M_\odot$, while the luminosity-distance median is 6127.9 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	6,196
Observed luminosity distance	6127.9 Mpc (median)
Detector-frame masses	$m_1 = 87.187 M_\odot$, $m_2 = 59.601 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.684$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9206$; $z_\pi = 0.9966$ (medians)
Post-handoff reorder fraction	12.02%

Table 665: Observational and coordinate summary for GW190805_211137. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	43.942 [31.279, 59.832]	29.784 [18.586, 43.251]
Physical compact-source mass M_c ($M_\odot$)	37.753 [24.670, 54.510]	21.511 [11.849, 33.800]
Geometric mass-equivalent M_{geom} ($M_\odot$)	4.523 [0.439, 17.860]	7.596 [0.864, 17.897]
Geometric share M_{geom}/M_∞ (%)	10.29 [1.04, 37.86]	26.62 [3.29, 49.47]
Exterior spin magnitude $ \hat{a}_\infty $	0.800 [0.306, 0.979]	0.502 [0.052, 0.935]
Polarization factor A_{pol}	1.115 [1.011, 1.609]	1.363 [1.034, 1.979]
Compact-source spin parameter $\hat{a}_c$	0.993 [0.793, 1.000]	0.932 [0.203, 0.999]
GR/CPTG matching radius r_m (km)	64.886 [46.188, 88.350]	43.980 [27.445, 63.865]
Structural Mode N_{BH}	1.5075 [1.4538, 1.7682]	1.6435 [1.4656, 1.9062]
Organization-shell radius r_N (km)	34.040 [24.166, 46.195]	23.147 [14.439, 33.535]
Inner trapping-surface radius r_- (km)	50.475 [34.767, 72.781]	30.466 [18.200, 45.499]
Structural- N lapse $\bar{\alpha}_N$	0.8053 [0.6016, 0.8484]	0.6998 [0.4877, 0.8388]
Structural- N clock $T_{24,N}$ (h)	19.328 [14.438, 20.361]	16.796 [11.704, 20.132]
Structural- N axis ω (rad s $^{-1}$)	963.52 [525.40, 2158.30]	2094.91 [762.68, 4156.04]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1454.68 [856.63, 2528.73]	2585.09 [939.17, 4695.58]
Structural- N equatorial ω (rad s $^{-1}$)	2055.99 [1364.18, 3013.66]	3141.63 [943.83, 5261.93]
Structural- N period P_ω (ms)	4.3193 [2.4847, 7.3347]	2.4305 [1.3381, 6.6902]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2860.12 [1928.42, 4102.67]	4101.41 [1030.59, 6796.92]
Trapping-surface period P_- (ms)	2.1968 [1.5315, 3.2582]	1.5320 [0.9244, 6.0967]

Table 666: CPTG compact-state, structural, clock, and geometric-rotation results for GW190805_211137. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.1411	6.4496	2833.90	4004.33
500 m inward	2.9015	4.5460	2845.57	4056.12
100 m inward	1.2872	2.0264	2857.33	4093.29
10 m inward	0.4064	0.6404	2859.76	4100.51
1 m inward	0.1285	0.2025	2860.09	4101.32
10 cm inward	0.0406	0.0640	2860.12	4101.40
1 cm inward	0.0128	0.0203	2860.12	4101.41
surface limit	0.0000	0.0000	2860.12	4101.41

Table 667: Near-inner-boundary profile for GW190805_211137. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $37.753 M_\odot$ and $21.511 M_\odot$, with completed geometric mass-equivalent shares of 10.29% and 26.62%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $34.040 < 50.475 < 64.886$ km, and component 2 gives $23.147 < 30.466 < 43.980$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5075$ and 1.6435 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.328 and 16.796 h. The determinant-averaged frame-dragging rates at those shells are 1454.68 and 2585.09 rad s $^{-1}$, approaching surface-limit values of 2860.12 and 4101.41 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.141/6.450 h at 1 km inward to 0.0128/0.0203 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

5.47 GW230928_215827

GW230928_215827 is an M3 upper observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 95.803 M_\odot$ and $m_2 = 53.158 M_\odot$, while the luminosity-distance median is 4934.9 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M3: Upper observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	55,512
Observed luminosity distance	4934.9 Mpc (median)
Detector-frame masses	$m_1 = 95.803 M_\odot$, $m_2 = 53.158 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.555$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7712$; $z_\pi = 0.8335$ (medians)
Post-handoff reorder fraction	4.43%

Table 668: Observational and coordinate summary for GW230928_215827. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	51.927 [35.339, 70.902]	28.308 [17.455, 42.463]
Physical compact-source mass M_c ($M_\odot$)	44.631 [27.934, 63.976]	19.952 [10.995, 33.414]
Geometric mass-equivalent M_{geom} ($M_\odot$)	5.676 [0.725, 19.210]	7.682 [0.896, 17.224]
Geometric share M_{geom}/M_∞ (%)	11.10 [1.46, 35.99]	28.40 [3.46, 49.55]
Exterior spin magnitude $ \hat{a}_\infty $	0.784 [0.339, 0.971]	0.471 [0.048, 0.931]
Polarization factor A_{pol}	1.125 [1.015, 1.562]	1.397 [1.036, 1.982]
Compact-source spin parameter $\hat{a}_c$	0.992 [0.827, 1.000]	0.918 [0.188, 0.999]
GR/CPTG matching radius r_m (km)	76.677 [52.183, 104.696]	41.800 [25.774, 62.702]
Structural Mode N_{BH}	1.5130 [1.4559, 1.7460]	1.6617 [1.4665, 1.9071]
Organization-shell radius r_N (km)	40.186 [27.330, 54.880]	21.995 [13.561, 33.013]
Inner trapping-surface radius r_- (km)	59.542 [38.944, 85.214]	28.528 [16.830, 44.929]
Structural- N lapse $\bar{\alpha}_N$	0.8010 [0.6194, 0.8466]	0.6858 [0.4870, 0.8381]
Structural- N clock $T_{24,N}$ (h)	19.224 [14.867, 20.319]	16.459 [11.687, 20.115]
Structural- N axis ω (rad s $^{-1}$)	845.96 [461.22, 1922.45]	2249.37 [760.10, 4541.54]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1253.48 [751.75, 2265.01]	2730.13 [928.99, 5093.25]
Structural- N equatorial ω (rad s $^{-1}$)	1754.06 [1189.73, 2705.69]	3281.41 [932.56, 5674.62]
Structural- N period P_ω (ms)	5.0126 [2.7740, 8.3581]	2.3014 [1.2336, 6.7634]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2446.52 [1685.80, 3661.51]	4259.84 [1017.74, 7315.09]
Trapping-surface period P_- (ms)	2.5682 [1.7160, 3.7271]	1.4750 [0.8589, 6.1737]

Table 669: CPTG compact-state, structural, clock, and geometric-rotation results for GW230928_215827. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.8584	6.7367	2424.76	4145.84
500 m inward	2.7089	4.7529	2435.86	4206.75
100 m inward	1.2042	2.1215	2444.68	4249.82
10 m inward	0.3803	0.6704	2446.26	4258.95
1 m inward	0.1202	0.2120	2446.50	4259.68
10 cm inward	0.0380	0.0670	2446.52	4259.82
1 cm inward	0.0120	0.0212	2446.52	4259.84
surface limit	0.0000	0.0000	2446.52	4259.84

Table 670: Near-inner-boundary profile for GW230928_215827. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $44.631 M_\odot$ and $19.952 M_\odot$, with completed geometric mass-equivalent shares of 11.10% and 28.40%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $40.186 < 59.542 < 76.677$ km, and component 2 gives $21.995 < 28.528 < 41.800$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5130$ and 1.6617 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.224 and 16.459 h. The determinant-averaged frame-dragging rates at those shells are 1253.48 and 2730.13 rad s $^{-1}$, approaching surface-limit values of 2446.52 and 4259.84 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.858/6.737 h at 1 km inward to 0.0120/0.0212 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M3 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M3 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6 M4: High Observed-Mass Black-Hole Binaries

The fourth candidate class contains 30 unique event-level numerical streams with detector-frame total masses constructed from released component marginal medians spanning $153.138\text{--}198.821 M_\odot$. The class boundary was fixed from the observational distribution before any CPTG output was examined. Within M4, 1 system falls in the strongly asymmetric mass-ratio cross-cut, 19 are moderately asymmetric, and 10 are near-equal in the same observational screen. The distance cross-cuts contain 0 nearer, 11 intermediate-distance, and 19 farther-distance systems; 9 candidates carry the predeclared broad-posterior flag. Candidate subsections are ordered by detector-frame total mass from lower to higher values. Neither Structural Mode nor any compact-state, clock, or rotation quantity controls class membership or ordering.

As in the preceding candidate classes, each subsection begins with the observational state and then gives the two post-handoff compact components separately. 8 M4 events have two retained release labels that are numerically identical under the declared transformation; their labels are preserved together for provenance but their identical numerical stream is reported once. The tables expose the physical compact-source mass, completed geometric mass-equivalent contribution, GR/CPTG matching radius, Structural-Mode organization shell, inner trapping surface, proper-time accumulation, and geometric frame dragging for every M4 binary. The descriptive comparisons in the closing paragraph of each subsection are made only after the fixed calculation has been completed and do not feed back into the classification or solve.

6.1 GW240621_214041

GW240621_214041 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 88.350 M_\odot$ and $m_2 = 64.788 M_\odot$, while the luminosity-distance median is 6778.7 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,901
Observed luminosity distance	6778.7 Mpc (median)
Detector-frame masses	$m_1 = 88.350 M_\odot$, $m_2 = 64.788 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.733$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9984$; $z_\pi = 1.0831$ (medians)
Post-handoff reorder fraction	19.26%

Table 671: Observational and coordinate summary for GW240621_214041. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	41.755 [29.017, 59.224]	31.181 [19.842, 46.278]
Physical compact-source mass $M_c (M_\odot)$	31.802 [20.318, 49.622]	20.482 [12.328, 32.898]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	8.136 [0.839, 22.733]	10.539 [1.522, 20.472]
Geometric share M_{geom}/M_∞ (%)	19.92 [2.17, 48.14]	35.49 [5.56, 49.84]
Exterior spin magnitude $ \hat{a}_\infty $	0.621 [0.100, 0.957]	0.347 [0.029, 0.890]
Polarization factor A_{pol}	1.249 [1.022, 1.928]	1.550 [1.059, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.969 [0.372, 1.000]	0.835 [0.114, 0.998]
GR/CPTG matching radius r_m (km)	61.656 [42.847, 87.452]	46.042 [29.300, 68.335]
Structural Mode N_{BH}	1.5808 [1.4596, 1.8909]	1.7403 [1.4782, 1.9104]
Organization-shell radius r_N (km)	32.444 [22.466, 45.968]	24.247 [15.435, 35.969]
Inner trapping-surface radius r_- (km)	43.845 [30.112, 66.226]	30.206 [19.026, 45.196]
Structural- N lapse $\bar{\alpha}_N$	0.7481 [0.4999, 0.8437]	0.6240 [0.4845, 0.8287]
Structural- N clock $T_{24,N}$ (h)	17.955 [11.998, 20.249]	14.977 [11.627, 19.888]
Structural- N axis ω (rad s $^{-1}$)	1361.64 [569.44, 2642.89]	2148.05 [484.15, 4058.63]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1786.98 [865.11, 3002.71]	2526.09 [484.52, 4533.51]
Structural- N equatorial ω (rad s $^{-1}$)	2265.23 [1116.77, 3407.44]	2936.43 [484.70, 5018.11]
Structural- N period P_ω (ms)	3.5161 [2.0925, 7.2628]	2.4873 [1.3859, 12.9680]
Trapping-surface limit Ω_- (rad s $^{-1}$)	3025.52 [1265.41, 4510.92]	3734.79 [528.43, 6466.73]
Trapping-surface period P_- (ms)	2.0767 [1.3929, 4.9653]	1.6823 [0.9716, 11.8903]

Table 672: CPTG compact-state, structural, clock, and geometric-rotation results for GW240621_214041. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0828	6.7788	2978.71	3637.40
500 m inward	3.5786	4.7940	3000.86	3687.37
100 m inward	1.5945	2.1427	3020.54	3725.75
10 m inward	0.5038	0.6776	3025.04	3733.88
1 m inward	0.1593	0.2143	3025.48	3734.68
10 cm inward	0.0504	0.0678	3025.52	3734.78
1 cm inward	0.0159	0.0214	3025.52	3734.79
surface limit	0.0000	0.0000	3025.52	3734.79

Table 673: Near-inner-boundary profile for GW240621_214041. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $31.802 M_\odot$ and $20.482 M_\odot$, with completed geometric mass-equivalent shares of 19.92% and 35.49%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $32.444 < 43.845 < 61.656$ km, and component 2 gives $24.247 < 30.206 < 46.042$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5808$ and 1.7403 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.955 and 14.977 h. The determinant-averaged frame-dragging rates at those shells are 1786.98 and 2526.09 rad s $^{-1}$, approaching surface-limit values of 3025.52 and 3734.79 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.083/6.779 h at 1 km inward to 0.0159/0.0214 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.2 GW240601_061200

GW240601_061200 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 94.484 M_\odot$ and $m_2 = 59.503 M_\odot$, while the luminosity-distance median is 5086.2 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	19,639
Observed luminosity distance	5086.2 Mpc (median)
Detector-frame masses	$m_1 = 94.484 M_\odot$, $m_2 = 59.503 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.630$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7904$; $z_\pi = 0.8546$ (medians)
Post-handoff reorder fraction	13.18%

Table 674: Observational and coordinate summary for GW240601_061200. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	49.685 [32.239, 70.966]	31.814 [20.995, 47.788]
Physical compact-source mass M_c ($M_\odot$)	35.927 [23.162, 54.635]	21.883 [13.161, 33.628]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.989 [1.144, 29.930]	9.908 [1.188, 21.167]
Geometric share M_{geom}/M_∞ (%)	24.67 [2.79, 48.99]	32.52 [4.20, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.536 [0.072, 0.945]	0.399 [0.035, 0.917]
Polarization factor A_{pol}	1.327 [1.029, 1.961]	1.482 [1.044, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.945 [0.278, 1.000]	0.876 [0.138, 0.999]
GR/CPTG matching radius r_m (km)	73.367 [47.605, 104.790]	46.978 [31.002, 70.564]
Structural Mode N_{BH}	1.6242 [1.4629, 1.9008]	1.7063 [1.4705, 1.9095]
Organization-shell radius r_N (km)	38.647 [24.999, 55.238]	24.730 [16.321, 37.183]
Inner trapping-surface radius r_- (km)	50.894 [33.953, 73.309]	31.593 [20.371, 46.558]
Structural- N lapse $\bar{\alpha}_N$	0.7147 [0.4920, 0.8410]	0.6509 [0.4851, 0.8348]
Structural- N clock $T_{24,N}$ (h)	17.152 [11.807, 20.184]	15.623 [11.643, 20.036]
Structural- N axis ω (rad s $^{-1}$)	1257.02 [506.38, 2312.19]	2024.60 [546.30, 3800.26]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1576.87 [666.94, 2633.14]	2451.05 [548.68, 4260.39]
Structural- N equatorial ω (rad s $^{-1}$)	1930.64 [692.21, 3027.77]	2921.69 [549.10, 4751.02]
Structural- N period P_ω (ms)	3.9846 [2.3862, 9.4209]	2.5635 [1.4748, 11.4516]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2540.17 [757.38, 4036.45]	3789.34 [598.50, 6123.13]
Trapping-surface period P_- (ms)	2.4735 [1.5566, 8.2959]	1.6581 [1.0261, 10.4982]

Table 675: CPTG compact-state, structural, clock, and geometric-rotation results for GW240601_061200. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.9023	6.5081	2505.52	3697.79
500 m inward	3.4585	4.5972	2524.41	3745.59
100 m inward	1.5432	2.0539	2537.31	3780.07
10 m inward	0.4877	0.6494	2539.77	3788.37
1 m inward	0.1542	0.2054	2540.15	3789.24
10 cm inward	0.0488	0.0649	2540.17	3789.33
1 cm inward	0.0154	0.0205	2540.17	3789.34
surface limit	0.0000	0.0000	2540.17	3789.34

Table 676: Near-inner-boundary profile for GW240601_061200. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $35.927 M_\odot$ and $21.883 M_\odot$, with completed geometric mass-equivalent shares of 24.67% and 32.52%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $38.647 < 50.894 < 73.367$ km, and component 2 gives $24.730 < 31.593 < 46.978$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6242$ and 1.7063 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.152 and 15.623 h. The determinant-averaged frame-dragging rates at those shells are 1576.87 and 2451.05 rad s $^{-1}$, approaching surface-limit values of 2540.17 and 3789.34 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.902/6.508 h at 1 km inward to 0.0154/0.0205 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M4 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.3 GW190519_153544

GW190519_153544 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 94.798 M_\odot$ and $m_2 = 59.880 M_\odot$, while the luminosity-distance median is 2601.7 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	11,339
Observed luminosity distance	2601.7 Mpc (median)
Detector-frame masses	$m_1 = 94.798 M_\odot$, $m_2 = 59.880 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.632$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4547$; $z_\pi = 0.4887$ (medians)
Post-handoff reorder fraction	10.66%

Table 677: Observational and coordinate summary for GW190519_153544. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	63.283 [44.401, 74.364]	40.199 [27.465, 59.456]
Physical compact-source mass $M_c (M_\odot)$	50.180 [35.615, 65.382]	29.645 [17.111, 42.614]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	10.830 [1.162, 26.697]	10.448 [1.348, 25.111]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	17.39 [2.11, 40.97]	26.90 [3.70, 49.47]
Exterior spin magnitude $ \hat{a}_\infty $	0.667 [0.251, 0.958]	0.497 [0.052, 0.927]
Polarization factor A_{pol}	1.210 [1.022, 1.694]	1.368 [1.038, 1.979]
Compact-source spin parameter $\hat{a}_c$	0.978 [0.720, 1.000]	0.930 [0.204, 0.999]
GR/CPTG matching radius r_m (km)	93.445 [65.563, 109.808]	59.360 [40.556, 87.794]
Structural Mode N_{BH}	1.5597 [1.4593, 1.8054]	1.6463 [1.4678, 1.9062]
Organization-shell radius r_N (km)	49.163 [34.295, 57.860]	31.219 [21.303, 46.318]
Inner trapping-surface radius r_- (km)	67.429 [51.707, 87.120]	41.393 [26.353, 57.902]
Structural- N lapse $\bar{\alpha}_N$	0.7645 [0.5710, 0.8439]	0.6977 [0.4877, 0.8371]
Structural- N clock $T_{24,N}$ (h)	18.348 [13.704, 20.254]	16.744 [11.705, 20.089]
Structural- N axis ω (rad s $^{-1}$)	894.39 [449.86, 1540.01]	1539.18 [609.93, 2927.74]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1230.01 [727.64, 1737.20]	1906.58 [680.51, 3273.84]
Structural- N equatorial ω (rad s $^{-1}$)	1569.20 [1148.00, 2024.18]	2338.84 [681.70, 3621.88]
Structural- N period $P_{\bar{\omega}}$ (ms)	5.1082 [3.6169, 8.6350]	3.2955 [1.9192, 9.2331]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2095.98 [1592.19, 2781.73]	3093.19 [745.18, 4688.11]
Trapping-surface period P_- (ms)	2.9977 [2.2587, 3.9462]	2.0313 [1.3402, 8.4317]

Table 678: CPTG compact-state, structural, clock, and geometric-rotation results for GW190519_153544. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.9723	5.4934	2071.63	3040.24
500 m inward	2.7979	3.8770	2084.19	3067.50
100 m inward	1.2478	1.7309	2093.86	3088.48
10 m inward	0.3943	0.5471	2095.85	3092.96
1 m inward	0.1247	0.1730	2095.95	3093.19
10 cm inward	0.0394	0.0547	2095.97	3093.19
1 cm inward	0.0125	0.0173	2095.98	3093.19
surface limit	0.0000	0.0000	2095.98	3093.19

Table 679: Near-inner-boundary profile for GW190519_153544. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $50.180 M_\odot$ and $29.645 M_\odot$, with completed geometric mass-equivalent shares of 17.39% and 26.90%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $49.163 < 67.429 < 93.445$ km, and component 2 gives $31.219 < 41.393 < 59.360$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5597$ and 1.6463; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.348 and 16.744 h. The determinant-averaged frame-dragging rates at those shells are 1230.01 and 1906.58 rad s $^{-1}$, approaching surface-limit values of 2095.98 and 3093.19 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.972/5.493 h at 1 km inward to 0.0125/0.0173 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M4 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M4 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.4 GW230824_033047

GW230824_033047 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 91.180 M_\odot$ and $m_2 = 64.529 M_\odot$, while the luminosity-distance median is 4726.9 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	40,809
Observed luminosity distance	4726.9 Mpc (median)
Detector-frame masses	$m_1 = 91.180 M_\odot$, $m_2 = 64.529 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.708$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7446$; $z_\pi = 0.8044$ (medians)
Post-handoff reorder fraction	18.23%

Table 680: Observational and coordinate summary for GW230824_033047. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	49.803 [35.436, 66.883]	35.722 [22.264, 51.917]
Physical compact-source mass M_c ($M_\odot$)	35.348 [24.144, 54.911]	23.107 [14.026, 35.069]
Geometric mass-equivalent M_{geom} ($M_\odot$)	12.604 [1.467, 27.593]	12.627 [1.895, 23.576]
Geometric share M_{geom}/M_∞ (%)	25.87 [3.21, 49.20]	37.04 [6.39, 49.85]
Exterior spin magnitude $ \hat{a}_\infty $	0.515 [0.064, 0.936]	0.320 [0.028, 0.874]
Polarization factor A_{pol}	1.349 [1.033, 1.969]	1.588 [1.068, 1.994]
Compact-source spin parameter $\hat{a}_c$	0.937 [0.249, 0.999]	0.809 [0.110, 0.998]
GR/CPTG matching radius r_m (km)	73.541 [52.326, 98.761]	52.749 [32.875, 76.662]
Structural Mode N_{BH}	1.6360 [1.4651, 1.9032]	1.7584 [1.4831, 1.9105]
Organization-shell radius r_N (km)	38.725 [27.483, 51.968]	27.764 [17.308, 40.344]
Inner trapping-surface radius r_- (km)	49.942 [36.400, 73.059]	34.251 [21.468, 49.035]
Structural- N lapse $\bar{\alpha}_N$	0.7056 [0.4901, 0.8392]	0.6094 [0.4843, 0.8248]
Structural- N clock $T_{24,N}$ (h)	16.935 [11.762, 20.141]	14.626 [11.624, 19.795]
Structural- N axis ω (rad s $^{-1}$)	1308.02 [479.63, 2230.41]	1937.01 [395.36, 3570.42]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1614.35 [636.91, 2512.14]	2253.53 [395.61, 4003.64]
Structural- N equatorial ω (rad s $^{-1}$)	1943.95 [658.63, 2832.88]	2597.24 [395.72, 4470.91]
Structural- N period P_ω (ms)	3.8921 [2.5011, 9.8651]	2.7882 [1.5694, 15.8824]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2544.65 [719.93, 3734.00]	3266.42 [431.36, 5769.06]
Trapping-surface period P_- (ms)	2.4692 [1.6827, 8.7274]	1.9236 [1.0891, 14.5661]

Table 681: CPTG compact-state, structural, clock, and geometric-rotation results for GW230824_033047. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0318	6.3992	2505.78	3188.94
500 m inward	3.5501	4.5269	2525.89	3229.11
100 m inward	1.5849	2.0243	2540.93	3259.67
10 m inward	0.5009	0.6402	2544.16	3265.66
1 m inward	0.1584	0.2024	2544.59	3266.34
10 cm inward	0.0501	0.0640	2544.64	3266.41
1 cm inward	0.0158	0.0202	2544.65	3266.42
surface limit	0.0000	0.0000	2544.65	3266.42

Table 682: Near-inner-boundary profile for GW230824_033047. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $35.348 M_\odot$ and $23.107 M_\odot$, with completed geometric mass-equivalent shares of 25.87% and 37.04%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $38.725 < 49.942 < 73.541$ km, and component 2 gives $27.764 < 34.251 < 52.749$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6360$ and 1.7584 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.935 and 14.626 h. The determinant-averaged frame-dragging rates at those shells are 1614.35 and 2253.53 rad s $^{-1}$, approaching surface-limit values of 2544.65 and 3266.42 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.032/6.399 h at 1 km inward to 0.0158/0.0202 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.5 GW230630_125806

GW230630_125806 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 93.755 M_\odot$ and $m_2 = 62.236 M_\odot$, while the luminosity-distance median is 5406.3 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	41,166
Observed luminosity distance	5406.3 Mpc (median)
Detector-frame masses	$m_1 = 93.755 M_\odot$, $m_2 = 62.236 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.664$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8308$; $z_\pi = 0.8987$ (medians)
Post-handoff reorder fraction	14.72%

Table 683: Observational and coordinate summary for GW230630_125806. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	48.620 [32.460, 69.620]	32.175 [19.176, 50.454]
Physical compact-source mass M_c ($M_\odot$)	37.729 [23.993, 61.080]	22.364 [12.196, 36.544]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.870 [0.806, 25.571]	9.415 [1.129, 21.638]
Geometric share M_{geom}/M_∞ (%)	16.69 [1.75, 47.58]	31.08 [4.06, 49.70]
Exterior spin magnitude $ \hat{a}_\infty $	0.680 [0.115, 0.965]	0.424 [0.039, 0.920]
Polarization factor A_{pol}	1.200 [1.018, 1.908]	1.451 [1.042, 1.988]
Compact-source spin parameter $\hat{a}_c$	0.980 [0.420, 1.000]	0.893 [0.153, 0.999]
GR/CPTG matching radius r_m (km)	71.794 [47.932, 102.803]	47.511 [28.315, 74.502]
Structural Mode N_{BH}	1.5541 [1.4574, 1.8844]	1.6904 [1.4698, 1.9089]
Organization-shell radius r_N (km)	37.732 [25.142, 53.952]	25.002 [14.909, 39.175]
Inner trapping-surface radius r_- (km)	51.779 [34.792, 81.334]	32.136 [18.583, 50.036]
Structural- N lapse $\bar{\alpha}_N$	0.7688 [0.5052, 0.8454]	0.6634 [0.4856, 0.8355]
Structural- N clock $T_{24,N}$ (h)	18.452 [12.126, 20.290]	15.922 [11.655, 20.051]
Structural- N axis ω (rad s $^{-1}$)	1091.50 [459.63, 2229.74]	1999.47 [602.73, 4076.73]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1488.90 [710.97, 2569.20]	2411.56 [624.85, 4586.83]
Structural- N equatorial ω (rad s $^{-1}$)	1933.49 [1010.91, 2981.59]	2878.57 [625.64, 5141.87]
Structural- N period P_ω (ms)	4.2200 [2.4456, 8.8375]	2.6054 [1.3698, 10.0555]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2606.01 [1243.52, 4006.26]	3722.39 [683.12, 6662.73]
Trapping-surface period P_- (ms)	2.4110 [1.5683, 5.0527]	1.6879 [0.9430, 9.1978]

Table 684: CPTG compact-state, structural, clock, and geometric-rotation results for GW230630_125806. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.5213	6.3923	2574.29	3634.55
500 m inward	3.1799	4.5139	2591.35	3680.51
100 m inward	1.4160	2.0153	2602.86	3714.51
10 m inward	0.4473	0.6371	2605.61	3721.76
1 m inward	0.1415	0.2015	2605.95	3722.33
10 cm inward	0.0447	0.0637	2606.00	3722.38
1 cm inward	0.0141	0.0201	2606.01	3722.39
surface limit	0.0000	0.0000	2606.01	3722.39

Table 685: Near-inner-boundary profile for GW230630_125806. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $37.729 M_\odot$ and $22.364 M_\odot$, with completed geometric mass-equivalent shares of 16.69% and 31.08%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $37.732 < 51.779 < 71.794$ km, and component 2 gives $25.002 < 32.136 < 47.511$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5541$ and 1.6904 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.452 and 15.922 h. The determinant-averaged frame-dragging rates at those shells are 1488.90 and 2411.56 rad s $^{-1}$, approaching surface-limit values of 2606.01 and 3722.39 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.521/6.392 h at 1 km inward to 0.0141/0.0201 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. At least one Structural-Mode shell retains comparatively more proper time than most M4 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.6 GW231113_150041

GW231113_150041 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 102.851 M_\odot$ and $m_2 = 54.929 M_\odot$, while the luminosity-distance median is 5150.2 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,941
Observed luminosity distance	5150.2 Mpc (median)
Detector-frame masses	$m_1 = 102.851 M_\odot$, $m_2 = 54.929 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.534$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7986$; $z_\pi = 0.8634$ (medians)
Post-handoff reorder fraction	9.29%

Table 686: Observational and coordinate summary for GW231113_150041. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	53.963 [30.577, 91.186]	28.560 [13.099, 48.155]
Physical compact-source mass M_c ($M_\odot$)	42.617 [22.444, 82.683]	19.587 [8.893, 35.397]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.666 [0.930, 24.919]	7.823 [0.860, 20.210]
Geometric share M_{geom}/M_∞ (%)	14.16 [1.69, 46.86]	30.69 [3.73, 49.69]
Exterior spin magnitude $ \hat{a}_\infty $	0.727 [0.133, 0.966]	0.431 [0.040, 0.926]
Polarization factor A_{pol}	1.165 [1.017, 1.882]	1.443 [1.039, 1.988]
Compact-source spin parameter $\hat{a}_c$	0.986 [0.471, 1.000]	0.897 [0.158, 0.999]
GR/CPTG matching radius r_m (km)	79.682 [45.150, 134.648]	42.173 [19.342, 71.107]
Structural Mode N_{BH}	1.5347 [1.4571, 1.8760]	1.6862 [1.4679, 1.9087]
Organization-shell radius r_N (km)	41.878 [23.746, 70.382]	22.185 [10.163, 37.403]
Inner trapping-surface radius r_- (km)	57.910 [32.575, 109.846]	28.341 [13.043, 48.425]
Structural- N lapse $\bar{\alpha}_N$	0.7839 [0.5121, 0.8457]	0.6667 [0.4858, 0.8369]
Structural- N clock $T_{24,N}$ (h)	18.814 [12.291, 20.297]	16.002 [11.658, 20.086]
Structural- N axis ω (rad s $^{-1}$)	909.22 [358.66, 2352.12]	2253.55 [648.78, 5506.49]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1274.39 [565.89, 2708.43]	2716.85 [695.17, 6342.23]
Structural- N equatorial ω (rad s $^{-1}$)	1688.73 [846.69, 3157.24]	3218.18 [695.53, 7301.23]
Structural- N period P_ω (ms)	4.9304 [2.3199, 11.1032]	2.3127 [0.9907, 9.0384]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2271.82 [1142.39, 4243.75]	4161.58 [759.63, 9478.87]
Trapping-surface period P_- (ms)	2.7657 [1.4806, 5.5000]	1.5098 [0.6629, 8.2714]

Table 687: CPTG compact-state, structural, clock, and geometric-rotation results for GW231113_150041. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.0909	6.8018	2245.76	4044.61
500 m inward	2.8710	4.7949	2260.01	4106.25
100 m inward	1.2779	2.1398	2269.82	4153.06
10 m inward	0.4035	0.6765	2271.73	4160.92
1 m inward	0.1276	0.2139	2271.81	4161.48
10 cm inward	0.0403	0.0676	2271.82	4161.57
1 cm inward	0.0128	0.0214	2271.82	4161.58
surface limit	0.0000	0.0000	2271.82	4161.58

Table 688: Near-inner-boundary profile for GW231113_150041. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $42.617 M_\odot$ and $19.587 M_\odot$, with completed geometric mass-equivalent shares of 14.16% and 30.69%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $41.878 < 57.910 < 79.682$ km, and component 2 gives $22.185 < 28.341 < 42.173$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5347$ and 1.6862 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.814 and 16.002 h. The determinant-averaged frame-dragging rates at those shells are 1274.39 and 2716.85 rad s $^{-1}$, approaching surface-limit values of 2271.82 and 4161.58 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.091/6.802 h at 1 km inward to 0.0128/0.0214 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. At least one Structural-Mode shell retains comparatively more proper time than most M4 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.7 GW230806_204041

GW230806_204041 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 93.205 M_\odot$ and $m_2 = 66.075 M_\odot$, while the luminosity-distance median is 5501.4 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	39,570
Observed luminosity distance	5501.4 Mpc (median)
Detector-frame masses	$m_1 = 93.205 M_\odot$, $m_2 = 66.075 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.709$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8427$; $z_\pi = 0.9118$ (medians)
Post-handoff reorder fraction	18.75%

Table 689: Observational and coordinate summary for GW230806_204041. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	47.810 [33.568, 66.312]	34.773 [21.680, 51.505]
Physical compact-source mass $M_c (M_\odot)$	34.919 [23.405, 53.614]	22.731 [13.532, 35.589]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	11.237 [1.294, 26.937]	11.779 [1.680, 23.026]
Geometric share M_{geom}/M_∞ (%)	23.98 [2.96, 48.91]	36.02 [5.72, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.548 [0.075, 0.941]	0.338 [0.031, 0.887]
Polarization factor A_{pol}	1.315 [1.031, 1.957]	1.563 [1.061, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.949 [0.289, 1.000]	0.827 [0.123, 0.998]
GR/CPTG matching radius r_m (km)	70.598 [49.567, 97.919]	51.347 [32.013, 76.054]
Structural Mode N_{BH}	1.6176 [1.4638, 1.8998]	1.7464 [1.4791, 1.9101]
Organization-shell radius r_N (km)	37.169 [26.010, 51.560]	27.024 [16.843, 40.083]
Inner trapping-surface radius r_- (km)	48.881 [34.893, 71.591]	33.576 [20.886, 49.441]
Structural- N lapse $\bar{\alpha}_N$	0.7198 [0.4927, 0.8403]	0.6192 [0.4847, 0.8279]
Structural- N clock $T_{24,N}$ (h)	17.274 [11.825, 20.167]	14.860 [11.633, 19.870]
Structural- N axis ω (rad s $^{-1}$)	1303.08 [516.67, 2308.87]	1954.85 [457.47, 3671.19]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1643.13 [719.02, 2607.38]	2297.80 [458.70, 4101.98]
Structural- N equatorial ω (rad s $^{-1}$)	2016.62 [795.72, 2966.07]	2675.63 [458.80, 4558.07]
Structural- N period P_ω (ms)	3.8239 [2.4098, 8.7386]	2.7344 [1.5317, 13.6979]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2658.36 [872.15, 3920.82]	3385.57 [499.94, 5876.72]
Trapping-surface period P_- (ms)	2.3636 [1.6025, 7.2043]	1.8559 [1.0692, 12.5680]

Table 690: CPTG compact-state, structural, clock, and geometric-rotation results for GW230806_204041. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.9936	6.4289	2619.92	3299.84
500 m inward	3.5223	4.5467	2640.26	3344.40
100 m inward	1.5719	2.0325	2654.84	3377.63
10 m inward	0.4968	0.6428	2658.05	3384.78
1 m inward	0.1571	0.2032	2658.31	3385.52
10 cm inward	0.0497	0.0643	2658.35	3385.56
1 cm inward	0.0157	0.0203	2658.36	3385.57
surface limit	0.0000	0.0000	2658.36	3385.57

Table 691: Near-inner-boundary profile for GW230806_204041. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $34.919 M_\odot$ and $22.731 M_\odot$, with completed geometric mass-equivalent shares of 23.98% and 36.02%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $37.169 < 48.881 < 70.598$ km, and component 2 gives $27.024 < 33.576 < 51.347$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6176$ and 1.7464 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.274 and 14.860 h. The determinant-averaged frame-dragging rates at those shells are 1643.13 and 2297.80 rad s $^{-1}$, approaching surface-limit values of 2658.36 and 3385.57 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.994/6.429 h at 1 km inward to 0.0157/0.0203 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.8 GW230820_212515

GW230820_212515 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 102.762 M_\odot$ and $m_2 = 58.821 M_\odot$, while the luminosity-distance median is 4037.7 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	31,622
Observed luminosity distance	4037.7 Mpc (median)
Detector-frame masses	$m_1 = 102.762 M_\odot$, $m_2 = 58.821 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.572$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6543$; $z_\pi = 0.7059$ (medians)
Post-handoff reorder fraction	12.14%

Table 692: Observational and coordinate summary for GW230820_212515. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	59.424 [40.584, 82.908]	33.446 [16.086, 54.595]
Physical compact-source mass $M_c (M_\odot)$	44.134 [29.200, 68.688]	23.170 [10.456, 39.653]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	12.503 [1.286, 33.642]	8.744 [0.976, 23.453]
Geometric share M_{geom}/M_∞ (%)	21.45 [2.42, 48.73]	29.86 [3.60, 49.64]
Exterior spin magnitude $ \hat{a}_\infty $	0.594 [0.082, 0.952]	0.445 [0.043, 0.929]
Polarization factor A_{pol}	1.273 [1.025, 1.951]	1.426 [1.037, 1.986]
Compact-source spin parameter $\hat{a}_c$	0.962 [0.310, 1.000]	0.905 [0.170, 0.999]
GR/CPTG matching radius r_m (km)	87.746 [59.927, 122.424]	49.387 [23.753, 80.617]
Structural Mode N_{BH}	1.5943 [1.4609, 1.8978]	1.6772 [1.4672, 1.9081]
Organization-shell radius r_N (km)	46.161 [31.398, 64.506]	25.972 [12.494, 42.463]
Inner trapping-surface radius r_- (km)	61.560 [43.054, 91.258]	33.609 [15.679, 54.082]
Structural- N lapse $\bar{\alpha}_N$	0.7378 [0.4944, 0.8426]	0.6738 [0.4862, 0.8375]
Structural- N clock $T_{24,N}$ (h)	17.706 [11.864, 20.223]	16.170 [11.669, 20.101]
Structural- N axis ω (rad s $^{-1}$)	990.85 [402.08, 1831.54]	1906.43 [595.78, 4783.09]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1282.62 [567.38, 2088.69]	2308.30 [636.50, 5393.35]
Structural- N equatorial ω (rad s $^{-1}$)	1605.16 [662.21, 2409.22]	2747.57 [636.76, 6067.83]
Structural- N period $P_{\bar{\omega}}$ (ms)	4.8987 [3.0082, 11.0740]	2.7220 [1.1650, 9.8715]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2135.50 [729.29, 3212.75]	3563.47 [694.55, 7845.03]
Trapping-surface period P_- (ms)	2.9423 [1.9557, 8.6155]	1.7632 [0.8009, 9.0464]

Table 693: CPTG compact-state, structural, clock, and geometric-rotation results for GW230820_212515. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.3682	6.2020	2112.55	3482.90
500 m inward	3.0792	4.3762	2123.86	3526.38
100 m inward	1.3737	1.9534	2133.12	3556.71
10 m inward	0.4341	0.6174	2135.27	3563.04
1 m inward	0.1373	0.1952	2135.48	3563.43
10 cm inward	0.0434	0.0617	2135.50	3563.46
1 cm inward	0.0137	0.0195	2135.50	3563.47
surface limit	0.0000	0.0000	2135.50	3563.47

Table 694: Near-inner-boundary profile for GW230820_212515. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $44.134 M_\odot$ and $23.170 M_\odot$, with completed geometric mass-equivalent shares of 21.45% and 29.86%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $46.161 < 61.560 < 87.746$ km, and component 2 gives $25.972 < 33.609 < 49.387$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5943$ and 1.6772; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.706 and 16.170 h. The determinant-averaged frame-dragging rates at those shells are 1282.62 and 2308.30 rad s $^{-1}$, approaching surface-limit values of 2135.50 and 3563.47 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.368/6.202 h at 1 km inward to 0.0137/0.0195 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M4 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.9 GW230708_230935

GW230708_230935 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 101.014 M_\odot$ and $m_2 = 63.098 M_\odot$, while the luminosity-distance median is 3526.8 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	41,226
Observed luminosity distance	3526.8 Mpc (median)
Detector-frame masses	$m_1 = 101.014 M_\odot$, $m_2 = 63.098 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.625$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5852$; $z_\pi = 0.6308$ (medians)
Post-handoff reorder fraction	13.52%

Table 695: Observational and coordinate summary for GW230708_230935. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	61.357 [42.457, 82.033]	38.800 [23.829, 58.121]
Physical compact-source mass $M_c (M_\odot)$	42.286 [29.343, 66.787]	26.122 [15.155, 39.905]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	16.525 [1.817, 34.442]	12.474 [1.598, 26.250]
Geometric share M_{geom}/M_∞ (%)	27.56 [3.36, 49.41]	34.29 [4.86, 49.79]
Exterior spin magnitude $ \hat{a}_\infty $	0.485 [0.055, 0.933]	0.368 [0.032, 0.904]
Polarization factor A_{pol}	1.380 [1.035, 1.977]	1.522 [1.051, 1.992]
Compact-source spin parameter $\hat{a}_c$	0.925 [0.214, 0.999]	0.853 [0.129, 0.999]
GR/CPTG matching radius r_m (km)	90.602 [62.693, 121.132]	57.293 [35.187, 85.823]
Structural Mode N_{BH}	1.6530 [1.4659, 1.9056]	1.7264 [1.4742, 1.9099]
Organization-shell radius r_N (km)	47.702 [32.945, 63.781]	30.166 [18.513, 45.175]
Inner trapping-surface radius r_- (km)	60.425 [44.221, 88.920]	38.142 [23.117, 55.636]
Structural- N lapse $\bar{\alpha}_N$	0.6925 [0.4882, 0.8386]	0.6351 [0.4848, 0.8319]
Structural- N clock $T_{24,N}$ (h)	16.619 [11.716, 20.126]	15.242 [11.636, 19.965]
Structural- N axis ω (rad s $^{-1}$)	1072.45 [372.74, 1811.92]	1727.50 [423.80, 3306.89]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1315.19 [451.30, 2047.64]	2049.61 [424.26, 3709.47]
Structural- N equatorial ω (rad s $^{-1}$)	1579.59 [459.82, 2337.02]	2408.46 [424.43, 4151.03]
Structural- N period $P_{\bar{\omega}}$ (ms)	4.7774 [3.0685, 13.9223]	3.0656 [1.6938, 14.8097]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2050.25 [501.89, 3094.71]	3092.31 [462.90, 5352.46]
Trapping-surface period P_- (ms)	3.0646 [2.0303, 12.5191]	2.0319 [1.1739, 13.5736]

Table 696: CPTG compact-state, structural, clock, and geometric-rotation results for GW230708_230935. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.6231	5.9777	2025.38	3029.78
500 m inward	3.2651	4.2241	2038.58	3062.36
100 m inward	1.4586	1.8882	2048.18	3086.23
10 m inward	0.4611	0.5970	2050.07	3091.66
1 m inward	0.1458	0.1888	2050.21	3092.25
10 cm inward	0.0461	0.0597	2050.24	3092.31
1 cm inward	0.0146	0.0189	2050.25	3092.31
surface limit	0.0000	0.0000	2050.25	3092.31

Table 697: Near-inner-boundary profile for GW230708_230935. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $42.286 M_\odot$ and $26.122 M_\odot$, with completed geometric mass-equivalent shares of 27.56% and 34.29%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $47.702 < 60.425 < 90.602$ km, and component 2 gives $30.166 < 38.142 < 57.293$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6530$ and 1.7264 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.619 and 15.242 h. The determinant-averaged frame-dragging rates at those shells are 1315.19 and 2049.61 rad s $^{-1}$, approaching surface-limit values of 2050.25 and 3092.31 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.623/5.978 h at 1 km inward to 0.0146/0.0189 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M4 descriptive distribution, making it a useful representative high-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.10 GW231119_075248

GW231119_075248 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 96.664 M_\odot$ and $m_2 = 67.468 M_\odot$, while the luminosity-distance median is 6680.2 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	37,473
Observed luminosity distance	6680.2 Mpc (median)
Detector-frame masses	$m_1 = 96.664 M_\odot$, $m_2 = 67.468 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.698$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9867$; $z_\pi = 1.0701$ (medians)
Post-handoff reorder fraction	18.13%

Table 698: Observational and coordinate summary for GW231119_075248. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	45.807 [31.079, 68.768]	32.846 [19.643, 50.611]
Physical compact-source mass $M_c (M_\odot)$	33.505 [21.637, 54.014]	21.534 [12.541, 34.512]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	10.948 [1.181, 27.460]	11.129 [1.477, 22.487]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	24.41 [2.82, 49.02]	35.96 [5.36, 49.81]
Exterior spin magnitude $ \hat{a}_\infty $	0.541 [0.072, 0.944]	0.339 [0.031, 0.894]
Polarization factor A_{pol}	1.323 [1.029, 1.961]	1.562 [1.057, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.946 [0.275, 1.000]	0.827 [0.122, 0.998]
GR/CPTG matching radius r_m (km)	67.639 [45.893, 101.544]	48.502 [29.005, 74.734]
Structural Mode N_{BH}	1.6218 [1.4630, 1.9010]	1.7458 [1.4771, 1.9101]
Organization-shell radius r_N (km)	35.608 [24.137, 53.492]	25.530 [15.265, 39.320]
Inner trapping-surface radius r_- (km)	46.975 [32.132, 72.816]	31.715 [19.121, 48.451]
Structural- N lapse $\bar{\alpha}_N$	0.7166 [0.4918, 0.8409]	0.6196 [0.4847, 0.8296]
Structural- N clock $T_{24,N}$ (h)	17.198 [11.802, 20.182]	14.871 [11.632, 19.910]
Structural- N axis ω (rad s $^{-1}$)	1338.92 [503.60, 2494.91]	2066.65 [471.88, 3986.04]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1690.70 [687.38, 2826.61]	2422.78 [472.26, 4494.53]
Structural- N equatorial ω (rad s $^{-1}$)	2084.75 [741.60, 3206.75]	2809.60 [472.47, 5017.83]
Structural- N period P_ω (ms)	3.7163 [2.2229, 9.1408]	2.5934 [1.3980, 13.3044]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2751.82 [817.45, 4235.14]	3548.56 [515.05, 6499.52]
Trapping-surface period P_- (ms)	2.2833 [1.4836, 7.6863]	1.7706 [0.9667, 12.1993]

Table 699: CPTG compact-state, structural, clock, and geometric-rotation results for GW231119_075248. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	5.0925	6.5894	2710.39	3455.34
500 m inward	3.5903	4.6579	2731.13	3503.54
100 m inward	1.6021	2.0822	2747.23	3540.25
10 m inward	0.5063	0.6583	2751.32	3547.75
1 m inward	0.1601	0.2082	2751.78	3548.48
10 cm inward	0.0506	0.0658	2751.82	3548.55
1 cm inward	0.0160	0.0208	2751.82	3548.56
surface limit	0.0000	0.0000	2751.82	3548.56

Table 700: Near-inner-boundary profile for GW231119_075248. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $33.505 M_\odot$ and $21.534 M_\odot$, with completed geometric mass-equivalent shares of 24.41% and 35.96%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $35.608 < 46.975 < 67.639$ km, and component 2 gives $25.530 < 31.715 < 48.502$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6218$ and 1.7458; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.198 and 14.871 h. The determinant-averaged frame-dragging rates at those shells are 1690.70 and 2422.78 rad s $^{-1}$, approaching surface-limit values of 2751.82 and 3548.56 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 5.093/6.589 h at 1 km inward to 0.0160/0.0208 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.11 GW241201_055758

GW241201_055758 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 95.809 M_\odot$ and $m_2 = 68.442 M_\odot$, while the luminosity-distance median is 7071.0 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,742
Observed luminosity distance	7071.0 Mpc (median)
Detector-frame masses	$m_1 = 95.809 M_\odot$, $m_2 = 68.442 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.714$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 1.0332$; $z_\pi = 1.1214$ (medians)
Post-handoff reorder fraction	16.66%

Table 701: Observational and coordinate summary for GW241201_055758. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	44.698 [31.275, 64.171]	32.169 [18.654, 47.813]
Physical compact-source mass $M_c (M_\odot)$	35.502 [22.560, 55.837]	21.733 [11.848, 35.021]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	7.320 [0.804, 22.272]	9.823 [1.114, 20.610]
Geometric share M_{geom}/M_∞ (%)	16.70 [1.87, 46.58]	32.65 [4.24, 49.72]
Exterior spin magnitude $ \hat{a}_\infty $	0.680 [0.140, 0.963]	0.397 [0.038, 0.916]
Polarization factor A_{pol}	1.200 [1.019, 1.872]	1.485 [1.044, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.980 [0.489, 1.000]	0.875 [0.149, 0.999]
GR/CPTG matching radius r_m (km)	66.002 [46.181, 94.756]	47.502 [27.544, 70.602]
Structural Mode N_{BH}	1.5542 [1.4580, 1.8727]	1.7077 [1.4708, 1.9090]
Organization-shell radius r_N (km)	34.702 [24.252, 49.833]	25.012 [14.474, 37.201]
Inner trapping-surface radius r_- (km)	48.279 [33.131, 74.370]	31.661 [18.028, 47.618]
Structural- N lapse $\bar{\alpha}_N$	0.7688 [0.5149, 0.8449]	0.6498 [0.4855, 0.8347]
Structural- N clock $T_{24,N}$ (h)	18.451 [12.357, 20.279]	15.596 [11.652, 20.032]
Structural- N axis ω (rad s $^{-1}$)	1174.53 [528.66, 2404.37]	2066.41 [603.62, 4148.42]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1612.23 [822.95, 2736.01]	2467.23 [616.07, 4698.74]
Structural- N equatorial ω (rad s $^{-1}$)	2102.72 [1190.53, 3141.36]	2896.32 [616.19, 5264.05]
Structural- N period P_ω (ms)	3.8972 [2.2965, 7.6349]	2.5467 [1.3372, 10.1989]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2844.80 [1535.76, 4172.42]	3725.12 [672.39, 6733.43]
Trapping-surface period P_- (ms)	2.2087 [1.5059, 4.0912]	1.6867 [0.9331, 9.3446]

Table 702: CPTG compact-state, structural, clock, and geometric-rotation results for GW241201_055758. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.6521	6.5349	2807.10	3632.98
500 m inward	3.2711	4.6191	2826.63	3681.47
100 m inward	1.4563	2.0637	2841.58	3717.99
10 m inward	0.4601	0.6524	2844.40	3724.40
1 m inward	0.1455	0.2063	2844.77	3725.07
10 cm inward	0.0460	0.0652	2844.80	3725.11
1 cm inward	0.0145	0.0206	2844.80	3725.12
surface limit	0.0000	0.0000	2844.80	3725.12

Table 703: Near-inner-boundary profile for GW241201_055758. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $35.502 M_\odot$ and $21.733 M_\odot$, with completed geometric mass-equivalent shares of 16.70% and 32.65%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $34.702 < 48.279 < 66.002$ km, and component 2 gives $25.012 < 31.661 < 47.502$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5542$ and 1.7077 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.451 and 15.596 h. The determinant-averaged frame-dragging rates at those shells are 1612.23 and 2467.23 rad s $^{-1}$, approaching surface-limit values of 2844.80 and 3725.12 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.652/6.535 h at 1 km inward to 0.0145/0.0206 h at 1 cm inward for components 1/2. At least one Structural-Mode shell retains comparatively more proper time than most M4 systems. Its Structural-Mode frame dragging is in the upper part of the M4 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.12 GW231029_111508

GW231029_111508 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 98.449 M_\odot$ and $m_2 = 66.097 M_\odot$, while the luminosity-distance median is 3111.6 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	35,469
Observed luminosity distance	3111.6 Mpc (median)
Detector-frame masses	$m_1 = 98.449 M_\odot$, $m_2 = 66.097 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.671$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5275$; $z_\pi = 0.5681$ (medians)
Post-handoff reorder fraction	17.68%

Table 704: Observational and coordinate summary for GW231029_111508. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	61.787 [42.728, 80.543]	42.081 [26.274, 62.843]
Physical compact-source mass $M_c (M_\odot)$	44.845 [31.395, 65.623]	28.651 [16.591, 43.692]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	13.845 [1.447, 33.774]	13.190 [1.745, 28.195]
Geometric share M_{geom}/M_∞ (%)	22.92 [2.68, 48.77]	33.27 [4.94, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.567 [0.081, 0.947]	0.386 [0.035, 0.902]
Polarization factor A_{pol}	1.297 [1.028, 1.952]	1.499 [1.052, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.955 [0.307, 1.000]	0.867 [0.139, 0.999]
GR/CPTG matching radius r_m (km)	91.237 [63.093, 118.931]	62.138 [38.797, 92.796]
Structural Mode N_{BH}	1.6077 [1.4623, 1.8981]	1.7147 [1.4747, 1.9095]
Organization-shell radius r_N (km)	48.055 [33.065, 62.639]	32.722 [20.396, 48.841]
Inner trapping-surface radius r_- (km)	62.810 [46.255, 87.300]	41.627 [25.360, 60.311]
Structural- N lapse $\bar{\alpha}_N$	0.7274 [0.4941, 0.8415]	0.6443 [0.4852, 0.8315]
Structural- N clock $T_{24,N}$ (h)	17.458 [11.857, 20.196]	15.464 [11.644, 19.957]
Structural- N axis ω (rad s $^{-1}$)	1005.75 [417.56, 1714.50]	1582.17 [420.54, 3038.77]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1285.83 [594.71, 1949.13]	1888.68 [421.46, 3392.45]
Structural- N equatorial ω (rad s $^{-1}$)	1579.22 [644.07, 2255.10]	2226.33 [421.66, 3779.42]
Structural- N period $P_{\bar{\omega}}$ (ms)	4.8865 [3.2236, 10.5652]	3.3268 [1.8521, 14.9082]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2087.51 [707.48, 3042.76]	2860.42 [459.89, 4886.53]
Trapping-surface period P_- (ms)	3.0099 [2.0650, 8.8811]	2.1966 [1.2858, 13.6623]

Table 705: CPTG compact-state, structural, clock, and geometric-rotation results for GW231029_111508. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.3700	5.6903	2063.93	2807.91
500 m inward	3.0823	4.0211	2076.49	2834.98
100 m inward	1.3755	1.7972	2085.41	2855.35
10 m inward	0.4347	0.5683	2087.21	2860.04
1 m inward	0.1375	0.1797	2087.49	2860.39
10 cm inward	0.0435	0.0568	2087.51	2860.42
1 cm inward	0.0137	0.0180	2087.51	2860.42
surface limit	0.0000	0.0000	2087.51	2860.42

Table 706: Near-inner-boundary profile for GW231029_111508. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $44.845 M_\odot$ and $28.651 M_\odot$, with completed geometric mass-equivalent shares of 22.92% and 33.27%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $48.055 < 62.810 < 91.237$ km, and component 2 gives $32.722 < 41.627 < 62.138$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6077$ and 1.7147; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.458 and 15.464 h. The determinant-averaged frame-dragging rates at those shells are 1285.83 and 1888.68 rad s $^{-1}$, approaching surface-limit values of 2087.51 and 2860.42 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.370/5.690 h at 1 km inward to 0.0137/0.0180 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M4. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.13 GW240527_183429

GW240527_183429 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 99.457 M_\odot$ and $m_2 = 66.805 M_\odot$, while the luminosity-distance median is 6438.3 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,363
Observed luminosity distance	6438.3 Mpc (median)
Detector-frame masses	$m_1 = 99.457 M_\odot$, $m_2 = 66.805 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.672$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9576$; $z_\pi = 1.0380$ (medians)
Post-handoff reorder fraction	13.12%

Table 707: Observational and coordinate summary for GW240527_183429. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	49.111 [35.421, 66.299]	32.038 [17.579, 49.146]
Physical compact-source mass $M_c (M_\odot)$	40.447 [26.675, 58.814]	22.410 [11.492, 37.090]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	6.785 [0.766, 22.170]	8.643 [1.025, 20.545]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	13.87 [1.65, 42.44]	28.92 [3.70, 49.55]
Exterior spin magnitude $ \hat{a}_\infty $	0.732 [0.224, 0.967]	0.462 [0.048, 0.927]
Polarization factor A_{pol}	1.161 [1.017, 1.737]	1.407 [1.038, 1.982]
Compact-source spin parameter $\hat{a}_c$	0.987 [0.676, 1.000]	0.914 [0.188, 0.999]
GR/CPTG matching radius r_m (km)	72.519 [52.303, 97.899]	47.308 [25.957, 72.571]
Structural Mode N_{BH}	1.5326 [1.4569, 1.8231]	1.6672 [1.4678, 1.9071]
Organization-shell radius r_N (km)	38.110 [27.364, 51.411]	24.925 [13.647, 38.201]
Inner trapping-surface radius r_- (km)	54.426 [38.289, 78.150]	32.264 [17.351, 50.028]
Structural- N lapse $\bar{\alpha}_N$	0.7856 [0.5563, 0.8459]	0.6815 [0.4870, 0.8371]
Structural- N clock $T_{24,N}$ (h)	18.855 [13.351, 20.301]	16.357 [11.687, 20.090]
Structural- N axis ω (rad s $^{-1}$)	987.51 [496.30, 2047.45]	2018.68 [694.35, 4398.86]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1415.30 [801.10, 2358.31]	2439.84 [810.52, 4969.70]
Structural- N equatorial ω (rad s $^{-1}$)	1898.37 [1232.38, 2735.72]	2907.34 [811.04, 5591.62]
Structural- N period P_ω (ms)	4.4395 [2.6643, 7.8432]	2.5752 [1.2643, 7.7520]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2601.31 [1699.54, 3663.86]	3775.49 [886.04, 7279.25]
Trapping-surface period P_- (ms)	2.4154 [1.7149, 3.6970]	1.6642 [0.8632, 7.0913]

Table 708: CPTG compact-state, structural, clock, and geometric-rotation results for GW240527_183429. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.2352	6.3523	2574.00	3691.45
500 m inward	2.9767	4.4842	2588.94	3736.70
100 m inward	1.3235	2.0010	2598.56	3769.21
10 m inward	0.4180	0.6325	2601.20	3774.64
1 m inward	0.1322	0.2000	2601.30	3775.41
10 cm inward	0.0418	0.0632	2601.30	3775.48
1 cm inward	0.0132	0.0200	2601.31	3775.49
surface limit	0.0000	0.0000	2601.31	3775.49

Table 709: Near-inner-boundary profile for GW240527_183429. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $40.447 M_\odot$ and $22.410 M_\odot$, with completed geometric mass-equivalent shares of 13.87% and 28.92%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $38.110 < 54.426 < 72.519$ km, and component 2 gives $24.925 < 32.264 < 47.308$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5326$ and 1.6672 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.855 and 16.357 h. The determinant-averaged frame-dragging rates at those shells are 1415.30 and 2439.84 rad s $^{-1}$, approaching surface-limit values of 2601.31 and 3775.49 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.235/6.352 h at 1 km inward to 0.0132/0.0200 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M4 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M4 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.14 GW231230_170116

GW231230_170116 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 99.846 M_\odot$ and $m_2 = 67.689 M_\odot$, while the luminosity-distance median is 5749.7 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	28,970
Observed luminosity distance	5749.7 Mpc (median)
Detector-frame masses	$m_1 = 99.846 M_\odot$, $m_2 = 67.689 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.678$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8735$; $z_\pi = 0.9456$ (medians)
Post-handoff reorder fraction	14.84%

Table 710: Observational and coordinate summary for GW231230_170116. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	50.504 [33.919, 102.712]	34.036 [20.148, 52.011]
Physical compact-source mass M_c ($M_\odot$)	39.922 [25.082, 75.774]	23.403 [12.956, 37.523]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.998 [0.920, 31.668]	10.233 [1.236, 22.630]
Geometric share M_{geom}/M_∞ (%)	17.95 [1.96, 47.23]	31.99 [4.23, 49.74]
Exterior spin magnitude $ \hat{a}_\infty $	0.657 [0.124, 0.961]	0.408 [0.036, 0.916]
Polarization factor A_{pol}	1.219 [1.020, 1.895]	1.470 [1.044, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.976 [0.446, 1.000]	0.882 [0.143, 0.999]
GR/CPTG matching radius r_m (km)	74.576 [50.086, 151.667]	50.259 [29.751, 76.801]
Structural Mode N_{BH}	1.5643 [1.4585, 1.8804]	1.7004 [1.4707, 1.9093]
Organization-shell radius r_N (km)	39.197 [26.277, 79.735]	26.456 [15.669, 40.432]
Inner trapping-surface radius r_- (km)	54.517 [36.122, 106.044]	33.824 [19.763, 51.553]
Structural- N lapse $\bar{\alpha}_N$	0.7609 [0.5085, 0.8446]	0.6556 [0.4853, 0.8347]
Structural- N clock $T_{24,N}$ (h)	18.262 [12.205, 20.269]	15.734 [11.646, 20.033]
Structural- N axis ω (rad s $^{-1}$)	1033.57 [416.62, 2175.13]	1916.05 [531.67, 3834.35]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1416.37 [583.72, 2490.90]	2297.34 [541.66, 4323.94]
Structural- N equatorial ω (rad s $^{-1}$)	1857.81 [702.34, 2878.98]	2723.95 [541.93, 4863.42]
Structural- N period P_ω (ms)	4.4361 [2.5225, 10.7641]	2.7350 [1.4531, 11.5998]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2515.54 [871.59, 3842.84]	3521.59 [591.27, 6286.37]
Trapping-surface period P_- (ms)	2.4977 [1.6350, 7.2089]	1.7842 [0.9995, 10.6266]

Table 711: CPTG compact-state, structural, clock, and geometric-rotation results for GW231230_170116. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.3736	6.2662	2487.22	3440.60
500 m inward	3.0776	4.4260	2501.67	3482.24
100 m inward	1.3701	1.9772	2512.78	3514.07
10 m inward	0.4328	0.6251	2515.26	3520.65
1 m inward	0.1369	0.1977	2515.51	3521.51
10 cm inward	0.0433	0.0625	2515.54	3521.58
1 cm inward	0.0137	0.0198	2515.54	3521.59
surface limit	0.0000	0.0000	2515.54	3521.59

Table 712: Near-inner-boundary profile for GW231230_170116. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $39.922 M_\odot$ and $23.403 M_\odot$, with completed geometric mass-equivalent shares of 17.95% and 31.99%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $39.197 < 54.517 < 74.576$ km, and component 2 gives $26.456 < 33.824 < 50.259$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5643$ and 1.7004; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.262 and 15.734 h. The determinant-averaged frame-dragging rates at those shells are 1416.37 and 2297.34 rad s $^{-1}$, approaching surface-limit values of 2515.54 and 3521.59 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.374/6.266 h at 1 km inward to 0.0137/0.0198 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.15 GW231102_071736

GW231102_071736 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 98.517 M_\odot$ and $m_2 = 69.800 M_\odot$, while the luminosity-distance median is 3844.6 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	41,826
Observed luminosity distance	3844.6 Mpc (median)
Detector-frame masses	$m_1 = 98.517 M_\odot$, $m_2 = 69.800 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.709$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6284$; $z_\pi = 0.6778$ (medians)
Post-handoff reorder fraction	18.06%

Table 713: Observational and coordinate summary for GW231102_071736. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	58.062 [41.771, 73.128]	42.130 [28.909, 60.232]
Physical compact-source mass $M_c (M_\odot)$	40.254 [28.775, 58.952]	27.499 [17.630, 41.029]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	15.610 [1.714, 32.021]	15.011 [2.304, 27.501]
Geometric share M_{geom}/M_∞ (%)	27.47 [3.36, 49.33]	37.02 [6.39, 49.83]
Exterior spin magnitude $ \hat{a}_\infty $	0.487 [0.059, 0.933]	0.321 [0.029, 0.874]
Polarization factor A_{pol}	1.379 [1.035, 1.974]	1.588 [1.068, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.926 [0.228, 0.999]	0.809 [0.115, 0.998]
GR/CPTG matching radius r_m (km)	85.735 [61.679, 107.983]	62.211 [42.688, 88.940]
Structural Mode N_{BH}	1.6521 [1.4659, 1.9046]	1.7582 [1.4831, 1.9103]
Organization-shell radius r_N (km)	45.132 [32.357, 56.849]	32.762 [22.446, 46.793]
Inner trapping-surface radius r_- (km)	57.397 [43.541, 78.408]	40.622 [27.572, 57.138]
Structural- N lapse $\bar{\alpha}_N$	0.6932 [0.4889, 0.8386]	0.6096 [0.4845, 0.8248]
Structural- N clock $T_{24,N}$ (h)	16.636 [11.734, 20.125]	14.631 [11.628, 19.795]
Structural- N axis ω (rad s $^{-1}$)	1156.32 [431.76, 1859.11]	1642.40 [350.32, 2855.01]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1413.70 [519.51, 2098.33]	1910.96 [350.56, 3163.99]
Structural- N equatorial ω (rad s $^{-1}$)	1687.97 [522.34, 2377.41]	2206.19 [350.62, 3491.04]
Structural- N period $P_{\bar{\omega}}$ (ms)	4.4445 [2.9944, 12.0944]	3.2880 [1.9858, 17.9234]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2198.48 [570.54, 3159.41]	2784.93 [382.11, 4485.47]
Trapping-surface period P_- (ms)	2.8580 [1.9887, 11.0127]	2.2561 [1.4008, 16.4432]

Table 714: CPTG compact-state, structural, clock, and geometric-rotation results for GW231102_071736. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.7597	5.8812	2170.53	2728.12
500 m inward	3.3611	4.1596	2185.59	2757.84
100 m inward	1.5013	1.8601	2196.18	2779.50
10 m inward	0.4746	0.5882	2198.13	2784.37
1 m inward	0.1501	0.1860	2198.44	2784.88
10 cm inward	0.0475	0.0588	2198.48	2784.93
1 cm inward	0.0150	0.0186	2198.48	2784.93
surface limit	0.0000	0.0000	2198.48	2784.93

Table 715: Near-inner-boundary profile for GW231102_071736. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $40.254 M_\odot$ and $27.499 M_\odot$, with completed geometric mass-equivalent shares of 27.47% and 37.02%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $45.132 < 57.397 < 85.735$ km, and component 2 gives $32.762 < 40.622 < 62.211$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6521$ and 1.7582 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.636 and 14.631 h. The determinant-averaged frame-dragging rates at those shells are 1413.70 and 1910.96 rad s $^{-1}$, approaching surface-limit values of 2198.48 and 2784.93 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.760/5.881 h at 1 km inward to 0.0150/0.0186 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.16 GW231004_232346

GW231004_232346 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 109.486 M_\odot$ and $m_2 = 60.247 M_\odot$, while the luminosity-distance median is 4360.3 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	28,888
Observed luminosity distance	4360.3 Mpc (median)
Detector-frame masses	$m_1 = 109.486 M_\odot$, $m_2 = 60.247 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.550$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6969$; $z_\pi = 0.7524$ (medians)
Post-handoff reorder fraction	9.32%

Table 716: Observational and coordinate summary for GW231004_232346. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	62.006 [40.648, 87.219]	33.783 [20.263, 53.921]
Physical compact-source mass $M_c (M_\odot)$	43.551 [28.496, 71.440]	23.560 [12.957, 38.066]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	15.963 [1.581, 34.851]	9.915 [1.189, 23.610]
Geometric share M_{geom}/M_∞ (%)	25.86 [2.96, 49.39]	31.27 [3.98, 49.69]
Exterior spin magnitude $ \hat{a}_\infty $	0.515 [0.056, 0.941]	0.421 [0.040, 0.921]
Polarization factor A_{pol}	1.349 [1.031, 1.976]	1.455 [1.041, 1.988]
Compact-source spin parameter $\hat{a}_c$	0.937 [0.219, 1.000]	0.891 [0.157, 0.999]
GR/CPTG matching radius r_m (km)	91.560 [60.022, 128.790]	49.885 [29.921, 79.621]
Structural Mode N_{BH}	1.6359 [1.4638, 1.9053]	1.6924 [1.4693, 1.9087]
Organization-shell radius r_N (km)	48.213 [31.468, 67.863]	26.257 [15.762, 41.886]
Inner trapping-surface radius r_- (km)	61.818 [42.520, 95.361]	33.854 [19.744, 52.802]
Structural- N lapse $\bar{\alpha}_N$	0.7057 [0.4884, 0.8403]	0.6619 [0.4857, 0.8358]
Structural- N clock $T_{24,N}$ (h)	16.936 [11.722, 20.167]	15.884 [11.658, 20.060]
Structural- N axis ω (rad s $^{-1}$)	1024.29 [365.49, 1845.21]	1890.32 [560.43, 3830.90]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1277.46 [464.56, 2090.35]	2289.56 [575.72, 4323.70]
Structural- N equatorial ω (rad s $^{-1}$)	1547.17 [477.94, 2410.42]	2736.94 [576.03, 4859.15]
Structural- N period $P_{\bar{\omega}}$ (ms)	4.9185 [3.0058, 13.5249]	2.7443 [1.4532, 10.9136]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2016.70 [522.87, 3200.35]	3548.16 [628.90, 6274.01]
Trapping-surface period P_- (ms)	3.1156 [1.9633, 12.0167]	1.7708 [1.0015, 9.9908]

Table 717: CPTG compact-state, structural, clock, and geometric-rotation results for GW231004_232346. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.5031	6.2306	1992.80	3472.53
500 m inward	3.1788	4.3994	2005.28	3511.66
100 m inward	1.4192	1.9647	2014.75	3541.35
10 m inward	0.4486	0.6212	2016.65	3547.40
1 m inward	0.1419	0.1964	2016.69	3548.08
10 cm inward	0.0449	0.0621	2016.70	3548.15
1 cm inward	0.0142	0.0196	2016.70	3548.16
surface limit	0.0000	0.0000	2016.70	3548.16

Table 718: Near-inner-boundary profile for GW231004_232346. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $43.551 M_\odot$ and $23.560 M_\odot$, with completed geometric mass-equivalent shares of 25.86% and 31.27%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $48.213 < 61.818 < 91.560$ km, and component 2 gives $26.257 < 33.854 < 49.885$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6359$ and 1.6924 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.936 and 15.884 h. The determinant-averaged frame-dragging rates at those shells are 1277.46 and 2289.56 rad s $^{-1}$, approaching surface-limit values of 2016.70 and 3548.16 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.503/6.231 h at 1 km inward to 0.0142/0.0196 h at 1 cm inward for components 1/2. Its interior readouts lie near the central part of the M4 descriptive distribution, making it a useful representative high-mass case. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.17 GW241229_155844

GW241229_155844 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 105.768 M_\odot$ and $m_2 = 66.826 M_\odot$, while the luminosity-distance median is 6777.3 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	16,352
Observed luminosity distance	6777.3 Mpc (median)
Detector-frame masses	$m_1 = 105.768 M_\odot$, $m_2 = 66.826 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.632$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9983$; $z_\pi = 1.0829$ (medians)
Post-handoff reorder fraction	15.01%

Table 719: Observational and coordinate summary for GW241229_155844. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	49.213 [32.969, 71.419]	31.822 [17.435, 50.689]
Physical compact-source mass M_c ($M_\odot$)	36.102 [23.226, 58.186]	21.530 [11.394, 35.472]
Geometric mass-equivalent M_{geom} ($M_\odot$)	11.089 [1.164, 28.544]	9.869 [1.141, 22.169]
Geometric share M_{geom}/M_∞ (%)	23.08 [2.60, 48.96]	33.33 [4.43, 49.75]
Exterior spin magnitude $ \hat{a}_\infty $	0.564 [0.074, 0.948]	0.385 [0.036, 0.912]
Polarization factor A_{pol}	1.300 [1.027, 1.959]	1.500 [1.046, 1.990]
Compact-source spin parameter $\hat{a}_c$	0.954 [0.283, 1.000]	0.866 [0.142, 0.999]
GR/CPTG matching radius r_m (km)	72.669 [48.683, 105.459]	46.989 [25.745, 74.849]
Structural Mode N_{BH}	1.6091 [1.4618, 1.9004]	1.7154 [1.4718, 1.9093]
Organization-shell radius r_N (km)	38.242 [25.582, 55.530]	24.740 [13.534, 39.501]
Inner trapping-surface radius r_- (km)	50.651 [34.365, 77.560]	31.337 [17.149, 49.280]
Structural- N lapse $\bar{\alpha}_N$	0.7263 [0.4923, 0.8418]	0.6438 [0.4852, 0.8338]
Structural- N clock $T_{24,N}$ (h)	17.431 [11.815, 20.204]	15.450 [11.646, 20.012]
Structural- N axis ω (rad s $^{-1}$)	1228.29 [466.22, 2299.84]	2061.09 [545.45, 4365.09]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1562.81 [643.61, 2609.08]	2466.26 [551.84, 4918.95]
Structural- N equatorial ω (rad s $^{-1}$)	1927.38 [728.78, 2994.48]	2908.52 [552.08, 5581.30]
Structural- N period P_ω (ms)	4.0205 [2.4082, 9.7624]	2.5477 [1.2773, 11.3859]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2553.53 [812.32, 3989.79]	3724.07 [601.82, 7263.04]
Trapping-surface period P_- (ms)	2.4606 [1.5748, 7.7348]	1.6872 [0.8651, 10.4403]

Table 720: CPTG compact-state, structural, clock, and geometric-rotation results for GW241229_155844. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.8778	6.5354	2518.52	3633.01
500 m inward	3.4384	4.6163	2536.69	3682.10
100 m inward	1.5343	2.0620	2550.21	3715.12
10 m inward	0.4849	0.6518	2553.09	3723.27
1 m inward	0.1533	0.2061	2553.48	3723.89
10 cm inward	0.0485	0.0652	2553.52	3724.05
1 cm inward	0.0153	0.0206	2553.53	3724.07
surface limit	0.0000	0.0000	2553.53	3724.07

Table 721: Near-inner-boundary profile for GW241229_155844. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $36.102 M_\odot$ and $21.530 M_\odot$, with completed geometric mass-equivalent shares of 23.08% and 33.33%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $38.242 < 50.651 < 72.669$ km, and component 2 gives $24.740 < 31.337 < 46.989$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6091$ and 1.7154 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.431 and 15.450 h. The determinant-averaged frame-dragging rates at those shells are 1562.81 and 2466.26 rad s $^{-1}$, approaching surface-limit values of 2553.53 and 3724.07 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.878/6.535 h at 1 km inward to 0.0153/0.0206 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is in the upper part of the M4 rotation distribution for at least one component. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.18 GW230601_224134

GW230601_224134 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 102.266 M_\odot$ and $m_2 = 70.652 M_\odot$, while the luminosity-distance median is 3669.4 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	42,066
Observed luminosity distance	3669.4 Mpc (median)
Detector-frame masses	$m_1 = 102.266 M_\odot$, $m_2 = 70.652 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.691$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6047$; $z_\pi = 0.6520$ (medians)
Post-handoff reorder fraction	16.72%

Table 722: Observational and coordinate summary for GW230601_224134. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	61.107 [43.799, 79.178]	43.249 [27.960, 62.136]
Physical compact-source mass $M_c (M_\odot)$	44.143 [30.639, 65.398]	28.641 [17.361, 42.966]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	14.351 [1.533, 33.401]	14.661 [2.070, 28.241]
Geometric share M_{geom}/M_∞ (%)	24.00 [2.74, 49.07]	35.76 [5.53, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.548 [0.070, 0.945]	0.343 [0.030, 0.891]
Polarization factor A_{pol}	1.316 [1.028, 1.963]	1.557 [1.059, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.949 [0.268, 1.000]	0.831 [0.118, 0.998]
GR/CPTG matching radius r_m (km)	90.232 [64.675, 116.916]	63.863 [41.287, 91.751]
Structural Mode N_{BH}	1.6178 [1.4626, 1.9016]	1.7433 [1.4780, 1.9102]
Organization-shell radius r_N (km)	47.514 [33.939, 61.584]	33.641 [21.728, 48.308]
Inner trapping-surface radius r_- (km)	61.884 [46.121, 87.007]	42.083 [26.772, 59.299]
Structural- N lapse $\bar{\alpha}_N$	0.7196 [0.4913, 0.8412]	0.6216 [0.4846, 0.8288]
Structural- N clock $T_{24,N}$ (h)	17.271 [11.791, 20.189]	14.918 [11.629, 19.892]
Structural- N axis ω (rad s $^{-1}$)	1027.09 [406.92, 1747.94]	1580.10 [358.42, 2871.76]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1297.54 [560.36, 1976.51]	1857.02 [358.59, 3203.59]
Structural- N equatorial ω (rad s $^{-1}$)	1587.37 [579.51, 2255.05]	2159.65 [358.67, 3568.47]
Structural- N period $P_{\bar{\omega}}$ (ms)	4.8424 [3.1789, 11.2128]	3.3835 [1.9613, 17.5221]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2090.93 [633.80, 2999.97]	2744.44 [390.88, 4576.23]
Trapping-surface period P_- (ms)	3.0050 [2.0944, 9.9135]	2.2894 [1.3730, 16.0743]

Table 723: CPTG compact-state, structural, clock, and geometric-rotation results for GW230601_224134. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.4567	5.7407	2066.95	2695.09
500 m inward	3.1448	4.0593	2079.47	2720.51
100 m inward	1.4039	1.8149	2088.93	2740.29
10 m inward	0.4438	0.5739	2090.65	2744.04
1 m inward	0.1403	0.1815	2090.91	2744.41
10 cm inward	0.0444	0.0574	2090.93	2744.44
1 cm inward	0.0140	0.0181	2090.93	2744.44
surface limit	0.0000	0.0000	2090.93	2744.44

Table 724: Near-inner-boundary profile for GW230601_224134. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $44.143 M_\odot$ and $28.641 M_\odot$, with completed geometric mass-equivalent shares of 24.00% and 35.76%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $47.514 < 61.884 < 90.232$ km, and component 2 gives $33.641 < 42.083 < 63.863$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6178$ and 1.7433 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.271 and 14.918 h. The determinant-averaged frame-dragging rates at those shells are 1297.54 and 1857.02 rad s $^{-1}$, approaching surface-limit values of 2090.93 and 2744.44 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.457/5.741 h at 1 km inward to 0.0140/0.0181 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.19 GW241230_233618

GW241230_233618 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 100.078 M_\odot$ and $m_2 = 73.837 M_\odot$, while the luminosity-distance median is 2715.2 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,967
Observed luminosity distance	2715.2 Mpc (median)
Detector-frame masses	$m_1 = 100.078 M_\odot$, $m_2 = 73.837 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.738$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4707$; $z_\pi = 0.5067$ (medians)
Post-handoff reorder fraction	19.00%

Table 725: Observational and coordinate summary for GW241230_233618. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	65.129 [47.890, 82.944]	49.150 [29.145, 69.028]
Physical compact-source mass $M_c (M_\odot)$	50.344 [33.942, 70.538]	32.389 [18.930, 49.698]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	12.123 [1.282, 33.684]	15.857 [2.081, 30.963]
Geometric share M_{geom}/M_∞ (%)	19.18 [2.11, 47.47]	34.31 [5.20, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.634 [0.118, 0.958]	0.368 [0.034, 0.897]
Polarization factor A_{pol}	1.237 [1.022, 1.904]	1.522 [1.055, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.971 [0.428, 1.000]	0.853 [0.136, 0.998]
GR/CPTG matching radius r_m (km)	96.171 [70.716, 122.477]	72.576 [43.036, 101.928]
Structural Mode N_{BH}	1.5745 [1.4593, 1.8832]	1.7266 [1.4761, 1.9096]
Organization-shell radius r_N (km)	50.601 [37.098, 64.526]	38.194 [22.654, 53.713]
Inner trapping-surface radius r_- (km)	68.834 [50.368, 93.901]	47.725 [28.804, 67.356]
Structural- N lapse $\bar{\alpha}_N$	0.7530 [0.5062, 0.8439]	0.6349 [0.4851, 0.8303]
Structural- N clock $T_{24,N}$ (h)	18.072 [12.149, 20.254]	15.238 [11.641, 19.928]
Structural- N axis ω (rad s $^{-1}$)	874.04 [401.50, 1607.98]	1388.11 [371.11, 2609.05]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1162.69 [624.84, 1821.49]	1638.66 [371.73, 2930.70]
Structural- N equatorial ω (rad s $^{-1}$)	1472.41 [853.44, 2067.51]	1911.47 [371.85, 3322.97]
Structural- N period $P_{\bar{\omega}}$ (ms)	5.4040 [3.4495, 10.0556]	3.8343 [2.1439, 16.9027]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1974.93 [960.85, 2745.88]	2439.16 [405.23, 4349.62]
Trapping-surface period P_- (ms)	3.1815 [2.2882, 6.5392]	2.5760 [1.4445, 15.5054]

Table 726: CPTG compact-state, structural, clock, and geometric-rotation results for GW241230_233618. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.0324	5.3740	1956.56	2397.40
500 m inward	2.8420	3.7999	1965.85	2419.35
100 m inward	1.2680	1.6983	1973.19	2434.76
10 m inward	0.4007	0.5370	1974.73	2438.73
1 m inward	0.1267	0.1698	1974.92	2439.16
10 cm inward	0.0401	0.0537	1974.93	2439.16
1 cm inward	0.0127	0.0170	1974.93	2439.16
surface limit	0.0000	0.0000	1974.93	2439.16

Table 727: Near-inner-boundary profile for GW241230_233618. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $50.344 M_\odot$ and $32.389 M_\odot$, with completed geometric mass-equivalent shares of 19.18% and 34.31%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $50.601 < 68.834 < 96.171$ km, and component 2 gives $38.194 < 47.725 < 72.576$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5745$ and 1.7266 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.072 and 15.238 h. The determinant-averaged frame-dragging rates at those shells are 1162.69 and 1638.66 rad s $^{-1}$, approaching surface-limit values of 1974.93 and 2439.16 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.032/5.374 h at 1 km inward to 0.0127/0.0170 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M4. The post-handoff component ordering changes for a comparatively large fraction of posterior draws, so preserving quantity attachment through the handoff is particularly important here. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.20 GW190602_175927

GW190602_175927 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 106.911 M_\odot$ and $m_2 = 67.604 M_\odot$, while the luminosity-distance median is 2835.5 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	34,485
Observed luminosity distance	2835.5 Mpc (median)
Detector-frame masses	$m_1 = 106.911 M_\odot$, $m_2 = 67.604 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.632$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.4887$; $z_\pi = 0.5255$ (medians)
Post-handoff reorder fraction	14.67%

Table 728: Observational and coordinate summary for GW190602_175927. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	69.536 [48.657, 88.545]	44.173 [24.243, 66.322]
Physical compact-source mass $M_c (M_\odot)$	50.245 [35.196, 76.219]	30.264 [15.759, 46.723]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	15.266 [1.531, 36.954]	13.073 [1.559, 29.926]
Geometric share M_{geom}/M_∞ (%)	22.55 [2.46, 48.88]	32.31 [4.33, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.574 [0.077, 0.951]	0.403 [0.037, 0.914]
Polarization factor A_{pol}	1.291 [1.025, 1.956]	1.477 [1.045, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.957 [0.293, 1.000]	0.879 [0.146, 0.999]
GR/CPTG matching radius r_m (km)	102.679 [71.848, 130.748]	65.226 [35.797, 97.933]
Structural Mode N_{BH}	1.6043 [1.4611, 1.8994]	1.7040 [1.4712, 1.9092]
Organization-shell radius r_N (km)	54.033 [37.641, 68.735]	34.336 [18.850, 51.568]
Inner trapping-surface radius r_- (km)	70.097 [52.202, 101.174]	43.926 [23.748, 64.058]
Structural- N lapse $\bar{\alpha}_N$	0.7300 [0.4930, 0.8424]	0.6528 [0.4854, 0.8343]
Structural- N clock $T_{24,N}$ (h)	17.521 [11.833, 20.219]	15.668 [11.649, 20.023]
Structural- N axis ω (rad s $^{-1}$)	889.37 [351.02, 1525.55]	1494.04 [425.11, 3113.05]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1143.75 [506.05, 1735.17]	1785.94 [426.90, 3524.59]
Structural- N equatorial ω (rad s $^{-1}$)	1407.84 [563.04, 2003.04]	2114.72 [427.24, 3979.31]
Structural- N period $P_{\bar{\omega}}$ (ms)	5.4935 [3.6211, 12.4162]	3.5181 [1.7827, 14.7183]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1850.04 [616.48, 2678.50]	2726.11 [466.40, 5175.22]
Trapping-surface period P_- (ms)	3.3962 [2.3458, 10.1920]	2.3048 [1.2141, 13.4715]

Table 729: CPTG compact-state, structural, clock, and geometric-rotation results for GW190602_175927. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.1343	5.5063	1831.91	2679.50
500 m inward	2.9166	3.8903	1841.11	2702.84
100 m inward	1.3019	1.7386	1848.30	2721.92
10 m inward	0.4116	0.5497	1849.85	2725.70
1 m inward	0.1302	0.1738	1850.03	2726.08
10 cm inward	0.0412	0.0550	1850.04	2726.11
1 cm inward	0.0130	0.0174	1850.04	2726.11
surface limit	0.0000	0.0000	1850.04	2726.11

Table 730: Near-inner-boundary profile for GW190602_175927. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $50.245 M_\odot$ and $30.264 M_\odot$, with completed geometric mass-equivalent shares of 22.55% and 32.31%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $54.033 < 70.097 < 102.679$ km, and component 2 gives $34.336 < 43.926 < 65.226$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6043$ and 1.7040 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.521 and 15.668 h. The determinant-averaged frame-dragging rates at those shells are 1143.75 and 1785.94 rad s $^{-1}$, approaching surface-limit values of 1850.04 and 2726.11 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.134/5.506 h at 1 km inward to 0.0130/0.0174 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M4. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.21 GW240107_013215

GW240107_013215 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 109.863 M_\odot$ and $m_2 = 65.360 M_\odot$, while the luminosity-distance median is 5817.5 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	32,438
Observed luminosity distance	5817.5 Mpc (median)
Detector-frame masses	$m_1 = 109.863 M_\odot$, $m_2 = 65.360 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.595$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8818$; $z_\pi = 0.9548$ (medians)
Post-handoff reorder fraction	10.57%

Table 731: Observational and coordinate summary for GW240107_013215. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	55.835 [37.339, 82.827]	32.104 [15.439, 53.632]
Physical compact-source mass M_c ($M_\odot$)	46.635 [28.720, 73.713]	22.648 [10.082, 41.509]
Geometric mass-equivalent M_{geom} ($M_\odot$)	7.161 [0.835, 23.820]	7.875 [0.878, 21.363]
Geometric share M_{geom}/M_∞ (%)	12.87 [1.57, 40.81]	27.48 [3.21, 49.55]
Exterior spin magnitude $ \hat{a}_\infty $	0.751 [0.254, 0.969]	0.487 [0.048, 0.936]
Polarization factor A_{pol}	1.148 [1.016, 1.690]	1.379 [1.033, 1.982]
Compact-source spin parameter $\hat{a}_c$	0.989 [0.724, 1.000]	0.925 [0.187, 0.999]
GR/CPTG matching radius r_m (km)	82.447 [55.135, 122.305]	47.406 [22.797, 79.195]
Structural Mode N_{BH}	1.5253 [1.4565, 1.8035]	1.6522 [1.4651, 1.9072]
Organization-shell radius r_N (km)	43.311 [28.901, 64.209]	24.952 [11.997, 41.711]
Inner trapping-surface radius r_- (km)	62.607 [40.602, 98.064]	32.453 [15.155, 56.015]
Structural- N lapse $\bar{\alpha}_N$	0.7913 [0.5726, 0.8462]	0.6931 [0.4869, 0.8392]
Structural- N clock $T_{24,N}$ (h)	18.991 [13.742, 20.308]	16.634 [11.686, 20.141]
Structural- N axis ω (rad s $^{-1}$)	832.56 [409.25, 1865.99]	1936.41 [633.16, 4909.32]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1208.80 [652.95, 2184.52]	2374.60 [780.19, 5560.73]
Structural- N equatorial ω (rad s $^{-1}$)	1653.01 [1001.11, 2574.50]	2850.49 [787.04, 6336.12]
Structural- N period P_ω (ms)	5.1979 [2.8762, 9.6228]	2.6460 [1.1299, 8.0535]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2274.40 [1395.04, 3469.28]	3703.72 [860.95, 8203.40]
Trapping-surface period P_- (ms)	2.7626 [1.8111, 4.5040]	1.6965 [0.7659, 7.2980]

Table 732: CPTG compact-state, structural, clock, and geometric-rotation results for GW240107_013215. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.8496	6.2433	2254.29	3614.26
500 m inward	2.7045	4.3988	2264.97	3662.69
100 m inward	1.2034	1.9622	2272.50	3697.16
10 m inward	0.3801	0.6200	2274.21	3702.84
1 m inward	0.1202	0.1961	2274.38	3703.64
10 cm inward	0.0380	0.0620	2274.39	3703.72
1 cm inward	0.0120	0.0196	2274.40	3703.72
surface limit	0.0000	0.0000	2274.40	3703.72

Table 733: Near-inner-boundary profile for GW240107_013215. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $46.635 M_\odot$ and $22.648 M_\odot$, with completed geometric mass-equivalent shares of 12.87% and 27.48%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $43.311 < 62.607 < 82.447$ km, and component 2 gives $24.952 < 32.453 < 47.406$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5253$ and 1.6522 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.991 and 16.634 h. The determinant-averaged frame-dragging rates at those shells are 1208.80 and 2374.60 rad s $^{-1}$, approaching surface-limit values of 2274.40 and 3703.72 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.850/6.243 h at 1 km inward to 0.0120/0.0196 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M4 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.22 GW230819_171910

GW230819_171910 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 116.488 M_\odot$ and $m_2 = 60.477 M_\odot$, while the luminosity-distance median is 4058.7 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed
Posterior samples	31,794
Observed luminosity distance	4058.7 Mpc (median)
Detector-frame masses	$m_1 = 116.488 M_\odot$, $m_2 = 60.477 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.519$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6571$; $z_\pi = 0.7090$ (medians)
Post-handoff reorder fraction	8.82%

Table 734: Observational and coordinate summary for GW230819_171910. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	67.551 [43.671, 110.352]	34.069 [16.124, 56.983]
Physical compact-source mass $M_c (M_\odot)$	50.579 [31.021, 94.876]	23.673 [10.786, 40.971]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	12.828 [1.276, 38.320]	9.148 [0.994, 24.476]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	19.48 [1.93, 48.76]	30.33 [3.59, 49.67]
Exterior spin magnitude $ \hat{a}_\infty $	0.629 [0.081, 0.962]	0.437 [0.041, 0.929]
Polarization factor A_{pol}	1.242 [1.020, 1.952]	1.435 [1.037, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.308, 1.000]	0.900 [0.161, 0.999]
GR/CPTG matching radius r_m (km)	99.748 [64.486, 162.949]	50.307 [23.810, 84.143]
Structural Mode N_{BH}	1.5771 [1.4583, 1.8981]	1.6822 [1.4671, 1.9085]
Organization-shell radius r_N (km)	52.438 [33.867, 85.566]	26.487 [12.528, 44.335]
Inner trapping-surface radius r_- (km)	70.200 [45.674, 128.206]	34.171 [16.006, 56.532]
Structural- N lapse $\bar{\alpha}_N$	0.7510 [0.4941, 0.8447]	0.6699 [0.4859, 0.8376]
Structural- N clock $T_{24,N}$ (h)	18.025 [11.859, 20.272]	16.077 [11.661, 20.102]
Structural- N axis ω (rad s $^{-1}$)	809.10 [298.32, 1706.49]	1860.76 [559.26, 4543.61]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1082.62 [435.99, 1949.98]	2261.50 [604.15, 5189.05]
Structural- N equatorial ω (rad s $^{-1}$)	1383.60 [549.94, 2242.51]	2700.24 [604.50, 5955.94]
Structural- N period P_ω (ms)	5.8037 [3.2222, 14.4114]	2.7783 [1.2109, 10.4001]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1835.90 [651.39, 2987.12]	3495.93 [659.27, 7770.89]
Trapping-surface period P_- (ms)	3.4224 [2.1034, 9.6458]	1.7973 [0.8086, 9.5305]

Table 735: CPTG compact-state, structural, clock, and geometric-rotation results for GW230819_171910. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.9715	6.1465	1818.86	3418.75
500 m inward	2.7987	4.3362	1828.05	3459.50
100 m inward	1.2483	1.9355	1834.61	3489.13
10 m inward	0.3945	0.6117	1835.75	3495.20
1 m inward	0.1247	0.1934	1835.88	3495.86
10 cm inward	0.0394	0.0612	1835.89	3495.92
1 cm inward	0.0125	0.0193	1835.89	3495.92
surface limit	0.0000	0.0000	1835.90	3495.93

Table 736: Near-inner-boundary profile for GW230819_171910. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $50.579 M_\odot$ and $23.673 M_\odot$, with completed geometric mass-equivalent shares of 19.48% and 30.33%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $52.438 < 70.200 < 99.748$ km, and component 2 gives $26.487 < 34.171 < 50.307$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5771$ and 1.6822; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.025 and 16.077 h. The determinant-averaged frame-dragging rates at those shells are 1082.62 and 2261.50 rad s $^{-1}$, approaching surface-limit values of 1835.90 and 3495.93 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.971/6.146 h at 1 km inward to 0.0125/0.0193 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M4 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.23 GW240612_081540

GW240612_081540 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 105.133 M_\odot$ and $m_2 = 74.706 M_\odot$, while the luminosity-distance median is 6225.7 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	14,937
Observed luminosity distance	6225.7 Mpc (median)
Detector-frame masses	$m_1 = 105.133 M_\odot$, $m_2 = 74.706 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.711$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9318$; $z_\pi = 1.0097$ (medians)
Post-handoff reorder fraction	18.32%

Table 737: Observational and coordinate summary for GW240612_081540. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	51.362 [36.142, 72.900]	37.074 [20.795, 55.851]
Physical compact-source mass $M_c (M_\odot)$	39.309 [25.725, 61.568]	24.647 [13.362, 38.853]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	9.973 [1.070, 27.671]	12.024 [1.458, 24.680]
Geometric share M_{geom}/M_∞ (%)	19.59 [2.19, 47.95]	34.67 [4.92, 49.76]
Exterior spin magnitude $ \hat{a}_\infty $	0.627 [0.105, 0.957]	0.362 [0.035, 0.903]
Polarization factor A_{pol}	1.244 [1.022, 1.921]	1.531 [1.052, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.389, 1.000]	0.848 [0.137, 0.999]
GR/CPTG matching radius r_m (km)	75.842 [53.368, 107.646]	54.744 [30.707, 82.471]
Structural Mode N_{BH}	1.5780 [1.4597, 1.8888]	1.7307 [1.4745, 1.9095]
Organization-shell radius r_N (km)	39.918 [28.005, 56.683]	28.819 [16.085, 43.419]
Inner trapping-surface radius r_- (km)	54.019 [38.156, 82.017]	36.238 [20.145, 53.869]
Structural- N lapse $\bar{\alpha}_N$	0.7503 [0.5016, 0.8436]	0.6317 [0.4851, 0.8316]
Structural- N clock $T_{24,N}$ (h)	18.008 [12.039, 20.247]	15.160 [11.642, 19.959]
Structural- N axis ω (rad s $^{-1}$)	1107.01 [462.63, 2111.30]	1828.39 [472.15, 3638.06]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1459.57 [692.13, 2390.91]	2161.67 [473.34, 4117.81]
Structural- N equatorial ω (rad s $^{-1}$)	1846.79 [913.76, 2728.15]	2508.08 [473.47, 4686.43]
Structural- N period P_ω (ms)	4.3048 [2.6280, 9.0781]	2.9066 [1.5259, 13.2741]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2474.86 [1048.73, 3638.77]	3207.09 [516.09, 6029.28]
Trapping-surface period P_- (ms)	2.5388 [1.7267, 5.9913]	1.9592 [1.0421, 12.1746]

Table 738: CPTG compact-state, structural, clock, and geometric-rotation results for GW240612_081540. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.5612	6.1563	2445.52	3136.74
500 m inward	3.2121	4.3505	2461.10	3172.87
100 m inward	1.4323	1.9440	2472.08	3200.51
10 m inward	0.4526	0.6146	2474.52	3206.62
1 m inward	0.1431	0.1944	2474.83	3207.09
10 cm inward	0.0453	0.0615	2474.86	3207.09
1 cm inward	0.0143	0.0194	2474.86	3207.09
surface limit	0.0000	0.0000	2474.86	3207.09

Table 739: Near-inner-boundary profile for GW240612_081540. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $39.309 M_\odot$ and $24.647 M_\odot$, with completed geometric mass-equivalent shares of 19.59% and 34.67%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $39.918 < 54.019 < 75.842$ km, and component 2 gives $28.819 < 36.238 < 54.744$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5780$ and 1.7307 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.008 and 15.160 h. The determinant-averaged frame-dragging rates at those shells are 1459.57 and 2161.67 rad s $^{-1}$, approaching surface-limit values of 2474.86 and 3207.09 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.561/6.156 h at 1 km inward to 0.0143/0.0194 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.24 GW190706_222641

GW190706_222641 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 117.530 M_\odot$ and $m_2 = 63.736 M_\odot$, while the luminosity-distance median is 3625.2 Mpc. One retained release label defines this numerical stream. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	6,316
Observed luminosity distance	3625.2 Mpc (median)
Detector-frame masses	$m_1 = 117.530 M_\odot$, $m_2 = 63.736 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.542$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5994$; $z_\pi = 0.6454$ (medians)
Post-handoff reorder fraction	5.18%

Table 740: Observational and coordinate summary for GW190706_222641. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	71.735 [51.450, 92.335]	38.320 [23.318, 59.770]
Physical compact-source mass $M_c (M_\odot)$	56.816 [37.605, 80.786]	27.510 [14.582, 45.490]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	12.263 [1.376, 33.648]	10.071 [1.194, 24.437]
Geometric share $M_{\text{geom}}/M_\infty (\%)$	17.33 [2.05, 45.24]	28.22 [3.31, 49.55]
Exterior spin magnitude $ \hat{a}_\infty $	0.668 [0.169, 0.959]	0.474 [0.048, 0.934]
Polarization factor A_{pol}	1.210 [1.021, 1.826]	1.393 [1.034, 1.982]
Compact-source spin parameter $\hat{a}_c$	0.978 [0.564, 1.000]	0.919 [0.188, 0.999]
GR/CPTG matching radius r_m (km)	105.926 [75.972, 136.344]	56.584 [34.432, 88.258]
Structural Mode N_{BH}	1.5593 [1.4590, 1.8567]	1.6598 [1.4657, 1.9071]
Organization-shell radius r_N (km)	55.713 [39.695, 71.601]	29.769 [18.066, 46.572]
Inner trapping-surface radius r_- (km)	76.606 [55.575, 107.300]	39.022 [22.358, 61.614]
Structural- N lapse $\bar{\alpha}_N$	0.7648 [0.5282, 0.8442]	0.6872 [0.4870, 0.8388]
Structural- N clock $T_{24,N}$ (h)	18.356 [12.676, 20.260]	16.493 [11.687, 20.130]
Structural- N axis ω (rad s $^{-1}$)	771.62 [363.28, 1429.61]	1612.05 [544.80, 3399.43]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1054.30 [581.57, 1629.55]	1985.22 [655.77, 3816.38]
Structural- N equatorial ω (rad s $^{-1}$)	1345.83 [863.08, 1873.82]	2402.69 [656.62, 4260.28]
Structural- N period P_ω (ms)	5.9596 [3.8558, 10.8037]	3.1650 [1.6464, 9.5814]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1809.86 [1137.67, 2506.83]	3145.02 [717.85, 5444.75]
Trapping-surface period P_- (ms)	3.4716 [2.5064, 5.5228]	1.9978 [1.1540, 8.7527]

Table 741: CPTG compact-state, structural, clock, and geometric-rotation results for GW190706_222641. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.7349	5.7092	1793.16	3085.18
500 m inward	2.6301	4.0283	1803.33	3113.96
100 m inward	1.1736	1.7984	1808.53	3137.62
10 m inward	0.3709	0.5687	1809.76	3144.41
1 m inward	0.1173	0.1798	1809.85	3144.95
10 cm inward	0.0371	0.0569	1809.86	3145.01
1 cm inward	0.0117	0.0180	1809.86	3145.02
surface limit	0.0000	0.0000	1809.86	3145.02

Table 742: Near-inner-boundary profile for GW190706_222641. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $56.816 M_\odot$ and $27.510 M_\odot$, with completed geometric mass-equivalent shares of 17.33% and 28.22%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $55.713 < 76.606 < 105.926$ km, and component 2 gives $29.769 < 39.022 < 56.584$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5593$ and 1.6598; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.356 and 16.493 h. The determinant-averaged frame-dragging rates at those shells are 1054.30 and 1985.22 rad s $^{-1}$, approaching surface-limit values of 1809.86 and 3145.02 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.735/5.709 h at 1 km inward to 0.0117/0.0180 h at 1 cm inward for components 1/2. Both components remain toward the lower-share side of the M4 geometric mass-equivalent distribution. At least one Structural-Mode shell retains comparatively more proper time than most M4 systems. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.25 GW241116_151753

GW241116_151753 is an M4 high observed-mass binary with a strongly asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 134.414 M_\odot$ and $m_2 = 48.473 M_\odot$, while the luminosity-distance median is 5738.0 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Strongly asymmetric (Q1)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	21,504
Observed luminosity distance	5738.0 Mpc (median)
Detector-frame masses	$m_1 = 134.414 M_\odot$, $m_2 = 48.473 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.361$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8720$; $z_\pi = 0.9440$ (medians)
Post-handoff reorder fraction	6.07%

Table 743: Observational and coordinate summary for GW241116_151753. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	67.287 [38.147, 123.548]	23.401 [10.861, 50.937]
Physical compact-source mass M_c ($M_\odot$)	54.446 [27.813, 114.825]	16.697 [6.848, 38.194]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.675 [1.051, 28.312]	5.768 [0.640, 20.008]
Geometric share M_{geom}/M_∞ (%)	12.60 [1.50, 45.20]	28.37 [3.23, 49.58]
Exterior spin magnitude $ \hat{a}_\infty $	0.756 [0.170, 0.970]	0.471 [0.046, 0.936]
Polarization factor A_{pol}	1.144 [1.015, 1.825]	1.396 [1.033, 1.984]
Compact-source spin parameter $\hat{a}_c$	0.990 [0.565, 1.000]	0.918 [0.181, 0.999]
GR/CPTG matching radius r_m (km)	99.358 [56.328, 182.434]	34.555 [16.038, 75.215]
Structural Mode N_{BH}	1.5234 [1.4561, 1.8563]	1.6615 [1.4652, 1.9075]
Organization-shell radius r_N (km)	52.258 [29.580, 95.201]	18.155 [8.444, 39.617]
Inner trapping-surface radius r_- (km)	73.492 [40.204, 152.596]	23.867 [10.441, 52.352]
Structural- N lapse $\bar{\alpha}_N$	0.7928 [0.5286, 0.8465]	0.6859 [0.4867, 0.8391]
Structural- N clock $T_{24,N}$ (h)	19.028 [12.685, 20.315]	16.463 [11.680, 20.139]
Structural- N axis ω (rad s $^{-1}$)	698.76 [254.92, 1870.17]	2475.45 [690.88, 7288.11]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1009.02 [409.17, 2178.76]	3090.85 [876.36, 8169.51]
Structural- N equatorial ω (rad s $^{-1}$)	1356.21 [637.41, 2551.04]	3714.95 [916.94, 9113.22]
Structural- N period P_ω (ms)	6.2270 [2.8838, 15.3560]	2.0328 [0.7691, 7.1697]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1829.79 [899.49, 3409.79]	4757.62 [1005.62, 11730.46]
Trapping-surface period P_- (ms)	3.4338 [1.8427, 6.9853]	1.3207 [0.5356, 6.2481]

Table 744: CPTG compact-state, structural, clock, and geometric-rotation results for GW241116_151753. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.5133	7.1085	1814.52	4593.90
500 m inward	2.4702	5.0047	1822.09	4675.81
100 m inward	1.0989	2.2239	1828.49	4743.10
10 m inward	0.3471	0.7024	1829.73	4756.48
1 m inward	0.1097	0.2221	1829.78	4757.51
10 cm inward	0.0347	0.0702	1829.79	4757.61
1 cm inward	0.0110	0.0222	1829.79	4757.62
surface limit	0.0000	0.0000	1829.79	4757.62

Table 745: Near-inner-boundary profile for GW241116_151753. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $54.446 M_\odot$ and $16.697 M_\odot$, with completed geometric mass-equivalent shares of 12.60% and 28.37%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $52.258 < 73.492 < 99.358$ km, and component 2 gives $18.155 < 23.867 < 34.555$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5234$ and 1.6615 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 19.028 and 16.463 h. The determinant-averaged frame-dragging rates at those shells are 1009.02 and 3090.85 rad s $^{-1}$, approaching surface-limit values of 1829.79 and 4757.62 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 3.513/7.109 h at 1 km inward to 0.0110/0.0222 h at 1 cm inward for components 1/2. Its strong observational mass asymmetry makes the contrast between the two solved compact interiors particularly useful for component-by-component comparison. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.26 GW230814_061920

GW230814_061920 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 114.366 M_\odot$ and $m_2 = 70.112 M_\odot$, while the luminosity-distance median is 3967.8 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the intermediate-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	43,680
Observed luminosity distance	3967.8 Mpc (median)
Detector-frame masses	$m_1 = 114.366 M_\odot$, $m_2 = 70.112 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.613$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6450$; $z_\pi = 0.6958$ (medians)
Post-handoff reorder fraction	12.22%

Table 746: Observational and coordinate summary for GW230814_061920. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	66.498 [44.716, 87.217]	40.782 [24.538, 63.672]
Physical compact-source mass $M_c (M_\odot)$	45.684 [31.222, 70.501]	27.684 [15.582, 43.483]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	18.302 [2.098, 36.538]	12.910 [1.581, 28.635]
Geometric share M_{geom}/M_∞ (%)	27.87 [3.67, 49.37]	33.91 [4.55, 49.77]
Exterior spin magnitude $ \hat{a}_\infty $	0.480 [0.057, 0.927]	0.375 [0.034, 0.910]
Polarization factor A_{pol}	1.386 [1.038, 1.975]	1.513 [1.048, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.922 [0.222, 0.999]	0.858 [0.134, 0.999]
GR/CPTG matching radius r_m (km)	98.193 [66.029, 128.786]	60.220 [36.234, 94.019]
Structural Mode N_{BH}	1.6562 [1.4676, 1.9050]	1.7221 [1.4725, 1.9097]
Organization-shell radius r_N (km)	51.729 [34.682, 67.824]	31.698 [19.034, 49.500]
Inner trapping-surface radius r_- (km)	65.336 [46.761, 94.016]	40.176 [23.738, 60.956]
Structural- N lapse $\bar{\alpha}_N$	0.6900 [0.4886, 0.8372]	0.6385 [0.4850, 0.8333]
Structural- N clock $T_{24,N}$ (h)	16.560 [11.726, 20.092]	15.325 [11.640, 19.999]
Structural- N axis ω (rad s $^{-1}$)	1012.69 [366.54, 1694.65]	1611.46 [408.84, 3196.85]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1237.26 [453.26, 1913.38]	1926.17 [409.47, 3589.46]
Structural- N equatorial ω (rad s $^{-1}$)	1466.68 [455.81, 2190.38]	2275.46 [409.61, 4004.49]
Structural- N period $P_{\bar{\omega}}$ (ms)	5.0783 [3.2838, 13.8622]	3.2620 [1.7505, 15.3446]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1897.73 [498.31, 2915.85]	2922.61 [446.68, 5165.05]
Trapping-surface period P_- (ms)	3.3109 [2.1548, 12.6089]	2.1499 [1.2165, 14.0664]

Table 747: CPTG compact-state, structural, clock, and geometric-rotation results for GW230814_061920. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.4490	5.7824	1875.21	2865.40
500 m inward	3.1419	4.0861	1886.61	2896.19
100 m inward	1.4036	1.8263	1895.51	2917.31
10 m inward	0.4438	0.5775	1897.52	2922.10
1 m inward	0.1403	0.1826	1897.72	2922.56
10 cm inward	0.0444	0.0577	1897.73	2922.61
1 cm inward	0.0140	0.0183	1897.73	2922.61
surface limit	0.0000	0.0000	1897.73	2922.61

Table 748: Near-inner-boundary profile for GW230814_061920. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $45.684 M_\odot$ and $27.684 M_\odot$, with completed geometric mass-equivalent shares of 27.87% and 33.91%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $51.729 < 65.336 < 98.193$ km, and component 2 gives $31.698 < 40.176 < 60.220$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6562$ and 1.7221 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.560 and 15.325 h. The determinant-averaged frame-dragging rates at those shells are 1237.26 and 1926.17 rad s $^{-1}$, approaching surface-limit values of 1897.73 and 2922.61 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.449/5.782 h at 1 km inward to 0.0140/0.0183 h at 1 cm inward for components 1/2. Its Structural-Mode frame dragging is comparatively slow within M4. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.27 GW241125_010116

GW241125_010116 is an M4 high observed-mass binary with a near-equal-mass detector-frame mass ratio. The released marginal medians are $m_1 = 103.699 M_\odot$ and $m_2 = 82.696 M_\odot$, while the luminosity-distance median is 4722.5 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut.

Observational class	M4: High observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,275
Observed luminosity distance	4722.5 Mpc (median)
Detector-frame masses	$m_1 = 103.699 M_\odot$, $m_2 = 82.696 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.797$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7440$; $z_\pi = 0.8037$ (medians)
Post-handoff reorder fraction	23.94%

Table 749: Observational and coordinate summary for GW241125_010116. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	56.128 [41.116, 71.440]	46.750 [28.948, 63.171]
Physical compact-source mass $M_c (M_\odot)$	43.060 [28.818, 61.145]	29.773 [18.222, 44.992]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	10.633 [1.160, 28.996]	16.219 [2.762, 28.654]
Geometric share M_{geom}/M_∞ (%)	19.50 [2.23, 47.75]	36.80 [7.24, 49.82]
Exterior spin magnitude $ \hat{a}_\infty $	0.629 [0.111, 0.956]	0.325 [0.030, 0.858]
Polarization factor A_{pol}	1.242 [1.023, 1.914]	1.582 [1.078, 1.993]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.406, 1.000]	0.813 [0.121, 0.997]
GR/CPTG matching radius r_m (km)	82.881 [60.713, 105.491]	69.032 [42.745, 93.280]
Structural Mode N_{BH}	1.5773 [1.4599, 1.8864]	1.7555 [1.4881, 1.9102]
Organization-shell radius r_N (km)	43.566 [31.894, 55.483]	36.383 [22.521, 49.094]
Inner trapping-surface radius r_- (km)	58.941 [42.941, 81.362]	44.418 [27.942, 61.439]
Structural- N lapse $\bar{\alpha}_N$	0.7509 [0.5036, 0.8434]	0.6118 [0.4846, 0.8207]
Structural- N clock $T_{24,N}$ (h)	18.021 [12.086, 20.242]	14.683 [11.631, 19.697]
Structural- N axis ω (rad s $^{-1}$)	1028.73 [464.70, 1885.44]	1544.46 [345.79, 2747.54]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1360.26 [710.65, 2137.86]	1781.78 [345.95, 3069.78]
Structural- N equatorial ω (rad s $^{-1}$)	1712.84 [969.18, 2414.69]	2028.57 [346.03, 3415.00]
Structural- N period P_ω (ms)	4.6191 [2.9390, 8.8414]	3.5264 [2.0468, 18.1620]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2293.25 [1079.58, 3198.32]	2549.52 [377.06, 4396.32]
Trapping-surface period P_- (ms)	2.7399 [1.9645, 5.8200]	2.4645 [1.4292, 16.6636]

Table 750: CPTG compact-state, structural, clock, and geometric-rotation results for GW241125_010116. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.3775	5.6533	2268.02	2503.20
500 m inward	3.0849	3.9996	2281.85	2527.53
100 m inward	1.3754	1.7889	2291.04	2545.35
10 m inward	0.4346	0.5658	2292.91	2549.05
1 m inward	0.1374	0.1789	2293.21	2549.47
10 cm inward	0.0435	0.0566	2293.25	2549.52
1 cm inward	0.0137	0.0179	2293.25	2549.52
surface limit	0.0000	0.0000	2293.25	2549.52

Table 751: Near-inner-boundary profile for GW241125_010116. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $43.060 M_\odot$ and $29.773 M_\odot$, with completed geometric mass-equivalent shares of 19.50% and 36.80%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $43.566 < 58.941 < 82.881$ km, and component 2 gives $36.383 < 44.418 < 69.032$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.5773$ and 1.7555; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 18.021 and 14.683 h. The determinant-averaged frame-dragging rates at those shells are 1360.26 and 1781.78 rad s $^{-1}$, approaching surface-limit values of 2293.25 and 2549.52 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.378/5.653 h at 1 km inward to 0.0137/0.0179 h at 1 cm inward for components 1/2. At least one component lies in the upper quartile of M4 geometric mass-equivalent share. At least one Structural-Mode shell is among the more strongly time-dilated M4 cases. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.28 GW240618_071627

GW240618_071627 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 120.181 M_\odot$ and $m_2 = 75.241 M_\odot$, while the luminosity-distance median is 5479.3 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	15,339
Observed luminosity distance	5479.3 Mpc (median)
Detector-frame masses	$m_1 = 120.181 M_\odot$, $m_2 = 75.241 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.626$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8399$; $z_\pi = 0.9087$ (medians)
Post-handoff reorder fraction	11.59%

Table 752: Observational and coordinate summary for GW240618_071627. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	62.137 [40.573, 86.525]	38.604 [23.534, 59.545]
Physical compact-source mass M_c ($M_\odot$)	45.686 [28.738, 71.500]	26.336 [14.798, 42.317]
Geometric mass-equivalent M_{geom} ($M_\odot$)	13.540 [1.507, 34.478]	11.781 [1.495, 25.968]
Geometric share M_{geom}/M_∞ (%)	22.28 [2.76, 48.88]	32.46 [4.45, 49.78]
Exterior spin magnitude $ \hat{a}_\infty $	0.579 [0.077, 0.945]	0.400 [0.033, 0.912]
Polarization factor A_{pol}	1.287 [1.028, 1.956]	1.481 [1.047, 1.991]
Compact-source spin parameter $\hat{a}_c$	0.958 [0.293, 1.000]	0.877 [0.132, 0.999]
GR/CPTG matching radius r_m (km)	91.752 [59.912, 127.765]	57.004 [34.750, 87.926]
Structural Mode N_{BH}	1.6018 [1.4627, 1.8994]	1.7056 [1.4719, 1.9097]
Organization-shell radius r_N (km)	48.269 [31.494, 67.272]	29.996 [18.277, 46.318]
Inner trapping-surface radius r_- (km)	63.994 [42.343, 95.146]	38.215 [22.669, 58.548]
Structural- N lapse $\bar{\alpha}_N$	0.7319 [0.4930, 0.8412]	0.6515 [0.4849, 0.8338]
Structural- N clock $T_{24,N}$ (h)	17.566 [11.833, 20.188]	15.636 [11.638, 20.010]
Structural- N axis ω (rad s $^{-1}$)	981.11 [387.23, 1839.54]	1679.68 [437.18, 3342.41]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1251.27 [556.10, 2101.49]	2020.09 [439.74, 3759.78]
Structural- N equatorial ω (rad s $^{-1}$)	1539.04 [619.09, 2430.50]	2394.98 [439.86, 4208.00]
Structural- N period P_ω (ms)	5.0215 [2.9899, 11.2987]	3.1104 [1.6712, 14.2884]
Trapping-surface limit Ω_- (rad s $^{-1}$)	2040.24 [681.59, 3237.78]	3080.94 [479.36, 5414.30]
Trapping-surface period P_- (ms)	3.0796 [1.9406, 9.2185]	2.0394 [1.1605, 13.1074]

Table 753: CPTG compact-state, structural, clock, and geometric-rotation results for GW240618_071627. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.3169	5.9141	2017.07	3020.40
500 m inward	3.0442	4.1783	2028.40	3053.69
100 m inward	1.3576	1.8665	2037.88	3075.78
10 m inward	0.4291	0.5900	2039.98	3080.71
1 m inward	0.1357	0.1866	2040.22	3080.92
10 cm inward	0.0429	0.0590	2040.24	3080.93
1 cm inward	0.0136	0.0187	2040.24	3080.94
surface limit	0.0000	0.0000	2040.24	3080.94

Table 754: Near-inner-boundary profile for GW240618_071627. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $45.686 M_\odot$ and $26.336 M_\odot$, with completed geometric mass-equivalent shares of 22.28% and 32.46%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $48.269 < 63.994 < 91.752$ km, and component 2 gives $29.996 < 38.215 < 57.004$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6018$ and 1.7056 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.566 and 15.636 h. The determinant-averaged frame-dragging rates at those shells are 1251.27 and 2020.09 rad s $^{-1}$, approaching surface-limit values of 2040.24 and 3080.94 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.317/5.914 h at 1 km inward to 0.0136/0.0187 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.29 GW240824_205609

GW240824_205609 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 128.190 M_\odot$ and $m_2 = 69.451 M_\odot$, while the luminosity-distance median is 5277.1 Mpc. One retained release label defines this numerical stream. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:IMRPhenomXPHM-SpinTaylor
Posterior samples	18,553
Observed luminosity distance	5277.1 Mpc (median)
Detector-frame masses	$m_1 = 128.190 M_\odot$, $m_2 = 69.451 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.542$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.8146$; $z_\pi = 0.8810$ (medians)
Post-handoff reorder fraction	9.31%

Table 755: Observational and coordinate summary for GW240824_205609. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	67.266 [41.454, 99.655]	35.911 [18.936, 59.781]
Physical compact-source mass M_c ($M_\odot$)	48.413 [29.292, 80.879]	25.000 [12.583, 42.729]
Geometric mass-equivalent M_{geom} ($M_\odot$)	15.701 [1.528, 39.399]	9.984 [1.102, 25.591]
Geometric share M_{geom}/M_∞ (%)	24.06 [2.57, 49.23]	30.18 [3.68, 49.68]
Exterior spin magnitude $ \hat{a}_\infty $	0.547 [0.063, 0.949]	0.440 [0.040, 0.927]
Polarization factor A_{pol}	1.317 [1.026, 1.970]	1.432 [1.038, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.948 [0.245, 1.000]	0.902 [0.159, 0.999]
GR/CPTG matching radius r_m (km)	99.327 [61.212, 147.153]	53.028 [27.962, 88.274]
Structural Mode N_{BH}	1.6184 [1.4617, 1.9034]	1.6806 [1.4676, 1.9086]
Organization-shell radius r_N (km)	52.226 [32.133, 77.477]	27.914 [14.706, 46.520]
Inner trapping-surface radius r_- (km)	68.394 [43.163, 108.326]	36.006 [18.725, 59.476]
Structural- N lapse $\bar{\alpha}_N$	0.7192 [0.4899, 0.8420]	0.6711 [0.4858, 0.8372]
Structural- N clock $T_{24,N}$ (h)	17.260 [11.757, 20.207]	16.107 [11.660, 20.092]
Structural- N axis ω (rad s $^{-1}$)	905.29 [331.31, 1792.52]	1760.35 [523.97, 3951.02]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1138.80 [448.39, 2049.69]	2149.11 [549.37, 4504.01]
Structural- N equatorial ω (rad s $^{-1}$)	1396.70 [487.11, 2365.05]	2566.61 [549.52, 5117.97]
Structural- N period P_ω (ms)	5.5174 [3.0654, 14.0128]	2.9236 [1.3950, 11.4370]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1832.57 [532.62, 3140.19]	3319.82 [599.48, 6660.08]
Trapping-surface period P_- (ms)	3.4286 [2.0009, 11.7968]	1.8926 [0.9434, 10.4810]

Table 756: CPTG compact-state, structural, clock, and geometric-rotation results for GW240824_205609. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.2072	5.9937	1813.82	3248.31
500 m inward	2.9690	4.2281	1823.01	3285.20
100 m inward	1.3256	1.8879	1830.82	3313.21
10 m inward	0.4190	0.5967	1832.48	3319.06
1 m inward	0.1325	0.1887	1832.55	3319.73
10 cm inward	0.0419	0.0597	1832.56	3319.81
1 cm inward	0.0132	0.0189	1832.57	3319.82
surface limit	0.0000	0.0000	1832.57	3319.82

Table 757: Near-inner-boundary profile for GW240824_205609. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $48.413 M_\odot$ and $25.000 M_\odot$, with completed geometric mass-equivalent shares of 24.06% and 30.18%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $52.226 < 68.394 < 99.327$ km, and component 2 gives $27.914 < 36.006 < 53.028$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6184$ and 1.6806 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 17.260 and 16.107 h. The determinant-averaged frame-dragging rates at those shells are 1138.80 and 2149.11 rad s $^{-1}$, approaching surface-limit values of 1832.57 and 3319.82 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.207/5.994 h at 1 km inward to 0.0132/0.0189 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. Both components remain toward the lower-share side of the M4 geometric mass-equivalent distribution. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

6.30 GW231001_140220

GW231001_140220 is an M4 high observed-mass binary with a moderately asymmetric detector-frame mass ratio. The released marginal medians are $m_1 = 128.513 M_\odot$ and $m_2 = 70.307 M_\odot$, while the luminosity-distance median is 4421.4 Mpc. 2 retained release labels are preserved for provenance and are numerically identical under the declared transformation. It lies in the farther-distance observational cross-cut. It carries the predeclared broad-posterior flag, so its candidate-level interpretation retains the full reported uncertainty.

Observational class	M4: High observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	29,438
Observed luminosity distance	4421.4 Mpc (median)
Detector-frame masses	$m_1 = 128.513 M_\odot$, $m_2 = 70.307 M_\odot$ (marginal medians)
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.547$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.7049$; $z_\pi = 0.7611$ (medians)
Post-handoff reorder fraction	8.66%

Table 758: Observational and coordinate summary for GW231001_140220. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	72.687 [46.710, 98.643]	39.388 [24.171, 62.602]
Physical compact-source mass M_c ($M_\odot$)	49.709 [33.016, 77.664]	27.404 [15.379, 43.952]
Geometric mass-equivalent M_{geom} ($M_\odot$)	20.207 [2.076, 41.611]	11.717 [1.338, 27.302]
Geometric share M_{geom}/M_∞ (%)	28.36 [3.39, 49.44]	31.59 [4.03, 49.73]
Exterior spin magnitude $ \hat{a}_\infty $	0.471 [0.054, 0.933]	0.415 [0.037, 0.920]
Polarization factor A_{pol}	1.396 [1.035, 1.978]	1.462 [1.042, 1.989]
Compact-source spin parameter $\hat{a}_c$	0.918 [0.209, 0.999]	0.887 [0.146, 0.999]
GR/CPTG matching radius r_m (km)	107.332 [68.972, 145.658]	58.161 [35.691, 92.440]
Structural Mode N_{BH}	1.6613 [1.4661, 1.9059]	1.6960 [1.4696, 1.9092]
Organization-shell radius r_N (km)	56.542 [36.225, 76.708]	30.627 [18.771, 48.687]
Inner trapping-surface radius r_- (km)	71.600 [48.930, 104.135]	39.396 [23.509, 61.405]
Structural- N lapse $\bar{\alpha}_N$	0.6861 [0.4880, 0.8384]	0.6591 [0.4854, 0.8356]
Structural- N clock $T_{24,N}$ (h)	16.466 [11.711, 20.122]	15.818 [11.649, 20.055]
Structural- N axis ω (rad s $^{-1}$)	918.98 [316.36, 1591.28]	1623.73 [461.62, 3247.26]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	1119.80 [381.81, 1805.97]	1964.34 [468.73, 3672.89]
Structural- N equatorial ω (rad s $^{-1}$)	1333.14 [384.71, 2092.12]	2355.52 [469.00, 4085.24]
Structural- N period P_ω (ms)	5.6110 [3.4791, 16.4563]	3.1986 [1.7107, 13.4046]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1722.05 [420.11, 2791.97]	3055.79 [511.62, 5301.27]
Trapping-surface period P_- (ms)	3.6487 [2.2504, 14.9562]	2.0562 [1.1852, 12.2809]

Table 759: CPTG compact-state, structural, clock, and geometric-rotation results for GW231001_140220. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical compact-source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	4.2727	5.7834	1704.23	2997.98
500 m inward	3.0181	4.0859	1713.31	3026.63
100 m inward	1.3484	1.8251	1720.18	3050.99
10 m inward	0.4263	0.5770	1721.87	3055.29
1 m inward	0.1348	0.1825	1722.03	3055.72
10 cm inward	0.0426	0.0577	1722.05	3055.79
1 cm inward	0.0135	0.0182	1722.05	3055.79
surface limit	0.0000	0.0000	1722.05	3055.79

Table 760: Near-inner-boundary profile for GW231001_140220. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $49.709 M_\odot$ and $27.404 M_\odot$, with completed geometric mass-equivalent shares of 28.36% and 31.59%. The median radial ordering is $r_N < r_- < r_m$ for both components: component 1 gives $56.542 < 71.600 < 107.332$ km, and component 2 gives $30.627 < 39.396 < 58.161$ km. Their Structural-Mode fingerprints are $N_{\text{BH}} = 1.6613$ and 1.6960 ; during 24 source-local exterior Kerr hours, the corresponding organization-shell clocks accumulate 16.466 and 15.818 h. The determinant-averaged frame-dragging rates at those shells are 1119.80 and 1964.34 rad s $^{-1}$, approaching surface-limit values of 1722.05 and 3055.79 rad s $^{-1}$. Across the registered radial sequence, the median clock falls from 4.273/5.783 h at 1 km inward to 0.0135/0.0182 h at 1 cm inward for components 1/2. The broad observational posterior is retained rather than filtered, so the quoted intervals are especially important for interpreting this system. These rotation values describe geometric ZAMO frame dragging and are not material angular velocities.

7 M5: Extreme Observed-Mass Black-Hole Binaries

The final observational class contains seven unique numerical streams representing six astrophysical events. Their detector-frame total masses, constructed from the released component marginal medians, span 242.018 – $327.048 M_\odot$. The lowest M5 stream therefore begins more than $43 M_\odot$ above the upper edge of M4 at $198.821 M_\odot$, producing a naturally isolated extreme-mass tail in the present observational distribution. This gap was found before the CPTG interior quantities were examined; no compact-state, Structural-Mode, clock, or rotation result is used to create the class.

Across the seven numerical streams, 0 are strongly asymmetric, 4 are moderately asymmetric, and 3 are near-equal in the detector-frame mass-ratio screen. The distance cross-cuts contain 2 nearer-distance streams, 2 intermediate-distance streams, and 3 farther-distance streams. 4 streams carry the predeclared broad-posterior flag. Five events have one retained numerical stream. GW231123_135430 is treated differently because its C00:Mixed and C00:Mixed+X04a release branches remain numerically distinct after the fixed transformation. They therefore remain separate compact reconstructions inside one event subsection and are never averaged.

Because M5 contains only six events, the candidate discussions are intentionally more detailed than in the larger classes. Each event is still evaluated with the same fixed operators used everywhere else: observational coordinate reconstruction, compact mass–spin closure, GR/CPTG matching radius, Structural Mode, inner trapping geometry, proper-time readout, and geometric frame dragging. The additional prose is descriptive only and does not introduce an event-specific fit.

7.1 GW190521_030229

GW190521_030229 is the 254th of 260 unique numerical streams when ordered by the detector-frame total constructed from the released component marginal medians, placing it in the isolated M5 extreme-mass tail. Its detector-frame medians are $m_1 = 152.364 M_\odot$ and $m_2 = 89.654 M_\odot$, with $q_{\text{det}} = 0.588$ and an observed luminosity-distance median of 3310.9 Mpc. The event is moderately asymmetric in the mass-ratio screen and lies in the intermediate-distance observational cross-cut. One retained release label defines this numerical stream. It carries the predeclared broad-posterior flag, so the complete posterior interval remains central to the interpretation.

Observational class	M5: Extreme observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C01:Mixed
Posterior samples	24,545
Observed luminosity distance	3310.9 Mpc (median)
Detector-frame masses	$m_1 = 152.364 M_\odot$, $m_2 = 89.654 M_\odot$ (marginal medians)
Detector-frame total from marginal medians	$242.018 M_\odot$
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.588$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.5560$; $z_\pi = 0.5984$ (medians)
Post-handoff reorder fraction	8.85%

Table 761: Observational and coordinate summary for GW190521_030229. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass $M_\infty (M_\odot)$	95.611 [65.952, 129.398]	55.892 [26.620, 90.996]
Physical compact-source mass $M_c (M_\odot)$	79.201 [46.305, 123.315]	39.412 [19.545, 69.827]
Geometric mass-equivalent $M_{\text{geom}} (M_\odot)$	11.900 [1.317, 42.200]	14.916 [1.604, 35.252]
Geometric share M_{geom}/M_∞ (%)	12.73 [1.32, 46.57]	28.76 [3.03, 49.67]
Exterior spin magnitude $ \hat{a}_\infty $	0.754 [0.140, 0.974]	0.464 [0.041, 0.940]
Polarization factor A_{pol}	1.146 [1.013, 1.872]	1.404 [1.031, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.989 [0.490, 1.000]	0.915 [0.161, 1.000]
GR/CPTG matching radius r_m (km)	141.182 [97.386, 191.072]	82.532 [39.308, 134.367]
Structural Mode N_{BH}	1.5243 [1.4552, 1.8726]	1.6655 [1.4641, 1.9085]
Organization-shell radius r_N (km)	74.099 [51.147, 99.670]	43.415 [20.666, 70.530]
Inner trapping-surface radius r_- (km)	105.427 [68.934, 165.377]	56.090 [27.594, 94.388]
Structural- N lapse $\bar{\alpha}_N$	0.7921 [0.5149, 0.8472]	0.6828 [0.4859, 0.8400]
Structural- N clock $T_{24,N}$ (h)	19.010 [12.359, 20.333]	16.388 [11.661, 20.160]
Structural- N axis ω (rad s $^{-1}$)	482.28 [223.52, 1141.12]	1132.64 [292.59, 2341.73]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	697.62 [361.90, 1290.81]	1374.36 [381.85, 2765.06]
Structural- N equatorial ω (rad s $^{-1}$)	951.10 [572.40, 1482.20]	1664.33 [386.84, 3367.54]
Structural- N period P_ω (ms)	9.0066 [4.8676, 17.3619]	4.5717 [2.2723, 16.4546]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1303.96 [782.02, 1981.53]	2174.70 [422.11, 4476.46]
Trapping-surface period P_- (ms)	4.8185 [3.1709, 8.0346]	2.8892 [1.4036, 14.8852]

Table 762: Full CPTG compact-state and finite-core results for GW190521_030229. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	2.9922	4.8358	1297.13	2148.73
500 m inward	2.1085	3.4154	1300.89	2161.68
100 m inward	0.9399	1.5258	1303.56	2172.00
10 m inward	0.2970	0.4824	1303.93	2174.37
1 m inward	0.0939	0.1525	1303.96	2174.66
10 cm inward	0.0297	0.0482	1303.96	2174.69
1 cm inward	0.0094	0.0153	1303.96	2174.70
surface limit	0.0000	0.0000	1303.96	2174.70

Table 763: Near-inner-boundary profile for GW190521_030229. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $79.201 M_\odot$ and $39.412 M_\odot$, with completed geometric mass-equivalent shares of 12.73% and 28.76%. The median radial ordering remains $r_N < r_- < r_m$ in both components: $74.099 < 105.427 < 141.182 \text{ km}$ for component 1 and $43.415 < 56.090 < 82.532 \text{ km}$ for component 2. The Structural-Mode fingerprints are $N_{\text{BH}} = 1.5243$ and 1.6655 , with organization-shell clock accumulations of 19.010 and 16.388 h per 24 exterior Kerr hours. The corresponding determinant-averaged frame-dragging rates are 697.62 and $1374.36 \text{ rad s}^{-1}$, approaching trapping-surface limits of 1303.96 and $2174.70 \text{ rad s}^{-1}$. Across the registered near-boundary sequence, the median clock falls from $2.992/4.836 \text{ h}$ at 1 km inward to $0.0094/0.0153 \text{ h}$ at 1 cm inward for components 1/2 before reaching the zero-lapse trapping-surface limit. This system sits at the lower edge of the observationally isolated M5 tail, yet its primary compact solution carries the largest exterior-spin median among the five single-stream M5 events. The two components consequently separate strongly in geometric share and Structural-Mode rotation, making the binary useful for contrasting two different interior organizations inside the same extreme-mass event. These rotation values are geometric ZAMO frame-dragging rates and are not material angular velocities.

7.2 GW230922_040658

GW230922_040658 is the 255th of 260 unique numerical streams when ordered by the detector-frame total constructed from the released component marginal medians, placing it in the isolated M5 extreme-mass tail. Its detector-frame medians are $m_1 = 146.265 M_\odot$ and $m_2 = 103.373 M_\odot$, with $q_{\text{det}} = 0.707$ and an observed luminosity-distance median of 6462.7 Mpc . The event is near-equal-mass in the mass-ratio screen and lies in the farther-distance observational cross-cut. One retained release label defines this numerical stream.

Observational class	M5: Extreme observed-mass regime; Near-equal mass (Q3)
Retained release label(s)	C00:Mixed
Posterior samples	40,731
Observed luminosity distance	6462.7 Mpc (median)
Detector-frame masses	$m_1 = 146.265 M_\odot$, $m_2 = 103.373 M_\odot$ (marginal medians)
Detector-frame total from marginal medians	$249.638 M_\odot$
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.707$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9605$; $z_\pi = 1.0413$ (medians)
Post-handoff reorder fraction	16.24%

Table 764: Observational and coordinate summary for GW230922_040658. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	71.399 [51.059, 100.754]	49.774 [25.846, 75.762]
Physical compact-source mass M_c ($M_\odot$)	58.634 [39.470, 85.413]	34.362 [17.289, 55.437]
Geometric mass-equivalent M_{geom} ($M_\odot$)	10.737 [1.241, 33.238]	14.046 [1.659, 32.300]
Geometric share M_{geom}/M_∞ (%)	15.11 [1.93, 41.71]	30.63 [4.16, 49.59]
Exterior spin magnitude $ \hat{a}_\infty $	0.709 [0.237, 0.962]	0.432 [0.045, 0.918]
Polarization factor A_{pol}	1.178 [1.020, 1.716]	1.442 [1.043, 1.984]
Compact-source spin parameter $\hat{a}_c$	0.984 [0.698, 1.000]	0.897 [0.179, 0.999]
GR/CPTG matching radius r_m (km)	105.430 [75.395, 148.776]	73.497 [38.164, 111.872]
Structural Mode N_{BH}	1.5419 [1.4584, 1.8144]	1.6855 [1.4703, 1.9076]
Organization-shell radius r_N (km)	55.404 [39.489, 78.276]	38.685 [20.084, 58.925]
Inner trapping-surface radius r_- (km)	78.933 [55.845, 113.669]	49.671 [25.668, 75.629]
Structural- N lapse $\bar{\alpha}_N$	0.7784 [0.5636, 0.8447]	0.6673 [0.4866, 0.8350]
Structural- N clock $T_{24,N}$ (h)	18.681 [13.526, 20.272]	16.014 [11.678, 20.041]
Structural- N axis ω (rad s $^{-1}$)	705.37 [361.20, 1412.05]	1335.58 [435.38, 2885.22]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	991.39 [574.75, 1630.08]	1598.91 [473.34, 3290.36]
Structural- N equatorial ω (rad s $^{-1}$)	1317.81 [862.13, 1890.52]	1890.23 [473.72, 3762.26]
Structural- N period P_ω (ms)	6.3378 [3.8545, 10.9320]	3.9297 [1.9096, 13.2743]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1802.21 [1172.41, 2545.58]	2442.10 [516.64, 4870.27]
Trapping-surface period P_- (ms)	3.4864 [2.4683, 5.3592]	2.5729 [1.2901, 12.1617]

Table 765: Full CPTG compact-state and finite-core results for GW230922_040658. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.5412	5.1451	1788.49	2405.53
500 m inward	2.4943	3.6346	1795.52	2425.09
100 m inward	1.1119	1.6238	1800.65	2438.88
10 m inward	0.3514	0.5133	1802.10	2441.71
1 m inward	0.1111	0.1623	1802.21	2442.07
10 cm inward	0.0351	0.0513	1802.21	2442.10
1 cm inward	0.0111	0.0162	1802.21	2442.10
surface limit	0.0000	0.0000	1802.21	2442.10

Table 766: Near-inner-boundary profile for GW230922_040658. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $58.634 M_\odot$ and $34.362 M_\odot$, with completed geometric mass-equivalent shares of 15.11% and 30.63%. The median radial ordering remains $r_N < r_- < r_m$ in both components: $55.404 < 78.933 < 105.430$ km for component 1 and $38.685 < 49.671 < 73.497$ km for component 2. The Structural-Mode fingerprints are $N_{\text{BH}} = 1.5419$ and 1.6855 , with organization-shell clock accumulations of 18.681 and 16.014 h per 24 exterior Kerr hours. The corresponding determinant-averaged frame-dragging rates are 991.39 and 1598.91 rad s $^{-1}$, approaching trapping-surface limits of 1802.21 and 2442.10 rad s $^{-1}$. Across the registered near-boundary sequence, the median clock falls from 3.541/5.145 h at 1 km inward to 0.0111/0.0162 h at 1 cm inward for components 1/2 before reaching the zero-lapse trapping-surface limit. The system combines a near-equal detector-frame mass ratio with a farther-distance observational classification. Its secondary component carries a substantially larger geometric mass-equivalent fraction than its primary while the two organization-shell clocks remain separated by more than two exterior-frame hours. These rotation values are geometric ZAMO frame-dragging rates and are not material angular velocities.

7.3 GW231005_021030

GW231005_021030 is the 256th of 260 unique numerical streams when ordered by the detector-frame total constructed from the released component marginal medians, placing it in the isolated M5 extreme-mass tail. Its detector-frame medians are $m_1 = 160.565 M_\odot$ and $m_2 = 98.878 M_\odot$, with $q_{\text{det}} = 0.616$ and an observed luminosity-distance median of 6251.1 Mpc. The event is moderately asymmetric in the mass-ratio screen and lies in the farther-distance observational cross-cut. One retained release label defines this numerical stream.

Observational class	M5: Extreme observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	27,256
Observed luminosity distance	6251.1 Mpc (median)
Detector-frame masses	$m_1 = 160.565 M_\odot$, $m_2 = 98.878 M_\odot$ (marginal medians)
Detector-frame total from marginal medians	$259.443 M_\odot$
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.616$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.9349$; $z_\pi = 1.0131$ (medians)
Post-handoff reorder fraction	11.40%

Table 767: Observational and coordinate summary for GW231005_021030. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	79.529 [53.022, 109.678]	48.285 [28.979, 75.127]
Physical compact-source mass M_c ($M_\odot$)	60.035 [37.769, 94.050]	33.185 [18.521, 53.982]
Geometric mass-equivalent M_{geom} ($M_\odot$)	15.191 [1.677, 41.886]	14.490 [1.766, 32.679]
Geometric share M_{geom}/M_∞ (%)	19.55 [2.25, 48.41]	31.94 [4.29, 49.71]
Exterior spin magnitude $ \hat{a}_\infty $	0.628 [0.092, 0.955]	0.409 [0.039, 0.915]
Polarization factor A_{pol}	1.243 [1.023, 1.938]	1.469 [1.045, 1.988]
Compact-source spin parameter $\hat{a}_c$	0.970 [0.346, 1.000]	0.883 [0.153, 0.999]
GR/CPTG matching radius r_m (km)	117.435 [78.293, 161.953]	71.299 [42.791, 110.935]
Structural Mode N_{BH}	1.5777 [1.4600, 1.8940]	1.6998 [1.4710, 1.9089]
Organization-shell radius r_N (km)	61.771 [41.074, 85.188]	37.527 [22.525, 58.465]
Inner trapping-surface radius r_- (km)	82.836 [55.692, 125.030]	47.967 [28.154, 74.371]
Structural- N lapse $\bar{\alpha}_N$	0.7506 [0.4974, 0.8433]	0.6561 [0.4856, 0.8345]
Structural- N clock $T_{24,N}$ (h)	18.014 [11.937, 20.240]	15.746 [11.654, 20.027]
Structural- N axis ω (rad s $^{-1}$)	716.95 [300.20, 1411.77]	1344.04 [396.43, 2693.79]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	945.08 [449.93, 1606.12]	1615.44 [407.03, 3039.44]
Structural- N equatorial ω (rad s $^{-1}$)	1192.69 [578.59, 1847.20]	1920.39 [407.27, 3422.43]
Structural- N period P_ω (ms)	6.6483 [3.9120, 13.9647]	3.8894 [2.0672, 15.4368]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1589.72 [646.72, 2470.10]	2477.15 [444.03, 4411.68]
Trapping-surface period P_- (ms)	3.9524 [2.5437, 9.7155]	2.5365 [1.4242, 14.1505]

Table 768: Full CPTG compact-state and finite-core results for GW231005_021030. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.6800	5.2614	1576.63	2438.27
500 m inward	2.5956	3.7157	1583.64	2458.41
100 m inward	1.1583	1.6604	1588.50	2473.58
10 m inward	0.3661	0.5250	1589.63	2476.81
1 m inward	0.1158	0.1660	1589.70	2477.11
10 cm inward	0.0366	0.0525	1589.72	2477.15
1 cm inward	0.0116	0.0166	1589.72	2477.15
surface limit	0.0000	0.0000	1589.72	2477.15

Table 769: Near-inner-boundary profile for GW231005_021030. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $60.035 M_\odot$ and $33.185 M_\odot$, with completed geometric mass-equivalent shares of 19.55% and 31.94%. The median radial ordering remains $r_N < r_- < r_m$ in both components: $61.771 < 82.836 < 117.435$ km for component 1 and $37.527 < 47.967 < 71.299$ km for component 2. The Structural-Mode fingerprints are $N_{\text{BH}} = 1.5777$ and 1.6998 , with organization-shell clock accumulations of 18.014 and 15.746 h per 24 exterior Kerr hours. The corresponding determinant-averaged frame-dragging rates are 945.08 and 1615.44 rad s $^{-1}$, approaching trapping-surface limits of 1589.72 and 2477.15 rad s $^{-1}$. Across the registered near-boundary sequence, the median clock falls from 3.680/5.261 h at 1 km inward to 0.0116/0.0166 h at 1 cm inward for components 1/2 before reaching the zero-lapse trapping-surface limit. The secondary component has the largest median geometric mass-equivalent share among the single-stream M5 systems, while the primary remains appreciably more source-mass dominated. That contrast propagates into distinct Structural-Mode fingerprints and frame-dragging rates without changing the common radial ordering of the solved interior surfaces. These rotation values are geometric ZAMO frame-dragging rates and are not material angular velocities.

7.4 GW231028_153006

GW231028_153006 is the 257th of 260 unique numerical streams when ordered by the detector-frame total constructed from the released component marginal medians, placing it in the isolated M5 extreme-mass tail. Its detector-frame medians are $m_1 = 158.640 M_\odot$ and $m_2 = 102.005 M_\odot$, with $q_{\text{det}} = 0.643$ and an observed luminosity-distance median of 4233.0 Mpc.

The event is moderately asymmetric in the mass-ratio screen and lies in the intermediate-distance observational cross-cut. One retained release label defines this numerical stream.

Observational class	M5: Extreme observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	45,063
Observed luminosity distance	4233.0 Mpc (median)
Detector-frame masses	$m_1 = 158.640 M_\odot$, $m_2 = 102.005 M_\odot$ (marginal medians)
Detector-frame total from marginal medians	$260.645 M_\odot$
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.643$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 0.6802$; $z_\pi = 0.7341$ (medians)
Post-handoff reorder fraction	8.24%

Table 770: Observational and coordinate summary for GW231028_153006. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	91.013 [70.004, 123.301]	57.467 [34.133, 81.004]
Physical compact-source mass M_c ($M_\odot$)	81.405 [61.022, 112.002]	41.322 [21.427, 66.017]
Geometric mass-equivalent M_{geom} ($M_\odot$)	8.652 [1.206, 23.822]	13.870 [1.698, 30.945]
Geometric share M_{geom}/M_∞ (%)	9.33 [1.41, 24.26]	25.28 [3.30, 49.19]
Exterior spin magnitude $ \hat{a}_\infty $	0.818 [0.543, 0.972]	0.525 [0.065, 0.935]
Polarization factor A_{pol}	1.103 [1.014, 1.320]	1.338 [1.034, 1.968]
Compact-source spin parameter $\hat{a}_c$	0.995 [0.947, 1.000]	0.941 [0.251, 0.999]
GR/CPTG matching radius r_m (km)	134.392 [103.370, 182.070]	84.857 [50.401, 119.613]
Structural Mode N_{BH}	1.5013 [1.4557, 1.6203]	1.6302 [1.4656, 1.9030]
Organization-shell radius r_N (km)	70.378 [54.026, 95.342]	44.637 [26.520, 63.111]
Inner trapping-surface radius r_- (km)	108.153 [81.439, 148.386]	58.375 [32.788, 87.967]
Structural- N lapse $\bar{\alpha}_N$	0.8103 [0.7177, 0.8468]	0.7101 [0.4902, 0.8388]
Structural- N clock $T_{24,N}$ (h)	19.447 [17.225, 20.324]	17.042 [11.765, 20.132]
Structural- N axis ω (rad s $^{-1}$)	433.22 [271.45, 824.98]	1107.32 [431.55, 2359.09]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	662.74 [443.13, 1058.16]	1363.63 [615.87, 2639.20]
Structural- N equatorial ω (rad s $^{-1}$)	971.18 [698.54, 1315.75]	1630.38 [667.33, 2936.50]
Structural- N period P_ω (ms)	9.4806 [5.9379, 14.1791]	4.6077 [2.3807, 10.2021]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1377.66 [1008.24, 1814.06]	2129.88 [729.72, 3775.08]
Trapping-surface period P_- (ms)	4.5608 [3.4636, 6.2318]	2.9500 [1.6644, 8.6104]

Table 771: Full CPTG compact-state and finite-core results for GW231028_153006. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	2.7187	4.5849	1372.87	2104.43
500 m inward	1.9129	3.2344	1375.34	2117.68
100 m inward	0.8523	1.4441	1377.19	2127.43
10 m inward	0.2693	0.4565	1377.67	2129.60
1 m inward	0.0851	0.1443	1377.67	2129.85
10 cm inward	0.0269	0.0456	1377.66	2129.88
1 cm inward	0.0085	0.0144	1377.66	2129.88
surface limit	0.0000	0.0000	1377.66	2129.88

Table 772: Near-inner-boundary profile for GW231028_153006. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of $81.405 M_\odot$ and $41.322 M_\odot$, with completed geometric mass-equivalent shares of 9.33% and 25.28%. The median radial ordering remains $r_N < r_- < r_m$ in both components: $70.378 < 108.153 < 134.392$ km for component 1 and $44.637 < 58.375 < 84.857$ km for component 2. The Structural-Mode fingerprints are $N_{\text{BH}} = 1.5013$ and 1.6302 , with organization-shell clock accumulations of 19.447 and 17.042 h per 24 exterior Kerr hours. The corresponding determinant-averaged frame-dragging rates are 662.74 and 1363.63 rad s $^{-1}$, approaching trapping-surface limits of 1377.66 and 2129.88 rad s $^{-1}$. Across the registered near-boundary sequence, the median clock falls from 2.719/4.585 h at 1 km inward to 0.0085/0.0144 h at 1 cm inward for components 1/2 before reaching the zero-lapse trapping-surface limit. This is the most source-mass-dominated single-stream M5 primary: its high exterior-spin median drives the component-1 geometric share below ten percent. The resulting organization-shell clock retains more proper time and its frame dragging is slower than in the lower-spin secondary, providing a particularly clear within-event comparison of the registered spin-controlled compact handoff. These rotation values are geometric ZAMO frame-dragging rates and are not material angular velocities.

7.5 GW230704_212616

GW230704_212616 is the 258th of 260 unique numerical streams when ordered by the detector-frame total constructed from the released component marginal medians, placing it in the isolated M5 extreme-mass tail. Its detector-frame medians are $m_1 = 181.281 M_\odot$ and $m_2 = 108.062 M_\odot$, with $q_{\text{det}} = 0.596$ and an observed luminosity-distance median of 7285.1 Mpc. The event is moderately asymmetric in the mass-ratio screen and lies in the farther-distance observational cross-cut. One retained release label defines this numerical stream. It carries the predeclared broad-posterior flag, so the complete posterior interval remains central to the interpretation.

Observational class	M5: Extreme observed-mass regime; Moderately asymmetric (Q2)
Retained release label(s)	C00:Mixed C00:Mixed+XO4a
Posterior samples	29,540
Observed luminosity distance	7285.1 Mpc (median)
Detector-frame masses	$m_1 = 181.281 M_\odot$, $m_2 = 108.062 M_\odot$ (marginal medians)
Detector-frame total from marginal medians	289.343 $M_\odot$
Detector-frame mass-ratio screen	$q_{\text{det}} = 0.596$ from marginal medians
Released / CPTG native phase	$z_{\text{rel}} = 1.0585$; $z_\pi = 1.1494$ (medians)
Post-handoff reorder fraction	10.58%

Table 773: Observational and coordinate summary for GW230704_212616. Classification is assigned only from pre-CPTG observational descriptors.

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	84.136 [54.957, 125.474]	48.466 [23.912, 82.324]
Physical compact-source mass M_c ($M_\odot$)	67.386 [40.599, 108.407]	33.890 [15.915, 59.639]
Geometric mass-equivalent M_{geom} ($M_\odot$)	13.317 [1.534, 41.285]	13.030 [1.423, 34.782]
Geometric share M_{geom}/M_∞ (%)	15.93 [1.99, 46.02]	30.00 [3.56, 49.66]
Exterior spin magnitude $ \hat{a}_\infty $	0.694 [0.152, 0.960]	0.443 [0.041, 0.929]
Polarization factor A_{pol}	1.189 [1.020, 1.852]	1.428 [1.037, 1.987]
Compact-source spin parameter $\hat{a}_c$	0.982 [0.523, 1.000]	0.904 [0.163, 0.999]
GR/CPTG matching radius r_m (km)	124.237 [81.150, 185.278]	71.566 [35.309, 121.562]
Structural Mode N_{BH}	1.5481 [1.4587, 1.8660]	1.6786 [1.4670, 1.9084]
Organization-shell radius r_N (km)	65.298 [42.587, 97.397]	37.663 [18.552, 63.966]
Inner trapping-surface radius r_- (km)	91.329 [58.774, 143.811]	48.699 [23.650, 82.149]
Structural- N lapse $\bar{\alpha}_N$	0.7735 [0.5204, 0.8444]	0.6726 [0.4860, 0.8377]
Structural- N clock $T_{24,N}$ (h)	18.563 [12.491, 20.266]	16.143 [11.663, 20.104]
Structural- N axis ω (rad s $^{-1}$)	610.47 [279.32, 1316.71]	1293.34 [386.71, 3099.01]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	846.69 [434.29, 1517.54]	1580.13 [416.30, 3523.55]
Structural- N equatorial ω (rad s $^{-1}$)	1111.71 [632.84, 1761.34]	1895.51 [416.55, 4011.47]
Structural- N period P_ω (ms)	7.4209 [4.1404, 14.4676]	3.9764 [1.7832, 15.0930]
Trapping-surface limit Ω_- (rad s $^{-1}$)	1507.21 [836.99, 2367.74]	2455.03 [454.90, 5194.25]
Trapping-surface period P_- (ms)	4.1688 [2.6537, 7.5069]	2.5593 [1.2096, 13.8123]

Table 774: Full CPTG compact-state and finite-core results for GW230704_212616. Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	3.3211	5.1305	1497.19	2417.46
500 m inward	2.3406	3.6225	1502.10	2437.76
100 m inward	1.0436	1.6172	1506.08	2451.43
10 m inward	0.3298	0.5112	1507.05	2454.64
1 m inward	0.1043	0.1616	1507.19	2455.00
10 cm inward	0.0330	0.0511	1507.21	2455.03
1 cm inward	0.0104	0.0162	1507.21	2455.03
surface limit	0.0000	0.0000	1507.21	2455.03

Table 775: Near-inner-boundary profile for GW230704_212616. Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

The CPTG mass accounting assigns physical compact-source medians of 67.386 $M_\odot$ and 33.890 $M_\odot$, with completed geometric mass-equivalent shares of 15.93% and 30.00%. The median radial ordering remains $r_N < r_- < r_m$ in both components: 65.298 < 91.329 < 124.237 km for component 1 and 37.663 < 48.699 < 71.566 km for component 2. The Structural-Mode fingerprints are $N_{\text{BH}} = 1.5481$ and 1.6786, with organization-shell clock accumulations of 18.563 and 16.143 h per 24 exterior Kerr hours. The corresponding determinant-averaged frame-dragging rates are 846.69 and 1580.13 rad s $^{-1}$, approaching trapping-surface limits of 1507.21 and 2455.03 rad s $^{-1}$. Across the registered near-boundary sequence, the median clock falls from 3.321/5.131 h at 1 km inward to 0.0104/0.0162 h at 1 cm inward for components 1/2 before reaching the zero-lapse trapping-surface limit. This is the most massive single-stream event before the branch-sensitive GW231123_135430 pair and it also carries the broad-posterior flag. The secondary again has the larger geometric share and faster Structural-Mode frame dragging, while both components preserve the same organization-shell,

trapping-surface, and matching-radius hierarchy across the posterior support. These rotation values are geometric ZAMO frame-dragging rates and are not material angular velocities.

7.6 GW231123_135430

GW231123_135430 occupies the extreme upper end of the observational distribution and is the only event in the retained sample for which two release labels remain numerically distinct after the declared transformation. The C00:Mixed+X04a stream has a detector-frame total of $325.204 M_\odot$ from the component marginal medians, while C00:Mixed gives $327.048 M_\odot$; these are the 259th and 260th numerical streams, respectively, in the observational ordering. Both are near-equal-mass, nearer-distance reconstructions and both carry the broad-posterior flag. They are therefore reported as two complete compact reconstructions of one astrophysical event rather than collapsed, selected between, or averaged.

Quantity	C00:Mixed	C00:Mixed+X04a
Posterior samples	71,756	83,280
Observed D_L median (Mpc)	1957.8	2237.8
Detector m_1 median ($M_\odot$)	180.848	185.766
Detector m_2 median ($M_\odot$)	146.200	139.438
Detector total from marginal medians ($M_\odot$)	327.048	325.204
q_{det} from marginal medians	0.808	0.751
Released z median	0.3568	0.3998
CPTG native z_π median	0.3837	0.4301
Post-handoff reorder fraction	5.19%	4.14%

Table 776: Observed and coordinate-level comparison of the two retained GW231123_135430 numerical streams. The branches are retained independently because their transformed samples are not identical.

C00:Mixed reconstruction

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	132.487 [113.269, 155.959]	103.357 [77.320, 124.196]
Physical compact-source mass M_c ($M_\odot$)	123.581 [106.112, 145.043]	92.294 [54.164, 112.945]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.529 [1.081, 26.928]	9.467 [1.124, 38.826]
Geometric share M_{geom}/M_∞ (%)	4.98 [0.85, 18.51]	9.26 [1.12, 40.20]
Exterior spin magnitude $ \hat{a}_\infty $	0.902 [0.647, 0.983]	0.819 [0.265, 0.978]
Polarization factor A_{pol}	1.052 [1.009, 1.227]	1.102 [1.011, 1.672]
Compact-source spin parameter $\hat{a}_c$	0.999 [0.974, 1.000]	0.995 [0.740, 1.000]
GR/CPTG matching radius r_m (km)	195.634 [167.256, 230.293]	152.620 [114.172, 183.390]
Structural Mode N_{BH}	1.4749 [1.4528, 1.5689]	1.5008 [1.4542, 1.7962]
Organization-shell radius r_N (km)	101.992 [87.035, 120.657]	79.850 [59.840, 96.267]
Inner trapping-surface radius r_- (km)	164.734 [141.030, 194.957]	122.607 [79.724, 151.206]
Structural- N lapse $\bar{\alpha}_N$	0.8313 [0.7574, 0.8492]	0.8106 [0.5786, 0.8480]
Structural- N clock $T_{24,N}$ (h)	19.952 [18.177, 20.380]	19.455 [13.887, 20.353]
Structural- N axis ω (rad s^{-1})	248.42 [191.23, 416.20]	379.46 [247.18, 969.33]
Structural- N determinant-average $\bar{\omega}$ (rad s^{-1})	400.68 [319.30, 567.22]	578.27 [408.96, 1106.73]
Structural- N equatorial ω (rad s^{-1})	630.66 [530.28, 747.29]	843.68 [666.80, 1258.03]
Structural- N period P_ω (ms)	15.6811 [11.0771, 19.6781]	10.8656 [5.6773, 15.3639]
Trapping-surface limit Ω_- (rad s^{-1})	907.37 [767.47, 1059.14]	1191.27 [946.94, 1660.45]
Trapping-surface period P_- (ms)	6.9246 [5.9323, 8.1869]	5.2744 [3.7840, 6.6353]

Table 777: Full CPTG compact-state and finite-core results for GW231123_135430 (C00:Mixed). Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	1.9250	2.5727	906.40	1187.62
500 m inward	1.3540	1.8115	906.96	1189.58
100 m inward	0.6029	0.8072	907.29	1190.92
10 m inward	0.1905	0.2550	907.36	1191.23
1 m inward	0.0602	0.0806	907.37	1191.26
10 cm inward	0.0190	0.0255	907.37	1191.27
1 cm inward	0.0060	0.0081	907.37	1191.27
surface limit	0.0000	0.0000	907.37	1191.27

Table 778: Near-inner-boundary profile for GW231123_135430 (C00:Mixed). Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s^{-1} . Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

Quantity	CPTG component 1	CPTG component 2
Exterior gravitational mass M_∞ ($M_\odot$)	134.016 [114.389, 158.148]	98.998 [49.260, 122.743]
Physical compact-source mass M_c ($M_\odot$)	125.547 [107.082, 149.549]	86.691 [30.451, 111.649]
Geometric mass-equivalent M_{geom} ($M_\odot$)	6.280 [1.126, 25.096]	10.532 [1.180, 36.511]
Geometric share M_{geom}/M_∞ (%)	4.67 [0.87, 17.35]	11.40 [1.24, 46.41]
Exterior spin magnitude $ \hat{a}_\infty $	0.908 [0.668, 0.983]	0.779 [0.144, 0.975]
Polarization factor A_{pol}	1.049 [1.009, 1.210]	1.129 [1.013, 1.866]
Compact-source spin parameter $\hat{a}_c$	0.999 [0.978, 1.000]	0.992 [0.500, 1.000]
GR/CPTG matching radius r_m (km)	197.891 [168.910, 233.526]	146.183 [72.739, 181.246]
Structural Mode N_{BH}	1.4731 [1.4530, 1.5593]	1.5150 [1.4548, 1.8707]
Organization-shell radius r_N (km)	103.162 [87.928, 122.136]	76.488 [38.272, 95.098]
Inner trapping-surface radius r_- (km)	167.324 [142.308, 200.804]	115.305 [46.737, 149.440]
Structural- N lapse $\bar{\alpha}_N$	0.8327 [0.7648, 0.8490]	0.7994 [0.5165, 0.8475]
Structural- N clock $T_{24,N}$ (h)	19.986 [18.354, 20.377]	19.186 [12.397, 20.341]
Structural- N axis ω (rad s $^{-1}$)	240.53 [186.67, 399.53]	438.97 [249.45, 1641.79]
Structural- N determinant-average $\bar{\omega}$ (rad s $^{-1}$)	389.69 [310.84, 552.51]	638.21 [410.62, 1844.28]
Structural- N equatorial ω (rad s $^{-1}$)	619.12 [513.79, 739.71]	888.19 [662.96, 2054.64]
Structural- N period P_ω (ms)	16.1236 [11.3722, 20.2136]	9.8450 [3.4068, 15.3016]
Trapping-surface limit Ω_- (rad s $^{-1}$)	893.80 [745.10, 1049.44]	1238.52 [927.58, 2640.84]
Trapping-surface period P_- (ms)	7.0297 [5.9872, 8.4327]	5.0731 [2.3792, 6.7738]

Table 779: Full CPTG compact-state and finite-core results for GW231123_135430 (C00:Mixed+X04a). Values are posterior median [5th, 95th percentile]. Component numbering is imposed after the compact handoff by physical source mass.

Location	Comp. 1 T_{24} (h)	Comp. 2 T_{24} (h)	Comp. 1 $\bar{\omega}$	Comp. 2 $\bar{\omega}$
1 km inward	1.8682	2.7979	893.09	1233.48
500 m inward	1.3140	1.9709	893.45	1235.98
100 m inward	0.5851	0.8788	893.71	1238.04
10 m inward	0.1848	0.2777	893.79	1238.51
1 m inward	0.0584	0.0878	893.80	1238.52
10 cm inward	0.0185	0.0278	893.80	1238.52
1 cm inward	0.0058	0.0088	893.80	1238.52
surface limit	0.0000	0.0000	893.80	1238.52

Table 780: Near-inner-boundary profile for GW231123_135430 (C00:Mixed+X04a). Time entries are median proper-time accumulation per 24 source-local exterior Kerr hours; rotation entries are median determinant-measure ZAMO frame-dragging rates in rad s $^{-1}$. Distances are Boyer–Lindquist radial-coordinate offsets inward from r_- .

Median quantity	Mixed	Mixed+XO4a	Δ (XO4a–Mixed)
Component 1 M_∞ ($M_\odot$)	132.487	134.016	+1.529
Component 1 M_c ($M_\odot$)	123.581	125.547	+1.966
Component 1 geometric share (%)	4.98	4.67	-0.31
Component 1 N_{BH}	1.4749	1.4731	-0.0017
Component 1 $T_{24,N}$ (h)	19.952	19.986	+0.034
Component 1 $\bar{\omega}_N$ (rad s $^{-1}$)	400.68	389.69	-11.00
Component 2 M_∞ ($M_\odot$)	103.357	98.998	-4.359
Component 2 M_c ($M_\odot$)	92.294	86.691	-5.603
Component 2 geometric share (%)	9.26	11.40	+2.14
Component 2 N_{BH}	1.5008	1.5150	+0.0142
Component 2 $T_{24,N}$ (h)	19.455	19.186	-0.269
Component 2 $\bar{\omega}_N$ (rad s $^{-1}$)	578.27	638.21	+59.95

Table 781: Branch-sensitivity summary for GW231123_135430. Differences compare separately propagated posterior medians and are descriptive; no branch average is formed.

The two reconstructions agree on the qualitative compact-state picture but preserve measurable branch dependence. For component 1, the physical compact-source median changes from 123.581 to 125.547 $M_\odot$, while the geometric share changes from 4.98% to 4.67%. The component-1 organization-shell clock changes only from 19.952 to 19.986 h, and its determinant-averaged Structural-Mode frame dragging changes from 400.68 to 389.69 rad s $^{-1}$. The secondary component shows the larger branch sensitivity: its geometric share changes from 9.26% to 11.40%, and $\bar{\omega}_N$ changes from 578.27 to 638.21 rad s $^{-1}$. Both branches nevertheless remain strongly source-mass dominated compared with the lower-spin M5 binaries, and both place the organization shells inside the inner trapping surfaces and the trapping surfaces inside the GR/CPTG matching radii. The branch dependence is therefore retained as part of the observational reconstruction rather than treated as a competing physical law. No Structural-Mode, clock, mass decomposition, or rotation value is averaged across the two reconstructions.

8 Mass and Mass-Equivalent Structure Across the Full Sample

The candidate-by-candidate sections establish the compact state separately for every retained event realization. This section asks a different question: what does the complete catalog reveal about how the exterior gravitational mass parameter is partitioned into physical compact-source mass and completed geometric mass-equivalent response? The synthesis is descriptive and uses the same solved quantities already reported for the individual candidates. No new fit is introduced.

Unless otherwise stated, catalog summaries below are calculated over the 260 unique numerical streams representing 259 astrophysical events. GW231123_135430 contributes its two retained reconstructions separately because they are numerically distinct and are not averaged. Quantities explicitly labeled “from marginal medians” are algebraic combinations of separately reported component marginal medians and are not claimed to be posterior medians of the corresponding combined random variable.

8.1 The compact closure fixes the fractional mass partition

For each component,

$$M_\infty = M_c + M_{\text{geom}}, \quad (48)$$

with

$$M_c = \frac{M_\infty}{A_{\text{pol}}}, \quad \frac{M_{\text{geom}}}{M_\infty} = 1 - \frac{1}{A_{\text{pol}}}. \quad (49)$$

The compact factor is determined by

$$q^2 A_{\text{pol}}^3 + A_{\text{pol}} - 2 = 0, \quad q = |\hat{a}_\infty|. \quad (50)$$

The fractional partition is therefore controlled by exterior spin magnitude in the registered compact handoff; exterior mass sets the absolute scale but does not independently alter the fraction. Differentiating Eq. (50) gives

$$\frac{dA_{\text{pol}}}{dq} = -\frac{2qA_{\text{pol}}^3}{1 + 3q^2A_{\text{pol}}^2}, \quad (51)$$

so that

$$\frac{d}{dq} \left(\frac{M_{\text{geom}}}{M_\infty} \right) = -\frac{2qA_{\text{pol}}}{1 + 3q^2A_{\text{pol}}^2} < 0 \quad (0 < q \leq 1). \quad (52)$$

The endpoints are fixed:

$$q = 0 : \quad A_{\text{pol}} = 2, \quad M_c = \frac{1}{2}M_\infty, \quad M_{\text{geom}} = \frac{1}{2}M_\infty, \quad (53)$$

whereas

$$q = 1 : \quad A_{\text{pol}} = 1, \quad M_c = M_\infty, \quad M_{\text{geom}} = 0. \quad (54)$$

The catalog occupies the interior of these exact bounds rather than defining them. Figure 1 shows all 520 compact components. The five observational mass classes fall on the same closure curve: the apparent change in geometric share from one mass class to another is therefore attributable to the spin distribution represented in that class, not to an additional mass-dependent coefficient.

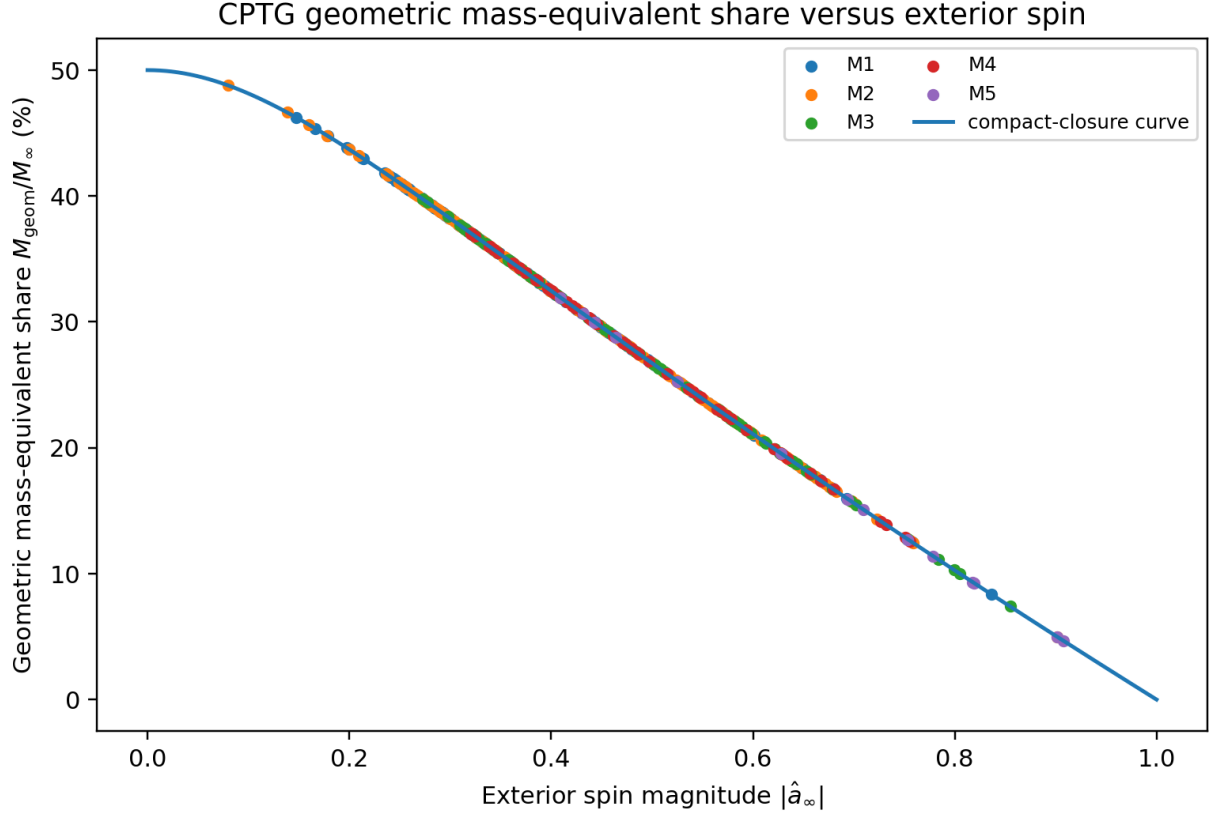


Figure 1: Geometric mass-equivalent share versus exterior spin magnitude for all 520 compact components. Points are grouped by the pre-CPTG observational mass classes M1–M5. The continuous curve is the registered compact-closure relation, not a fit to the catalog.

8.2 Catalog-wide mass accounting

Across the numerical streams, the component-1 geometric-share medians span 4.67%–46.23%, while component 2 spans 9.26%–48.79%. The corresponding exterior-spin medians span 0.148–0.908 for component 1 and 0.080–0.819 for component 2. These catalog ranges are narrower than the exact 0–50% closure interval because no retained component median sits exactly at $q = 1$ or $q = 0$.

The stream-level combined geometric share constructed from the two component marginal medians spans 6.78%–46.92%, with a median of 28.79%. This combined quantity is useful for catalog description but should not be confused with a posterior median of a system-level random variable.

Table 782 shows how the five observational mass classes populate the fixed compact relation. M5 has the smallest median combined geometric share, 17.70%, because its primary components have a substantially higher median exterior spin than those in the lower classes. The same table also shows why this trend cannot be interpreted as an independent dependence on total mass: the fractional share follows spin at the component level.

Class	Streams / events	Comp. 1 median $ \hat{a}_\infty $	Comp. 1 median geom. share (%)	Comp. 2 median $ \hat{a}_\infty $	Comp. 2 median geom. share (%)	Median combined geom. share* (%)
M1	82 / 82	0.396	32.70	0.334	36.24	34.66
M2	94 / 94	0.560	23.34	0.326	36.74	29.04
M3	47 / 47	0.590	21.66	0.360	34.78	26.32
M4	30 / 30	0.586	21.87	0.400	32.49	25.03
M5	7 / 6	0.754	12.73	0.464	28.76	17.70

Table 782: Mass-partition summaries by observational class. *The combined share is constructed from separately reported component marginal medians. M5 contains seven numerical streams representing six events because GW231123_135430 retains two distinct reconstructions.

8.3 Physical compact mass versus exterior gravitational mass

The same closure gives

$$\frac{1}{2}M_\infty \leq M_c \leq M_\infty. \quad (55)$$

Figure 2 places every component inside these exact limits. The upper boundary is approached by rapidly rotating compact states, for which the geometric mass-equivalent fraction becomes small. The lower boundary is approached by slowly rotating states, for which the geometric and physical-source contributions become comparable.

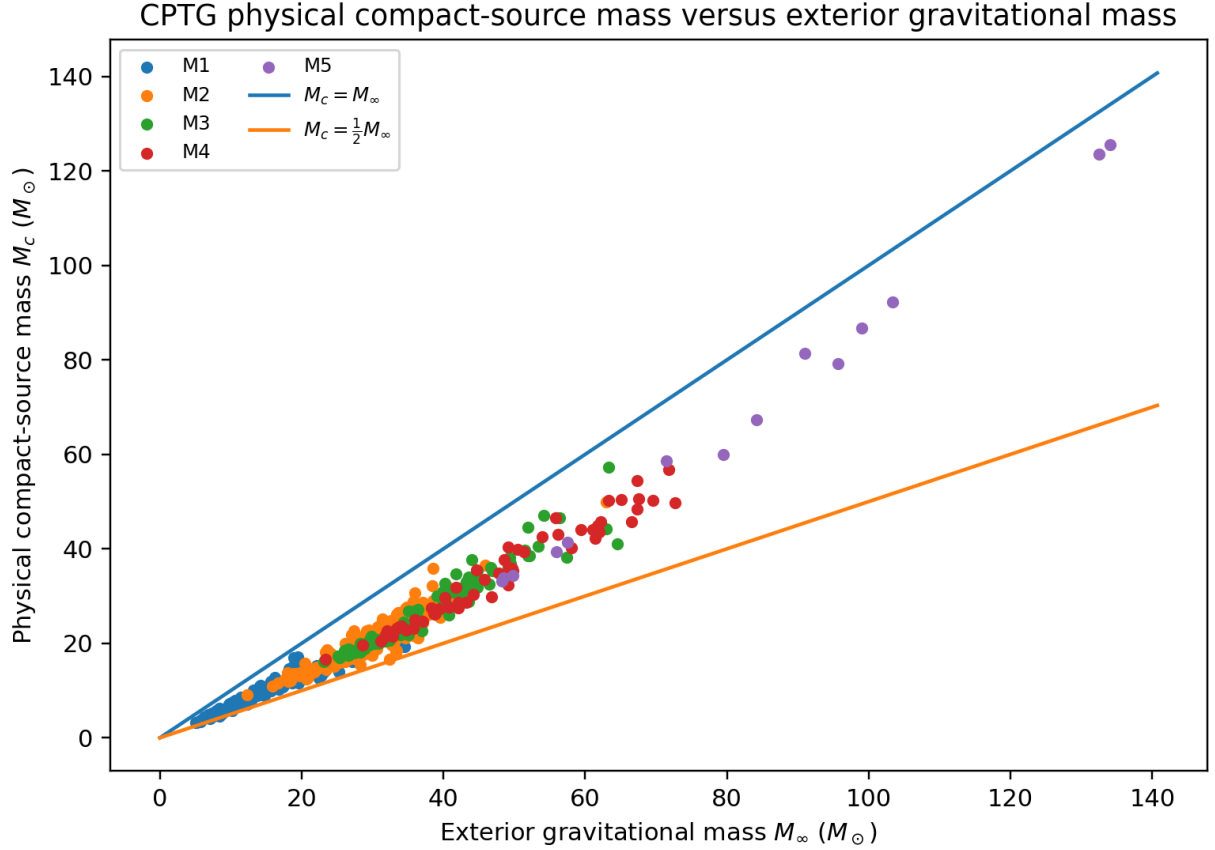


Figure 2: Physical compact-source mass versus exterior gravitational mass for all 520 components. The lines $M_c = M_\infty$ and $M_c = M_\infty/2$ are exact bounds of the registered compact handoff. The points are observationally grouped into M1–M5 only for context.

The distinction between fractional and absolute geometric contribution is important. The largest combined geometric mass-equivalent constructed from component marginal medians occurs for GW231001_140220 (M4), at $31.924 M_\odot$. By contrast, the largest combined exterior mass occurs in the C00:Mixed reconstruction of GW231123_135430, at $235.844 M_\odot$, while its combined geometric share is only 6.78%. Its physical compact-source total constructed from the component marginal medians is consequently $215.875 M_\odot$.

The highest combined geometric fraction occurs instead in GW250114_082203 (M2), where the constructed share reaches 46.92%. Its two exterior-spin medians are only 0.159 and 0.080. Thus the event with the largest exterior mass is not the event with the largest absolute geometric contribution, and neither is the event with the largest geometric fraction. The three quantities answer different questions:

$$M_\infty, \quad f_{\text{geom}}(q), \quad M_{\text{geom}} = f_{\text{geom}}(q)M_\infty. \quad (56)$$

8.4 Catalog extremes illustrate the closure

The component-level extrema provide direct checks on the same spin-controlled behavior. GW230627_015337 has the largest component-1 geometric-share median, 46.23%, with $|\hat{a}_\infty| = 0.148$. The C00:Mixed+X04a reconstruction of GW231123_135430 has the smallest component-1 share, 4.67%, with $|\hat{a}_\infty| = 0.908$. For component 2, GW250114_082203 reaches 48.79% at $|\hat{a}_\infty| = 0.080$, whereas the C00:Mixed reconstruction of GW231123_135430 gives 9.26% at $|\hat{a}_\infty| = 0.819$.

The absolute geometric mass-equivalent behaves differently because it carries the exterior mass scale. GW190929_012149 has the largest component-1 median M_{geom} at $22.238 M_\odot$, while GW241125_010116 has the largest component-2 median at $16.219 M_\odot$. At the system level constructed from marginal medians, GW231001_140220 reaches the catalog maximum of $31.924 M_\odot$ even though its combined geometric fraction is only 28.48%.

These extrema are not fitted anomalies. They are the expected consequence of multiplying a spin-controlled fraction by a varying exterior mass scale.

8.5 Why component identity is imposed after the compact handoff

Because $M_c = M_\infty/A_{\text{pol}}$ depends on spin as well as exterior mass, the component with the larger released or exterior mass need not remain the component with the larger physical compact-source mass on every posterior draw. The registered procedure therefore solves both raw component states first and reorders the complete attached state only afterward.

Across the 260 numerical streams, the sample-wise post-handoff reorder fraction spans 0.00%–41.36%, with a stream median of 16.73%. The maximum occurs for GW231226_101520, where 41.36% of retained draws require the physical-source ordering to differ from the raw source ordering. This is not a numerical ambiguity; it is a direct consequence of solving physical compact mass rather than assuming that the exterior-mass ordering must survive the handoff.

For this reason every attached quantity— M_∞ , $\hat{a}_\infty$, A_{pol} , M_c , M_{geom} , $\hat{a}_c$, r_m , N_{BH} , r_N , r_- , clock, and rotation—is moved together when the post-handoff component order changes. The catalog synthesis would be internally inconsistent if mass were reordered without its structural and rotational state.

8.6 Interpretation of the mass decomposition

The compact handoff changes the meaning of “black-hole mass” inside the CPTG description. M_∞ is the exterior gravitational charge that controls the retained Kerr field. M_c is the physical compact-source mass of the finite interior. M_{geom} is the mass-equivalent completed geometric response required by the same compact closure. They are not three independent fitted reservoirs; they are linked by Eqs. (48)–(50).

This separation is what permits the later Structural-Mode, clock, and rotation readouts to be tied to a finite physical compact source while preserving the exterior gravitational field. It also explains why CPTG can assign less physical source mass than the exterior gravitational mass parameter without changing the exterior Kerr charge seen by a distant observer. The event-by-event sections establish that accounting individually; the catalog synthesis shows that the same rule survives across the full mass range from M1 through the extreme M5 systems.

The next synthesis section turns from mass accounting to the post-solution organization of those finite compact states through Structural Mode N_{BH} .

9 Structural Mode and the Internal Organization of the Solved Compact States

The mass synthesis separates the physical compact source from the exterior gravitational charge. Structural Mode asks a different question: how is that already solved finite compact state organized internally? The answer is carried object by object. No Structural-Mode value is averaged, pooled, or used to define an observational class. The 520 component entries in this section therefore remain a registry of individual post-solution fingerprints, including both independently retained GW231123_135430 reconstructions.

9.1 Structural Mode is a scale-free post-solution readout

For the registered compact realization [4, 3],

$$N_{\text{BH}}(q) = \frac{W_0}{\int_0^1 w(x) \xi(x, q) dx}, \quad q = |\hat{a}_\infty|, \quad (57)$$

where

$$\xi(x, q) = \left[\frac{x(x^2 + q^2)}{1 + q^2} \right]^{1/3}, \quad (58)$$

and the structural weight $w(x)$ is fixed by the finite-core profile declared in Sec. 2.5. The numerator W_0 and the entire integration domain are dimensionless. Mass therefore cancels before N_{BH} is evaluated.

The spin dependence is monotonic. For $0 < x < 1$,

$$\frac{\partial \ln \xi}{\partial q} = \frac{2q(1 - x^2)}{3(x^2 + q^2)(1 + q^2)} > 0 \quad (q > 0), \quad (59)$$

so the denominator of Eq. (57) increases with q and

$$\frac{dN_{\text{BH}}}{dq} < 0. \quad (60)$$

The spherical endpoint is the registered value

$$N_{\text{BH}}(0) = 1.912233171442. \quad (61)$$

As $q \rightarrow 1^-$, the same registered construction approaches $N_{\text{BH}} \simeq 1.448629$. Figure 3 shows the exact registered curve together with every component median. The maximum catalog point-to-curve residual at the stored precision is 1.73×10^{-8} ; the catalog therefore samples one common structural law rather than a set of object-specific fitted coefficients.

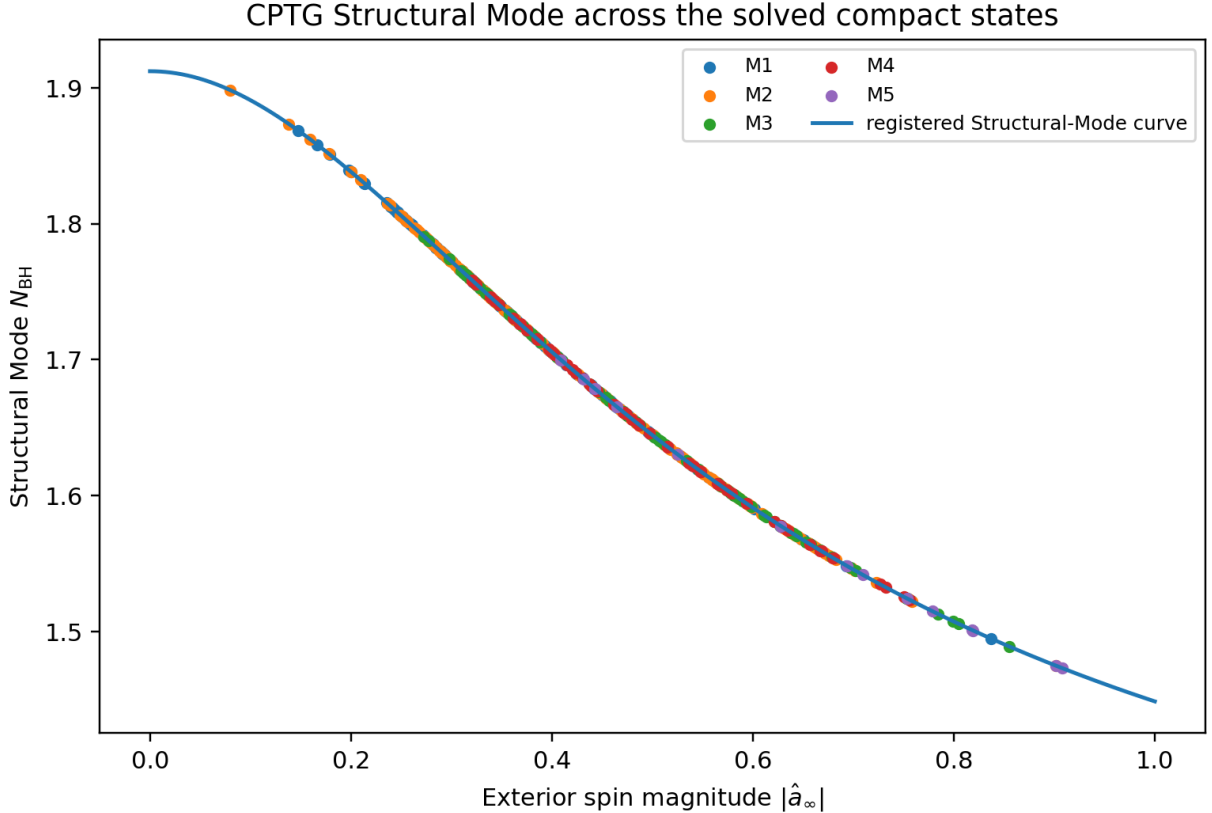


Figure 3: Structural Mode versus exterior spin magnitude for all 520 solved compact components. Each point remains an individual component fingerprint. The continuous curve is the registered post-solution Structural-Mode construction, not a fit to the catalog.

Component 2 of GW250114_082203 gives the largest retained component median, $N_{\text{BH}} = 1.898389$, at $|\hat{a}_\infty| = 0.079772$. At the opposite end of the retained support, component 1 of GW231123_135430 in the C00:Mixed+XO4a reconstruction gives $N_{\text{BH}} = 1.473148$ at $|\hat{a}_\infty| = 0.907663$. These remain separate object-level fingerprints and are not combined into a catalog-mean N .

9.2 The organization shell separates dimensionless shape from physical size

The Structural-Mode organization shell is fixed by

$$x_N(x_N^2 + q^2) = \frac{1 + q^2}{N_{\text{BH}}^3}. \quad (62)$$

Once Eq. (57) has been evaluated, x_N depends only on the solved exterior spin magnitude. The inner trapping coordinate x_- is likewise determined by q , while the GR/CPTG matching boundary is $x = 1$. The normalized internal architecture can therefore be written compactly as

$$(N_{\text{BH}}, x_N, x_-) = \mathcal{S}(q), \quad (63)$$

whereas the physical radii recover the mass scale through

$$(r_N, r_-, r_m) = \frac{GM_\infty}{c^2} (x_N, x_-, 1). \quad (64)$$

This separation is one of the useful consequences of the normalized finite-core construction: spin fixes the dimensionless organization, and exterior gravitational mass fixes its physical size.

Figure 4 shows the normalized geometry. Across the full registered spin domain $0 \leq q < 1$, x_N remains tightly confined between approximately 0.518502 and 0.530110. Its shallow maximum, $x_N \simeq 0.530110$, occurs near $q \simeq 0.419$, where $N_{\text{BH}} \simeq 1.6936$. The trapping surface behaves very differently: x_- begins at 0.572357 in the spherical limit and moves outward toward the matching boundary as $q \rightarrow 1^-$.

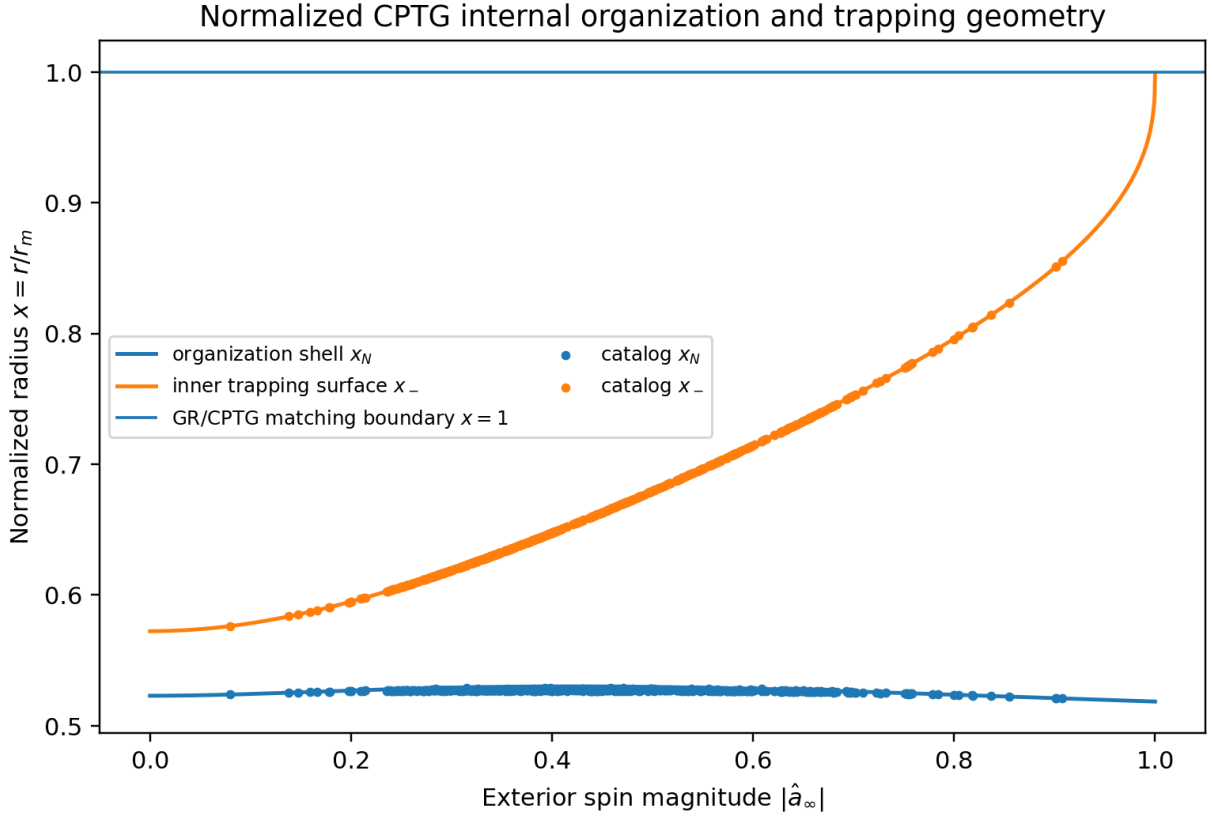


Figure 4: Normalized internal architecture of the registered finite compact state. The Structural-Mode organization shell x_N remains near one half of the matching radius over the full spin domain, while the inner trapping surface x_- moves strongly outward with increasing spin. The outer GR/CPTG matching boundary is fixed at $x = 1$.

The organization shell is therefore not simply another name for the trapping surface. At the catalog medians, x_N spans 0.520906–0.529182, whereas x_- spans 0.576235–0.855322. The component with the largest retained x_N median is component 1 of GW190412_053044, with $x_N = 0.529182$ at $|\hat{a}_\infty| = 0.314853$.

9.3 The Structural-Mode shell remains inside the trapping surface

The radial ordering is not imposed after the fact. It follows from the registered structural and trapping constructions and is checked on every retained posterior stream. Over the full allowed spin domain,

$$x_N < x_- < 1. \quad (65)$$

The minimum registered spin-domain separation is

$$\min_{0 \leq q < 1} (x_- - x_N) = 0.049408417286901, \quad (66)$$

which occurs in the spherical limit. The complete posterior audit over all 7,626,239 retained draws independently finds the same strict ordering, with minimum sample-wise separation

$$(x_- - x_N)_{\text{min,posterior}} = 0.049408417286901. \quad (67)$$

Thus no retained Structural-Mode shell reaches or crosses the inner trapping surface.

The physical shell separation does not remain fixed because Eq. (64) multiplies the normalized geometry by r_m . At the reported component medians, the smallest physical $r_- - r_N$ separation is 0.758 km for component 2 of GW230627_015337; the largest is 64.162 km for component 1 of the C00:Mixed+X04a reconstruction of GW231123_135430.

9.4 Physical organization spans the full compact-object mass scale

Although N_{BH} and the normalized shell coordinates are scale-free, their physical realization is not. Figure 5 shows r_N and r_- against the matching radius r_m for all 520 component medians. The smallest physical organization-shell radius in the catalog is $r_N = 3.949$ km for component 2 of GW190924_021846; its trapping and matching radii are 4.793 and 7.492 km. The largest is $r_N = 103.162$ km for component 1 of the C00:Mixed+X04a reconstruction of GW231123_135430, with $r_- = 167.324$ km and $r_m = 197.891$ km.

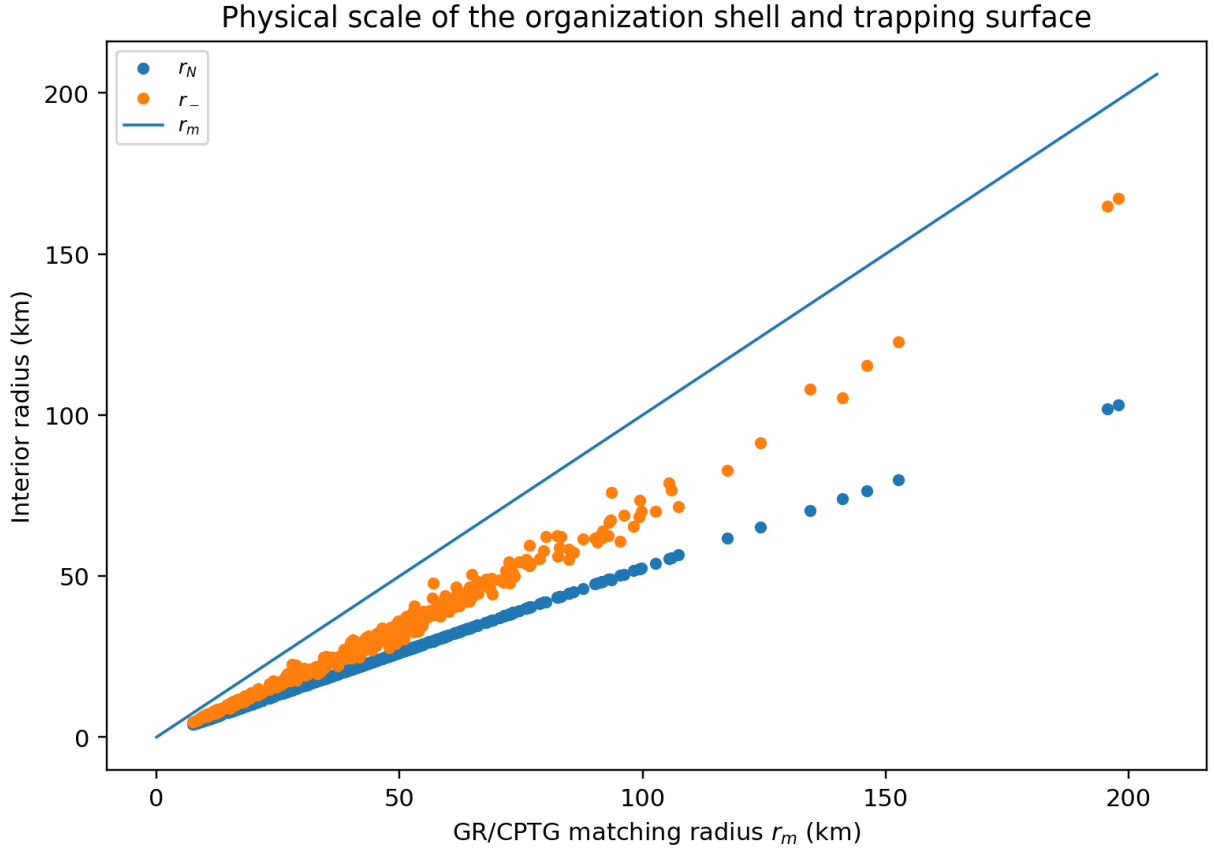


Figure 5: Physical organization-shell and trapping-surface radii against the GR/CPTG matching radius for all 520 components. The mass scale enters only after the normalized structural state is solved, through multiplication by $r_m = GM_\infty/c^2$.

Fingerprint	Event	Comp.	$ \hat{a}_\infty $	N_{BH}	x_N	x_-	r_N (km)
largest retained N_{BH}	GW250114_082203	2	0.080	1.8984	0.5239	0.5762	25.083
smallest retained N_{BH}	GW231123_135430 (XO4a)	1	0.908	1.4731	0.5209	0.8553	103.162
largest retained x_N	GW190412_053044	1	0.315	1.7622	0.5292	0.6230	21.468
smallest physical r_N	GW190924_021846	2	0.291	1.7785	0.5269	0.6165	3.949
largest physical r_N	GW231123_135430 (XO4a)	1	0.908	1.4731	0.5209	0.8553	103.162
smallest physical $r_- - r_N$	GW230627_015337	2	0.198	1.8397	0.5264	0.5945	4.463
largest physical $r_- - r_N$	GW231123_135430 (XO4a)	1	0.908	1.4731	0.5209	0.8553	103.162

Table 783: Selected individual Structural-Mode fingerprints illustrating the retained structural range. Rows identify actual component medians; they are not averages or pooled Structural-Mode estimates.

9.5 Branch sensitivity remains object-specific

The two retained GW231123_135430 reconstructions illustrate why Structural Mode is never averaged across independent reconstructions. For component 1, the C00:Mixed branch gives

$$|\hat{a}_\infty| = 0.901643, \quad N_{\text{BH}} = 1.474891, \quad x_N = 0.521064, \quad x_- = 0.851345, \quad (68)$$

whereas C00:Mixed+X04a gives

$$|\hat{a}_\infty| = 0.907663, \quad N_{\text{BH}} = 1.473148, \quad x_N = 0.520906, \quad x_- = 0.855322. \quad (69)$$

The secondary component shifts from $N_{\text{BH}} = 1.500824$ to 1.515017 between the same two reconstructions. Each value remains attached to its own posterior branch and its own solved compact geometry.

9.6 Interpretation of Structural Mode

Structural Mode does not supply the compact mass, choose a branch, strengthen gravity, or classify a candidate. Those quantities are already fixed before Eq. (57) is evaluated. In the present compact realization, N_{BH} is the dimensionless organization readout associated with the solved spin state, while r_N places that organization inside the physical finite core after the exterior mass scale is restored.

The synthesis therefore exposes a useful separation. The dimensionless internal shape is controlled by the solved rotational state and is common to every object through the same function $\mathcal{S}(q)$. The physical realization of that shape retains the individuality of each compact source through M_∞ , branch provenance, and the full posterior state. The organization shell remains well inside the inner trapping surface across the entire registered spin domain and across every retained posterior draw, so the Structural-Mode clock and rotation readouts used elsewhere in this paper are evaluated on a geometrically distinct interior shell rather than at a trapping boundary.

The next synthesis section uses that shell to examine how proper time accumulates inside the finite compact states.

10 Time Inside the Finite Compact Core

The Structural-Mode synthesis identifies a geometrically distinct organization shell inside every solved finite compact state. The clock analysis now asks how much proper time accumulates there, and how that accumulation changes as a locally nonrotating observer approaches the inner trapping surface. All times in this section are measured relative to 24 source-local asymptotic exterior Kerr hours. They are not detector-frame, Earth-frame, or cosmological clock intervals.

Two different kinds of clock readout must be kept separate. The Structural- N clock is evaluated at a dimensionless shell whose location is fixed by the solved spin state; it is therefore scale-free. The registered near-boundary profiles instead use fixed physical Boyer–Lindquist radial-coordinate offsets from r_- . Those offsets reintroduce the compact object’s physical length scale through $r_m = GM_\infty/c^2$.

10.1 The Structural-Mode clock is scale-free

At the organization shell,

$$T_{24,N}(q) = 24 \bar{\alpha}(x_N(q); q) \text{ h}. \quad (70)$$

Both x_N and the determinant-averaged lapse are dimensionless functions of the solved exterior spin magnitude. No mass scale appears in Eq. (70). In the registered realization, numerical evaluation over $0 \leq q < 1$ gives a monotonic increase from

$$T_{24,N}(0) = 11.592884 \text{ h} \quad (71)$$

to

$$T_{24,N} \rightarrow 20.461495 \text{ h} \quad (q \rightarrow 1^-). \quad (72)$$

Thus the Structural-Mode shell experiences stronger time dilation at low exterior spin and retains a larger fraction of the exterior 24-hour interval as the solved spin approaches the upper end of the allowed range. Figure 6 shows the registered curve and all 520 component medians. The maximum point-to-curve difference at stored median precision is 4.84e-07 h.

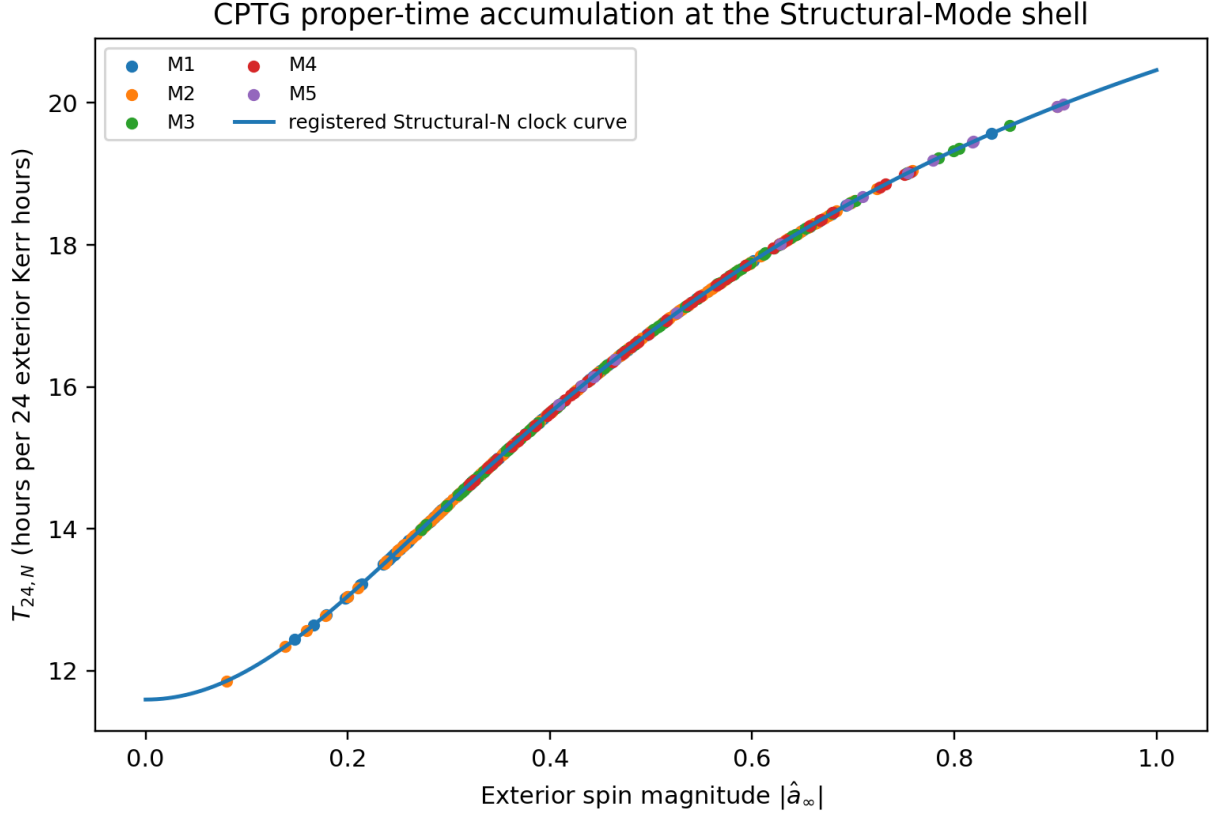


Figure 6: Proper-time accumulation at the Structural-Mode organization shell versus exterior spin magnitude. The continuous line is the registered finite-core clock relation; the points are the 520 individual component medians grouped only by the pre-CPTG observational mass classes.

The retained component medians span 11.852622–19.985616 h. The smallest is component 2 of GW250114_082203, with $|\hat{a}_\infty| = 0.079772$. The largest is component 1 of the C00:Mixed+X04a reconstruction of GW231123_135430, with $|\hat{a}_\infty| = 0.907663$. These extremes follow the same spin-controlled curve; they do not define separate clock laws for low- and high-mass black holes.

10.2 Approaching the inner trapping surface

Let

$$d = r_- - r > 0, \quad \epsilon = \frac{d}{r_m}, \quad x = x_- - \epsilon. \quad (73)$$

Here d is the declared Boyer–Lindquist radial-coordinate offset inward from the inner trapping surface, not a proper distance. For a nonextremal inner root, expand

$$\delta(x, q) \simeq -\delta'(x_-, q) \epsilon \equiv \kappa(q) \epsilon, \quad \kappa(q) > 0. \quad (74)$$

At $x = x_-$,

$$\mathcal{A}_d = (x_-^2 + q^2)^2, \quad \sigma_-(\theta) = x_-^2 + q^2 \cos^2 \theta. \quad (75)$$

Substituting these into the determinant-measure lapse gives the leading near-surface form

$$\bar{\alpha} \simeq \mathcal{C}(q) \sqrt{\frac{d}{r_m}}, \quad (76)$$

with

$$\mathcal{C}(q) = \frac{\sqrt{\kappa(q)} \int_0^\pi \sigma_-^{3/2} \sin \theta \, d\theta}{x_-^2 + q^2 \int_0^\pi \sigma_- \sin \theta \, d\theta}. \quad (77)$$

Therefore

$$T_{24}(d) \simeq 24 \mathcal{C}(q) \sqrt{\frac{d}{r_m}} \text{ h.} \quad (78)$$

This exposes two pieces of the near-boundary behavior directly. First, at fixed compact state, proper time approaches zero as $d^{1/2}$. Second, at fixed spin and fixed physical coordinate offset, the same readout scales as $r_m^{-1/2}$, so the mass scale returns even though the Structural- N clock itself is scale-free.

10.3 The full catalog follows the square-root boundary law

Every retained component was evaluated at 1 km, 500 m, 100 m, 10 m, 1 m, 10 cm, and 1 cm inward from r_- , followed by the exact trapping-surface limit. All 520 median profiles decrease monotonically toward zero at the surface, and the stored identity

$$T_{24} = 24 \bar{\alpha} \quad (79)$$

closes to $3.55\text{e-}15$ h at median precision.

Figure 7 summarizes the seven finite offsets. The central curve is the median across the 520 component medians; the shaded band contains the 5th–95th percentile range across those component medians. These are descriptive catalog summaries, not posterior intervals for a single object.

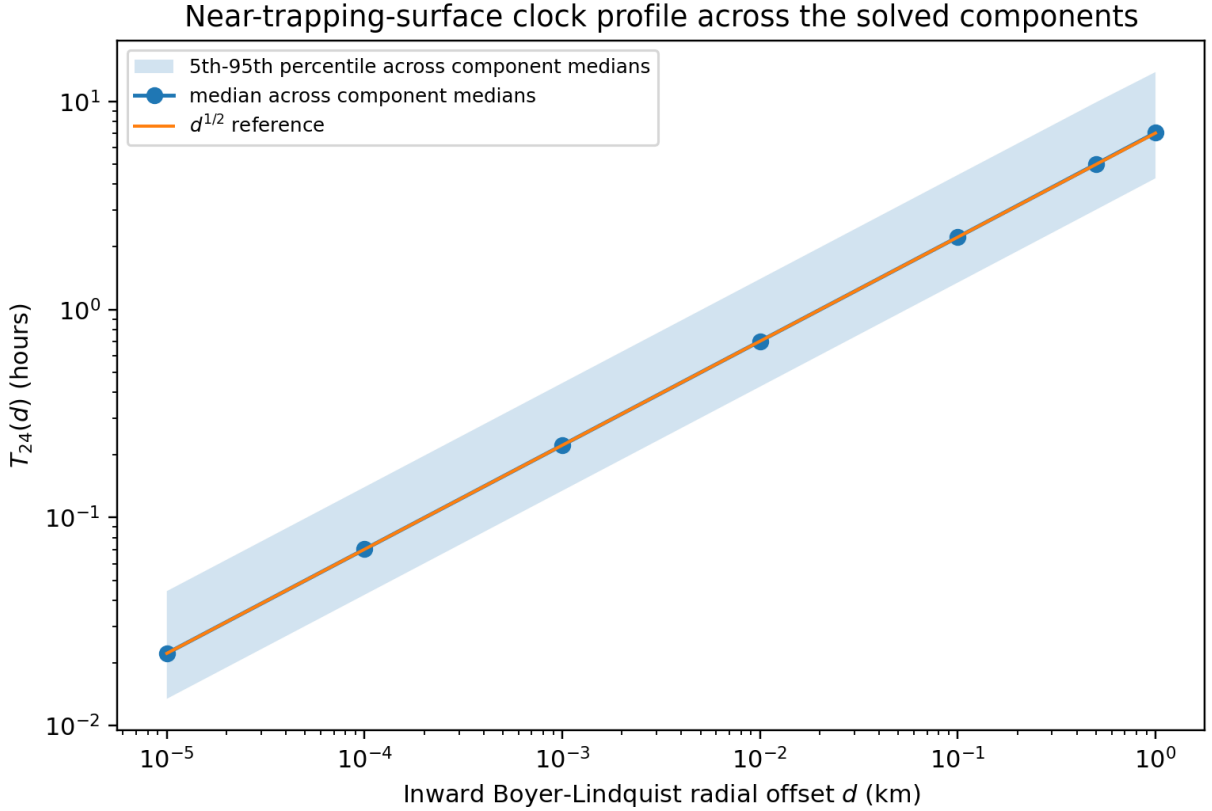


Figure 7: Near-trapping-surface proper-time profiles. The median and 5th–95th percentile band are taken across the 520 component medians at each registered Boyer–Lindquist coordinate offset. The $d^{1/2}$ line is a reference normalized at the 1 cm catalog median and illustrates the simple-root asymptotic behavior.

A direct log–log fit to the 1 cm, 10 cm, and 1 m median readouts of each component gives

$$T_{24} \propto d^\beta, \quad (80)$$

with

$$0.49999464 \leq \beta \leq 0.50001636, \quad \text{median}(\beta) = 0.50000089. \quad (81)$$

The entire retained set therefore reproduces the predicted $\beta = 1/2$ boundary exponent to within 1.64×10^{-5} in the fitted exponent at the component-median level.

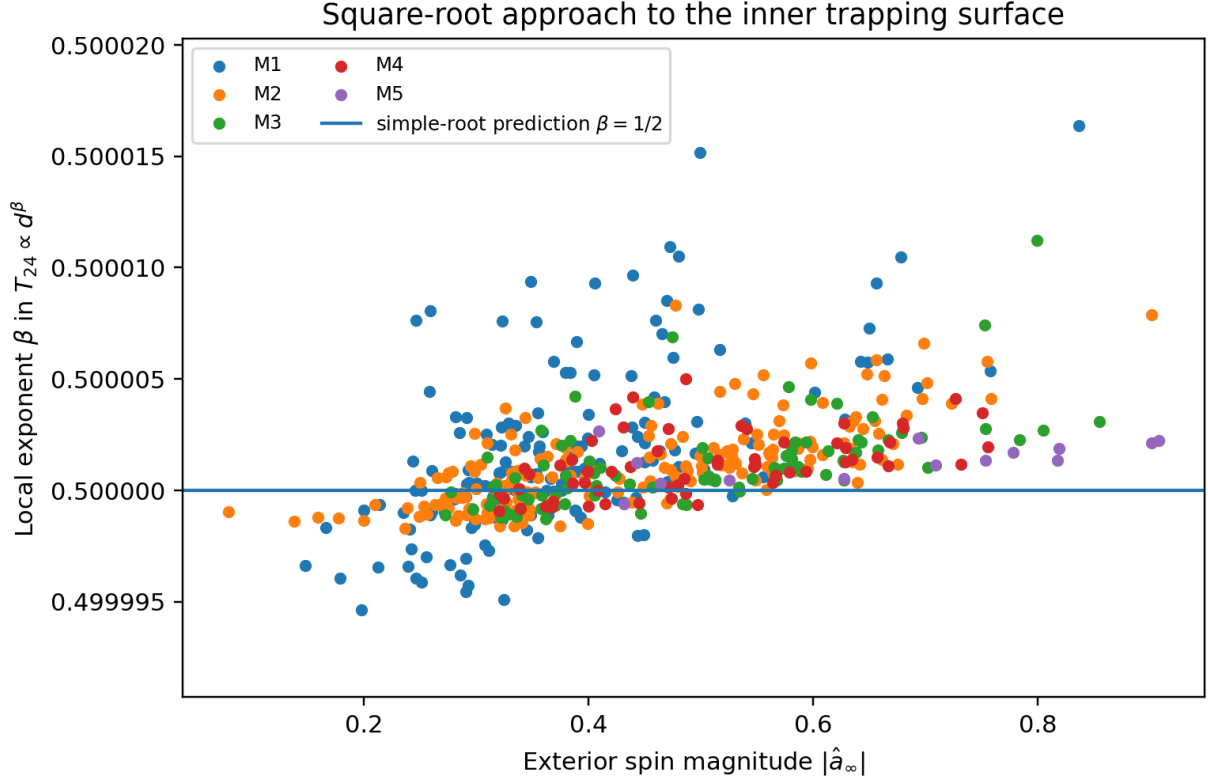


Figure 8: Local near-surface exponent obtained independently for each component from the 1 cm, 10 cm, and 1 m median clock readouts. The horizontal line marks the simple-root prediction $\beta = 1/2$.

Offset	Minimum	5th pct.	Median	95th pct.	Maximum
1 km	1.86824	4.27082	7.07368	13.82619	16.79920
500 m	1.31402	3.01599	4.99079	9.88883	12.19039
100 m	0.58510	1.34727	2.22648	4.42933	5.51764
10 m	0.18484	0.42587	0.70398	1.40060	1.74739
1 m	0.05845	0.13465	0.22261	0.44295	0.55263
10 cm	0.01848	0.04258	0.07040	0.14008	0.17475
1 cm	0.00584	0.01346	0.02226	0.04430	0.05526

Table 784: Catalog distribution of median proper-time accumulation, in hours per 24 source-local exterior Kerr hours, at the seven registered inward coordinate offsets. Percentiles are taken across the 520 component medians.

At 1 km inward, the component medians span 1.868244–16.799197 h. By 1 cm inward the range has fallen to 0.005844–0.055261 h, or approximately 21.04–198.94 s. The trapping-surface limiting value is zero by construction.

10.4 Fixed physical offsets restore the mass scale

The distinction between Eqs. (70) and (78) is visible when the observational mass classes are used only as descriptive groupings. Table 785 lists medians taken across the component medians in each class.

Class	Components	Median $T_{24,N}$	Median 1 km	Median 100 m	Median 1 m	Median 1 cm
M1	164	15.067	11.436	3.636	0.3636	0.03636
M2	188	16.030	6.950	2.196	0.2196	0.02196
M3	94	16.620	5.667	1.782	0.1781	0.01781
M4	60	16.589	5.233	1.650	0.1650	0.01650
M5	14	18.622	3.431	1.078	0.1077	0.01077

Table 785: Descriptive clock summaries by observational mass class, in hours. Each entry is the median across the component medians in that class; none is used to define the class or replace the individual posterior results.

The Structural- N column is governed by the spin distribution because the organization-shell clock is scale-free. The fixed-offset columns behave differently. Their characteristic time decreases toward the higher-mass classes because 1 km,

100 m, or 1 cm represents a progressively smaller fraction of the larger matching radius. Spin still contributes through $\mathcal{C}(q)$, so the class trend is descriptive rather than a one-parameter mass law.

This distinction produces an instructive reversal at the extreme end of the catalog. Component 1 of the C00:Mixed+X04a reconstruction of GW231123_135430 has the *largest* Structural- N clock median, 19.985616 h, because its spin is high. The same component has the *smallest* 1 km and 1 cm fixed-offset clock medians, 1.868244 h and 0.005844 h, because its $r_m = 197.891$ km scale makes those physical offsets place the readout exceptionally close to the trapping surface in normalized coordinates. Conversely, the largest 1 km clock median, 16.799197 h, occurs for component 2 of GW190924_021846 with $r_m = 7.492$ km.

10.5 Branch sensitivity remains attached to the clock state

GW231123_135430 again illustrates why independent reconstructions remain separate. In component 1, the C00:Mixed branch gives

$$T_{24,N} = 19.951987 \text{ h}, \quad T_{24}(1 \text{ km}) = 1.924975 \text{ h}, \quad T_{24}(1 \text{ cm}) = 0.006023 \text{ h}, \quad (82)$$

whereas C00:Mixed+X04a gives

$$T_{24,N} = 19.985616 \text{ h}, \quad T_{24}(1 \text{ km}) = 1.868244 \text{ h}, \quad T_{24}(1 \text{ cm}) = 0.005844 \text{ h}. \quad (83)$$

For component 2, the Structural- N clock changes from 19.455133 to 19.185739 h between the same branches. Each clock remains attached to its own mass, spin, Structural Mode, and trapping geometry; no cross-branch average is formed.

10.6 Interpretation of the interior clock

The clock construction provides a local geometric readout rather than a global “black-hole time.” The Structural- N value measures proper-time accumulation on the post-solution organization shell. The near-boundary sequence measures the same local ZAMO lapse at explicitly declared Boyer–Lindquist radial-coordinate offsets from the inner trapping surface. A whole-core average is neither required nor substituted.

The synthesis reveals two complementary pieces of the finite-core geometry. At the organization shell, the 24-hour proper-time fraction is controlled by the dimensionless spin state and is independent of the overall mass scale. Near the trapping surface, the simple root of δ forces the clock toward zero with the universal square-root law, while the conversion of a physical offset into the normalized coordinate d/r_m makes the readout sensitive to compact size. This is why a very massive, rapidly rotating component can retain comparatively large proper time at its organization shell while showing exceptionally little accumulated time only 1 km inside its trapping surface.

The next synthesis section combines the same geometry with the internal ZAMO angular velocity to examine geometric rotation and its relation to the clock field.

11 Internal Geometric Rotation and Its Relation to the Clock Field

The same finite geometry that fixes the local clock also fixes the angular velocity of a zero-angular-momentum frame. The quantity reported here is geometric frame dragging relative to source-local asymptotic exterior Kerr coordinate time. It is not a material spin law, rigid-body rotation rate, fluid angular velocity, or tangential matter speed. The retained parameter-estimation inputs provide spin magnitudes, so only angular-speed magnitudes are reconstructed.

11.1 Normalized rotation separates shape from physical scale

Define the dimensionless frame-dragging rate

$$\hat{\omega}(x, \theta; q) \equiv \frac{r_m}{c} \omega(x, \theta; q), \quad (84)$$

where $r_m = r_g = GM_\infty/c^2$. At the Structural-Mode organization shell,

$$\bar{\omega}_N = \frac{c}{r_m} \hat{\omega}_N(q), \quad P_N = \frac{2\pi r_m}{c \hat{\omega}_N(q)}. \quad (85)$$

At the inner trapping surface,

$$\Omega_- = \frac{c}{r_m} \hat{\Omega}_-(q), \quad \hat{\Omega}_-(q) = \frac{q}{x_-^2 + q^2}, \quad (86)$$

and

$$P_- = \frac{2\pi r_m}{c \hat{\Omega}_-(q)}. \quad (87)$$

The dimensionless rotational geometry is therefore fixed by the solved spin state, while the physical angular rate scales as r_m^{-1} and the coordinate period scales as r_m . This is the rotational counterpart of the Structural-Mode shape/scale separation developed in the preceding sections.

11.2 Clock and rotation respond differently to spin

Figure 9 compares three dimensionless functions evaluated in the same registered compact geometry: the Structural- N lapse $\bar{\alpha}_N$, the determinant-averaged Structural- N rate $\hat{\omega}_N$, and the trapping-surface rate $\hat{\Omega}_-$. The clock increases monotonically with q , but the dimensionless frame-dragging rates do not.

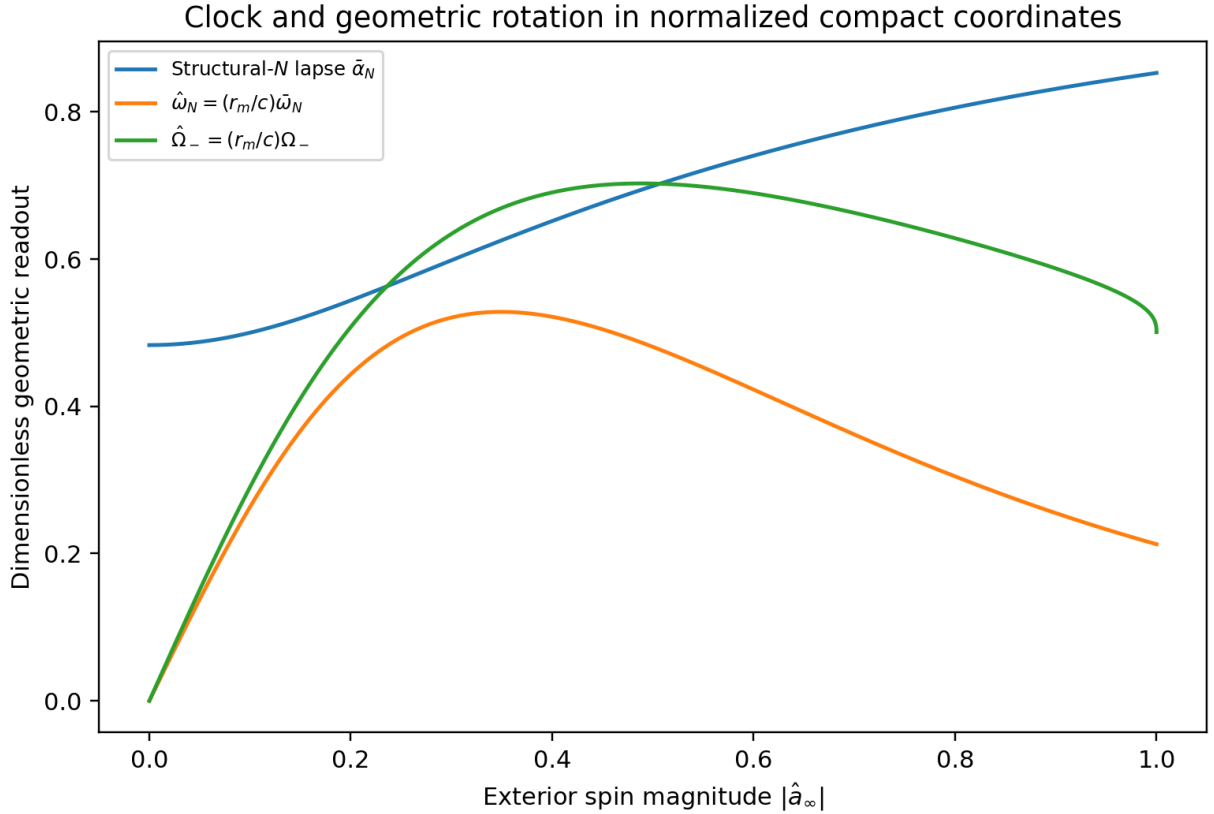


Figure 9: Dimensionless clock and geometric-rotation readouts across the registered spin domain. The Structural- N lapse increases monotonically, whereas both dimensionless frame-dragging rates peak at intermediate spin before declining toward their high-spin limits.

The registered Structural- N frame-dragging factor reaches

$$\hat{\omega}_{N,\max} = 0.528156778 \quad (88)$$

near

$$q = 0.3492, \quad (89)$$

while the surface factor reaches

$$\hat{\Omega}_{-, \max} = 0.702527632 \quad (90)$$

near

$$q = 0.4883. \quad (91)$$

As $q \rightarrow 1^-$, the two factors approach 0.212681158 and 0.500663199, respectively. Thus increasing spin does not imply a monotonically increasing normalized frame-dragging rate at a fixed structural location. The interior geometry itself changes with spin, including the outward motion of the trapping surface and the spin-dependent organization shell.

For comparison of the two determinant-averaged reporting operators, define

$$\mathcal{R}_N(q) \equiv \frac{\hat{\omega}_N(q)}{\bar{\alpha}_N(q)}. \quad (92)$$

This clock-normalized ratio has an intermediate-spin maximum, 0.872500414 near $q = 0.2807$. Because $\hat{\omega}_N$ and $\bar{\alpha}_N$ are averaged separately, $\mathcal{R}_N$ is not an angular average of a local $d\phi/d\tau$ and does not define a material angular velocity. At any fixed polar angle, the corresponding unaveraged geometric ratio is ω/α .

11.3 Physical frame-dragging rates are controlled by both spin and compact size

The retained component medians make the $1/r_m$ scaling directly visible. Across all 520 components, the determinant-averaged Structural- N rate spans

$$389.688 \leq \bar{\omega}_N \leq 16330.834 \quad \text{rad s}^{-1}, \quad (93)$$

with coordinate periods

$$0.384744 \leq P_N \leq 16.123627 \quad \text{ms}. \quad (94)$$

The trapping-surface medians span

$$893.801 \leq \Omega_- \leq 23276.013 \quad \text{rad s}^{-1}, \quad (95)$$

with

$$0.269943 \leq P_- \leq 7.029734 \quad \text{ms}. \quad (96)$$

The fastest Structural- N frame rotation belongs to component 2 of GW190924_021846, with $r_m = 7.492$ km and $\bar{\omega}_N = 16330.834$ rad s⁻¹. The slowest belongs to component 1 of the C00:Mixed+X04a reconstruction of GW231123_135430, with $r_m = 197.891$ km and $\bar{\omega}_N = 389.688$ rad s⁻¹. The latter has a much higher exterior spin, but its substantially larger compact length scale dominates the physical angular rate.

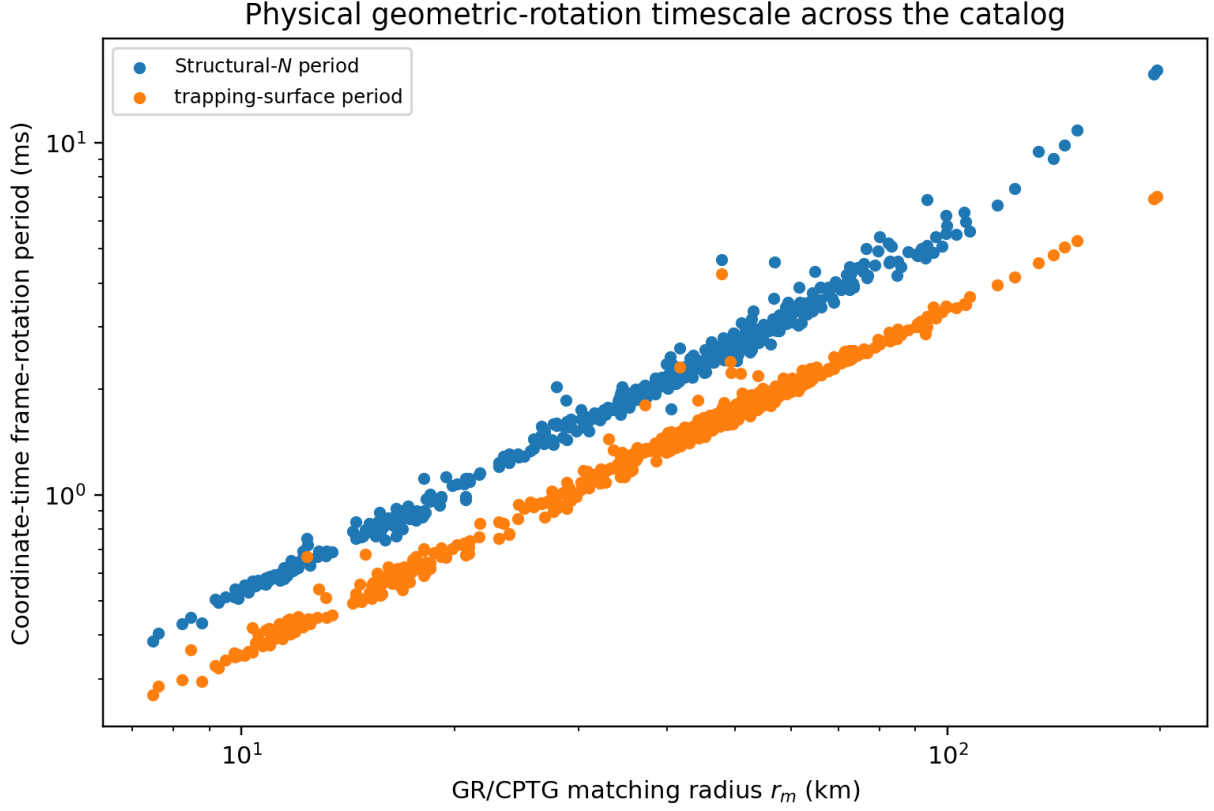


Figure 10: Coordinate-time geometric-rotation periods at the Structural-Mode shell and inner trapping surface versus the GR/CPTG matching radius. Spin controls the dimensionless factors, while the physical period carries the overall r_m scale.

Class	Components	Median r_m (km)	Median $\bar{\omega}_N$ (rad s $^{-1}$)	Median P_N (ms)	Median Ω_- (rad s $^{-1}$)	Median P_- (ms)
M1	164	15.84	7696.92	0.8163	11144.89	0.5638
M2	188	42.99	2731.62	2.3002	4161.37	1.5099
M3	94	57.68	2026.24	3.1009	3217.49	1.9528
M4	60	66.82	1736.24	3.6213	2735.28	2.2971
M5	14	120.84	895.88	7.0346	1548.47	4.0606

Table 786: Descriptive geometric-rotation summaries by observational mass class. The entries are medians across component medians and do not replace the individual event-level posterior results.

11.4 Angular anisotropy disappears at the trapping surface

Away from r_- , the registered rotating geometry gives

$$\omega_{\text{axis}} \leq \bar{\omega} \leq \omega_{\text{equator}}. \quad (97)$$

At the Structural- N shell, the catalog-median ratio $\omega_{\text{equator}}/\omega_{\text{axis}}$ spans 1.006443–2.574044. The largest angular contrast occurs for component 1 of the C00:Mixed+X04a reconstruction of GW231123_135430. At the inner trapping surface, however, $\delta = 0$ removes the angular dependence of $\mathcal{A}_d$, and the three angular readouts collapse exactly to the single value Ω_- .

This collapse is a geometric boundary property. It does not imply rigid material rotation of the compact source.

11.5 The clock vanishes while geometric frame rotation remains finite

The near-surface clock result from Sec. 10.2 gives

$$\bar{\alpha} = \mathcal{O}\left(\sqrt{\frac{d}{r_m}}\right). \quad (98)$$

The frame-dragging rate has a different limiting structure. Since $\omega(x, \theta; q)$ is regular at the simple inner root,

$$\bar{\omega}(d) = \Omega_- + \mathcal{O}\left(\frac{d}{r_m}\right). \quad (99)$$

Combining the two expansions gives

$$\bar{\omega} - \Omega_- = \mathcal{O}(\bar{\alpha}^2). \quad (100)$$

Thus the local clock collapses to zero while the geometric frame-dragging rate approaches a finite surface value. Figure 11 shows this behavior directly for all seven finite offsets and all 520 component medians.

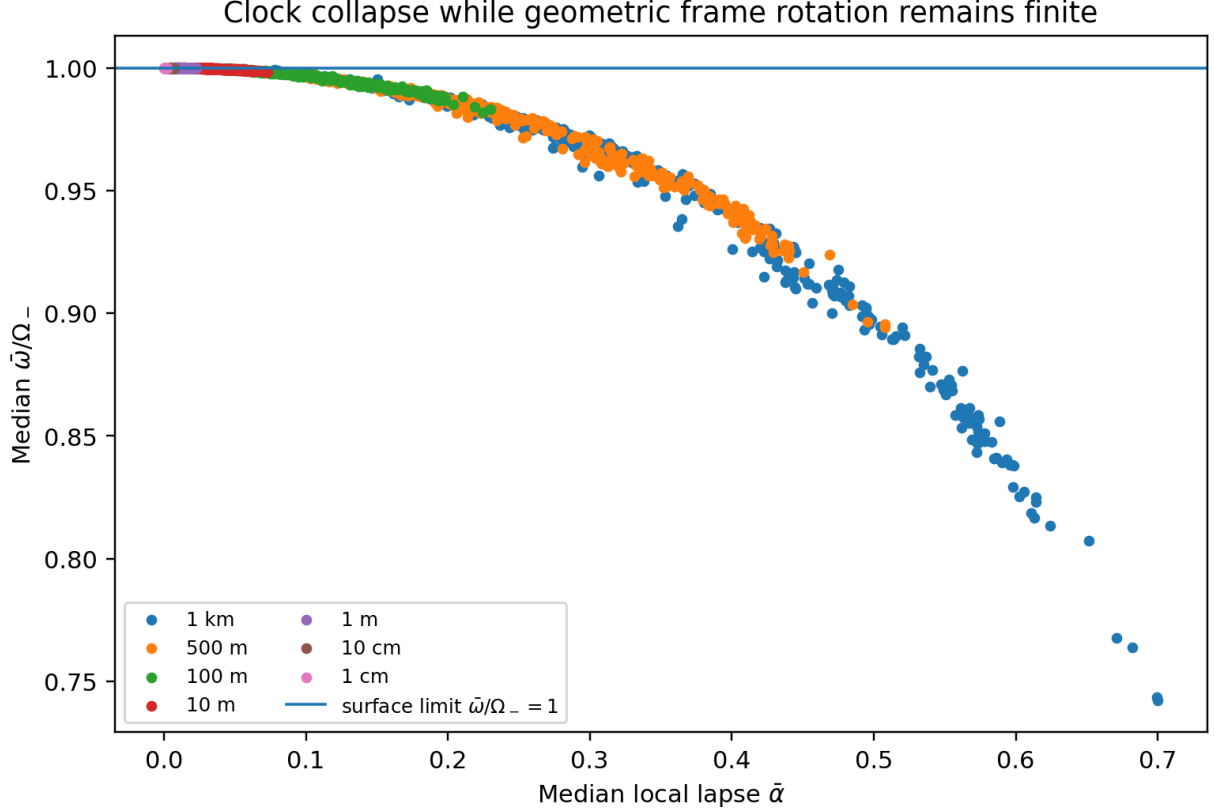


Figure 11: Relation between the local lapse and the determinant-averaged geometric frame-dragging rate normalized by its trapping-surface limit. Each point is a component median at one registered inward coordinate offset. The clock approaches zero while $\bar{\omega}/\Omega_-$ approaches unity.

At 1 km inward, the median across components is

$$\text{median}\left(\frac{\bar{\omega}}{\Omega_-}\right) = 0.97031862, \quad (101)$$

while at 100 m, 1 m, and 1 cm it is 0.99735416, 0.99997421, and 0.99999974, respectively. By 1 cm, the 5th–95th percentile range across the component medians is only 0.99999888–0.99999999.

A log-log fit of $|\bar{\omega}/\Omega_- - 1|$ against $\bar{\alpha}$ over the 1 cm, 10 cm, and 1 m medians gives a catalog-median exponent of 2.000041, consistent with the quadratic relation in Eq. (100). The 5th–95th percentile span, 1.757509–2.284134, is broader because some profiles pass very close to a shallow local turnover, suppressing the leading coefficient.

At the component-median level, 25 of 520 profiles show such a shallow nonmonotonic approach to the surface. The largest overshoot above Ω_- is only 2.40900037e-05 in relative terms. No radial-monotonicity condition is therefore imposed: the required closure is finite rotation, angular collapse at the surface, and convergence to Eq. (86).

The ratio of the two determinant-averaged reporting operators consequently has the inward-limit behavior

$$\frac{\bar{\omega}}{\bar{\alpha}} \sim \frac{\Omega_-}{\mathcal{C}(q)\sqrt{d/r_m}}. \quad (102)$$

This ratio diverges as the trapping surface is approached even though Ω_- itself remains finite. It is a comparison of separately averaged geometric readouts, not an angular average of a local $d\phi/d\tau$. At any fixed polar angle the unaveraged ratio ω/α has the same square-root divergence. Equation (102) is an inward limiting statement only; it does not assign a stationary timelike ZAMO to the trapping surface and does not describe material rotation.

11.6 Branch sensitivity remains attached to the same compact state

For GW231123_135430 component 1, the C00:Mixed reconstruction gives $\bar{\omega}_N = 400.685 \text{ rad s}^{-1}$ and $\Omega_- = 907.369 \text{ rad s}^{-1}$, while C00:Mixed+X04a gives 389.688 and 893.801 rad s^{-1} . Component 2 changes from 578.265 to 638.213 rad s^{-1} at the Structural- N shell. The corresponding clock, mass, Structural-Mode, and radius values remain attached to the same branch throughout; no cross-branch rotational average is formed.

11.7 Interpretation of the rotation field

The combined clock and rotation synthesis shows why a single phrase such as “faster-spinning black hole” is insufficient inside the finite CPTG compact state. Exterior spin fixes the dimensionless geometry, but the normalized frame-dragging functions themselves peak at intermediate spin. Physical angular rates then acquire an additional inverse dependence on compact size. The clock follows a third response: at the Structural-Mode shell its lapse increases monotonically with spin, whereas near the trapping surface it collapses as the square root of coordinate distance.

The result is a finite and internally structured rotating geometry. The organization shell has its own clock and angular-frame rate; the trapping surface has zero lapse but a finite, angle-independent frame-dragging limit; and the GR/CPTG matching radius sets the overall physical time and length scale. None of these readouts requires a material rotation law.

The next section returns to the population-level push-forward, where the catalog-scale consequences of the same compact mass and coordinate transformation are evaluated statistically.

12 The CPTG–GR Boundary and the Finite-to-Kerr Matching Structure

The preceding synthesis sections resolve the finite compact state into mass accounting, Structural Mode, local clock behavior, and geometric frame dragging. Those interior quantities acquire a common physical interpretation only when the outer handoff is stated clearly. The CPTG compact state used here occupies $0 \leq r \leq r_m$. At r_m , the registered finite realization joins the retained Kerr exterior carrying the same exterior gravitational charge and angular momentum. The organization shell r_N and the inner trapping surface r_- are both interior structures and are not alternative names for the exterior matching boundary.

CPTG compact-object anatomy (schematic, not to scale)

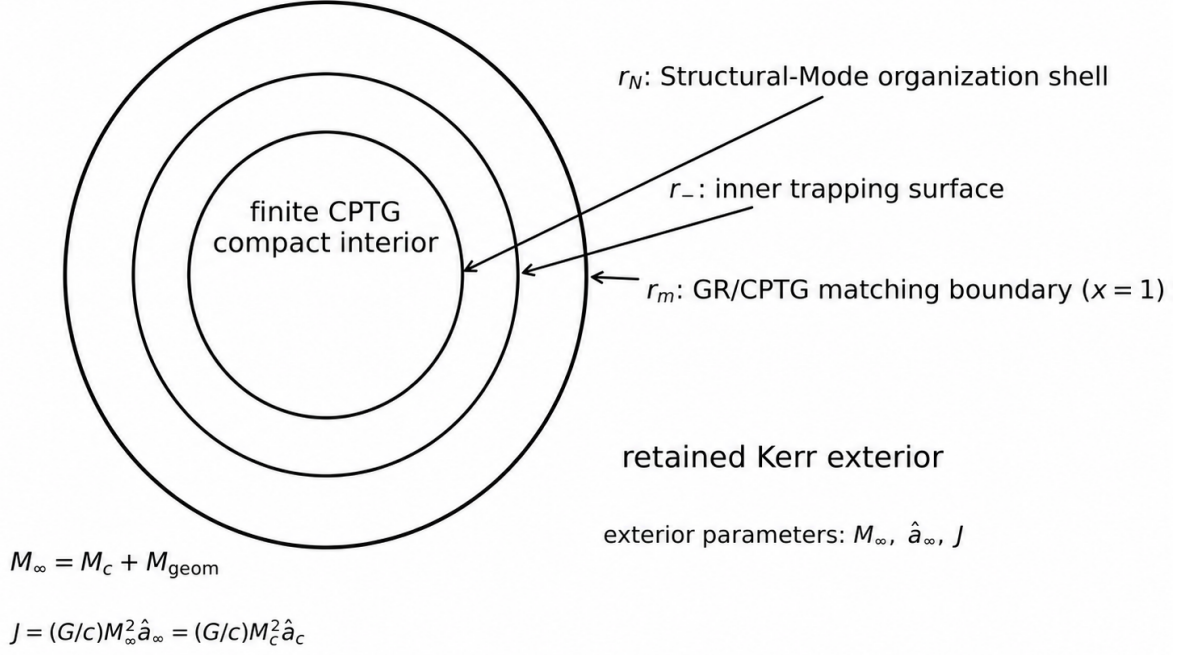


Figure 12: Schematic anatomy of the registered rotating CPTG compact state. The Structural-Mode organization shell r_N lies inside the inner trapping surface r_- , and both lie inside the GR/CPTG matching radius r_m . The finite interior resolves the exterior gravitational charge into physical compact-source and geometric mass-equivalent contributions, while the retained Kerr exterior continues to carry the exterior parameters M_∞ , $\hat{a}_\infty$, and total angular momentum J . The drawing is schematic and not to scale.

12.1 Three radii with three different roles

The radial hierarchy is

$$r_N < r_- < r_m \quad \Longleftrightarrow \quad x_N < x_- < 1. \quad (103)$$

Each radius answers a different question. r_N locates the post-solution Structural-Mode organization shell on which the principal clock and frame-dragging readouts are evaluated. r_- is the inner trapping surface of the declared rotating finite-core realization, where the ZAMO lapse tends to zero and the frame-dragging field tends to the finite angle-independent limit Ω_- . r_m is the outer compact boundary: the point at which the finite CPTG state hands off to the retained Kerr exterior.

The 520 reported component medians occupy

$$0.520906 \leq x_N \leq 0.529182, \quad 0.576235 \leq x_- \leq 0.855322. \quad (104)$$

Their median physical matching radii span 7.492–197.891 km. The corresponding organization-shell radii span 3.949–103.162 km and the inner trapping radii span 4.793–167.324 km. These physical ranges are broad because the dimensionless internal architecture is rescaled by $r_m = GM_\infty/c^2$.

The complete sample-wise audit over all 7,626,239 retained posterior draws independently verifies the interior part of Eq. (103). The minimum recovered separation is

$$\min(x_- - x_N) = 0.049408417286901, \quad (105)$$

with maximum inner-root and Structural-Mode-shell closure residuals of 5.55e-15 and 3.33e-16, respectively. The outer inequality $x_- < 1$ follows from the physical nonextremal root of the registered trapping polynomial for $0 \leq q < 1$; at the extremal endpoint $q = 1$, the root approaches the matching boundary.

12.2 The finite radial profile closes onto the Kerr radial block

The outer edge of the normalized finite interior is $x = 1$. The registered profile is

$$F(x) = \frac{1}{8} (63x^5 - 90x^7 + 35x^9). \quad (106)$$

It satisfies

$$F(1) = 1, \quad F'(1) = 0, \quad F''(1) = 0. \quad (107)$$

The interior radial Kerr-like factor is

$$\delta_{\text{int}}(x, q) = x^2 - 2xF(x) + q^2, \quad (108)$$

while the retained exterior Kerr factor in the same normalized coordinate is

$$\Delta_K(x, q) = x^2 - 2x + q^2. \quad (109)$$

Equation (107) therefore gives

$$\delta_{\text{int}}(1, q) = \Delta_K(1, q) = q^2 - 1, \quad (110)$$

and also

$$\left. \frac{\partial \delta_{\text{int}}}{\partial x} \right|_1 = \left. \frac{\partial \Delta_K}{\partial x} \right|_1 = 0, \quad \left. \frac{\partial^2 \delta_{\text{int}}}{\partial x^2} \right|_1 = \left. \frac{\partial^2 \Delta_K}{\partial x^2} \right|_1 = 2. \quad (111)$$

The other building blocks used by the clock and frame-dragging operators share the same boundary values because

$$\sigma = x^2 + q^2 \cos^2 \theta \quad (112)$$

is common to the interior and exterior construction and

$$\mathcal{A}_d = (x^2 + q^2)^2 - q^2 \delta \sin^2 \theta \quad (113)$$

inherits the closure of δ . Thus the radial functions actually used in the finite-core clock and rotation calculation reach the Kerr boundary with matching value and the declared smooth radial behavior. The third radial derivative of the interior interpolation need not equal the exterior one; the matching demonstrated here is the registered finite-to-Kerr interface, not the introduction of an additional exterior layer.

12.3 Exterior charge and angular momentum are preserved through the handoff

The exterior observer is governed by the conserved exterior state

$$(M_\infty, \hat{a}_\infty, J), \quad (114)$$

with

$$J = \frac{G}{c} M_\infty^2 \hat{a}_\infty = \frac{G}{c} M_c^2 \hat{a}_c. \quad (115)$$

The compact handoff resolves the exterior gravitational charge according to

$$M_\infty = M_c + M_{\text{geom}}, \quad (116)$$

without replacing M_∞ in the retained exterior solution. Likewise, the compact-source spin $\hat{a}_c$ is fixed by angular-momentum conservation rather than substituted for the exterior Kerr parameter:

$$\hat{a}_c = A_{\text{pol}}^2 \hat{a}_\infty. \quad (117)$$

The matching radius itself closes the same relation,

$$r_m = \frac{GM_\infty}{c^2} = \frac{GM_c}{c^2} \left(1 + \sqrt{1 - \hat{a}_c^2}\right). \quad (118)$$

The finite interior can therefore contain less physical compact-source mass than the exterior gravitational mass parameter while presenting the same registered exterior gravitational charge and total angular momentum at the handoff.

Location / quantity	Role in the compact construction	What is carried forward
r_N	Interior Structural-Mode organization shell	N_{BH} , local lapse, and geometric frame-dragging readouts
r_-	Interior trapping surface	$\bar{\alpha} \rightarrow 0$ and finite angle-independent Ω_-
r_m	Finite CPTG / Kerr handoff boundary	Exterior charge M_∞ , exterior spin $\hat{a}_\infty$, and total J
M_c	Physical finite compact-source mass	Remains an interior compact-state quantity
M_{geom}	Completed geometric mass-equivalent response	Completes $M_\infty = M_c + M_{\text{geom}}$ inside the compact accounting
Kerr exterior	Retained exterior relativistic description	Governed by the exterior state rather than by an independent refit of the interior readouts

Table 787: Separation of the three characteristic radii and the quantities attached to the finite-to-Kerr handoff.

12.4 Catalog-wide boundary closure

The catalog does not require candidate-specific boundary tuning. Once M_∞ and $\hat{a}_\infty$ are reconstructed, r_m follows from the same compact closure for every component, while x_N and x_- follow from the same dimensionless structural functions. The smallest median $x_- - x_N$ separation in the 520-component registry is 0.052379, and the largest is 0.334416. The smallest median outer separation $1 - x_-$ is 0.144678. It occurs for component 1 of GW231123_135430, the high-spin end of the retained sample, where the trapping surface moves closest to the matching boundary without being identified with it.

The full architecture can therefore be summarized as

$$\underbrace{r_N}_{\text{organization}} < \underbrace{r_-}_{\text{trapping}} < \underbrace{r_m}_{\text{finite-to-Kerr handoff}} < \underbrace{r}_{\text{retained Kerr exterior}}. \quad (119)$$

This hierarchy is important for interpreting the preceding clock and rotation results. The zero-lapse limit occurs at r_- , not at the GR/CPTG handoff. The Structural-Mode clock is read still farther inward at r_N . The exterior gravitational-wave description is attached at r_m , outside both of those interior structures.

12.5 Scope of the matching statement

The boundary result established here is the operational closure of the registered minimum finite compact realization used throughout this paper. The exterior Kerr parameters are preserved, the finite radial building blocks close onto their Kerr counterparts at $x = 1$, angular momentum is conserved, and the full posterior population preserves the interior ordering required by the declared geometry.

This is the matching statement needed by the present observational calculation. A derivation of a unique rotating interior directly from a more fundamental parent action, or a separate junction-condition analysis formulated independently of the registered compact construction, would be a further theoretical step rather than an additional parameter in the current solve.

With the finite interior and the retained exterior now joined explicitly, the next section returns to the selection-corrected population posterior and asks how the same compact transformation reorganizes the catalog statistically.

13 Population-Level Results

The candidate sections and interior syntheses characterize the retained gravitational-wave events individually. The population calculation is deliberately separate. It does not average the event-level Structural-Mode, clock, rotation, or boundary results and it does not feed any of those quantities back into the statistical population model. Instead, it takes the official GWTC-5.0 selection-corrected `default_mmax` hyperposterior as a separate observational authority and pushes that posterior through the same native-coordinate and compact-state maps defined in the methodology.

The active population calculation in this paper is the final frozen native-coordinate correction. All 8200 official hyperposterior rows are retained and each row is propagated with 512 joint realizations, giving

$$8200 \times 512 = 4,198,400 \quad (120)$$

population realizations. No event-side posterior is changed by this calculation.

13.1 Native-coordinate contract

The released population redshift is not treated as a CPTG-native observable. It is first decoded through the reporting convention carried by the GWTC product, using Eqs. (12) and (13), to recover the luminosity-distance coordinate that the released redshift represents. That external convention,

$$H_0 = 67.9 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_m = 0.3065, \quad (121)$$

is therefore used only as a decoder.

The decoded luminosity distance is then held fixed and Eq. (10) is inverted for the CPTG native phase z_π . The exact population mass-coordinate scale is

$$s_M \equiv \frac{1 + z_{\text{release}}}{1 + z_\pi}, \quad M_\infty = s_M M_{\text{source,release}}. \quad (122)$$

This is equivalent to reconstructing the detector-frame mass represented by the released source-coordinate pair and then dividing by $1 + z_\pi$. The released source-coordinate mass is thus an externally encoded comparison quantity, not a native CPTG mass input.

Across the 8200 hyperposterior-row medians, the released redshift-coordinate median is 1.400153 and the recovered CPTG native phase is 1.528041. The mass-coordinate scale has median 0.949412. Figure 13 shows the row-by-row phase reconstruction.

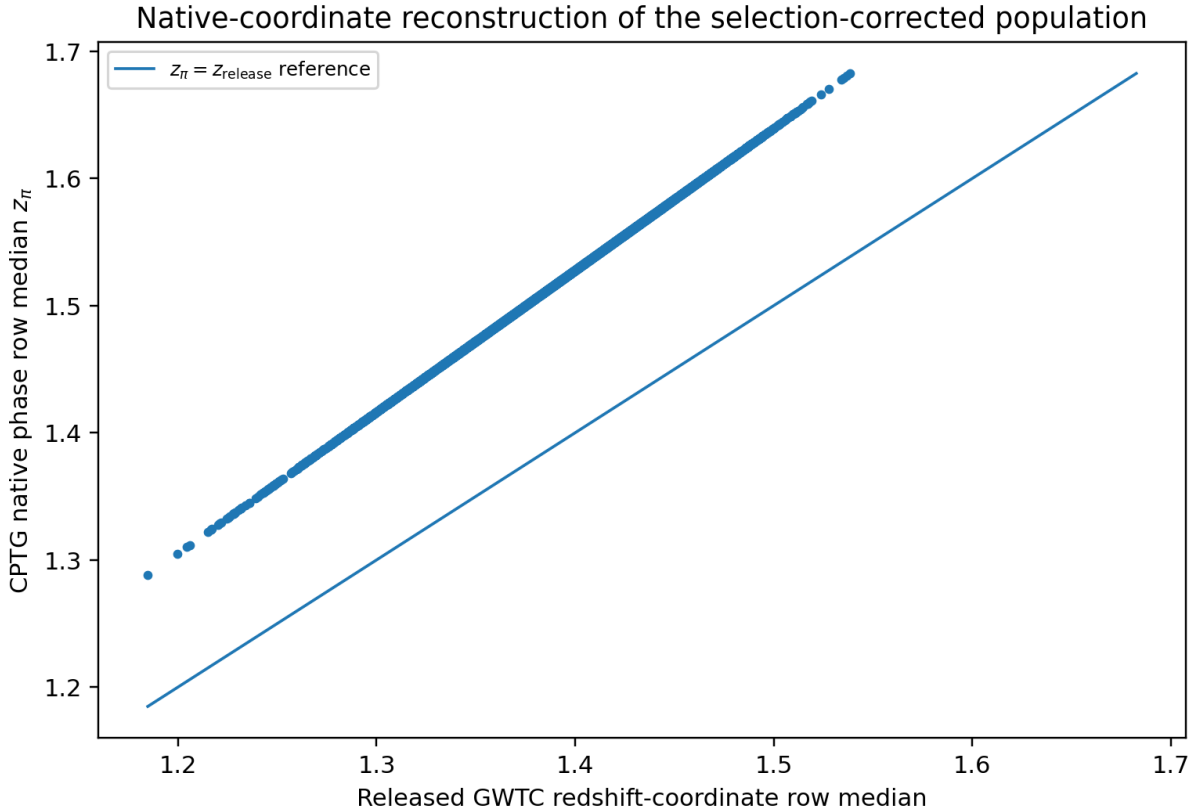


Figure 13: Released GWTC redshift-coordinate row medians versus the CPTG native phase recovered after decoding to luminosity distance and inverting the registered affine luminosity observation map. The diagonal is an identity reference only. The external GWTC reporting convention is used to decode the released coordinate and does not enter the CPTG compact solve as a native parameter.

Per-row median coordinate readout	5th percentile	Median	95th percentile
Released GWTC redshift coordinate	1.310551	1.400153	1.468137
Decoded GWTC luminosity distance (Mpc)	9475.892	10281.484	10900.332
CPTG native phase z_π	1.428172	1.528041	1.603969
CPTG native-branch $D_{L,\pi}$ (Mpc)	9988.467	10837.635	11489.959
Mass-coordinate factor	0.947837	0.949412	0.951560
Equivalent-standard reporting coordinate ($\text{km s}^{-1} \text{Mpc}^{-1}$)	71.6204	71.7414	71.8298

Table 788: Coordinate audit across the 8200 hyperposterior-row medians. The decoded GWTC luminosity distance is the observation-coordinate distance held fixed during inversion. The CPTG native-branch distance $D_{L,\pi}$ is a downstream readout of the solved native phase under the native normalization and is therefore not required to equal that held observation-coordinate distance.

The native GWTC-style comparison coordinate is evaluated only after z_π has been solved:

$$H_{0,\text{GWTC}}^{\text{eq},\pi} = 71.7414_{-0.1209}^{+0.0884} \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (123)$$

The interval in Eq. (123) is the 5th–95th percentile range across the row-median reporting coordinates. It is not a fitted CPTG H_0 , is not used to reconstruct z_π , and never feeds back into the masses, spins, or compact-state handoff. The frozen audit explicitly confirms this double-projection guard.

13.2 Population mass transformation

After the native coordinate has been reconstructed, each joint population draw is passed through the same compact mass–spin closure used for the individual events. Table 789 reports percentiles across the 8200 hyperposterior-row medians.

Per-hyperposterior-row median quantity	5th percentile	Median	95th percentile
Released component-1 mass coordinate	9.875	10.243	10.612
CPTG exterior component-1 M_∞	9.239	9.623	10.003
CPTG physical component-1 M_c	6.146	6.507	6.882
Released component-2 mass coordinate	8.007	8.507	8.947
CPTG exterior component-2 M_∞	7.865	8.407	8.879
CPTG physical component-2 M_c	4.560	4.837	5.087
Component-1 geometric share (%)	31.27	34.15	36.69
Component-2 geometric share (%)	42.54	44.04	45.19
Post-handoff reorder fraction (%)	17.19	22.46	27.54

Table 789: Native-coordinate population push-forward. Masses are in $M_\odot$. Released component masses retain the external release ordering. CPTG components are assigned only after the compact handoff by physical compact-source mass, so the three mass rows for a component are population summary coordinates rather than a claim that one release label remains attached to the same component on every draw.

At the median of the row medians, component 1 follows the summary-coordinate sequence

$$10.243 M_\odot \longrightarrow 9.623 M_\odot \longrightarrow 6.507 M_\odot, \quad (124)$$

from the released mass coordinate, to the CPTG exterior gravitational mass, to the physical compact-source mass. Component 2 gives

$$8.507 M_\odot \longrightarrow 8.407 M_\odot \longrightarrow 4.837 M_\odot. \quad (125)$$

The first shift in each sequence is coordinate reconstruction; the second is the compact mass–spin decomposition. The median geometric shares across the row medians are 34.15% for component 1 and 44.04% for component 2. The median row-level post-handoff reorder fraction is 22.46%, so the population analysis preserves the same rule used on the event side: solve both component states first and move every attached quantity together only after physical compact-source mass is known.

13.3 Distribution-level consequence

The push-forward changes the inferred distribution of localized physical compact-source mass without removing the corresponding high-mass exterior gravitational systems from the input posterior. Figure 14 compares the released primary-mass density with the CPTG physical component-1 density. Each curve is the pointwise median of the 8200 row-level density estimates; the surrounding bands show the pointwise 5th–95th percentile range.

Selection-corrected primary-mass density before and after the CPTG push-forward

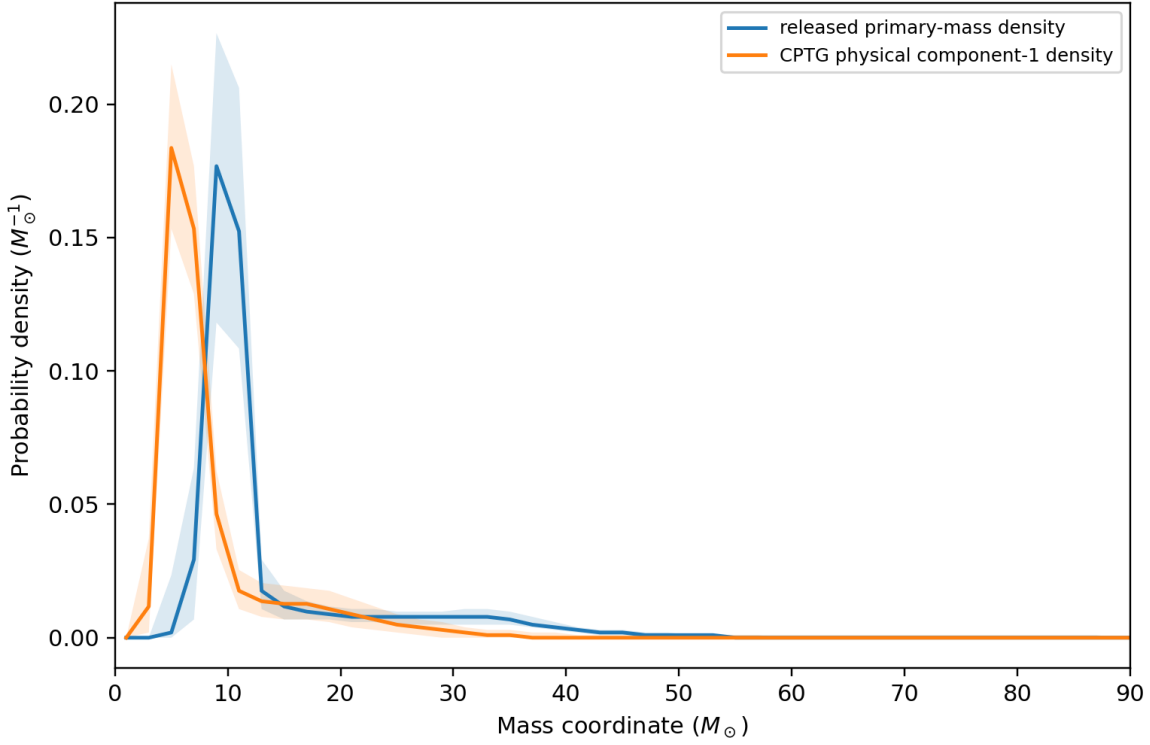


Figure 14: Selection-corrected primary-mass density before and after the native-coordinate CPTG push-forward. The released curve is expressed in the release source-mass coordinate; the CPTG curve uses the post-handoff physical component-1 mass. Bands are pointwise 5th–95th percentile ranges across the 8200 hyperposterior rows.

The pointwise median released density reaches its maximum in the 8–10 $M_{\odot}$ bin, whereas the CPTG physical component-1 density reaches its maximum in the 4–6 $M_{\odot}$ bin. This is a coordinate-plus-compact-state transformation of the same selection-corrected posterior, not a second population fit and not evidence that the exterior high-mass systems have disappeared. The low physical-source tail is interpreted subject to the same observational selection cautions discussed later in Sec. 14.

13.4 Predeclared mass-anchor diagnostics

The 10, 35, 40, and 45 $M_{\odot}$ anchors remain downstream diagnostics only. Table 790 reports the mean row fraction above each mass scale, with the 5th–95th percentile range across the 8200 hyperposterior rows.

Threshold	Released m_1	CPTG comp. 1 M_c	Released m_2	CPTG comp. 2 M_c
10 $M_{\odot}$	57.529% [46.680, 69.150]	20.391% [16.016, 25.195]	27.070% [21.484, 33.008]	11.689% [8.594, 15.234]
35 $M_{\odot}$	5.184% [3.711, 7.031]	0.631% [0.195, 1.172]	1.504% [0.781, 2.344]	0.117% [0.000, 0.391]
40 $M_{\odot}$	2.609% [1.758, 3.516]	0.332% [0.000, 0.586]	0.751% [0.195, 1.367]	0.057% [0.000, 0.195]
45 $M_{\odot}$	1.397% [0.977, 1.953]	0.189% [0.000, 0.391]	0.416% [0.000, 0.781]	0.027% [0.000, 0.195]

Table 790: Mean row fraction above each predeclared diagnostic mass scale, with 5th–95th percentile ranges across the 8200 hyperposterior rows. Each row contains 512 joint realizations, so one sampled realization corresponds to 0.1953125%. Zero entries in the rare tails can therefore be finite-resolution outcomes rather than exact zero-probability statements.

For the physical compact-source primary, the mean row fraction above 35 $M_{\odot}$ is 0.631%, compared with 5.184% in the released primary-mass coordinate. At 45 $M_{\odot}$ the corresponding values are 0.189% and 1.397%. These comparisons quantify how much of the high exterior gravitational charge is reassigned to the completed geometric contribution under the compact closure; they do not alter the existence of the high-mass exterior systems in the posterior-inferred catalog.

13.5 Numerical closure and authority

The final population product is frozen to the official selection-corrected population file with SHA-256

$$\text{aa159c4259ccdbb9140d4ab65382095f4744ef42d7531dc727f19e75d19ef048.} \quad (126)$$

The deterministic primary-mass stratification closure is

$$9.7655039816 \times 10^{-4}. \quad (127)$$

The frozen validation also regresses the physical population result against the decoder-based cross-check calculation. Across the shared row-level physical population quantities, the maximum absolute difference is

$$1.1212 \times 10^{-9}, \quad (128)$$

and the mass-anchor posterior tables agree exactly at the stored precision. The event-side calculation is unchanged, and the equivalent-standard reporting coordinate remains downstream only.

The population result therefore closes the candidate-first analysis at the statistical level. The same native coordinate reconstruction and compact mass-spin handoff that were applied draw by draw to the event catalog also propagate through the separate selection-corrected population posterior, while the event-specific Structural-Mode, clock, rotation, and boundary readouts remain individual compact-state quantities rather than ingredients of the population fit.

14 Physical Interpretation

The preceding sections resolve the same gravitational-wave systems at several levels: exterior gravitational state, physical compact-source mass, internal organization, local clock behavior, geometric frame dragging, the finite-to-Kerr boundary, and the selection-corrected population posterior. These are not independent fits. They are successive readouts of one registered compact-state construction. The physical interpretation therefore depends on keeping the quantities at those levels distinct.

14.1 Exterior gravitational mass and physical compact-source mass are not the same quantity

The central compact-state distinction is

$$M_\infty = M_c + M_{\text{geom}}. \quad (129)$$

M_∞ is the gravitational mass parameter carried by the retained exterior Kerr state. M_c is the localized physical compact-source mass of the finite interior. M_{geom} is the mass-equivalent completed geometric response required by the same compact closure. The three terms are therefore not independently adjustable reservoirs.

The compact relation fixes

$$\frac{1}{2}M_\infty \leq M_c \leq M_\infty, \quad (130)$$

with the fractional partition controlled by the solved exterior spin magnitude through A_{pol} . The full catalog makes the consequence visible: an object can remain a high- M_∞ exterior system while carrying substantially less localized compact-source mass. This is why the M5 systems are not removed or reclassified by the CPTG solve. Their exterior gravitational charge remains large; the compact handoff changes how that charge is partitioned inside the finite state.

The same distinction explains why the absolute geometric mass-equivalent and the geometric fraction need not rank systems in the same order. A large M_∞ can produce a large M_{geom} even when the fractional geometric share is modest, whereas a lower-mass low-spin object can have a larger fractional geometric share. The event catalog therefore supports three separate questions: how large the exterior gravitational charge is, what fraction of it is assigned to the geometric contribution, and what absolute mass-equivalent that fraction represents.

14.2 Structural Mode describes organization, not gravity strength

Structural Mode is evaluated only after the compact mass-spin state has been solved. In the registered rotating compact realization,

$$(N_{\text{BH}}, x_N, x_-) = \mathcal{S}(|\hat{a}_\infty|), \quad (131)$$

while the physical radii are obtained by restoring the exterior mass scale,

$$(r_N, r_-, r_m) = \frac{GM_\infty}{c^2}(x_N, x_-, 1). \quad (132)$$

This separates dimensionless organization from physical size. The 520 retained component fingerprints span $N_{\text{BH}} = 1.473148\text{--}1.898389$, but no catalog-average Structural Mode is assigned. Each value remains attached to its own solved component and posterior reconstruction.

The organization shell is also geometrically distinct from the trapping surface. The complete posterior audit gives

$$x_N < x_- < 1, \quad (133)$$

with a minimum recovered sample-wise separation $x_- - x_N = 0.049408417286901$. The Structural-Mode readout is therefore located on an interior organization shell rather than on a trapping boundary or on the GR/CPTG handoff itself.

14.3 The interior clock has both a scale-free and a scale-dependent reading

The Structural- N clock is a dimensionless-geometry readout. At r_N ,

$$T_{24,N} = 24 \bar{\alpha}_N \text{ h}, \quad (134)$$

and the retained component medians span 11.852622–19.985616 hours per 24 source-local exterior Kerr hours. Because both x_N and $\bar{\alpha}_N$ are fixed by the solved dimensionless spin state, this clock does not depend directly on the overall compact mass scale.

The fixed physical offset readouts behave differently. Near the inner trapping surface,

$$T_{24}(d) \simeq 24 \mathcal{C}(q) \sqrt{\frac{d}{r_m}} \text{ h}. \quad (135)$$

The catalog independently recovers the expected square-root exponent: fitting the 1 cm, 10 cm, and 1 m medians gives $\beta = 0.49999464\text{--}0.50001636$, with median 0.50000089 for $T_{24} \propto d^\beta$. A fixed 1 km offset is therefore not the same normalized location in a 7.5 km compact state and a 198 km compact state. The large-scale M5 objects make this distinction particularly visible: they can retain comparatively large proper time at the Structural-Mode shell while accumulating very little proper time at a fixed physical distance from r_- .

This behavior should be read locally. The paper does not replace the interior lapse field with a whole-core average, nor does it use M_c/M_∞ or the geometric mass fraction as a surrogate clock. The time coordinate throughout is the source-local asymptotic exterior Kerr coordinate used by the same solved compact state.

14.4 Geometric rotation is a frame property, not a matter-rotation law

The internal rotation result concerns the angular velocity of the local zero-angular-momentum frame. In normalized form,

$$\bar{\omega}_N = \frac{c}{r_m} \hat{\omega}_N(q), \quad \Omega_- = \frac{c}{r_m} \hat{\Omega}_-(q). \quad (136)$$

Spin fixes the dimensionless rotational geometry, while the matching radius sets the physical angular-rate scale. This is why normalized frame dragging peaks at intermediate spin even though the physical catalog rates also depend strongly on compact size. Across the retained component medians, $\bar{\omega}_N$ spans 389.688–16330.834 rad s^{−1}, with coordinate-time periods of 0.384744–16.123627 ms.

The clock and rotation fields also separate sharply at the inner trapping surface. The lapse vanishes as $\sqrt{d/r_m}$, whereas the frame-dragging field approaches the finite angle-independent value Ω_- . The catalog reproduces

$$\bar{\omega} - \Omega_- = \mathcal{O}(\bar{\alpha}^2), \quad (137)$$

with a median fitted exponent 2.000041. The finite Ω_- is a property of the rotating geometry. It does not determine a fluid rotation law, rigid-body spin rate, or local material tangential velocity.

14.5 The finite interior and the retained Kerr exterior describe one compact state

The three characteristic radii now have a single consistent interpretation:

$$\underbrace{r_N}_{\text{organization}} < \underbrace{r_-}_{\text{trapping}} < \underbrace{r_m}_{\text{finite-to-Kerr handoff}}. \quad (138)$$

The zero-lapse limit is at r_- ; it is not the exterior matching surface. At r_m , the registered finite radial profile reaches the retained Kerr radial block with matching value, first radial derivative, and second radial derivative. At the same handoff, the exterior state continues to carry M_∞ , $\hat{a}_\infty$, and

$$J = \frac{G}{c} M_\infty^2 \hat{a}_\infty = \frac{G}{c} M_c^2 \hat{a}_c. \quad (139)$$

The finite-state decomposition therefore does not require the distant exterior to be reassigned the smaller localized mass M_c . The exterior remains the Kerr state associated with the conserved exterior gravitational charge and angular momentum; the CPTG compact sector specifies how that state is continued inward in the registered finite realization.

14.6 The population result is a transformation of an existing posterior, not a second event catalog

The selection-corrected population calculation extends the same distinction to the statistical level. The released redshift coordinate is first decoded to the luminosity distance represented by the GWTC reporting convention, that observation-coordinate distance is held fixed, and the CPTG native phase is reconstructed. Only then is the source-mass coordinate transformed and the compact handoff applied.

At the median of the 8200 hyperposterior-row medians, component 1 follows

$$10.243 \rightarrow 9.623 \rightarrow 6.507 M_\odot, \quad (140)$$

for released source coordinate, CPTG exterior gravitational mass, and physical compact-source mass, respectively. The corresponding sequence for component 2 is

$$8.507 \rightarrow 8.407 \rightarrow 4.837 M_\odot. \quad (141)$$

The selection-corrected physical component-1 density consequently shifts toward lower localized compact-source mass while the input posterior continues to contain the same high-mass exterior systems. The low-mass end should be read with the published GWTC-5.0 population caution: the population analysis advises additional care below approximately $8 M_\odot$ because of the event-classification criterion used to define the BBH sample [8]. The present push-forward inherits that selection context rather than removing it.

The equivalent-standard coordinate reported after the native solve is likewise only a downstream comparison quantity. It is not the normalization used to obtain z_π and does not feed back into the compact masses. That ordering is essential: the population result is a native-coordinate push-forward of the released posterior rather than a re-expression of CPTG in the external reporting coordinates it is intended to decode.

15 Discussion

The full calculation now contains two complementary kinds of evidence. The event side carries 7,626,239 retained posterior draws through the native luminosity-coordinate reconstruction and compact closure, preserving 260 distinct within-event numerical streams for 259 astrophysical events. The parallel population side propagates all 8200 rows of the official selection-corrected hyperposterior with 512 joint realizations per row. Agreement of these two routes is important because they test different parts of the inference chain: one resolves individual compact states, while the other tests how the same transformation acts on the released population posterior.

15.1 What is fixed by the present calculation

Within the registered compact realization used in this paper, several relationships are no longer object-by-object choices. The mass fraction is fixed by the compact mass–spin closure; Structural Mode and the normalized shell geometry are fixed

by the solved spin state; the physical radii are obtained by restoring the exterior mass scale; the near-trapping clock obeys the square-root boundary law; the geometric rotation field approaches a finite surface rate; and the outer finite profile closes onto the retained Kerr radial block at $x = 1$.

This is why the 259-event sample is more informative than a small collection of demonstrations. No event receives a separate compact coefficient, Structural-Mode rule, clock exponent, rotation prescription, or matching radius normalization. Object-to-object variation enters through the posterior-inferred observational state and the registered maps applied to it.

The observational mass classes M1–M5 should be understood in the same limited way. They organize a very large catalog for presentation using pre-CPTG detector-frame mass information. They are not branches of the gravitational law, and none of the later CPTG quantities is allowed to move an event from one class to another.

15.2 Dependence on the released gravitational-wave inference

The present study begins from released gravitational-wave parameter-estimation and population products rather than refitting detector strain with a fully native CPTG waveform likelihood. That choice preserves the observational information supplied by the public release, but it also means that some reported release coordinates carry the assumptions under which the external analysis constructed them [7, 9]. The procedure used here is designed to avoid promoting those derived coordinates to native CPTG inputs: event luminosity distance and detector-frame mass are used before source-frame reconstruction, and the population redshift coordinate is decoded back to luminosity distance before the native phase is solved.

This distinction matters most for quantities such as source-frame mass and redshift. The paper does not assume that an externally reported source coordinate is automatically a CPTG source coordinate. Instead, it preserves the closest available observation-coordinate information and rebuilds the native phase before applying the compact law. The regression of the frozen population result against the decoder-based cross-check calculation provides a numerical check on that translation, but the broader observational uncertainties of the public inference products remain part of the input evidence.

15.3 Branch structure and posterior uncertainty are retained rather than compressed away

The release contains multiple retained labels for some events. Twenty-eight paired labels are numerical duplicates under the declared transformation and therefore collapse only for descriptive catalog counting. GW231123_135430 is different: its two retained reconstructions remain numerically distinct and are carried separately through mass decomposition, Structural Mode, clock, rotation, and boundary summaries.

That treatment illustrates the uncertainty policy of the analysis. A branch difference is not turned into an average compact object, and Structural Mode fingerprints from independent reconstructions are never pooled. Similarly, the candidate subsections report posterior intervals rather than reducing each event to a single best-fitting compact state. The catalog-wide synthesis uses medians and percentile summaries for description, while the numerical closure audits continue to operate on the full retained posterior draws.

15.4 The finite-core realization is registered, but further theoretical derivation remains possible

The rotating interior used here is the registered minimum finite compact realization carried consistently through the observational test. Its radial profile, compact handoff, Structural Mode, clock operator, frame-dragging operator, and finite-to-Kerr boundary are therefore part of the tested construction. The present results show that this construction is mutually compatible across the declared event and population evidence without introducing event-specific corrections.

A stronger theoretical result would derive the same rotating finite interior uniquely from the parent CPTG action and independently recover the same boundary structure. That would deepen the origin of the registered realization rather than change the observational bookkeeping performed here. Likewise, determining a material angular-velocity field would require source dynamics beyond the zero-angular-momentum-frame geometry evaluated in this paper. These are natural extensions because the current calculation has already isolated which quantities belong to geometry and which would require additional matter dynamics.

15.5 Statistical resolution and population tails

The population push-forward uses 512 joint realizations per hyperposterior row. A single sampled realization therefore represents 0.1953125% of one row. Rare high-mass threshold entries can consequently reach zero at this sampling resolution without implying an exact zero underlying probability. For that reason the 10, 35, 40, and 45 $M_{\odot}$ anchors are retained as diagnostics with row-to-row percentile ranges rather than interpreted as hard population cutoffs.

The same caution applies at the low-mass end for a different reason. The transformed physical-source density extends into a region where the published GWTC population study already advises care because of the BBH event-selection boundary [8]. The shift of the CPTG physical-source distribution is therefore a result of the registered coordinate and compact transformations, while the detailed astrophysical interpretation of the lowest-mass tail remains tied to the selection properties of the underlying catalog.

15.6 Next observational and theoretical tests

The most direct observational extension is a fully native inference in which the CPTG luminosity coordinate and compact-state map are incorporated before posterior construction rather than applied to released posterior products. That would test the coordinate translation at the likelihood level and remove the need to decode externally reported source coordinates after the fact. Repeating the present fail-closed workflow on later gravitational-wave releases would then test whether the same compact closure continues to hold as the event sample and selection model change.

On the theory side, a parent-action derivation of the rotating finite interior and a corresponding native waveform treatment would connect the registered interior more directly to detector-level observables. Any extension to material rotation should remain separate from the geometric ZAMO frame-dragging field established here. The current result provides the necessary bookkeeping for those tests: exterior charge, physical source mass, Structural Mode, local clock, geometric rotation, and the finite-to-Kerr boundary are already distinguished and attached to the same posterior-resolved object state.

The conclusion therefore does not rest on a single catalog average. It rests on the repeated closure of one compact construction across individual event posteriors, the full retained finite-core draw set, and a separate selection-corrected population posterior.

16 Conclusion

This paper asked whether one registered CPTG compact-state construction could be carried through the complete declared GWTC-5.0 binary-black-hole analysis set without introducing event-specific compact coefficients, changing the structural rule from object to object, or treating externally derived source coordinates as though they were native CPTG observables. Within the stated observational inputs and the registered finite compact realization, the answer is affirmative.

On the event side, all 259 astrophysical events remain in the analysis. The 288 retained release-label rows reduce to 260 unique numerical streams only after exact duplicate reconstructions are identified; the genuinely distinct GW231123_135430 branches remain separate. In total, 7,626,239 posterior draws pass through the same luminosity-coordinate reconstruction, compact mass-spin closure, Structural-Mode construction, finite-core clock, and geometric-rotation operators.

The compact closure consistently separates the exterior gravitational charge from the physical finite source,

$$M_{\infty} = M_c + M_{\text{geom}}, \quad (142)$$

while preserving total angular momentum. The fractional geometric contribution is fixed by exterior spin through the same compact relation for all 520 solved components, rather than by a mass-class-dependent fit. The observational M1–M5 classes therefore organize the catalog but do not define different CPTG branches.

Structural Mode remains a post-solution fingerprint. Across the retained component medians, N_{BH} spans approximately 1.473–1.898, while the normalized organization shell remains confined near one half of the matching radius. The full posterior audit preserves

$$x_N < x_- < 1, \quad (143)$$

with no retained draw in which the organization shell reaches or crosses the inner trapping surface. These radii have distinct meanings throughout the calculation: r_N is the organization shell, r_- is the inner trapping surface, and r_m is the

finite-to-Kerr handoff boundary.

The clock and rotation results expose two complementary properties of the finite interior. At the Structural-Mode shell, the retained median proper-time accumulation spans about 11.85–19.99 hours for every 24 source-local exterior Kerr hours. Approaching the inner trapping surface at fixed Boyer–Lindquist radial-coordinate offset,

$$T_{24}(d) \propto \sqrt{\frac{d}{r_m}}, \quad (144)$$

and all 520 component-median profiles independently recover the expected square-root exponent while tending to zero at the surface. The geometric frame-dragging field behaves differently: it approaches a finite, angle-independent trapping-surface limit even as the lapse vanishes. The reported angular rates are ZAMO frame-dragging rates, not material rotation laws.

The outer boundary closes the finite compact state onto the retained Kerr exterior at $x = 1$. For the registered radial profile,

$$F(1) = 1, \quad F'(1) = 0, \quad F''(1) = 0, \quad (145)$$

so the interior and Kerr radial blocks agree in value and in the first two radial derivatives at the declared handoff. The exterior continues to carry M_∞ , $\hat{a}_\infty$, and the conserved total angular momentum, while M_c , M_{geom} , N_{BH} , the interior clock, and internal frame dragging remain resolved properties of the finite compact state. This is the matching statement established by the registered realization; a unique derivation of that rotating interior from the parent CPTG action would be a further theoretical result.

The parallel population calculation reaches the same compact-state map through a different route. All 8200 rows of the official selection-corrected `default_mmax` hyperposterior are propagated with 512 joint realizations per row, for 4,198,400 population realizations. The final frozen population calculation first decodes the released population redshift coordinate to luminosity distance, holds that observation-coordinate distance fixed, solves the CPTG native phase z_π , and only then reconstructs the exterior and physical compact masses. At the median of the row medians, the component-1 mass coordinate follows

$$10.243 M_\odot \longrightarrow 9.623 M_\odot \longrightarrow 6.507 M_\odot, \quad (146)$$

from the released source coordinate to CPTG exterior gravitational mass to physical compact-source mass. The equivalent-standard H_0 -like quantity is retained only as a downstream reporting coordinate and never enters the native solve.

Taken together, these results establish a catalog-wide closure result for the declared GWTC-5.0 analysis: one registered compact construction simultaneously preserves the exterior relativistic state, resolves a finite physical source and completed geometric contribution, assigns an object-specific Structural-Mode organization, produces a regular interior clock and frame-dragging field, closes onto the retained Kerr exterior, and propagates consistently through the separate selection-corrected population posterior. The result is not a claim that every further strong-field question is already solved. The next decisive tests are a detector-level inference using native CPTG waveform and luminosity coordinates from the likelihood stage onward, and a parent-action derivation of the rotating finite interior that independently recovers the registered boundary structure.

Data, Code, and Evidence Availability

The complete numerical evidence used for the compact-state, finite-core time, geometric-rotation, and population results is collected in the standalone publication archive

`CPTG_GWTC5_COMPACT_OBJECT_POPULATION_PUBLICATION_EVIDENCE_FINAL.zip`

The archive contains the exact ordered 259-event BBH source registry and checksums; retained release-label identities and sample counts; event-level compact-state and Structural-Mode summaries; the complete branch-resolved finite-core time and geometric-rotation tables and audits over 7,626,239 posterior draws; the frozen 8200-row native-coordinate population push-forward and coordinate diagnostic; frozen protocols; implementation source; execution logs; numerical audits; and SHA-256 manifests.

The supporting CPTG papers, source code, and publication resources are maintained in the public CPTG repository at:

<https://github.com/CLG2025/CPTG>.

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